

Supply Chain Management

Text and Cases



V. V. Sople

SUPPLY CHAIN MANAGEMENT

Vinod V. Sople

Director

ITM Business School

Dombivli Campus

PEARSON

Delhi • Chennai • Chandigarh

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*This book is dedicated to my beloved parents,
Vasant Sople and Vimal Sople*

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PREFACE

Most of the companies have identified supply chain management as a route to competitive advantage. Through supply chain (SC), firms aim to achieve two polemic goals: cost-effectiveness and customer satisfaction. In fact, supply chain serves as a road map to enterprise success. Supply chain encompasses all the functional areas of an enterprise. Integration of areas such as procurement, processing and distribution is possible through proper management and control of supply chain operations.

Technology plays a vital role in managing supply chain and making it efficient and effective in all its operations. Information Technology (IT) has helped in seamless integration of supply chain. Decisions in supply chain are mostly information-based and IT contributes to speed in collection, processing and distribution of information at various decision-making points across the supply chain.

Some of the supply chain strategies are based on the physical assets and industry in which the companies operate and dictated by the firm's place in the value chain, namely, upstream, mid-stream or downstream. There are four generic supply chain strategies for gaining competitive advantage: rationalization, synchronization, customization and innovation. In today's context, it is not the product or enterprise that competes, but rather it is the supply chain of one company that is pitted against the supply chain of another. Hence, all business managers and those aspiring for careers in supply chain should have an in-depth knowledge about its functioning.

Spread over 29 chapters, the book is divided into seven parts as follows:

1. Supply Chain Basics
2. Supply Chain Sub-systems
3. Tactical and Operational Decisions
4. Strategic Approach
5. Measurement and Controls
6. Sustainability Processes and Practices
7. Cases

Each chapter is amply supported by cases, illustrations and an exhaustive pedagogy of summary, review questions, internet exercises, video links, project assignments and bibliography. A comprehensive Glossary of SC terms and Sources on SC Information are included as Annexures. In addition Web supplements for the book comprising of Instructor's Manual, PowerPoint Slides, Teaching Notes and Multiple-Choice Question Banks can be accessed at www.pearsoned.co.in/vinodvsople.

This book is targeted at students of MBA, PGDM and PGPM who have 'Supply Chain Management' as one of their main or optional subjects in their curriculum. Comments and suggestions to improve upon the quality of the contents may be mailed to me at vvvsople@itm.edu.

Vinod V. Sople

ABOUT THE AUTHOR

Vinod V. Sople is Director at the Dombivli Campus of ITM Business School. An active researcher in the areas of marketing, intellectual property rights, logistics and supply chain management, he was awarded a Ph.D. in Management by the University of Pune, from where he had earlier obtained a Master's Degree in Business Administration. He graduated in Mechanical Engineering from Visvesvaraya Regional College of Engineering, Nagpur.

Professor Sople has a rich industrial experience spanning twenty-five years, having worked with reputed companies such as Grindwell Norton, Crompton Greaves, Chicago Pneumatics, Amtek Projects and Gujarat Perstorp in various capacities handling assignments in functional areas of marketing, distribution, customer service, projects and product management. He has over a decade of experience in teaching various management subjects within the country and abroad and has contributed several research papers and articles in leading management journals and magazines. He has authored five books entitled *Logistics Management*, *Managing Intellectual Property*, *Legal Aspects of Marketing in India*, *Business Process Outsourcing* and *E-Marketing*. He is the chief editor of *International Journal of Business Insights and Transformation* and *Supply Chain Pulse*.

PART-I

SUPPLY CHAIN BASICS

1 Understanding Supply Chain

This chapter covers the basic concepts of supply, supply chain and supply chain management. The value chain and extended supply chain concepts are also covered along with its various linkages. Further role of three flows (inventory, information and cash), four cycles (order fulfillment, replenishment, manufacturing and procurement) and capabilities of service providers are discussed.

2 Role of Logistics in Supply Chain

In this chapter the logistics is defined as an activity, which is essentially based on the information. The chapter covers the detailed discussion on logistics mix such as warehousing, transportation, inventory, material handling & storage, packaging and information flow. Further included is logistics decision hierarchy for making supply chain more cost effective and efficient. The various business processes impacted by logistics are also discussed in details in this chapter.

3 Supply Chain vs. Demand Chain

This chapter covers the concept of demand chain management as a model, which combines the strengths of marketing and SCM by shifting the focus to the customer and designing customer-centered supply chain. Here the concept of demand chain has been addressed from marketing perspective by outlining the roles of marketing and understanding the gaps in value offering processes.

4 Customer Value Through Supply Chain

This chapter starts with the discussion on various dimensions of customer value creation, customer service attributes and phases. The role of customer service in supply chain with service design parameters based on customer service perception is included in this chapter. The chapter also covers the various value added services provided by the business firms through their supply chains.

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UNDERSTANDING SUPPLY CHAIN

1

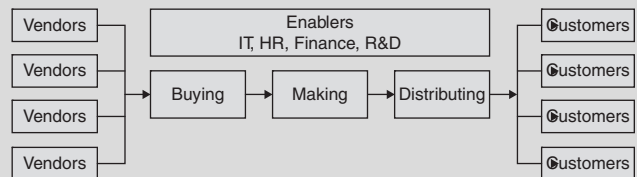
“Supply chain management can be compared with a well-balanced and well-practiced relay team

—Cooper, M. C. and L. M. Ellram¹”

Chapter Objectives

To understand:

- Evolution of the supply chain concept
- Supply, supply chain, and supply chain management
- Supply chain elements and extended supply chain
- Supply chain linkages and integration
- Supply chain decisions
- Prevailing practices in supply chain management



Generic Supply Chain

Supply chain management (SCM) is becoming an important concept not only in manufacturing but also in the service industry. In today's dynamic markets, consumer demands are changing fast, consumerism is on the rise, and the consumer is more knowledgeable, informed and demanding. This calls for enhanced product quality, customer-centric service and speedy delivery beyond time and place boundaries to achieve consumer delight. Today, traditional business strategies do not hold good. Competition is no more between products, but between supply chains. Hence, supply chain has added a new dimension to business strategies. For sustainable competitive advantage, the supply chain process needs seamless integration of its sub-systems.

Kitchen Supply Chain

The kitchen supply chain is similar to the one in the industry. The demands in terms of volumes and varieties (choices of family members) for eatables are typically fulfilled by the housewife. She procures grocery and vegetables in an exact quantity for daily, weekly, or monthly consumption. In Indian villages, the grains are procured for a year's consumption at the appropriate season at the lowest prices. The materials are stored (in a warehouse at the back of the house) and drawn for daily consumption in small quantities. The daily consumption inventory is kept on the kitchen shelf in small containers within arm's reach.

She, over a period, develops trusted vendors for her supply of grocery and vegetables or procures them from a retail chain outlet that she chooses after thoroughly researching the available sourcing points.

The kitchen is a processing (conversion) place like the shop-floor in a manufacturing unit, where cooking schedules are planned and implemented as per the requirements (demands) of family members. The process is flexible to take care of varieties and volumes. The housewife also makes use of the latest technology. Modern kitchen equipments are installed and deployed by her to cut short the processing time and to minimize the hardship of manual operations.

The waste is disposed in a dustbin, which is typically kept below the kitchen sink, similar to practices in the industry where the place of operations is kept neat and clean. The housewife ensures that wastages are at the minimum. The extra demand of family members for entertaining guests at dinner is met with emergency purchases from markets. On holidays and festivals, customized dishes are prepared with advance planning on inputs (ingredients and raw materials). This speaks of end-to-end planning and implementation in the kitchen supply chain.

For the occasional celebrations, the housewife, like a supply chain manager, outsources the place of celebration and awards the catering contract to a third party. For this, she assesses the capacity requirements and plans them in advance keeping in mind a fixed budget that is within the salary limits of her husband. Don't you think that the housewife is a great supply chain manager around without any formal degree or training in SCM?

INTRODUCTION

Supply chain management (SCM) has emerged as an important business process concept that links the supplier to the customer through manufacturing. The capabilities of a supply chain can be leveraged to gain competitive advantages. Over the years, business firms have recognized that the processes whereby the business firms satisfy customer demand are of critical importance to any organization. Using these processes products are developed, manufactured, and delivered to customers, meeting their continuing service needs. The process that links the various elements of business operations is called as the supply chain process. A supply chain is a network of facilities and distribution options. This network performs the functions of materials procurement, conversion of these materials into finished goods, and finally distribution of the finished goods. Supply chains exist in both service and manufacturing organizations. However, the complexity of the chain depends on its product-market configuration.

SCM controls the physical flow (capacity and speed) of goods from source to point-of-use by aligning the capabilities of supply chain partners (suppliers, manufacturers, channel partners, and customers). SCM supports both differentiation- and cost-based strategies. The SCM approach contributes to world-class performance, which is beyond functional integration. SCM integrates the activities of all members of the value chain. Integration results in higher levels of performance than what can be achieved individually. SCM practices create supply chain integration that leads to superior business performance.

In a firm with a functional approach (marketing, distribution, planning, manufacturing, and purchasing) to its supply chain operation, the functional areas have their own objectives that are often conflicting, resulting in loss in productivity and profitability. For example, the marketing department's objectives of high customer service and maximization of sales volumes through product varieties conflict with manufacturing department's objective of producing standard products in volumes.

In a manufacturing organization, the operations are designed to enhance output and minimize costs. However, this agenda has little consideration for optimisation of inventory level and exploitation of capabilities of company's distribution channel. The procurement department often negotiates with suppliers with very little information beyond past buying patterns. This shows absence of an integrated plan for the organization. Hence, there is a need for integrating the different functions in the organization. However, integration is possible only through proper supply chain initiative.

For integration of the supply chain, a variety of business practices such as just-in-time manufacturing, quick response, and continuous replenishment are used. The integration will increase value at all stages of the chain of activities, from source to delivery. Integration of the supply chain raises customer satisfaction and provides increased value to customers. This is done by coordinating activities to reduce costs for all participants and creating value by eliminating non-value-added functions. Use of appropriate SCM practices can help in increasing the productivity and profitability of an organization. The supply chain approach recognizes that customers evaluate suppliers more than they do product attributes and availability. Value extends beyond price and includes total cost and service. Innovation can create new markets, while quality permits long-term presence. However, differentiation from competition can be achieved only through a capability that increases value to customers and protects margins.

EVOLUTION OF SUPPLY CHAIN CONCEPT

In the past, the functions of logistics, transportation, purchasing and supplies were referred to as SCM. But today, SCM looks into process integration, inventory visibility, cycle time reduction and channel coordination. The activities that integration covers are as follows:

- Procurement and inventory movement
- Supplier development and integration
- Operation management
- Channel and distribution management

Logistics activities have been in practice since the early 1900s. These activities were predominantly associated with operations of the military during wars for the movement of food and arms to its personnel. The use of logistics was to ensure the availability of the required material at the right place and at the right time. Logistics is used by the military even today supported by modern science and technology.

After the 1900s, industries started adopting SCM methods. The production, manufacturing and service sectors were the first to apply these methods. The marketing, accounting and production sectors were responsible for inventory management. SCM was being used as a tool for business transformation and was considered as a road map to a company's success, providing operational efficiency and cost effectiveness and enhancing customer service. Further, with advancement in technology, industries adopted Internet-based applications in SCM.

DEFINING SCM

Due to the globalization of markets, shortening lifecycle of products, spiralling costs, and increasing demands of customers, businesses the world over are faced with formidable challenges. Today, businesses are constantly under threat because the speed and manner in which business is conducted has changed radically. In the light of highly competitive and dynamic markets, business organizations, today, consider SCM to be the key area that can significantly impact the bottom-line. SCM may be defined as follows:

A **supply** is the quantity of inventory/goods available (in required quantity) for use or the actual replenishment of a product or component.

A **supply chain** is the link connecting a set of facilities, companies, demand and supply points, and service providers. This chain links the upstream suppliers and downstream customers with the flows of products, services, finances and information from a source to a customer.

SCM is the systematic and strategic coordination of all business functions within a company and across businesses within the supply chain to improve the long-term performance of individual companies and the supply chain, as a whole.

The primary focus of SCM is to serve consumers with excellent goods and services against optimum costs and quick response time. The purpose of SCM is to improve customer value and satisfaction.

In SCM, the following activities can be identified:

- Integration of customers and suppliers to business processes.
- Information flow among channel members for planning, implementation and monitoring processes.
- Sharing channel risks amongst partners to gain a competitive advantage.
- Co-operation among channel members to effectively manage the distribution process.
- Integration of processes for synergistic effects.
- Building and maintaining long-term relationships with partners.

The term “supply chain management” was first coined by consultant Keith Oliver, of the strategy consulting firm Booz Allen Hamilton, in 1982. As per the Council of Supply Chain Management Professionals,² SCM is defined as follows:

Supply chain management encompasses the planning and management of all activities involved in sourcing and procurement, conversion, and all logistics management activities. Importantly, it also includes coordination and collaboration with channel partners, which can be suppliers, intermediaries, third party service providers, and customers. In essence, supply chain management integrates supply and demand management within and across companies.

More comprehensively, SCM is defined by Mohanty R. P. and Deshmuk S. G.³ as follows:

A supply chain is a network of facilities and distribution options that performs the functions of procurement of materials, transformation of these materials into intermediate and finished products and distribution of these finished products to customers. Supply chain management is the integration of these activities through improved supply chain relationships to achieve a competitive advantage.

In essence, SCM can be understood by what it can achieve, such as:

- It manages three flows
 - Inventory, information and cash
- It manages four cycles
 - Procurement, manufacturing, replenishment and order fulfilment
- It aligns capabilities of
 - Suppliers and service providers

EXTENDED SUPPLY CHAIN

There are basically two types of supply chains: generic and customized. The generic supply chain links supplier, manufacturing and distribution divisions in a sequential flow. The customized supply chain, on the other hand, eliminates some of the processes that are covered in the traditional supply chain. Take, for example, the SCM of “Dell” (Figure 1.1), wherein the manufacturing process is eliminated and substituted by assembly centres and the distribution is made directly to customers, eliminating any intermediaries. Dell SC design cuts cost by eliminating non-value-added processes in the supply chain.

The supply chain of a manufacturing firm has three main functions to integrate and manage: buy, make and distribute. However, if the supply chain is further extended to look into suppliers’ suppliers and downstream to customers’ customers, it is called an extended supply chain. Extended supply chains can become too complex to control, but if managed properly, they result in better control of costs, delivery and end-product quality, which in turn translates to higher customer satisfaction and profitability. Extended supply chain (Figures 1.2 and 1.3) is one wherein everyone contributes to a product. For example, if a company makes garments, then its extended supply chain would include

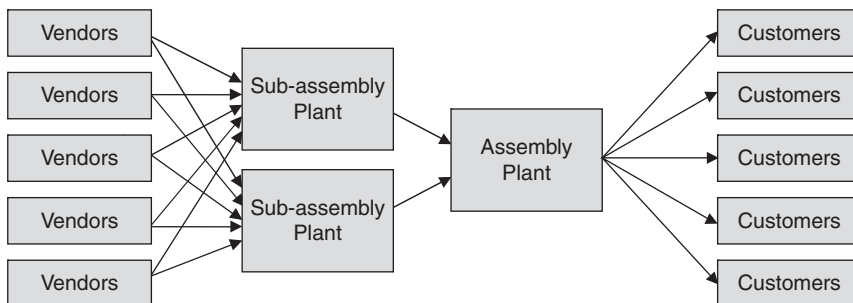


Figure 1.1 Supply Chain—Dell Model

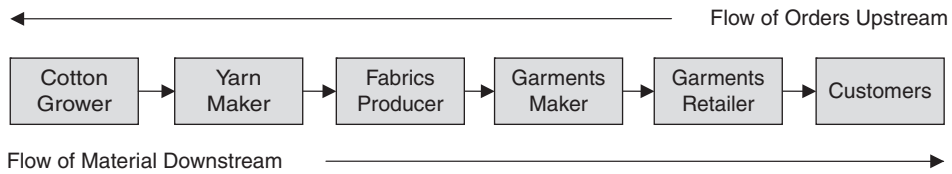


Figure 1.2 *Extended Supply Chain of Garment Manufacturers*

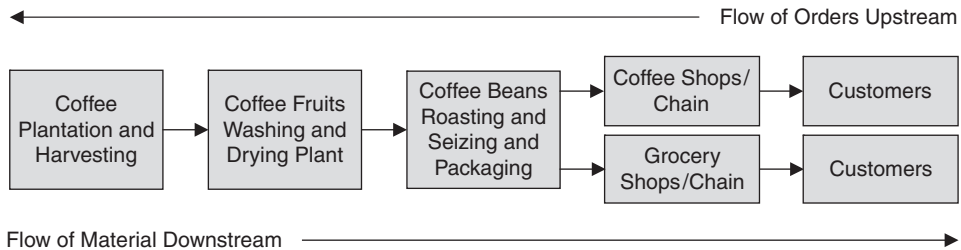


Figure 1.3 *Extended Supply Chain of Coffee Suppliers*

Box 1.1

Extended Supply Chain—A Cost Burden on Berry Interlining

Berry Interlining (BI) is the manufacturer of “interlinings,” supplying its products to garment manufacturers and tailoring firms. In their extended supply chain, BI procures yarn from multiple yarn producers, sends the inspected yarn to weavers for producing fabrics, which are later sent for bleaching to third-party processors, and finally uses coloured fabric for producing interlinings. In this yarn-to-interlining manufacturing cycle, there was multiple material handling, inspection and transportation of material. This results in a high percentage of damage during transit, huge rejection during quality inspections and delay in the manufacturing cycle. The working capital requirement obviously increases year by year, putting pressure on the profit margins. BI re-engineered its extended supply chain in a bid to reduce costs and improve delivery. It started procuring fabrics duly inspected and certified by the suppliers, thus eliminating all non-core processes. Thus, by eliminating the extended part of its supply chain, BI reduced costs, improved on the manufacturing cycle, reduced working capital requirements and enhanced profit margins.

Eliminating Extended Supply Chain—VP Luggage

VP Luggage (VPL), a leading moulded luggage manufacturer, used to procure components (springs and levers) for assembly of locks used in hard luggage (briefcases, trunks, bags, etc.). They had a separate lock assembly and quality checking department engaging 25 workers. During pre-assembly quality checks, lock component rejections accounted for 3.5 per cent whereas post-assembly quality checks revealed 1.5 per cent rejections. The field rejections with entire luggage (due to lock problems) were to the extent of 0.5 per cent. To reduce this and to use the 25 workers for other productive purposes, VPL stopped procuring lock components and assembling them thereafter. They asked the vendors to supply complete locks that were duly tested and certified. Thus, by eliminating the extended supply chain, VPL could contain the rejections within 0.1 per cent.

factories where the cloth is made and printed, factories where the yarn is produced, factories of other raw material producers and so on. In the extended supply chain, it is critical for a company to keep track of what is happening in its chain because a supplier or a supplier's supplier could end up having an impact on you.

For example, a problem in a manufacturing unit that supplies ball-bearings to an automobile manufacturing company may cause the latter to run out of its inventory of ball bearings. If the unit knows about the shortfall in its extended supply chain in advance, it can find another supplier to keep its operations running.

SUPPLY CHAIN INTEGRATION

In today's competitive world, a company cannot survive on its own; it can only grow with the help of suppliers and their co-operation. Therefore, SCM should deal with the process of co-ordinating the flow of information on the one hand and the flow of goods on the other, across a network of suppliers, manufacturers and distributors.

There is a great deal of uncertainty about demand in the market. Consumer loyalty is at its lowest; even the products are offered in great variety and features with attractive prices. Therefore, information regarding changes in customer demand pattern should flow back to manufacturer. This feedback should be timely to reschedule the production programme.

With timely and accurate market feedback, a manufacturer can bring about a change in production and requirements of raw material at the production plant itself. Otherwise, inventory will pile up, which will be either slow moving or dead. Therefore, it would not be wrong to say that the success of SCM depends on the co-ordination with the service providers and marketing channels.

Customer requirement is determined at the retail level. This determines production need, which in turn, helps to chalk out the need for raw materials, parts and components. Thus, supply chain integration acts as a planning process and an information-based activity, which helps to determine market requirement and convert demand into supply and supply into goods through a well-defined planning process.

Supply chain integration calls upon decision-making nodes in the chain to join hands to solve problems and improve quality of work, reduce costs, reduce lead times and enhance information sharing. Thus, integration would help reduce the element of uncertainty and increase quality and market acceptance of the product. The new products could be co-designed with suppliers and customers to achieve economies of scale with standardization of parts.

For example, McDonald works in close co-operation with all its vendors, helping them set up their operations, arranging equipment for them, providing technical inputs and training them to meet its tough standards. This "to help, to get helped" strategy is at the base of supply chain integration.

Today, supply chain integration is practised in a broad range of industries with varying techniques to suit specific issues and business characteristics of suppliers, manufacturers, channel partners, customers and market-product configuration. The benefits of supply chain integration are available to all. However, the success of supply chain integration can be evaluated against the objectives of:

- **Service:** Actual against desired with quality, place and timing.
- **Cost:** Cost to customer in meeting his/her service and time requirements.
- **Assets:** Effectiveness of assets to meet variability of consumer demand and to simultaneously create operational efficiencies.
- **Time:** The shortest timeframe to convert customer order into cash realization.

SCM is fundamentally based on the idea of partnership with suppliers, marketing channel members and other service providers. Traditional models of business organization are based on the philosophy of maximizing the revenues and minimizing the costs. In many cases, these goals can be achieved by disadvantaging another entity in the chain. However, under the SCM model, the goal is to maximize profit through enhanced competitiveness, imparting benefits to all partners in the supply chain. These goals can be achieved only if the entire supply chain is closely integrated so that total channel inventory is minimized, bottlenecks are avoided, timeframes are compressed and quality problems are eliminated.

Today, in the mega-competitive age, individual companies compete, not as companies, but as one supply chain against another. Hence, successful companies will be those whose supply chains are more cost-effective than those of their competitors.

SUPPLY CHAIN LINKAGES

The key linkages in a supply chain are between procurement and manufacturing and between manufacturing and distribution. Figure 1.4 outlines the critical linkages that connect the marketplace to the supply chain. The three sub-systems—procurement, manufacturing and distribution—have a number of critical elements.

Procurement

Companies with supply chain initiatives have adopted the philosophy of co-makership. This is based on the idea of a mutually beneficial relationship between the supplier and the buyer. It is a partnership approach to identify opportunities for taking costs out of the supply chain instead of simply pushing them upstream or downstream. In co-makership, both the buyer and the seller are at a win-win situation. The information is shared to jointly solve the problem and improve upon the quality. Co-makership will normally involve long-term relationships based on single sourcing rather than multiple sourcing. Mahindra & Mahindra, in India, has adopted the co-makership philosophy, which has resulted in their supplier base falling to nearly one-fifth. Here, the approach is to consider the supplier as your younger brother and his supply chain as an extension of your supply chain. You help him financially and technically to reduce cost and enhance productivity. The profits gained or cost reduced will be equally shared. This is possible only with smaller supplier base which will enable the firm to have better control on performance and its measurement.

A fundamental feature of this integrated approach to SCM is the adoption of some form of materials requirement planning (MRP) linked to schedule coordination.

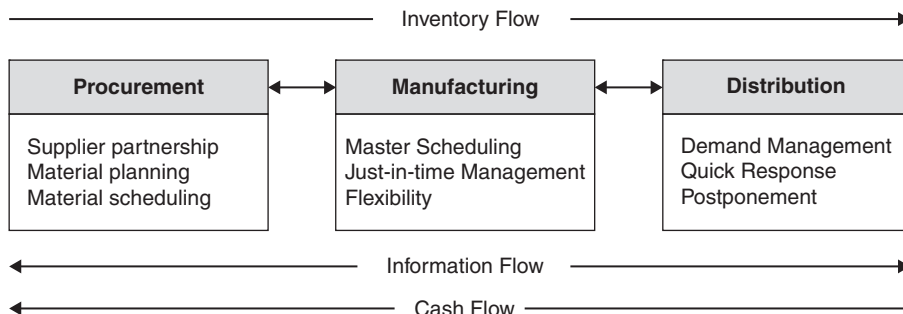


Figure 1.4 Supply Chain Linkages

Procurement is none but MRP to manage the inbound flow of materials by linking the factory to its suppliers. In the recent past, there has been a number of developments in procurement practices, such as e-procurement, reverse auction, and vendor management inventory (VMI), which have enabled a more flexible demand-based approach in the procurement process.

MRP requires the linking of suppliers' production schedules with those of their customers. The objective is to consider the suppliers' operations as an extension of the customer's. Companies, through Internet-based communication systems, develop links with all of their suppliers, so as to have full visibility of the production schedules. By using electronic data interchange (EDI), corporations can now reduce lead times, eliminate multiple data entries and take costs out of the supply chain.

Manufacturing

Keeping in mind the volatile market demands, there is a need for flexibility in manufacturing capability and capacity in terms of the ability to produce variety and volumes without affecting the cost. In the past, manufacturing philosophy was dominated by 'economies of scale'. This led to the setting up of large capacity plants that were capable of producing large quantities of a standard product at very low unit production costs. Many MNCs adopted this strategy in producing a limited range of products for global market consumption.

On the other hand, companies went to the extent of building up large inventories of finished products ahead of demand to cover up the inability to respond rapidly to changing customer requirements. This resulted in offering limited product varieties to the customer. In such situations, they looked out for strategies to reduce total supply chain costs (not just manufacturing costs) that would offer maximum flexibility against customer requirements.

The necessity forced the companies to trace the route to flexibility in manufacturing, which does not necessarily lie in using new technology, for example, robotics. The Japanese have shown that flexibility can be achieved through focusing upon the time a company takes to plan, schedule, set up, change over and document its manufacturing process. These activities are the classic barriers to flexibility. If these barriers are removed, then manufacturing can respond far more rapidly to market requirements. Ideally, with zero lead times, total flexibility can be achieved with no forecasts and inventory. However, zero lead times are impossible to achieve. The Japanese, however, have shown that impressive reductions in lead times can be achieved by questioning everything we do and the way in which we do it.

Distribution

The role of distribution in SCM is beyond transport and warehousing. The success in distribution today depends on managing the demand.

In the demand management process, the company anticipates and fulfils the orders against defined customer service goals. Information flow is the key to success in this process. The categories of information required are:

- Information from the marketplace in the form of medium-term forecasts.
- Information from customers preferably based upon actual usage and consumption.
- Information on production schedules and inventory status, and
- Information on marketing activities such as promotions that may cause demand to fluctuate.

In reality, forecasting accuracy will rarely be achieved. Instead, the objective should be to reduce the dependence upon the forecast by improved information on demand and by creating systems capable of rapid response to such demand. This is the principle behind the quick response supply chain,

which in turn, reduces cost and enhances customer service. In this case, items that are consumed or purchased are transmitted to the supplier triggering an immediate response. Today, customers prefer faster and frequent deliveries in smaller quantities. The higher transportation cost will be offset by the cost of reduced inventory in the pipeline. Information technology has been a major enabling factor in quick response supply chain, linking the point of sale or consumption with the point of supply.

A further trend that is visible in distribution is postponement. Postponement refers to the process where the final configuration or form of the product is delayed until the last possible moment resulting in flexibility and reduction in finished goods inventory. For example, at Asian Paints, using 'paint-dispensing machines' at retail outlets, the company postponed the making of the final 'paint shade' (as required by the customer) to the dealer's place. This method reduced the inventory of finished goods at the factory level.

In essence, distribution in the integrated supply chain is an information-based, value-added activity, providing a critical link between the marketplace and the factory.

DECISIONS IN SUPPLY CHAIN

The decisions in SCM are taken at three levels: strategic, operational and tactical. Strategic decisions have a longer time horizon. These are always in line with corporate strategy, which guide supply chain policies. Strategic decisions influence supply chain network designs. Operational decisions are short-term and focus on activities on a day-to-day basis. Tactical decisions are for enhancing the efficiency and effectiveness of the supply chain system. The major decision areas with strategic, tactical and operational elements in SCM are:

- Network
- Vendors
- Production
- Inventory
- Distribution
- Information

Network Related

This relates to the choice of location of supply points near demand points, including production facilities, stocking points and sourcing points in creating a supply chain. The facilities location decision involves commitment of resources to a long-term planning horizon. The location of production facilities in the network decides the path of goods movement. The network decision is of great significance to a firm for reaching the market. It impacts revenue, cost and level of service. In general, network-related decisions are based on manufacturing costs, taxes, duties and duty drawback, tariffs, local content, distribution costs, and production limitations. The location decisions are strategic in nature; however, they have implications on operational-level activities.

Vendor Related

The suppliers of raw materials, parts and components play an important role in the supply chain due to the fact that the planning of production schedules depends on the supplier's dispatch schedules. For a production schedule to run on time, firms should have reliable suppliers, conforming to delivery

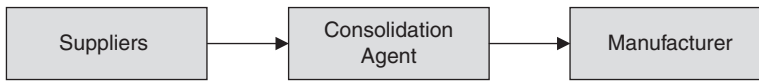


Figure 1.5 *Single Sourcing*

schedules and lead times. Reducing suppliers' base normally enhances the buyer's control on the procurement process. It also enhances reliability and loyalty of suppliers.

Therefore, the buyer should go in for a single source (Figure 1.5) of supply rather than depend on multiple vendors. Smaller vendor bases help in developing long-term relationship based upon trust, mutual dependence and sincerity.

Reduction in the number of suppliers would enhance coordination of flow of information and raw material and improve reliability of delivery in terms of time, resulting in reduction in procurement costs. TELCO, while developing the Nano car, squeezed its supplier base to keep the cost low, the quality high and to ensure delivery at short notice. Sona Steering, Telco's single supplier of steering, developed and supplied steering assembly in the shortest possible time. This indicates the benefits derived from maintaining a single source of supply.

For reducing supplier base, firms can employ distributors who will collect raw material or components from various suppliers and dispatch them to the manufacturer. With this, the procurement process will be under tight control as the manufacturer would have to contact only a single source for supply.

For example, at McDonalds, mutton comes from Hyderabad-based Al-kabeer, lettuce is flown in from Pune and Ooty, buns are from Cremica Industries, Ludhiana, and cheese comes from Dynamix Industries, Baramati (Maharashtra). The hubs of the cold chain are located in the North and the West and are managed by Radhakrishna Foodland that dispatches the supplies to the restaurants, processes information and coordinates the work.

The hub concept is fast gaining acceptance as a distribution hub cuts inventory costs due to economies of scales, control on inventory handling and upkeep, and helps in enhancing competitiveness of the distribution process. Hindustan Unilever Ltd dispatches trainload of goods periodically from Mumbai to their distribution hub located at Nagpur to take care of secondary distribution in central India. Similarly, Ford chose Nagpur as its distribution hub in India mainly because of its convenient location.

Production Related

Production-related decisions are about product mix, location of plant and suppliers base to that plant. In other words, these decisions are about allocation of plants to distribution centres (DCs) and allocation of DCs to markets. Production-related decisions impact revenues, costs and customer service levels of the firm. The other key consideration in decision-making is the capacity of the manufacturing facilities to serve the assigned market. Operation-related decisions include production planning, manufacturing and equipment maintenance. Other factors that affect these decisions are workload balancing and quality control measures at a production facility.

In today's competitive markets, manufacturers are required to offer a variety of products. This requires manufacturers to establish flexible manufacturing systems to enable them to provide a broad range of products and also to develop and introduce new products. Hence, manufacturers should develop flexible manufacturing systems to respond to orders from customer in terms of volumes and varieties without any significant difference in cost. For example, SKF Ltd manufactures 20,000 varieties of bearings for different applications. In high-precision products like bearings, manufacturing demands continuous adjustment in process standardization. This helps in designing various types of products to cater to the changing requirements of customers.

Offering a variety of products will be of little value if the market demands quick response. Therefore, an organization needs to change its strategy, depending upon what form of flexibility the company needs from its plants.

For customized products, manufacturers would need to develop the ability to carry out a large range of jobs in the plant. However, to develop quick response as its primary competitive weapon, firms would need to focus on building flexible manufacturing processes. For manufacturing flexibility, measurement methods as well as personnel employed should be aligned to provide form to flexibility in the entire system including its sub-systems.

Inventory Related

Inventories in the supply chain include raw materials and semi-finished or finished goods. These can also be processed between multiple locations. The purpose of inventory is to reduce the level of uncertainty in the supply chain. Inventory holding can cost anywhere between 20 and 25 per cent of the material value. To cut this holding cost, efficient management is important in supply chain operations. Inventory management is an important activity in the supply chain as excess inventory eats into profits. The inventory policy decisions have long-term implications. Hence, inventory decisions are strategic in nature. Normally, the top management sets goals for level of holding the inventory. These include inventory control policies such as deciding optimal levels of order quantities and reorder points, and safety stock levels at each stocking location. These decisions are critical in nature as they are determinants of customer service levels and customer satisfaction levels thereof.

Distribution Related

Selection of mode of the transportation and the route are critical decisions in controlling the cost and speed to the market. For example, for low unit price products, the logistics cost, as a percentage of product price, is high. Here, the selection of transportation mix for reaching the markets, that too in bulk quantity, is important to reduce the incidence of logistics cost on product price. These decisions are closely linked to the inventory decisions. The best choice of transportation mode is decided by trading off the cost of using the transportation mode with the cost of inventory being moved by that mode. For example, air will be preferred as a transportation mode because it is faster, reliable and warrants lesser safety stocks—but it is too expensive. Sea or rail transportation may be much cheaper, but they require holding inventory in bulk quantity. In cases wherein transportation is more than 30 per cent of the total logistics costs, operating efficiently makes perfect economic sense. Hence, shipment size, vehicle routes and dispatch scheduling are important areas that need to be covered in a firm's transport strategy.

Information Flow Related

Today, it is not one enterprise competing against another; rather it is one supply chain competing against another. Hence, for the supply chain to be efficient and cost effective, it is necessary to have better integration among the entities in the supply chain. In competitive markets, responsiveness to changes with speed and accuracy is vital for success. As entire supply chain activities are information based, a company's continued success depends upon its ability to build information architectures that are responsive to dynamic markets. Therefore, effective online integration of a company with its buyers and suppliers is a must. Use of Internet technologies facilitates meaningful communication between a firm's internal and external partners. Enterprise resource planning can also be adopted as an integrated

planning and control tool to connect customers, suppliers and enterprise activities in line with the value chain of the firm.

Today, the complexities involved in sustaining, surviving and succeeding in a volatile environment have increased substantially. Organizations will be able to cope with the turbulent environment and with the complexities of operations only if they develop an efficient supply chain system that is integrated, have reduced suppliers' base, are flexible in manufacturing and make optimum use of information technology.

PREVAILING PRACTICES

Over the years, SCM has emerged as one of the most powerful business improvement processes. Smart companies have created networks for raw material suppliers, product manufacturing units, storage points, DCs of the products, and finally, they have started delivering the products and services to customers.

The objective of supply chain networking is to encourage product and service innovation with an objective to satisfy the customers. These organizations see value enhancement through integration of systems and supply chain operations across all business functions. Today, organizations are moving towards a total system of supply and delivery linking directly to the market demands.

Business organizations with supply chain integration have succeeded in increasing their revenue and profits. They have been able to cut supply cost through improvement of their purchasing function as per the market demand. Today, companies apply information technology to their supply chain to gain advantage of the firm's operational improvements. As customer demands are ever changing and markets are becoming more and more volatile, business firms have formidable challenges ahead in managing their supply chains.

PRTM⁴ Management Consultants Inc., USA, conducted a study on 350 manufacturing and service companies across the globe and published its report on 22 June 2010. The study reveals the following key supply chain challenges that companies will be up against:

- Increased supply chain volatility and uncertainty
- Creating global customer and supplier networks
- Creating cost-optimized supply chain configurations
- Risk management in the end-to-end supply chains
- Integrating and empowering existing supply chains

The main challenge for many companies is not to redefine their organization models, but to transition and manage the organizational change. To make a truly empowered supply chain organization work, companies must first determine what their target models should look like and persuade the senior management to make the required changes. To make an integrated supply chain work, it is essential to train and acquire top talent with end-to-end supply chain knowledge.

Successful firms with supply chain initiatives have developed cross-functional trust and also cross-organizational trust with distributors and suppliers. They have succeeded in improving co-operation amongst internal functions and have gained significant operational benefits. Many companies benefited through flexible manufacturing systems resulting in reduced inventory and operation costs. Using demand management tools in their supply chains, companies can manage the process of acquiring the right raw materials more efficiently, utilizing the necessary machines and human resources available within the appropriate timeframe, thereby scheduling output to meet demand without excess inventories.

Firms with supply chain initiatives are now focusing on linking the demand chain with the supply chain in a holistic manner to achieve cost reductions throughout the chain. This has resulted in substantial savings by way of efficient handling, storage and delivery of products in the supply chain.

Automobile manufacturers integrate their suppliers to have just-in-time (JIT) deliveries at the assembly plant to make the supply chain lean. The assembly plant receives only the components needed for the specific production schedule resulting in very low, or no buffer inventories. It then passes on the information on production schedule to suppliers only hours/days before assembly. The suppliers finalize their production according to the models to be produced by the automobile manufacturers. Thus, the flow of product between suppliers and manufacturers is integrated to eliminate costs while maintaining service to production. Supply chain integration is practised in both consumer good and retailer industries. These industries practise supply chain integration to better their response to consumer demand, with low investments in finished goods inventory. FMCG companies have designed programmes of continuous replenishment to retailers' DCs to provide high store service at lower cost. The replenishment is based on daily sales information as well as store order and DC inventory levels provided by the retailer. Here, the manufacturer takes responsibility for supply chain integration.

The concept of SCM is universally accepted and practised. However, the cutting edge lies in speed to market, cost controls and customer satisfaction. A focus on the objectives of service, cost, assets and time, when coupled with creative thinking, brings forth new approaches to business practices. In conclusion, well-managed supply chains create profitable growth for all members of the chain.

SUMMARY

SCM is becoming an important concept in modern business systems. It is a continuous process of shaping and reshaping performance within the company. This is done by using information technology tools, products and services, and organizational and personal excellence to exploit the ever-changing context of customer opportunities. It is about managing three flows (inventory, information and cash), four cycles (procurement, manufacturing, replenishment and order fulfilment) and aligning the capabilities of the service providers. The decisions in supply chain are about network, vendors, production, inventory, distribution and information. This requires integration, and the success of supply chain integration can be evaluated against the objectives of service, cost, assets and time. The complexities in supply chain depend on the organization's product-market configurations. Extended supply chains are complex in nature. Due to multiple decision-making points, integration of all the elements of a supply chain is a must to achieve the objectives of productivity and cost effectiveness.

REVIEW QUESTIONS

1. Explain the concept of supply, supply chain and supply chain management.
2. Discuss the critical linkages in a supply chain.
3. Explain the concept of "extended supply chain" and the issues therein.
4. Which are the critical decision-making areas in the supply chain?
5. "Supply chain management is a road map to success for a business". Explain.

INTERNET EXERCISES

1. Council of Supply Chain Management Professionals is world's leading source for supply chain profession. Visit <http://cscmp.org/aboutcscmp/definition.asp> and study various basic definitions on supply chain topics and also look into the supply chain trend studies published by the CSCMP.
2. The Infosys global supply chain management blog enables leaner supply chains through process and IT-related interventions. Visit <http://www.infosysblogs.com/supply-chain/> and study the latest trends and solutions across the supply chain management landscape.
3. For further reading on introduction to supply chain management visit http://logistics.about.com/od/supplychainintroduction/a/into_scm.html

VIDEO LINKS

1. What is Supply Chain Management?
<http://www.youtube.com/watch?v=MilQBxVjZAw&feature=related>
2. Supply Chain - 3 Key Things to Know, <http://www.youtube.com/watch?v=OaxJ4oEuBO8&feature=related>
3. Preview for Supply Chain Management by CSCM (Council on Supply Chain Management), <http://vimeo.com/1905919>

PROJECT ASSIGNMENTS

1. "E-Chaupal" is an innovative business model implemented by ITC, India. Prepare a flowchart for this supply chain. Compare and contrast the working of the traditional and the E-Chaupal supply chain model bringing out the cost and productivity aspects (www.itcportal.com).
2. "Starbucks" is a leading chain of coffee shops across the world. They have their own coffee farms and procure coffee beans from the markets. The distribution is through their coffee shops and other organized grocery chains. Prepare a report on their extended supply chain covering the supply chain strategies that they have successfully implemented.

NOTES

1. Cooper, M. C., and Ellram, L. M. (1993), 'Characteristics of Supply Chain Management and the Implications for Purchasing and Logistics Strategy', *The International Journal of Logistics Management*, 4, 2, pp. 13–24.
2. See <http://www.cscmp.org/aboutcscmp/definitions.asp>.
3. Mohanty, R. P. and Deshmukh, S. G. (2005), *Supply Chain Management: Theories and Practices*, Biztantra: New Delhi.
4. See <http://www.supplychainstandard.com/PRTM-study>

ROLE OF LOGISTICS IN SUPPLY CHAIN

2

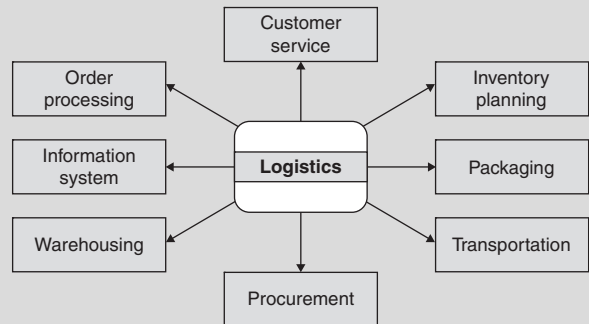
“Most skills and competencies needed to excel in logistics and supply-chain management are the same skills and competencies needed to excel at disaster relief operations

—Matt Waller”

Chapter Objectives

To understand:

- Concepts, scope and objectives of logistics
- Logistics mix
- Logistics role
- Logistics strategies



Logistics System Components

Logistics is a supply chain enabler. Logistics basically connects the source of supply with the source of demands. It bridges the gaps between the market demands and the capabilities of supply sources. Logistics helps in bridging these gaps so as to make them invisible at the demand points in the supply chain. To accomplish this, all the components of logistics system such as the warehousing network, the transportation network, the inventory control system and the supporting information system come into operation with the objective of delivering the right product at the right place and at the right time at the least cost. The logistics system has to strike a balance between the two polemic objectives of customer service and cost. Logistics makes it possible to deliver products to the customer anywhere irrespective of its manufacturing location.

Logistics for 'Works of Art'

United Artlogistics Pvt. Ltd. (UAPL) has pioneered the concept of art logistics in India by launching these services in 2009. Art logistics is a special service that manages the movement of arts in a very specialized manner. It covers the following:

- Art crating and packaging in the right manner
- Services of experts in art handling
- Paperwork related to transportation
- Certification of art for authenticity
- Art security from beginning to end
- Art warehousing and preservation

India is rich in art and culture with various artworks and artefacts amounting to a value of Rs 15,000 mn (\$300 mn). These artefacts are moved from one exhibition to another, but a insignificant percentage (5%) of revenue is utilized for logistics services involved in movements of artworks. In fact, art is a very precious asset and manifestation of an artist's passion, in the context of the prevailing culture and its expression. The art pieces require specialized services to preserve their originality during movement from one place to another. Art logistics caters to the needs of art galleries and individuals for movement of priceless pieces across the globe.

UAPL is experiencing slow growth in art logistics in India. The reason is the lack of awareness for such services and absence of service providers in this specialized area. In India, barring a few, neither art galleries nor the buyers insist on expert services for movement of artefacts and artworks. UAPL faces the following four challenges in movement of artworks and artefacts to their global destinations:

- Packaging conforming to international standards
- Use of right transportation or line haul
- Customs clearance with right documentation
- Onward forwarding to certified art dealers

UAPL provides comprehensive art logistics solutions to galleries and individuals in moving artworks to desired domestic or global destinations. Art insurance is another value-added service provided by UAPL for insurance coverage of items like framed and unframed paintings; drawings and photographs; sculptures of marble, wood, metal, stone, or glass; ceramics and pottery; art glass; architectural models; figurines and collectibles; antiques and movie props; museum exhibitions; gallery exhibitions and historic artefacts of any size. Today art galleries are increasingly using the services of UAPL for both domestic and global art exhibitions.

INTRODUCTION

Logistics fills the gap between supply and demand. However, when the gaps tend to be larger and the risk of dilution of service level is higher, there arises a need for an integrated system to make the product and information flow seamless. Supply chain management is the process of integration to bridge

the gap between supply and demand. The supply chain tries to understand the demand signals and the profile of the target customer base and adjust itself for planning and execution of the customer requirements with the desired service level.

The efficiency and effectiveness of inventory movement across the supply chain is greatly dependent on the capability of the logistics management. Hence, the integration of supply chain is not possible without the capability and reliability of logistics operation. Cost reduction and customer satisfaction are not possible in the supply chain without efficient logistics operations like warehousing, material handling, inventory control, packaging and transportation. In fact, logistics and the supply chain management cannot be separated from each other and they are part of the same customer service solution. Logistics operations may continue irrespective of whether enterprises follow the supply chain philosophy or not. The inventory movement needs to take place in order to bridge the gap between supply and demand. Efficient logistics management takes care of the gap, and their effectiveness depends on the enablers. In other words, the success of the supply chain depends to a great extent on logistics. However, for logistics and supply chain to be successful, the following need to be planned and managed properly:

- Close coordination with the suppliers
- Reduction of inventory levels
- Speed, reliability and consistency in inventory movement
- Faster replenishment cycle
- Shorter performance cycle
- Flexible manufacturing cycle
- Asset utilization and productivity
- Innovations for value additions in customer service offerings

In a nutshell, logistics is a key to the success of supply chain management. The degree of success depends on the level of integration between them using the enablers like information and communication technology. In the new millennium, globalization is making more and more national economies integrated into global economy, world trade is expanding at an exponential rate, technologies are advancing and becoming integrating forces, customers are demanding value for their money, marketers are experiencing competitive pressure and businesses are struggling not only for growth but also for survival. Products today are sourced from different parts of the world, assembled at different locations and shipped to various destinations to ensure greater customer satisfaction and squeezing out unproductive time and cost from the supply chain cycle. These trends have resulted in numerous marketing, distribution and logistics opportunities across the global markets through expanded global supply chains and enhanced competitiveness.

DEFINING LOGISTICS

In the olden days, logistics was referred to as the art of moving the army and supplies thereof. The origin of logistics can be traced to the Greek word 'logistikos' and the Latin word 'logisticus'. Both words mean 'the science of computing and calculating'. In recent times, the word 'logistics' was first used in the seventeenth century by the French army. However, logistics gained importance in army operations during the Second World War for the movement of supplies, men and equipment to the war front. The US army officially started using 'logistics' after the Second World War. Today logistics is used in

business for movement of inventory from vendors to the manufacturers and finally the finished products to the consumers.

According to the American Council of Logistics Management,¹ logistics is defined as follows:

The process of planning, implementing and controlling the efficient, cost effective flow and storage of raw materials, in-process inventory, finished goods and related information from point of origin to point of consumption for the purpose of conforming to customers' requirements.

Philip Kotler² defines logistics as follows:

Planning, implementing, and controlling the physical flows of materials and finished goods from point of origin to point of use to meet the customer's need at a profit.

Logistics is a supply chain enabler coordinating the inventory movement process by optimizing the flow of materials and supplies through the business operations to finally reaching to the customers.

Martin Christopher³ defines logistics as follows:

Logistics is essentially a planning process and an information based activity.

Philip Scary has the following definition:

The corporate traffic cop that directs flow of material from source through production and distribution to the final customer.

The function of logistics ultimately supports corporate goals by delivering products to the consumer at a time and place of his choice, at an affordable cost .

There is a marked difference between 'logistics' and 'supply chain'. However, many times these two words are used synonymously. Logistics is a physical bond to achieve free flow of inventory, information and cash. As opposed to this, 'supply chain' is an emotional bond which develops capacity in the chain to allow this flow. Logistics is a bundle of activities which gets integrated in the supply chain process. It ensures cost-effectiveness, and creates competitiveness through supply chain management.

It is important for an organization to have a streamlined process to deliver products and services to the customers. The effective and efficient physical movement of the tangible product will speak about the intangible services associated with it and about the organization which is delivering it. In case of intangible products, the delivery of tangibles at the right place at the right time will speak about its quality. Logistics is one of the major enablers for the growth of trade and commerce activity in the country. At a macro level, logistical infrastructure such as modes of transportation, transportation equipment, storage facilities, connectivity and information processing contribute to a large extent in the physical movement of goods produced in various sectors such as manufacturing, mining and agriculture. The speed and reliability in distribution from place of production to place of consumption will contribute to a great extent to the growth of a country's domestic and international trade.

At a micro level, logistics has a great role to play in delivering value-added services to the customer by providing differentiated services. The major objective of logistics management is to achieve the desired level of service and quality at the lowest possible cost. The scope of logistics covers management of material procurement through manufacturing to delivery of finished products to the consumer. The improvement in productivity achieved in any part of logistics systems would help in cost reduction and optimum usage of resources for productive purpose.

Logistics integrates activities of material movement across business processes that are interdependent and need close coordination. It covers the following functional areas, and is termed as 'Logistics Mix'⁴ by Martin Christopher:

- Information Flow
 - Order registration
 - Order checking and editing
 - Order processing
 - Coordination
- Warehousing
 - Material storage
 - Load unitizing and material handling
 - Site selection and network planning
 - Order picking and filling
 - Dispatch documentation
- Inventory Control
 - Material requirement planning
 - Inventory level decisions for customer service objectives
- Packaging
 - For handling and damage prevention
 - For communication
 - For inter-modal transportation
- Transportation
 - Route planning
 - Mode selection
 - Vehicle scheduling

The objective of logistics is to ensure the flow of material across the supply chain cost-effectively and making available the product at the right place and at the right time. Logistics has to achieve two opposite goals: customer satisfaction and cost-effectiveness.

According to Michel Porter, businesses gain a competitive edge either by reducing their cost or by offering differentiated products and services. The cost can be reduced by effective logistics management, which will also enhance the differentiation through superior customer service. As shown in Figure 2.1, the value-acquiring activities of the firm can be categorized into inbound logistics, in-process logistics and outbound logistics.

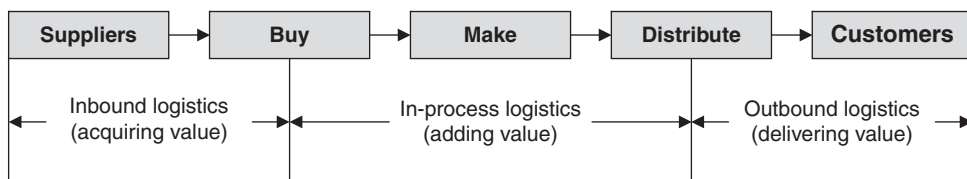


Figure 2.1 *Customer Value Delivery Chain*

Logistics delivers the value to the customer through the following three stages in the supply chain:

- **Inbound logistics:** Includes movement of raw materials and components for processing from suppliers.
- **In-process logistics:** Includes storage and movement of raw materials and components within the manufacturing premises as per the manufacturing schedules.
- **Outbound logistics:** Includes the warehousing, transportation and inventory management and distribution of finished products to customers.

The total logistics cost as percentage of the sales differs with the categories of products. With proper control, the logistical cost can be reduced by way of value addition through a combination of price, quality and service.

GOALS OF LOGISTICS

The mission of the logistical system is to effectively and efficiently move the inventory in supply chain so as to extend desired level of customer service at the least cost. The key elements of effective logistics which are core to supply chain management are as follows:

- Product movement
- Information movement
- Time and service
- Cost
- Internal integration

Supply chain management requires logistics support so as to meet customer demands. To meet these demands, organizations need to be flexible both internally and externally, need to develop teamwork to meet the organizational objectives and need to make use of technology to manage the requirements. The inefficiencies in supply chain are due to excess inventories and high costs in buying and moving the inventory. The following are subsets of logistics objectives under its broader objectives of cost reduction and customer satisfaction through service.

Responsiveness

Responsiveness refers to the capability of the organization to quickly respond to market changes and customer demands by way of extending services in the shortest time period. The latest technologies in information collection, processing and communication enhance the degree of responsiveness in terms of accuracy, timing and flexibility to fulfil customer requirements, irrespective of the volume and variety, in the shortest possible time period. For example, the supplier can deliver small-size consignments rapidly and frequently at consumption points.

Reducing Inventory Burden

Inventory in the supply chain adversely affects the bottom line of the business. Generally, organizations carry excess inventory for the purpose of providing excellent customer service. The funds invested in inventory are blocked and cannot be used for any other productive purpose. There is always capital cost associated with inventory. The cost of carrying inventory is equated to the interest rates on the funds loaned by the banks at that time. The carrying cost is a drain on the profit of the enterprise.

Hence, the major objective of logistics is to maintain the inventory at a minimum level to keep the inventory carrying cost lower. With frequent and small size supplies, the customer service goal can be managed effectively. In such cases, the higher transportation cost will be offset by the inventory carrying cost which is much higher. This, in turn, enhances the profit margins.

Delivery Reliability and Consistency

The customer will always prefer on-time delivery, which is crucial to keep his production schedule on. The delivery ahead or beyond of agreed dates will spoil their relations with the customers. Faster delivery will increase inventory level at the customer end because he has to organize payments in advance. Late delivery will disturb his production schedules. However, as a result of proper planning, transportation modes and inventory availability will reduce the variance. In addition, the other important objective of logistics is consistency in delivery performance which will help in building customer confidence for keeping long-term relationship.

Lowering Freight Cost

In logistics, freight cost is the major contributor. The control measures to reduce freight cost are freight consolidation, selection of transport mode, route planning, load unitizing and long-distance shipments.

Minimum Transit Damages

Products may get damaged due to improper logistical packaging, absence of load unitizing and frequent handling of consignments. All this adds to the logistical cost. The solution is to use mechanized material handling equipment and suitable logistical packaging.

FUNCTIONS OF LOGISTICS

Logistics is an information-based activity involving movement of goods in the supply chain, mostly from suppliers to customers via manufacturing. However, this activity covers a variety of functions, which need to be managed properly in order to ensure effectiveness and efficiency in the supply chain of the enterprise. The logistical functions are shown in Figure 2.2.

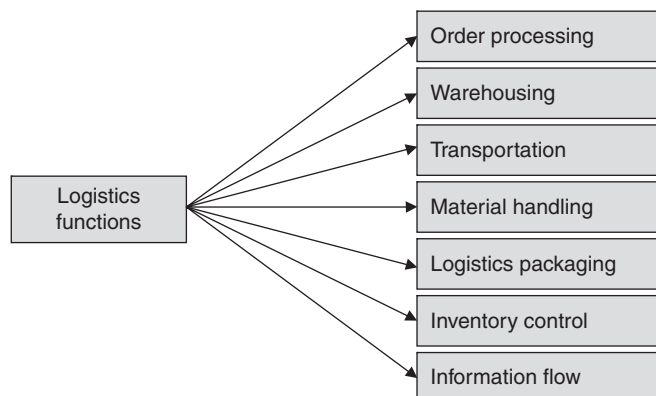


Figure 2.2 Logistics Functions

Order Processing

The purchase order placed by the buyer on the supplier is the base document to effect transaction between the two parties. This document covers details of the product (specifications) to be supplied, price, delivery period, payment terms, taxes and other commercial terms as negotiated and agreed. This document is processed further to make the ordered goods available to the customer at the right time and the place of his choice. The order processing activity includes the following steps:

- Checking the order (against the agreed or negotiated terms)
- Prices, payment and delivery terms
- Checking the availability of materials in stocks
- Production and material scheduling if stocks are not available
- Acknowledging the order indicating deviations, if any

The order processing time will have a direct effect on the committed delivery period and cost of transportation. The transportation cost to deliver the material fast will be higher so as to compensate for the time lapsed at the order processing stage.

In large-sized organizations where thousands of orders are received per day, the manual processing of the orders will create time delays and lot of errors. To avoid this, automated systems are always preferred. An electronic or IT-based system will register and process the order quickly and correctly.

Controlling Flow of Inventory

This function has two opposite goals. The first one is to keep enough inventory stocks to meet customer requirements; the second one is to keep the inventory carrying cost at a minimum. This exercise is to strike a balance between not losing market opportunity and not to increase the inventory cost so as to meet the same. The inventory carrying cost indirectly eats away the profits. The inventory carrying cost consists of cost of financing the inventory, insurance, storage, losses, damages and pilferages. In general, depending on the type of product, the average cost of carrying the inventory varies from 10 to 25 per cent of the total inventory per year. In case of perishable products, it is on the higher side.

To manage inventory there are two approaches. The first one is cost approach and other is customer satisfaction. Organizations try to strike a balance between the two. Companies adopt different strategies such as operating on zero inventory level by adopting JIT (Just-in-time) technique. But this is only possible with co-partnership between the purchaser and the supplier and communication between them on a real-time basis.

Storage or Warehousing

Warehousing is required because the demand–supply cycles never match. The inventory is kept in a warehouse till the order is received. Warehousing plays an important role in logistics operations of the organization. Warehousing can be used as a tool for competitive advantage. The effectiveness of the marketing strategies is often based on the decision of the warehousing. Warehousing is not merely a storage place in today's context; it is treated as a switching facility in logistics network. The delays in delivery of finished goods are an outcome of inefficient warehousing management. Therefore, warehousing is an important decision area in logistics. The major decision areas with regards to warehousing are as follows:

- Location of warehousing facilities
- Number of warehouses

- Size of warehouses
- Layout of warehouses
- Design of the building
- Type of warehouses (private, public or contractual)

The warehousing capability exhibits the ability of the firm to deliver a desired level of service to the customer. The ownership of the warehouse is private, public or contractual, and after looking into the advantages and disadvantages associated with each one of them, the organization has to decide on the best option. The decision will be based on the objectives of the business and the resources available.

Transportation

Transportation is at the heart of logistics. Transportation makes the movement of inventory happen from the supplier to the customer. Transportation refers to the physical movement of goods to the customers by using the various transportation modes available. The transportation cost as percentage of product price varies with the unit price of the product. For products with low unit price value, it is 20 per cent of the product price. In some cases, logistics costs varies up to 50 per cent such as in the case of mass-market products like salt.

Transportation modes are chosen by the company depending on the availability, reach and freight cost. Cost is the most important factor in the selection of a particular transport mode. However, under exceptional circumstances timely delivery of goods overrides cost considerations and the goods are sent through the fastest mode—although this is an expensive alternative.

In today's context, transportation being a non-core activity, firms generally do not own fleet but go in for outsourcing. The consideration may be based on initial investment, maintenance cost and the freight. Finally, depending on customer requirements, availability of transportation, infrastructure and its reach, and cost, firms decide on the mode with optimum cost under the given product-market conditions.

Material Handling and Storage System

Material handling plays a crucial role in speeding up of the inventory movement across the supply chain. Non-availability of proper material handling methods or equipment will add to product damages and delays in deliveries and incidental overheads. The productivity in logistics system can be enhanced by using mechanisation or automation in material handling equipments system. The other considerations for the selection of material handling systems are the volumes to be handled, the speed required for material movement and the level of service to be offered to the customer.

Storage system is important for maximum space utilization (floor and vertical) in the given size of a warehouse. The material handling system should support the storage system for speedy movement (storage and retrieval) of goods in and out of the warehouse.

Logistical Packaging

The packaging used to withstand varying conditions of warehousing, transportation and handling is called logistical packaging. It plays an important role in damage protection, ease in material handling and saves storage space. In logistical packaging, 'load unitization' has a major role to play to economize the packaging cost.

Box 2.1

Adding Value to Airport Logistics—Baggage Handling System

For flight passengers the greatest worry is their baggage; damage-free loading and unloading without delay. At Frankfurt Airport,⁵ the baggage handling system with its automatic baggage conveyor facility is unique in the world. It differentiates itself from the baggage handling systems in any other airports in the world in terms of its size, functionality and performance. It guarantees the airport staff a minimum connecting time of 45 minutes. Hence, the Frankfurt hub is at a true competitive advantage.

The conveyors are automated and are having 70 kilometres of conveyor routes. During peak hours, the system handles more than 99,000 baggage items per day. These items are transported reliably with speed to their destination points. On an hourly basis the system takes care of 18,000 baggages per hour with a transport speed of 2.5 metres per second. The automated system ensures very short transfer times keeping reliability rating of 99.65 per cent. It is a unique system, which differentiates itself in terms of scale, speed and reliability. The luggage is safely transported in coded plastic tray-type containers which are subjected to 100 per cent X-ray screening of non-EU baggage into the baggage conveyor system. The end result is customer satisfaction by eliminating baggage worries of air travellers.

Information

Logistics is an information-based activity of goods movement across the supply chain. Information system plays an important role in delivering superior service to customers by way of delivering the right product at the right place and at the right time. The use of IT tools for information flow support logistics and also enhances its competitiveness.

BUSINESS PROCESSES IMPACTED BY LOGISTICS

Logistics encompasses almost all areas of management. It deals with movements of inventory, information and also cash. The decision in logistics has implications in many management areas. The areas covered are product development, customer service, inventory management, transportation, warehousing, information systems and material handling. Ultimately, logistics is a supply chain enabler to achieve cost effectiveness and customer satisfaction. The following are the areas where logistics is vital to the success of supply chain operations:

Customer Order Fulfilment

Order fulfilment involves closing the customer order as per the terms and conditions negotiated and accepted by both the buyer and the seller. The order fulfilment is a process originating from suppliers and ending with customer via the company's manufacturing plant. The order fulfilment process includes traditional logistics activities such as filling and delivering orders, and it is the function of logistics that contributes resources to perform these activities. Logistics as a service function greatly contributes to the value created for customers by the order fulfilment process. Today technology such as WMS, EDI, TMS and RFID are being used to enhance the effectiveness and efficiency of the order fulfilment process.

Manufacturing Flexibility

In-process logistics management deals with the flow of inventory to and between manufacturing facilities of the company. To meet the market demand, the manufacturing activities of the firm should coordinate with its suppliers with the help of logistics. The degree of manufacturing flexibility in the supply chain is decided by the logistic capabilities of the firm. If the company wants to be truly flexible and if the degree of manufacturing is large, the organization will require appropriate logistics capabilities which should be implemented and managed accordingly. Logistics capabilities that fall short create hindrance in manufacturing flexibility. For example, for 'postponement strategy' the organization should have manufacturing flexibility and appropriate logistical capability to handle the customer service. For inventory decoupling points in postponement strategy what is required is trade-off analysis that includes manufacturing flexibility, logistics capabilities and costs.

Product Development Process

A new product development process requires close coordination with material suppliers, R&D and the key customers. Here the material flow is from the suppliers to customers via the company's R&D and pilot plant. The logistics process enables management to coordinate the efficient flow of ingredients and finally new products across the supply chain. Logistics plays an important role in the development of new products. In today's context, the product development process needs to be completed in a very short time frame. The new product should reach the market with speed to have the first mover advantage. All the elements of logistics such as transportation, warehousing, material handling and packaging are involved in the product development process. Logistics costs associated with sourcing, manufacturing and distribution of new products needs to be assessed for product profitability. Logistics plays a key role in successful product introduction after it is commercialized. It helps in product availability during the early stages of the life-cycle of products.

Reverse Logistics

In general, organizations design their supply chain for forward (flow from manufacturer to the end user) flow of goods. Most of the manufacturers forget their product responsibility after it is delivered to the customer. The manufacturers consider that their responsibility is limited to the extent of replacement of defective products covered under warranty period or products damaged during transit. They do not think about the used materials, left out materials, packaging waste or disposable wastes. The packaging waste and leftover material cause environmental pollution. Due to growing consumerism and concern about environment in the developed countries, they have regulations on the product wastages. They have passed legislation asking the manufacturers to take care of products discarded by their customers after usage. Many leading companies are taking this as an opportunity to develop competitive advantage. They are setting up system for reverse material flow. They are designing supply chains to take care of both forward and reverse flow of material. Reverse Logistics is becoming an integral component of retailers' and manufacturers' profitability and competitive position.

Logistics plays a major role in reverse flow of materials. Reverse logistics network can be used for various purposes such as refilling, repairs, refurbishing and remanufacturing. BMW (German auto-maker) has installed a factory to dismantle the used cars and dispose of the parts and components for recycling and refurbishing. They have established a system for taking back used cars from customers through the car dealers. In Germany, the FMCG companies have come together and established a system to take back the packaging wastes from the customers through the retail chains. Logistics plays a vital

role in reusing of packaging material in industries such as soft drinks, wines, oil and LPG distribution. Reverse logistics is integrated to their regular supply chain because of the reusable nature of the packages such as glass bottles, tin containers and metal cylinders. No extra transportation costs are involved in the process as the delivery van originates and terminates its journey at factory where these reusable bottles are refilled for re-delivery to the customers. The arrangement is like the hub and spoke distribution system. For consumer durables, the manufacturers establish the reverse logistics system not only for offering the free service during warranty period of the product, but also for extending the services beyond the warranty period on a chargeable basis. The other important area for reverse logistics is product recall in an emergency situation wherein the products distributed in the markets are called back to the factory due to various reasons. These may be: quality complaints, warranty claims, product not giving the guaranteed performance, product causing harm to human life, products crossing the expiry date, defective design to save the company image. In general, companies do not have reverse logistics system designed in advance. But, many firms have shown great deal of organizing ability to mobilize company's resource so as to achieve time-bound objectives. Business firms have always a forward logistics network in place. But they have to create reverse logistics network to take care of backward movement of used, unwanted, discarded and damaged goods. However, forward logistics provide the data and analytical skills necessary for performing a network analysis associated with the reverse flow.

Customer Relationship

Customer relations are developed by extending value added service to the customers. The decisions on customer relationships management are critical to the firm's success at any point of time. Management categorizes their customers based on their value over time and designs products and services to increase customer loyalty. The logistics function contributes to the customer relationship management process by providing tailor-made and value added services to key customers and channel members. The cost and revenue potential of each deliverable needs to be estimated in advance and suitable logistics service components are offered to each customer depending on their strategic importance and value. Logistics is an important source of competitive advantage and hence differentiating logistics capabilities needs to be identified to avail of the market opportunities and to develop continued customer relationships.

Supplier Relationship

A company needs to develop strong relationships with key suppliers, and manage transactional relationships with them. Logistics cost has an impact on the total cost of purchased materials. The procurement cost includes logistics costs which covers the expenses on transportation, inventory carrying, ordering, receiving and quality control. In addition, there will be a cost associated with service failures from suppliers. Thus, logistics plays a role in deciding the delivery performance of the suppliers. With the appropriate logistics strategy, there is great scope for improvement of supplier's performance and maintain good relationships with key suppliers. For superior performance on input side firms need excellent suppliers' capabilities with lower costs implications. Appropriate logistics involvement contributes to cost savings in procurement to the vendors' satisfaction.

Customer Service

Logistics is an activity impacting the customer service. For sellers to remain competitive it is necessary to respond to customer requests with speed. The delivery of orders to the customers at the right time

and place is a responsibility of the logistics function. However, this requires visibility in logistics, which depends on information systems for tracking product flow, detecting deviations and correcting the errors in failure of service. Hence, logistics capabilities of the organization should be considered, while designing appropriate response strategy for each standardized event to extend differentiated customer service. Regardless of the type of events, logistics frequently is involved in service-recovery operation. The examples are handling non-standard order, speeding of trans-shipping. Hence, by involving logistics in the customer service delivery process, the customer satisfaction can be gained even into fewer exceptions, emergency situations and faster error recovery.

Managing the Demand

The demand management process covers activities such as forecasting, synchronizing supplies, reducing demand variability, managing supply chain flexibility and developing contingency plans to take care of interruptions to supply or unexpected changes in demand. Logistics play an important role in the demand management process. It provides input to the forecasting process. Logistics communicates current inventory levels, existing logistics capabilities, and any known system constraints. Logistics also provides input on cost and recovery alternatives when products are in short supplies and requires allocations.

LOGISTICS DECISION HIERARCHY

The decisions in logistics impact the supply chain efficiency and effectiveness. The decisions in logistics are taken at four levels as shown in Figure 2.3.

The strategic level decisions are based on the services the firms are going to provide to the customers. For example, a pizza company may offer '30 Minutes Pizza' or 'Available 24x365'. To cater to these service levels the firm will have to design the distribution network so as to have access to the markets and customers. This includes the decisions on locations of distribution centres, network of channel members to ensure the proximity of supply points to demand points. The third one is the decisions on

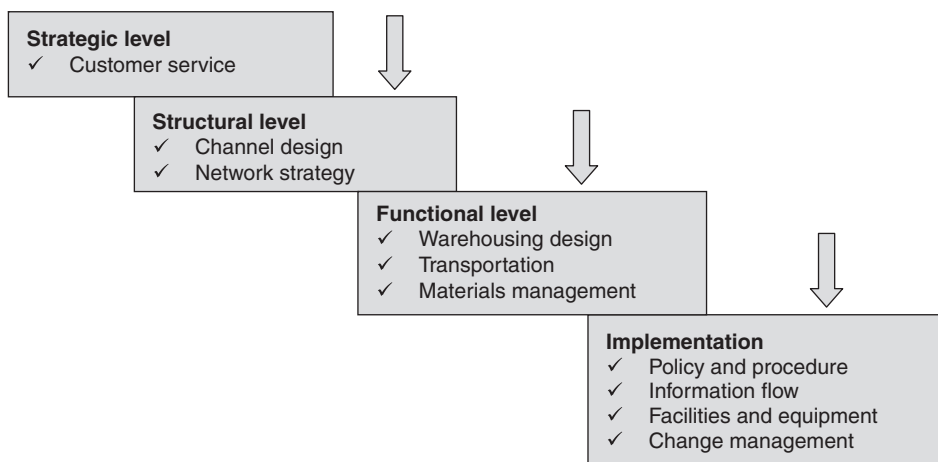


Figure 2.3 Logistics Decision Hierarchy⁶

functional areas of logistics, that is, warehousing (size, storage facility, material handling equipments and so on), transportation (freight consolidation, transportation mode, freight charges, route planning and so on) and materials management. The last one is the implementation phase, that is, who will do it? And how they will do it? Hence, in the organization culture, employee training and motivation are the critical areas in this phase. The areas in implementations phase are decisions regarding equipments and facilities, information flow procedures and policies to run the show.

LOGISTICS IN THE INDIAN CONTEXT

At present, 'outsourced logistics' accounts for only one-third of the total logistics market in India, which is a significantly lower proportion vis-à-vis the developed markets. Growth in this industry in India is currently being driven by over \$300 billion worth of infrastructure investments, phased introduction of VAT, the development of organized retail and agro-processing industries, along with a strong manufacturing growth. In addition, strong foreign direct investment inflows expected in the Indian markets, would lead to increased market opportunities for providers of logistics services in India. The primary growth drivers of this industry are as follows:

- **Investments in the infrastructure sector:** Government activities ranging from building of Golden Quadrilateral to roads and bridges in rural India has sent a wave of positive sentiment in the industry.
- **Streamlining of the indirect tax structure:** The introduction of value added tax (VAT) and the proposed introduction of a singular goods and services tax (GST) are expected to significantly reduce the number of warehouses. Manufacturers are required to maintain warehouses in different states, thereby resulting in a substantial increase in demand for integrated logistics solutions.
- **Robust trade growth:** Strong economic growth and liberalization have led to considerable increase in domestic and international trade volumes over the past 5 years. Consequently, the requirement for transportation, handling and warehousing is growing at a robust pace and is driving the demand for integrated logistics solutions.
- **Globalization of manufacturing systems:** Globalization of manufacturing systems coupled with advancements in technology are increasingly compelling companies across verticals to concentrate on their core competencies and avail the cost saving potential of outsourcing. This is expected to contribute to an increase in the need for integrated logistics solutions, which is the niche of every Third Party Logistics Service (3PL Services) provider.

In the Indian logistics industry, logistics managers of user firms need to realize that, with supply chains getting more and more complex, outsourcing part or all of their logistical activities to experienced 3PLs (Third Party Logistics Service Providers) will help reduce their overheads, streamline supply chains, reduce costs and improve service delivery.

India's spend on logistics activities—equivalent to 13 per cent of its GDP is higher than that of the developed nations (US—9.9%, Europe—10% and Japan—11.4%). The key reason for this is the relatively higher level of inefficiencies in the system, with lower average trucking speeds, higher turn-around time at ports and high cost of administrative delays, being just a few of the examples.

These inefficiencies have arisen over the years from a combination of a non-conducive policy environment, extensive industry fragmentation and lack of good basic infrastructure. India's indirect tax regime discouraged large centralized warehouses and led, over time, to fragmentation in the warehousing sector. At the same time, the absence of a single logistics 'champion' (whether in form of a

ministry or otherwise) in the government (or industry) led to a disintegrated approach to development of the sector.

Extensive fragmentation meant the incapacity of industry players to develop the industry as a whole and poor support infrastructure, such as roads, ports and telecom, led to a situation where the opportunity to create value is limited.

However, much of this is changing with the government now demonstrating a strong commitment towards providing an enabling infrastructure and creating conducive regulations. There is significant current and planned investment in infrastructure to the tune of (Rs 15 trillion) over the next few years and an increased emphasis on public–private partnership. At the same time, regulations around rationalization of tax structures and prevention of overloading, for example, are creating an environment of positive change. Players now have the opportunity to leverage economies of scale, complemented with better infrastructure, to provide integrated logistics solutions which are cost-effective.

In addition, the evolving business landscape and increasing competition across industries, is creating the need for more efficient and reliable logistics services than what exist today. For example, rapid growth of organized retail and the need to reach out to the large untapped rural markets in India are necessitating development of strong back end and front end supply networks.

In Indian logistics sector, major sector investors are FMCG, aviation, metal and mining, and consumer durables. Despite shortfalls, the Indian logistics industry is growing at 20 per cent vis-à-vis the average world logistics industry growth of 10 per cent. Since the organized sector accounts for merely 1 per cent of the annual logistics cost, there is immense potential for growth of the sector.

Many large Indian business houses such as Tata and Reliance Industries have been attracted by the potential of this sector and have established logistics divisions. They started providing in-house logistics services, and soon sensing the growth of the market, have started providing services to others as well.

Large express cargo and courier companies such as Transport Corporation of India (TCI) and Blue Dart have also started logistics operations. These companies enjoy the advantage of already having a large asset base and an all-India distribution network. Some large distributors have also forayed into the logistics business for their clients. There are also logistics service providers without assets; there is growing interest among entrepreneurs to venture into this business.

Indian shippers are gradually becoming more aware of the benefits of logistics outsourcing. They are now realizing that customer service and delivery performance are equally important as much as cost to remain competitive in this global economy.

The Indian government has focused on infrastructure development. Examples include the golden quadrilateral project, east-west and north-south corridors (connecting four major metros), Free Trade and Warehousing Zones (FTWZ) in line with special economic zones (SEZ) with 100 per cent foreign direct investment (FDI) limit and public–private partnerships (PPP) in infrastructure development. It is expected that infrastructure development would boost investments in the logistics sector.

In India, 100 per cent FDI is allowed in logistics whereas in China, until recently, foreign investment was not allowed in domestic logistics. Almost all large global logistics companies have their presence in India, mainly involved in freight forwarding. They have tie-ups with Indian companies for domestic transportation and warehousing. As the Indian logistics scenario looks promising, these MNCs are expected to play a bigger role, probably forming wholly-owned subsidiaries or taking the acquisition route. The latter may be the preferred route of investment since the target company is readily acquired with its asset base and distribution network, and the need for building everything from scratch can thus be avoided. The benefits for the acquired company include the patronage of an MNC and access to the MNC's global network. As an example, DHL Danzas, the biggest logistics company in the world, has taken over Blue Dart.

In spite of the challenges, Indian Logistics industry is continuously improving its performance in the global logistics industry by improvement of customs, trade-related infrastructure, inland transit, logistics services, information systems, and port efficiency, which helps to trade goods and services on time and at a low cost. The future holds opportunities for a better tomorrow.

SUMMARY

Today, logistics is widely used in every area. It can form the basis for a grocery inventory in mom and pop shop on one extreme, to complex activities in manufacturing of aircraft, on the other. Logistics is an information-based activity of moving goods from one place to another. Logistics solution needs to be tuned to the customer needs. Logistics optimizes the material flow in the company from suppliers to manufacturing unit and finally to the customers through the physical distribution system. It encompasses three functional areas of the business units: buying, manufacturing and finally distribution. The objectives of logistics management are optimizing the material flow, reduce the goods movement cost, customer service and improve the profitability of the manufacturing enterprise. At a macro level, it has an important role to play in the country's economy. However, at a micro level, logistics acts as a bridge between the customer demand and the goods supply. The efforts to improve productivity in logistic system will result in reduced operating costs. Logistics covers the functions such as order processing, warehousing, inventory control, transportation, packaging and information flow. The above elements are called as 'logistics mix'. The logistics management is concerned with the integration of the above functions to get system efficiency and cost-effectiveness in material flow across the supply chain. Logistics ensures the time and place utility of product to the consumer. Logistics is used as a tool for developing the competitive advantage in supply chain. Hence, to gain competitive advantage logistical system should focus on reduction in inventory, enhance responsiveness, minimize product damages, keep consistency in delivery performance and achieve freight economy. Integrated logistics helps in effective deployment of available resources. Logistics helps in enhancing the system productivity resulting in profitability and improvement in ROI. Today, to face competition, business firms are finding ways to develop competitive edge through logistics. Logistics has a very pervasive role in supply chain. Logistics is a supply chain enabler. It is present in material procurement (inbound logistics), manufacturing (in-process logistics), distribution (outbound logistics) and product returns (reverse logistics) management.

REVIEW QUESTIONS

1. Explain the domain of logistics.
2. Discuss the role of logistics in supply chain.
3. How is logistics used for gaining competitive advantage? Explain.
4. Discuss how logistics impacts the various business processes in an enterprise.
5. Differentiate between 'logistics' and 'supply chain management'.

INTERNET EXERCISES

1. Study the latest trends in logistics management at <http://www.logisticsmgmt.com>
2. <http://findarticles.com> is site publishing latest articles on logistics and supply chain management. Visit and study how logistics is impacting supply chain performance in various industry sectors.

VIDEO LINKS

1. Future Vision on Logistics, <http://www.youtube.com/watch?v=R4HPYYR5iLw&feature=related>
2. Transport & Logistics Management Introductory, <http://www.youtube.com/watch?v=fNGUnbtRHvA>
3. Nike logistics Center, <http://www.youtube.com/watch?v=NkfHVYv5nUo&feature=related>

PROJECT ASSIGNMENTS

1. Logistics cost as a percentage of product price is very high in case of commodity products with low unit price. Study the logistics system for cements and soft drinks and write a report on the strategies used for cost reduction in logistics activities therein.
2. Study the logistics system for the high value fashion garments for overseas markets.

NOTES

1. This definition was developed through Council of Logistics Management, Oak Books, Illinois.
2. Philip Kotler (2001), *Marketing Management*. Prentice Hall, Millennium edition, p. 537.
3. Christopher Martin (1994), 'New Direction in Logistics', *Logistics Distribution Planning*. Ed. by James Cooper, Kogan Page, p. 17.
4. Martin Christopher (1986), *The Strategy of Distribution Management*. Gower Publishing Co., England.
5. http://www.fraport-grounds-services.com/cms/default/rubrik/16/16165.baggage_handling_system-flash.html.
6. Anderson Consultants.

SUPPLY CHAIN VS. DEMAND CHAIN

3

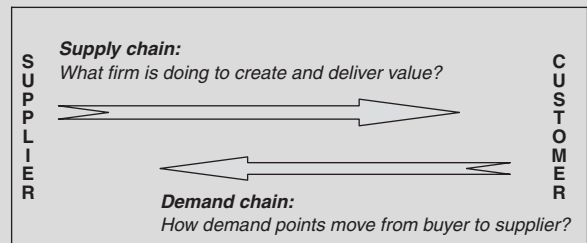
“The most recently introduced approach of demand chain management (DCM) seems to capture the proposed synergies between SCM and marketing by starting with the specific customer needs and designing the chain to satisfy these needs, instead of starting with the supplier/manufacturer and working forwards

—Heikkilä.”

Chapter Objectives

To understand:

- Concepts of value, demand and supply chains
- Value creation process
- Tuning of supply chain to have a proper fit with demand chain
- Supply chain strategies for competitiveness



Value chain

What Should the Firm Ideally Do to Create Value?

The value chain is the blueprint of how an enterprise can deliver value to the customer. It is the sequence of activities, which, if done properly, can make the entire delivery process cost-effective and productive. The supply chain of an enterprise is a 'value chain' in practice. Hence, effectiveness in supply chain is considered critical for companies to serve their customers better. The suppliers focus on re-engineering their supply chains for better operational efficiencies to add value. However, more value can be added if they also focus on the demand chain that transfers the demand from the market to the suppliers.

Supply Chain Innovation

In Practice...

Edu Value Publishers (EVP)—Novel Value Proposition

EVP is one of the educational book publishers in India. EVP is well known for its textbooks in the field of economics, commerce, management, computers, science, psychology and history. EVP has its presence in India and its neighbouring countries for the past 25 years. With an excellent infrastructure, committed staff, good management and branches all over the country, EVP enjoys premium position in the Indian publishing industry today.

The customers of EVP are typical bookstores and they decide which books to sell and, thus, go for planning. They estimate how many of them it needs and accordingly manage their inventory by placing its orders for appropriate quantities with publishers. EVP's value offering point is their warehouse or distribution centre, wherein they offer books for purchasing and then ship-to-order. This system may seem to work smoothly for them, but for students it works less well as books are expensive and some of the high-priced books are not necessary as teachers often assign only parts of the books. In this event, the bookstore underestimates demand; some students must wait for the publisher to make extra shipments. In such a situation, how should a publisher help its customers (bookstores) to help their end consumers, that is, the students?

EVP, in its bid to help students, examined the demand chain and shifted the value offering point. Although the retailer's planning process appears to respond to students' demand, it is really shaped by the teachers, who choose the reading lists for courses. EVP therefore moved its value offering point back to instructors, offering to tailor collections of reading materials (syllabus) for each of them. This is done by moving the order penetration point out of the warehouse and forward to the retailer. EVP publishing system (arrangements with authors) allowed instructors to choose standard EVP textbook chapters from a database and to add complementary materials (such as course objectives, instructions, test questions and cases) from a variety of sources. All of the readings are then combined into a single package that is printed, bound and delivered to the educational institutes.

With this strategy EVP is associated with various educational institutes in the country for the various programmes in developing their content. This assemble-on-demand system benefits both students and publishers in several ways. For students, most tailored textbooks are cheaper than their standard text books. Furthermore, printing technology makes the publisher's unit costs independent of batch sizes. In this deal publisher can avoid the cost of returns, which can amount to 30 per cent of sales. EVP has used this new value offering strategy to win over teachers on many college campuses.

INTRODUCTION

The pursuit of growth and the need to access new markets have been propelling business firms across the world to search for sustainable competitive advantage. This has led companies to greater customer consciousness and need for cost-effectiveness. The global nature of business has forced business organizations to recognize the critical role of back end operations, in today's dynamic business environment. As the business firms are focusing on production, marketing and finance, a greater attention is required

in achieving customer satisfaction through effective and efficient operations of various processes in the business. At the firm level, profitability, cost, productivity and market share are all indicators of competitiveness. Profitability is a major indicator of current competitiveness, although profitability is best measured over an extended period. The other parameter is customer satisfaction. But it is one of the cost drivers. To address the issue of competitiveness, the firm has to look into its value creation process (value chain) and see ways to translate it into effective and productive supply chain.

This is not enough. The firm has to look into the demand chain too, that is, the process that transfers the demand from market and customer to the supplier. The firm has to look into fine-tuning of supply chain to the requirement of demand chain to achieve a process fit (of supply chain) to achieve a sustainable competitive advantage.

VALUE CHAIN

Michael Porter first developed the concept of value chains in his work on competitive advantage. Value chain analysis, along with supply and demand chain analysis, is a gift for modern business management worldwide. Porter has proposed the value chain as a tool for identifying ways to create (Figure 3.1) more customer value. According to this model, every process is a synthesis of acquiring, adding and delivering value to its products. In other words, the value chain consists of vertically aligned activities that add value to the goods or services in moving from basic supplies to the finished product. In typical marketing value chain (Figure 3.4), the value is acquired through procurement activity, the value is added through manufacturing or processing, and value to the customer is delivered through logistics and sales distribution. The value to the customer is created through processes, activities, organizations and structures, all of which have to be aligned to get the changing perception of the value to the customer in the dynamic markets.

Porter suggested that the firms should continuously evaluate their business strategy to remain competitive. He further said that the firms should focus on their core competencies and should outsource the non-core activities to the experts. The experts can do the same job at lower cost and with speed. Here, Dell is the classic example. Dell asks their customers to configure the computing machine as per their requirements. The product is assembled at the nearest location of the demand and delivered within the stipulated or agreed time period. The parts are obtained from the suppliers. As the customer places the order for the machine, the suppliers get the bill of material simultaneously. With this outsourcing model, Dell has eliminated many business processes and reduced the costs and, as a result, offers its products at a competitive price.

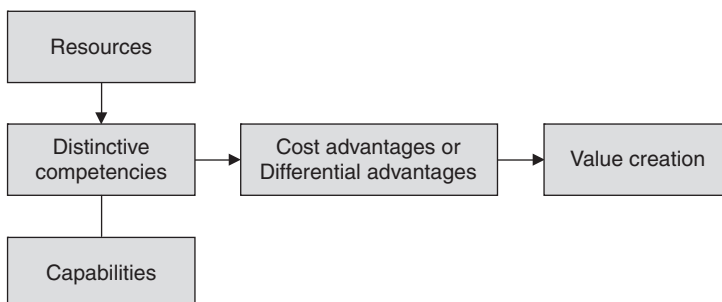
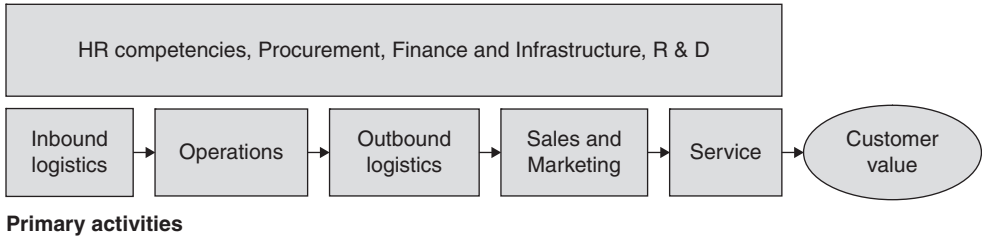


Figure 3.1 *M. Porter's Model of Value Creation*

Secondary activities**Figure 3.2** *M. Porter's Value Chain Model¹*

As per Porter's value chain model there are two types of enterprise activities: primary activities and support activities (Figure 3.2). With the help of primary activities there is direct creation of product or service and the delivery of the same thereafter. The primary activities include inbound logistics, operations, outbound logistics, marketing and sales, and service. The primary activities need supports for their effectiveness and inefficiency. The support activities are called secondary activities and these are procurement, technology development (including R & D), human resource management, and infrastructure (systems for planning), finance, quality and information management.

The primary and secondary activities together create value. The margin means the some total of all the value delivered by the business firm to the customers. The 'margin' indicates the degree of differentiation in the way a business is conducted and value is delivered.

The product differentiation and value creation strategies depend on the policy decisions of an enterprise. For example, the pizza companies offer '30 minutes pizza' strategy for competitiveness. Thus specialization depends on the ways the same activities are performed differently. The firms can create competitive advantage through its own internal activities. As per Porter, these activities are categorized into primary and secondary activities. However, the importance and criticality of the activities depends on the industry in which the firm is operating.

All the activities indicated in the value chain helps to create customer value. However, the value is assessed by the customer and his willingness to pay the price is the measure of value. A business firm can earn the profits only when its selling price exceeds the cost of doing all the activities to produce a product. For a firm there are two options for being competitive. The first one is to provide comparable (with competition) buyer value at lower cost but perform activities more efficiently than its competitors. The second one is to do the activities in a unique way that generates greater buyer value and customer is willing to pay a higher price than a competitor.

The competitive advantage can be gained by way of doing things differently than the competitors. The firms may use new technology, new processes or new procedures which will ultimately reduce the cost and time. The firm may reorganize its value chain and its linkages to incorporate the effectiveness and efficiency to deliver the differentiated value to the customer. It may have to optimize the activities and processes and create trade-offs in performing different activities to create value. For example, product design with expensive components and quality inspection may reduce after-sale service costs as the breakdowns and repairs are rare. The trade-off decisions are dependent on the strategy to achieve competitive advantage. In fact, strategy is a guide map for a firm to perform individual activities and organizes its entire value chain.

To reduce the cost in chain the activities need to be coordinated. The transaction cost gets substantially reduced if the activities are coordinated. The coordination of activities allows better flow

of information for control purposes. The management of activities and their linkages are in fact at the base of competitive advantages. Thus, for creating competitive advantage the management of the entire value chain as a system rather than management of its parts is a must. To bring about the major improvements in the value chain what is required is relocating, reordering, regrouping or even eliminating activities.

The value chain is the tool for the company to know the sources of reducing the cost and enhancing the value simultaneously. It gives a fair idea of various value chain linkages wherein the optimization is possible. The value chain also reveals sources of differentiation.

In essence, 'customer value' is the satisfaction of customer requirements at the lowest total cost of acquisition, ownership and use. It is achieved when requirements are met fully, reliably and cost effectively. The primary focus in value chain is on benefits that accrue to customers. These are interdependent processes, which generate value and finally generate demand. Effective value chain generates profits.

Value chain analysis² is a useful tool for working out how organizations can create the greatest possible value for the customers. Value chain analysis helps the businesses identify the ways in which they can create value for their customers, and then helps them think through how they can maximize the value. The available alternatives are superb products, great services or jobs well done. Value chain analysis is a three-step process:

- **Activity analysis:** To identify the activities which should be undertaken to deliver product or service.
- **Value analysis:** For each activity, to think through what the business would do to add the greatest value for its customer.
- **Evaluation and planning:** To evaluate whether it is worth making changes, and then plan for action.

By using value chain analysis and by following it through the action, organizations can achieve excellence in the things which are demanded by the customers.

DEMAND CHAIN

It relates to the activities that transfer demand from the market to the suppliers. The demand generation and the subsequent fulfilment resulting into placement of an order on supplier in the customer activity chain is called a demand chain (Figure 3.3). Ideally the suppliers should keep close partnership with customers to create and deliver value. But this does not happen with everyone. The reason is the absence of integration between suppliers and customers. The integration process involves investments in information technology.

However, the emergence of Internet technology and the adoption of the ERP system helped firms exchange data with speed and ease.

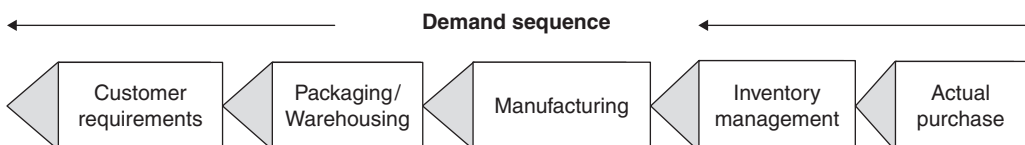


Figure 3.3 *Customer Demand Chain*

Demand chain management is the management of upstream and downstream relationship between supplier and customers to deliver the best value to the customer at the least possible cost to the demand chain as a whole. Demand chain is referred to as customer pull. Demand chain management as a concept aims to integrate supply and demand processes in order to ensure customer value creation and delivery. DCM helps to improve an organization's processes by advancing the coordination between the supply chain and demand-driven activities. The role of marketing within demand chain management is to facilitate process integration by providing customer and market information.

For example, to order a customized car, customers would be enabled to configure their own vehicles, add options, select colours and accessories, and understand how these decisions have an impact on product price.

SUPPLY CHAIN

Supply chain is basically a value delivery chain. It is a process with sequential activities to deliver value to the customer. In supply chain, value is acquired from vendors, then value is added through firms processing/manufacturing and value is delivered through firm's distribution system (Figure 3.4). Supply chain management is an approach for controlling the physical flow of products from source to point of use by aligning the capabilities of suppliers, manufacturers, channel partners and customers. Supply chain management is also a tool to achieve sustainable competitive advantage. It supports both differentiation-based and cost-based strategies. The supply chain management approach contributes to world-class performance by progressing beyond functional excellence and cross functional integration. It focuses on the practices used by the different businesses that together produce the product and service for the customer. It integrates the activities of all members of the value-added chain to produce higher levels of performance than can be achieved individually. Supply chain management practices create supply chain integration that yields superior business performance.

Supply chain integration utilizes a variety of business practices such as just-in-time manufacturing, quick response and continuous replenishment. With the supply chain initiatives and implementation of strategies, the firms increase productivity and value at all stages of the chain, from source to delivery. The integration of the supply chain helps in cost reduction, speed to market and, ultimately, customer satisfaction. The integration process helps in reducing wastages and delays and enhances the value to customers by coordinating activities to reduce costs for all participants in the supply chain. The supply chain approach adds value through innovations. It creates new markets and builds trusted long-term relationships with customers.

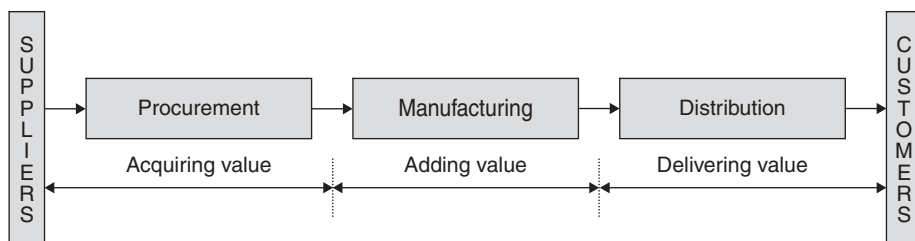


Figure 3.4 Supply Chain—Value Addition

Supply chain management focuses on operational efficiencies, and tends to be cost-orientated while marketing is more concerned with revenue by focusing on the demand side of the company. Both marketing and supply chain management together determine the company's profitability. From a marketing perspective, effective marketing strategy implementation demands efficient supply chain management, because it includes the distribution component of a marketing strategy. In addition, in the customer-centric marketing approach there is a need for marketing to become responsible for supply management. In markets with increasing diversity in customer needs, companies will have to rapidly adjust their supply chains to meet demand leading to a more sustainable competitive advantage. This may not be possible by just offering superior customer value propositions, but the firms will have to adopt a unique business system to support it.

SUPPLIER-CUSTOMER CONNECT

Ideally, the suppliers should look into the performance enhancement of their client's operation. But how many suppliers are doing that? The answer is only a few. All efforts with regards to re-engineering of the supply chain are done by suppliers by reducing obsolete inventory or inventory in general, cutting through non-value added activities and times associated with them. A supplier can adjust its supply chains in ways that improve the services to be delivered to its customers and thus enhance the customer's performance. This is a difficult re-engineering task.

The changes normally suppliers implement at his end do not add much value from the customer's point of view. A supplier may reduce its inventory level by pruning the product mix or product portfolio or evolving some new inventory strategy. This may not help the customer. Even changing strategy from 'make to stock' to 'assembling to order' increases delivery times. The beneficiary here is the supplier and not the customer.

The supply chain of a packaged consumer goods manufacturer, for instance, comprises manufacturing, packaging, distribution, warehousing and retailing. The concept of the customer's demand chain, which transfers demand from markets to suppliers, is significantly less familiar. To give one example, a retailer's demand chain would consist of assortment planning (deciding what to sell), inventory management (deciding the quantity of supplies needed), and the actual purchase. Together, these two chains (Figure 3.5) form the supply-demand chain. They are connected at two places: one the 'order placement point' and two the 'value-offering point'.

The order placement point (OPP) is the place in the supply chain where the supplier allocates the goods ordered by the customer. The manufacturer may manufacture the goods after the order is received in case of a 'make-to-order' policy. The manufacturer may allocate the inventory from a warehouse once the orders have been received if the inventory policy of the company is 'deliver-to-order'. Each order placement point has different costs and benefits for the supplier and its customer. When the supplier allocates orders from its distribution centre, it can deliver them quickly if they are in stock. Rapid delivery (a benefit for the customer) therefore depends on holding a large inventory (a cost for the supplier). Of course, the wider the product range, the bigger the inventory; so the supplier either incurs large inventory costs to minimize delivery times or cuts inventory and risks delays in fulfilling orders.

To add value to customers operation, it is possible to move the order placement point back to packaging or assembly, wherein the goods are turned into finished products.

With 'pack-to-order' policy the supplier will have a benefit of lower inventory expenses. However, the customer will have to wait for the goods to be packaged. The time lost in waiting is cost to customer.

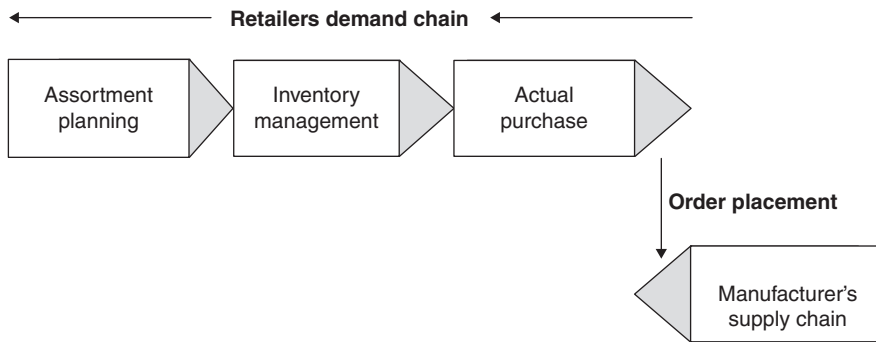


Figure 3.5 *Retailer Demand Chain*

To reduce that delay and extending benefit to the customer, the supplier must absorb the cost of additional packaging capacity. The further advancement in demand chain is to produce (for the supplier) to meet the specifications of individual customers. It is beneficial to the customer but the delivery time goes up. It may be a costly proposition to the customer due to increased delivery time. The supplier's process efficiency declines each time a customized design replaces a standard one and that results into a cost for the supplier. In this case, both supplier and buyer will have similar cost implications.

For the supplier it becomes difficult to keep up the delivery time if the order placement point is moved backward in the supply chain. It may save the cost and lower the inventory level but the result may be reduction in sales volumes, longer delivery time and higher total costs for customers. Customers and suppliers never benefit equally.

The supplier can add value to customer operations from various points in their supply chain, where the supplier fulfils demand in the customer's demand chain. In general, seller can add value at purchasing department to offer inventory. He can further move in buyer's demand chain and can offer to manage buyer's inventory, that is, the vendor management inventory (VMI). In this case, buyer will have an advantage of having no inventory burden. It may be vendor managed inventory with JIT supplies. The seller can further move and plan the inventory requirements of the buyer. Suppliers are expected to use this kind of collaboration to improve their delivery performance. The result is a more profitable

Exhibit 3.1 *Supplier Inventory Strategies*³

<i>Supplier's Inventory Policies</i>	<i>Customer</i>	<i>Supplier</i>
Make-to-order	• Delivery as per lead time • Higher cost to customer	• Keeps no inventory • Process efficiency goes down • Finished goods 'turns' less • Cost to supplier
Assemble-to-order	• Customer to wait for delivery	• Lower finished goods inventory • Flexibility in producing product mix
Make-to-stocks	• Gets quick delivery • Benefit to customer	• Keeps large inventory • Cost to supplier

use of warehousing space (converted to productive use) by the buyer. But in such a case, the supplier may ask for quick payment for the inventory consumed or used in production.

The last one, 'offer to end user', is like the Dell strategy, wherein the seller eliminates all transaction-related costs and channel costs. Dell went all the way back in the demand chain to the end consumer by fulfilling orders with direct-sales model for clients.

The suppliers can manipulate the supply-demand chain to improve customers' performance and benefit them. The suppliers can also use this approach to discover completely new value propositions for customers. Suppliers can thus extract new value from current accounts by escaping the commodity trap and also can find new customers.

To benefit both the seller and buyer the value offering points should be moved back in demand chain of the customer and order placement points should be moved forward in supply chain. Moving order placement points will result in benefits to one party only. However, by movements in both the demand and the supply chains, suppliers can improve their customers' performance and bring efficiency in their own operations.

In vendor managed inventory situation, the supplier is moving value offering point to reduce inventory burden on the buyer. Here the order placement point is the manufacturing operation of the customer. The seller gets a place in the customer's plant to stock the inventory in small quantities. He may incur more expenses in transportation in small consignment and integration, but he gets his payment at the end of the day—a win-win situation. In fact, managing the customer's inventory gives the manufacturer much earlier access to information about that customer's demand. Such access permits the manufacturer to cut its own inventory costs by packaging products on order rather than stocking all its products at a distribution centre; none of this comes at the expense of the customer's delivery times.

By pushing value offering point to customer, Dell profitably offers excellent service to end users and simultaneously it has pushed the order placement point back in the supply chain by assembling PCs to order. In this process, Dell gets prior information about customer demand. This gives the company time to assemble PCs out of the front end of the process. In this strategy, Dell has eliminated inventory costs and can buy components later than its conventional competitors by taking advantage of falling prices.

Thus by manipulating supply-demand points in the chain, suppliers can offer their customers completely new value propositions and improve their supply chains performance. This can improve the customer service and also its own operating cost. The re-positioning of the value offering point at the customer's end of the supply chain and the exploitation of better information about customer demand extend benefits to both buyer and seller.

SUMMARY

In today's markets, understanding the customer and responding effectively to his changing needs through the coordination of marketing and SCM can be a source of superior value creation. Demand chain management as a model which combines the strengths of marketing and SCM by shifting the focus to the customer and designing customer-centred supply chain. Marketing is, traditionally, externally focused and creates customer value, while SCM is inwardly focused and concentrates on the efficient use of resources in implementing marketing decisions. Marketing and SCM integration is between those that define demand with those who fulfil it.

Until today, the concept of demand chain has been addressed from SCM and operations perspectives; however, despite its clear relevance, no marketing contribution can be cited. By outlining the roles of marketing in demand chain, gaps in value offering processes can be addressed. Widely cited examples of successful companies following the principles of DCM are Dell in the computer industry and Zara in the fashion industry. This leads us to believe that more companies will adopt demand chain in their quest to gain competitive advantage. These companies increase profitability through product availability, delivery accuracy, responsiveness and flexibility by closely linking customers and suppliers. Within demand chain, marketing and supply functions work together to develop suitable relationships for different customers, develop joint customer prioritization strategies, process accurate customer information and match value requirements with operational capabilities.

REVIEW QUESTIONS

1. Discuss the concept of 'value chain' and how it helps enterprises to gain competitiveness?
2. Supply chain is 'value chain' in practice. Explain
3. Dell reversed the 'value chain', that is, Dell started with customers rather than their suppliers. Study the implication of this value chain model on the supply chain of Dell.
4. Explain the concept of 'order placement point' and 'value offering point' and how these are manipulated to offer superior customer service leading to competitive advantage.

INTERNET EXERCISES

1. To study the Porters value chain concept and also value creation index of a business enterprise, visit http://www.valuebasedmanagement.net/methods_porter_value_chain.html
2. Visit <http://www.marketingexperiments.com/improving-website-conversion/value-proposition.html> and search for value proposition

VIDEO LINKS

1. Demand Chain, <http://www.youtube.com/watch?v=WJFTfOPLf-8&feature=related>
2. Supply and Demand Chain Management, <http://www8.hp.com/us/en/solutions/solutions-detail.html?compURI=tcn:245-341953>

PROJECT ASSIGNMENTS

1. Arvind Mills had offered the scheme of 'Stitch-a-Trouser' within an hours time. Study the supply-demand chain and prepare the report on how the order placement points and value offering points are adding value to customer offerings.
2. Study the automobile industry supply-demand chain and prepare a plan to enhance customer value offerings by selecting proper order placement and value offering points in the chain.

NOTES

1. Porter, Michael E. (1985). *Competitive Advantage Creating and Sustaining Superior Performance*, The Free Press, p. 37.
2. 'Value Chain Analysis, Achieving Excellence in the Things that Really Matter', http://www.mindtools.com/pages/article/newTMC_10.html
3. Dixon R. David, *Lean in the Job Shop*, <http://www.technicalchange.com/Lean-in-the-job-shop-tracks.html>

VALUE CREATION THROUGH SUPPLY CHAIN

4

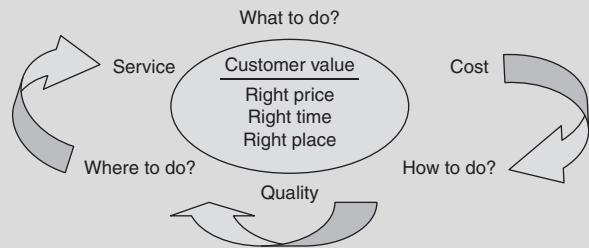
“You can have it in any colour as long as it is black

—Henry Ford”

Chapter Objectives

To understand:

- Dimensions of customer service
- Customer service attributes and phases
- Value-added services
- Role of customer service in supply chain strategy
- Customer service perception and service design



Customer Value Variants

For the manufacturing and distribution companies in the past, the need for excellence in supply chain management (SCM) has never been greater. However, the increasing complexity of markets and greater competitive pressure are forcing companies to take a new view of SCM. The days of looking at SCM as a cost centre are gone. Today SCM is a strategically important way to create a continuous new customer value. It is a paradigm shift that companies can no longer ignore customer value proposition through SCM. Supply chain is a source of value creation and gaining competitiveness thereof.

Lijjat Papad—Creating Value Without Technology Support

Shri Mahila Griha Udyog Lijjat Papad Cooperative Society is recognized by the Khadi and Village Industries Commission of India. It is a unique organization of the woman, by the woman, for the woman started in 1959 by seven women. Today it boasts of being a 42,000 women member organization across the country. They make papad, a crisp Indian bread. The cooperative now has annual sales exceeding Rs 3,000 mn (\$64 mn) with 63 branches and 40 divisions spread across 17 Indian states.

Procurement

The raw material is centrally procured at the head office. Dal is procured from Myanmar, heeng (asfoetida) is imported from Iran, black pepper is purchased from Cochin, dal is processed at Lijjat's own mills at Vashi and Nashik. They work on economies of scale.

Processing

In a Lijjat branch, the activities of the groups start early in morning. One group of women goes to the branch at 4:30 AM every morning to knead dough. At 7:00 AM dough is collected by a group of other women, who roll it into papads. When these women come in to collect the dough, they also give in the previous day's production. Every day the papads produced the previous day are sent for a quality test at 10:00 AM. The tested papads are packed by another team. Every member gets her share of vanai (rolling or labour charge) every day for the work she does and this is possible only because the rest of the system is geared to support it.

Packaging

Three core teams: kneading (dough), rolling and packaging convert the raw material into packed finished products. Two pack sizes are used for packing papads—50 and 100 pieces. The boxes full of 13.5 kg papad packets are sent to the depot. Papads are made according to demand to avoid inventory storage. Strict quality checks are done on raw material and final products for maintaining consistency in the quality of the product.

Distribution

For the final distribution, around 400 boxes from 3 to 4 branches reach the depot every day. The material is sold through independent distributors with allocated geographical areas in the country. Lijjat Papads are exported to more than 15 countries. In 2003–04, their export revenue was Rs 120 mn.

The cost of production for papads is very low and the production is decentralized. The use of technology in the production of Lijjat Papad would go against the basic principle of low skill production technique and decentralized production.

Lijjat was formed with the 'value system' of encouraging self-employment, dignity of work, earning capacity of women working from home and Mahatma Gandhi's concept of trusteeship. Lijjat is a unique supply chain creating value without using any technology.

INTRODUCTION

The value perceived by the customer is always in terms of cost and/or service. If the value perceived by the customer is high, he will be willing to pay more. The marketer should have the focus on the market where the value of the product/service is decided.

According to Peter Drucker, the purpose of any business is to create and maintain satisfied customers. The firm can earn profits only if it can create value in its products and deliver the same to the customer. Profit is the price paid by the customer for the product minus the cost of producing the goods. Hence, the price should always be greater than the cost. Thus, it is imperative for the firm to create value for its survival in the market. A market-oriented company always follows this concept. Customer value can be created through efficient and cost-effective operations of supply chains.

In olden days of mass production, people were just happy to buy products. This was true until Model-T was introduced by Henry Ford in 1937. The concept of choice was unknown to both customers and the manufacturers. However, today consumers demand more choices, higher quality, better price points and anytime anywhere delivery. Customers are spoilt for choices today and want value for money. Hence, value proposition through supply chain is important today.

M. Porter evolved the concept of value chain, which is the combination of nine generic value-added activities of the company. These activities work together and provide value to customers. The net result of value chain activities is profits if activities are performed cost effectively.

CUSTOMER VALUE

In today's dynamic markets, ensuring customer satisfaction demands the most visible and flexible supply chain. Customer value is defined as the benefits derived by the customer from the product or service at the lowest cost of acquiring, possessing and using of the product.

In other words, customer value is understanding of customers' desire and aligning internal resources to respond to those needs effectively and efficiently. In simple words, it should have customer focus and subsequently customer support. To deliver customer value, organizations need to focus on innovation, quality in product and service, maximum utilization of available resources, skill sets of the employees and infrastructure to create the value. In general, value is a combination of a company's image, quality of product or service, and the price at which it is available. Quality service is the most important of all the factors which create value. For example, Nike created superior customer value by evolving unique footwear designs, maintaining consistency in quality, creating right image through its brand value, and filtering spurious goods by distributing the products through Nike outlets.

From the customer perspective value is based on two main dimensions: (1) solution to his problem and (2) his experience of solving the problem. From the problem-solving dimension, the value starts at a point, which is beyond the augmented product. From the experience dimension, value creation starts at a point beyond confidence building based on a safe and trustworthy experience that relies on respect, honesty and fairness. The customer value if managed properly will maximize the business prospective through customer satisfaction by catering to the needs of the customers. Customer service is a value producing activity in supply chain process. The value is intrinsically related with all activities in a supply chain and customer service is one of the vital and important activities in the firm's value chain.

The 'bundle' is made up of customer services, logistics and other value-added ancillary services or products attached to the core product, and this is the 'bundle' which determines the 'value' of the product. By looking at the product in this way one can recognize that 'customer service' is definitely a value adding activity, which makes significant contribution towards the total value attributed to a product.

Box 4.1

DECATHLON: A Global Supply Chain Creating Customer Value in Sports Goods

Decathlon is a sport hypermarket in Europe. It was founded in 1976 in France by Michel Leclerc with the objective to offer sport goods at the most competitive price. In 1986, the company had gone international in order to find a broader potential market and to decrease costs through the economies of scales and for cheaper cost of labour forces. Decathlon decided to create its own products, through the 'Decathlon Production' subsidiary. Decathlon's products represent today 50 per cent of the hypermarkets' turnovers but the firm has not broken with its traditional profession that is the distribution of other brands like Nike or Adidas.

In 1996, Decathlon launched 'Tribord' the brand for water sports, 'Kipsta' for collective sports, 'Quechua' for mountaineering, 'Inesis' for tennis and golf, etc. This strategy of specialization is a continuous process. These they call as 'Passion Brands.' Decathlon's value chain comprises of the main activities and support activities of the company. Within the main activities are the internal logistics (procurement, storage), the manufacturing (includes the packaging), the external logistics (collection, storage and physical distribution of the products to the customers), sales and marketing, and the services (equipment, maintenance), whose total costs include the labour force, the machinery and its running, and the service costs. The support activities include the firm's infrastructure, the human resources management, and the technological development (Research and Development).

Manufacturing remains currently one of the activities, which is subcontracted. Decathlon used to subcontract production to small industrial firms and businesses in the North of France in the past, but today is relocated to countries where the labour cost is cheap such as Morocco, Tunisia and Taiwan. Under 'Decathlon Production International', more than 52 per cent of the products are manufactured abroad. However, the designing of the product remains Decathlon's specialty. Distribution is entrusted to the hypermarkets, often located near highway exits or in the heart of commercial areas.

Decathlon banks on internalizing all steps of the supply chain including design process. Decathlon, however, subcontracts the production, in order to avoid heavy investments. The so-called 'vertical integration' also enables the company to limit the suppliers' pressure (Nike, Adidas) and to distinguish itself from the competitors.

Decathlon invested (2008) in an Information Technology system called 'FlexPLM' in order to radically optimize the development process, from the designing of the product to its distribution. 'FlexPLM' system with its high-performance web platform improved the interoperability and communication between the various working groups. This gave a higher visibility in the supply chain process and a considerable gain of time in terms of information flow. Decathlon is truly a global supply chain company maintaining its exclusivity in sport goods markets with its specialized supply chain strategies. It is a well-integrated supply chain with synchronized processes on technology platform and implementing specialized supply chain strategies to keep its leadership position in the sports goods market.

Customer value in the context of SCM is of fundamental significance. This is because SCM is focused on providing system wide optimization of resources while producing superior customer value. The traditional concept of logistics management, was focused on reducing the transportation and distribution costs of the firm. But today, the logistics initiatives are concerned with customer service and

Box 4.2

Telematics: Adding Value by Capturing 'Customer-Life-Cycle' and 'Vehicle-Life-Cycle'

This is a new wave of technological breakthrough and it is applied in automotive industry. This is a customer management on demand side. In the automotive industry, this is an enhanced value delivery system through innovative customer service using advanced technology–telematics. Over the years, customer preferences and changing lifestyle are demanding a shift in the needs of car features. The telematics devices are on in growing in demand. These devices offer navigational, rescue and emergency services, remote diagnostics and so on. The value delivered through telematics would achieve 'customer lockin'.

its impacts on value. The concept of value needs to be appreciated in a much broader context in the supply chain framework where optimization and value creation extend beyond the firm's four walls to include other partners in the supply chain.

The concept of customized logistics service has been developed to emphasize the integration of logistics services with the core product in order to give each customer a value-added tailor-made product to suit his needs. Hence, the logistics service has become a key differentiator of a commodity product, wherein there is very limited scope of differentiation.

'Value' is a subjective concept. It depends on the context too. For different customers the understanding of value differs. Thus, 'value' is determined by customers in the complex trade-off calculations. The challenges involved with creating a logistically distinct business are many and the success depends on the understanding of the customers and their needs. Hence, effective SCM requires customer segmentation based on their requirements and to implement a service strategy to these segments.

CUSTOMER SERVICE FOR COMPETITIVENESS

Customer service is a set of activities and programmes initiated by the firm to make the buying experience more rewarding. These activities increase the value of the product and services offered to the customer from the seller. Customer service is the most important dimension of the product or service offered to the customer. Value-added customer service is leveraged to gain competitive advantage. Good customer service builds customer loyalty amongst the existing customers and generates positive word-to-mouth communication, which will attract new set of customers. In other words, customer service is the basis for 'Customer Relationship Management' programmes, which many leading firms are implementing.

Customers evaluate services based on their perceptions. These perceptions are fulfilled by the actual service provided by the firm. These perceptions are affected by a high degree of intangibility and are therefore, very hard to evaluate accurately. The firms who can identify these gaps accurately and bridge them with value-added services will succeed in enhancing customer satisfaction level and remain competitive. Exhibit 4.1 lists the few cues affecting the customer perception of service quality. These cues on customer perception on the quality of service stimulate the satisfactory or unsatisfactory evaluations of the firm's services. Hence, the firm needs to bridge the service gaps (between the perception and actual) with its services to improve the customer satisfaction. These gaps may rise due to communication, standards adopted, service delivery and customer knowledge (Exhibit 4.2).

Exhibit 4.1 *Cues Affecting Customer Perception of Service Quality²*

Competence	Information about the firm reaching to customers through product brochures, manuals, Internet websites, sales talk and the product or services offerings of the firms.
Reliability	Product or service delivery in terms of place, time and quality.
Responsiveness	Responding to customer telephone calls, emails, faxes and letters within the shortest possible time frame and resolving the customer problems or complaints in shortest possible time period.
Transaction security	This relates to the confidentiality of the transaction and customer information
Trustworthiness	This can be built up with evolving policies on product return, warranty and guarantee, and honouring commitments,
Access	Ease with which customer has an access to information on firm's product or services, his order status, product complaints, claims, damages in after-sales phase.

Exhibit 4.2 *Customer Service Gaps³*

Communication	Difference between actual service provided and what is communicated through the media.
Standards adopted	Difference in service standards adopted and the standard as per customer's expectations.
Service delivery	Difference in actual services provided and service standards of the firm.
Customer Knowledge	Lack of knowledge about customer's perception of the service standard.

The firms should make a note of these gaps and make efforts to fill them in order to satisfy customers and have a large customer base. Supply chain is the process whereby a large share in an organization's success can be achieved by delivering the product at the right place and at the right time. For physical products, it may be a manufacturing plant, a wholesaler, a retailer or an end user. For service products, it may be a point of sale or service like hotels, banks or hospitals. However, the focus here is on customer service, irrespective of the different ways, means and methods to achieve this. This approach requires the marketing orientation to all the functional areas of the organization. The logistics helps in creating the time and place utility of the product, which satisfies the customer need, which has both time and place as its dimensions. In the today's competitive markets, the success of the firm is judged by, how efficiently and effectively the firm manages these (time and place) dimensions by availing the existing opportunity and creating new opportunities for repeated sales to the same customers by providing superior service. Hence in supply chain process, logistics competency is critical to the customer service planning and needs to be developed as a core competency for developing sustainable competitive advantage.

SERVICE ATTRIBUTES

Efficient distribution is making the right product available at the right place and at the right time. This should be followed by after-sales service, if necessary, depending on the product and as per industry practices. In general, the product distribution is dependent on the length and breadth of the channel. The logistics programmes should be designed (customized) to the requirements of the channel partners.

For these programmes to be successful, the channel members need to be serviced and motivated by the manufacturers. The services extended to this trading community require different attributes than those required for satisfying the end customers. The customer service attributes have two variants, one distribution and second the trading. The distribution aspects cover: order processing time, delivery consistency, delivery frequency and stock availability. The trading aspect covers credit facility, stock financing, service support and handling convenience.

Order Process Time

It is an important measure of customer service in SCM. It is the time between the placement of an order by the buyer and the supply of the material by the seller against order. This involves the supply of all the materials against the order placed in the agreed time frame without any error either in documentation or physical supply. This attribute helps in building a long-term buyer-seller relationship. The order processing time consists of the time required for registration of order in supplier's system after thorough technical and commercial scrutiny, material allocation and pickup from the work-in-progress inventory, warehouse, or distribution centres, packing of material, documentation and dispatch of material. A well integrated activities based on information flow will shorten the order process time and help in speedy delivery to the market.

Delivery Consistency

Customers do not want any deviation in delivery period irrespective of the number of trips or transactions. Delivery consistency means maintaining the same delivery period for delivering materials to the buyer over a period of time. For example, against the 50 repeat orders over the year, the supplier dispatches the material as per the agreed delivery time for 48 orders, in this case, the delivery consistency is 96 per cent. This is an excellent delivery performance. The delivery consistency speaks about efficiency and effectiveness of the seller's supply chain. In fact, to cover up inconsistency in deliveries, the buyer may carry excess inventory of raw materials and components and block more funds resulting into costs.

Delivery Frequency

The customer does not want to carry excess inventory but wants his operations to run without interruptions. As a result, the customer prefers frequent deliveries in small lots. This may increase the transportation cost, but reduces the inventory-related cost drastically, with a net effect of overall reduction in overall supply chain cost. The additional transportation cost may be borne by the supplier to compensate for the buyer's inconvenience. This practice of frequent deliveries of small lots is quite common in retail chains.

Stock Availability

With the excess stocks, the supplier may extend excellent service to the customer, but inventory-related cost reduces the profit margin of the business operations. Hence, the firm needs to strike a balance between the inventory level and the desired customer service level through the integrated logistics operations. The reduction in stock holdings may be exercised through a centralized inventory control from a single mother warehouse, by reducing the field distribution warehouses at multiple locations.

Credit Facility

Offering credit facility to the buyer for payment against material dispatched is one of the value-added services. This is done for building long-term relations with the customer and to get repeat bulk business from the buyer.

Stock Financing

In competitive logistics industry, the warehouse service supplier finances to the goods depositors by way of loan to an extent of 30–50 per cent of the stocks deposited. The loan is made available at lower than the bank interest rates. In Indian context, warehouses attached to cooperative sector industry, extend loan facilities to farmers to an extent of 70 per cent of the value of the food grains deposited in the warehouse.

Service Support

This may be in terms of the technical support for product installation, commissioning, process stabilizing, spare parts supply, routine equipment check-up as part of the annual service contract. Service support is required to keep the downtime of the equipment very low, as it ultimately has a significant effect on the productivity at customer's end.

Handling Convenience

The supplier, for convenience of material handling and storage at customer end, may develop novel methods for product unitizing such as pallets, boxes, or cartons. These methods are customized to suit the existing product handling arrangement at the customer end.

CUSTOMER SERVICE DIMENSIONS

The buyer will look for value for money that he is spending, while the seller, in delivering superior customer service, will look for a trade-off between cost and customer satisfaction. Hence, customer service depends on the phase of the transaction it is passing through. There are three phases associated with the goods exchange process. The degree of importance to each phase varies with organizations and depends on the product, and the customer requirements (Figure 4.1).

Pre-transaction Phase

Policy formulation for defining the service level comes under the purview of this phase. It gives guidelines to the operational staff regarding the dimensions and limitations on the customer requirement. In this phase, there is a creation of a service platform to the customer, so as to build credibility in the market and create positive image amongst the existing and the prospective customers. This phase will help in moulding the organization towards customer orientation and in turn will influence the image of the firm in the minds of the customer. The following are the important elements of the pre-transaction phase.

Formulating service policy This will indicate the service standards of the firm. For example, Pizza Hut makes a policy commitment to deliver pizza to their customer within 30 minutes of the order. Further, they commit for free supply, in case the pizza reaches after 30 minutes. In this phase, the firm will

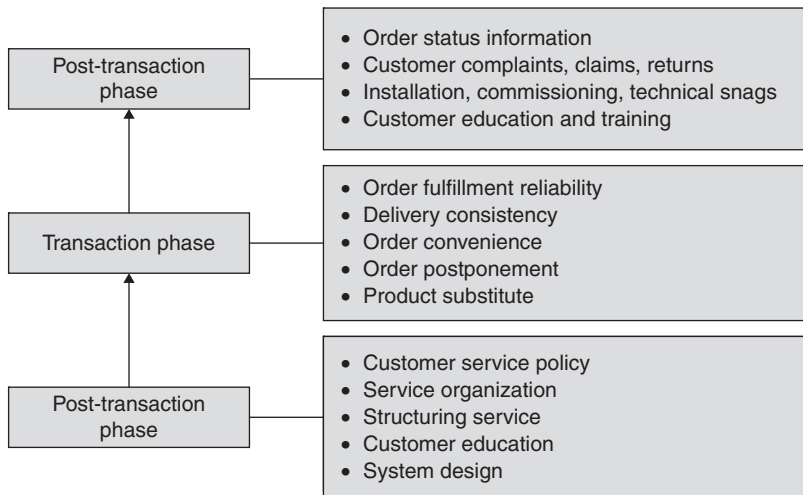


Figure 4.1 *Service Requirements in Different Transaction Phases⁴*

have to evolve a policy framework for performance measures, evaluation methods, reporting structure and rewarding structure.

Service organization For implementing the policy on customer service, the firm then formalizes the reporting structure, delegates the authority and allocates responsibility. The contact person's name and the contact number needs to be informed to the customers for getting information on order status, dispatch details, warranty claims and so on. A proper rewarding system will motivate the employees involved in customer service to effectively and efficiently interface with the customer.

Service structuring The structure of the service delivery depends on the customer expectation, industry standard and the service standard of the firm. The capital goods marketing firm may extend a free periodic product check-up to gain competitive edge. In such case, he may absorb all service-related cost as a value-added free service to the client. For a sustainable competitive advantage, innovation in service is a must. Innovation adds value to the services provided to customers. These services may be offered as a complete package along with the product, which is rather difficult for the competition to meet or emulate. The importance of service structure is its delivery. This may vary with the product and the client's need. The delivery has two dimensions, that is, time and place. The firm may have to allocate and coordinate with its resources to deliver the services at the time and place desired by the customer. The Excide Industry Ltd, a leading automotive battery manufacturer in India, introduced 'Bat-Mobile Service' to gain competitive advantage. They are offering a free van service for battery-related problems (of any battery brand) on the vehicle stranded anywhere on the highways within the major metro-city limits. They have their own fleet of vans with equipment installed in it and their technicians accompanying and carrying inventory of new batteries dedicated to this service. The communication is through a centralized wireless network. This value-added service has resulted in customer loyalty and growth in the business.

Educating customers This is done to minimize customer complaints on delivery of products, product operations and maintenance, spare parts inventory requirements and maintenance, freight charges, transit damages and so on. Customer education is done through manuals, training, seminars or workshops.

System design There should exist a system to answer all the possible queries in the customer's mind, before placing an order. The system may be manual or fully automatic like in an e-commerce. The queries can be answered through FAQs or call centres. However, the pre-requisite of the system is the responsiveness of the organization to the customer requirements and the flexibility to take care of any unplanned events.

Transaction Phase

In this phase, the customer service is associated with routine tasks performed in the supply chain. All the activities in this phase require coordination in delivering the service to the customer as per the desired standard. The following are service variants associated with the transaction phase:

Reliability in order fulfillment The most important aspect here is the reliability to fulfill the order within the agreed time frame covering the quantity and quality of the material ordered. This depends on close coordination and management of the various components of order cycle such as order processing, material planning, processing, allocation, picking, packing and transportation.

Consistency in delivery The other important factor in transaction phase is the consistency of delivery in repeat orders. For example, out of 50 deliveries, say 30 are on time while 20 are with large deviations in delivery time. This will lead to production interruptions at customer end, who will be dissatisfied with such inconsistencies. The selling firm needs to improve upon this service element.

Order convenience The barriers to order convenience are the paper work required by the supplier, compliance to various procedures, complex payment terms, poor communication network at suppliers end and poor coordination in marketing network of suppliers. In competitive markets, the above barriers may lead to opportunity loss and waning of customer base.

Order postponement The customer, due to some reason may require the entire order or in parts to be postponed for execution in parts and in a phased manner. In yet another case, due to availability of certain product in future, the seller may allow the buyer to place the order now and ship the product when it is available on future dates.

Product substitute In the event the product ordered can not be shipped due to certain manufacturing or quality problems, the seller may extend the service by offering a substitute product of similar or better quality in different sizes or from the available brands in the market at the same terms and conditions. He may do this with the consent of the customer to build a long-term relationship with the customer. Once the customer consents for the substituted product, the seller needs to closely interact and communicate clearly with customer.

Post-transaction Phase

In this phase, the seller tries to build a long-term relationship with the customer. In other words, it is the 'CRM' (customer relationship management) phase. For service-based products this is an important phase. The relationship depends on the service quality which may make or break the image of the company in the minds of the customer.

Order status The customer after payment of part value (some times full value) of the product as advance, requires feed back on the status of the shipment on a periodical basis. Many leading firms

have a consignment tracking and tracing system installed on their website, for their clients to have an online access to the information of the consignment.

Customer complaints, claims and returns In many cases, the customer may get products damaged during transit, product may not perform as per functional requirement, or he may get a wrong consignment. To resolve these issues, the manufacturers have product return policies and implement these through the reverse logistics system.

Product installation, commissioning and technical snags Technically complex products need installation, commissioning and stabilization services from suppliers, or product may develop technical snags during warranty period. To handle these issues, firms normally have separate departments only for after-sales service. The after-sales department takes care of the documentations, customers' technical complaints, product installation, commissioning, stabilization and handing over.

Education and training to customers This is an important service element in post transaction phase. In case of technically complex products, it is necessary for the seller to train or educate the user regarding the operation for getting the desired functional output. This may be done through the product manuals, training workshops or demonstrations.

SUPPLY CHAIN VALUE DELIVERY

The supply chain is a value chain in practice. The value is acquired from suppliers, value is added through company's manufacturing process and value is delivered through the company's distribution network (Figure 4.2). The company thus gains competitive advantage through supply chain by creating customer value at optimum cost. If costs are cut too much, the company's ability to service the customer diminishes drastically, and service is the key ingredient behind customer retention. If a firm's costs are too high, then a competitor may gain an advantage by efficiently balancing costs with customer value.

The objective of SCM is to have all links in the chain working together, delivering products and services to customers when, where, and how they want it. At the same time, SCM must focus on minimizing costs and resources so that value is enhanced. Most value chains will consist of the following three links:

- **Resources:** Acquiring the materials, people, and other resources to produce the required product or service.
- **Production:** Converting resources into finished products and services according to the demands of the customer.
- **Distribution:** Delivering products and services to customers.

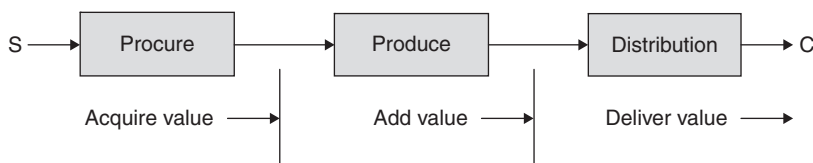


Figure 4.2 Value Chain Process

The entire supply chain process is customer-driven and distribution is closest to the customer. Traditionally, distribution had several links: manufacturer to agent, agent to distributor, distributor to retailer and retailer to customer. In today's global e-commerce world, it is quite common to see only visible link such as manufacturer selling directly to the customer. By removing the links, lead time and transaction are greatly reduced within the supply chain.

A good example of supply chain with lot of 'value addition' is IKEA, a leading global ready-to-assemble furniture product retailer. The first part in the value-chain is product design, IKEA uses simple designs and parts. Secondly, IKEA keeps costs down by having the customer transport the product to the home and assemble the product. This eliminates non-value-added links in the value-chain. Third, IKEA sells products of extremely high quality. IKEA also leverages technology and its Scandinavian image to create a competitive advantage through inventory management and marketing. The leveraging of core competencies is critical to creating value out of the supply chain.

Thus, the entire supply-chain should be evaluated, from suppliers to end-users of the product. Supply chains must be externally focused in a highly competitive environment. This requires that firm should work very closely with suppliers, customers, and everyone involved in the supply chain.

A supply chain and a value chain are complementary to each other. The supply chain is nothing but value chain in operation. IKEA is integrated supply chain enabling the flows of products and services in such a way that value for customer is created. The customer perceives the value which triggers the demand. That flow process of demand is also called as a 'demand chain' and is exhibited through the flows of orders and cash that parallels the flow of value. The main task in supply chain is to focus upstream to integrate procurement and manufacturing activities and value chain task is to focus downstream to create image in the eyes of the customer.

The key to customer value is effective SCM, which links customer's engineering, production, financial and information requirements to supplier resources and capabilities. Therefore company should involve in designing the network, optimizing the resources, integrating the processes and synchronizing the activities.

Customer value is mostly dependent on supply chain performance. The failure to add value will reduce the customer satisfaction, which can be determined by service attributes such as on time delivery or cost. However, the value is determined by the performance of the entire supply chain, which extends over multiple partners and functional areas.

The emergence of information technology in the last 30 years has changed the perception of value to a great extent. Web-based shopping, banking and services not only have changed the patterns of human activities, but have also changed the perception of value. As information has become available with much greater ease, so has the ability of firms to provide a level of service, or a 'bundle' of product, which was quite impossible in the pre-'IT era'. The value-generating effect of improvements in transportation and distribution has resulted in reduced costs and greater service. These are possible now with the support of improved information technology and capabilities.

VALUE-ADDED SERVICES IN SUPPLY CHAIN

The purpose of providing services is to deliver value to the customer for the money he is spending in acquiring that product. For tangible products, to add value, intangibles like after-sales services are provided to the customer; similarly for intangible products, to add value, the help of tangibles is taken. For example, the ambience in a star hotel is supported by the presence of tangibles around, which

speaks about the quality of service being delivered. The service can be categorized into three types as described in the following.

Basic service For the business to get going minimum basic customer service is necessary. The basic customer service means, treating all customer equally and extending services to build fundamental business relationship. It is a bare minimum for survival of the business. The supply of spare parts for a trouble-free operation of machineries is the basic service a machinery supplier can provide for his product to be worth buying and build future relationship with the buyers.

Zero-defect service The second one is the zero-defect service. This involves gearing up of business process for perfect operational performance. In zero-defect service, systems have significant role to play. The repetitive operations are performed without errors by automated systems. The essential activities such as speed, accuracy, reliability are well coordinated, monitored and controlled through the systems approach. The scope for errors is eliminated as manual intervention in performing the activities is minimal.

Value-added service The third one is value-added service which is unique and add efficiency and effectiveness to the basic service capabilities of a firm. With value-added services, firms can deliver superior customer service which leverage competitiveness. These are basically tailor made or customized service packages for prestigious customers. These services are in addition to the basic services provided to customers by the seller. For example, a supplier may take up the responsibility of delivering a high value inventory at the customers end for their manufacturing operation. He may open a small shop in the buyer's premises to keep minimum inventory to meet customer's daily requirements. The supplier gets paid on a daily basis for the inventory consumed. Thus the supplier gets business on a continuous basis due to the value-added service he is providing to the customer. The customer is satisfied as his logistics-related cost for the item is zero. The supplier is happy as he gets repeat business and immediate realization of money for the material supplied.

In another case, a transporter may extend services of product mixing and labelling of packs in his transshipment warehouse before delivery of final product to the customer. He may also collect money from the customer on behalf of the seller against delivery of consignment. The typical value-added logistical services extended by the logistics service providers are as follows:

- Customized transportation (vehicles, perishables)
- Product mixing/packaging (products from two firms for co-promotion)
- Cross-docking
- Inventory management
- Web-based consignment tracking and tracing
- Frequent deliveries in small lots
- Reverse logistics

The other value-added services are time bound ones, such as inventory on demand for manufacturing assembly line, which may involve implementation of JIT (just-in-time) system to lower the inventory-related costs. VMI (vendor managed inventory) is another value-added service. The evolution of value-added services is an outcome of innovation. Innovation will result into differentiated services in competitive markets.

SUMMARY

SCM targets two opposite goals. The first one is customer satisfaction and the other one is the cost reduction. The supply chain aims at making available products at the place and time required by the customer. A company with SCM initiative can cut down the cost and price of the product/service in such a way that a customer is willing to pay. The customer value is measured by the customer satisfaction. The customer satisfaction comes from various segments of activities such as transportation, logistics, marketing, sales, product design, reverse logistics and promotional incentives.

Customer service is an important element in marketing irrespective of product being tangible or intangible. In fact, effective customer service is leveraged for competitiveness. Internally for an organization, customer service is every body's business; therefore the organization needs to be customer-oriented or customer-focused. In supply chain strategy, the customer service is a key element. The customer service requirements differ in different phases of the transaction. In pre-transaction phase, the emphasis is on policy enunciation for defining the service level and related activities in qualitative and quantitative terms. In transaction phase, customer service is associated with efficiency of routine tasks performed in the supply chain. In post-transaction phase, it is related more towards customer satisfaction and building of long-term relationship with the customer. It involves commitment of resources to offer the desired level of service. The customer service attributes have two aspects to look into, the first is the distribution aspects and the second one is the trading aspect. The physical distribution aspect focuses on place and time utility of the product while the trading aspects focuses on motivation of channel members and customers. Due to competition, competitiveness, the manufacturers adopt innovative approaches in service offerings through customization of services to meet the needs of strategic customers. The unique or specific services are designed to increase the efficiency and effectiveness of the supply chain operations.

REVIEW QUESTIONS

1. Discuss how supply chain helps in extending differentiated customer service and also delivering value to the customer.
2. What are the various customer service attributes in supply chain system?
3. Explain with few illustrations the concepts of extending 'value-added services through supply chain operations for gaining competitive edge'.
4. Discuss the various phases of customer service requirements in different stages of goods transaction process.

INTERNET EXERCISES

1. Visit <http://www.supply-chain.org> and study how SCOR model may be used to create and deliver customer value.
2. <http://www.supplychainvalue.com/> is devoted to supply chain development and implementation of SC strategies. Read articles on customer value through SCM.
3. Visit the following Internet websites and find out more about the customer service and how it can be used as a tool for competitive advantage <http://www.sbinfocanada.about.com> and <http://www.customerservicemanager.com>.

VIDEO LINKS

1. Supply Chain Types & Value Streams, <http://www.logisticsbureau.com.au/video-supply-chain-logistics.html>
2. Cost to Serve - An Introduction analysis of the Supply Chain and the benefits it can reveal. <http://www.logisticsbureau.com.au/video-supply-chain-logistics.html>

PROJECT ASSIGNMENTS

1. All courier companies are struggling for getting a larger pie of the market through competitive product offerings. Study the supply chain of any one of the companies: Fedex, Blue Dart, DHL or AFL and prepare a report on how they are creating value for their customers.
2. Study supply chain of any national dailies like Times of India, Hindustan Times or Hindu. Find out how they are deploying their supply chain backed by marketing strategies to keep their customer satisfied adding value element in their products.

NOTES

1. Henry Ford's famous comment about the Model 'T' introduced in 1930.
2. Sople V.V. (2010), *Logistics Management*, Pearson, New Delhi, 2nd edition, pp. 24.
3. Fitzsimmons James A. and Mona J. Fitzsimmons (2006), *Service Management*, Tata McGraw Hill, New Delhi, pp. 131.
4. Sople V.V. (2010), *Logistics Management*, Pearson, New Delhi, 2nd edition, pp. 26.

PART-II

SUPPLY CHAIN SUB-SYSTEMS

5 Supply Chain Planning

This chapter starts with a discussion on the role and levels of planning process in managing supply chain. Here, supply chain planning matrix covering capacity, network, demand, manufacturing, procurement, distribution, logistics planning and the role of IT in planning process are discussed in detail.

6 Procurement Methods

This chapter begins with the modern approach to procurement and the methods in practice. The procurement methods covered are EOQ, just-in-time, vendor managed inventory, online procurement etc. and their applications and limitations.

7 E-procurement

This chapter compares the traditional and e-procurement practices, along with the framework and prevailing models practiced in the industry. The chapter further discusses the benefits and limitations of reverse auction process, outsourcing of e-procurement activity and open trading environment.

8 Strategic Sourcing

This chapter details the prevailing trends in the procurement process and the role of strategic sourcing in managing the supply chain. The chapter also covers various purchase orientations and the success factors in strategic sourcing process.

9 Lean Manufacturing

The chapter starts with the basic principles of lean manufacturing. It covers the various elements in lean manufacturing process, their benefits and role in managing the supply chain. This chapter describes the various methods of waste reduction leading to productivity enhancement. It further includes the success factors for leanness.

10 Distribution Decisions

This chapter is all about role of distribution process and distribution net working in success of supply chain. The chapter further discusses structure of distribution and prevailing practices and distribution strategies evolved by the business firms to support their supply chain.

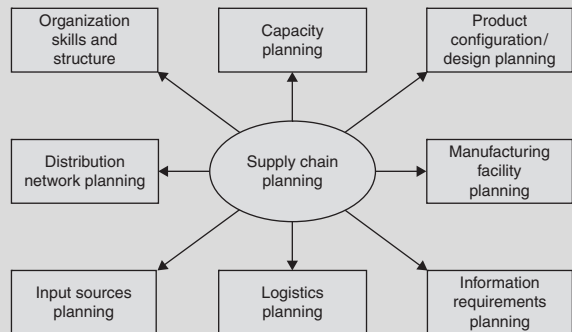
SUPPLY CHAIN PLANNING

“Good fortune is what happens when opportunity meets planning
—Thomas Alva Edison”

Chapter Objectives

To understand:

- Role and levels of SC planning
- Demand–supply planning
- Capacity planning
- Collaborative planning
- Role of IT in planning



Supply Chain Planning Components

Supply chain planning is the major component of supply chain management (SCM). It involves in predicting future requirements of inputs and infrastructure support to balance supply and demand. Domain of supply chain planning covers: supply chain modelling, designing of distribution network, identifying the supply locations close to demand points, logistics mix and information flow system. The planning process is made easier by using the latest technologies in information and communication. The technology gives the real-time input on the information to work closer to the material suppliers, process vendors, distributors and customers. Without supply chain planning firms may lose their direction towards their corporate goal.

Gillette: Demand Planning to Improve SC Health

Gillette India Limited is the Indian subsidiary of a US based-MNC. They are the leaders in personal grooming products (for men, razors, blades and shaving preparations and for female, wet shaving products and hair epilation devices), alkaline batteries and oral care products. Local manufacturing comprises of a factory in Bhiwadi, Rajasthan and a number of third party vendors. Products move from a central stock point (CSP) in New Delhi to 24 clearing and forwarding agents (C&FAs) across the country. From these locations, the products are invoiced to over 700 distributors and direct retail customers.

Gillette noticed that inaccurate demand planning in overall supply chain planning process would result in poor customer service levels and high inventory level. They experienced that poor demand planning exercise caused poor customer service and high or imbalanced inventories resulting in distortion in financial planning. This means there is a need to hold high safety stocks as buffer for low demand predictability. The steps taken to institutionalize the planning process across the company wide supply chain were: appointment of a sales forecasting manager, monthly measurement of forecast accuracy and the formation of a series of structured reviews. Gillette considered demand planning as the first step in the supply chain. They started the process through the creation of the role of a demand and supply planner. They started conducting regional demand planning meetings. This step was to improve demand-planning capability to ensure adequate organizational resources to support the implementation of this process.

Gillette India decided that to have an accurate demand plan, it needed to create a dedicated position, with the person-in-charge capable of developing an objective and statistical demand plan by incorporating his best business intelligence. Thus, to make a single individual responsible for the complete process, the position of demand and supply planner was created. The planner was given complete responsibility for demand planning and maintenance of the optimum level of inventories and to deliver high level of customer service. The move to have a dedicated resource was also borne out by Gillette India benchmarking its processes vis-à-vis its peer group. The new process consisted of the steps such as development of a baseline (statistical) forecast by the SKU, incorporating 'business intelligence' to influence demand patterns, conducting periodically a formal demand-planning meeting involving sales, marketing and finance. The demand planning was supported by organizational changes such as a sales forecasting manager was appointed within the sales function creating a new role of the demand and supply planner for demand planning and maintenance of the optimum level of inventory to deliver customer service and a regional value chain manager to generate a bottoms-up forecast.

The most tangible benefit of this initiative has been the enhanced forecast accuracy across business units. The new organizational process has helped in the forecast accuracy moving up from 40 per cent to over 65 per cent, service levels rose to 98 per cent and total inventory reduced by 25 per cent, despite growing sales. The impact on the organization has been massive, driving not only process efficiency, but also bringing about a true cultural change. This ensured that the initiative was sustainable. The organization has clearly seen the benefits of this initiative and key value chain meetings are well attended by the senior management, including the managing director. The result was every manager at Gillette India becoming a forecaster, exhibiting a cultural change that is generating its rewards in top and bottom line.

INTRODUCTION

Businesses have tried to achieve specific purposes, and in this effort some sort of planning has always found a place. Planning brings the following benefits to organizations:

- Leads to a better position, or standing for the organization.
- Helps the organization progress as expected by the management.
- Helps employees to think, decide, and act effectively in the desired direction.
- Helps keep the organization flexible.
- Stimulates a cooperative, integrated, enthusiastic approach to problems.
- Helps to evaluate and check upon the progress towards the objectives.
- Leads to socially and economically useful results.

Today companies attach significant importance to planning. They insist on some sort of planning exercise to meet the rapidly changing business environment. Growth is the objective of every firm. However, growth does not happen by itself. Growth must be carefully planned. Unplanned growth will be haphazard and may fail to provide desired levels of profit. Therefore, for a business firm to realize orderly growth it should plan for products, markets, facilities, personnel and financial resources. These must be evaluated and selected wisely.

Planning is required in making a choice among the many attractive alternative opportunities a firm may have. Planning, thus, helps in essentially making the right selection. Planning for future action has been called by many different names: long-term planning, corporate planning, comprehensive planning, and formal planning. However, planning always refers to the future.

Planning is essentially a process directed toward making today's decisions with tomorrow in mind and a means of preparing for future decisions so that they may be made rapidly, economically, and with as little disruption to the business as possible.¹

In all plannings, the common thread behind the exercise is the emphasis on the future. Today's fast-paced and rapidly changing business environment is placing increasing pressure on companies to perform consistently and reliably. This requires planning. We are working in an environment where the balance of a push and a pull requires an adaptive supply chain, that is, one driven by customer's demand and supply needs beyond time and place limitations.

The supply chain must be able to meet the highest levels of demand accurately, order fulfilment and customer satisfaction through product, order execution, tracking and logistics. Transportation and fulfilment operations need to be smoothly synchronized. The supply chain must also provide robust planning capabilities that allow firms to maximize returns on assets and ensure a profitable balance of supply and demand.

Planning perspectives developed in response to needs that arise internally are those that impinge on the organization from outside. Top management focuses on reviewing major investment proposals and approving annual operational budgets. Long-term corporate plans are put together and they are used for strategic decision-making.

The value of effective strategic planning is virtually unchallenged in today's business world. A majority of the Fortune firms across the globe, for instance, have senior executives responsible for spearheading the strategic planning efforts. Strategic planning requires that company assets (i.e., resources) be managed to maximize financial returns through selection of a viable business in accordance with the changing environment. One very important component of strategic planning is the establishment

of the product/market scope of a business. It is within this scope that strategic planning becomes relevant for business. Thus, as companies adopted and made progress in their planning capabilities, a new strategic role for supply chain emerged. In this strategic role, companies concentrate on supply chain to survive and growth with differentiation.

WHY SUPPLY CHAIN PLANNING?

In supply chain—which covers movement of information, inventory and cash, from suppliers to the end customers via firm's manufacturing—there are large number of individual decision-making points. The right decisions have to be taken and coordinated. Planning supports the decision-making process by identifying the alternatives of future activities and selects some good ones. According to the time frame, planning exercise is classified into three different levels.

Long-term planning The decisions taken at this level have long-term effect and involve resources which decides investment decisions. The long-range planning in supply chain includes facility locations and system implementations. The effects of these are noticeable over several years.

Mid-term planning This in general is done within the scope of strategic decisions. It determines an outline of regular operations in terms of quantities and time frame for their flows in supply chain. The planning horizon varies from 6–24 months.

Short-term planning This planning is for activities at operation level and restricted to short span of time. It is a lowest planning level platform and calls for highest degree of details and accuracy of data or information. Short-term planning is limited by the decisions on structure and quantitative scope from mid-term planning. Here the planning horizon is between a few days and 3 months.

In general, the entire planning process is sub-divided into phases like recognition and analysis of a problem, definition of objectives, visioning of future development, identification of alternative solutions and finally selection of feasible solution.

The supply Chain Planning Matrix (Rhode et al., 2000) classifies the planning activity in two tasks, the first one is 'planning horizon' and the second one is 'supply chain processes'. The planning horizon is divided into long-term, mid-term and short-term. The supply chain processes are procurement, production and distribution.

SUPPLY CHAIN PLANNING MATRIX

Supply chain is basically a chain linking procurement, manufacturing and distribution. These three linkages should be integrated properly to achieve business objectives. Hence, planning on the above three stages is a must. In supply chain planning, there are various tasks. These can be broadly categorized into the following planning exercises.

Planning of capacity The capacity planning means the throughput the supply chain system can handle. The supply chain may be planned on mega capacity basis just like Wal-Mart where the per day transaction level is in millions. Any distribution centre in Wal-Mart retail chain in general handles 0.1 million packets varying in weight up to 20 kgs. In the manufacturing, capacity planning shows how to use the available production capacity of one or more facilities in a cost-efficient manner.

Master production scheduling has to deal with seasonal fluctuations of demand and to calculate a framework for necessary amount of overtime capacity. The plan is based on product mix and their

manufacturing schedule; therefore, multiple production processes need to be planned. Here, planning objective is to balance the cost of capacity against the cost of inventories. If more than one production facility is considered, the transportation costs between the locations have to be included in the capacity planning exercise.

The manhour planning provides the necessary working time for converting inputs into outputs. The personnel planning has to calculate the personnel capacity required at the different stages of production before the product reaches the final configuration. This planning stage takes into consideration the skill sets required and availability of required manpower to do the job. If required, personnel need to be hired for temporary basis, that is, casual labours. This planning exercise helps to have additional part time employees.

Material requirement planning based on the master production schedule allows to select the vendors considering their capacity to supply raw materials, components and sub-assemblies, etc. The ERP system in the firm may help to work out capacity planning for suppliers. The components can be classified into A, B and C categories. The classification helps to plan capacity of vendors for each category of material.

Planning for network The supply chain network covers suppliers, manufacturing plants and distribution centres. As a thumb rule, the supply points should be near to the demand points. For inputs to the manufacturing plants, the raw material sources should be near to the plant. Hence, for continuous and uninterrupted supply of raw materials, the locations of vendors/suppliers should be near to the plants. This will help in reducing the inbound logistics costs. The manufacturer will have control on the suppliers due to their proximity to plant. Dell has planned its suppliers network around its assembling centres which are region-specific.

The other consideration in planning of manufacturing facility is proximity to markets and related cost-effectiveness in outbound logistics so as to reach end customer with ease and speed. In pharmaceutical/FMCG sectors in India, the companies are planning for local manufacturing vendors (through sub-contracting the manufacturing). These are regional-specific vendors to cater to that particular region. In India, Ambuja Cements has planned its manufacturing locations on the coastal area of Gujarat for two reasons. The first reason is proximity to raw material. The second one is using sea route for dispatch of cement (finished product) to various markets (Maharashtra, Kerala, Karnataka, Tamil Nadu). The large portion of their cement is transported through the sea route, which is cheapest amongst all transportation modes (road, rail, etc.)

Purchasing, manufacturing and distribution have to be integrated through transportation planning. Hence, planning the location of mother warehouses (distribution centre) and field warehouses occupies significant importance in network planning exercise. This planning exercise can be done using simulation to optimize the logistics cost and enhance the customer service by way of finished goods availability at the point of consumption. Here, the planning should have two objectives, that is, material availability and lowest cost of service. This can be done by organizing transportation networks (for details refer to Chapter 11) as listed in the following:

- Point-to-point network
- Multiple delivery points
- Trans-shipment points
- Nodal network
- Hub-and-spoke network

The important element in distribution is warehousing, which is a switching facility in logistics network. It is a strategic element in logistics system. The effectiveness of marketing operations of a company can be considerably enhanced by proper decision on warehousing. Warehousing is an important linkage in the supply chain of a manufacturing company. The objective of warehouse planning should be in line with the overall strategy framework of a supply chain. For example, for a supply chain of fresh vegetables and fruits, the location of warehouses are at the point of production (farms) and at the point of consumption (markets). This is because the products are highly perishable and need to be stored in temperature-controlled warehouses at both ends. The transportation modes required here are also temperature-controlled.

The warehousing functionality covers operations such as holding, consolidation, bulk break, cross docking, postponement, mixing, packaging, material handling, information handling and order fulfilment. Companies may plan for their own warehouse (investment is high), contract warehouse (product-specific) or public warehouse (less investment and less controls). The choice of warehouse depends on product, customer service and cost.

The planning for warehouse layout is an important exercise. This covers workflow layout, storage arrangement and material handling equipment. The choice of all these depends on the nature of product, product mix, and throughput.

Planning for transportation Transportation is an importation element in supply chain planning. It is a cost spinner. The transportation cost covers inbound logistics, in-plant logistics and outbound logistics. It is a most visible element in logistics operation and has significant share in logistics cost. In logistics planning, the choice differs between generalized transportation mode or customized mode depending on the nature of product and its unit price. For example, today automobiles manufacturers use customized vehicles for transporting their vehicles (cars). A truck or trailer is designed to carry maximum number of cars. In other cases, products like cements, cotton, coal, fuel and grains are transported in bulk through rail which is cheaper than road. However, for white goods, manufacturers prefer road transportation even though the freight is high. The reasons are door-to-door service, less material handling and less transportation damages. For example, the glass manufacturing companies (Saint Gobain, Float Glass) use road transportation with no trans-shipment. The material loaded at factories gets unloaded at the dealer's place. Companies do not maintain any field warehouse for glass storage.

Today in the age of multimodal transportation, companies also have choice for container transportation. The advantages are obvious: door-to-door and damage-free transportation.

Planning for inventory Inventory planning helps to develop a strategic and tactical inventory policy that is mapped to company-specific service objectives and cost constraints. Using an analytical process, it applies leading decisions on science and technology to implement optimal inventory placement, reorder quantities and multi-tiered product flow that ultimately improve customer service levels and return on assets. Inventory planning includes the following:

- Inventory optimization planning
- Inventory deployment planning
- Service level planning

Inventory planning can help to set optimal inventory levels of finished goods, spare parts, work-in-process and raw materials throughout the supply chain network. By focusing on the right item, in the right quantity, in the right place, at the right time, a company will profitably satisfy customer demand and service expectations with minimal overall inventory investment.

The inventory deployment planning can help effective positioning of finished goods and aftermarket parts across multi-tiered distribution networks by determining optimal stocking locations and item-wise inventory policies. Quantitative analysis determines how much to stock at what location, and where to source from, while meeting the customer's service level and response time expectations at the lowest possible cost. In conjunction with its inventory policy and network design, it helps to position inventory, consider unique delivery alternatives with different service times and costs, and revisit their distribution network structure.

Service level planning can help to maximize the margin contribution of inventory to the bottom line by recognizing that not all items need to be managed at the same service level. Given the working capital and space constraints, this will help to determine the most profitable service level of each item in relation to other items in the network.

Planning for information system This is a critical area in planning exercise. The competitiveness of supply chain can be considerably enhanced through its information flow planning. On-time information will reduce the cost of delays in decision-making. It is observed that the inventory pile ups or shortages happen due to delays in reaching right and timely information at the decision-making points. As inventory is the greatest culprit in supply chain, it needs to be managed properly. Otherwise, it will lead to bull-whip effect which is detrimental to the health of supply chain. The company should also plan for supply chain information system. This will help in collecting the information, analysing it and disseminating at various decision-making points in right formats. The information system requires investment. The company can plan for information system to suit its organizational requirements. There are standard software packages on ERM (enterprise resource planning), MRP (material requirement planning), DRP (distribution requirement planning) and SCP (supply chain planning). However, a word of caution here is that these software systems are standard packages and should be tuned to the company's requirements. In a nutshell, the information should help in supply chain coordination, integration and synchronization amongst its processes and partners. It should help in enhancing the visibility in supply chain.

Planning for demand and supply A distinction should also be made between planning and forecasting. Forecasting considers future changes in areas of importance to a company and tries to assess the impact of these changes on operations of the company. Planning takes over from there to set the objectives and goals and develop strategies. Briefly, no business, however small or poorly managed, can do without planning. Although planning may be nothing new for an organization, the current emphasis on it is indeed different. No longer just one of several important functions of the organization, planning's new role demands linking of various parts of an organization into an integrated system. The emphasis has shifted from planning as an aspect of the organization to planning as the basis of all efforts and decisions, the building of an entire organization to achieve its designated objectives.

Today companies know how difficult it can be to synchronize volatile demand and supply factors throughout the supply chain. Sudden changes in customer demand and supply disruptions can leave once solid MRP plans in tatters. The firms need to minimize overtime, optimize plant utilization, avoid parts shortages, and streamline production lines to respond to the changing demands on a daily or weekly basis. How do companies rise to meet these strategic challenges and achieve an optimally synchronized multi-tier supply chain to successfully fulfil every individual order?

For demand planning and forecasting, firms can use state-of-the-art forecasting algorithms in order to anticipate demand for products or product characteristics. Firms can also model and plan new product introductions, trade promotions, or causal events that will significantly impact demand. For safety stock planning, firms can assign optimal safety stock—stock kept on hand to satisfy unexpected

demand—and target stock levels across the entire supply network. This means you can meet your target customer service levels while maintaining a minimum amount of safety stock.

By using collaboration in demand planning, firms can improve their visibility in supply and demand (leading to reduction in inventory buffers), increase the velocity of raw materials and finished goods through the pipeline, improve customer service, and increase revenues.

The spare parts and component demand planning can improve accuracy of forecasts through better modelling of demand for finished goods quantities with their deviations. Firms can select sophisticated forecast models and optimize model parameters to improve forecasting for slow-moving parts or for parts with irregular demands.

Firms can reduce inventory levels and improve retail services by providing precise demand modelling. Firms can distribute inventory optimally within the multi-echelon supply chain to ensure high service levels while keeping inventory levels at a minimum.

Firms can reduce inventory in the supply chain by improving their alignment with the suppliers, increasing automation and developing accurate supply plans. Firms can also reduce operational costs through efficient purchasing practices.

COLLABORATIVE PLANNING

In recent years, industries have faced numerous challenges that have made it increasingly difficult for carriers, shippers, manufacturers, distributors and retailers to unite in support of a true capacity-planning process. Given these challenges, both carriers and shippers have been exploring collaborative processes and technologies that will enable them to:

- Reduce costs.
- Ensure that more shipments arrive on time.
- Improve both capacity coverage and asset utilization.

Manufacturers, distributors and retailers have increasingly expanded using an alternative means of transportation modes. The conflicting objectives of carriers and shippers exacerbate the problems. Motivated by factors like inventory positions and turns, shippers are driven to shrink lead times and condense their peaks. Transporter on the other hand, would prefer more ample lead times and more evenly distributed demand. Many shippers have not clearly understood, or at least fully appreciated, the implications of these conflicting objectives on carriers. Without an adequate shipment/capacity forecasting process, shippers often force their carriers into a reactive mode.

To overcome the significant challenges that inhibit the ability of vendors, manufacturers, shippers and carriers to improve planning, they must improve their collaborative processes. The most fundamental data necessary to drive a collaborative planning process is historical information. At a minimum, partners should try to leverage their own historical data.

However, given the recent power shift resulting from costs and capacity within the industry, manufacturer must be more willing to share a complete and accurate shipment history with their vendors and carriers. In a perfect world, this history would supply them with details and information on seasonal peaks, promotions and trends.

The following three levels of collaborative planning can help counteract the demand fluctuations and other roadblocks to a more efficient and reliable shipments:

- At a strategic level, new or projected demand volumes and/or patterns derived from statistical methods.
- At a tactical level, partners can synchronize planning with master planning cycles (in a manufacturing environment) or replenishment planning cycles (in a retail or distribution environment).

The tactical plan can be determined by leveraging actual orders (or allocations) and/or projected demand volumes.

- At an operational level, planning typically takes place on actual orders (whether they are inbound purchase orders, outbound customer orders or inter-facility stock transfers). The operational plan communicates the actual tender, booking or manifest depending on the transportation mode, manufacturing schedules, etc.

ROLE OF TECHNOLOGY IN PLANNING

Technology is the essential enabler of successful collaborative planning. From the manufacturer's perspective, technology should enable the company to review shipment history and incorporate that

Box 5.1

Dairy Farm Supply Chain—Technology Planning

Though India is one of the world's largest milk-producing nations, the yields here are very low when compared to other nations. The American dairy industry has an average yield of 10000-11000 liters per lactation cycle, compared to 1000-1200 liters for buffaloes in India. The research revealed that technology planning in dairy-farm supply chain is one of the major elements in contributing to the enhancement in milk yields.

The tiny microchip or radio frequency identification (RFID) tag punched on buffalo's ear sends information about her daily needs and feeding details, amongst other information to a radio sensor located inside the milk-farm premises. The data collected by system is then accessed real time by dairy managers for carrying out specific activities such as monitoring the diet (over and under eating), nutritional mix, diseases and yield.

Unlike plastic plates with a unique 15 digit number (a traditional way), RFID tag is a more intelligent device, which helps in enhancing the productivity on milk-farm supply chain. The milk farm can be compared with a manufacturing unit wherein modern techniques and technologies are planned and used to enhance the supply chain efficiency, cost effectiveness, assets utilisation and productivity potential. Today RFID becomes an essential tool for evolving a new better breed at individual animal level. RFID provides complete life history of a cattle across generations and gives breeders necessary knowledge to do selective breeding. As RFID monitors health and food habits cattle farmers can change the nutrition mix according to needs. A cattle (with RFID tag) approaching a feeding machine kept in the farm house which refuses to offer her feed because it has already completed quota of diet for the day. Overeating among the cattle may lead to fatal diseases and insufficient diet may cause weight loss.

A RFID tag with 4 kb of memory space is sufficient to store information about the entire lifecycle of cattle. With controlled feeding process early diagnosis of illness and pedigree analysis of cows and buffaloes has resulted into 3-4 times better milk yield than the average. In addition to above, the insurance companies are using RFID technology to redefine the cattle insurance industry. The cattle insurance process is now done on same day (with RFID) as compared to 30 days earlier done in traditional way. To name a few in India, Chitale Dairy, Tamil Nadu Dairy Farms and many others are using RFID technology to in Milk-Dairy Supply Chain to get productivity benefits. (Source: Pankaj Mishra, Farmers milking tech for big yield, Economic Times, Mumbai, April 01, 2010.)

history into demand planning. It is just not enough to look purely at purchase orders; the technology must show data at the discrete shipment level, because the relationship between order and shipment is frequently 'one-to-many'. And order management systems often don't have complete, accurate information with regard to actual shipment location/addresses.

From a true collaborative planning perspective, it is important that shippers are able to share with each partner the historic business they have done with everyone. The shared data can be limited to a specific business and/or geographic area of operations. But for shipping data to be meaningful to partners, it must be complete and detailed.

The collaborative process often fails when shippers change or add sources for a specific line of business without sharing the information with their carriers. With this in mind, shippers need to work with their purchasing/procurement organizations to ensure that they are aware of these types of changes to their business. With this transparency, they can accurately incorporate these adjustments in their shipment-level forecasts to carriers.

On the partner side, the technology that enables collaborative planning should at least capture the partner's complete business history for each of its customers, including, if possible, the business that the carrier has rejected.

By collaborative planning, shippers and partners can achieve tremendous results. Shippers can ensure that more shipments arrive on time. They can also improve their capacity coverage by reducing tender rejects and minimizing last-minute fire drills to find capacity with contract vendors.

Supply chain partners can ensure that necessary assets are in place to meet demand, without overextending themselves. Most importantly, costs for both shippers and partners can be reduced. Ultimately, the collaborative planning process will enable shippers and partners to build strong partnerships that foster trust, open communication and dependability—all leading to end-customer satisfaction.

SUMMARY

Today's business environment is more complex than ever. In addition to the keen competition that firms face from both domestic and overseas companies, a variety of other concerns, including environmental protection and legal compliances. Thus, it is desirable for a firm to be cautious in undertaking risks, which again calls for a planned effort. Many firms pursue growth internally through research and development. This route to growth is not only time-consuming but also requires a heavy commitment of resources with a high degree of risk. In such a context, planning is needed to choose the right type of risk. Today competition is between supply chains and not in products. Hence supply chain has gained significant importance.

Though planning is conceptually rather simple, implementing it is far from easy. Successful planning requires a blend of many forces in different areas, not the least of which are behavioural, intellectual, structural, philosophical and managerial. Achieving the proper blend of these forces requires making difficult decisions. The broad areas of planning in supply chain are demand planning, network planning, capacity planning, inventory planning, etc. The role of technology is crucial in supply chain planning. It is rather an enabler for successful operations of supply chain. Of late, collaborative planning has gained importance in planning exercise in supply chain. It involves the collaborative efforts of all supply chain partners (suppliers, vendors, customers, service providers) in planning process to mitigate the risks in supply chain.

REVIEW QUESTIONS

1. What is planning? Explain the role of planning in supply chain.
2. What are the critical areas of planning in supply chain?
3. Discuss the role of technology in supply chain planning.
4. Explain 'collaborative planning', its prerequisites and benefits.

INTERNET EXERCISES

1. For more details on supply chain planning process visit <http://searchmanufacturingerp.techtarget.com/definition/supply-chain-planning-SCP>
2. Epiq Technologies is a leading provider of supply management solutions for enterprise and mid-sized customers. Epiq Supply Management Solutions provide advanced tools and intelligence to drastically reduce supply management costs. Visit site to study SC Planning Process http://www.epiqtech.com/supply_chain-Planning-Systems.html

VIDEO LINKS

1. Supply Chain Planning Insights, http://www.scdigest.com/scm_planning_TV.php
2. Director of Supply Chain Planning, <http://www.youtube.com/watch?v=0d7CQzRoFUA>

PROJECT ASSIGNMENT

1. A new retail chain for fresh fruits and vegetable is in the process of setting up supply and distribution network in the State of Maharashtra. The supplies are to be sourced from Nashik, Jalgaon, Pune and Kolhapur. The distribution outlets are to be opened in A and B grade cities in the State. Prepare a complete blue print for network of suppliers, vendors, cold storages and outlets. Work out a plan for integration of the entire network. List out the assumptions, if any, to complete this planning exercise.

NOTE

1. Scott Arnold (2007), *Marketing and the Concept of Planning and Strategy*, <http://www.e-articles.info/e/a/title/marketing-and-the-concept-of-planning-and-strategy>.

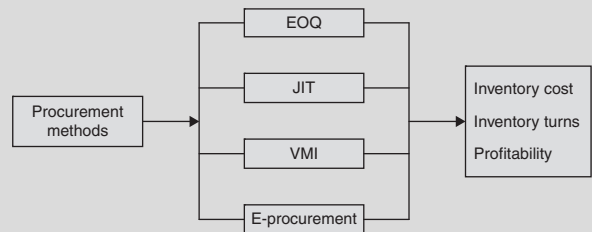
PROCUREMENT METHODS

“If a man will make a purchase of a chance he must abide by the consequences
—Richard Richards”

Chapter Objectives

To understand:

- New paradigms in inventory and purchase management
- Economic order quantity (EOQ) inventory model
- Just-in-time practice
- Vendor-managed inventory
- Online procurement



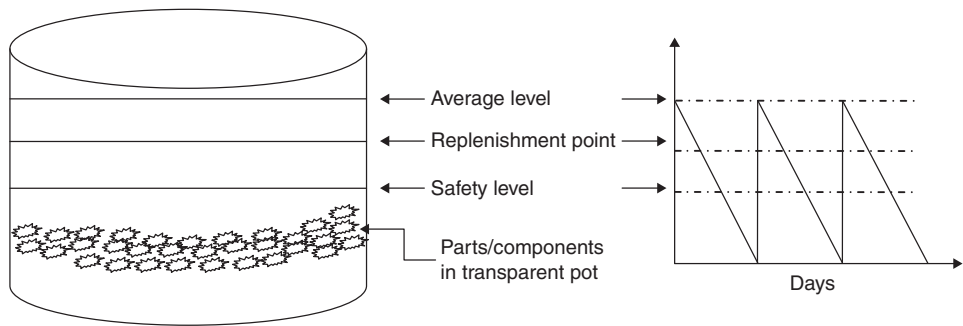
Procurement Methods

Procurement methods can impact inventory levels. The EOQ calculation is traditional model to control inventory. Accurate input is required in the EOQ equation. However, based on current operational cost, EOQ method presents the most cost-effective quantity. For further reduction in the ordering cost firms need to increase inventory turns. This can be done by adopting the new methods like e-procurement, vendor-managed inventories, bar coding and RFID. The technology can reduce the costs associated with processing an order. Further enhancement in accuracy level of forecasting and reduction in lead time will help in reduced safety stocks resulting in reduced inventory levels.

Vendor-Managed Inventory at ZICO Ltd

ZICO Ltd is a leading auto component manufacturer in India producing fuel injection pump assembly (FIPA). At their plant, they have implemented VMI in a unique manner. VMI is the emerging and alternative method of managing demand. In this model, the customer no longer places orders but instead shares information with the vendor. On the basis of this information, the supplier takes responsibility for replenishment of the customer's requirement. It is the responsibility of the supplier to fulfill the customer's requirement within the specified stock bands.

FIPA consists of 6–7 very small parts such as screws, spring diaphragms apart from the main body (bigger in size) of the pump. They have installed 6–7 huge transparent pots in the shop. The small parts are poured into the pots by the supplier at a specific interval (days) so that they are always full with the components/parts. Each pot is allocated for different parts/components.



The assembly line worker draws the parts/components from the transparent pot for completing the FIPA. At the end of the day, the total FIPA manufactured are displayed on the electronic sign board and accordingly the parts/components used during the day are automatically calculated by the computer. The payment to the supplier is made accordingly at the end of the day.

In this VMI system, ZICO does not own or maintain inventory in the pot. The responsibility of replenishment lies with supplier. There are three lines on the pot. The green line (top) indicates the average level of inventory to be maintained by the supplier. The blue line (middle) indicates the replenishment point and the red line (bottom) indicates the safety stock level. The performance of the supplier is judged on how many time in a year the level touches/goes below the red line, which normally should not happen. Otherwise, with VMI it is a win-win situation. With VMI, ZICO benefits by shortening the supply chain, the inventory is replenished in priority, there are no inventory-related costs. The supplier could plan his manufacturing in advance and get his payment against consumption on the same day.

INTRODUCTION

Inventory is a critical element in the supply chain. Inventories are maintained for meeting production requirements, supporting the operations, extending customer service and hedge against future uncertainties. The cost associated with inventory was always ignored in the past. However, the investment in inventory has implications on the profitability of the firm. The inventory carrying cost consists of the cost of the funds invested in the inventory, its handling cost, damage and obsolescence cost and opportunity cost. The carrying cost is approximately 18–20 per cent of the investment in the inventory. In supply chain system, inventory is viewed as a liability which reduces both the profits and returns on investment. For making the supply chain leaner, the firms are using selective control techniques such as EOQ, JIT, VMI and inventory control models like MRP, DRP and AITS. For managing the inventories across the supply chain, the policy guidelines have to be framed for inventory procurement, maintenance, positioning and placement with respect to quantity, time and the customer service. The procurement policy decisions are also based on managing the inventory independently or interdependently across the various distribution networks. In a nutshell, inventories should be held only when the benefits of holding the inventory exceeds the cost of carrying the inventory.

The modern approach to control inventories is focused on the following factors:¹

- **Flow:** It is concerned with the movement of product from supplier to the distribution centres and ultimately to the customer. Here comes the role of logistics operation for planning the material flow across the distribution networks based on the market demand. This is supported by the production planning at the factory.
- **Flexibility:** It is related to the flexibility in frequency and volumes in delivery of the products to the customers. The role of warehouse management is crucial to manage operations by fine-tuning to the demands of the target customers.
- **Balancing:** The inventory levels have bearing on the customer satisfaction levels. An inadequate inventory level will create customer dissatisfaction and, on the other hand, an excess inventory will increase the carrying cost. The latest inventory models focus on maintaining a delicate balance between the two polemic goals.
- **Integration:** The supply chain efficiencies and effectiveness is very much dependent on integration. The inventory models are developed for strategic fit for both forward and backward integration to meet the objectives of lower inventory carrying cost and enhanced customer satisfaction.

PROCUREMENT

To support manufacturing, the support of purchasing department is required. Procurement is buying of raw material or component that a company needs as an input to produce finished goods. The raw material in general constitutes 60–70 per cent of the cost of a product. Therefore, the firm should give greater focus to purchasing. The inventory-related cost can be best managed through better management of procurement. Business firms are looking to procurement as a strategic function to control costs. Cost is also attached to the method of procurement and hence, companies are trying to improve purchasing processes, which is ultimately improving the level of customer satisfaction.

The purchasing is a process covering many steps such as sending enquiries, request for quotations, soliciting quotations, negotiations, issuing purchase orders, shipping advice, invoice and finally issuing payments. This is a traditional process and is time-consuming and not cost-effective. The procedure is same for any item irrespective of its unit price. In this approach, the objective is to get the product whenever it is required at any cost.

The purchasing methods are responsible for inventory cost control. Over the years, there is dramatic change in using the purchasing methods. More sophisticated method are being evolved and practised. The suppliers have come forward to suggest to buyers to reduce their inventory burden. They also look into customer satisfaction and profitability of the buyer. Over the years, many companies have changed their outlook towards purchasing process. Due to competitive pressure, for many companies procurement has become a strategic function and source of cost reduction.

Today companies are focusing on total cost of inventory acquisition and are concerned more with purchasing decisions. To lower total cost of ownership companies are taking a number of steps to improve purchasing methods.

ECONOMIC ORDER QUANTITY (EOQ)

EOQ is an inventory decision model, based on differential calculus. It determines the optimum order size for purchasing an inventory item of stock. The optimum order quantity is that which equates the total ordering cost and total inventory cost. EOQ is the number of units that a company should add to the inventory with each order to minimize the total costs of inventory which covers holding costs, order costs and shortage costs. EOQ is used as part of a continuous review inventory system, in which the level of inventory is monitored at all times, and a fixed quantity is ordered each time the inventory level reaches a specific reorder point. The EOQ provides a model for calculating the appropriate reorder point and the optimal reorder quantity to ensure the instantaneous replenishment of inventory with no shortages. The basic EOQ formula is as follows:

$$EOQ = \sqrt{\frac{2(\text{Order cost})(\text{Annual usage in units})}{(\text{Annual carrying cost per unit})}}$$

$$Q = \sqrt{(2 \text{ } cd/h)}$$

where

- Q is the quantity to be purchased
- c is the cost of processing an order for delivery
- d is the demand in the period for that stock item
- h is the cost of holding a unit of stock

For calculating optimal order quantities and reorder points, manufacturing companies are using the EOQ model for a long time. This model helps in reducing inventory level and related cost. EOQ will sometimes change as a result of quantity discounts, which are provided by some suppliers as an incentive for customers to place larger orders.

The EOQ model is based on the following two assumptions:

- Demand is constant
- Inventory is depleted at a fixed rate until it reaches zero

Box 6.1

EOQ Example

A small scale engineering unit uses 50 litres of furnace oil per day at Rs 32 per litre and works 300 days per year. Under this scenario, the firm's annual oil consumption (demand) is 15,000 litres. The firm incurs a holding cost of Rs 20 per litre every year and order cost of Rs 150 per order. In this firm's optimal order, quantity can be found out as follows:

$$EOQ = \sqrt{2cd/h}$$

Where; $c = 150$ (cost of processing order)
 $d = 15,000$ (demand of oil per year)
 $h = 20$ (cost of holding a unit stock)

$$\begin{aligned} EOQ &= \sqrt{2 \times 150 \times 15,000 / 20} \\ &= 512 \end{aligned}$$

Thus firm should order 512 litres 29 times per year or every 2 weeks, in order to minimize the inventory cost.

At the replenishment point, a specific number of items arrive to return the inventory to its beginning level. Since the model assumes instantaneous replenishment, there are no inventory shortages or associated costs. Therefore, the cost of inventory under the EOQ model involves a tradeoff between inventory holding costs (the cost of storage, as well as the cost of tying up capital in inventory rather than investing it or using it for other purposes) and order costs (any fees associated with placing orders, such as delivery charges). Ordering a large amount at one time will increase inventory holding costs, while making more frequent orders of fewer items will reduce holding costs but increase order costs. The EOQ model finds a quantity that minimizes the sum of these costs.

To go as per the formula will be rarely possible because of the following reasons:

- The ordered quantity figure may be modified to take into account the standard pack size available in the market.
- To avail of the quantity discount offered by the suppliers, the order quantity figure may be modified.
- The availability of funds will force the buyer to go for the less than minimum ordered quantity.
- To take care of the anticipated shortage of material in the market, the higher quantities may be ordered.

The EOQ formula is normally used as a guideline rather than deciding on the exact material requirements. EOQ inventory model works well with accurate product costs, activity costs, forecasts, history and lead times. Today ERP packages (software) automatically calculate EOQ. However, the users do not understand how it is calculated and therefore do not understand the data inputs and system setup which controls the output.

In general, the firms in order to enhance their bottom lines focus on increasing inventory turns. This may clash with EOQ approach. Thus, corporate goals and business strategies clash with each other. Calculating EOQ is a mathematical approach to decide the order quantity at the point, cost related to inventory carrying an order placement are lowest. For every inventory situation, calculating the optimum level of order quantity using EOQ formula may not be always suitable. But the EOQ formula will give some indication on order quantity to proceed for placement of order. EOQ is useful for repetitive purchasing of standard inventory items under make-to-stock manufacturing policy. EOQ gives better results under the situations wherein the demand for item is steady. However, for item with seasonal demand, EOQ calculations are not recommended because they exist for a shorter time frame.

If all the costs associated with purchasing and receiving are included in order cost and all costs associated with storage and material handling are included in inventory carrying cost, the resultant EOQ is likely to be inaccurate. Thus, EOQ may be inaccurate if inputs costs are wrong.

The order cost comprises of all fixed costs applicable on each order. The fixed cost are associated with the frequency of the orders and not on the quantities ordered. The order cost, in other words, is the labour associated with processing the order. In many cases, the other costs such as phone calls, faxes, postage and envelopes are also included.

The carrying cost is the holding cost associated with inventory on hand. The carrying cost covers storage cost, interest on the money borrowed from the bank, insurance cost and the taxes on the value of the inventory. Other costs that are covered in carrying cost are risk cost on transit/handling damages, obsolescence and pilferages.

There are many variations on the basic EOQ model as listed below:

- Quantity discount on inventory ordered
- Fixing quantities for items subject to spoilage or obsolescence
- Determining lot sizes for longer production runs
- Safety stock calculation based on order cycle time

EOQ can be implemented manually. It is done by taking one item at a time. However, if the SKU numbers exceeds 1000, EOQ needs to be programmed in the computers to get the calculation with speed.

JUST-IN-TIME PROCUREMENT (JIT)

In 1990, JIT caught up the attention of manufacturing companies and became one of the biggest trends in the manufacturing industry. With JIT initiatives, companies maintained just enough inventories they need for immediate manufacturing. Parts/components are continuously ordered and reach assembly line immediately after delivered to the manufacturing shop floor. The JIT system has many advantages such as reduction in inventory level, improved quality, reduction in lead time and improved asset utilization. A company adopting JIT manufacturing system, must shift to JIT procurement system. The objective of JIT is to speed up customer response and minimizing inventory. The larger inventories help to respond quickly to changing customer demands, but they also increase inventory carrying cost resulting in increased requirements of working capital.

'Just-in-time', is a manufacturing organization's philosophy, pioneered by Taiichi Ohno in Japan at the Toyota car assembly plants in the early 1970s. JIT cuts waste by supplying parts just at the time of assembly process. At the heart of JIT lies the *KANBAN* which means a card in Japanese language. The *KANBAN* card is sent to the warehouse to reorder a standard quantity of parts as and when those have been used up in the assembly/manufacturing process. JIT is used primarily for high-volume repetitive flow of manufacturing processes.

The American Production and Inventory Control Society (APICS) define JIT as follows:²

A philosophy of manufacturing based on planned elimination of all waste and continuous improvement of productivity. It encompasses the successful execution of all manufacturing activities required to produce a final product, from design engineering to delivery and including all stages of conversion from raw material onward. The primary elements include having only the required inventory when needed; to improve quality to zero defects; to reduce lead time by reducing setup times, queue lengths and lot sizes; to incrementally revise the operations themselves; and to accomplish these things at minimum cost.

This is a pull type production process based on a good communication between the suppliers and production department and it is initiated by actual demand, rather than by plans based on forecasts. In contrast in a push philosophy, large volumes of materials and components are produced, transported and stored ahead of demand and requirements. The idea behind this is the reduction or elimination of work in progress, that is, large stocks of goods and materials that have been produced and are waiting for transportation or further completion. Typical attention areas of JIT implementations include:

- Inventory reduction
- Smaller production lots/batch sizes
- Quality control
- Reduced complexity
- Transparency in movement
- Flat delegation
- Waste minimization

With the advent of the Internet and availability of the supply chain planning software, companies extended JIT manufacturing externally. They asked suppliers to deliver inventory to the factory only when it is needed for assembly, making JIT manufacturing, ordering and delivery processes even speedier, more flexible and more efficient. They formed integrated supply networks (demand networks) or electronic supply chains resulting into 'lean production' system.

Just in time is a 'pull' system of production. Here, the actual orders provide a signal when a product should be manufactured. Demand-pull enables a firm to produce only what is required in the desired quantity and time.

This means that stock levels of raw materials, components, work in progress and finished goods can be kept to a minimum. This requires a carefully planned scheduling and flow of resources through the production process. Modern manufacturing firms use sophisticated production scheduling software to plan production for each period of time, which includes ordering the correct stock. Information is exchanged with suppliers and customers through EDI (electronic data interchange) to ensure that data is correct.

Suppliers deliver right quantity of material to the production line only when it is needed. For example, a car manufacturing plant might receive exactly the right number and type of tyres for one day's production, and the supplier would be expected to deliver them to the correct loading bay on the production line within a short time slot.

With the JIT adoption, companies will have lower stock holding resulting in reduction in storage space and consequently save on rent and insurance costs. The working capital tied up in stocks is also reduced. The risk of inventory getting obsolete due to change in demand or technology is reduced. The time spent on checking and re-working the product of others is reduced as the emphasis is on getting the work right the first time.

The disadvantages associated with JIT system are there is little room for mistakes as minimal stock is kept for re-working faulty products, production schedule get delayed if stocks not supplied in time and there is no spare finished product available to meet unexpected orders from the market.

JIT is a philosophy of continuous improvement in which non-value-adding activities (or wastes) are identified and removed for the purposes of:

- Reducing cost
- Improving quality
- Improving performance
- Improving delivery
- Adding flexibility
- Increasing innovativeness

JIT technique eliminates waste thus protecting the environment and simplifying the processes to improve operations. When JIT technique is implemented correctly, significant benefits are accrued leading to competitive advantages. JIT can have applications in all functional areas of management of an organization such as order processing, procurement, production, distribution, sales, accounting, design, etc. JIT usually identifies seven prominent types of waste to be eliminated:

- Waste from overproduction
- Waste of idle time/waiting time
- Transportation waste
- Handling/motion waste
- Inventory waste
- Waste from product defects
- Processing waste

In JIT, material-related costs are reduced by finalizing on the suppliers and developing long-term contracts with few suppliers. JIT implementation eliminates excess material handling, inspections and storage of parts. The primary goal in JIT is to eliminate non-value-adding activities.

JIT purchasing approach is preferred to the EOQ approach as the JIT purchasing approach helps in space reduction in the manufacturing plant. JIT purchasing requires a nearly total change in purchasing philosophy and company culture. JIT provides reduction in suppliers' base and zeros down on a few who can deliver high-quality products as and when required. To make JIT a success story, the JIT supplier should demonstrate excellent product quality, make frequent on-time deliveries and provide very large volume commitments.

To implement JIT, it is essential to organize small (quantity), frequent but on-time deliveries. This is possible if suppliers are located in same geographical region as buyers. The example is automobile industry wherein most of the ancillary industries are located near to the main auto assembly plant. The location proximity helps responsiveness of suppliers. In a single sourcing arrangement, the buyer can have a lot of influence on the manufacturing process. In such arrangement, the buyer helps supplier in technical and commercial aspects of suppliers business and in turn expects value-added or extra services from him.

VENDOR-MANAGED INVENTORY (VMI)

Today, most of the firms are concentrating on the 'core competences'. They want to outsource minor tasks and activities, when it is cost-effective to do so. For a distributor, an example of one of these tasks is the replenishment of less-expensive products. For a manufacturer, it may be the procurement of MRO (maintenance, repairs, and operations) inventory. A popular way to outsource these procurement activities is a vendor-managed inventory (VMI) agreement. Under a VMI agreement, a supplier takes full responsibility for maintaining stock of products at a customer's facility. VMI differs from traditional inventory management, in that the customer is billed for material when it is delivered, not when it is consumed or issued.

Vendor-managed inventory, just-in-time distribution and efficient consumer response (ECR) all refer to similar concepts, but applied to different industries. For example, the grocery and apparel industries tend to use ECR, whereas the automobile industry tends to use VMI and JIT distribution. VMI reduces stock outs and reduces inventory in the supply chain. Some features of VMI include the following:

- Shortening of the supply chain
- Centralized forecasting
- Frequent communication on inventory position
- Inventory fill up in a prioritized order
- Relationship with downstream distribution channels
- Result inventory and stock out reduction

In general, the supply chain is not synchronized to consumer demands. Hence, to secure high service levels, both the wholesaler and the retail chain feel the need to buffer against supply disruption. This in turn, distorts the demand that is communicated to the supply factories. The objective of VMI is to find an effective way for the vendor to take responsibility of the wholesaler's inventory. This way the need for double buffering against supply disruptions could be eliminated and the basis for planning supply request from manufacturers could be improved.

In its simplest form, VMI is the process where the vendor assumes the task of generating purchase orders to replenish a customer's inventory. VMI covers many types of supply chain initiatives. These different VMI activities can vary substantially in purpose and application. VMI is about improving visibility of demand and product flow in a supply chain, facilitating a more timely and accurate replenishment process between a supplier (vendor) and an inventory site (customer, distributor, distribution centre, etc.).

VMI can be applied at any point within a supply chain:

- Manufacturer—Wholesale distributor/retailer
- Manufacturer—End customer/OEM
- Manufacturer—Internal inventory sites

For VMI partnership to be successful and fruitful, the supplier and customer when entering an agreement must agree on the following:

- The specific products that will be covered under the VMI agreement.
- 'Acceptable availability' of these products at the customer's site and the corresponding investment required by the customer.

Box 6.2

'VMI' System in Retail Industry

In organized retail chain industry, VMI is a process where the supplier (manufacturer) maintains the inventory and fulfills orders of the retail stores based on demand information sent by the retail store. During this process the supplier is guided by mutually agreed objectives for inventory levels, fill rates and transaction costs. The inventory data is typically segmented into various groups such as inventory on hand, on order, committed, back ordered and so forth. This transaction is the back bone of VMI and is sent by the retail stores on a prearranged daily schedule. The decision to order is based on this data. The business process supported by this data is relatively simple. The manufacturer (supplier) reviews the information that has been sent in by the retail store to determine if an order is needed. This review of the data varies by supplier and the software being used. The supplies (manufacturers) undergo the following steps:

- *Data verification for accuracy and meaningfulness (Software verifies it automatically).*
- *Software calculates a reorder point for each item based on its movement data and any overrides contributed by the retail store or manufacturer. These overrides might include information such as projects, seasonality, new items and so forth.*
- *Software compares the quantity available at the retail store with the reorder point for each item at each location to determine whether material is needed.*
- *Order quantities are then calculated taking into considerations the carton quantities and transaction costs in completing the order process.*

Further VMI transaction informs the retail stores what product to expect from the supplier (manufacturer). There are two transactions being used for this function. The most frequently used is the purchase order acknowledgment. This document contains the product numbers and quantities ordered by the supplier on the retailer's behalf. A few distributors skip this and rely on the advance shipping notice to alert them to the order and shipment. This document differs in both timing and content. This document is sent after the shipment has been made and contains information on the part numbers shipped as well as additional information such as carrier and waybill information.

The VMI system imparts advantages to both manufacturer and the retailer stores in terms of lower inventories levels, better planning and lower administrative costs.

- Mutually acceptable 'service level' agreement.
- How often the stock of these products will be replenished.
- Automatic return of material not required by the customer.

The advantages for a customer participating in a VMI programme cover the following:

- Eliminating of transaction cost.
- Reduced inventory capital costs.
- Establishing an extremely reliable source of supply for products.
- Customer does not pay for the inventory until it is sold or used.

- Elimination in inventory stock outs.
- Better financial planning.
- Elimination in ordering errors.

The advantages for a supplier in a VMI programme cover the following:

- Securing the customer's overall business for products it supplies.
- Better planning of its own inventory replenishment needs.
- Better capacity utilization of its plant.
- Better demand management.
- Customer-focused approach leading to competitive advantage.
- Supply chain visibility.
- Better planning.

The risks involved for suppliers in participating in VMI programmes are high administrative cost as it is assuming responsibility for replenishment activity that was previously carried out by the customer's buyers.

The real world implementations of VMI can be broken into three main categories:

- Collaboration
- Automation
- Cost transfer

A collaborative planning model consists of sharing data, and jointly developing forecasts and/or production schedules amongst supplier chain partners. This collaborative process occurs at the tactical or item level. The 'buyer' collaborates with the supplier on demand/usage plans in order to develop an agreed upon consensus forecast of future demand that both companies will use to drive their business. This strictly collaborative model is applicable to supply chains where a few, distinct items (SKUs) generate substantial volumes of business. In this environment, it is valuable for people to review and arrive at consensus on forecasting and replenishment plans for each SKU.

In business environments, where thousands of SKUs have to be managed daily, collaboration at the tactical (item) level is impractical, costly and error-prone. The more effective collaborative process is at the strategic level, where overall service and inventory investment goals are agreed upon, along with the constraints within each company. The collaborative stage is critical in establishing the goals and key performance indicators for the VMI relationship. Periodically, this stage is reviewed for current performance and adjusted or reconfirmed with the goals and constraints. When the collaboration and planning stages are done properly, the execution stage becomes automated with very few exceptions, requiring less human interaction on a daily basis. Furthermore, the execution stage can provide suppliers with valuable information beyond the quantity in the purchase order, enabling improvements to the order fulfilment and inventory allocation processes.

In many instances, a VMI relationship is the first time in supply chain where both partners have access to, and are measuring performance using the same metrics. When two companies are focused on the same goals and have access to the same key performance metrics, a true supply chain partnership emerges, resulting in a better performing supply chain.

VMI delivers tangible results throughout the supply chain. As the concepts and practices of 'lean' extend beyond the manufacturing floor down through the supply chain, VMI is the enabling process to drive out costs and time. For success of VMI implementation the following are the keys:

- Set, review and maintain performance goals.
- Manage all SKUs through VMI to minimize transactions.
- Ensure data accuracy.
- Utilize automated replenishment system.
- Organize periodic performance reviews.
- Use the metrics to find cost and inefficiencies.

VMI can be made to work, but the problem is not just one of logistics. VMI often encounters resistance from the sales force and distributors. The issues are roles, skills, trust and power shifts.

Effective implementation of VMI depends on smoothly addressing the concerns of various stake holders. Some of the concerns can be addressed as explained below:

- Defining incentive programmes based on partnership and not on sales volume.
- Building partnerships with management commitment.
- Conduct simulations and pilots before actual implementation.
- Organize training sessions before launching VMI programme.
- Getting into service level agreements (SLA)

Box 6.3

Marico Industries

Marico Industries, a leading FMCG company developed an in-house distribution automation software and an Internet-based system called MI-NET; allowing distributors to log in and feed the data online. Using this software Marico knows the inventory level at distributor's place. Marico accordingly dispatches inventory to distributors as per inventory levels at distributors stocks. The implementation of the VMI with key distributors resulted in a declining sales skew within a month, and reduction in excess stocks and stock-outs at the distributor's end.

P&G and Wal-Mart

Wal-Mart uses satellite-based communication system. Using this communication system P&G replenishes Wal-Mart's inventory, based on inventory data received from Wal-Mart's distribution centres (DCs). Online data allows P&G to manage the inventory levels to insure that P&G products are in stock at all times. This helps P&G to reduce the order cycle time and increase inventory turns which results in a reduction in the inventory of the entire system. This VMI system by P&G at Wal-Mart is working successfully for the last many years.

ONLINE PROCUREMENT

E-procurement system allows businesses to use the Internet for acquiring the necessary goods and services. There are three main categories of e-procurement systems. One type focuses on improving the transactions and the decision-making capabilities of the company. Businesses may deal with hundreds of transactions on a weekly basis, but these applications simplify the process and help to build stronger relationships between buyers and suppliers. The second category of e-procurement system involves managing assets. Systems in this category provide inventory management, maintenance scheduling, in-house product availability, as well as other similar services. These applications are useful for businesses that need to keep a close watch on the quality of materials in stock. The third category includes systems designed to optimize a company's production operations. Many of these applications deal with the entire production cycle, including the procurement of materials when the inventory runs low, the management of supplier contracts and the production scheduling. Because of the differences between the systems, it is important for companies to choose the one that is most appropriate for their industry. For example, an automobile manufacturing company is likely to use systems from the third category. The applications would allow them to maintain minimum level of inventory but they also need to have a system which helps them plan and forecast their production. However, companies such as automotive repair shops, would be more likely to use e-procurement systems from the second category. Since they need to keep track not only of their inventory of car parts, but also in helping them to set repair schedules.

E-procurement system in businesses imparts benefits such as saving money on purchases, improving the timeliness of the purchasing process and eliminating waste. In addition to these benefits, companies can also improve the efficiency of their supply chain. Additionally, using e-procurement to enhance supply chain relationships can make it easier for accounting departments to track and keep a record of payments and invoices.

E-procurement systems do not automatically boost supply chain efficiency. However, the company must select a system that has the capabilities necessary to achieve those benefits first. For example, the system must include applications to assist with contract management, including pricing information, maintaining sales terms and helping negotiations. By having all of this information in one place, the purchasing process is expedited. E-procurement allows buyer to easily compare suppliers (on price) so that the best one can be chosen to meet that company's particular needs. However, choosing the right supplier depends on more than just price: it also involves product availability, customer service, industry reputation and quality.

E-procurement is more than just a system for making purchases online. A properly implemented system connects companies and their business processes directly with suppliers while managing all interactions between them. This includes management of correspondence, on tender bids, questions and answers, previous pricing, and multiple emails sent to multiple participants. A good e-procurement system helps a firm organize its interactions with its most crucial suppliers. It provides those who use it with a set of built-in monitoring tools to help control costs and assure maximum supplier performance. It provides an organized way to keep an open line of communication with potential suppliers during a business process. The system allows managers to confirm pricing, and leverage previous agreements to assure each new price quote is more competitive than the last.

E-procurement helps with the decision-making process by keeping relevant information neatly organized. E-procurement process is template-driven, which makes all transactions standardized and traceable. Keeping track of all tender bids means leveraging firm's knowledge to obtain better pricing. Well-

managed e-procurement helps reduce inventory levels. E-procurement system allows multiple access levels and permissions and help managers to organize administrative users by roles, groups or tasks.

PURCHASING CARDS

As transaction costs goes up, companies are looking to buy smarter and cut costs. In recent years, many companies are providing to their key employees in procurement department with corporate credit cards/purchase cards that can be used for purchasing materials.

The purchase cards are similar to credit cards. The purchase cards are issued by leading card companies such as VISA, Master-Card and American Express. Generally, the purchase cards are used to speed up the procurement of low unit priced items in small quantities. Like other credit cards, the bill is forwarded to the purchasing department of the company for payments. Sometimes the cards work only between a buyer and suppliers eliminating the bank. The purchase card reduces the amount of paper work. The cards also contain information on sales tax data, customer code, taxpayer PAN and transaction type. The vendors are also benefited by cards at customer end as the vendor receives payment within a short period. There is no transaction cost at supplier end. The supplier prior to effecting transaction through credit card knows the credit worthiness of the customer. Hence his risk is minimized.

SUMMARY

Traditionally enterprises were using EOQ model for procurement of inventory. EOQ is an inventory decision model, based on differential calculus. It determines the optimum order size for purchasing an inventory item of stock. The EOQ provides a model for calculating the appropriate reorder point and the optimal reorder quantity to ensure the instantaneous replenishment of inventory with no shortages. 'Just-in-time', is a manufacturing organization's philosophy, pioneered by Taiichi Ohno in Japan at the Toyota car assembly plants in the early 1970s. JIT cuts waste by supplying parts just at the time of assembly process. At the heart of JIT lies the KANBAN which means a card in Japanese language. The KANBAN card is sent to the warehouse to reorder a standard quantity of parts as and when those have been used up in the assembly/manufacturing process. JIT is used primarily for high-volume repetitive flow manufacturing processes.

Under a VMI, a supplier takes full responsibility for maintaining stock of its products at a customer's facility. VMI differs from traditional inventory management. In VMI, the customer is billed for material when consumed and not when delivered. In VMI both supplier and manufacturer get benefited by way of immediate payment realization and reduction in inventory, respectively. E-procurement method of purchase makes use of the Internet. E-procurement system is beneficial in saving money on purchases, improving the timeliness of the purchasing process and eliminating waste. A well-managed e-procurement helps reduce inventory levels.

REVIEW QUESTIONS

1. Explain EOQ model of procurement and its relevance in today's context.
2. Explain JIT system of procurement and its issues in implementation.
3. Compare and contrast EOQ and JIT system purchasing.

4. Explain VMI and its advantages.
5. Discuss challenges in implementation of VMI.
6. Explain the concept of 'purchase card' and its utility.

INTERNET EXERCISES

1. Website, [http://Home.Ubalt.Edu/Ntsbarsh/Business-Stat/Otherapplets/ Inventory. html](http://Home.Ubalt.Edu/Ntsbarsh/Business-Stat/Otherapplets/Inventory.html);, is a part of e-Lab learning with objects for inventory decision-making. Visit the above site to study EOQ Models for Inventory Management.
2. Inventory Solutions Logistics Corporation is a full service warehousing, distribution and logistics corporation. Visit [http://www.inventorysolutions.org /def_jit.html](http://www.inventorysolutions.org/def_jit.html) and study the various inventory solution in practice.

VIDEO LINKS

1. Toyota JIT Concept, <http://www.youtube.com/watch?v=OBK2KM0Gmq0&feature=related>
2. Global Procurement, <http://www.logisticsbureau.com.au/video-supply-chain-logistics.html>

PROJECT ASSIGNMENTS

1. Study the 'JIT System' of procurement in any automobile manufacturing company and find out the problems they faced initially in implementation of the same. Further study how they have re-engineered their business processes to suit JIT system.
2. VMI is mostly practiced in automobile industry for high value and large consumption SKUs. Take any auto manufacturing firms and study their VMI system in details.

NOTES

1. Sople V. V. (2009), *Logistics Management*, Pearson Education (2 edn), pp. 112.
2. APICS (1992), *APICS Dictionary*, American Production and Inventory Control Society Inc., Falls Church, VA.

E-PROCUREMENT

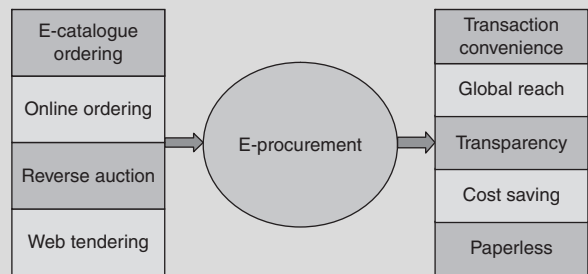
“E-procurement can deliver significant ‘buy-side’ cost savings to businesses, companies and marketplaces. They are now looking at other appropriate areas of their business to drive forward of performance throughout the value chain based on supply and demand

—Lusiana”

Chapter Objectives

To understand:

- Traditional vs. e-procurement
- E-procurement framework and models
- Reverse auction and its benefits
- Outsourcing e-procurement
- Open trading environment



E-procurement Model

Organizations allocate significant percentage of the purchasing budget to non-productive items like travel, office supplies and follow-ups. The optimization of supply chain cost requires streamlined, collaborative business processes and a supporting technology infrastructure that enables synchronization of information and material flow across the supply chain. Today's enterprises are increasingly reliant on their business partners for competitive advantage through cost leadership. E-procurement can reduce significant 'procurement' cost of the business. A number of innovative companies are now looking at procurement as an important area of their business to drive forward their business performance through this component of the value chain. Automating the purchasing (of goods and services) through implementing an e-procurement system, organizations can make significant reductions on purchasing expenditures resulting in significant increase in the bottom line.

Tata Motors Benefiting Through E-Procurement

Tata Motors is India's largest and the fully integrated automobile company. It ranks top 6 among manufacturers of medium and heavy vehicles in the world. In late the 1990s and early 2000s, Tata Motors were facing financial losses due to cost pressures. They also lost market share to the competition. Tata Motors identified the problem in their cost structure. Materials cost amounted to 65 per cent of the entire cost in FY2000. Tata Motors decided to address their problem starting from direct materials procurement. They used e-sourcing and e-procurement to attain precision and cost cutting. Tata Motors has employed SAP and Oracle-based IT systems to support its database and transactions management. They went into outsourcing of e-procurement process to Free Markets, Tata Technologies Ltd. and Dynamic Logistics to enhance the efficiency of the supply chain. Systematic implementation of e-sourcing ensured savings to a tune of Rs 2000 mn (\$50mn) in FY2001. Tata Motors embraced e-sourcing with an objective of cost reduction, smooth transactions with lower cost, transparent processes and global reach for material sourcing.

In 2001, Tata Motors employed FreeMarkets.com, a global supply management solutions company as partner in running the e-Sourcing programme. The first reverse auction under this initiative was conducted in June 2001. E-procurement in the company revolved around reverse auctions for on-line negotiation of prices for a variety of products and services procured by the company. Tata Motors extended the technique to the procurement of maintenance, annual contracts, tool engineering, administration, canteen purchases, medicines and hotel bookings. They extended the newfound success of e-sourcing to tier-II sourcing, etc. Using e-procurement, Tata Motors covered around Rs 18,500 mn (\$370 mn) of spend from 2002 to 2004. They have covered 25 per cent of direct material purchasing and 55 per cent of indirect material purchasing through e-procurement in that period. E-procurement has reduced purchasing cycle time, enhanced flow of information, consolidated supplier base, reduced procurement overheads and rationalized supply conditions, quality norms and made provision for multiple locations procurement possible.

Tata Motors suppliers were also happy with the e-procurement technique since it allowed them to reduce their marketing cost, getting assured contracts, fair and transparent transactions and clarity of supply conditions. The e-procurement project brought discipline and rigour to the sourcing process and resulted in the creation of a cost-competitive and cost-conscious supply base. The benefits accrued are reduced procurement time, reduced procurement cost, realized better value for money spent, standardized the procurement process and allowed equal opportunity to all vendors.

INTRODUCTION

Procurement is a critical function in the supply chain. The procurement function includes vendor development, material ordering, issuing purchase orders, preparing procurement schedules, follow up with vendors and releasing payment. As the size of the business firm increases, the number of transactions goes up and so is the cost of transaction along with the time taken to complete the purchase order. The transaction-related cost is hidden and never taken into consideration in traditional costing

Exhibit 7.1 *Traditional vs. E-procurement*

<i>Traditional Buying Process</i>	<i>E-procurement Process</i>
<i>Paper based</i> Floating enquiry, inviting quotations, evaluation, negotiations, purchase order, follow-ups, etc.	<i>Paperless</i> Online process without using paper.
<i>Biased</i> The terms and conditions are favourable to either seller or buyer depending on his strength.	<i>Non-biased</i> Terms and conditions mutually acceptable and varies with each contract.
<i>Secretive</i> Bidder not allowed to know the competitors' names as well as their offerings.	<i>Transparent</i> Each of the competitors know others prices (not the names) as auction is online.
<i>Time-consuming</i> Takes sometimes months together for the final placement of order.	<i>Quick process</i> Can be finished within a few hours.

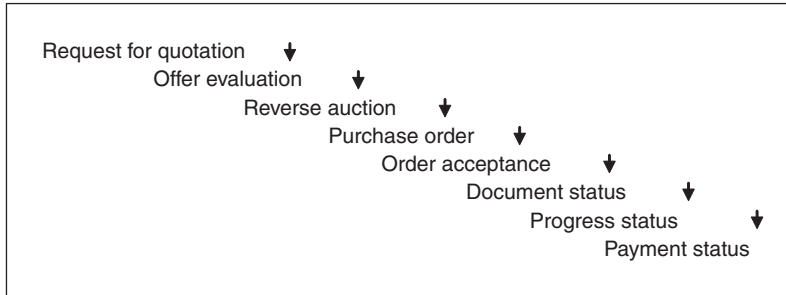
methods. In the traditional procurement, (Exhibit 7.1) the time required for processing an order varies from 15–45 days. In the competitive business environment, organizations are in lookout for cost and time reductions. E-procurement, which is done for the standard products (procured in volumes) helps in reducing time and the cost of transaction. The buying process is completed within one/two hours in reverse auction method. E-procurement process is paperless resulting into tremendous time and cost saving. However, the buying organization has to develop suppliers for calling for on-line offer submission and negotiation process. E-procurement process is done with the help of specialized software solution. E-procurement services are provided by service providers or large organizations have their own software to perform e-procurement process. Indian firms like Carrier Aircon, Telco, Blue Star, Hindustan Motors and Wipro are outsourcing the e-procurement for standard items. On the other hand, L&T, Asian Paints, Marico Industries, Maruti, Grasim, LML, HDFC, Amtrex, Blue Star and Goodlass Nerolac Paints have their own software for carrying the e-procurement. With e-procurement many companies had an effective saving of 2 per cent to 22 per cent on procurement costs. Many large companies in various industry sectors are leveraging on e-procurement to increase their margins.

E-PROCUREMENT

It is a method of the purchase of goods or services electronically and is an integral part of an overall strategic procurement plan in the current business environment. The plan includes, but is not limited to strategic sourcing or supplier rationalization, supply chain automation, and participation in one or more marketplaces. The commodities purchased can be operational resources (non-production) or production resources (raw materials).

E-procurement is perhaps the most direct and effective way for an organization to leverage the Web to reduce costs, improve productivity and boost profits. It automates and streamlines the process by creating a Web-based, self-service environment that pushes product selection and order initiation to the computer for a buyer while maintaining corporate trading agreements, workflows and authorization rules. With adoption of e-procurement, the purchase department is relieved of time-consuming traditional procurement activities. They now focus on upstream activities such as supply source development, negotiation and vendor management.

For organizations procurement is a most critical decision-making process. Procuring goods at the right price and from the right vendor takes a long period. However, e-procurement provides an on-line platform for both buyers and suppliers to interact and reduce the cycle time of procurement. With

Exhibit 7.2 *Building Blocks of E-procurement*

e-sourcing the buyers analyse, purchase and identify opportunities at a better price and moreover, they have direct access to goods and services. The building blocks of e-procurement are shown in Exhibit 7.2. E-procurement helps in compression of procurement cycle. It is a transparent ordering process as the supplier can see the lowest price quoted without the vendor's name being revealed. If e-procurement is outsourced to a third party service provider, there is no possibility for dummy quotes to creep in.

E-procurement process reduces costs and increases efficiency. Thus, businesses can increase efficiencies and maximize savings on procurement. This helps them to make better use of their resources and capital. In India, procurement services and solutions segments such as Wipro's 01markets and neoIT are the leading players. At the global level, for e-procurement services FreeMarkets, Commerce One and Ariba Technologies are the most popular names.

E-procurement is done through the Internet, electronic data interchange (EDI) or enterprise resource planning (ERP). It is done in B2B and B2C business as well. E-procurement Web site brings buyers and sellers on the same platform to perform transactions. The buyers and sellers are subscribers to the site. Buyers specify the price for goods and invite the bids from the sellers. E-procurement software helps to automate buying and selling. E-procurement process reduces the procurement (transaction) cost and shortens procurement cycle.

REVERSE AUCTION (RA)

RA is a process of bidding by suppliers through the Internet using reverse auction engine (a software programme). All the bidders can participate simultaneously in the bidding process. The buyer fixes the specifications and commercial terms, and fixes purchase price below which the bidders have to quote. The bidding starts at pre-fixed start time. The bidders can see their prices quoted by others so that they can reduce the prices step by step. However, buyers can see all the bids.

In the bid, no increase in price is allowed. The entire bidding process finishes within an hour or so. Normally hectic bid changes are seen just before closing time (Figure 7.1). Reverse auction gives good price reduction benefits to buyers.

E-procurement techniques like bidding on reverse auctions are one wherein companies will find they have everything to gain. It's a win-win situation for both the buyers and the sellers. The following are the features of e-reverse auction process that cater to the needs of global outsourcing:

Easy process Taking part in a reverse auction as a bidder does not require much efforts as required in a standard RFP (request for procurement) process. Paperwork is greatly reduced in the online setting. In RA process, the interface is easy between the buyers and the sellers.

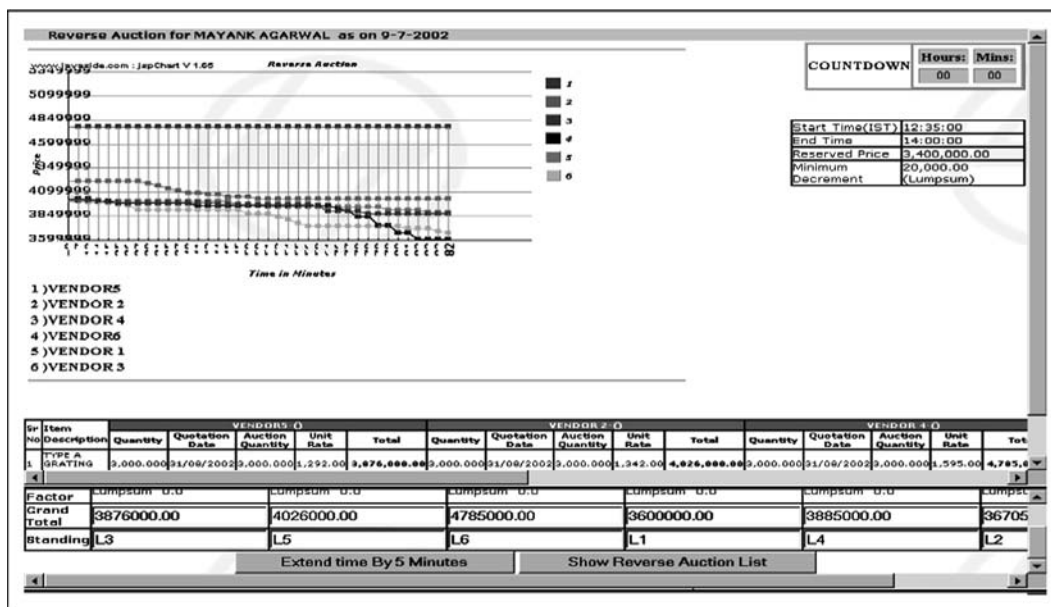


Figure 7.1 Reverse Auction in Progress

Global reach In RA, bidders will find requests for products and services from all over the world. Companies can access new markets across the globe without having their physical offices at the locations.

Transparency Reverse auctions are set up to be very transparent. The bidders can see what others are offering. This is an excellent way for companies to actually see if they are as competitive as they need to be.

BENEFITS OF E-PROCUREMENT

With e-procurement process, the buyer benefits as a result a significant reduction in the procurement cost (Exhibit 7.3) and procurement cycle time. For example, Larsen & Toubro could procure a 1.6 lakh inch-metre piping (bulk) valued at Rs 80 mn within 22 days of receipt of final MTO to achieve the milestone date in one of the projects. The other benefits L&T got was a substantial reduction in procurement cost. This is because of reduction in skilled manpower requirement, reduced photocopying, courier and faxes of large files, reduced phone calls, etc. E-procurement process can be initiated from any place and at any time as per the convenience of the buyer and seller. The entire process is transparent and it is a common platform for all suppliers who are competing.

An e-procurement solution supports buyer-seller collaboration. It enables timely communication of the buying plan to the suppliers. This helps in reducing procurement cost and supply chain wastage.

There are three process cycles in procurement such as identifying, evaluating vendors and calling the suppliers for submitting bids. All these processes are simplified in an e-procurement process.

With reduction in procurement time using e-procurement process, the procurement personnel could devote more time for procurement action on complex and critical items other than the standard

Box 7.1

Reverse Auction at Blue Star...

Blue Star is India's largest company manufacturing central air-conditioning systems and having an annual turnover of Rs 4760 mn (\$100 mn) with network of offices in 29 cities and three modern manufacturing facilities. Blue Star manufactures and markets a wide range of air-conditioning and refrigeration systems and products. Their other businesses include marketing and maintenance of hi-tech electronic and industrial products such as machines, data communication products, medical and analytical instruments, and special control valves.

Using the 01markets reverse auction tool, Blue Star sought to procure galvanized steel. Blue Star wanted to optimize its costs by increasing the base of suppliers, thus creating a transparent process online and cutting cycle time drastically for the organization. Due to the nature of steel, getting every manufacturer on the same base level for bidding was a crucial activity. Due to spread of the suppliers across different geographies, coordination was another crucial activity. 01markets.com understood the requirements through a comprehensive study. 01markets.com ensured compliance of suppliers, standardizing supplier's bids and products in order to create a level playing field. The bidders were called on Web on a particular day. RA continued as follows:

At 15.00 hours, the cost of the galvanized steel coils stood at Rs 9.6 mn. But as the RA continues the cost fell, to Rs 8.8 mn. In 2 hours, four manufacturers had submitted 21 bids to move the price-line past the reserve price to create a final savings of Rs 0.8 mn as detailed below:

Supply Category: Galvanized steel

Quantity: 1 lot

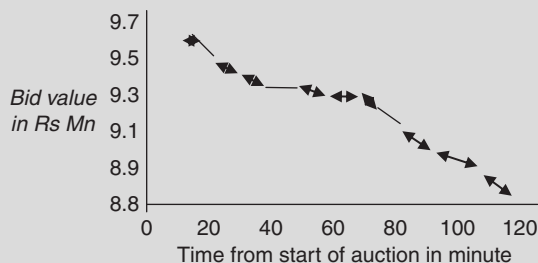
Mode: Reverse auction

No of suppliers: 4

Pre-auction buying price: Rs 9.6mn

Pos-auction price: Rs 8.8 mn

The reverse auction process just finished within 120 minutes



Source: 01 Markets–Wipro

ones. The bill of material consisting of more standard items can derive excellent benefits on cost front by reverse auction. The organization needs to invest in the E-procurement software package, as it may not get economies of scales, which the service provider may get to serve a large customer base. It is then prudent to outsource the e-procurement to a service provider.

Exhibit 7.3 E-procurement Benefits—Case Studies

<i>Buyer: Carrier Aircon</i>	<i>Buyer: Hindustan Motors</i>
Material—Compressor Oil	Material—Automotive Paints
Purchase value—Rs 120 lakh/pa	Annual purchase—Rs 12.25 Cr
E-Sourcing provider—Freemarkets India	E-Sourcing provider— 01 markets (WIPRO)
Bidders—9 Large oil companies including PUCs	Participant suppliers—4 Nos
Results—12% saving (When oil prices go up worldwide)	Results—Rs 77.69 lakh (6.34% Savings)

Source: BT, Feb 2002

CONSTRAINTS OF E-PROCUREMENT

As against the benefits, e-procurement has certain constraints, which are as follows:

- **Infrastructure:** It includes lack of continuous Internet facilities at the supplier's end. There is difficulty in downloading large files due to bandwidth/speed limitation. Sometimes there are problems in continuous access to e-procurement site. In few occasions, dates of auction are rescheduled once or twice.
- **Reluctance for acceptance of new system:** Lack of knowledge and reluctance to use e-procurement at supplier's end will pose a major problem in utility of the system. Reluctance on the part of buyers to get feedback from suppliers and thereafter will be implemented with the help of the e-commerce team.
- **Different time zones:** Vendors at various time zones in reverse auction.

E-PROCUREMENT TYPES AND MODELS

There are six main types of e-procurement:²

- **Web-based ERP (electronic resource planning):** Creating and approving purchasing requisitions, placing purchase orders and receiving goods and services by using a software system based on the Internet.
- **E-MRO (maintenance, repair and operating):** Similar to Web-based ERP except that the goods and services ordered are non-product related to MRO supplies.
- **E-sourcing:** Identifying new suppliers for a specific category of purchasing requirements using the Internet.
- **E-Tendering:** Sending requests for information and prices to suppliers and receiving the responses of suppliers using the Internet.
- **E-Reverse auctioning:** Using the Internet to buy goods and services from a number of known or unknown suppliers.
- **E-Informing:** Gathering and distributing purchasing information both from and to internal and external parties using the Internet.

A variety of practices are used in e-procurement. Some common purchasing methods are as follows:

- **Private trading exchanges** are owned or operated by a single company in order to trade or purchase materials with the owner's suppliers and customers.
- **Public trading exchanges** offer a common technology platform for auctions, spot buying and catalogue ordering. They are either owned by independent investors or industry consortia.

Exhibit 7.4 *E-procurement Models*³

<i>Models</i>	<i>Advantages</i>	<i>Disadvantages</i>
Buyer-centric	– Tight control over the buying process – Solution to meet the buyer's specific needs – Integration of solution into buyer's financial control system – Speedier procurement – Lower transaction cost to suppliers	– Restricted purchasing – Higher maintenance cost – costs to buyer – Issues of fresh data to suppliers – High system investments – Limited choice to suppliers
Seller-centric	– Lower transaction cost to buyer – Wider access to market	– Limited support to the buyers. – Buyer have no control on data
Third party-managed	– Both buyers and sellers have wider access to market – Useful to volume purchasers – Solutions are cost-effective	– Higher transaction cost – Solutions have huge going costs – Buyers and suppliers need to subscribe for access

- An on-line reverse auction is a process by which suppliers are asked to place competitively lower bids to provide materials or services.

There are two types of E-procurement models. The first one is buying application software and other one is electronic catalogue(s). The buying application software allows an organization to carry out purchasing processes and to perform a variety of procurement functions. Three basic models (Exhibit 7.4) can be broadly described as follows:

Buyer-centric e-procurement model A buying organization has its own software to manage the procurement process. The data from suppliers is fed into the system and can be used by the purchasing officers. For example, all major auto manufacturers have Web-based e-procurement models for the auto parts, components and accessories. In 2000, Ford, GM and Chrysler released a co-operative e-procurement site 'COVISINT' for joint procurement action to achieve huge cost reduction from suppliers based on economies of scale.

Seller-centric e-procurement model In a seller-centric procurement model, the seller is at the centre. Buyers use the supplier's system which is accessible over the Internet. The buyer can go through supplier's catalogue which is available on Web site and place orders. Examples are eBay, Yahoo, Rediff, etc.

Third party-managed e-procurement model Here the procurement software and system is provided by a third party which is a service provider. The buyer and suppliers are brought together to initiate and complete the transaction. Both the buyer and seller have to pay the fees to a third party for the procurement service provided.

These models falls under 'closed system' and have their pluses and minus through the perspective of buyers and sellers.

OPEN TRADING ENVIRONMENT

Looking into the limitations of closed system, an open trading system is getting evolved. This allows buyers and sellers to establish systems that meet their requirements. Under an open trading model,

buyers implement software that allows them to establish and maintain local internal catalogues (voluntary), but also to access e-marketplaces and supplier's stand-alone catalogues.

Suppliers have choice to log their own catalogues or use marketplaces. These arrangements benefit suppliers, who are able to make their catalogues visible to the large number of customers. The buyer can access supplier's Web site or direct to supplier in the open market.

There are two most prominent standards on open buying available on the Internet: standard and XML (extensible markup language). The open-trading architecture is based on the premise that process owners should be responsible for maintaining information associated with their business processes and procedures.

GOVERNMENT INITIATIVES

In India, the Government of Andhra Pradesh has leveraged information technology (IT) to attain a position of leadership and excellence in the information age and to transform itself into a knowledge society. The government envisioned providing good governance by establishing a committed, accountable, responsive, inspiring, nationalist, genuine administration—that is, a caring government. E-procurement is one of the vehicles that has been gainfully used in reaching the goal of caring governance by the government.

The www.e-procurement.gov.in (Govt. of Andhra Pradesh) e-platform provides its members with access to several trading suppliers. E-procurement auction services offer government departments easy-to-use, Web-based solutions for conducting dynamic exchanges in an on-line environment. It provides real-time bidding solutions for buyers and sellers that bring in an unprecedented level of profitability, control, and simplicity to corporate procurement and liquidation processes.

SUMMARY

In the new economy, those companies that can effectively leverage the Web to redesign, automate and integrate all business operations will remain competitive and gain success. The new e-businesses will be built on a dynamic supply chain or series of supply chains—collectively known as a 'supply net'—that can be assembled and adjusted as needed to provide rapid and optimal response to customer demand. Responsiveness will only be possible if all participants of an e-business supply net can communicate, analyse and collaborate on requirements, plans, schedules and other business initiatives as a single, integrated, extended enterprise. E-procurement is a panacea to most of the above-mentioned issues. The advantages and benefits of E-procurement will far outweigh the minimal risks involved in the implementation of this solution.

Procurement is an important organization function in the supply chain. With E-procurement process the chances of finding a right supplier that matches the firm's requirements are more. E-procurement enables companies to reduce material costs, slash overhead and improve supplier relationships. Both government and private sector companies in India are using E-procurement to gain cost advantage. There are three basic electronic procurement models, namely, buyer-centric, seller-centric and e-marketplace. In these models, one dominant player establishes the trading rules for the system. However, these rules limit other participants' flexibility. The above limitations can be overcome in an open-trading environment, which allow buyers and sellers to establish systems that meet their requirements.

REVIEW QUESTIONS

1. Compare and contrast traditional and E-procurement processes of buying.
2. Discuss the various E-procurement models prevailing in the market.
3. What are the major benefits of E-procurement? Support your answer with illustrations.
4. What is 'Reverse Auction' and how does it differ from 'Forward Auction'?
5. What are the major constraints of E-procurement process?

INTERNET EXERCISES

1. The Office of Government Commerce (OGC) is an independent office of HM treasury, established to help government deliver best value from its spending. Find out from various cases how e-procurement process could deliver value from it. Visit http://www.ogc.gov.uk/case_studies_eprocurement_case_studies.asp
2. Online reverse auctions are an ideal way for buyers to reduce purchasing costs and streamline procurement when purchasing capital equipment, materials and services from suppliers around the world. Visit <http://www.reverseauctionsblog.com/e-procurement> and study various benefits of E-procurement process to both the buyer and seller.

VIDEO LINKS

1. E-procurement, <http://www.youtube.com/watch?v=cVtPU5qsYXM>
2. Improving productivity through e-procurement, <http://www.supplychainbrain.com/content/videos/biotech-09/improving-the-bottom-line-through-e-procurement/>

PROJECT ASSIGNMENTS

1. 'Web Tendering' is quite common with Indian government departments and public sector units in India. Conduct a survey of these establishments and write a case report on 'Web tendering' process, its pros and cons, products for tendering, and terms and conditions.
2. Visit online 'B2B Hub' in any industry sector (construction, steel, auto parts, chemicals, etc.) and write a report on its functioning, strategies for attracting memberships, transaction policies and procedures.

NOTES

1. 'E-procurement for Communications: Complex Requirements, Unique Solutions', <http://www.busmanagement.com/pastissue/article.asp?art=26809&issue=165>
2. <http://www.wikipedia.com>
3. <http://www.agimo.gov.au>

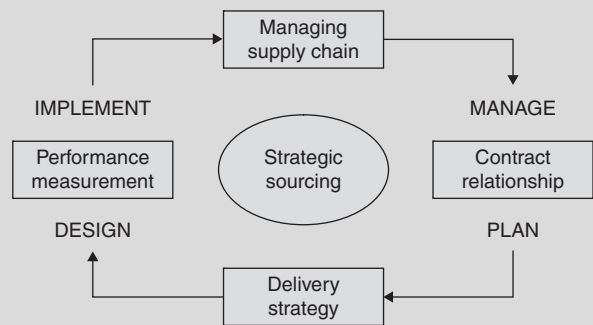
STRATEGIC SOURCING

“Purchasing should be concerned with the strategic planning aspects of procurement process
—Ben Lapner”

Chapter Objectives

To understand:

- Role of procurement in supply chain
- Prevailing procurement practices
- Strategic procurement process
- Strategic sourcing vs. traditional sourcing
- Success factors in strategic sourcing



Framework of Strategic Procurement Process

Strategic sourcing is a process aimed at obtaining maximum advantage on cost, technology, process and quality by leveraging the company's buying power. Strategic sourcing is one of the key methods that purchasing departments are using to lower costs and improve quality. Strategic sourcing recognizes that people, including innovative suppliers, are a valuable part of an organization and focuses on reducing waste or non-value-added costs. It is a disciplined, systematic process for reducing the total costs of externally purchased materials, products and services while maintaining or improving levels of quality, service and technology. Strategic sourcing studies are tedious and time-consuming initiatives.

Strategic Sourcing at Speed Auto²

In Practice...

Speed Auto (SA)—a leading two-wheeler manufacturer—has a large supplier base with a high level of technical knowledge in their respective areas. SA decided to involve the suppliers in the manufacturing process and increase R&D capability. SA wants their suppliers to be more integrated in the product development process in their supply chain. They mapped out the product development process at SA to determine at which point each supplier is inducted into the process and identify where in the supplier should be included.

After process mapping, a list of select suppliers were invited to the SA headquarters to introduce the concept and find out which supplier would best fit in SA product development process. The suppliers were from various industries such as forging, electrical, tyres, glass, plastics, spark plug, fuel injection pump, stamping, metal pressing, aluminium, steel, castings, gears, rubber, etc. The executives of SA explained to the suppliers that the company is shifting from its internally focused process to suppliers' focused process to become more involved in its product development. To emphasize the point, suppliers were given a full walk through of the production plant and were given a brief on manufacturing process. While most of the suppliers made materials that went into two-wheelers, not all of them were fully aware of the actual process at SA. During the plant visit, some suppliers provided valuable feedback and process improvement ideas. SA also prepared a list of 15 specific challenges and it wanted the suppliers' input on the same. Suppliers were advised to study the challenges and submit their solutions to SA at the earliest. SA got hundreds of ideas from suppliers on the above 15 areas. The ideas ranged from materials management to product development to cost reduction. Some were directly focused on addressing the challenges.

Suppliers appointed specific people to focus on the SA challenges. SA's procurement department fixed up facilitators to coordinate with the suppliers. SA has taken care of providing access to suppliers on certain proprietary information through legal contracts. For SA, this strategic sourcing project served to bring the procurement and the technology organizations closer. SA found that suppliers were thrilled with the openness of SA to share their new technologies in development. With the great benefits from the above strategic sourcing move, SA continued supplier's involvement in innovation process in future.

INTRODUCTION

Today the procurement function of most of the high-technology companies covers interaction with important business processes that goes into planning, sourcing, making and delivering a final product. Since material cost accounts for over 50–70 per cent of the total cost of production of product, efficient functioning of material management has received greater attention. Hence, the sourcing has been identified as one of the thrust areas for improvement in most of the manufacturing companies.

Strategic procurement exists whenever an organization seeks to acquire a project, a service, a product, or a commodity that has high value significance or importance to it. The need to procure a strategic project or service gives rise to a challenging business exercise. Such exercises should be approached with seriousness and well-structured processes.

The process used for strategic procurement is designed to ensure that the organization's needs, issues and risks are effectively managed. For achieving this, the best qualified supplier is selected for the purpose. Strategic procurement often gives rise to strategic relationships. A strategic relationship may exist either within an organization or between organizations. They exist even when a product, service or commodity is being supplied that has significant potential to damage the organization in some form if the supply is delayed, not affected or increases in cost.

Strategic sourcing is a process which starts with the thorough analysis of the input material. The analysis covers the volumes the firm buys, economies in purchasing, marketplace of the products and usage pattern. The relationship with suppliers is highly integrated in strategic procurement as compared to traditional buying process.

In the strategic procurement process, more focus is given on the products which are purchased in large quantities. These products justify the time and expense, which the buying company would spend on the study of the product and the industry to further develop a long-term relationship with the supplier. The research would be based on how the key suppliers operate, their business practices, and where the money is added to the final cost of the product. The buyer will work with the supplier to redesign processes and lower production costs to maximize on margins that suppliers make.

Box 8.1

Dell—Leveraging Supplier Expertise for Green Packaging Solution

Dell when decided to make its packaging more environmentally friendly, decided to get out-of-the-box packaging solutions. Dell collaborated with packaging suppliers to get the reach of the desired goal.

Dell wanted to reduce the size of its boxes. The aim was to use more green materials in packaging to make it recyclable. Dell wanted a packaging material other than hardwoods and softwoods, which takes a long time to grow.

Dell invited suppliers for brainstorming sessions on new ideas on packaging problem. One of those suppliers was Unisource Global Solutions. The material that Unisource suggested for use in Dell's packaging was bamboo. Being a member of the grass family, bamboo grows back much faster than hardwoods, making it more environmentally friendly. The process of turning bamboo into pulp is mechanical and uses fewer chemicals than paper pulp. The green feature is that bamboo has a smaller carbon footprint as compared to other materials. So from an environmental perspective, bamboo was a final choice for Dell. Unisource provided samples of the materials it would use and suggested supply sources for the raw bamboo. The cost analysis showed bamboo would reduce Dell's packaging costs and the idea moved from the research phase to the production phase. Today, Dell is shipping Inspiron Mini 10 and 10v Netbook with product cushions made of bamboo. According to Dell, this type of project is only possible by adopting strategic sourcing approach. This approach builds a trust with suppliers and makes them feel like they are able to suggest new ideas and solutions at any time. In addition, Dell's packaging suppliers do business in different industries, Dell introduced some best practices from beyond the computer market.

Source: David Hannon, Dell leverages supplier expertise for Green packaging solution by Purchasing, 14/1/2010

In the strategic sourcing approach, the buyer after knowing the market and costs of production at supplier end will set the target price and go backward. The buyer further studies the manufacturing plant and process at supplier's end and suggests improvements to reduce the cost of the product. The buyer's engineer helps the supplier rearrange the production layout to make it more efficient. The collaboration with vendor helps in improving buyer/seller communication, planning lead time, engineering and customer service. The supplier collaboration sometimes leads to significant improvements in quality of the product. The buyer-supplier collaborative efforts gives many benefits such as reduction in inventory levels, speed to market, large cost savings and reduction in product development costs. However, with the strategic sourcing approach, companies reduce the suppliers base, but the supplier need to have commitment to continuous improvement, cost consciousness, customer orientation, employee involvement, flexibility in operation, and financial stability.

PURCHASING ORIENTATIONS

Invariably in the industry there are three orientations observed to procurement. The first one is buying orientation which exhibits discreet transaction. This means that buyer-seller relationship does not exist after the transaction is over. The second one is the procurement orientation wherein the buyer assures the relationship, provided the supplies are as per buyer's requirements (specifications). The supplier has to upgrade his product and process quality as per the requirements of the buyer. There is no technical or financial help to the seller from the buyer's side. The third one is the supply chain orientation, wherein the buyer considers the supplier's supply chain as an extension of his supply chain. The supply chain orientation exhibits strategic sourcing process.

For improving the procurement, companies can go in for innovative methods in sourcing. Many firms build cross-functional teams and purchasing personnel as part of its team building exercise. His involvement is right from product design to commercializing the product. This is contrary to what was practiced earlier wherein purchase people were not involved in product design process and were simply asked to buy the materials after the product design is complete. Today even suppliers are involved in product designing at buyer's end. The suppliers are experts in their field and hence, they bring along with them a large knowledge base to design product that would otherwise be absent. This helps in preventing poor product designs and also manufacturing errors. In many companies, the cross-functional teams include personnel from purchasing, manufacturing, engineering, and sales and marketing.

Procurement is the development of a true partnership between a company and a supplier of value. The arrangement is usually long-term, single-source in nature, and addresses not only the buying of parts, products, or services, but also product design and supplier capacity. Partnering agreements allow mutual exchange of confidential information, proprietary restrictions, non-disclosure, non-competitive, certification of processes, routine audits, delivery performance, quality performance, as well as the transfer of control of a process or product.

The relationship between partners is unlike the traditional adversarial relationship between buyer and seller, where the buyer pits one supplier against another frequently during the year, focusing entirely on unit cost. Under these conditions exists a mutual distrust between parties that is counter-productive to both. For the buying company, there generally exists an unleveraged multiple supply base that results in competitive cost disadvantage. For the seller, he is continually negotiating price

reductions at a disadvantage, kept in the dark, expected to 'jump through hoops' in an instance, and generally kept subservient to the buyer.

In a strategic procurement agreement, both companies view each other as equal partners, assisting each other with training, technical support, and developing complete trust. The buyer involves the supplier in the new product design from the beginning and thus allows the supplier to bring its expertise to the table. The cost savings of the product or process innovations are shared between the buyer and the supplier.

Within the buyer's company, procurement is elevated to the strategic level and included in the long-term planning process. Strategic objectives such as part number reduction, supply base reduction and new strategic agreements are assigned to the senior procurement officer and the procurement group for achievement. The idea is to leverage the procurement for a competitive advantage.

The biggest single issue in strategic procurement is trust and reliability on a group of single suppliers. In such cases, each supplier/partner should have adequate capacity and should be equipped to recover quickly from a sizeable catastrophic event (fire, storm, flood, etc), minimizing the risk and exposure of losses to the buyer.

STRATEGIC SOURCING

Today many companies have started realizing the importance of strategic sourcing as it is a source of cost leadership leading to competitive advantage. Many companies have now switched to E-procurement and further to strategic sourcing.

The strategic sourcing can be defined as an effort on the part of buyers to carefully evaluate and to form long-lasting relationships with suppliers. The relationship between buyers and suppliers is beneficial to both and has an impact on their attitude. In the strategic sourcing initiatives (Exhibit 8.1), buyers consider their suppliers as partners instead of commodity providers and this change in attitude has a positive impact on the success of both the buyer and supplier.

In the manufacturing firm, there is a large spent on purchasing raw materials, parts and components. The purchases are to an extent of 75 per cent of the company's budget to create finished products and services. The flow of input materials has to be continuous without delays to run a company's production cycle without stoppage.

Exhibit 8.1 *Traditional vs Strategic Sourcing*

<i>Traditional Sourcing</i>	<i>Strategic Sourcing</i>
Transactional buying	Knowledge-based buying
Negotiation-based buying	Integrated procurement
Objective avoid stock-outs	Life-time cost approach
Commercial and follow-up skills	Leverage volume buy
Individualistic style	Cross-functional approach
Emphasis on lowering price	Emphasis on value creation
Discreet buying behaviour	Collaborative working style
Authoritative approach	Interpersonal and coaching skills
Discreet contract	Life-time contracting

Hence, companies go for strategic sourcing initiatives to build a stable relationship with key suppliers to ensure that they have a steady supply of quality materials without frequent price hikes. By finalizing a proper purchasing arrangement, the buyer can secure a lower price on those items because the vendor is assured for buying those materials continuously for a longer time.

The supplier selection process should be based on certain parameters. The suppliers should be chosen based on their experience, reputation, price, quality and reliability. The contract terms between buyer and seller should be mutually beneficial. This will help in strengthening the relationships between them.

To cut cycle time, the buyer and seller can use software solutions in procurement process. If a company is interested in strategic sourcing, using e-sourcing software and changing the traditional way of purchasing can enhance the procurement process.

In today's competitive business scenario, the market decides the price of the product and hence the profit of any organization depends on controlling the cost of operations. In most of the manufacturing organizations, 60 to 70 per cent of the budget is spent on the procurement function. Hence, there is a significant opportunity in improving the bottom line through strategic sourcing. Thus 'strategic sourcing' is a comprehensive process designed to pursue all avenues of creating value by leveraging a company's buying power with select suppliers, conducting best price evaluations, sourcing globally and conducting company/supplier joint process improvements.

The strategy is to shift the company's focus from the current 'transaction'-oriented independent buying of goods and services to a 'product'-oriented strategic approach. The strategic sourcing approach is designed to:

- Reduce total cost of acquisition of inputs
- Overview supply market and internal requirements
- Enhance bottom line
- Create value

Many organizations have executed successful strategic sourcing programmes with significant results. Strategic sourcing provides guidelines to handle key sourcing decisions as follows:

- Outsourcing
- Low cost country sourcing
- Accelerated sourcing
- Collaborative sourcing

For high value procurement opportunities, a company should identify the highest impact cost reduction opportunities. They must understand what they buy and from whom they buy it. With strategic sourcing approach the buying organization can avail the following benefits:

- Increased customer leverage
- Increase in buying power
- Rationalization in product specifications
- Better financial controls
- Improved contract compliance

KEY FACTORS LEADING TO SUCCESS

Strategic sourcing is an important activity, the success of which is dependent on the following factors:

- Commitment from top management.
- Involvement in terms of money and manpower.
- Empowering the sourcing teams.
- Structured procedures and practices.
- Continuous monitoring of sourcing process.
- Use of IT and communication technology.
- Rationalization and development of suppliers.

Organizations spend large percentage of the cost of goods in purchasing. This makes purchasing an important area to focus to gain competitiveness. This fact has been realized by some of the progressive companies in India along with its counterparts in the Western world. Tata Steel, HUL and Maruti are a few amongst them.

In today's context, an organization's success could well be tied to the ability of its suppliers to be collaborative in new product development, responsive to a change in demand and achieve improvements in product design and manufacturing efficiency. Supplier collaboration can reduce the need for large inventories. It can quickly introduce goods and services to accommodate new product designs as well as changes in existing ones. Thus strategic sourcing process focuses on the following:

- Knowledge-based buying
- Systems development for world-class purchasing organization
- Value creating relationships with suppliers

In the strategic sourcing process, spend analysis is at the front. It identifies, consolidate and standardize information from a wide range of data sources. With the help of spend analysis, firms can create 'commodity matrix' as shown in Figure 8.1. It is a methodology for placing commodities in four quadrants. It shows expenditure on two dimensions: Complexity of the product and Saving potential of the product. Analysing these dimensions of the product/service reveals the commodities that are to be considered for strategic sourcing study.

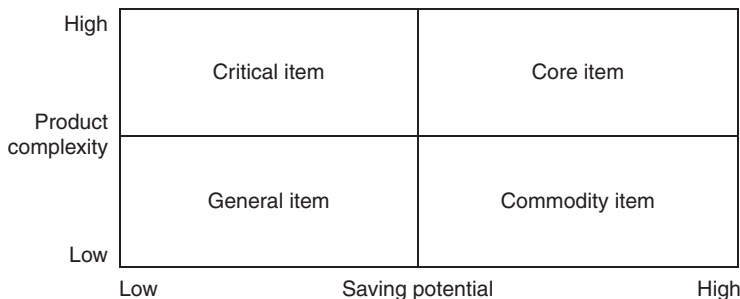


Figure 8.1 *Commodity Matrix*

Core items are those that directly impact the company's throughput. The value of these items is high and so is the complexity of the product. These items are generally manufactured by medium to large companies; the thrust is on developing alliances.

Company's throughput is not affected by the commodity items, but they play an important role in reducing the downtime of the plant and machinery. These items are less complex and saving the potential of these items is also less compared to the core items. These items are typically bought throughout the company under various applications (wide variety) and their demand too fluctuates. Hence, the challenge lies in consolidation and standardization of these items (cement, electrical and mechanical spares).

General items, like commodity items, do not impact the throughput of the company but they facilitate the creation of a working environment. These items are also bought throughout an organization. But unlike commodity items they are less standardized. So, the challenge lies in consolidating company-wide buy so as to maximize the volume. Examples are personal computer peripherals and other office supplies.

Critical items are important from the safety, health and ergonomics point of view of employees. The complexities of these items are high. The requirement is critical items with the right specification.

Strategic sourcing is a proven approach to understand and deliver significant cost reduction and building a sustained value-creating relationship with the suppliers. The fundamental objective of any organization is lowering the cost of goods manufactured. This can be achieved through reduction in prices of purchased products and services and reduction in their specific consumption. Strategic sourcing does exactly the same by addressing the total cost.

Supply chain organization has an active role in improving the bottom line of the company by taking advantage of its systematic influence upon the three types of costs categories, that is, material cost, working capital cost and overhead cost. The first two are linked to procurement policies and methodologies. To achieve competitive edge, firms need to look into two aspects. The first one is the sourcing effectiveness, which covers strategic relationship management, spent aggregation, supplier consideration and category expertise. The second one is procurement efficiency, which encompasses process and policy, automation, centralization, consolidation and outsourcing.

Supply chain management concept eliminates functional silos by accepting the whole supply chain as a 'single entity'. All functions along the chain share the common objective of 'supply' and are involved in strategic decision-making. The emphasis is on integration rather than on interfaces between the different functions. The supply chain becomes responsible for total inventories in the supply pipe line and also for satisfying customer needs and expectations while meeting all agreed customer service parameters and company targets.

MAJOR TRENDS IN PROCUREMENT

The important trends observed in procurement practices across the various industries are as follows:

- Focus on global sourcing.
- Compressing supply chain cycle.
- Eliminating middlemen.
- Green procurement to address environmental issues.
- Real-time value added partnerships.
- Emphasis on outsourcing to focus on core competencies.

Today companies are using advanced supply management techniques such as collaborative cost reduction, end-to-end integration, tiered sourcing and design-to-cost. These techniques generate nearly twice the rate of savings on their procurement of direct and indirect materials and services than companies solely relying on traditional sourcing methods. To capture cost savings through traditional sourcing strategies, procurement needs to drive other operational benefits such as improved working capital or increased quality of products and services.

In addition, companies are doing more in-depth analyses of comparative risks (in offshoring) of various low cost countries (socio-political, environmental and skill sets) as opposed to a rush to a particular low cost country. They are engaged in development of domestic value-added partners. In a nutshell, the following market trends are currently impacting the way procurement is done:

- Increased multi-sourcing.
- Greater significance to governance.
- Continued rise of global sourcing.
- Consolidation in supplier market.
- Emphasis on value creation.

However, the mantra is to keep track of current suppliers, evaluate their pricing and keep competition high while going for global sourcing.

SUMMARY

Strategic sourcing redefines the traditional approach to buying and using materials and services. Strategic procurement is the development of a true partnership between a company and a supplier of strategic value. The arrangement is usually long-term single-source in nature. It addresses not only the buying of parts, products, or services, but product design and supplier capacity. Partnering agreements allow the exchange of confidential information, proprietary restrictions, non-disclosure, non-compete, certification of processes, routine audits, delivery performance, quality performance, as well as the transfer of control of a process or product between buyer and seller.

The relationship between partners is unlike the traditional adversarial relationship between buyer and seller, where the buyer pits one supplier against another focusing on unit cost. Under these conditions exists a mutual distrust between parties that is not productive to both. For the buying company, there exists multiple supply bases that results in a competitive cost disadvantage. For the seller, the buyer is continually negotiating price reductions, which is a disadvantage.

In a strategic procurement agreement both companies view each other as equal partners, assisting each other with training, technical support and developing complete trust. The buying company involves the supplier right from the new product design cycle, allowing the supplier to bring its expertise to the table. The cost savings, in general, are mutually shared for product or process innovations.

Within the buying company, procurement is elevated to the strategic level and included in the long-term planning process. Strategic objectives such as part number reduction, supply base

reduction and new strategic agreements are assigned to the procurement group for achievement. The idea is to leverage the procurement talent and assets for a competitive advantage of the organization.

The strategic procurement is an operational strategy that, if implemented properly, will provide a new dimension to competing: quickly introducing new customized high-quality products and delivering them with unprecedented lead times, swift decisions, and manufacturing products with high velocity.

REVIEW QUESTIONS

1. Discuss the various purchasing orientation in various organizations.
2. Explain the strategic procurement process with its benefits and its limitations on the organizations to practice it.
3. How does strategic sourcing differ from traditional purchasing process? Explain.
4. What are the key factors leading to the success in strategic sourcing process?
5. Discuss the latest trends in procurement process with their implication.

INTERNET EXERCISES

1. Visit <http://www.purchasingpractice.com/services/sourcing-mgt/> to learn about strategic sourcing. Purchasing Practice was founded by 'Unique Consultancy Services' and they have developed tools that deliver dramatic results in the area of procurement.
2. <http://www.eNotes.com>, is a comprehensive online educational resource which is used daily by thousands of students, teachers, professors and researchers. E-notes provide innovative services and online learning environment. Visit <http://www.enotes.com> to study further on strategic sourcing.

VIDEO LINKS

1. Strategic Procurement, <http://www.logisticsbureau.com.au/video-supply-chain-logistics.html>
2. Global Sourcing, <http://www.logisticsbureau.com.au/video-supply-chain-logistics.html>

PROJECT ASSIGNMENT

1. Select a leading firm manufacturing steel (Tata Steels) or consumer durables (LG) or electrical equipments (ABB) and study its strategic procurement process, procedures and policies.

NOTES

1. <http://www.mercorgroup.com.au/What+We+Do/Mercor+Consulting/default.aspx>
2. Based on the summer project report of Suresh Bhatnagar, ITM PGDM student. The name of the company is changed.

LEAN MANUFACTURING

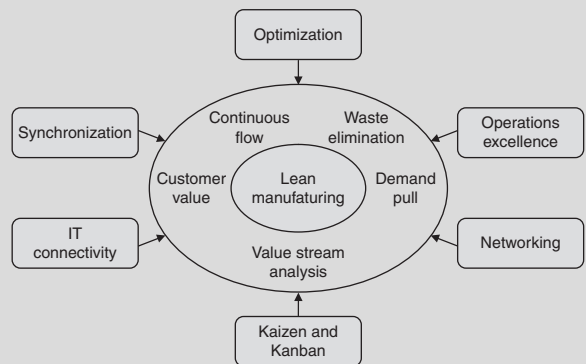
“The most dangerous kind of waste is the waste we do not recognize

—Shigeo Shingo”

Chapter Objectives

To understand:

- Basic elements in lean manufacturing
- Benefits of lean manufacturing
- Integration of lean manufacturing into SCM
- Success factors for leanness



The significant innovations in supply chain management (SCM) are quickly moving away from interfacing with customers and suppliers to integrating with customers and suppliers. The very foundation on which demand and supply business relationships were established is rapidly disappearing in favour of more effective supply chain management methods. An added dimension of rapid response to customer requests and absolute dependability in meeting commitments takes more and more flexibility as the speed of material flow changes. In other words, from the early manufacturing focus, manufacturers started looking at their overall organization in the process of value creation. The lean manufacturers are extending their boundaries to their customers and suppliers in order to create value. Lean manufacturing is all about eliminating wastes from the system and doing more with less time, inventory, space, labour and money.

Lean Manufacturing at Arya Motors

In Practice...

Arya Motors Limited, a manufacturing company in an automotive industry decided to adopt lean manufacturing process. The key challenges before them were to cut lead time for components, increase flexibility in production, improve inventory management and forecasting accuracy and implement lean manufacturing practices. They have accomplished a customized SCM software to help reduce stock volumes and long shipment times. They have selected and implemented 'SAP custom development', an SAP service. They choose SAP because it provided one-stop solution for all elements of consultancy, design, development, and implementation, helped minimize risk and protect existing investment and provided ongoing support and maintenance for a customized solution. The key benefits they could get within next 6 months of implementations were; reduction in lead time for components from 55 days to 35 days, enabled accurate prediction of changeover time for components on production line, provided ability to make changes to production schedule on the day of production, improved planning and forecasting. They could get real-time visibility of stock levels which enabled company to provide accurate bills of materials with shipped products.

Arya Motors for manufacturing tailor-made vehicles, adopted the strategy of a build-to-order which resulted into improved profitability over the years. For the standard models the inventory policy was make-to-stock. With SAP implementation the company would integrate its demand management system with its manufacturing systems to achieve availability of product to fulfill the customer order. The integration helped Arya Motors to get its products with speed to the market. It also helped them to enhance their ability towards incorporating changes late into the order. With SAP system, Arya Motors could take decisions faster because of the information and inventory visibility. The faster and timely information flow could help them to exercise better control on order for 'late changes'. They could gain lot of flexibility in the production system to incorporate change till the order is put into the production cycle. They could resort to reactive production planning to use limited capacities in-house with short lead time. With the web-based information system the vendors could enhance their responsiveness, thus bringing total flexibility into Arya's end-to-end supply chain process. The overall result was drastic reduction in inventory, increased customer response and speed to market. It was a win-win situation both at upstream and downstream of Arya Motor's supply chain.

INTRODUCTION

It is a fact that achieving low cost and high quality no longer guarantees a business its success. In light of stiff global competition, companies are concentrating more than ever on reducing lead-times as a way of achieving operational flexibility. This is because as lead-times decrease, production time falls, quality improves and cost reduces. The methodology that an increasing number of companies are using to accomplish such a business strategy is the implementation of the Toyota production system (TPS), also called lean manufacturing, a strategy for lean supply chain.

Lean manufacturing initiative focuses on cost reduction and increase in turnover by planned elimination of non-value-adding activities. In today's competitive market, lean manufacturing is the solution for survival and success. Lean manufacturing helps organizations to achieve enhancement in productivity by introduction of the latest techniques and tools. It focuses on waste reduction and elimination of the same thereafter. Lean thus becomes part of organization culture and turns every process into a profit centre. Many leading companies across the globe have achieved tremendous productivity and ROI (return on investments) by adopting lean manufacturing practices and techniques. Indian industry has also a few success stories in automotive, process and other industries.

Lean manufacturing is best defined as a management philosophy that focuses on reducing waste in all business processes. It originated from a time when the Toyota of today was a manufacturer of power looms. In its modern form, lean was pioneered by Toyota Motor Co. in the 1950s, when it competed with the U.S.-based dominant and formidable automakers, in U.S. Automobile Market.

The National Institute of Standards and Technology (NIST) manufacturing extension partnership's lean network offers the following definition of lean manufacturing:

A systematic approach to identifying and eliminating waste through continuous improvement, flowing the product at the pull of the customer in pursuit of perfection.¹

The lean manufacturing can also be defined as a systematic approach to reduce and eliminate waste in the assembly line while manufacturing products. It involves continuous improvement with the help of process reviews to reduce costs by means of doing it right at the first time. The basic principles of lean manufacturing are less processing time, zero waiting time and zero inventory. The goal of lean manufacturing is to reduce or eliminate the wastes of the following items:²

- Underutilized workforce
- More processing time
- Overproduction
- Inventory
- Transportation delays
- Product/material defects
- Space
- Money, etc

In simple words, lean manufacturing eliminates waste by means of identifying non-value-adding activities which leads to poor product quality in the production process that will not be able to meet the customer's expectations in the marketplace. Thus, lean manufacturing is driven by five areas: cost, quality, delivery, safety and morale.

MANUFACTURING TRANSFORMATION

The manufacturing industry is witnessing continuous change over the years. The changes are due to many factors such as competitive pressure, changing needs of the customers, technological evolution, use of information technology and environmental regulations.

The two major factors which are playing a prominent role in bringing about changes in the manufacturing strategies of companies are the Internet and removal of global trade barriers. Companies are setting up cost-effective production facilities across the globe to serve global markets. They are going

in for mass customization and are taking initiatives to reach the markets with new products speedily. They are achieving this through adoption of lean manufacturing processes, supply chain integration and collaborative design and development.

To face the changing environment and remain competitive, companies have resorted to lean manufacturing strategy. However, for the implementation of lean manufacturing, companies need to look into the following:

- Organizational structure
- Processes and systems to gain flexibility
- Production capacities for volumes
- Delivery lead-times
- Product quality
- Reduction in inventory

Lean strategy can be manifested in terms of localized manufacturing: cost economics, concepts of customization and postponement, flexible manufacturing systems and large production facilities based on economies of scale.

The new wave of 'digital manufacturing (DM)' is sweeping into industry. DM is an integrated information model spanning across the entire enterprise regardless of size, geography or complexity. DM has three tiers of applications in the manufacturing firm, as follows:

- Applications such as ERP, SCM, CRM, SRM, DRP, MRP, etc.
- Plant level systems of shop floor automation for WIP tracking, job scheduling, plant maintenance, etc.
- Plant automation and controls for tracking and monitoring of various process and equipment parameters.

Today, most of the companies are having IT applications in their infrastructure processes. However, very few are having tier-to-tier connectivity. IT tools or softwares are used for collecting, analysing and distribution of information to decision-makers. The IT brings in speed for on-time information availability. IT applications thus support strategic, tactical and operational decision-making across the extended supply chain. IT-enabled information model supports plant automation, production execution and other enterprise process applications. This helps to have planning and execution together as there is no time gap in the two processes because of speedy information flow in the integrated supply chain. Due to IT enablement, data sharing and exchange is done on real-time basis.

Companies are today shifting to the digital manufacturing model. This model enables efficient exchange of information between multiple manufacturing applications through an integrated information flow system. Digital manufacturing provides information infrastructure for data movement across the various process in the enterprise. The data inputs are from various systems such as forecasting, inventory, plant maintenance and many others. The other advantage of digital manufacturing is its support in scaling up of more applications through minimal interfacing needs.

'Digital manufacturing' works on an IT platform, supporting all the functions of manufacturing of an organization from product conceptualization to after-sales service to the customers. Digital manufacturing helps in creation of a knowledge management infrastructure that would facilitate learning and reuse, internal and external business process automation through business process workflows,

collaboration and informational exchange through EDI with supply chain partners and operational excellence through a synchronization of planning and execution to have a supply chain with agility and leanness.

PRINCIPLES OF LEAN MANUFACTURING

The lean principles are basically customer value driven and suitable for manufacturing environments. There are five basic principles of lean manufacturing:

- **Customer value:** Understanding the value perceived by the customer and the same should be externally focused.
- **Value stream analysis:** Mapping the business process and analysing the steps to determine which ones actually add value. The non-value-adding steps should be modified or removed from the process.
- **Continuous flow:** Production should flow continuously from raw materials to finished goods.
- **Demand pull:** Make products only if is required by customer.
- **Waste elimination:** Eliminate waste (time, material, space, mistakes) from processes and allow flow of inventory continuously according to the market demands.

GAINING COMPETITIVENESS

Lean supply chain is the operational and strategic management philosophy that utilizes IT enabling to effect the continuous regeneration of supplier and service partner networks. A lean supply chain network is designed to deliver differential customer value at the lowest cost. This is achieved through the collaborative, real product/service transfer, demand priorities, real-time marketplace information and logistics delivery capabilities. The continuous regeneration of networks of supply chain partner concept implies that companies succeed by generating constantly from new networks of supply chain partners to achieve specific objectives. In general, the continuously evolving networks can respond to the dynamic nature of customer demand. This can be done through new forms of customer/supplier collaboration and scaleable product and information delivery flows.

The lean initiatives can be implemented manually. However, lean initiative would be more effective through IT application. IT would improve data visibility in the areas of problem resolution, design changes, quality issues and customer demand. There are three main functional areas, wherein software supports lean manufacturing initiative.

Managing Performance

The performance matrix is designed on a software platform. It maps the current processes in relation to the benchmarked ones. The right matrix helps in minimizing the wastes and enhances manufacturing system efficiency and its overall effectiveness. It also helps the enterprise to integrate processes and people leading to differential performance for gaining competitive edge.

Planning and Execution

The planning process is initiated followed by necessary steps for the execution as per the schedules.

Waste Reduction

Waste is commonly defined as non-value-added activity, as follows:

- **Overproduction:** It means the enterprise produces more than the market demands. In some cases, the goods are ready very much in advance of the demand or before need arise. The effect of overproduction is building up excess inventory blocking, the money which otherwise would have put to productive use.
- **Delays:** It means waiting resources such as materials, tools, information, equipment, etc. The reasons may be poor planning, late deliveries by suppliers, lack of communication, overbooking of equipment or erratic demand.
- **Transportation:** The movement of material from and to processes is a non-value-added activity which involves incidence of costs. The movement should be minimal otherwise it is a waste.
- **Inventory:** The money is blocked in inventory. Hence, storing more material is a waste in terms of space and cash.
- **Undesirable processing:** The inspection and reworking is nothing but doing more work on a part than is necessary. This results in waste of time and money.
- **Defects:** Defects consume precious resources. To fix the defects extra labour and machine time are required.
- **Excess motions:** The undesirable or non-value-added motions of employees in picking parts, getting tools or moving from place to place makes the worker tired, resulting in loss of productivity. This may be because of poorly planned work layout and workflow.

VALUE STREAM ANALYSIS

Value stream mapping is an excellent method to study and find scope for lean implementation in the business enterprise. The value mapping is studying of the existing process to know the value-added and non-value-added activities performed. The process of mapping starts right from product design phase to delivery of product to the customer. In the mapping process, the manufacturing company gets a clearer picture of the place of waste and sources of waste. With this background information, the firm can create the roadmap of a desired future state of the company. This is an interactive process. The value mapping is not a one-time activity, but it is a continuous process to find out ways of improving the flow of material and information.

ESSENTIALS OF LEAN SUPPLY CHAIN

To achieve objective of leanness in supply chain, the organization should have a major focus on its manufacturing operations and follow the following:

1. The organization should eliminate the wastes generation, that is, anything that consumes material or labour and that does not add value to the end customer. The waste includes rework, less productivity, overtime, slow motions, bottleneck in material flow and over capacity.
2. The manufacturing facility should have the equipments which are reliable to run when production requires it to run. The down time should be low. The manufacturing process should be latest and be capable of always making defect-free parts. The flow of the inventory needs

to be continuous. This is possible when the defects are reduced, adequate process capability introduced and lead time is reduced for raw material and components.

3. The manufacturing process should be designed to prevent the product from being built incorrectly.
4. The system should be such that the production line would be stopped when bad quality is being produced.
5. The use of *Kanban* technique in the production system ensures leanness in the supply chain. In *Kanban*, the material is pulled through the production process by customer demand. *Kanban* involves use of cards to move material along the value stream.
6. Adoption of standard work system is one of the key elements in lean supply chain which is a system of organizing work steps and documenting them.
7. Achieving leanness in supply chain, each workstation should have the information and equipment for the worker to inspect and produce good quality parts.
8. The requirement is levelling the production to the customer demand. This is possible with quick change over from one product to another one.

SUCCESS FACTORS FOR LEANNESS

The lean manufacturing techniques will help the manufacturing organization to cut cost and gain a competitive advantage. It may also help the organization remain viable for survival. The benefits with lean manufacturing are reduction in work-in process, increased inventory turns, release from tied up unproductive capacity, reduction in cycle-time and improved customer satisfaction. In general, with the lean techniques the following improvements are possible:

1. Operational improvements such as reduction in production cycle time, increase in productivity, reduction in work-in-process inventory, improvement in quality and reduction in space utilization.
2. Administrative improvements such as reduction in order processing errors, streamlining of customer service functions so that customers are no longer placed on hold, reduction of paperwork in office areas, reduced work force requirement, documentation and streamlining of processing steps, reduction in turnover and the resulting costs of attrition, implementation of job standards and pre-employment profiling.
3. At the strategic level the improvements are: reduction in working capital, reduction in costs. This provides opportunities for marketing to gain market share from competitors.

To achieve leanness in the supply chain, the following are the key elements.

Proper Information System

Lean SCM is a supply chain operational and strategic management philosophy that utilizes Internet-enabling technologies to effect the continuous regeneration of supplier and service partner networks. A lean supply chain network generates unique customer-winning value at the lowest cost through the collaborative, real product/service transfer, demand priorities, vital marketplace information and logistics delivery capabilities.

Information gathering, tracking, monitoring and harnessing process should use IT platform. The prime driver of SCM is information—then the faster information can be gathered, analysed and

disseminated through the channel network. Internet technology provides the means to make real-time demand management, manufacturing, logistics and storage data visible to all trading partners. Getting timely information about actual consumer sales electronically from the point-of-sale systems enables companies to truly move towards a demand-driven environment.

Supply chain partners need visibility on internal and external logistics functions so that they can allocate resources to meet a specific demand flow. Logistics planners need accessible real-time information in order to configure a value network. To be successful, planners will require web-based applications that enable cross-channel system to provide information on the status of production capacities, storage capabilities, inventory availability, velocity, replenishment processes and transportation resources at each node in the supply network.

Synchronization

Synchronization of demand/supply information minimizes work-in-process and finished goods inventories up and down the channel. This reduces the impact of 'bullwhip effect'. A synchronized supply chain will consist of a unified business strategy, common measurements for product and performance excellence, and the selection of enabling technologies.

Collaboration

The most critical component of lean SCM is the willingness of channel members to engage in and constantly enhance collaborative relationships. Many companies have come to realize that short-term benefits brought about by logistics optimization and technology automation are incapable of producing the radical competitive breakthroughs that can be attained when channel partners strive to build long-term collaborative relationships. As a supply network, the dominant characteristics of lean SCM are collaboration and synchronization. Collaboration is essentially the ability to share and interact upon critical data. Synchronization means possessing the channel intelligence to access the right product and the right service in the supply chain to satisfy the customer.

Optimization

Optimization requires the pursuit and merger of three levels of improvement. First is focus on eliminating waste and engineering 'perfect' processes on the company level. It aims to reduce costs while at the same time providing for the establishment of new sources of customer value. Second is the pursuit of optimization across trading partner relationships and systems. This initiative is much harder to accomplish. Channel nodes must closely collaborate, synchronize and integrate processes. The goal is to bring about value in chain-wide processes that promote cost savings, efficiencies and productivities that make the entire supply chain ecosystem more competitive. Finally, lean SCM optimization requires the standardization and rationalization of all supply channel processes.

Operations Excellence

The foundation for effective lean SCM lies in the ability of entire supply chain networks to achieve collaborative operations excellence. The SCM standard of excellence requires trading partners to coordinate technology and processes in order to provide the highest level of customer fulfilment and service, while managing multiple levels of external relationships. Establishing both the framework and content of channel synchronization requires commitment, trust, and willingness and capacity to acquire and expand new skills.

Connectivity and Networking

Connectivity and networking are lean supply chain fundamentals. Connectivity presupposes the availability of a technical infrastructure that links computer systems and people. The word commonly used for this process is integration—a term often erroneously used as a synonym to connectivity and interfacing. The second fundamental technology dimension is networking, that is, connecting different computers and their databases together in a peer-to-peer network. Networked systems enable people to communicate directly both within the enterprise and across the entire value chain. Connectivity is the process of linking business functions together and networking activates those links.

Kanban

The word ‘*Kan*’ means ‘visual’ in Japanese and the word ‘*ban*’ means ‘card’. So *Kanban* refers to ‘visual cards’. *Kanban* system is used for continuous supply of components, parts and supplies to production assembly line. *Kanban* reduces inventory, product obsolescence, waste and scrap. With *Kanban*, products and components are only manufactured when they are needed. This eliminates overproduction. In *Kanban*, the raw materials are not delivered until it is required. Thus, it reduces waste and storage costs. *Kanban* gives firms the flexibility to rapidly respond to a changing demand with regards to product mix. In *Kanban*, the supply of raw materials and components is eliminated as a bottleneck.

In this system, the material flow will stop if there is a production problem. This makes problems visible immediately to take corrective actions. Thus, *Kanban* reduces total cost by preventing overproduction, reducing waste and scrap, minimizing wait times and logistics costs, reducing stock levels, etc.

Kaizen

In Japanese ‘*Kaizen*’ means continuous improvements in product, process and procedures. The improvements may be in all areas of the enterprise such as quality, technology, processes, company culture, productivity, safety and leadership. The word *Kaizen* is a combination of two Japanese words ‘*kai*’ meaning ‘change’ or ‘to correct’ and ‘*zen*’ means ‘good’. *Kaizen* can be applied in every aspect of business processes.

In ‘*Kaizen*’ initiative, employees of all levels of management are involved. They are encouraged or motivated to come up with small improvement suggestions on a regular basis. Japanese companies, Canon and Toyota are the pioneers of the concept of *Kaizen* in Japan. In *Kaizen*, they could get 60 to 70 suggestions per employee per year which were written down, shared and implemented. These suggestions are ideas for changes in product, process and procedures on a regular basis. These small changes help in improving productivity, safety and effectiveness resulting into waste reduction.

Kaizen in Japan is considered as the road to business success. *Kaizen* also covers quality circles, automation, suggestion systems, just-in-time, *Kanban* and 5S for running a business. *Kaizen* sets and supports standard. In Japan, Toyota is well-known as one of the leaders in using *Kaizen*. In 1999, at one U.S. plant, 7,000 Toyota employees submitted over 75,000 suggestions, of which 99 per cent were implemented.

Kaizen is a critical concept in lean manufacturing. *Kaizen* differs from what most companies implemented in the United States as continuous improvement in the 1990s. *Kaizen* is often linked with the word ‘event’, to signify a short time period. *Kaizen* events are often just a few days long, with a specific group of employees working on a very specific challenge.

The implementation of *Kaizen* involves all employees in solving real problems. As per *Kaizen* philosophy, 'Instead of making large changes that may require significant amounts of investment and risk, small incremental changes are better options.' Thus *Kaizen* advocates the small incremental improvements that have a large positive impact over time. *Kaizen* is one of the best concepts to involve employees in the business and help them to feel part of the reason for the success of the business and the lean manufacturing system.

LEAN—ROADMAP TO SUCCESS

The manufacturers in general do not built their entire supply chains to respond in a synchronized way to variability in customer demand. This means production is centred on making products to forecast. These products inventory is then pushed into the supply chain to manage variability. These companies push products through pricing discounts or promotional campaigns to create demand to move the excess inventories. Real-time demand information coupled with real-time inventory visibility and availability for production is the biggest hurdle to lean production system. However, solutions are needed to manage and optimize the changes in customer demand to transaction in the most effective way for making the lean enterprise.

In today's increasingly fast-moving and highly-complex environments, enterprises must continually manage large amounts of data while executing complex workflows within and outside the enterprise. These supply chains require enabling software and a robust service-oriented technology platform. Without IT, extended enterprises cannot be truly lean or Six Sigma enabled.

Leading manufacturers will need interdependent supply networks which combine back-end business planning and operations in conjunction with lean methodology and practices across the supply chain. Traditional pull systems and ERP fall short when applied to a networked supply chain model for the following reasons:

- Lack of visibility across the entire network.
- Rigid, slower and error-prone ERP systems.
- Legacy systems which are not scalable and unable to be supported by customers, suppliers and partners.
- ERP and legacy systems struggle in managing continuous demand variability or sudden spikes in demand.

In conclusion, lean supply chain with support from lean manufacturing will help organization to gain competitive advantage through cost leadership.

SUMMARY

Lean manufacturing practices have many benefits to offer in all manufacturing environments, regardless of the product mix and volume. Supply chain management challenges are unique in very high-mix, low-volume and volatile demand environments compared to very high-volume and low-mix environments. More and more, manufacturers are confronted with this problem today.

The lack of proper education and training in an organization has been shown to be the biggest barrier to a successful lean manufacturing implementation. Lead time in different areas of the supply chain is a main cause of inefficiencies. Through proper training, encouragement and

participation at all levels, excellent results in supply chain optimization in high-mix, low-volume manufacturing environments can be attained. Value stream mapping has proved to be a very simple but extremely effective tool for identifying issues affecting the key metrics of a supply chain. To achieve leanness in supply chain, firms should use postponement techniques.

To succeed and lead, the entire supply chain system must continuously adapt and adopt information system (information accumulated, tracked, monitored and harnessed over the internet), supply chain synchronization, supply chain collaboration, optimization, operations excellence and networking, and connectivity.

The challenge to supply chain management (SCM) is to seek to develop 'lean' strategies up to the de-coupling point, but 'agile' strategies beyond that point. In other words, by using generic or modular inventory to postpone the final commitment, it should be possible to achieve economies of scale through product standardization. In lean supply chain, collaboration is critical. Collaborative relationships enable greater flexibility, responsiveness and modularity which is critical to reducing the trade-offs of customizing products.

IT applications complement lean manufacturing with improved data visibility, speeding information flow for problem resolution, design changes, quality issues, and changing customer demand.

REVIEW QUESTIONS

1. Explain the principles of lean manufacturing.
2. Explain the role of 'information technology' in lean supply chain.
3. Compare and contrast the 'lean' and 'agile' supply chain designs.
4. Discuss the key factors contributing to the success of lean supply chain.
5. Explain 'Kanban' and 'Kaizen' and their role in lean supply chain.

INTERNET EXERCISES

1. The lean supply chain blog discusses lean manufacturing topics, including articles, links to articles, and discussion. Visit <http://lean-supply-chain.blogspot.com/> to study concepts of lean supply chain and its applications.
2. Visit, http://www.newvectors.net/.../pdfs/esd_LeanSupplyChainGuideBook.pdf, to read "*Developing Lean Supply Chains*" —A Guidebook, December 2002, Developed by T. Phelps, Altarum, T. Hoenes, Boeing M. Smith, Messier-Dowty.
3. Ultriva is the leading provider of Lean software solutions. With over 100 successful deployments under them they guide to make the transition to lean and help the firms to realize substantial savings. Visit <http://www.kanban.com/> to study the various aspects of lean manufacturing. Ultriva helps the firms to move Kanban in practical and incremental steps.

VIDEO LINKS

1. Introduction to Lean Manufacturing, <http://www.youtube.com/watch?v=ZdHGTCXcJQU&feature=fvwm>
2. Lean Supply Chain Basics, <http://www.youtube.com/watch?v=8RlA0D6cjDc>
3. Boeing 737 Lean Manufacturing, <http://www.youtube.com/watch?v=r5pgKKiNORA&feature=related>

PROJECT ASSIGNMENT

1. Many leading companies in India like Siemens, Maruti Motors, M&M adopted Japanese techniques in their lean manufacturing initiatives. Choose a company and study how they have put the lean initiatives in practice, right from planning to implementation stage, problem faced and finally the benefits accrued.

NOTES

1. Jerry Kilpatrick, (2003), *Lean Principles*, Utah Manufacturing Extension Partnership.
2. <http://www.toyotaproductionsystem.com>

DISTRIBUTION DECISIONS

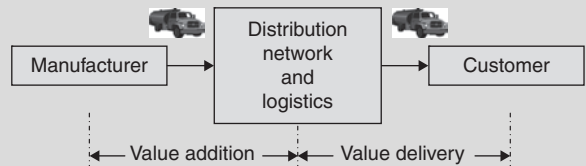
“Distribution provides a number of opportunities for the marketer that may normally be associated with other elements of the marketing mix

—Sam Walton”

Chapter Objectives

To understand:

- Role of distribution in supply chain
- Structure of distribution channel
- Designing distribution channel
- Distribution network in practice
- Logistical requirement of channel partners



Distribution and Logistics Role

Distribution involves getting the product from the manufacturer to the ultimate consumer. Distribution is often a much underestimated factor in marketing. Channel decisions are among the most important decisions that management faces and will directly affect every other marketing decision. A marketing channel describes the movement of a product or commodity from the site of production to the place of consumption. Intermediaries are interdependent organizations that connect between market and manufacturer. These organizations make a product or service available for use or consumption by the consumer. Distribution logistics is somewhat broader in scope, by incorporating the selection and management of the institutional channel (s) of distribution for a company's products as well as the physical facilities required.

Skylak: Moulded Luggage Distribution Strategy

Distribution network gives access to the market. However, the product specific distribution creates brand differentiation leading to competitive advantage. For making significant presence in the international market, a leading Indian luggage major Skylak Ltd, introduced a range of luggage brands to cater to the requirements of all segments of markets. The company purchased the UK-based world's fourth largest luggage manufacturer 'Diana', which was in the state of insolvency. This gave Skylak access to a well-known global brand and also access to award-winning designs, IPRs, mould plants and machinery. Skylak started manufacturing 'Diana' product range in their factory located in Western India. This acquisition is a step towards company's goal of becoming the world's leader in moulded luggage. Diana served to counter the growing appeal of rival Samsonite. It also plugged a fairly big gap in Skylak luggage product mix. Diana is the premium luggage brand to draw top-end consumers.

'Skylak' had 'Royal' brand catering to mid segment clocking 70 per cent revenue. It is also competing with regionally strong brands such as Aristocrat and Safari. What holds promise for Skylak in the economy segment is the overall market trend of consumer's preference for soft luggage over hard luggage. The hard luggage market is growing at a rate of 2–5 per cent while the soft luggage market is growing at an all time high of 30–35 per cent during the last 3 years. The maximum growth for soft luggage is in the economy segment. This is because the Chinese imports of soft luggage started flooding Indian market with low-cost and semi-stylish products. Skylak has 'Aura' a brand serving the low end luggage market. The firm is planning to extend the 'Aura' range into soft luggage in this segment.

For all brands, the distribution strategy lies in differentiating value proposition in all three segments. Skylak evolved three different distribution networks for three distinct brands catering to three different markets.

The top end brand 'Diana' was sold through exclusive 'Skylak Lounge' located at business centres at airports and five star hotels within and outside the country. The pricing for 'Diana' was up in the range of Rs 5,000–15,000. The 'Aura' product range was available below the price range of Rs 1,000 and was available at multi-brand outlets and large format stores such as Big Bazaar. However, 'Royal' brand was retailed in 'Skylak' showroom chains.

Obviously, the marketing and promotional strategy for these brands are different. 'Diana' was led by PR and events. 'Royal' was supported by mass media and was promoted through innovative offers such as the recent exchange offers. 'Aura' had mass media support, but the focus was rural India. Today the company has 35 Skylak Lounges, 150 Skylak showrooms and over 2,500 retail outlets across the country.

INTRODUCTION

The distribution strategy is concerned with the product or service placement at the right time to customer at the right place. In distribution there are two categories of issues and decisions which need to be handled while designing the distribution strategy. These are management of marketing channels and management of physical supplies.

In supply chain channel decisions refer to the managerial decisions on the selection of best routes or paths for moving goods from the producer to the consumer. Channels of distribution are concerned not only with the physical movement of goods but also with their promotion, selling and marketing control. An example is Colgate-Palmolive whose toothpaste is sold in large quantities in the Indian market has distributor's network of 1,00,000 spread over 3,500 towns and cities in India.

Channels of distribution help movement of goods from one place to another to create place utility, time utility and create convenience value. They make it possible for the consumer to obtain goods at a price he is willing to pay and under conditions which bring him satisfaction and pride of ownership and thus create possession utility.

The alternative to indirect distribution is to establish own retail stores, for example, Bata Ltd. that has established its own retail stores throughout the country, while Raymond's has franchised a number of retailers to sell their products to the consumers.

For a business firm it is important to make a wise, informed decision about how extensive the business's distribution channel should be. For achieving wider distribution, services of intermediaries are used by a majority of manufacturers as they are unable to sell directly to customers cost-effectively. As a thumb rule, for low unit-priced mass distributed products, the services of middlemen are used but for high priced technically complex products, the manufacturers use their own sales force to market the products. For a soft drinks marketing company, distribution of products beyond 120 km will be a costly proposition. This is because the cost of distribution of soft drinks increases exponentially beyond 120 km distance.

Hence, most of the concepts surrounding distribution are related to costs. Many of the practical matters relating to distribution channels, however, have to do with customer control. In general, business firms think that by selling their product through the distribution channel, their role has come to an end. But in order to be effective, it is vital for businesses to take a market-oriented approach and manage every level of their product's distribution until it arrives at the end user.

Depending on the product and markets there are three levels of channel distribution structures as follows:

- **Intensive distribution:** Wherein a business firm engages large majority of the resellers are stockists. The products are having low unit prices and used for daily consumptions. The availability, rather than brand loyalty is a major criteria for purchasing the product.
- **Selective distribution:** Wherein suitable and selected resellers are selling the product. The unit price of the product is higher and customer takes time in arriving at final buying decision. Normally consumer durables (TVs, freezers, washing machines and air conditioners) are sold through selective outlets.
- **Exclusive distribution:** In which only selected resellers are permitted to sell the product—that is, usually one seller per geographical region. The unit price is considerably high. The customer may purchase the product once or twice in his lifetime. Examples are expensive cars, diamond jewellery, high-priced wristwatches, etc.

An important element of supply chain management is the management and monitoring of the distribution channels. Just like a company's own sales and distribution departments have to be taken care of, each level of the distribution chain will have to be managed in a similar way. However, in practice, the companies use a mixture of different distribution channels. The mixture of channel, in fact help a direct sales team, call on bigger accounts, work with agents, or help to take care of smaller accounts and prospects.

The trend in distribution management is the concept of vertical marketing wherein the producers, wholesalers and retailers are united into one integrated channel. This happens as a result of one of the members of the distribution chain owning the others outright. It may be due to company's business strategy of forward or backward integration. For example, McDonald operates via franchising model; similarly Wal-Mart have integration with their suppliers. The other approach is via a contractual system often dictated by retail or wholesale cooperatives.

A firm's marketing goals and distribution objectives are highly related. The exclusive and high end market distribution will generally have less intensity and lesser reach. Hence, the firm has to look for trade-off against speed of delivery and intensity. The consumer behaviour study shows that consumers will prefer to switch soft drink brands rather than walking from a vending machine to a convenience store. This shows the importance of intensity of distribution. For consumer durables, consumers are willing to travel few departmental/discount stores/dealers to choose amongst the premium brands. Decisions on channels are dependent on the buying behaviour of the customers. Hence, the marketers should take into consideration the following factors such as:

- Customer involvement (time devoted) in purchasing given category of product
- Number of brands compared
- Comparable substitutes
- Availability of similar products nearby

DISTRIBUTION: CHANNELS AND LOGISTICS

Distribution involves getting the product from the manufacturer to the ultimate consumer. Thus, distributors add efficiency by:

- Breaking bulk
- Consolidation and distribution
- Carrying inventory
- Financing
- Promotion

Distribution channel helps in creating time and place utilities of the product, which are important for customer satisfaction. The channel will help the firm to make the product available at the right place and the success of channel efficiency and effectiveness is greatly dependent on the logistics operation of the firm to fulfil the demand. However, the logistics will help in movement of the product to make it available in right quantity whenever there is a demand by customers. Hence, the logistics network needs to be integrated with the distribution channels of the firm. The distribution channel is the place of transaction for the firm's products and it acts as an interface between the customer and the firm. However, the effectiveness of this interface is greatly dependent on efficiency of the logistics operations to move the products in required quantity and variety at speed. In fact, logistics programmes have to be tuned to the needs of the channel members to reduce the loss of sales opportunity. The basic role of logistics in distribution channel is to ensure smooth movement of products to make it available at distribution outlets as and when, it is required by the customer. It is not a so easy task as it involves great degree of planning, for material availability and movement at the lowest cost and with speed.

The channel structure is referred to the group of channel members to which a set of distribution task has been allocated. The channel structure plays a vital role in the physical distribution of the products.

Box 10.1

HUL's Focus on Rural Distribution Network

With need of the hour and evolution of market, FMCG leader HUL modified its distribution networks. HUL noticed that urban market for its products is almost saturated and growth is reducing continuously over the years. HUL along with other FMCG firms started concentrating on rural markets. HUL evolved a new strategy and structure for its distribution network for rural market. HUL has 'star stores' in its distribution network. For rural market, they removed this layer. HUL also reduced the margins of the rural distributors by 2 per cent from 6.76 per cent to 4.76 per cent. With the new system, the rural distribution channel became shorter, efficient and cost-effective as compared to the old system. In the old network, RDs (rural distributors) place orders on depots to get the stocks. The RDs then supply to the star sellers based on a fixed route schedule. The star sellers in turn service the retail outlets in each village. In the new distribution, star sellers were removed. HUL has introduced the concept of 'Milk Run'. In this, the RD's vans (loaded) will start from a base town (where the RD is located). The van will take particular route selling to retail outlets on the way. The van (fully empty) will end at the base town itself. However, for village clusters (which are few) that are far away from the base town, the RD may retain the star seller. RDs will replenish the star seller's stocks on a regular basis (once a week or once a fortnight). Here, the star seller would continue to service the retailers as before. HUL had in some places wholesalers and the RD may sell to wholesalers in that area. The retail outlets would come to the wholesaler to buy their stocks.

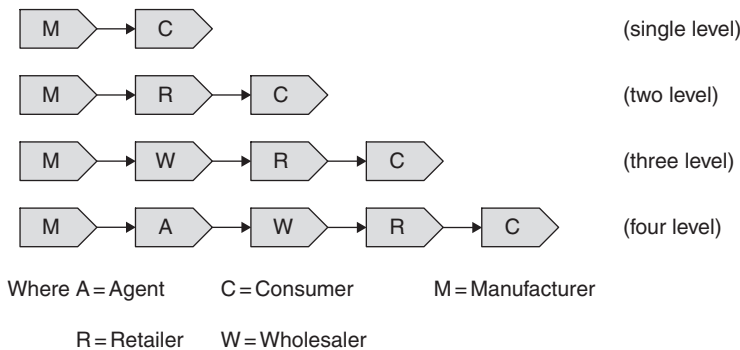
With RDs the service quality and coverage to rural market improved a lot as the rural outlets are served directly. HUL worked on new credit policy, visibility of stocks and the area coverage, which helped in increase of rural sales. With larger area coverage by RDs the sales turnover increased, resulting in reduced fixed cost and improvement in RD's margins.

The channel structure (Figure 10.1) primarily facilitates the product flow from the manufacturer to the end consumer through hierarchy of channel members. The typical distribution channel for consumer goods company is a multilevel structure, while a company manufacturing capital goods may sell its products directly to the user without involving any channel member. Depending on the product and the other marketing variables, the manufacturing company opts for or designs the particular structure (Table 10.1), which fits into its marketing philosophy and policy. But the major considerations are the cost of distribution, wider coverage and maximum market penetration. However, the most common channel structure being adopted by the manufacturers.

The consumer goods companies have a long and dense channel structure because the end customers are very large in numbers, they are spread over wide geographical areas, the unit price of the product is low and the frequency of purchase is more. Under these circumstances the manufacturer prefers to have more outlets for product distribution to meet the customer needs.

On the other hand, the industrial machinery manufacture prefers to deal directly with the customer because of the product complexities, high unit price, and service requirements both pre- and after-sales. In such circumstances, manufacturer may sometimes appoint an agent to strike the deal. In each of the above cases, different channel members have been allocated certain distribution tasks and have different roles to play. The width and the depth of the risk at each level is different and depends on the financial involvement and the control on the next lower level.

The channel structure facilitates product flow from manufacturer to the end customer. However, the physical movement of the product is taken care of by the logistics. The place and the time

**Figure 10.1** Channels Structures**Table 10.1** Distribution Channel for Different Products

<i>BHEL</i>	<i>LG</i>	<i>Wohkardt</i>	<i>Pepsico</i>	<i>HUL</i>
		Distributor	Depot/Warehouse	Distributors
		Stockist	Distributor	Sub-distributors
	Dealers	Retailer	Retailer	Retailers
Customers	Customers	Customers	Customer	Customer

utility of the product are not possible without the close coordination between channel task and the logistics operations.

The information flow in the channel structure is two-directional. The manufacturer sends the information on the order status like material availability, shipping dates, transportation mode, payment dues, etc, while channel members send the information on customer requirements like sale forecast, warranty claims, competition activities, requirements scheduling, etc. For the movement of goods from the manufacturer to the channel members, the logistics department acts as the traffic cop in supply chain. The logistics activities such as order registration, order processing, inventory planning, order picking, packaging, and the transportation requires continuous flow of information to service the channel members by making the ordered material available to them in the required quantity at the right time. In fact, the information flow in the channel structure is the basic input in logistics to activate the material flow from the manufacturer to the customer via channel members.

The channel structure facilitates the transfer of ownership title (Table 10.2) of the goods as it passed from the manufacturer to the customers. The communication for advertising, sales promotion, exhibitions, personal selling is part of the information flow through the channel structure. The timely flow of the funds from channel members to the manufacturers is important to keep the operation going.

There are several tasks that have to be accomplished as part of physical distribution. These are as follows:

Location of manufacturing facilities The basic decisional parameters would be the availability of basic raw material and the location of the market. It may be decided to locate the manufacturing facility nearer to the source of supply and ship the finished outputs to the markets or to erect the production facility near to the geographical market to be served and arrange the shipment of the inputs accordingly. Location of the Mathura Refinery was based on the latter's concept while the super-

Table 10.2 *Channel Member's Functions*

<i>Channel Member's Functions</i>	<i>Functional Details</i>
<i>Exchange</i>	Buying Selling Title transfer Negotiation
<i>Logistical</i>	Storage Transportation Requirement scheduling Inventory management
<i>Facilitating</i>	Credit financing Payment collection Promotion After-sales service Market information

thermal power plants of the National Thermal Power Corporation are based on the former. The basic consideration involved obviously is the relative costs of transporting inputs and outputs, including the rates of different modes of transportation which may be used to transport raw materials and finished products. For example, transporting gas via pipeline is more cost-effective than distributing fertilizer through the normal modes of surface transport. The decision regarding the number of facilities to be created will be influenced by the geographical distance among various sub-national markets to be served, the available product line, and the scale economies of production, among other factors.

Location of warehouses One important consideration in this context is the nature of the product being sold. If the product is a household consumption item, such as tea packets, soaps or toothpaste, the retail outlets will be at the bottom of the distribution channel. Success of the distribution system would depend upon ensuring the availability of the product at that level. This obviously would need an organizational structure with number of central/regional warehouses, a larger number of company depots, field sales staff and an elaborate system of delivery vans for transportation. A manufacturer of capital equipment, on the other hand, can have only one centralized warehouse for the main product but has to maintain a number of service centres to stock spare parts.

Transportation mode and method The choice is quite wide for optimal transportation which includes road, rail, air or even post parcel (in the case of mail-order) transport, which is becoming popular in India, especially in the field of designer garments and accessories. The factors to be considered include the relative costs of the alternative modes of transportation, speed, reliability and appropriateness, keeping in view the product attributes such as perishability, hazardousness, human consumptions (food products), gases, liquids, etc.

A company may decide to own a fleet of delivery vehicles, hire an outside transport company or lease the vehicles from a leasing company. Alternatives are to be evaluated taking into account the cost of capital in case of company-owned vehicles and the cost of managing the fleet, and the charges to be paid to the leasing firm or to the outside transporter.

Inventory decisions Bulk of the total distribution cost, which generally account for 5–25 per cent of the product costs, is incurred on the physical distribution and costs of holding inventories. Since inventory holding costs are always on the rise due to all round increase in prices. A careful attention

has to be paid on how much inventory should be maintained, of what items and where. Many of these decisions have to be taken keeping in view the broader corporate objective of service reliability, that is, the capacity of the firm to deliver on time.

FLOWS IN DISTRIBUTION CHANNEL

There are five flows involved in the distribution channel. These flows provide the links between channel members and other service providers together in the distribution of goods and services. These flows are:

- Product flow
- Negotiation flow
- Ownership flow
- Information flow
- Promotion flow

The product flow is the actual physical movement of the product from the manufacturer through all of the parties who take physical possession of that product, from its point of production to final consumers. This involves the usage of various modes of transportation such as road, rail, water, air and pipelines.

The negotiation flow represents the exchange in the buying and selling function associated with the transfer of title of products. The transportation firm is not included in this flow because it does not participate in the negotiations.

The ownerships flow shows the movement of the title to the product as it is passed along from the manufacturer to final consumers. Here again, the transportation firm is not included because it does not take title to the product or become actively involved in facilitating its transfer. It is only involved in the transportation of the physical product itself.

The information from the manufacturer to consumers is two-directional. All parties participate in the exchange of information, and the flow can be either up or down. Without proper information flow the products cannot be made available at the right place and at the right time.

Finally, the promotion flow refers to the flow of communication in the form of advertising, personal selling, sales promotion and publicity.

The concept of channel flows provides another basis for distinguishing between channel strategy and logistics management. In the context of the channel flows concept, channel strategy and management involve planning for and managing all of the flows, whereas logistics is concerned almost exclusively with the management of the product flow.

CHANNEL STRATEGY

Channel structures vary somewhat by the nature of the product. The technically complex and customs build products are shipped directly to the customers. The examples are aircrafts, heavy engineering machinery, higher capacity power plants, etc. The FMCG (fast moving consumer goods) are shipped through a chain of middlemen like distributors, wholesalers and retailers wherein the goods are handled more efficiently along with other products from many different suppliers. Several layers of middlemen may exist, depending on the product. In many cases, the middlemen provide a number of services— inspection, installation, insurance services, commissioning, warranty claim settlements, etc. The manufacturers would prefer to have their wider geographical distribution of their products.

This is especially the case for convenience products where the customer has little motivation to go to a less convenient retail outlet to get his preferred brand. Cooking salt would be an example here. However, different parties involved in the marketing of products tend to have different, and often conflicting, interests such as full service retailers dislike intensive distribution, manufacturers may be tempted toward intensive distribution and service requirements differ by product category, etc.

For the manufacturers it is generally more economical to go through intermediaries rather than selling directly to the customer. For FMCG products it is more appropriate because customers are dispersed over wider geographical areas. In case of business-to-business markets, buying enterprises prefer to buy goods from manufacturer because of large volumes and technical complexity.

The cost is not the only consideration in deciding the channels structure. For example, channels for perishable products are short, but the additional cost is temperature-controlled storage and transportation to keep the products in fresh condition.

The design of the logistical programme to serve the given channel structure depends on the channel strategy of the enterprise. The channel strategy is broad principles by which the firm expects to achieve its distribution objectives for its target market. Depending on the nature of the product, the industry practices and the market trends, the enterprise may formulate the channel strategies to serve the end customers. The channel strategy decision can be examined through the following views.

Channel Length

This is more related to the levels of the channel structure. The decision is whether the sales are directly made to the customer or through a number of intermediaries. The decision is based on the market reach, distribution cost and degree of control on the channel members. The firm may have its own channel outlets, wherein the investment on infrastructure is limited. However, there is full control on the distribution activities when firms own their own distribution channels. Showrooms (Raymonds, Nike) and retail chains (Wal Mart, Pantaloon) are examples of captive distribution channels where the activities are under full control. Here, the logistics can be pre-planned and executed in close coordination to control the inventory channel.

The other option is the contractual system like franchising, wherein high degree of control can be exercised on the channel activities and the logistics programmes can be organized to support them accordingly. In both of the above systems, the firm is directly in contact with the end customers. The limitations of the above systems are the investment in setting up the infrastructure and limited coverage of the market. The third option is to employ the independent intermediaries like the wholesalers,

Box 10.2

Eureka Forbes Distribution Strategy

Eureka Forbes Ltd, (EFL) an example of direct selling, which markets vacuum cleaners and water purifying equipment. It believes that if the market is in the customer's house, the best way to get there is to knock at the door. The company has clearly demonstrated that door-to-door selling can be effective in Indian conditions. Its sales force of over 500 people spread out over nearly 40 branch offices in 29 towns make it the country's largest commercial direct sales organization. They took the cue from Electrolux, the world's leading manufacturer of vacuum cleaners, and a firm believer in door-to-door selling. During the last few months EFL has tried to reach the customers through distributors also.

stockists, distributors, dealers, agents and retailers. The advantage is that the firm gets readymade infrastructure to operate from, without any investment. They are widespread and wider market coverage is possible. They are easily accessible but the firms will lose their degree of control. Firms do not have direct interface with the customers in this system and will have to think of a system for information linkage with the customers. The degree of control on the channel inventory is loose as compared to the earlier two systems. The failure probability of the logistical programmes designed for this distribution system is more, due to absence of cohesive linkages in channel structure.

Channel Breadth

This decision on the breadth of channel depends on the market coverage. Consumer goods, packed foods, newspaper, confectionary, garments are mass consumed items which need wider distribution channel for building a market share. The profits are generated through the volume sales, which is possible if products are available at the point of demand, which has both time and the place convenience.

The decision on the channel breadth depends on the frequency of purchase of product, ease with which customer wants to acquire the product, market coverage to build market share, and buyer behaviour, etc. The logistics programmes to support wider channel networks require logistics support to coordinate activities like order processing, warehousing, packaging, transportation, inventory management, etc. The control becomes more complex for mass distributed consumer products covering the national and international markets. Information technology and modern communication means play a significant role in efficient and effective control of such complex activities.

Marketing Strategy

This is another dimension to select channel, particularly the breadth of the channel. For building market share, wider market coverage is required and hence an intensive distribution will help support the objective. However, to support differentiation-based marketing strategies for specific customer groups, exclusivity in distribution channel will help the firm to exercise full control on the distribution task and logistical activities. It will also help in offering customized pre- and after-sales service to focused customer group to build trust and confidence in the product and the firm. In the exclusive channel structure, there is a need and scope of customized logistics programmes as the firm's marketing resources are deployed for the focused group of customers who bank on the service rather than the price.

The effectiveness of the channel depends on the following two things:

- Degree of control on intermediaries
- Backup supply efficiency

The degree of control is directly proportionate to the compensation the channel members are getting for the distribution task they are performing for the firm. This is to be backed up with efficient supply of goods by making goods available in desired quantity as and when required. This may involve frequent deliveries in small lots. This may increase the transportation cost but will be offset by significant reduction in incidence of inventory carrying cost on the system.

CHANNEL MEMBERS

Marketing channel structure consists of a variety of intermediaries performing distribution tasks, but they may be different in terms of financial involvement, relationship with manufacturer (place and time context) and the functions performed.

Brokers

They take neither possession nor acquire ownership of the goods but only serve to bring the buyers and sellers together. They negotiate purchase and sale of goods on behalf of other parties. Their task is over as soon as the buyer and the seller come to terms in respect of the purchase or sale of the goods. The broker works for a certain percentage of commission on the business transacted on behalf of his principal. Their role is to facilitate the sales/purchase deals. They do not permanently represent either the buyer or the seller; neither do they involve in handling the products. Their function is to organize the negotiation meetings of the two parties and facilitate the deal. They do not provide services like financial assistance, payment collection, inspection, etc. Their relations with buyers and sellers are limited to a particular transaction. For their services, both the parties have to pay the brokerage. Generally, services of brokers are quite common in stock markets, scrape selling/purchasing and organizing the auctions, etc.

Commission Agents

They sell goods on behalf of the sellers. They not only negotiate the sale of goods but also take possession of the goods and make arrangements for the transfer of title to the goods. The commission agent has to perform the functions of warehousing, grading, packing or sampling in addition to assembling and dispersion. For their services, the commission agents get a certain percentage of commission on the sales. A few commission agents are authorized to sell on credit and agree to bear the risk of bad debts for some additional commission.

Wholesalers

The wholesalers are quite common in consumer goods industry. They operate as full-fledged marketing organizations. They purchase the goods from the manufacturers in large quantities. They have network of marketing offices supported by their sales staff to market the products. They also have a network of the warehouses to store the goods at different locations. They collect orders from retailers and also large customers (institutional) and operate on volumes. They are important communication links for manufacturer about the market information. They are the dominant source of supply for the retailers in mass distributed consumer products. Their logistical needs are large quantity consignments at regular intervals as per schedules. They prefer freight consolidation for lower per unit transportation charges. They work on the scale economics.

Wholesalers are those middlemen between the primary producers, manufacturers or importers, on one side, and retailers or industrial consumers, on the other. They buy goods and commodities in large quantities with a view to selling them to retailers in smaller quantities. They assemble merchandise from many sources, warehouses and regroup the goods for convenient buying by retailers. Thus, wholesalers make it possible for the manufacturer to sell to a large number of retailers to whom the merchandise cannot be easily sold directly from the factory.

The wholesaler performs the following important functions of marketing, assembling, dispersion, warehousing, transportation, financing, risk-assuming, grading and packaging. In a nutshell, wholesalers render valuable services to both the manufacturers and the retailers.

Retailers

This is last link in the distribution network before consumer. They are quite common in the mass distributed consumer products. They buy the material form the wholesalers in small quantities. They have very limited financial capacity for investment in the business. They purchase the goods from

wholesaler on credit or against cash depending on the relationship and the prevailing trends in the industry. The logistics requirement is a frequent supply in small quantities. They do not want to stock the inventory because of their limited financial capacity.

The retailer is the last link and the most important intermediary in the chain of distribution. Mass production in the present day set-up is geared to the requirements of the ultimate consumer. Retailers are directly and personally in touch with the ultimate consumers and thus occupy a strategic position in the whole chain of distribution. The basic feature of retail trading is the purchase of goods from wholesalers and selling it in small lots to consumers. Thus retailing includes all activities directly related to the sale of goods to the ultimate consumers.

Van Dealers

A van dealer is totally a different type of concept in the distribution system. They are neither the authorized dealers nor the distributors of the manufacturers. They distribute different brands of the same product. They often buy the products from authorized wholesalers and sell the product in their areas. They some times sell the products at lower price than the company's authorized dealers. They manipulate the prices based on the discounts offered to them by the wholesalers. Van dealers are quite common in electrical industry in India. Their investments risk is distributed over several brands. They do not keep the inventory. They buy on daily basis on cash terms. They can reach to remote places where the company's authorized dealer finds it uneconomical.

Selling Agents

They are appointed on contractual basis for selling the company's products to the clients in lieu of an elaborate sales setup. They are contracted not to sell the competitor's products. However, they can sell the complementary products. They are deployed for collecting outstanding payments from the clients. They extend services like organizing loans and offering credit facilities to the clients. They are compensated for their services by way of commission on the sales. The sales agents are quite common in the textile industry.

Manufacturer's Agents

These intermediaries are appointed on a contractual basis for selling of industrial goods. They represent the manufacturer in creating awareness of the product, technical discussion, commercial negotiation and liaising with the clients. The manufacturing agents are former employees with technical background and know the technical intricacies of the product. They are compensated on case-to-case basis. They do not have any organization for marketing. They are mostly involved in pre-sales activities and finalizing on sale deals.

C&F Agent

C&FA is basically a carrying and forwarding agent. His job is to store the material on behalf of the company and forward it to further entities in the channel structure as per the delivery order of the company. He is a custodian of the material and not the owner because material from the head office will be stock transferred (without invoicing) to C&FA for further distribution to other channel members. The services expected from the C&FA are safe storage, proper dispatch, maintenance of inventory accounts, compliance of local taxes etc. He gets compensation for his services by way of commission from the company.

Dealers/Stockists

They carry the stocks/products of the company for sales on a consignment basis. The stocks are transferred to (with the time frame to sale) the dealers/stockists. The payment has to be made to the manufacturer at the end of the agreed credit period. They have a full-fledged marketing set up including backup facilities of warehousing. As the manufacturer realizes the payment of the inventory from dealers/stocks after the stipulated credit period this transaction is reflected in the accounts books of the company. Exclusive dealers do not deal with products of the competitors.

DISTRIBUTION CENTRES

Distribution is the heart of the logistical system and there are many obvious reasons why distribution centres (DCs) need a critical focus for management of its operations, as follows:

- Maximizing space occupancy.
- Operating cost-effectiveness.
- Enhancing the assets productivity.
- Optimizing secondary transportation costs.
- Sourcing for order replenishments.
- Inventory managements
- Vehicle scheduling.

Most often the distribution system will simply slip away from the optimum arrangements, as the market and the products change. For example, if a company has not looked at its distribution system for more than 4 years, it could well be paying 200 per cent more than what is expected. It is always worthwhile managing a distribution system, not only to control that part of the business but also to ensure that the system is serving the needs of the company efficiently.

The considerations of distribution site selection revolve around the two major factors: service and cost. The product availability can be greatly enhanced by locating the DC near to the market place. The smaller and frequent deliveries, which nowadays the customer prefers, can be organized. This will enhance customer confidence in the supplier. However, the transportation cost, which is the major element in logistical cost, is dependent on the location of the DC. The other factors affecting the selection of sites are as follows:

- **Infrastructure:** Availability of proper infrastructure such as connecting roads, utilities (water, electricity, communication), and labour has a great effect on the efficiency and the effectiveness of warehouse operations. For cold storage warehouses, availability of electric power is a major factor influencing the site selection. Non-availability of proper road or rail siding facility will have an effect on the operations of warehouses and as a result the transportation cost may go up considerably. Lack of infrastructure such as wagon platform, material-handling equipment may pose problem in loading and unloading of materials and special provisions may cost more.
- **Market:** Distribution warehouses are planned in proximity to the markets or consumption centres for offering better service to the customers. The frequent deliveries with small quantities as required by the customers can be organized due to limited geographical area coverage.
- **Access:** The location of the warehouse has greatest effect on the primary transportation cost. The difficulty in access will have an influence on the transportation cost.

Box 10.3

Polaroid–Shift to Centralized Distribution

Polaroid an American photographic company in 1990's for reducing distribution cost and enhancing customer satisfaction decided to move from a decentralized warehousing system to a centralized one. They had earlier 12 warehouses across Europe and had high inventory cost. They shifted to a single distribution warehouse located in Dutch. Polaroid adopted a system of direct distribution to retailers through Europe from the Dutch distribution centre. They outsourced entire logistics operation for European distribution to a third party logistics service provider. They established the satellite warehouses in the different regions to service the retailers. It was hub and spoke distribution system. Polaroid redesigned the distribution systems to tune to the requirements of retailers.

- **Primary transportation cost:** The transportation cost is the largest component in the product cost, particularly for low unit-priced products. The location of the warehouse will have greatest influence on the primary transportation cost. The product demand pattern will influence transportation cost in two ways, one on frequency of truck trips and two on loading patterns (full or partial) of the transportation vehicles.
- **Availability:** Availability of warehousing space in the urban area particularly in metros at cheaper rate is a remote possibility. In such cases, the site has to be shifted beyond the municipal city limits where the storage space is available at considerably cheaper rate. However, this arrangement may add to the transportation cost.
- **Product:** The type of product will have a profound effect on number of warehouses and their locations. For example, perishable products need to be delivered to the customer within its expiry period and hence, it should be located near the consumption centres. Warehouses with delivery limitation and geographical reach should be small and numerous. Warehouse service has a common storage room for many street shops fits into the above category.
- **Regulations:** For certain types of products (explosives, hazardous chemicals, and radioactive materials), which can cause damage to human life, the storage site selection is guided by the government regulations. In such cases, very little options are left with the firms to choose the site from.
- **Local levies:** Depending on the sales tax and the octroi charges in the region, the location of the warehouse is planned. Due to non-uniformity of sales tax across the Indian states, the marketers invariably plan the warehouse to avail the benefits of the local sales tax disparities.

DISTRIBUTION CENTRE SITE SELECTION

It will involve single depot or multiple depot systems for a given product-market configuration with the objective of wider area coverage and product availability at the retail outlets.

Single Depot Location

The distribution system of a firm normally consists of multiple warehouses located at various places. However, location of a depot for the particular geographical area is decided using the following principles:

Centre of gravity of area model A map of marketing areas may be drawn and the sheet representing the area may be balanced. The point where the sheet gets balance is the site for warehouse location. This point is expected to be at minimum average distance from all the locations in the area.

Centre of gravity of load model This may be illustrated visually by imagining a sheet having the holes representing the customer locations and their demand load of annual purchases. A weight is hung at each hole in proportion to the demand load. The point at which the model sheet gets balanced gives the centre of weights. This is the proposed point where the warehouse may be located.

Centre of gravity of tonnes-kilometres model This model takes into consideration both loads and distances to arrive at the optimum transportation cost. The final solution is arrived after lot of trial and error method.

Multiple Depot Network

The cost of warehousing per unit of inventory is directly proportional to the number of warehouses in the given marketing territory. If the total sales is divided into more number of warehouses, the sales handled by each individual warehouse decreases and warehousing cost per unit sold increases. This can be offset by increasing the sales per warehouse. For planning the chain of warehouses, the management will have to do the cost-benefit analysis. The factors influencing decision on planning optimum numbers of warehouse are:

- Market size
- Area coverage by each warehouse
- Product type
- Demand pattern—continuous or seasonal
- Customer service level
- Competition
- Establishment and operating cost

The warehousing cost is directly proportional to the number of warehouses, while the transportation (primary and secondary together) cost goes down with the increase in warehouse numbers as shown in Figure 10.2.

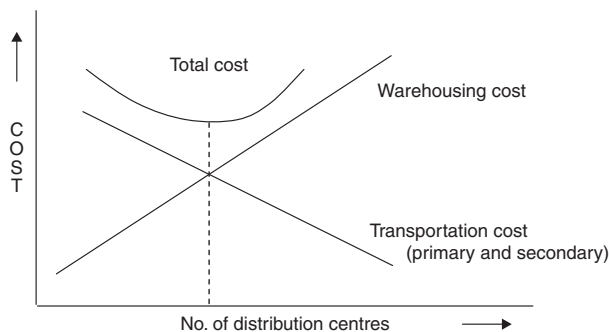


Figure 10.2 Number of DCs vs. Cost of Operations

Invariably, many manufacturing companies hold too many inventory stocks in many warehouses spread across wider geographical areas. The reasons are wrong procurement policies, complex transportation system, and historical ownership pattern, and leniency to customer satisfaction. However, due to cost pressure, the firms are trying to find out the trade-off. For cost reduction, firms may try to move towards the centralized warehouse system, but the decision is depending on the nature of the product and its movement through the supply chain. For auto manufacturing companies, the centralized warehouse for the spare parts is an ideal system but this is not applicable for FMCG products, which need to be moved in large quantities and with speed, wherein the system of scattered warehouse is the right solution.

The criticality of the demand and the cost of serving the customer determine the boundary served by the product. If the products are at par on quality and the customer is not deriving any specific advantages using the particular brand of product, then the choice of supplier is based on the landed cost he has to pay for.

The relation between the cost of DC operation and optimum area served by the same is best explained by the Bowman-Stewart formula. The major warehouse location criterion is its proximity to the market. However, there is no restriction for the area it may serve. The cost of delivery increases with the distance between warehouse and the place of delivery. Therefore the cost of delivery is proportional to the square root of the area covered by the stocking point. The choice of location is decided on the cost of delivery.

LOGISTICS: THE DISTRIBUTION IMPERATIVE

The most carefully designed and managed marketing channel rely on logistics to ideally deliver products to customers. The creation of time and place utilities by distribution channel and essential for customer satisfaction are mostly dependent upon logistics. The movement of the right products to the right place in right quantity at the right time, is the essence of the role of logistics in the marketing channel.

The channel structure is of critical importance in the physical distribution of the company's products. It has a deep impact on customer satisfaction and the logistical cost. Exploiting the distribution channel typically requires significant flexibility in logistical capability of the supply chain to take care of the changing needs and the problems of the channel members in the markets. Even the most carefully designed and managed marketing channel system must rely on the logistics to create time and place utilities of the product for customer satisfaction. The channel helps the firm to make the product available at the place and time required by the customer. The basic role of logistics in distribution channel is to ensure smooth movement of products to make it available at distribution outlets as and when, it is required by the customer. It involves great degree of planning for material availability and movement at the lowest cost with maximum speed. The channel strategy fits under the distribution variable of the marketing mix. Thus, channel strategy must be formulated even before logistics management can be considered.

Logistics as a systems has both fixed and variable cost components. The major cost spinners in logistics are transportation, warehousing, material handling, packaging, system integration and maintenance, and administration. These costs can be controlled through proper tuning of logistics to suit the requirements of various channel partners.

Channel management is a broader and more comprehensive element of distribution strategy than is logistics management. Channel management is involved with the administration of all of the major

Box 10.4

Channel Integration to Achieve Logistics Objective at HUL

Hindustan Unilever has undertaken a new e-enabled supply chain system, which is called 'Project Loop' to achieve the objective of seamless integration of 7,000 odd re-distributors. The project's main aim is to help correctly forecast the secondary sales to retailers. Thus, they are achieving lower inventory levels from 13 days to 10 days and finally to 5 days. This helps the company to bring down its working capital requirements by 50 per cent, also product availability up from 70 per cent to 90 per cent. This project also helps HUL to review its SKU policies based on fast and slow moving categories.

channel flows (product, negotiation, ownership, information and promotion), whereas logistics is concerned mainly with the product flow. For effective distribution, channel management and logistics management should go hand in hand. This requires a good degree of coordination between channel management and logistics management. To achieve this, the following major areas of interface between channel management and logistics management need to be looked into by the manufacturer:

Logistics Service Standards

Any logistics standard has its own cost. The higher the service standards offered, the higher are the costs involved. However, a well-designed logistics systems and modern technology can reduce the cost considerably. It is not possible to eliminate the cost completely. Domino Pizza is offering free pizza if the home delivery exceeds 30 minutes after placement of an order. The firm has organized its logistics based on this service standard so that not a single home deliver exceed beyond 30 minutes.

A manufacturer must cover the costs either indirectly in the price it charges for products, or by passing them along to channel members in the form of service charges. In either case, there is little point in offering logistics services that channel members do not want for higher levels of service than they desire. The key issue in offering logistics service standards is based on levels of logistics service desired by the channel members. It can be decided only after taking into consideration the views of channel partners.

Logistics Programme Designing

A logistics programme should be a part of channels strategy of the manufacturer. The following are variants and limitations (Table 10.3), which should be taken care of while designing the logistics programme for channel members:

Table 10.3 *Logistics Program Variants and Limitations*

Variants	Limitations
• Delivery frequency • Deliver time • Lot size • Packaging (loose/packed/unitized) • Transportation mode • Customer service norms	• Order processing cycle • Inventory norms • Transportation cost

The design of logistics programme can be done either by the company's executives or the job can be assigned to an outside expert. For example, Gillette Company was faced with a staggering assortment of changes in its operations. Among these were diversification into a broad range of toiletry products, a shift of its main distribution channels from drug and tobacco chains, and the introduction of stainless steel blades by competitors. Gillette sought to get 'one up' on the competition in providing logistics service to its thousands of channel members by emphasizing speedy deliveries of its new blades.

The logistics programme designed to accomplish this was dependent upon the fastest mode of transport—airfreight. But it was turned down by many of its own employees and channel members because of the high costs and resulting lower profit margins. Gillette quickly dropped this programme and developed a new one based on the use of lower-cost surface transport programmes that are carefully designed to focus on meeting customer service requirements. From the channel members' standpoint, the fast delivery provided by airfreight was not worth the added cost. Perhaps, if Gillette had been in a position to absorb all of these added costs by itself, the channel members might have been delighted with the system, but this would have been too expensive for Gillette. So what appeared to be a good idea on paper turned out to be unworkable in practice. Thus, programmes should be designed to be consistent with the service needs of channel members.

Adoption of Logistics Programme

Regardless of how good a manufacturer perceives its logistics programme, it still must convince the channel members of its value. The logistics programmes are appealing to channel members if it ensures the following:

- Reduction in out-of-stock frequent situations
- Ensures reduction in channel member inventories
- Ensures added manufacturer support for the channel

By minimizing out-of-stock situation through an improved logistics programme, lost sales will be reduced. If a manufacturer can convince channel members that the new logistics system will indeed help them to achieve growth, it will be a very potent argument for gaining an enthusiastic reception as their bottom-line and customer retention will improve. Computer-to-computer ordering arrangements and EDIs are good systems that do appear to be living up to the benefits promised. Already these systems are revolutionizing traditional ordering practices and reducing out-of-stock occurrences. The order could also pass through a third party with time-sharing capabilities. Manufacturers in 'just-in-time' environment if executed properly can bring dramatic results in inventory costs reduction, sometimes well over 50 per cent. JIT (just-in-time) system has been associated mainly with manufacturers, and particularly the automotive industry, but has a much wider potential application.

The logistics programme can be a potent marketing tool for building the support of channel members. Indeed, the role of better logistics management in strengthening the overall marketing efforts of channel members is growing in importance. A well-designed logistics programme offers great potential as a strategic marketing tool. But it will offer even greater potential to manufacturers who are able to extend their superior logistics capabilities to help channel members improve their logistics and marketing capabilities as well.

Cost-benefit Analysis

The logistics programme calls for deployment of firm's resources. The returns may be superior but may cause a drain on the profits. For example, for faster delivery, air transportation may be used, but

the transportation cost will be unaffordable for low unit-priced products. The logistics manager has to take into consideration the implications of the proposed logistical programme on the inventory levels to be maintained by the firm. A faster order fulfilment cycle will have to be supplemented by faster procurement, manufacturing and order processing, and cash generation cycle. This may require additional resources to tune the system to logistical programme. Hence the logistical programmes have to be designed and implemented in coordination to reap maximum benefits with optimum cost. A firm desirous of introducing a new product may think of developing an elaborate logistical network only after ascertaining the product acceptance by the consumers and demand for the product. Initially, he may rely on service providers network on limited counts and may pay little higher till the sales volumes are built. The logistical programme involving the use of air transportation for distribution of fresh flowers in the European market may be justified on the grounds that it is a high unit-priced it is fetching and its short shelf life.

Programme Implementation and Monitoring

The implementation of the logistical programmes requires managerial skills. The close monitoring of the programme through regular feedback on deviations in delivery cycle time, inventory turn over ratio, service level, cost and productivity is a must, to remove the bottlenecks which were not visualized at the time of planning. A leading automobile manufacturing company earlier operating on credit sales to its dealers, changed over to 'no credit system' and offered higher compensation for buy, stock and sell of the products have changed the logistics programme to suit the pull system. This helped them in improvement in their cash flow, inventory levels (both at factory and in channel). The deliveries are frequent but lot size is small. Even though the transportation cost has gone up, the inventory carrying cost has reduced drastically, resulting into overall gains.

Logistics systems must be continuously monitored, both in terms of how successfully they are performing for the manufacturer and, just as importantly, how well they are meeting the needs of changing channel members. Thus, as part of an overall attempt to learn about the needs and problems of channel members, the channel manager should continually monitor the channel members' reactions to logistics programmes.

SUMMARY

Distribution is the all-important link between a manufacturer and his customers. The concern is to design a distribution strategy to provide a smooth physical flow of products from the manufacturer to the place from where the customers can buy them. Channels of distribution refer to the alternative paths through which the goods can be routed. Direct selling and indirect selling through intermediaries such as wholesalers and retailers are the two alternative channels of distribution to choose from. To ensure quick, smooth and shortest possible route for the physical movement of goods, the manufacturer also has to decide the location of manufacturing facilities, warehouses, and the type of transportation to be used. Proper inventory management is also important so that there are always goods whenever demand arises.

Channel structure primarily provides place utility to the product. However, the logistics management takes care of the actual flow of the material and provides time utility to the product. The choice of channel structure has deep impact on customer satisfaction and on logistical costs.

The channel structure provides the outlets for product flow from the manufacturer to the end consumer through hierarchy of channel members. The success of channel efficiency and

effectiveness is greatly dependent on the logistics operation. And hence, the logistics network needs to be integrated with the distribution channels of the firm.

The channel structure is made up of the wholesalers, dealers, agents, brokers and retailers. The specificity of the channel members depends on the characteristics of product. For example, the consumer goods industry will have multilevel channels, while buyer of capital goods prefers to deal directly with the manufacturer. The channel members basically perform exchange, logistical and facilitating functions. The logistical requirement of channel members depends on the service standard decided by the manufacturer. In fact, the channel management and the logistics management should go hand in hand for efficient and effective distribution of the products.

To achieve the above objective, it is essential to fine-tune logistic operations to the requirements of channel members. The complexity of the logistic programmes will depend on the length and breadth of the channel. The information and communication technologies play a vital role in efficient and effective coordination of the channel and logistics management so as to achieve common objectives of customer satisfaction at lowest operating cost.

REVIEW QUESTIONS

1. Efficiency and effectiveness of supply chain very much depends on distribution network, Explain.
2. Discuss the various logistics roles in the distribution channel interfaces.
3. The distribution channel is a function of proper logistics programme. Discuss.
4. Do 'just-in-time' inventory systems have any effect on the firm's logistics and channel management?
5. 'Information technology' has a vital role in logistics and channel management. Discuss.

INTERNET EXERCISES

1. For studying decision approaches in channel management decisions visit <http://www.channelmarketing.com.au>
2. Visit <http://www.aberdeen.com/channels/scm.asp> and find out more information on logistics design for channel management
3. Tompkins Associates provides global supply chain services, distribution operations consulting, technology implementation, material handling integration, and benchmarking and best practices. Visit <http://www.tompkinsinc.com> and study the distribution solutions.

VIDEO LINKS

1. Supply Chain, Logistics, Distribution – Challenges, http://www.youtube.com/watch?v=_wWHfrBz104
2. Products Distribution Network, <http://www.youtube.com/watch?v=icpshe7Ywbq>

PROJECT ASSIGNMENTS

1. In the post merger scenario of Gillette and P&G, the distribution network of both the companies was merged and revamped. Accordingly the logistics programme to serve a new distribution structure was also evolved. Find out the major considerations for optimization of distribution network and also study the logistics strategy to support the 'merged' distribution network.
2. Find out a leading manufacturing company having wider product mix and serving both consumer and business markets. Study the distribution network and the logistics programmes to serve these two distinct markets.

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PART-III

TACTICAL AND OPERATIONAL DECISIONS

11 Transportation and Freight Management

This chapter talks about the role of freight management in supply chain, the factors influencing freight cost and transportation networking. Further the modes of transportation their influence on freight cost and goods movements are also discussed here. The importance of multimodal transportation and role of containerization in logistics productivity at both micro and micro level is also discussed.

12 Inventory Management

The chapter discusses the role of inventory in managing supply chain. Here inventory functions along with its classifications, inventory related costs aspects are also covered. The chapter further covers the risk pooling, policy guidelines in managing inventory, modern methods of managing inventory and the basics dimensions of inventory visibility.

13 Network Designing

In this chapter various factors influencing network design decision along with role of network in supply chain are covered. A framework for network design decision and prevailing models for facility location are also covered here.

14 Information System and IT Enabling

This chapter covers the role and characteristics of information system in managing supply chain. The factors in designing the cost effective and efficient supply chain system are also covered here. The chapter further discusses the importance of information system in supply chain integration and visibility.

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TRANSPORTATION AND FREIGHT MANAGEMENT¹

11

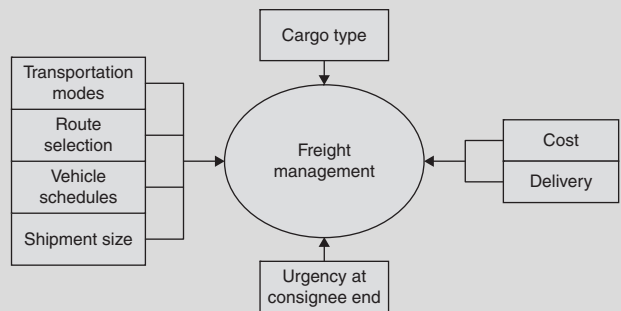
“ *Accessibility is the ultimate goal of most transportation and so is the best approach to use*

—Todd Litman”

Chapter Objectives

To understand:

- Freight management
- Factors impacting freight cost
- Transportation network
- Containerization



Freight Management

Transportation is an enabler to movement of products from one location to another. The movement starts at the beginning of a supply chain (i.e., suppliers) to the end of supply chain (i.e., customers). Transportation is the most visible element of logistics operation. It has significant share in logistics cost of the organization. It needs a thorough planning exercise to control it. Managing transportation cost presents many challenges. Hence, supply chain's success is closely linked to the appropriate use of transportation strategy. Transportation decisions should ensure that a firm's transportation strategy supports its competitive strategy.

Cargo Movement by Sea-Air Mode—A New Concept

With the change of the global scenario and to meet the logistics needs and challenges of the shippers and consignees. The NVOCC (non-vessel operating common carrier) companies were introduced which later came to be known as 'multimodal operators,' who created SCM with new concepts to meet these requirements associating themselves with liner shipping companies who normally prefer catering port to port shipments.

In this logistic change was born what is known as the 'sea-air movement.' This clearly means that a commodity moves from one port to another by two modes of transport, that is, making use of the SCM in multimodal transport. This proved to be a concept, which was able to meet the shipper/consignee deadlines not only with on-time deliveries but also at economical rates. The reason being, if it was moving by sea then it is slower but costs much cheaper and if it was moving through air, then it was faster but expensive. This proved to be disadvantageous for movement of textiles, automotive parts, high tech, leather items, etc.

The movement of cargo from the Far East and the South East to the United Kingdom and the United States had its own challenges with each customer having his own unique requirements. The new concept evolved was able to meet this challenge and proved to be 30–40 per cent cheaper. The sea-air cargo will always be the last to be loaded and first to be unloaded to reduce shipment time and lower the cost. Dubai became the main hub for this trade and all major logistics company set their offices there. A cost comparison will give an idea as to how this works out cheaper when compared to only air mode.

There is a consignment of 1,000 kilos of automotive parts to be moved from Mumbai to London. In the first case, moving by air mode (with cheapest airline) the costing will be Rs 1,55,000. The cargo is to be kept ready and delivered at the airport 24 h before the departure of the flight. In the second case, movement by sea-air mode for 1,000 kg equals to 3 cbm; the freight cost from Mumbai to Dubai by sea is zero using services of NVOCCs. The other charges by local transport, handling and CHA equals to Rs 7,250. Air freight from Dubai to Heathrow airport is Rs 70,000. Hence, the total freight is Rs 77,750. Since Dubai is the hub and volume cargo is always available the freight rate comes down. Today due to competitive market, Mumbai to Dubai is going on minus freight and shippers get a refund of \$20 per cbm from shipping companies. Therefore, it can be seen that sea-air mode has caught up with many of the shippers in this part of the world as it saves cost and time. The mode is well protected throughout by insurance companies.

INTRODUCTION

Transportation ensures the movement of people as well as goods (manufactured or produced) from one place to another and helps in growth of trade and commerce of the country. A good transportation system is one of the pillars of the whole structure of the industry and commerce in the country.

Along with the growth of industry and commerce, the transportation system has achieved economic and social objectives. The modes of transportation are changing as per the needs of the time and it is keeping pace with the growth of science and technology across the world. However, the degree of sophistication in modes of transportation equipments varies with level of economic condition and growth of

the particular region or nation. The transition from subsistence farming to commercial farming and growth of manufacturing activities widened the scope of development of transportation mode.

The result was development of four modes of transportation, that is, road, rail, air and water. The simultaneous upgradation of infrastructure took place which supported the transportation modes to achieve speed in movement. Even with the availability of sophisticated modes of transportation, the older modes still continued to serve the society though on a smaller scale.

In pre-independence India, the transportation system was mainly comprised of roads and railways, which was developed for smooth administration by the British government for communication with major ports and cities keeping in view the administrative, strategic and trade imperatives of the government ruling then. However, the focus changed after the independence to development of railways and road infrastructure to support the developmental needs of Indian economy and society. Later, slowly the seaports and airports were developed to support the industrialization in the country.

TRANSPORTATION INFRASTRUCTURE

In the movement of raw materials or products from the place of production to the place of consumption, transportation is the most important component of the logistical activity in the supply chain system. The transportation serves two purposes: one is the product movement and the second is the in-transit product storage. The movement of the product can be achieved through various modes such as road, rail, air, sea subject to the availability of and access provided by the infrastructure. The other function of transportation, that is, in-transit storage is not cost effective for a longer period of time. The guiding principle for choosing the transportation mode is the least cost per unit weight/volume of the product moved over the unit distance. However, selection of a particular mode is dependent on the availability of transportation infrastructure in the region.

In India, 60 per cent of the total cargo movement is done through road followed by rails which contributes 30 per cent of the movement; the remaining 10 per cent is shared by inland water, air and sea.

Road

Road transport is one of the promising means for agricultural and industrial advancement of the country. It is suitable for short and medium distances where other means are unable to reach. It provides door-to-door service, which is not possible by other means. Road transportation provides the basic infrastructure to bring the trade and commerce from remote rural areas to urban areas or vice versa and brings far-off villages into mainstream national life ensuring connectivity. With the manifold growth in industrial and agricultural activities and the outputs, road transport has assumed greater importance due to growing demand for making available the right product at the right place of consumption. Today road transportation is occupying a predominant position in the transport network in the country. Road transportation offers a number of advantages which are as follows:

- Door-to-door service
- Flexibility
- Reliability
- Reach to remote places
- Speed

As far as the trucking industry in the country is concerned, it is predominantly in the unorganized private sector and bulk of truck operators are single truck owners. The operators face many irritants, like levy of multi-point *octroi*, traffic police checks, which impedes the free flow of goods traffic. Currently 7.0 million trucks are plying on Indian roads. The Indian road conditions not being good, the average distance travelled by truck is 250–300 kilometres per day as against 550–600 kilometres by their counterparts in developed countries. The average operating cost of the Indian truck is Rs 1.7–1.9 per km–per tonne. The major domestic cargo traffic in the country is moved on national and state highways spreading over to 34,850 kilometres and 137,120 kilometers length, respectively. The national highways carry nearly 40 per cent of total road traffic in India. Out of the total national highways, 2 per cent of their length is four lanes, 34 per cent is two lanes, and 64 per cent is single lane. The share of national and state highways in total roads in the country is 1.42 and 5.56 per cent, respectively, which is grossly inadequate. However, India is better placed in per capita road availability compared to Brazil, Indonesia, China, Hungary as per the road index computed by Cooper and Lybrand—India 0.58, Brazil 0.47, Indonesia 0.47, Hungary 0.41, China 0.29 and Mexico 0.27.

Railways

Railways are the principal carriers of men and material and play a major role in country's trade and commerce activities. It is a main source of supply of essential commodities, by transporting through the length and breadth of the country. Across the globe, the railways have played an important role in industrialization and development of nations. It was the primary means available for movement of raw material and the finished goods and all countries in the initial stages of industrialization and development have heavily relied on railways.

The Indian Railways (IR) are operating through the network of 6,896 railway stations across 62,800 kilometres of route length with the route density of 138 km per 1000 sq mt of area, of which are 44,200 km is broad gauge lines which account for 85 per cent cargo movement by rail. Goods movement is done through 253,186 wagons with total carrying capacity of 10.6 million tonnes. The IR's 96 per cent of cargo consists of bulk items such as coal, iron ore, cement, fertilizers, raw materials for steel plants, finished steel products and petroleum. Bulk cargo is transported through the railways because of cost advantage over other modes. In 2009–10, Indian railways handled 833 million tonnes of freight traffic on the above network.

To handle the increased domestic and international trade, Indian Railways have introduced the container service and goods trains to carry the box containers way back in 1967. Today IR is handling 2.5 million TEU (tonnes equivalent units of 20 ft container) of box container traffic. To increase the share of rail traffic, IR is tying up with national road carriers to provide door-to-door service to their strategic clients. The Konkan Railways have introduced a unique scheme of transporting the complete trucks loaded with cargo to the destination. This will save precious fuel which other wise would have been used by the number of trucks carrying the trainload of cargo. Road traffic may be relieved to some extent and air pollution by those trucks will be eliminated.

Sea

The shipping industry is divided into the following parts:

- Liner service
- Tramp shipping

- Industrial services
- Tanker operations

The shipping fleet across the world comprises of tankers, dry bulk carriers, container ships and special vessels. The tanker fleet accounts for 33 per cent of the total fleet while dry bulk carriers contribute 32 per cent of total in terms of million gross registered tonnes (grt).

India has 55 shipping companies, of which 19 are exclusively in coastal trade, 29 in overseas trade while remaining are in both. Indian shipping has very high dependence on international trade with 95 per cent of overseas cargo by volume and 77 per cent by value moving by sea. Crude oil, oil products, iron ores, coal, food grains and steel are some major products transported through sea.

In shipping, India ranks sixth in Asia and fifteenth in world merchant shipping trade. At present, Indian shipping industry comprises of 510 ships of which 240 are overseas fleet and 270 are coastal ships including 70 offshore support vessels and about 25 supply vessels for offshore services and 12 dredges. The Indian shipping companies (SCI and Essar) are slowly moving from 'carrying business' to become a complete sea logistics solution providers, thereby maximizing shareholders wealth through focus on energy (oil) transportation business and moving upward in the value chain by providing one stop integrated logistics solution and value addition by providing their client web-based logistics solutions.

In India, 95 per cent of international trade and 85 per cent of domestic sea trade is handled by 12 major ports. Indian ports handled around 272 million tonnes of both domestic and international cargo in 2009–10 through 12 major and 187 small ports. This sector will continue to play a major role in Indian economy as India's export is likely to grow by 15 per cent and imports by 14 per cent. There will be major imports of POL, LNG and LPG in the near future and these will require better storage and handling facilities at the Indian ports.

Air

Compared to all other modes of transport, air transport is less hazardous in nature. Air transportation is a costly affair and it is used only for highly respiring and transpiring commodities and items, whose life is short and value is high. Even though the cargo handled by air is growing at the rate of 15–16 per cent, it is insignificant as compared to cargo handled by other modes of transportation. The air cargo traffic is handled by 8 international airports, 87 domestic airports and 28 civilian airports (shared with defence). After the domestic air services were liberalized in 1994, there was a sudden increase in the number of air transport operators to 7 scheduled private operators and 22 non-scheduled operators for passenger and cargo movement. Within two years of post liberalization, the market share of private operators went up to 40 per cent. The air cargo traffic is concentrated around the gateway airports such as Mumbai, Chennai, Delhi, Kolkatta and Bangalore. These airports handle 87 per cent of air cargo traffic in India. Due to the prospect of increase in air-cargo traffic in India, the government is inviting private participation in air transportation and the related infrastructure (air ports) to take care of growing passengers and cargo traffic in future.

Inland Water Transportation (IWT)

IWT is a eco-friendly transportation mode. IWT can play a significant role in augmenting the country's transportation infrastructure. India has a potential of 145,000 kilometres of inland water transportation infrastructure comprising of rivers, lakes, channel and it has recently started receiving attention from the government. IWT in many Western countries has already emerged as an alternative mode of

transportation. Cargo movement through IWT in India is meagre 1 per cent as against 10–12 per cent in the United Kingdom, Europe and China.

With globalization of Indian economy and implementation of WTO directives in India, there will be greater movement of goods to and fro. This will create a heavy pressure on already burdened transportation system of rails and roads in India. There is limit to expansion of road capacity because of available land, high cost and environmental considerations. The IWT is cost-effective as compared to road and rail infrastructure. A kilometre of highway construction can cost Rs 6 crores, while more than 100 kilometre waterways can be developed in the same cost at the current rate. As per the estimates, if Re 1 is spent for IWT development, the corresponding cost for development of roadways and railways would be Rs 2 and Rs 5, respectively. In case of energy conservation the ratio would be 1.5 and 4, respectively.

There are some hazardous commodities, which should not be allowed for transportation on road. In view of the above constraints and the advantages, the development of IWT has become relevant in today's context. The major advantage of IWT is doubling of load capacities for a small increase in depth and thereby providing an ageing flexibility and cost elasticity, which does not exist in other modes of transportation. Besides lower fuel consumption and construction cost, IWT has the advantage of ensuring minimum human loss against the very frequent accidents occurring in case of road and railways.

Pipelines

Pipeline as a mode of transportation was first developed by Samuel Van Syckel in 1870 to transport petroleum. Twenty years later, Standard Oil Company changed the face of transportation. Though petroleum was the first product transported in this mode, pipelines later became useful for several other commodities such as coal in slurry form, iron ore fines in slurry form, chemicals, natural gas, etc. The basic advantage of pipelines is that they reduce the operational cost, though the initial investment is high. In India, pipelines are used for oil transportation by all public and private sector petroleum refineries. For iron ore transportation, they are used by Kudremukh Iron Ore Project to transport iron ore over 67 kilometre distance along difficult terrains in the Western Ghat.

Pipeline is an eco-friendly transportation mode. The cost of moving oil by rail or road continues to rise over the years; however, pipeline heads towards downtrends. The transportation cost of moving oil is Rs 1.15 per tonne per kilometre on a new pipeline while it is Rs 0.87 on a depreciated pipeline. At the current rate, transportation of oil by road costs Rs 2.50 per tonne per kilometre and by rail costs Rs 2.00 per tonne per kilometre. Although laying of pipeline costs Rs 15 millions per kilometre.

Currently, 27 per cent of the petroleum products (petrol, kerosene and diesel) are moved by pipelines of 6,350 kilometre length in India, while in the United States 59 per cent via 180 pipelines and another 35 per cent via barges along the Pacific and Atlantic coast. Today in India, pipeline moves 58.7 million tonnes of oil per annum which is expected to rise to 85–90 million tonnes by 2015.

Ropeways

In India, more than 16 per cent of the geographical area is hilly. Communication to such areas is a problem because of the long circuitous routes. The transportation of goods and essential commodities to the hilly area is some times important because of their strategic location from the defence point of view. In view of oil shortages, ropeways can prove more economical and faster than road transport, particularly in hilly areas. The ropeways have the following advantages:

- Ropeways cause least damages to the ecology.
- Inaccessible hilly areas can be reached with shortest distance.

- Other modes of transportation are uneconomical.
- Bulk material can be moved faster over short distance.

This mode of transport is used in hilly areas in Sikkim, Meghalaya, Mizoram, Himachal Pradesh and Uttar Pradesh. Currently 175 kilometres of ropeways are in use in the above territory for transportation of both materials and human beings.

FREIGHT MANAGEMENT

Freight transport management includes various strategies for increasing the efficiency of freight and commercial transport. Logistics is a technical term for efficient freight management, including shipping practices (e.g., vehicle type, shipment size, frequency, etc.), facility visiting, and related activities. Logistics usually focuses on minimizing shipper costs, with little consideration for social cause such as congestion or pollution impacts. In general, transport management activities cover the following:

- Using transportation modes with lower social costs, such as rail and water transport rather than truck for longer-distance shipping as trucking uses much more energy per unit of transport than rail or water.
- Improve scheduling and routing to reduce freight vehicle mileage and increase load factors (e.g., avoiding empty trips).
- Organize delivery systems so fewer vehicle trips are needed to distribute goods by using load consolidation.
- Reduce total freight transport by relying on local products from facilities closer to their destination markets.
- Practice fleet management programmes that reduce vehicle mileage, use optimal-sized vehicles for each trip, and insure that fleet vehicles are maintained and operated in ways that reduce external costs (congestion, pollution, crash risk, etc.).
- Encourage minimizing excessive packaging and unnecessary delivery frequency, and relying on more local suppliers.
- Encourage efficient freight transport equipments.
- Improve vehicle operator training to encourage more efficient driving.

The selection of transportation mode depends on the characteristics of the product and customer service requirements. For example, the raw materials are invariably transported in bulk and through cheaper mode of transportation like rail or sea. The unit value being less, the transportation cost, as percentage of the value of material being transported is very high. Hence, freight reduction through scale economics becomes necessary. However, in case of high-value items, the deliveries required are faster and in smaller consignments with reliability. In this case, the unit product value being high, the transportation cost as a percentage of the product value is not so significant. Hence, faster and costly modes of transportation such as road or air are preferred.

For logistics management, cost, speed and reliability are the most important factors to select amongst the available transportation options based on the characteristics of products.

Speed and Availability of Service

The speed of the service is the elapsed time to move products from one facility to another and finally to the customer which is often more important than the cost of the service. The slower modes of

transportations involves lower transportation cost, they result in lower service levels. The availability depends on the existing transportation infrastructure and the ability of mode to serve the given pair of locations. This sometimes becomes major constraint in speedy delivery. It necessitates the usage of inter-modal transportation.

Reliability

This means ability of the carrier to deliver the shipment in good condition, within the stipulated delivery time frame to the customer.

Capability

It is the ability of the carrier to accommodate the cargo in size, weight and quantity for transportation to the destination. This is very important for transporting the odd size and heavy products over longer distances. In such cases, special trailers are required for product movement. The cost of transportation goes up in such cases. Along with transportation mode, the other considerations are availability of special material handling equipments.

Frequency

This refers to the number of schedule movements of the carrier between a pair of locations. Pipelines are preferred for movement of liquid products because of their continuous operations. However, for solid products road carrier is the best option.

The transportation decisions are dependent on a number of other factors like unit value of the product, predictability of demand, saving in transit time, cost of transport mode, its impact on inventory cost, and the desired level of customer service. The selection of modes and carrier type is done on the basis of delivery time, delivery reliability, freight rate, etc.

FACTORS INFLUENCING FREIGHT COST

The logistics manager has to achieve objectives of freight cost reduction, speed and reliability in delivery. However, the freight cost is influenced by the following factors.

Volume

With the economies of scale, the cost of operations is distributed over the large volumes resulting into the lower per unit cost of cargo movement. For example, a 9 tonne loading capacity truck will have same cost components of fuel consumption, loading and unloading charges, employee cost, etc. for the distance covered, irrespective of whether truck is fully or partially loaded.

Distance

The variable cost is directly proportional to the distance covered by the vehicle irrespective of the load carried. It includes fuel and maintenance cost. However, the relationship is not linear because the fixed cost element such as employee wages, loading and unloading charges are fixed irrespective of the distance travelled.

Product Density

The freight cost depends on the weight per unit volume of the product. It includes two variables of weight and volume. A 2-axle 9 tonnes capacity truck has a limitation on the volume for low-density products such as cotton bales, wood, plastics, etc. In such cases, the freight cost per unit weight assessed is relatively higher.

Similarly for high density products the limitations are on the weights, which should not exceed the allowable payload of the vehicle. In such cases, the truck is partly loaded with respect to the available loading space on the vehicle.

Product Shape

The size and shape of a product pose difficulty in handling and affect the speed of its movement. Such products require special arrangement for loading, unloading and towing. For such heavy payloads, vehicles with long trailer are deployed for movement. The operating charges for such special vehicles are obviously much higher because of higher operating cost, slow movement of the vehicles and in special cases escorts are deployed to accompany them. The large size chemical processing steel vessels, heavy-duty electrical transformers, long and odd shaped steel products attract higher transportation charges.

Product Handling

For handling heavy as well as odd-shaped cargo, special material handling equipments are deployed. These equipments are not available at loading or unloading points because of very low frequency (at the time of project installations only) of usage. Hence, these equipments have to be specially organized. This adds into the unit transportation charges of the product.

Product Type

Products such as fruits, food products, certain pharmaceutical formulations require special temperature and humidity condition for storage and transportation. These products need to be transported through refrigerated vehicles costing double and having operating cost three to four times more than that of a normal vehicle. The products are susceptible to explosion, evaporation, damage or theft require special transportation arrangement such as specially constructed vans or box containers. These products attract additional transportation charges.

Market Dynamics

In the Indian context, goods transportation business is in the hands of transport agents. The rates are fixed by the agents and not by the truck owners, except in case of very large fleet owners who operate across the country through their marketing infrastructure. However, depending on the demand and supply of trucks for cargo movement to particular destination, the freight rates are fluctuating. Even if more vehicles are available, there is a cut throat competition amongst the truck operators to get the business and the rates charged are sometimes below normal cost. For odd and problem routes the rates are on the higher side to cover the risk.

TRANSPORTATION-RELATED PROBLEMS IN SC

There are two perspectives of transportation and which give rise to the problems in supply chain design and operation. The first one is the shipper's perspective. The shipper always wants to balance inventory, facility and transportation costs while maintaining an excellent service level in the operations of the supply chain. The second one is the carrier's perspective, which calls for maximize profit by making best use of the available transportation capacity. From service providers' point of view, it includes costs, which covers, capital costs, maintenance cost, terminal-related: land/rent, equipment, personnel cost, fleet operational costs, trip-related (fuel, drivers, wear and tear) cost, freight handling cost and overhead covering planning and scheduling, customer service, etc.

Freight charging patterns for common carriers is another issue in transport-related problem. The rate varies in relations to the shipment size LTL (less than truck load), minimum total rate for quantities below a minimum threshold, several weight categories with different rates and time-volume rates. The other one is related to distance such as uniform rates, proportional rates, combination of fixed rate plus variable rate per distance (truckload rates), tapered rates (increase with distance) and blanket rates (constant rates for certain intervals of distance for certain time period), etc. The supply chain operations need to be planned in advance taking note of the above aspects.

TRANSPORTATION NETWORKS

The movement of goods from the point of production to the point of consumption is done through various modes of transportation. Different types of networks are in use depending on the transportation load, number of delivery points, existing distribution centres, product value, frequency of delivery, urgency and the cost economics.

Point-to-Point Network

This type of network is quite common for long distance hauls on the national highways. The point of origin and destinations are fixed. Full truck loading is assured both the ways (Figure 11.1).

Multiple Delivery Points

This network is used for round-the-trip operations, with multiple pick up and delivery points. For example, the delivery of filled bottles and pickup of empty bottles of soft drinks at multiple points (retailers) on the fixed route (i.e., starting and ending at bottling plant) is quite common (Figure 11.2).

Trans-shipment Points

The two local area networks (across national and state highways) having a common point where the loading and unloading takes place for freight consolidation or break bulk. The most of the national

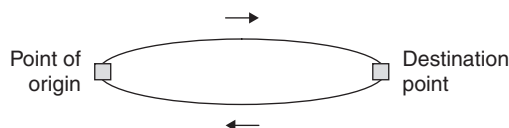


Figure 11.1 Point-to-Point Network

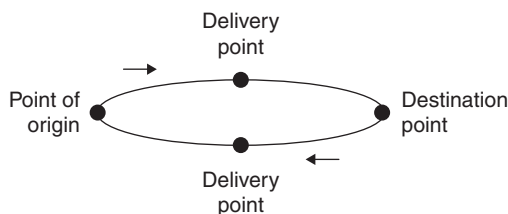


Figure 11.2 *Multiple Delivery Points*

transporters maintain two types of fleet: vehicles dedicated for national long distance haulage and other one for catering to local network. The consignment from long distance fleet is trans-shipped to local vehicles for distribution across the local areas (Figure 11.3).

Nodal Network

These networks are used for multi-modal transportation system and include multi-stop and trans-shipment pickup and delivery stations. For example, a box container truck may have predetermined multipoint pickup stations for freight consolidation on its way to rail terminal where from the box container may be taken to port terminal for loading onto the ship (Figure 11.4).

Hub-and-Spoke Network

The network arrangement is like a hub and spoke (Figure 11.5) of the wheel. The hub acts like a central feeder point to the distribution centres, which are at the strategic locations spread across

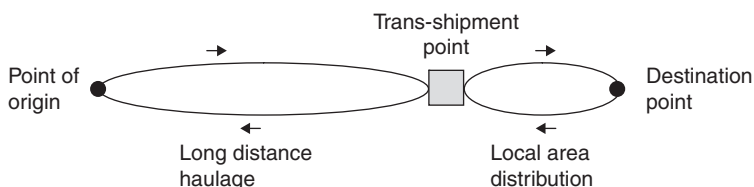


Figure 11.3 *Trans-shipment Point*

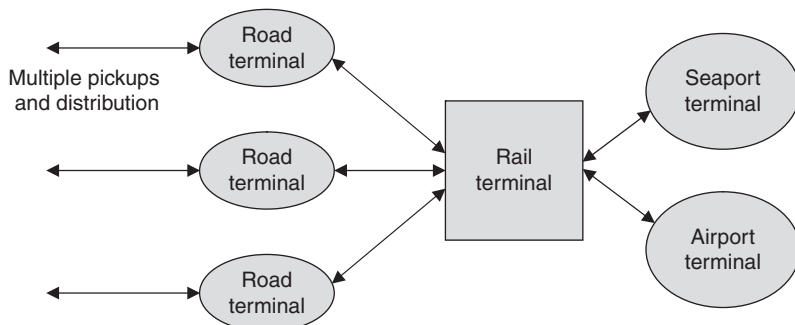


Figure 11.4 *Nodal Network*

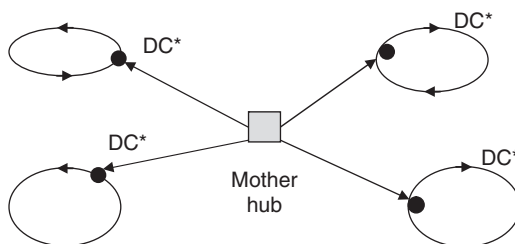


Figure 11.5 *Hub-and-Spoke Network*

*DC—Distribution Centre

the geographical area. The high volume and high-speed shipment takes place from the hub to the distribution centres through the predetermined shortest routes called spokes. The trans-shipment of consignment is done at distribution centres for distribution across the local area.

This arrangement optimizes the number of distribution centres in the network and ensures reduction in inventory and improvement in customer service.

MILK-RUN

In logistics the vehicle picks-up the material from vendor (s) and delivers to the factory/customer(s). This needs to be planned if the vehicle has to pick up inventory in small quantity from various vendors, who are on the route. This is similar to milk being collecting by a tanker-van en-route from various farmers at different point of time from different locations during the day or milk packets are to be delivered to various destinations en-route. In inbound process after the tanker is full, the milk is delivered to the factory for further processing. The milk collection process is time bound. Any delay beyond 6–8 hours to reach to factory will initiate the milk to spoil.

In the business context, milk-run (Figure 11.6) is a tested and proven method of optimizing a service provider's trucking operations in collection or dispatch activities. It is used in automobile industry for periodic collection of spare parts from a large number of vendors located in a particular industry cluster. Transportation within automotive spares today would require optimization in terms of cost and time. Milk-run can solve the problem of scheduling and dispatching vehicles in vehicle-based internal transport systems within warehouses and production facilities in automotive and electronic industries.

Milk-run is quite efficient and cost-effective where there is demand at multiple pick-up/delivery points for urgent services, hence there is the need to optimally prioritize. In milk-run the service provider using his vehicle to provide delivery/pick-up needs to optimize the routes. Milk-run is useful when precision electronics/automotive spares from certain manufacturing clients need to be urgently shipped on a JIT (Just-in-Time) basis to the next stage in the supply chain

In milk-run the nodal points of collection/distribution in a milk-run are established first. These might be collective pick-up/delivery points for the pick-up or delivery vehicles. The critical variables in milk-run design are distance, cost of delivery/pick-up, cycle-time and perishability of the product. At every point in a distribution network, milk-run calls for an optimization of the above variables. The service provider has to prioritize on the above variables. Factors to be considered during this practice are to increase vehicle capacity utilization, shortest routes and optimum transportation cost. The



ROUTE PLANNING

The main objective of route planning is to cover the distance between two points with shortest distance ensuring operating economy resulting into lowest transportation cost. To plan a route for a truck for multiple pickups and delivery points spread over the vast geographical area is a challenging task. The planning task becomes complicated in case of perishable commodity having short shelf life. Generating distance matrix, which is a one time exercise, and generating the route matrix with nodes for pickups and delivery can solve the problem. The other factor is the time of travel in case of perishable products.

CONTAINERIZATION

Containerization is making waves in the transportation of goods ensuring manifold benefits to users, transport companies, and the country's economy. A container can be defined as a transport device to move solids, or liquid material. It is a case or a tank with adequate strength to be repeatedly used for packing and transportation of the material. The container can be used on several transport modes without material being stuffed, de-stuffed or trans-shipped.

Today containerization is an accepted trend in shipping for international and domestic trade. Initially containers were in use for sea transportation, but now are being profusely used in inland transportation. It was Malcolm McLean, owner of a huge trucking company in the United States, who conceived the idea of containers in 1956. The containerization in shipping started in the 1960s. The first container shipping named 'Hawaiian Citizen' for port management with a vision and strategic planning was developed by sophisticated handling facilities. The container traffic has grown in leaps and bounds, and more than 200 million TEU is estimated to be handled the world over in 2010. In 2009 the major ports in the world (Exhibit 11.2) handled 140 TEUs of container cargo. This is expected to grow to 417–491 million TEUs by 2015 with the largest growth being planned in Asia. Globally containerization is expected to grow at 5 per cent in the next 10 years and by 8–10 per cent in Asia.

Ports and shipping industry the world over have accepted that containerization is the future of shipping and has initiated projects to keep pace with the developments. The ship builders have already started building huge container carrier cargo ships with capacity of more than 6,000 TEUs. The ports are also gearing up to meet these challenges. Some ports like Heliex, Canada, have already taken action to deepen the container birth to maintain the draft of at least 52 feet to receive ships with 8,000 TEU and above. The containers used for movement of solid cargo are box type construction with side or top opening with twist locking system. These are normally water and air tight constructions using steel or aluminium alloy and are built as per ISO (International Standard Organization) standards having common sizes as indicated in Exhibit 11.1.

The container services in India were started in 1966 and initially containers with payload of 5 tonnes were introduced. The Indian Railways first started container service between Mumbai and Ahmedabad in 1968. However, slowly the railway rakes were developed to adopt the large size containers of

Exhibit 11.1 Sea Freight Container Dimensions

20 ft Dry Volume: 31 cum (1079 cu ft)	40 ft Dry Volume: 67cum (2372cu ft)
20 ft Dry High Cube Volume: 33 cum (1178 cu ft)	40 ft Dry High Cube Volume: 76 cum (2675cu ft)
20 ft Refer (Refrigerated) Volume: 23 cum (803 cu ft)	40 ft Refer (Refrigerated) Volume: 58cum (2053 cu ft)
20 ft Open Top Volume: 32 cum (1116 cu ft)	40 ft Open Top Volume: 64 cum (2246 cu ft)
20 ft Flat Rack Volume: 32 cum (1116 cu ft)	40 ft Refer High Cube (Refrigerated) Volume: 62cum (2197cu ft)
20 ft Open Side / Top Volume: 31 cum (1079 cu ft)	40 ft Refer High Cube (Refrigerated) Volume: 69 cum (2419 cu ft)

Source: <http://www.concorindia.com>

Exhibit 11.2 *Container Traffic Handled at Various Major Ports (2009) Across the Globe*

<i>Port</i>	<i>TEUs (min)</i>
Singapore	25
Shanghai	28
Hong Kong	23
Shenzhen	21
Busan	12
Dubai	10.6
Taiwan	10.2
Rotterdam	9.8

Source: <http://www.container-transportation.com>

20 ft and 40 ft length. Until 1988, the growth of container traffic was very slow. However in 1988, the Container Corporation of India (CONCOR) under the Ministry of Railways was formed to look after the container traffic in India.

This has resulted into speedier development of container handling (cranes, trailers, reach stacker, heavy duty fork lifts, side loaders) and movement (special railway rakes, container trucks) facilities in the country at various locations. The growth rate during the last few years, observed in container traffic in India was 18–20 per cent per annum. In 2010, CONCOR handled 2.20 million TEUs of container traffic. CONCOR has a monopoly position in container handling with 60 per cent share in inland container traffic. CONCOR has a network of 17 inland container depots (ICDs), 7 container freight stations (CFSs), 5 port side container terminals (PSCTs) and 10 domestic container terminals (DCTs) to handle the growing container traffic in the country. To take care of the container traffic, CONCOR has its own inventory of various sizes of containers and in addition, Indian Container Leasing Company in the private sector is engaged in leasing of containers to transporters and corporates. Exhibit 11.2 gives an idea of container traffic handled at major ports across the globe. As against this, all Indian ports, together, handled 6.5 million TEUs of container traffic.

SUMMARY

Transportation is the barometer which measures the economic, social and commercial progress of the country. It is the most visible element of logistics operation. It has a significant share in overall logistical cost of the firm and needs a great deal of planning to control it. In the movement of raw materials or products from place of production to place of consumption, transportation is the most important component of the logistical system. The transportation infrastructure of the country plays a vital role in the speedy movement of goods within and outside the boundaries of the nation. Transportation modes such as roads, rail, inland water, sea-air and rope play a major role in cargo movement and trade development of the country. The regulatory environment in the country should be supportive to the growth of the logistics industry and simultaneously enforce discipline amongst the industry operators. At a micro level, the freight management is crucial for cost controls. For a logistics manager, selection of transportation

mode, carrier selection and route planning are the areas wherein his right decisions will result into an overall efficiency and effectiveness in supply chain. The transportation documentation supports the movement of the goods to reach the right place. The containerization is making waves in the transportation of goods ensuring manifold benefits to users, transport companies and the country's economy. It facilitates the usage of multi-modal transportation and safe door-to-door transportation of the cargo.

REVIEW QUESTIONS

1. 'Decisions on transportation modes are crucial to organization' Explain.
2. What are the various factors affecting the freight cost?
3. Discuss the different networks in transportation services.
4. What is containerization and what is its role in logistics?

INTERNET EXERCISES

1. Visit Texas Transportation Institute for more information on multimodal transportation, <http://tti.tamu.edu>
2. Study various facets of 'Transportation Lifecycle Management' and its role in 'Supply Chain Solutions' at <http://www.manh.com>
3. To understand on going trends in truck freight rates across various routes, lorry availability for a given period etc in India visit <http://www.tcil.com/irfi/index.asp>

VIDEO LINKS

1. Freight Optimization, <http://www.logisticsbureau.com.au/video-supply-chain-logistics.htm>
2. Transport Tips, <http://www.logisticsbureau.com.au/video-supply-chain-logistics.htm>

PROJECT ASSIGNMENT

1. Select any firm manufacturing the low unit-priced and mass consumed products like salt or cement and find out how they are organizing transportation-mix to achieve trade-off between cost and product availability.

NOTE

1. Sople V. V. (2010), *Logistics Management*, Chapter 8, Pearson 2nd edition, pp. 125–147.

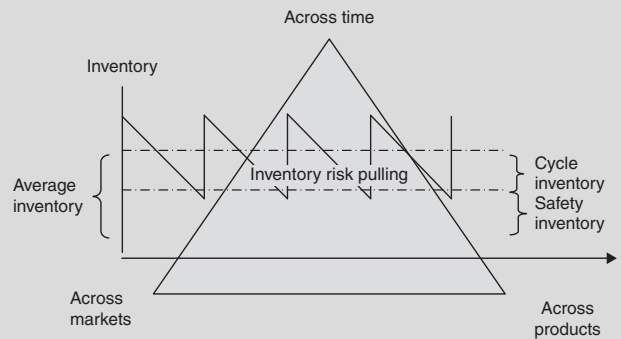
INVENTORY MANAGEMENT

“It’s easy to turn cash into inventory ... the challenge is to turn inventory back
—Jon Schreibfeder¹”

Chapter Objectives

To understand:

- Role of inventory in supply chain
- Inventory classification and functions
- Inventory-related costs and risk pooling
- Managing inventory
- Inventory policy guidelines



Inventory Planning

Inventory levels are affected by the level of customer service, demand variations and flexibility in business operations. Hence, managing the inventory is the most important function of supply chain to achieve trade-off between cost and customer service. Inventory decisions fall in high risk areas and have deep impact on the business operations. Without a proper inventory management, marketing may find the loss of sales and customer satisfaction. Hence, inventory planning is critical to both manufacturing and marketing. Shortages can disrupt marketing and manufacturing operations. But overstocking creates problem in working capital. Inventory increases cost and reduces profitability through added warehousing, working capital requirements, deterioration, insurance, taxes and obsolescence.

Priya Cosmetics: Seeking a Solution to Reduce Inventory

Priya Cosmetics (PC) is a leading firm dealing in cosmetics and personal care products. The company had a supply chain network of 4 factories (3 contract manufacturers and 1 its own), 4 mother depots and 40 satellite depots managed by CFAs (carrying and forwarding agents) for servicing distributors. Goods moved from the factory to mother depots to other depots and finally to distributors. The major problem Priya Cosmetics faced was high level of inventories at all depots. The management, in order to reduce inventories, appointed an outside consultant to find a solution to the problem.

The consultant was able to find out the following from his studies: The average inventory holding at mother depots was 8.2 sales/weeks and at the depots was 6.5 sales/weeks. The discrepancy in the book and physical stocks was over 30 per cent. The firm had no process in place to review inventory level and the demand forecast. They had push-system in place. The organization did not have a clear policy on periodic reconciliation of physical stocks with book records.

The action undertaken was to introduce the Bin Card System for each rack at the CFAs. An IT solution was identified and implemented for accounting the cycle count process, providing MIS on deviations and accounting the adjustment notes. The forecast was calculated based on consolidated orders with factoring for promotions and seasonality. They implemented FIFO (first-in-and-first-out) system in the depot and resorted to automated allocation of inventories. The demand planning and forecasting were made a periodic activity using the above IT solution. This helped in aligning forecasting with market orders and actual sales. Responsibility and accountability were set up in the organization to monitor and authorize activities.

The benefits reaped were an inventory record accuracy of 95 per cent within 2 months. The stock level was reduced from 8.2 weeks to 5.5 weeks at the mother depots and from 6.5 weeks to 4 weeks at the satellite depots. The inventory levels across all the stocking points were dropped by 30 per cent. PC has achieved complete visibility of information and inventory across the supply chain.

INTRODUCTION

Inventory is the total amount of goods and/or materials available in a store, factory and finished goods at warehouses together at any given time. Inventory covers raw materials, components, work-in-progress and finished products. Marketers need to know the precise number of items and storage areas in order to place orders or control losses. Efficiency in effective inventory management will always give a competitive edge to the business, regardless of its nature. With effective control and management over inventory stock, as well as accurate visibility and fast efficient fulfilments, competitive pricing can be offered to the customer. Adequate stocks of inventory are necessary to ensure uninterrupted business operations. Inventory management covers tasks like stocks count, maintaining inventory records, re-ordering time, re-ordering quantity, orders placing on the vendors, inspection, delivery of new orders and ensuring storage of inventory in adequate quantity. All the above are done with a purpose to reduce inventory cost burden on the company and extending desired service level to the customers.

Inventory generally constitutes the second largest item after fixed asset in the financial balance sheet of the manufacturing company. Under the financial perspective, this current asset is worth maximizing the value to the firm and shows no disadvantage in carrying more inventories. The funds invested in inventory are costing to the firm by way of interest on the working capital borrowed from the bank. Hence, the investments in inventory are cost to the company. Thus, the reduction in inventory will reduce the inventory-related cost (capital, handling, storage, transportation, insurance, obsolescence and exchange rate fluctuation costs). The benefits of inventory reduction are visible in profit margins, ROI (return on investment) and EVA (economic value addition) in a positive direction.

In today's competitive scenario, inventory-related investments are viewed as supply chain cost driver rather than the material asset. Therefore, lean supply chain operations are preferred. Firms are using MRP (material requirement planning), DRP (distribution requirement planning) or JIT (just-in time) systems to have maximum inventory turns (ratio of sales to average inventory) for reducing cost on inventory investments and enhancing profitability and return on investments.

INVENTORY FUNCTIONS

Inventory management is a strategic area in supply chain process. Inventory management helps in reducing purchasing and inventory cost, gain visibility to inventory process, improve customer satisfaction and reduce time to market. Inventory has an impact on efficiency and effectiveness of the overall supply chain system. As the cycle of production and consumption never matches, the goods have to be kept in stock to get over the fluctuations in demand and supply. To counter uncertainties, the higher inventory levels are maintained. But this affects the bottom line of the company. This is a high risk and high impact area, which has to strike a balance between two opposite goals of lower cost and higher level of customer service.

Inventories are maintained in the warehouses, which have accountability for the inventories in the stocks. The inventory levels in the company also affect the efficiencies of the other divisions. Inventory acts as protective cushion for continuous operation in customer supply chain. The top management views inventory as cash investment and expect to derive the profits from it through effective and efficient customer service. The poorly managed inventories can create cash flow problem. This

Box 12.1

Supply Chain of Bhel-Puriwala

There is a rare Indian who has not visited a Bhel-Puriwala—a hawker with his moving cart selling chat preparations. His inventory is highly perishable. For a day's (6 hours in the evening) requirements, he cuts onions, potatoes, etc. in the quantity which is sufficient to serve his customers. He knows the demand quite accurately and sees that there is no leftover perishable inventory (he has to carry home), which cannot be used the next day. He manages the inventory in such a way that even on weekends or festival days, when demand for chat is more, there are no shortages and no leftovers. The customer demand in volumes and varieties of chat is served within 10–15 minutes. Don't you think that he is an effective supply chain manager with a very limited finance, managing his supply chain ensuring customer delight and making his earnings?

may happen due to inaccurate forecast and excess production. The inventory management is basically concerned with the following:

- Proper level of inventory
- Balance between inventory cost and customer service
- Treating inventory as liability or asset

The industry will have to manage basically three types of inventories, which are held at the various stages of the supply chain of the company. These are (1) raw materials and components on the procurement side, (2) in-process or work in progress inventory and (3) the finished goods inventories (at source and distribution centres). The distribution side is primarily concerned with the finished goods inventories, which again are divided into three parts such as:

- Non-excite paid goods at plant warehouse
- Pipeline inventory (in transit)
- Channel inventory

As considerable amount of funds are blocked in the inventory, the investments in inventory cannot be used for other productive purpose or creating assets. Inventories carry the risk of damage, theft, pilferage or obsolesces. However, the nature of risk varies with product, pace of change of technology, severity in competition and enterprise's position in the distribution channel. The risk of carrying inventory also varies with its location as follows.

Manufacturer

Manufacturer maintains the inventories of raw materials, work in progress and the finished goods. He commits inventories for longer periods. Hence, his depth of risk is highest amongst the other members of the supply chain. In general, the manufacturer handles less number of product varieties as compared to wholesalers and retailers. The manufacturer's inventory commitments are closely related to the investments made in anticipation of the expected returns, lead time of the raw materials and components, complexity and width of distribution network, unit value of the product and the nature of demand.

Wholesaler

Wholesaler handles more product lines as compared to the manufacturer. His risk is spread over the different products. The cycling variations in demand of different product at different point of time keep wholesaler's risk factor limited to the non-performing product lines. For the seasonal products the wholesaler purchases the inventory in advance in anticipation of the future sales, which increases his risk factor. The inventory commitments of the wholesaler are not of longer duration as compared to manufacturer.

Retailer

Retailer faces shorter risk duration period as compared to manufacturer and wholesaler. His risk is spread over the range of products. Hence, his commitment to inventory is not deep. Retailer normally buys and sells and does not stock the material for a longer duration. His risk is more on the marketing side than on the inventory.

INVENTORY AS BUFFER

Manufacturers may exhibit their flexibility by quickly responding to changes in demand. For example, a company that ships to the customers in a single day of the customer order shows a greater flexibility. With its responsiveness it could easily attain a 100 per cent customer satisfaction. But this ideal situation can never be attained. This is due to building of lead time across the supply chain at production, transportation and warehousing. The time from the materials are ordered till the finished products reach the customer may be long. It may range from few days to weeks or months. This may lead to customer dissatisfaction. It is evident that customers will not wait this long period from order to delivery. The manufacturer needs to plan ahead by estimating future demand by making forecasts. If planning of production and inventories was perfect, the firm would be able to implement a pure JIT (just-in-time) strategy. In JIT scenario, the components are delivered to production as and when they are needed, but this is not as simple due to the many unforeseen events and uncertainties. These limitations need to be accounted for to implement 'JIT' successfully. These limitations arise out of late shipments from suppliers, defect incoming material, low production yield, production process breakdown, or uncertain product demands.

In the longer planning horizon, the accuracy level of the plans diminishes. In the high fashion apparel industry the planning horizon is typically in weeks. The long lead times make the manufacturer inflexible and vulnerable to unforeseen changes and inaccurate demand forecasts.

In such circumstances, a manufacturer will get over the uncertainties and unforeseen events by keeping safety stocks. The safety stocks act as buffers for the lack of flexibility in the supply chain. In the reduced lead times in the supply chain, the planning horizon is shortened resulting into increased visibility and flexibility in operations. Hence, with shorter lead times the need for a buffer in the form of inventory will also get reduced. This means that with higher flexibility the firm can maintain less inventory stocks to deliver the same level of customer service.

REASONS FOR CARRYING INVENTORY

Inventory is required as inputs to produce the finished goods, extending the service to the customers and to keep the customer's manufacturing operations running. Hence, inventory plays a crucial role in the supply chain of the organization. It helps in keeping the smooth flow of products across the supply chain. The excess inventory in the supply chain means more cost of holding the inventory, which does not balance the cost-benefit equation to get desired profits. The benefits and loss of carrying the inventory should be examined in light of the reasons for the carrying the inventory.

Balancing Supply and Demand

Raw material, components and parts are required for producing the finished goods. The manufacturing organization keeps stocks of this material to meet the continuous requirements of the production. The companies operating on JIT principle also keep some inventory in stocks to meet contingencies. However, the stocks are very less. The work in progress inventory is a major portion of the production-related inventory. The reduction in this category of inventory results reduction in inventory-related investments in the production process. The production and consumption cycles never match. However, the sudden requirement of a product in large quantities may not be fulfilled immediately as the production cannot be taken up in a short time. In such a case, the products are manufactured in advance in anticipation of demand and kept in stock for supply for the pick period.

Seasonal Variation

For certain products, the demand varies with the period/season in a year. In certain times of the year, it is the peak period, while it is lean for the rest of the year. The production runs in the factory are scheduled based on the average demand in the year. The excess production in the lean period is kept as an inventory to take care of the peak demand. In cases where raw material for manufacturing food products is available seasonally, the products are manufactured in advance and stocked as inventory to meet the demand of the finished product through the year.

Scale Economics

Products are manufactured in large quantities to get fixed cost benefits due to economies of scale. The produce is kept in stocks for distribution to consumption centres as and when it is required. The distribution is done in economical lot sizes for system efficiencies, speed and cost. In short, inventories provide demand utility for the products at time and the places it is required for consumption.

Operational Support

The inventories are also required for repairs, maintenance and operational support. These inventories include spare parts of the production machineries, consumables like lubrication oils and welding rods, chemicals, pallets, etc. The companies do not have the correct picture of their investments in this category of inventory. Many times excessive stock is held to avoid emergency rushing to the market for procuring little quantity.

Customer Service

The suppliers maintain the inventory of spare parts to extend the after-sales service to their valued clients. The availability of the parts at the time of requirements at customer end is crucial for customer satisfaction. This may be used as a tool for competitive advantage. However, maintaining the significant level of inventory at all the time requires major investment. This depends on to the level of service offered by the company to its clients.

Anticipated Future Requirements

The organisation sometimes keeps excess inventory to take care of the shortages of material availability in the market in the future. He anticipates increase in the prices of the products and hence he buys in excess and keeps in stocks the critical inventory to keep the operation running without interruptions. This results into increase in the inventory level for a short period. However, in many cases the benefits of keeping excess inventory offsets its carrying cost.

INVENTORY-RELATED COSTS

The major portion of the working capital of the firm is blocked in inventory. If the inventory is in excess of the optimum level, more funds will be blocked and it cannot be used for other productive purposes resulting into opportunity loss. Hence, the funds are tied up unnecessarily. There are other costs (Figure 12.1) related to the inventory. The incidence of those costs will also be more if inventories are in excess of the optimum level.

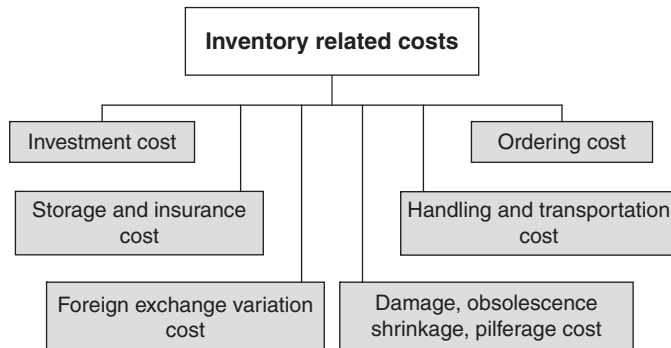


Figure 12.1 *Inventory-Related Cost*

Investment Cost

For producing inventory, funds are required. The funds once blocked cannot be invested in any other productive activities. However, the cost of blocked funds is income which the firm would have generated by way of investment in other productive activities. The fund invested in the inventory attracts the interest charges (12–18 per cent) on working capital borrowed from the bank. The interest charges on investment in excess inventory will erode the bottom line. Thus, investment cost or loss of opportunity cost is simply calculated by way of interest earned on the bank deposits of equivalent amount.

Ordering Cost

This refers to the cost involved in the ordering process. The cost on paper work faxes, phone calls, etc. will add to the inventory-related cost.

Storage Cost

This covers the cost related to product holding in warehouses. Depending on the warehouse (private, public or contract) cost is incurred on the space occupied depending on the duration of storage. This cost varies from 1.5 per cent to 4 per cent and may be taken into consideration while computing inventory-related costs.

Handling and Transportation Cost

The inventory procured needs to be transported at the place, where it is to be stored and then distributed. It has to be loaded and unloaded multiple times. All this adds to the cost.

Insurance Cost

To minimize the risk of holding the inventory, firm insures it for various reasons to recover its total or partial loss. Hence, firm may insure it for a desired period.

Damage, Pilferage and Obsolesce Cost

The material stored carries the risk of damage, shrinkage and loss of weight. The product also carries risk of pilferage or obsolescence due to technology change or availability of substitute. The percentage varies from 0.5 per cent to 2 per cent depending on the product.

Exchange Rate Fluctuations

In case of imported inventories, the valuation is done based on the current currency exchange rates in the market. Any fluctuation may increase or decrease the value of the inventory. Due to the fluctuations in the exchange rate, there is a risk of selling the material at prices lower than the landed cost.

INVENTORY AND CUSTOMER SERVICE

The inventory availability is very critical for enhancing the customer satisfaction. For the FMCG or daily consumed products, the brand loyalty of the customer very much depends on the product availability. For example, a brand loyal customer to Colgate toothpaste will opt for any other brand available if the Colgate is not available with the retailer as he cannot wait for brand for a mouthwash even for a short period of time. Hence, order fulfilment and customer satisfaction, thereafter, are solely dependent on the availability of inventory at the demand point. At the plant level, the stock-outs of raw material and parts will lead to production shut downs and ultimately the stock-outs in the finished inventory. Hence, it is prudent to keep adequate or more inventory than what is required to avoid the stock-out situations. However, the excess or more inventories will push the inventory-related cost upwards. The excess inventory will erode the profitability in the business. Hence, it is always beneficial for the company to balance inventory-related cost and the cost of customer satisfaction. Maintaining the trade-off between the two is always a major issue at the forefront of the supply chain management.

APPROACHES TO MANAGING INVENTORY

Inventory management has a critical role in firm's value chain. The firm may opt for either push or pull inventory approach through its distribution channel. Alternatively, a firm may go for planning approach. This approach prompts the firm to schedule inventory movement through the channel-based forecasted demand and product availability. A third approach is responding to product and market environments.

The controls on the inventory are based on the inventory policies of the company. In order to implement the desired inventory management policies, control procedures need to be devised. These procedures may cover review period, review frequency, reorder point, safety stock level, etc.

Box 12.2

Dell's Lean Supply Chain

In the computer industry, Dell carries fewer inventories than its competitors. Dell keeps inventory of less than 10 day's worth as compared to 80–100 days by competition. As Dell has an alliance with Intel, any new chip introduced by Intel allows Dell to go to market with a PC faster than the competition. Dell maintains very less inventory. The transportation company simply picks up the appropriate number of computers from Dell's Austin plant and monitors from Sony's factory in Mexico, matches them by customer order and delivers them to the customers. This strategy allows Dell to save time and money associated with the extra handling of monitors and ensures that defects are not introduced into large quantity of production. If there is a problem at the customer end, production is stopped and flaws are fixed in real-time resulting into minimum defective finished products.

With the pull approach, the firm responds to a channel member's inventory needs by moving the product through the distribution channel. Replenishment process is initiated when available stock levels fall below a predetermined minimum or order point. In pull approach, the firm waits for customer demand to pull the product through the system. In this retailers individually make a choice regarding when and how much to order from the distribution centre. Then, the distribution centre orders independently from its suppliers (manufacturers). These actions initiated from various demand points result in uncertainty at each reorder point throughout the distribution channel. To support uncertainty, adequate safety stocks are required to get the desired performance.

The other way of doing it is by going for advance planning where a common information base is used to coordinate inventory requirement at different demand points. The planning methods used are fair share allocation and distribution requirements planning (DRP).

In the fair share allocation method, inventory is allocated from mother warehouse depending on the indent received from distribution centres. In this method, inventory levels across multiple sites are coordinated. However, it does not consider factors like performance cycle time, economic order quantity, or safety stock requirements at each of the demand points. Hence, fair share allocation method has limited application. Through this method the problem of managing multistage inventories cannot be properly addressed.

The other way of managing multistage inventories is using distribution requirements planning (DRP) approach that considers multiple distribution stages and the characteristics of each stage. DRP operates in an independent environment where customer demand at various distribution points/locations determines inventory requirements. DRP takes care of the finished goods which are received in the plant warehouse. DRP coordinates inventory levels, plans inventory movement, and reschedules inventory between levels. DRP schedules for each SKU and distribution facility.

DRP (software) is an integrated inventory planning system. It offers a number of benefits such as improved service levels that increase on-time deliveries and decrease customer complaints, reduces distribution centre freight costs resulting from coordinated shipments, reduces inventory levels, reduces warehouse space requirements, reduces customer freight, and improves inventory visibility and coordination between logistics and manufacturing. DRP can effectively simulate inventory and transportation requirements under multiple planning scenarios.

For managing inventory and getting improvements in cost reduction and customer service, the initiatives such as policy definition, performance measures, integration of information, training and use of expert system are needed on the part of the firm. Policies and procedures related to inventory management need to be defined to guide inventory-related decisions at various management levels of the organization. The clear and consistent performance measures for inventory management process are key to its success. These measures must reflect the trade off between service and inventory level. Inventory management performance measures must clearly reflect the trade-offs desired by the enterprise, and the reward structure for individuals who maintain the system must be defined to reflect requirements. It should clearly define the goals and understanding of how goal achievement affects personnel performance evaluations.

Inventory management being a very complex discipline, it is important on the part of concerned employees to understand the nature and dynamics of relations between inventory management and other entities within the value chain. Hence, the importance of sophisticated training to employees to improve inventory management decision-making cannot be denied.

Effectiveness of inventory-related decisions and performance thereof can be substantially increased by integrating information requirements across the enterprise and amongst channel partners. Later, IT and communication technology can be used which enable the above.

The latest addition to inventory controls tools is use of expert systems to enhance inventory performance. The shared expertise/experience can supplement the training process. Expert system provides insight into the review period, inventory management logic, and strategies to employ with each product market group. It helps in improvements in productivity and inventory performance levels.

Inventory can be effectively managed through proper policy guidelines and its implementations. Inventory policy formulation is an exercise the management has to do to put down guidelines towards inventory procurement and storage to offer the desired custom service level with pre-decided inventory level.

RISK POOLING IN INVENTORY MANAGEMENT

Risk pooling is minimizing the risk in inventory overstocks or under stocks by way of taking appropriate inventory stocking and distribution policy. For a large company, there will be a network of field warehouses serving different markets. Alternatively, the firm may have centrally located warehouse to cover large geographical area. The centralized system can aggregate the demand and take care of the fluctuations in various markets by averaging out the variations. This is based on the fact that the decentralized facilities require inventories at each location and each manager would decide on replenishment on the basis of the inventory position of each individual facility. The average inventory level at each location is the safety stock. This safety stock represents idle capital for the firms. A logical approach to minimize the level of inventory throughout the system without sacrificing the quality of service is the centralization of inventory. The concept of risk pooling is based on the firm's inventory availability at fewer locations thereby reducing total inventory without reducing the service level. This will result in aggregating the total demand as high demand in one location will be offset by low demand in another location. This concept of risk pooling has been taken from square root theory of inventory.

Major challenge in supply chain management (SCM) is matching supply with uncertain demand. Risk pooling is an efficient and promising strategy to meet this challenge by reducing the underlying demand uncertainty through aggregation. With risk pooling, the demand variability is reduced if demands across the demand points in the markets are aggregated. The reduction in variability helps the management to manage the inventory in better ways and reduce the safety stocks to address the uncertainty. For managing the inventory guidelines for inventory procurement, maintenance, positioning and placement with respect to quantity, time and customer service are needed to be evolved. The policy decisions also concerned with managing the inventory independently or interdependently across the various distribution centres. In essence, the inventory-related policy helps in improving the inventory effectiveness across the supply chain.

Centralized/Decentralized System

In the centralized system, the inventories are stocked at one central location for distribution to the customers or depots. The centralized system helps to have tight control on the inventory movement both in replenishment and dispatches. In case the volume of transaction grows, the greatest disadvantage of centralized system is longer order performance cycle and delayed response time for the smaller but larger number of customers. On the other hand, the decentralized systems are closer to the customer and thus decisions are taken independently. The decentralized system offers better customer service with quick response and shorter performance cycle time. However, the controls are lax and the operating cost is more. The management has to make a policy decision to strike a balance between the customer service, controls and the cost.

Service Level

The service level target is decided by the management. The inventory function should gear up to achieve the service level target. The basis of service level is defined in terms of the following:

- Order cycle time
- Case fill rate
- Order fill rate

The order cycle time is the time required for completing from the date of release of an order by the customer till he receives the entire ordered material at his door step. The case fill rate means the percentage of material available for dispatch to the quantity ordered by the customer. For example, if a customer orders 10 items and only 9 are available for supply, then the case fill rate is 90 per cent. The order fill rate is percentage of orders that could be filled completely. Out of 100 orders received only 60 could be filled completely then the order fill rate is 60 per cent. The traditional approach to meet the targeted customer service level is to increase the inventory level. However, the modern approach considers the faster transportation mode, better communication means, and the alternative source of supply to meet the desired customer level.

Stock Levels

To meet the desired level of customer service, the average level of inventory stocks to be maintained at any point of time is determined in terms of number of days of sales value. The average inventory includes the base level inventory, safety stock inventory and in-transit inventory.

The stocks deplete every day after the orders are filled (Figure 12.2). To maintain the average level as decided earlier, the inventory has to be ordered and replenished before it further depletes to zero. The reorder level is called a base level inventory.

The other part of the average inventory is stock maintained to meet uncertainties or emergencies. The safety stocks are maintained to meet the inventory demand, which is in excess of the anticipated consumption. In fact, safety stock inventory covers the short-range variations in the demands. The average inventory includes the one half of ordered quantity and the safety stock inventory.

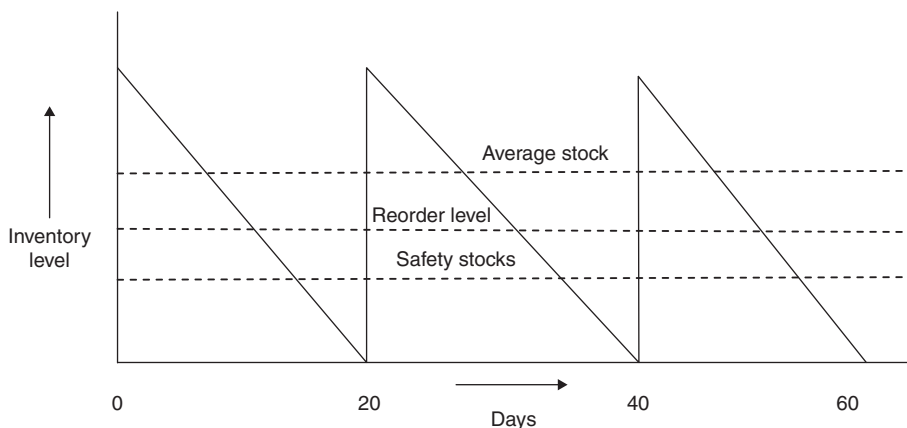


Figure 12.2 *Inventory Performance Cycle*

The in-transit or pipe inventory is of importance to consigner as the ownership of this inventory still lies with him till it reaches the destination and handed over to the consignee. There is great uncertainty associated with the transit inventory. This inventory is not accessible and cannot be used. The whereabouts of the transit inventory is not known unless there is a sophisticated vehicle-tracking system. This is an excise-paid inventory and it covers major portion of the value of the inventory of the company. In JIT system, the emphasis is on small consignment with frequent delivery schedules, resulting into the maximum portion of the inventory in transit phase.

A risk pool is one of the forms of risk management mostly practiced by insurance companies. Under this system, insurance companies come together to form a pool, which can provide protection to insurance companies against catastrophe risks such as floods, earthquakes, etc.

Risk pooling is an important concept and has an application in supply chain management. Risk pooling suggests that demand variability is reduced if one aggregates demand across locations because, as we aggregate demand across different locations, it becomes more likely that high demand from one customer will be offset by low demand from another. This reduction in variability allows a decrease in safety stock and therefore reduces average inventory. There are two primary policy decision roots in exercising controls on inventory level. These are centralization and decentralization. The companies adopt one or the other depending on their business philosophy. However, Mr. D. H. Maiser advocated risk pulling for centralization as against maintaining the inventory in many warehouses, which will cost more. He has proven this with the mathematical calculations and his law is known as the 'square root law of inventory' (SRL). This law is the basis of the concepts of risk pooling and inventory centralization. Square root law stipulates:

Total safety stock inventories in a future number of facilities can be approximated by multiplying the total amount of inventory at existing facilities by the square root of the number of future facilities divided by the number of existing facilities.

As per the above statement, the increase in number of warehouses of a company will result in an increase in certain costs. Thus, with lesser number of warehouses, the company will enhance its savings of certain costs. This increase, either in a positive or negative direction, can be mathematically computed using the square root law indicated below.

$$X_2 = (X_1)\sqrt{(n_2)/n_1}$$

Where;

- n_1 = number of existing facilities
- n_2 = number of future facilities
- x_1 = total inventory in existing facilities
- x_2 = total inventory in future facilities

For example, if a business firm is having a network of 25 field warehouses for stocking their inventory to serve the markets, then their costs would be likely to be reduced to one fifth if the firm starts operating from a single warehouse. Similarly, if a company originally had 4 warehouses and decided to increase their stocking points to 16, the firm would find the inventory cost doubled.

Table 12.1 indicates how average inventory level increases with the increase in number of stocking points. The inventory level indicated below is directly related with safety inventory.

Square root law is based on the following assumptions:

- Inter locations inventory transfers not a common practice
- Constant lead time

Table 12.1 *Average vs. Stocking Points*

<i>Number of Warehouses</i>	$\sqrt{n}$	<i>Total Average Inventory (Units)</i>	<i>Percent Change</i>
1	1.0000	3,885	—
2	1.4142	5,494	141%
3	1.7321	6,729	173%
4	2.0000	7,770	200%
5	2.2361	8,687	224%
10	3.1326	12,285	316%
15	3.8730	15,047	387%
20	4.4721	17,374	447%
25	5.0000	19,425	500%

Source: John J. Coyle, Edward J. Bardi and C. John Langley, 2001, *The Managements of Business Logistics*, p. 260.

- Customer service level uniform at all locations.
- 'Normal distribution' for all locations demand

Zinn et al. (1989) showed that SRL will be most effective when markets have demands which are negatively correlated and there is little or no benefit from consolidation when demands faced by the various stocking points are positively correlated.

As per the SRL, the multiplicity of warehouses increases the fixed costs in the same proportions. This also supports the fact that managing multiple warehouses becomes complex. This is because a firm has to keep track of the inventory at several locations: Watching stock amounts, handling orders and planning distribution are complicated with multiple stocking points.

One centralized warehouse may not be possible for a firm but certainly it can reduce the number of stocking points. The solution is to have alliance with third party logistics service providers to reduce their own stocking points. SRL cannot be used in isolation. The decisions on number of warehouses considering SRL should be weighed in light of other factors like transportation costs, lead time, inventory availability and warehouse proximity.

In general, inventories at multiple warehouses or stocking points are at decentralized locations. The advantage is to take inventory-related decisions locally based on local needs. They are in a position to anticipate the sudden requirements and can act with speed. They foster a sense of ownership and control that is desirable.

One of the major disadvantages of decentralized inventory control system is that local personnel may not be competent and may not be having inventory management skills. They may operate on a highly subjective basis. They may not be able to take into consideration the holistic view which exists in case of centralized inventory control system. Local management may overreact to events which are transient in nature, even when fairly sophisticated tools are available. In general, local management has a strong bias for high inventories being visually present. This approach may enhance customer service and inventory levels too which may offset the benefits.

Decentralization in business functions allows the people to take decision faster as they are closer to the problem. The speed in decision-making impacts their operations. If local management is having

Box 12.3

RPL—Tracking Hidden Cost of Inventory

RPL a leading computer equipment manufacturer faced a situation wherein the supply chain was not sustainable in spite of it being flexible. The cost driver was excess inventory cost due to a mismatch in demand and supply. The various costs associated with inventory were as follows:

- 1. Component devaluation cost—In the electronic component industry, the prices fall down quickly. The holding cost of the excess inventory with reduced price added up to devaluation cost.*
- 2. Price protection cost—To compete in the market, the dropping of component price after it is shipped to channel member resulted into reimbursement to channel partner for the difference of unit that is not yet sold.*
- 3. Return cost—The unsold goods returned by distributors constituted more than 10 per cent of the total product revenue. This cost is not included in the company's cost of goods sold. Hence, it is an additional burden on cost front.*
- 4. Obsolescence cost—These include the end-of-life product write-offs due to short life cycle of PCs.*
- 5. Holding cost—It is the holding cost of the existing inventory in the stocks.*

After analysis, RPL found that the highest contribution to the cost was due to price protection in a competitive environment. For controlling the first cost component driver as stated above, RPL reduced the number of nodes in supply chain, which were causing additions to inventory and resulting into higher costs. RPL consolidated their manufacturing facilities and resorted to JIT philosophy. Towards reduction in price protection cost, RPL had gone for curtailing channel partners inventory which could never exceed the minimum number of days. They kept their manufacturing turn-around time shorter and replenishment cycle frequent. For returns RPL worked closely with channel members by agreeing with them to keep specific inventory level and delivery expectations. RPL improved its service quality both towards channel members and customers. To control obsolescence, RPL launched new models just as the last remaining units of old models sold out. Thus, lowering inventory-driven cost could not only reduce the total costs and raise revenue but also lower the working capital requirements by reducing the number of days inventory outstanding. The lesson is to develop metrics to track the hidden costs.

technical strength and personnel disciplines to inventory management using a proper software system, then local controls are recommended.

Centralized purchasing and ordering has the added benefit as the company can invest in more highly trained purchasing and inventory management experts. In addition, with central control the management can more effectively study special buy considerations and other economic trade-offs.

SELECTIVE CONTROL TECHNIQUES

A policy is one that sets a boundary for any activity. A procedure, however, defines the precise activities taking place within the boundaries of the policies created. A procedure is usually sufficient to use as a guideline for an employee who needs to understand how a process works. When combined with the proper level of training, the policies and procedures manual is an effective way not only to increase control over inventory, but also to increase the efficiency with which it is handled. There are various

methods of inventory control and few ones which are commonly used in the industry are discussed as follows.

ABC Analysis

In this method, the degree of control varies with importance of the item in the supply chain. The ABC classification is an analysis of arranging of items as per consumption pattern and rupee value attached to it. They are usually classified into A, B and C categories.

- A – Outstanding importance
- B – Of average importance
- C – Relatively unimportant as a basis for a control scheme

These categories are handled in a different way. However, more attention is being given to category A, less to B and still less to C. The categorization is based on Pareto's law or known as '80/20' rule. ABC analysis is used in inventory management as a thumb rule. According to this about 80 per cent of the rupee value (consumption wise) of an inventory remains in about 20 per cent of the items.

In short, ABC analysis is financial evaluation, for ranking and comparison of inventories. The objective of the classification is for knowing which item should receive the most attention. 'A' class item should have perpetual attention while C class item may be reviewed with less periodicity than B class items.

Thus, ABC inventory classification method is used to categorize inventory into groups based upon certain activity characteristics. It is the process of classification of products as per the level of importance in terms of their relative criteria such as purchase or sales volume. Examples of ABC classification also would include:

- ABC by velocity (times sold)
- ABC by sales rupees
- ABC by quantity sold or consumed
- ABC by average inventory investment
- ABC by margin

ABC classifications (Exhibit 12.1) are used to develop inventory planning policies, set frequencies for cycle counting, slot inventory for optimized order picking, and other inventory management activities. It can be described as a technique that is used in a business for categorization of large volume of data into groups. These groups thereafter can be marked as A, B and C. This means that activities that are considered high on priority are classified as A, those with a lesser priority are grouped under B and the groups of activities that are last on the list of priority are classified as C.

ABC analysis is a technique for inventory control. The basic assumption is that different value is attached to different stocks. Hence, degree of management focus will differ with different stock items. ABC analysis of inventory helps the management to exercise a degree of control on a particular item and accordingly it can be monitored.

For example, A class items which are critically important require close monitoring and tight control. A class items account for large value but these are less in numbers or are in a small percentage of the total inventory items. These items are required to be continuously reviewed. B class items are less critical as compared to A class items and, hence, require standard controls and periodic reviews of usage. C class items carry very less value and are least critical. Hence, these items require the least

Exhibit 12.1 *Inventory Classification Matrix by Characteristics and Actions*

Type	Class A Items	Class B Items	Class C Items
Characteristics	High consumption value and low unit counts	Moderate consumption value and moderate unit counts	Low consumption value and high unit counts
Actions	• Tight control • Continuous follow-up • No safety stocks • Accurate forecast • Central procurement • Phased delivery • Responsibility of senior executives managers	• Moderate controls • Moderate follow up • Low safety stocks • Reasonable forecast • Decentralized purchases • Phased delivery • Responsibility of middle managers	• Loose control • Periodic/no follow up • High safety stocks • Bulk purchases • One time delivery • Local purchases • Routine purchase by lower level executives

controls. These items can be kept in large stocks also. ABC analysis forms the basis for inventory control processes to decide priorities for review, reorder and maintaining the safety for controlling the inventory stocks and usage.

VED Analysis

This is related to vital, essential and desirable (VED) status of the inventory items. As the term implies, certain parts and items are considered to be vital for meeting the operational requirements and this aspect is taken into consideration while making the forecast. The modified version of this is ABC-VED (Figure 12.3) analysis which takes into consideration both value and the criticality of the item. The high value and critical items are under continuous review and ordered in low quantities while low value least critical items are periodically reviewed and ordered in large quantities with lower safety stock requirements. The examples are car engine which is vital, followed by seats, which are essential and tail light which is desirable item in car assembly.

SAP Analysis

Scarce, available and plenty (SAP) allows to build into provision forecast. The ordered quantity is governed by the scarcity factor. The limitations in supply or obsolescence of the item in near future will be the guidelines for procurement policy decision.

FSN analysis

Fast, slow or normal (FSN) determines the consumption pattern of the item. However, a consumption pattern where the production run is slowed down for various reasons may not give a realistic picture for procurement action.

In all above control techniques the degree of control varies with the importance of items. For example, for the A class, vital, scarce and fast-moving items, the perpetual reviews are recommended; while for B class, essential, medium-moving items, the periodic reviews will be alright. In the case of C class, desirable, slow-moving items, the periodicity of review will be longer (Figure 12.3).

		Criticality of item		
		High		Low
		V	E	D
Consumption value	High	A Continuous review Procure as needed High safety stock Low order quantity		Continuous review Procure as per need Low safety stock Low order quantity
		B		
	Low	C Period review High safety stocks High order quantity		Periodic review Low safety stocks High order quantity

Figure 12.3 Inventory Decision Matrix

BULLWHIP EFFECT

The variability in customer demand if not conveyed properly, or if conveyed with distortions as it travels upstream in the supply chain, will lead to over-stocks or a stock-out situation. This phenomenon is called 'bullwhip effect'. The bullwhip effect is a deformation in information when it travels from the market to the vendors through the company's distribution network, manufacturing and procurement department upstream in the supply chain. The effect of demand volatility ripples up the supply chain, resulting into inventory problems. This finally affects profitability of the firm or its service to the customer. This happens due to lack of smooth and speedy information flow resulting into improper coordination and synchronization of actions of supply chain partners. In this process, the demand of the customer is distorted each time it goes from a one decision point to another. Problems are severe in supply chains where communication is minimal between supply nodes. In short, the bullwhip phenomenon is the amplification of demand variability from a downstream site to an upstream site resulting into inventory pile up or stock out. In essence, it is due to mismatch in demand and supply.

In 1958, Forrester identified that the action, reaction and interaction of the flows of information, inventory and cash amongst the self-interested parties in the supply chain would lead to delays, oscillation, and amplification in the flows of information and physical goods through the channel. As per Sterman (1989), the decision-makers commit errors in perceiving the dynamic environments that they were involved in. The Bullwhip effect is the result of their decision which is based on the misperceptions of the environment they live in. The bullwhip effect is a phenomenon observed in supply chain mostly because of incorrect assessment of the demand at various decision points in the supply chain. Moving up the supply chain from end-consumer to raw materials supplier, each supply chain participant observes variation in demand and tries to keep safety stock as per his own calculated requirements. His requirement may not be similar to other members of the SC partners. Hence, independent decisions taken by the supply chain partners at various points of time without having integration

amongst them lead to un-uniform decision-making causing the bullwhip effect. In each decision-making point in supply chain profit maximization is tried through forecasting, this as a result will have an effect on the overall profitability of the supply chain. The bullwhip effect is a problem in forecast-driven supply chains.

The severity of bullwhip effect increases as one moves closer to the source of production. The effect is costly because it causes excessive inventories, unsatisfactory customer service and uncertain production planning. There are two broad categories of causes for bullwhip effect such as operational causes and behavioural causes.

Operational causes cover errors in demand processing, inventory rationing, order batching and price variations. Behavioural causes lead to time delays in making ordering decisions and orders placed but not yet received at the source. Another behavioural cause based on coordination risk is uncertainty about the actions of other decision-makers which often triggers instability. There are few key reasons for the occurrence of the bullwhip effect. These are as follows:

- Frequent changes in demand forecasts
- Lack of coordination amongst SC partners
- Lead-time variability
- Order batching
- Price fluctuations
- Rationing
- Errors in demand forecasting

The updating of demand forecasts appears to be a major source of the bullwhip effect. The parties of the supply chain build their forecasts on the historical demand patterns of their customers. The retailers' build on the actual demand patterns of the customers, the other parties adjust to fluctuations in the ordering policies of those preceding them in the supply chain.

As everybody react to fluctuations with smoothing techniques, the fluctuations in upstream SC get amplified because the differentials in decisions at the various dependent decision-making points. Thus, the safety stocks as smoothing devices at various points amplify the bullwhip effect and which at the end erodes the competitive positioning of an organization.

Bullwhip Effect (BWE) is an internal (firm level) response to the environmental phenomena. It actually reflects when partners operate in isolation and optimize their individual objective functions. The most effective process for countering the oscillations of the 'bullwhip effect' will be customers and suppliers collaboratively working to improve information quality and compressing cycle times throughout the entire supply chain.

The methods intended to reduce uncertainty, variability and lead-time are vendor-managed inventory (VMI), just-in-time (JIT) replenishment and strategic partnership. Following are the few measures to counter bullwhip effect:

1. Timely information availability
2. Integration of all decision-making nodes
3. Use of IT for managing information analysis and flow
4. Reduction in demand variance upstream in supply chain
5. Reduction in the order size flexibility and implementing capacity
6. Minimum fluctuation in prices
7. Inventory visibility across the SC

It a nutshell, sharing of useful and undistorted information is essential at all levels to counter bullwhip effect.

MODERN INVENTORY CONTROL TOOLS

The following features are quite prominent in modern² inventory control tools such as flow of inventory, flexibility towards the market demand, balancing inventory level to customer satisfaction and opportunity loss. Excess inventory will increase the carrying cost. The latest inventory models address both the above problems to maintain the balance between two polemic goals and integration to bring efficiencies and effectiveness. Most of the inventory control tools which are available in the market are IT based using the computer software. The few ones commonly used are described in the following text.

Material Requirement Planning (MRP)

MRP concept was developed in the 1970s after the high-speed computers were introduced. This system does the work of materials manager to control the inventory of the items to lean the supply chain. The forecast of the inventory items is controlled by production item on which their demand is dependent. Application of MRP is typically done to the inbound material movement in the enterprise and is based on the production requirements and scheduling. The MRP system (Figure 12.4) is suitable for both the push and the pull types of supply chain systems. In the push system, information is required in the form of elaborate material requirement planning for master production schedule and roll back, creating schedules for suppliers with inventory part/component types, quantities and delivery dates.

In the pull system, information is required on actual demand, which needs to be transmitted extremely quickly throughout the entire supply chain so that production and distribution of parts or products can accurately reflect the real demand.

Manufacturing Resource Planning—MRP II

MRP II is defined by advancing productivity, innovation and competitive success (APICS)—the association for operations management, as a method for effective planning of all resources of a manufacturing company. It started out as material requirements planning and was concerned with ordering and scheduling materials based on the inventory. The large-scale introduction of computers led to extensions into a ‘closed loop MRPII’ system concerned with the whole manufacturing environment. MRPII works on ‘push’ where a forecast is generated, a manufacturing plan is derived and the plan then drives the manufacturing process via work orders or the purchasing process via purchase orders.

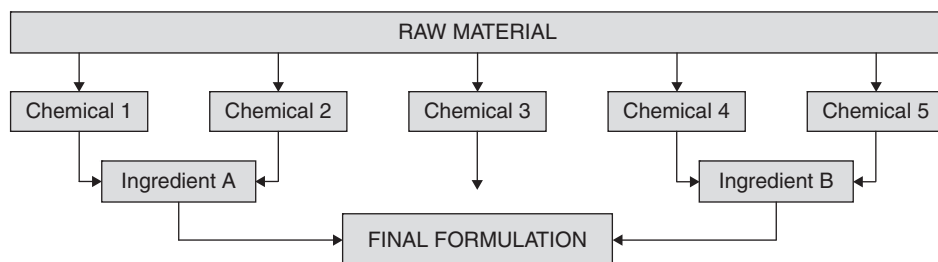


Figure 12.4 Framework of MRP System

MRPII systems begin with MRP (material requirements planning). Input to MRP is the sales forecast (from marketing), which determines the raw materials and parts requirements. Both MRP and MRPII systems work on master production schedule which is further broken down into specific plans for each product on a production line. MRP helps in coordination of raw materials purchasing. However, MRPII helps in production scheduling taking into consideration machine and labour capacity, scheduling the production runs according to the arrival of materials. MRPII system is quite elaborate and provides data on production cost, machine time, labour time and materials consumed and final output to accounting and finance. MRP primarily deals with raw materials required for manufacturing, while MRPII is concerned with the coordination of the entire manufacturing production raw material parts, finance and man power.

In short, MRPII is a long-term planning tool for complex products. It can give accurate completion date at the time of order. It fits in with conventional accounting and the progress of manufacture inventory size is available at all times. It tightly controls work orders and changes therein. However, MRPII is necessary to maintain accurate data base. Inventory accuracy requirement is vital to an extent of 99 per cent. It is a computer-based system, inflexible and relies on forecast.

Distribution Requirement Planning (DRP)

DRP is one of the latest IT tools for controlling the inventory in the distribution system of an organization. It is the logical extension of MRP. DRP (Figure 12.5) is guided by customer demand as against the MRP which is based on production requirement. It allocates the inventory from mother warehouse to the various distribution centres based on the following:

- Demand pattern
- Safety stock provision
- Order quantity
- Reorder point
- Average performance cycle length

DRP helps in coordinating the finished goods requirements across the distribution network. The DRP system success is dependent on the accuracy of the forecast with respect to location and time

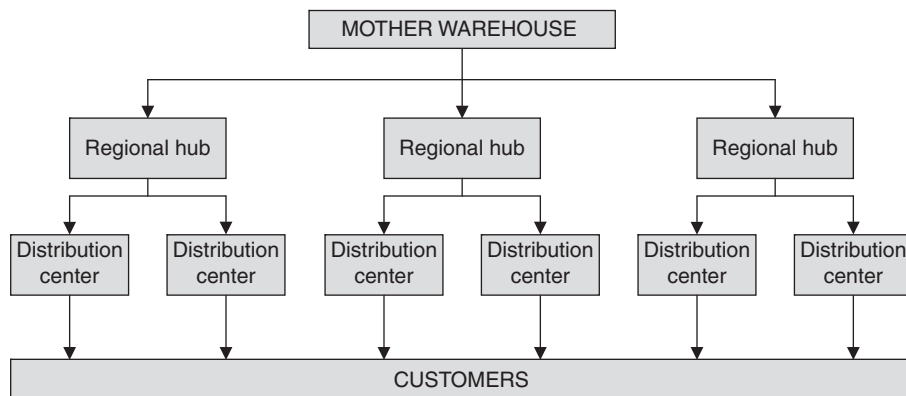


Figure 12.5 Framework of DRP System

of the requirement across the distribution centres. The consistency in performance cycle for timely movement of the goods across the system is a must for system effectiveness. The uncertainty in performance cycles may defeat the very purpose of DRP system. Just like MRP, DRP is used along with ERP (Enterprise Resource Planning) rather than a standalone system, to get full benefits of the ERP on both procurement and distribution side. The major benefits of using DRP are improvements in customer service level, effective marketing efforts for high stock items, decrease in inventory levels resulting in carrying costs and decrease in inventory thereby reducing warehouse space requirements. DRP helps in effective simulation of transportation and inventory requirements for shipment coordination to result into reduction in inventory carrying and transportation costs.

Just-in-Time (JIT) System

JIT is a concept based on the fact that an activity should not take place until there is need for it. Hence, the inventory item should not be brought into the system until it is required for making the final product. It is characterized by maintaining zero inventories of the raw materials and assemblies at the assembly plant. For this, JIT system involves the close coordination of the buyer and the supplier on real-time basis: this means frequent receipts of the materials from the suppliers. Following are the prerequisites of successful JIT system:

- Buyer-seller partnership
- Online communication and information sharing
- Commitment to zero defects from both the sides
- Frequent and small lot size shipments

The success of JIT system works on mutual trust and commitments from the both seller and purchaser side. The main barriers to the successful operation of JIT system are:

- Organization structure
- Organization culture
- Technology differentials at buyer and supplier ends
- Reluctance to information sharing
- Dispersed suppliers

JIT system has its success stories in Japan wherefrom it originated. The reason is the culture of the team. JIT could not bring benefits in American companies because of the individualistic decision-making culture prevailing there. The success of JIT technique depends on collaborative relationship and decisions reached by group consensus rather than individuals empowered in the organization based on their authority structure. To make JIT a success, a total cultural and structural change is needed in both buyer-seller organizations and they should operate from a mutually beneficial policy platform. Refer to Chapter 6 for more details on JIT.

Vendor-Managed Inventory

Vendor-managed inventory (VMI) was made popular by Bose Corporation. It is widely used in the industry and is proving results. In VMI, the supplier takes charge of the inventory management of the product and manages the replenishment process based on the consumption pattern of the customer.

They use EDI or other inter-organizational software packages or place supplier's representative at the customer's place.

In India, a leading commercial vehicle manufacturer allowed the tyre supplier to open up a small inventory shop in his manufacturing premises to supply tyres for the day's production of the commercial vehicles. The supplier keeps 3 days inventory in the area allocated to him and keeps his representative to manage the shop and hand over tyres to customers as per the assembly line requirements during the day. The inventory replenishing is done once in 2 days by direct truck-loads of tyres from the supplier's manufacturing plant. This is based on an advance production planning for the coming week. The customer pays daily by cheque to the supplier for the quantity of tyres drawn from the supplier's shop. The result is reduction in inventory liability of 15 days on vehicle manufacturer to the inventory level of 3 days, which is entirely managed by the vendor who gets the payment daily for the supplies made. This is a win-win situation in which both the buyer and the supplier benefit.

For VMI to be successful, three things are essential: right partner, right set of products, and mutual trust. The right partners imply those who proved themselves on consistency in quality, low cost transactions and reliability as far as on-time delivery is concerned. The benefits of VMI are reduction in inventory-related costs and risk of stock outs and for the vendor speedy payment realization, reduction in transaction cost and attaining the assured business goals. In Chapter 6, VMI is discussed in details.

Kanban

It is basically an information system to support JIT inventory for manufacturing operations. It signals supply of material when used. *Kanban* (Figure 12.6), meaning signboard or label, is used as a communication tool in the inventory system. A *Kanban* is attached to the each box of parts as they go to the assembly lines. Because these parts are funnelled to the line as needed, the *Kanban* can be returned as parts are used to serve as both record of work done and as order for new lot. *Kanban* coordinates the inflow of parts and components to the assembly line, minimizing the replenishment processing. *Kanban* is used in process logistics for movement of parts and components on the shop floor of the manufacturing plant.

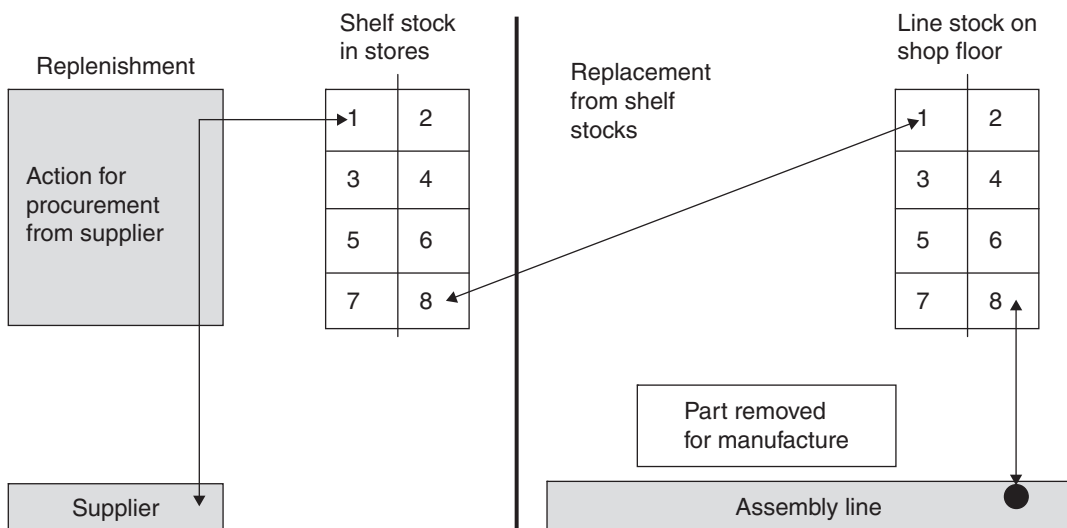


Figure 12.6 The Kanban System

The system was first evolved at TOTOYA in Japan to support the concept and practice of JIT in assembly line. The philosophy behind is 'use one buy one'. For more discussion on KANBAN refer to Chapter 6.

Automated Inventory Tracking System (AITS)

This system is adopted by the leading retail chain giant like Wal-Mart in the United States to track the inventory based on the sales generated. This system involves usage of the electronic data interchange (EDI) system to track the flow of inventory through the supply chain. The system connects suppliers, manufacturers, distribution hubs and retail stores and the institutional buyers. The system is based on IT integrating EDI and bar coding system to track inventory for controlling inventory investment through the supply chain.

SUMMARY

Inventory costs are of great concern to the management. The inventory requires investments and hence excess inventories will tend to lower the profit margins. The inventory shortage will close down the factory or lead to lose of market opportunities and customers. Hence, managing the inventories is a very important role for the management. Inventories are broadly classified as raw materials and components, work-in-progress, finished goods and operational support inventories. Inventories are maintained for meeting production requirements, supporting the operations, extending customer service and hedge against future uncertainties. Inventories are reflected as current assets in the balance sheet of the firm but from the point of view of the supply chain, inventories are a liability. The investment in inventory has great implications on the profitability of the firm. Inventory requires working capital which is available at cost. The inventory carrying cost is the sum total of all the costs such as cost of the funds invested in inventory, inventory handling cost, inventory storage cost, damage and obsolescence cost, opportunity lost cost, etc. In general, it trails between 15–20 per cent of the funds blocked in the inventory. For managing inventories, firms are using selective control techniques like EOQ, ABC, CED, FMS and inventory control models like MRP, DRP, JIT, VMI and AITS. For effective control on inventories across the supply chain, firms need to frame policy guidelines for inventory procurement, maintenance, positioning and placement with regards to quantity, time and customer service. Risk-pulling techniques have to be deployed for managing the inventory independently or inter-dependently across the various distribution networks. In a nutshell, it is prudent to hold inventory at a certain level, as the benefits derived exceed the inventory carrying cost.

REVIEW QUESTIONS

1. Explain the role of inventory in the supply chain system.
2. Discuss the phenomenon of 'bullwhip effect' with its causes and strategies to mitigate it.
3. How can inventory be used as a measurement system for performance evaluation of the supply chain?
4. Discuss the various modern techniques for inventory control with their merits and demerits.
5. What is 'Risk Pulling' and how is it applied in inventory control?
6. Discuss the various methods of inventory classification and the policies derived thereof to control the inventory levels.

INTERNET EXERCISES

1. Visit <http://www.effectiveinventory.com/> to study inventory management practices adopted by different companies.
2. For further reading on latest inventory management visit: http://www.iimm.org/knowledge_bank/6_purpose-of-inventory-management.htm
3. <http://www.inventoryops.com/articles.htm> is source for information on inventory management and warehouse operations. Visit for studying articles on the latest inventory management.

VIDEO LINKS

1. Inventory Management—An Introduction, <http://www.logisticsbureau.com.au/video-supply-chain-logistics.htm>
2. Modernizing Inventory Management, <http://www.youtube.com/watch?v=1d0O8MAMyAM>
3. Inventory Management Made Easy http://www.metacafe.com/watch/921652/microsoft_inventory_management_made_easy/

PROJECT ASSIGNMENT

1. Study the how inventory is controlled in a hospital or Hotel and suggest the measures to control the inventory to minimize the cost without compromising the service level.

NOTES

1. Jon Schreibfeder, 'A Realistic Approach to Inventory Reduction,' <http://www.effectiveinventory.com/article74.html>
2. Sople, V. V. (2010), *Logistics Management*, Pearson, New Delhi, 2nd edition, pp. 112.

NETWORK DESIGNING

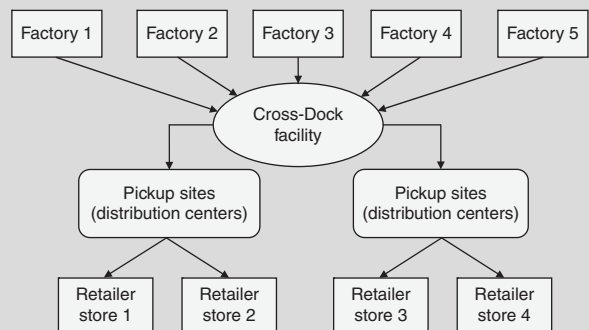
“As the trend to outsourcing continues to grow, so too does the complexity of the supply/demand network

—Anne Mulcahy¹”

Chapter Objectives

To understand:

- Role of network design in supply chain
- Factors influencing network design decision
- A framework of network design decisions
- Models for facility location and capacity utilization
- Network designs in practice
- Network design in supply chain mergers



Supply Chain Network

Network design decisions in the supply chain are very critical as it has great implications on the service (speed and quality) to the customer. Designing a supply chain network provides the basic structure for supply chain operations, where the network is a major element in a firm's competitiveness and a significant area of capital investment. To compete and succeed in today's demanding marketplace, companies must create and maintain the most efficient supply chain possible. Designing supply chain network involves numerous decisions like manufacturing, site locations, product mix, transportation, inventory deployment, suppliers, cost-service levels, and distribution in supply chain operations. And, each of these decisions will impact the firm's services, profitability and business potential. To ensure an optimal design, it is critical that key business data from information systems and intangible business insights are included.

Camy Petroleum Ltd: Network Design

Camy Petroleum Ltd (CPL) is the leading automotive lubricant manufacturing company in the country. CPL markets lubricants in 200 SKUs in different pack sizes. In the present setup, CPL hired warehouses (Figure 13.1) in every state to supply material to distributors. These state warehouses were fed from the mother warehouse, which is located in Mumbai. CPL appointed 5–6 distributors in each state. The average primary logistics cost for transporting material from Mumbai to state warehouses is Rs. 5 per litre. And the secondary logistics cost for transporting material from state warehouses to distributors is averaging to Rs. 3 per litre. The average sale of each distributor is 6 kl per month. In order to keep primary logistics cost lowest, CPL delivers material in 16 kl truck to state warehouses.

Chhattisgarh having low demand of lubricants, CPL appointed only one distributor, who gets the material from the state warehouse located at Raipur. CPL being cost conscious, wants minimum inventory at state warehouses and also availability of all SKUs at distributors to avoid sales loss. In Raipur, the average off-take of lubricants is 6 kl per month. If CPL dispatches 16 kl vehicle to Raipur warehouse, the inventory carrying cost is very high and also CPL finds it difficult to dispatch any particular SKU if it is in out of stock situation, resulting into sales loss. Alternatively, if smaller vehicle is used for Raipur, the logistics cost averages out to Rs. 15 per litre, which is a loss proposition. In both the above said cases, CPL was in a dilemma. But to grab the sales it was necessary to feed the state warehouse from mother warehouse as and when material is required. In experimentation, CPL dispatched the material to Raipur in 9 kl vehicle, but this could not meet the demand of certain SKUs in smaller quantities. There were both cost overruns and sales loss. Finally, CPL decided to route Raipur warehouse requirements through Nagpur warehouse (Figure 13.2), which was hired for this purpose only. The material is transported through 3 kl vehicle from Nagpur to Raipur. To make Nagpur warehouse viable three dealers were attached to it. The new arrangement made both the warehouses (Nagpur and Raipur) viable, keeping inventory carrying cost and logistics cost under control. CPL could now feed any SKU in small quantity to Raipur warehouse on demand by Chhattisgarh distributor.

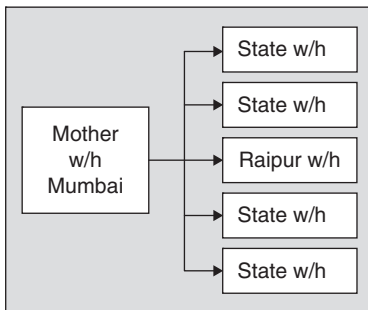


Figure 13.1 Existing Network

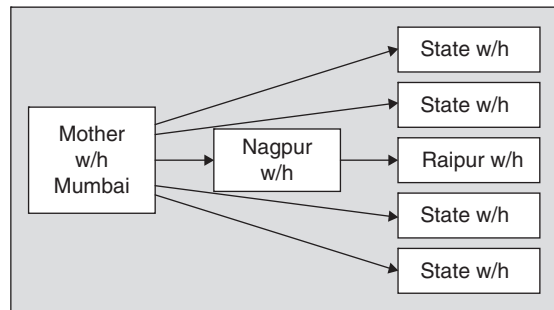


Figure 13.2 Modified Network

INTRODUCTION

Supply Chain Management (SCM) is the process of planning, implementing and controlling the operations of the supply chain efficiently. SCM spans all movements and storage of raw materials, work-in-process inventory, and finished goods from the point-of-origin to the point-of-consumption. Part of the planning processes in SCM aim at finding the best possible supply chain configuration so that all operations can be performed in an efficient way. This covers integrating facility location with other important functions of the supply chain such as procurement, production, inventory, distribution and routing.

In supply chain planning is done at three levels depending on the time horizon: strategic, tactical and operational. The strategic level deals with decisions that have a long-lasting effect on the firm. These include decisions regarding the number, location and capacities of warehouses and manufacturing plants, or the flow of material through the logistics network.

The terms 'network design' and 'supply chain network design' (SCND) are often considered synonymous of strategic supply chain planning. Although typically no location decisions are made on the tactical or even operational level, a number of issues are strongly related to them such as inventory control policies, the choice of transportation modes and capacities, warehouse layout and management, and vehicle routing. In today's competitive market, a company's distribution network must meet service goals at the lowest possible cost. In some instances, a company may be able to save millions of rupees in logistics costs and simultaneously improve service levels by redesigning its distribution network. To achieve this, an ideal network must have the optimum number, size, and location of warehouses to support the inventory replenishment activities of its distributor/retailers. This requires proper facility location models to determine the best supply chain configuration which underlines the interrelation between the strategic and the tactical/operational planning levels.

ROLE OF NETWORK DESIGN

Designing a supply chain network provides the basic structure for supply chain operations. The network is a major element in a firm's competitiveness and a significant area of capital investment. Network design focuses on the process, location and capacity aspects of a supply chain. This covers location of production and warehousing facilities to best meet customer demand at the right locations. Not only determining of location is included in network design analysis, but also the replenishment linkages from a facility to its downstream supply chain partners are covered.

SCND normally utilizes optimization technology, which develops least cost network efficiently. An optimization model can be constrained by capacity limitations on facilities, and also by limiting the length/time of replenishment sources. This addresses time-based aspects of a supply chain based on distance between the point of demand and the point of supply. However, this type of constraint is a representation of lead time and many other factors beyond just distance, which contribute to the ability of facility to fulfil the product/service demand.

The optimization technology on network design solution is capable of effectively selecting a suitable supply chain structure from amongst thousands of options. The macro nature of this analysis is best suited to work on structural and location problems.

The role of facility decisions in a supply chain includes the location of manufacturing, storage, or transportation-related facilities and the allocation of capacity and roles to each facility. Facility decisions are classified as indicated in Exhibit 13.1.

Exhibit 13.1 *Classification of Facility Decisions*

Facility role	Processes performed by each facility
Facility location	Facilities location sites (region/country/FTZ)
Capacity allocation	Capacity allocated to each facility
Market allocation	Markets each facility serve

All network designs are interrelated; any decisions made will affect each other. Therefore a decision must be made taking this fact into consideration. Decisions regarding the role of each facility are of significant importance because they determine the amount of flexibility the supply chain has in meeting the demand.

Facility location decisions have a long-term impact on a supply chain's performance. It is very expensive to shut down a facility or move it to a different location. Hence, organizations must live with their location decisions for a long time (decades). A good location decision can help a supply chain be responsive while keeping its costs low. Toyota, built its assembly plant in the United States in 1988. This plant proved very profitable for Toyota when the yen strengthened and cars produced in Japan were too expensive to be cost competitive with cars produced in the United States. The plant allowed Toyota to be responsive to the U.S. market while keeping costs low.

A poorly located facility makes it very difficult for a supply chain to be efficient. It is very difficult to be responsive and cost-effective too. Capacity allocation decisions also have a significant impact on supply chain performance. Capacity allocation to the facilities can be altered more easily than location. But, capacity decisions do tend to stay in place for several years. Allocation to a large capacity to a plant for location markets with less demand, results in poor capacity utilization and hence higher costs. Alternatively, too little capacity results in poor responsiveness if demand is not satisfied.

The allocation of supply sources and markets to facilities has a significant impact on performance because it affects total production, inventory, and transportation costs incurred by the supply chain to satisfy customer demand. This decision should be reconsidered on a regular basis so that the allocation can be changed as market conditions or plant capacities change. For example, a company has grown its customer base, the company has built new warehouses and changed the markets supplied by each warehouse. As a result, company will lower the costs and improve the responsiveness. The allocation of markets and supply sources can be changed only if the facilities are flexible enough to serve different markets and receive supply from different sources.

Decisions on network design have a significant impact on performance because they determine the supply chain configuration and set constraints within which inventory, transportation, and information can be used to either decrease supply chain cost or increase responsiveness. A company has to focus on network design decisions as its demand grows. Then its current configuration may not prove cost-effective and will exhibit poor responsiveness. For example, in India majority of pharmaceutical companies outsource their manufacturing to local vendors to serve local markets divided into zones. This arrangement is cost-effective from logistics point of view.

Network design decisions are also very important when two companies merge. Because the markets served by the combined firms are different from those served by either of the two separate firms. In mergers, consolidating some facilities and changing the location and role of others can help reduce cost and improve responsiveness.

STRATEGIC FACTORS IN NETWORK DECISION

A decision on the SCND is greatly influenced by the competitive strategy in service offerings to the customers. A company opting for cost leadership will decide on the lowest cost location for their manufacturing facilities. This may reduce the level of responsiveness of the firm. However, firms focusing on responsiveness will locate facilities near to the demand points. This may increase the cost location.

When designing a supply chain network the strategic factors such as speed to market, economic factors, infrastructure and network size are very important in determining a network that will deliver the desired level of customer service. However, achieving a trade-offs among these factors properly is the key to getting it right. Similarly, when designing a logistics network, strategic factors (Figure 13.3) such as speed to market, cost effectiveness and system effectiveness are important in determining a network that will deliver the intended level of customer service. Managing the trade-offs among these factors is the key to effectiveness of overall business.

- **Speed-to-market:** The term 'speed-to-market' mostly applies to how fast a company can create and launch a new product or service. Speed to market is how quickly a company can differentiate itself by providing an innovative or value-added product for a targeted customer segment. A major benefit is increased profitability through greater efficiency in revenue.
- **Cost-effectiveness:** As the number of facilities increase, facility costs, inventory costs and transportation costs will increase. Thus, these trade-offs must be analysed in order to arrive at a best solution. Sometimes near to the market doesn't mean the facility is system effective. Outsourcing is the one option available but economic factors such as taxes and exchange rates play a pivotal role in the location design decision. As a thumb rule, the following needs to be looked into before locating a facility on cost consideration:
 - Inventory cost increases with increase in number of facilities.
 - Facility cost increases with increase in facilities.
 - Transportation cost increases with increase in number of facilities.

Cost-effectiveness also very much depends on the choice of available production technology, which has a significant impact on network design decisions. With new technology, if economies of scale are possible to achieve, a few large capacity plants are more cost-effective. In many cases such as soft drinks bottling, a local production facility with lower fixed cost is preferred to reduce the transportation costs. The distribution cost of soft drink increases exponentially, if distribution is done beyond

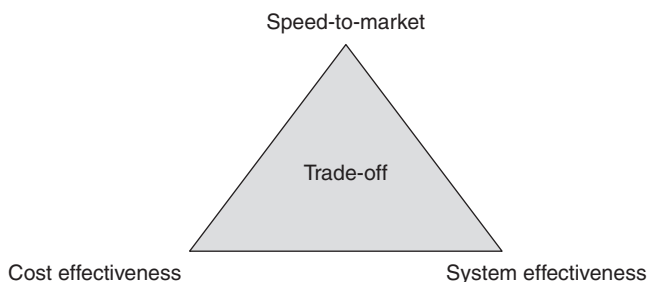


Figure 13.3 *Strategic Factors Trade-off*

120 kilometres. To reduce transportation costs, Coca-Cola/Pepsi have set up many bottling plants each serving its local market. In India, each one of them has 40–45 bottling plants located across in almost all states.

The lack of flexibility in production technology prompts the company to set up separate facility for each market. In such case, a firm has to set up local facilities to serve the market in each country. On the contrary, with flexible technology, manufacturing in a few large facilities is possible with significant cost-effectiveness.

- **System effectiveness:** Access to all the important variables such as man, machines, and materials is essential to work supply chain system effectively. Services such as water/drainage, consistent supply of electricity and the quality and proximity to main transportation routes are vital.

OPERATIONAL AND TACTICAL FACTORS

These factors have a major impact on network design decisions.

- **Market reach:** As a thumb rule, in supply chain the supply points should be near to the demand points. If the speed to the market (availability of product in FMCG industry) is a differentiator, then outsourcing production to the vendors near to demand points will be a prudent decision. Many organizations outsource production to low-cost regions/countries to have a competitive advantage, but it's important to consider what impact that has on customer service levels. If an organization is following a low-cost strategy, it makes sense to source where it is most cost-effective and undermining the speed to market, that is, sourcing facilities in close proximity to the consumer market would be a better strategy.
- **Product type:** The type or nature of product will decide the number of warehouses. FMCG products are purchased in small quantities and frequently for daily consumptions. Hence, to address the availability, large number of warehouses are required near the demand points (retailers). On the other hand, the customer is ready to wait for engineering products which are not frequently purchased. Here, the warehouses are limited in numbers located in metros/factory areas. The other example is perishable products. The cold warehouses are located near the

Box 13.1

Ambuja Cements: Lowest Logistics Cost

Ambuja Cements is the largest manufacturers of cement in India with 30 mn production capacity. The majority of their production facilities are located in the coastal areas of Gujarat. In cement marketing, the logistics cost is a major cost spinner. The logistics cost as a percentage of product cost varies from 14–22 per cent. Ambuja due to its plant locations, takes advantage of sea transportation to move its products to the major markets like Maharashtra, Goa, Karnataka, Kerala, Chennai, etc. The sea transportation forms the major part of its transportation mode. Large quantities of its cement production are dispatched through the sea route. As sea transportation is cheaper than both road and rail transportation, Ambuja is spending less on transportation cost. This is the reason, why logistics cost for Ambuja is 14–15 per cent as against their rivals, who are managing with 17–22 per cent as their plants are located in land-locked areas wherein they have no option but to use road or rail transport.

consumption centres and also at the place of production to keep the product (vegetable/fruits) in good condition before they are transported to the markets.

- **Regulations:** It is mandatory to follow the warehouse location regulations for certain products like explosives, hazardous chemicals and radioactive materials which can cause damage to human life. These facilities are located quite away from human habitation and outside municipal limits of cities/towns.
- **Political factors:** The political stability of the country under consideration plays a significant role in the location choice. Countries with independent and clear legal systems allow firms to feel that they have recourse in the courts should they need it. This makes it easier for companies to invest in facilities in these countries.
- **Tariff and tax incentives:** The factors such as taxes and exchange rates play an important role in the network (facility location) design decision. In cross-country trade, exchange rates are of prime concern for any company doing business overseas. Placing the facility in tax-free zone (FTZ) will be an added advantage to overcome fluctuations in tax rates. Many countries also offer tax incentives. Wherein, companies do not have to pay taxes for 10 years in an effort to attract them to invest in their country. In India, many states are offering tax exemptions or concession taxes (excise duty or sales tax) over a period of 5–10 years, if manufacturing facility is placed in the tax-free zone.
- **Infrastructure:** The availability of amenities like water, electricity and drainage comes under infrastructure. This must also be taken into account while choosing the facility location. For raw material and finished goods movement, quality and proximity to main transportation routes are vital. The availability of skilled labour is also an important factor. They will have a major impact on the cost.
- **Network size and logistics costs:** The inventory cost and administrative costs are directly proportional to the number of facilities (Figure 13.4). However, the secondary transportation costs will decrease. Hence, trade-offs must be analysed in order to arrive at the best solution. Once the solution is reached, it is also necessary to determine suitable sizing of the facilities.
- **Transportation costs:** Transportation is the most visible element of logistics operation. It forms a link between supply chain networking components. It has a significant share in the overall cost of the firm and needs a great deal of planning to control it. The choice of transportation from the available modes such as air, rail, road and sea is most important to

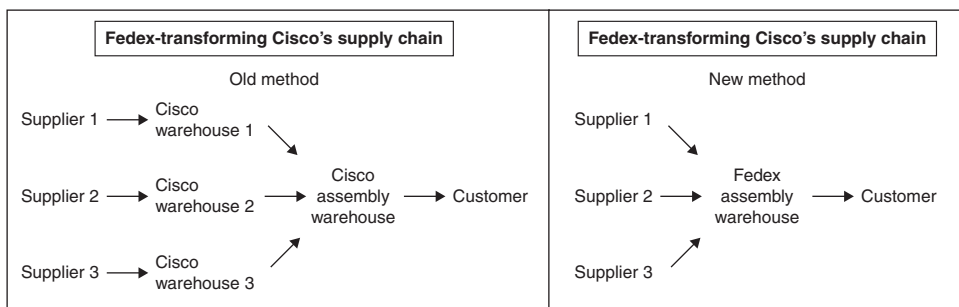


Figure 13.4 Old and New Supply Chain Networking at CISCO

optimize the cost. The selection of transportation mode depends upon several factors. The manufacturer decides upon the transportation mode by taking into consideration the following criteria: speed, frequency, availability of service, reliability, consistency in delivery, capability and finally the cost.

Warehousing

Warehousing is an important switching facility in the supply chain network. It will facilitate a smooth speedy distribution. Hence, an important step in supply chain network design is deciding on the centralized or decentralized distribution system. The field distribution centre/warehouse can act as a mother feeding/replenishing facility to the demand points. Centralized distribution ensures tight control on inventories and can operate on economies of scale, resulting in lower operating cost. However, decentralized warehouse operations, with multiple distribution centres, ensures speedier services to customers. Product availability can be greatly enhanced by locating the warehouse near the market. With this arrangement, smaller and frequent deliveries, which customers prefer nowadays, can be organized. This will enhance customer confidence in the supplier. However, transportation cost, which is the major element in logistics cost, depends on the location of the warehouse. For planning a chain of warehouses, the management will have to undertake a cost-benefit analysis. The factors influencing decisions in planning an optimum number of warehouses are:

- Size of the market
- Nature of demand (seasonal/continuous)
- Product type
- Level of customer service
- Intensity of competition
- Establishment and operating cost

Site selection for a warehouse is an important step in network design. It can affect the overall costs of the operations impacting the margins. Thus site selection is a critical step. Following are the techniques which will help in selecting the site in a scientific way.

Single depot selection The distribution system of the firm normally consists of multiple warehouses located at various places. However, the location of a depot for a particular geographical area is decided using the following principles:

- **Centre of gravity of area model:** A map of marketing areas may be drawn and the sheet representing the area may be balanced. The centre of balance is the site for warehouse location. This point is expected to be at a minimum average distance from all locations in the area.
- **Centre of gravity of load model:** This may be illustrated visually by imagining a sheet having holes representing customer locations and their demand load of annual purchases. A weight, in proportion to the demand load, is hung at each hole. The point at which the model sheet is balanced gives the centre of weights. This is the proposed site where the warehouse can be located.
- **Centre of gravity of tonnes—kilometres model:** This model takes into consideration both load and distance to arrive at the optimum transportation cost. The final solution is arrived at through a trial and error method.

Multiple depot network As a thumb rule, the cost of storage per unit of inventory is directly proportional to the number of warehouses. If the total sales is divided into more warehouses, the sales handled by each individual warehouse decreases and warehousing cost per unit sold increases. The increased cost can be offset by increasing sales per warehouse or decreasing number of warehouses.

STEPS IN NETWORK DESIGN PROCESS

The goal while designing supply chain network is to maximize the firms' profits while satisfying customer needs in terms of demand and supply. To design an effective network a manager must consider all the factors described above. The framework of supply chain network consists of the following seven phases:

- Phase I: Define the network objectives.
- Phase II: Define a firm's broad supply chain design.
- Phase III: Evolve a supply chain strategy (based on competition, internal constraints, capital and growth Strategy).
- Phase IV: Define the regional facility configuration.
- Phase V: Identify facilities locations and their potential roles and capacities.
- Phase VI: Select a set of potential and desirable sites.
- Phase VII: Select a precise location and capacity allocation for each facility.

Attention is restricted to the desirable potential sights selected in phase 3. The network is designed to maximize total profits taking into account the expected margins and demand in each market, various logistics and facility cost and the taxes and tariffs at each location.

FACILITY LOCATION AND CAPACITY ALLOCATION

Whatever the situation, the ultimate aim of network design is to maximize the profits while satisfying customer needs. The following information ideally should be available in making the design decisions:

- Location of supply points (sources)
- Location of demand points (markets)
- Site location of potential manufacturing facility
- Market potential and demand forecast
- Capital and operating cost of facility
- Freight (inbound and outbound logistics)
- Inventory costs by site
- Sale price realization
- Taxes and tariffs
- Desired response time and other service factors

Different models are used during different phases of network design framework. These are as follows:

- **Network optimization models:** This pertains to decisions regarding which regions should be chosen to locate the facilities. The major factors to be considered by a manager are: regional demand, tariffs, economies of scale and aggregate cost factors. One of the network optimization models used is 'the capacitated plant location model'.

The capacitated plant location network optimization model requires the following inputs:

- Number of potential plant locations with their capacities
- Number of markets or demand points
- Annual demand from assigned market
- Potential capacity of plant
- Annualized fixed cost of keeping assigned factory open
- Cost of producing and shipping from assigned factory and market

The following 'decision variables' are defined:

- Plant is open or otherwise full
- Quantity shipped from assigned plant to market

The model is solved using an Excel sheet with all the requisite data. The model can also be modified to account for strategic imperatives that require locating a plant in some other region. This can be done by restricting one plant to be located in a particular region.

Network optimization models are also used for deciding on the location and the capacity allocation for each facility. Besides this, the manager also has to decide how the markets are allocated to facilities. This allocation must account for customer service constraints in terms of response time. Both location and allocation decisions are made jointly.

The problem of selecting the optimal location and capacity allocation is very similar to the regional network configuration problem. The only difference is location-specific costs and duties. In some cases, the company designs a supply chain network to serve separate market with one single factory/source. The rationale behind this is to minimize the complexity of coordinating the network and reduce flexibility from each facility. However, to achieve the capacitated plant location the model needs modification.

In cases where plants and warehouses are to be located simultaneously in the entire supply chain network from the supplier to the customer, a general form of the plant location model needs to be considered. In general, companies consider a model in which location and capacity allocation decisions have to be made for both factories and warehouses at the same time.

This model assumes that one unit of input from the supply source produces one unit of finished product. The model requires the following inputs:

- Number of demand points
- Number of potential factory locations
- Number of suppliers
- Number of warehouse locations
- Annual demand
- Potential capacity of factory
- Supply chain at suppliers
- Potential warehouse capacity at site
- Cost of locating a plant at desired site
- Fixed cost of locating a warehouse at desired site
- Shipping cost from different factories to warehouses to markets

The goal is to identify the plant and warehouse location as well as the quantities shipped between various points that minimize the total fixed and variable cost. The following decision variables need to be defined:

- If factory is located at a desired site or otherwise
- If warehouse is located at desired site or otherwise
- Quantity shipped from factories to warehouses to markets

Gravity Location Models

Gravity location models are very useful as they can be used to identify suitable geographic locations within a region. These models are used to minimize the cost of transporting raw materials from suppliers and finished goods to the markets. Gravity models assume that both the markets and the supply sources can be located as grid points on a plane. All distances are calculated as per the geometric distance between two points on the plane. This model also assumes that the transportation cost grows linearly with the quantity shipped.

The basic inputs required for the model are as follows:

- Coordinated location of either a particular market or supply source.
- Shipping cost for one unit for unit distance between the facility and market.
- Quantity to be shipped between facility, warehouse and market.

The equation can be solved using an Excel sheet provided the requisite information for the model is available.

NETWORK DESIGN DECISION ISSUES

With the passage of time, environmental factors change continuously; the demand changes too. With technological evolution, demand for old generation products disappears and finally the company's product mix changes. Each facility has its own working culture. The employees have typical way to work and exhibit their capability and capacity. In addition, quality differentials may be observed across the facilities. Hence, while deciding on a supply chain network, the following factors need to be kept in mind by the management.

Facility's Life Span

Facilities last a longer time and have an impact on a firm's performance. Therefore, it is very important that long-term consequences are considered when making facility decisions. The demand, costs and technology may change in future. Failure to take a note of these changing factors makes the facilities useless within a few years and become a financial burden to the firm.

Within most supply chains, production facilities are harder to change than storage facilities. Supply chain network designers must consider that any factory that they put in place will stay there for an extended period of a decade or more. Warehouses or storage facilities, particularly those that are not owned by the company, can be changed within a year of making the decision.

Cultural

Network design decisions regarding facility location and facility role have a significant impact on the culture of each facility and the firm. The culture at a facility will be influenced by other facilities in its vicinity. Network designers can use this fact to influence the role of the new facility and the focus of the people working there. A new facility in industrial area with unionized workforce will often have to face labour problems.

The location of a facility has a significant impact on the extent and form of communication that develops in the supply chain network. Locating a facility far away from headquarters will likely give it more of a culture of autonomy. This may be beneficial if a firm is starting a new division that needs to function in a manner different from the rest of the company. In contrast, locating two facilities closer together is likely to encourage communication between them. Extensive communication can be very useful if decision made at either facility have a strong impact on the performance of the other facility.

Quality of Life

The quality of life at a selected facility location has a significant impact on performance because it influences the workforce and their morale. In many instances, a firm may be better of selecting a higher cost location if it provides a much better quality of life. Failure to do so can have dire consequences. For example, an aerospace supplier decided to relocate to an entire newly area with a lower standard of living to reduce costs and most of the marketing team, however, refused to relocate. As a result, customer relation deteriorated and the company had a very difficult transition. The effort to save costs hurt the company and effectively curtailed firm's status as a major player in its market.

Tariffs and Tax Incentives

Facility location decisions should carefully be done on tariffs and tax incentives. When considering international locations, it is amazing how often tax incentives drive at the choice of location, often overcoming all of the other cost factors combined. For instance, Ireland has developed large high-tech industry by enticing companies with their low taxes to locate their European facilities there. Even within nations, local governments may offer generous packages of low to no taxes and free land when firms decide to locate facility within their jurisdiction. Leading pharmaceutical companies in India have their manufacturing facility in Baddi (Himachal Pradesh), to avail of the tax incentives offered by the state government.

DISTRIBUTION NETWORK DESIGN

Distribution refers to the steps taken to move and store a product from the supplier stage to a customer stage in the supply chain. Distribution occurs at every pair of stages in a supply chain. Raw materials and components are moved from suppliers to manufacturers, whereas finished products are moved from the manufacturer to the end customer. Distribution is the key driver of the overall profitability of a firm because it directly impacts both the supply chain cost and the customer experience.

Although, customer service consists of many components, measures that are influenced by the structure of distribution network include the following:

Box 13.2

Adani: Networking in End-to-End Agricultural Supply Chain

The Adani has designed an end-to end supply-chain for agricultural produce which is practical and quite close to an ideal supply-chain. Adani has taken care of the complete supply-chain from grain storage to transportation to the end distribution connectivity. Adani has built grain silos where grains will be stored depending on the grade. It reduces loss due to rodents and insects attack. The silos have mechanized rapid loading facilities to load wagons rapidly. Adani got custom- made wagons, BCBFG, to carry grains. These wagons are like covered BOBRN wagons used for carrying coal. They are covered on top and they have bottom opening doors for rapid discharge into a hopper. They have built smaller silos in the cities where the grain has to be distributed. These are similar to depots or RDCs (Regional Distribution Centres). Adani's model can be replicated for other products. The liquids, like acids, edible oils can use this model. Tanks could be setup in ports and specialized railway wagons can transport the material from source to destination, that is, port to plant or customer. Similar supply-chains can be setup for other raw materials in powdered form.

Exhibit 13.2 Delivery Performance–Related Network

Retail storage with customer pickup	Response time, order visibility, information, returnability, speedier transportation
Manufacturer storage with direct shipping	Product variety, product availability, inventory
Manufacturing storage with in-transit merge	Product variety, product availability, inventory
Manufacturing storage with pickup	Product variety, product availability, inventory, speedier transportation

- Response time is the time between when a customer places an order and receives delivery.
- Products variety is the number of different products/configuration that a customer desires from the network.
- Availability is the probability of having a product in stock when a customer order arrives.
- Customer experience includes ease with which the customer can place and receive their orders.
- Order visibility is the ability of the customer to track their order from placement to delivery.
- Returnability is the ease with which a customer can return unsatisfactory merchandise and the ability of the network to handle such returns.

A distribution network design needs to consider product characteristics as well as network requirements when deciding on the appropriate reliable delivery. The various networks considered earlier have different strengths and weaknesses. In Exhibit 13.2, the various delivery networks are ranked relative to each other along with different performance dimensions. A ranking indicates the best performance along a given dimension; as the relative performance worsens, the ranking gets higher. Only few companies will end up using a single distribution network. Most companies are best served by a combination of delivery networks. The combination used will depend on product characteristics as well as the strategic position that the firm is targeting.

- Response time
- Product variety
- Product availability
- Customer expectations
- Inventory cost
- Transportation
- Information

SUPPLY CHAIN MERGER

It is known that profitable growth is influenced by the size, function, and location of their manufacturing and distribution facilities. Even the technology for designing supply chain networks has been around since the early 1980s, when such tools were basically spreadsheets linked to inputs on a map.

But today's business world is altogether different than it was two decades ago. Faced with a more fluid, complex, global, and unpredictable environment, companies must reassess the composition, structure, and mission of their supply chain networks with greater diligence and frequency. And while many business changes legitimately warrant a reassessment of a supply chain network, the need to revisit and redesign that network is particularly critical when an organization goes for merger or acquisition.

Supply chain economies are usually the largest source of savings following a merger, acquisition, or sale of a division or business unit. However, most merged companies shortchange themselves by performing their network (re)designs only at a high level, and only during the post-merger integration process. One exception is a European manufacturer that purchased six companies in 10 years. As part of each acquisition, the company conducted a supply chain-network analysis that revealed new opportunities to:

- Curtail redundancy
- Increase delivery density
- Improve service by rationalizing deliveries within and across regions
- Cut travel times and increase truckload utilization
- Rationalize inventories
- Streamline order management and fulfillment
- Combine inbound flows and renegotiate supplier contracts
- Raise quantities of economic orders

The important network design questions in supply chain mergers are as follows:

- Suppliers sourcing and purchasing
 - Which, what and how much?
- Manufacturing
 - Where, what and how much?
- Transportation
 - DC locations

- Customer service
 - Responsiveness, flexibility and error recovery
- Channel network
 - Numbers, capacity and location

In mergers, the management mainly focuses on the financial and cultural aspects. In many cases, they fail to perform detailed supply chain network analyses of the merged company. Then they face the risk of going forward with duplication of processes, incompatible processes, higher inventory levels and higher cost of customer service. In such cases, customers do not view the organization as a unified business as they do not get uniform responses as in a single company.

SUMMARY

SCND decisions include the location of manufacturing, storage, or transportation-related facilities and the allocation of capacity and roles to each facility. Supply chain network design decisions are classified as facility role (Role of facility and processes performed at each facility), facility location (Where should facilities be located?), capacity allocation (Capacity allocated to each facility) and market and supply allocation (What market should each facility serve? Which supply sources should feed each facility?)

To achieve competitiveness in supply chain a focus on network design, management and internal processes is a must. The co-operation between network partners leads to improved performance in general. For network to be truly effective, it requires a significantly higher level of joint strategy development. Hence, network partners must collectively agree strategic goals for the network and the means of attaining them. With effective network design an enterprise and networked partners will have a win-win situation leading to customer satisfaction.

The main aim of decisions on facilities location and capacity allocations is always to maximize the profitability of the supply chain network coupled with enhancement in overall responsiveness towards the customers and markets requirements. The profitability of the firm is also impacted by the taxes and tariffs and hence the network must be designed to maximize profits after taxes and tariff. The trade-offs that must be considered while designing the network are that a larger number of facilities might reduce cost of transportation but will increase the facility and inventory costs of the firm simultaneously.

Factors to be considered for network design are product characteristics as well as delivery mechanism. The network, however, is tailored to match the characteristics of the product or the needs of the customer. Fast moving and emergency items are stocked locally and customers can either pick them up directly or have them shipped depending on the urgency. Slower moving items are stocked at a national DC from where they are shipped to the customer within a day or two.

Capacity allocation decision also has a significant impact on supply chain performance. Allocation of too much capacity to a location results in poor utilization and, as a result, higher costs. Allocation of too little capacity results in poor responsiveness if demand is not satisfied or high cost if demand is filled from a distant facility. The allocation of markets and supply sources can only be changed if the facilities are flexible enough to serve different markets and receive supply from different sources.

REVIEW QUESTIONS

1. What is the role of network design in supply chain?
2. Discuss the various factors influencing network design.
3. Explain the framework for network design decisions.
4. In mergers, what are the network optimization issues?

INTERNET EXERCISE

1. Supply chain consultants (SCC) delivers strategic business solutions that enable clients in collecting, processing and sharing information across the supply chain network, visit <http://www.supplychain.com/> and study some of the network solution they have offered.

VIDEO LINKS

1. Network Design—How it can help you, <http://www.logisticsbureau.com.au/video-supply-chain-logistics.htm>.
2. Warehousing and distribution network, <http://www.logisticsbureau.com.au/video-supply-chain-logistics.htm>.

PROJECT ASSIGNMENT

1. Ambuja Cements Ltd. have lowest logistics cost and wider market reach amongst all the cement manufacturing companies in India. Study their supply chain network and their distribution strategy, which is largely contributing to their cost leadership.

NOTE

1. Anne Mulcahy, Chairman and CEO, Xerox.

INFORMATION SYSTEM AND IT ENABLEMENT

14

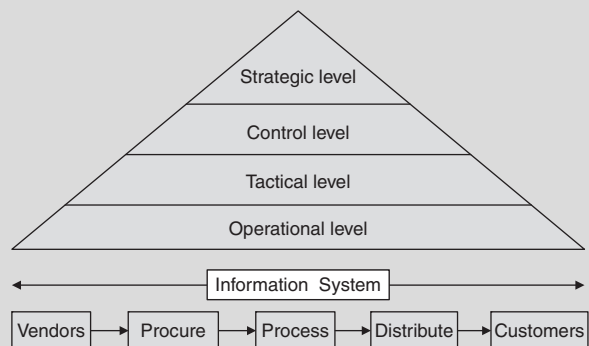
“Information is a source of learning. But unless it is organized, processed, and available to the right people in a format for decision making, it is a burden, not a benefit

—William Pollard”

Chapter Objectives

To understand:

- Designing supply chain information system
- Characteristics of information system
- Effective forecasting
- Information for coordination of system
- Information and SC integration



The day-to-day running of any business depends on the information systems it deploys for collecting, analysing and presenting data from both internal and external sources. It is known to all that 'information is power' to remain competitive and alert. Hence, for the successful running of a business, the organization needs information systems that can make available the right information at the right time in the right format for decision-making which will help the organization to sustain and grow. The entire supply chain operation is an information-based process of network planning, organizing, deployment and controlling. Hence, using information is the single important source of competitive advantage.

Ambuja Cement—IT Initiatives

Ambuja Cement is India's leading cement manufacturing company producing 30 million tonnes of cement per annum. The firm is in the forefront for deploying IT and other manufacturing technologies into its entire supply chain process. In the year 2001, Ambuja amongst all cement companies was the first to deploy Red Hat Linux at its manufacturing plants for various business process applications.

In 2004, Ambuja Cements launched 'Connect India Plus' a software system to get connected within all its remotely located plants and field offices to collect, analyse and disseminate real-time information across all the decision-makers at the different location in the country. This was one of the most significant large-scale IT deployment within the company and also in the industry. However, as the technology developed over the years, in the year 2006, Ambuja Cements had gone for an ERP system called SAP in 2007 for all group companies as a standard system.

SAP with all its modules covered the company's 200 locations across India and 2,500 users with a single instance on a server in Mumbai. SAP implementation was a great challenge for Ambuja. The IT team faced immense problems while implementing this enterprise-wide ERP at multiple plants each with their own computer systems and processes. They deployed large number of trained IT professionals to integrate individuals (existing employees) with diverse background to be able to work as a focused team to use the ERP.

The company formed the core team of 75 members. They also involved additional people (indirectly or directly) for the purpose of data migration (from eight different legacy systems) and training of personnel, etc. The HR team was quite active to organize enormous amount of man-management skills. IT team took on the task of setting up an adequate and reliable WAN using MPLS and VSATS. By end of 2008, all 200 locations including factories, bulk cement terminals, grinding units, regional offices and warehouses were connected through SAP.

After SAP, Ambuja had gone in for a smart-card based vehicle tracking system to improve the operational efficiency. This helped the company determine the exact cycle for a vehicle carrying cement from the factory to a destination and carrying raw material as a return load-back to the factory. SAP implementation accrued lot of benefits to Ambuja by significantly reducing the cost of operations and maintenance of the IT system. The responsiveness of the company to the changing business environment increased manifold.

INTRODUCTION

Supply chain management is a cross-functional approach to manage the movement of raw materials into an organization, a full or part internal processing of materials into finished goods, and finally movement of finished goods to the end-consumer. The purpose of supply chain management is to improve trust and collaboration among supply chain partners, and to improve inventory visibility and velocity. Supply chain management is essentially an information-based process of material movement,

wherein the flow of information is key to the success of the entire operations. The starting point of the information in supply chain is the customer order. The customer spells out his needs and passes it on to the suppliers in the form of either paper or electronic documents. Depending on his needs in terms of product quantity, product variety, place of use and urgency, the logistical operation chain at the supplier's end fulfil his requirements at the right place, at the right time with least cost. Thus, the material flow in the supply chain of the enterprise is essentially dependent on the information flow across the system. The information flow starts with the enterprise's customers and end with suppliers via distribution, manufacturing and procurement centres. At each of the above places, the decisions are taken based on the information available. The degree of success of the decisions largely depends on the information quality, timeliness and the form. With advancement in information and communication technologies, the speed of information analysis and flow has increased tremendously. As a result efficiency and effectiveness of supply chain operations improved considerably, resulting in increasing the level of customer satisfaction. The electronic movement of the information helps reducing the cost of logistics operations due to increased coordination between the various activities across the supply chain and, moreover, decisions are taken on real-time basis, minimizing the risk element.

In the supply chain, information is required for operation planning and control, system development, and also for strategic decision-making process at various levels. The operational planning covers, customer order registration, order processing, procurement actions, storage, order picking, packing and transportation. The inventory management provides coordination between the supply chain operations, procurement and the manufacturing.

Box 14.1

Developing a Bottling Hub that Increased Efficiency by 75 per cent

The world's most popular soft drinks marketer had 1,400 bottlers in one country. For bottling systems, the company was highly dependent on contract bottlers, mostly Mom and Pop stores that did not have the resources to modernize their process. The bottlers, who set the pricing, sent voluminous faxes (up to 70,000 a year) to the soft drink marketer's main office with pricing, confirmation updates and profitability reports. The customer's office, would, in turn, direct them to the correct recipients. Retailers complained about this approach to faxing weekly price changes. To end the time and labour-intensive problem, the customer decided to put in a centralized system that would make the information available online to retail outlets. Before designing the perfect solution, the IT consulting team studied the bottling systems of the customer, the industry and the operations of major retail chains. The result is the bottler hub/extranet that automates the distribution of pricing data, and is made available to the customer's entire group of registered bottlers and retailers. Now, the 1,400 bottlers log onto the system with just a web browser, input information, like price changes negotiated with retailers, and send it to their customers, who in turn conveys it to the respective retail outlet. After the deployment of the bottler hub, the customer is today the one face for all retailers instead of the 1,400 different bottlers, earlier. The bottler hub/extranet was developed in Java using IBM's VisualAge developer tools. The Java 2 Enterprise Edition components link to the customer's SAP financial, manufacturing and ERP applications. The application is connected to Unix-based IBM WebSphere application servers. The bottler hub/extranet has made the soft drink company, 75 per cent more efficient than the industry standards in cycle time of announcing a pricing change, getting it approved by retailers and fixing errors.

The supply chain capacity and capability planning and network design requires allocations and commitment of resources. These fall under the purview of strategic planning wherein the information needs are quite different than what is required for operational planning. There is high degree of risk element in strategic planning exercise which can be minimized through designing proper information system using latest technologies, ensuring speed and reliability in information availability, processing and movement.

Manufacturing environments today have lean machines and optimized manufacturing processes. However, to achieve this, the company should have a visibility of inventory into the product distribution. This is possible only with availability of real-time information updates which will help to optimize scheduling and planning costs; this is possible only when supply chain is backed up with information system. Due to the large number of transactions in the business process, it is impossible to achieve the effectiveness in supply chain without IT.

DESIGNING INFORMATION SYSTEM

Similar to management information system (MIS) which is for general application, the supply chain for its efficient and effective operation needs information system for decision-making. The information system is an interactive structure of people, equipment, methods and control designed to create information flow in the required format for the user to take decisions to reduce the risk element. The information system basically consists of the following elements.

- **Information sources:** There are two basic resources for the information, that is, external customers and the internal departments or functional areas of the enterprise.
- **Information collection system:** This information from the customer in the form of purchase order may be fed into the system manually if it is in the paper form. The information in the electronic form will be directly decoded and entered into the information system. The customer order may be assigned with a particular code (bar code) to track and trace the order by various internal information users. From internal sources the information may be collected and punched manually or fed into system electronically by using automatic identification technology such as bar coding, radio frequency tags, and voice interactive systems.
- **Storage:** The information is stored into magnetic tapes, CDs or the hard disk of the centralized computers. With the advanced data storage systems based on magnetic principle, it is possible to store the large data.
- **Processing:** With the computer technology (using the microchips), information is processed instantly with great accuracy. However, selection of the right computer system for the given application needs to be done cautiously, keeping in view the objectives, the investment cost and the maintenance capabilities.
- **Retrieval:** Data warehousing and mining system will facilitate instant data retrieval at the users' terminals spread across the system.
- **Report formatting:** If suitable software is installed in the system, then reports can be generated in the required formats for decision-makers at various levels.

The above elements need to be configured as per requirements of various users of the system. However, the over-designed system will have the various capabilities but the cost will be prohibitive, while the low cost system will not be capable of satisfying all the users of the system. Hence, the trade-off may be thought of for cost and the system capabilities for the given applications.

DECISION-MAKING LEVELS

The information needs of the organization for planning and implementing the supply chain (SC) decisions depends on objectives. The quality of the management decisions in SC depends on the quality of the information generated by the information system (IS). The criticality of decision will be decided on the speed and accuracy needs of the desired information. However, for reducing the risk involved in decision-making, the following principles need to be followed while designing the IS:

- Accuracy
- Availability
- Timeliness
- Flexibility
- Format
- Interactive

Information is required at all levels (Exhibit 14.1) in the supply chain management for effective decision-making. The type of information depends on the authority level and the criticality of the decision. Depending on the requirements, the supply chain information system is designed around the following four levels of decision-making.

- **Operating level decisions:** This starts with receipt of the order from the customer. The operating level components, which need to be coordinated, are order registration, order processing, procurement, order filling, inventory planning, packaging, warehousing, transportation and delivery. To handle voluminous activities, information system is needed for efficiency.
- **Tactical level decisions:** At the tactical level, the information system is required to provide information for taking the decisions on the formulation and implementation of supply chain strategies. The decision process here requires information for bringing in the cost-effectiveness in the logistic system. The typical decision areas wherein the information analysis is required for evaluating the tactical alternatives are facility planning, vehicle routing and scheduling, inventory management, outsourcing, channel integration and customization.
- **Control level decisions:** Supply chain information system's (SCIS) major application is in the area of management control. Based on the operating level data, SCIS is required to generate the

Exhibit 14.1 *Decision-Making Levels in Supply Chain*

Operating Level (1) Order registration Order processing Inventory planning Warehousing and distribution Transportation and delivery	Tactical Level (2) Inventory management Facility planning Channel integration Vehicle route planning/scheduling Outsourcing
Control Level (3) Customer service System productivity Cost control Asset utilization	Strategic Level (4) Capability planning Capacity planning Alliances and partnership Customizing

reports on the performance of the various supply chain operations. SCIS is required to generate reports on a regular basis on the operating cost of the supply chain system. This includes the cost of warehousing operation, operating cost of material handling equipment, wage bill, freight, and utility cost. The deviations can be probed for the timely corrective action. The major area of management control which needs the SCIS support are asset management, customer service, cost control and system productivity.

- **Strategic level decisions:** Strategic level decision requires information support for developing supply chain strategy for gaining the competitive edge over the competitors. These decisions are having long-term objectives. The frequency of decisions is very low. The decision-making requires the analysis of the variety of data collected at the operating level to build up the models to take strategic decisions. The strategic decisions in this area are capability and capacity planning, alliance and partnership, etc.

DECISION SUPPORT SYSTEM

Decision support systems (DSS) are a specific class of computerized information system that supports business and organizational decision-making activities. A properly-designed DSS is an interactive software-based system intended to help decision-makers compile useful information from raw data, documents, personal knowledge, and/or business models to identify and solve problems and make decisions.

Supply chain decision support system is an information system that helps with decision-making in the formation of a supply chain plans and strategies. A supply chain, which is also referred to as a value chain or value network, consists of suppliers, manufacturing centres, warehouses, distribution centres and retail outlets, as well as materials and goods, information and money that flow between the facilities. There are several decision areas within supply chain and it has been classified into two broad areas: structure and control. Structural decision areas are typically localization of production plants, warehouses and choice of suppliers and transporters. For controls, the structure of the supply chain is considered as given and focus is on effectively managing the supply chain. In its broadest sense, any mental or physical construct that supports and improves decision-making process constitutes a decision support method. A computer-based system is called as decision support systems (DSS).

DSS is not a new concept. In the good olden days the companies were having DSS but with very poor and delayed responsiveness. But in today's business environment, companies have adopted the technology-based DSS to have real-time decision based on real-time data. Today's DSS exhibits three capabilities: dialogues, data and modelling. The importance of each in supply chain varies with product-market configurations. DSS consists of a number of systems, tools and technologies. It may include ERP, DRP, EDI, GIS, GPS, data warehousing, mining tools, etc. DSS used for the entire enterprise to cover all business processes are linked to large data warehouses. ERP (enterprise resource planning) is also a DSS. ERP helps in seamless integration of all functions in an organization.

ROLE OF IT IN MANAGING SUPPLY CHAIN

Information technology and, in particular, the Internet, plays a key role in achieving goals of supply chain integration. As the management of information flow has always been a key aspect of supply chain management, the web-based information transfer between companies, their suppliers, and their customers has decidedly increased the importance of information management in creating effective

supply chains. Today, the Internet has emerged as a most cost-effective means of driving supply chain integration. The most visible manifestation of the Internet has been in the emergence of electronic commerce as a new retail channel. The Internet has a profound impact on business-to-business interaction, especially in the area of supply chain integration. The Internet has redefined how back-end operations—product design and development, procurement, production, inventory, distribution, after-sales service support and even marketing—are conducted and the relationships between various parties are fostered in the networks, services and business models.

Before IT was enablement in the supply chain, the company had to follow the processes enumerated below in order to purchase an item:

- Determination of goods and process needs.
- Needs determination for delivery.
- Entering the order into the computer.
- Checking on information that the correct product is being bought.
- Receive three bids from vendors.
- Evaluation of bids and ordering.
- Physical receipt of goods.
- Matching freight bill, packing list with order.
- Processing receipt.
- Sending paperwork to accounts payable.
- Entering paperwork into computer.
- Checking for payment.
- Receipt of invoice.
- Matching invoice to receipt.
- Receiving payment authorization.
- Cutting the cheque and Mailing the cheque.

Having learned the costs involved in these processes, companies used IT to eliminate lots of non-value-adding processes, and the result was that a typical purchase operation would be IT enabled by using the following e-tools:

- EDI for ordering and invoicing.
- Bar coding for tracking the items being ordered.
- Automated clearing house for payment.
- Electronic fund transfer for payment.

For optimizing performance of supply chains the business organizations are using software technology to get product information. The software technologies are used in Materials Requirement Planning (MRP), Manufacturing Resources Planning (MRP), and Enterprise Resource Planning (ERP) in their day-to-day planning activities. Most of applications are for capturing of data from internal and external sources. The emergence of the Internet technology has given rise to new business models and increased competitive pressures. IT has helped in enhancing supply chain efficiency and responsiveness by sharing real-time information regarding inventory, shipment status and other key information

like product design, product availability and demand between the partners. The IT-enabled supply chain seamlessly integrates all its internal and external processes. With the speed of information sharing, it became possible for the business organization to adopt pull strategy as against push strategy. The benefits of IT enablement in supply chain are as follows:

- Product acquisition costs reduced
- Procurement-related transaction cost reduced
- Inventory level drastically reduced
- Quality in decision-making enhanced
- Cycle times considerably reduced
- Wastages reduced
- Customer service improved

IT-enabled planning application focuses on demand forecasting, inventory simulation, distribution, transportation, and manufacturing planning and scheduling. Planning software is designed to improve forecast accuracy, optimize production scheduling, reduce inventory costs, decrease order cycle times, reduce transportation costs, and improve customer costs, decrease order cycle times, reduce transportation costs and improve customer service. IT-enabled execution process addresses procuring, manufacturing and distributing products throughout the value chain. Supply chain execution applications are designed to manage the flow of products through distribution centres and warehouses and help ensure that products are delivered to the right location using the best transportation alternative available. There are four key dimensions (Exhibit 14.2) in which the impacts of IT on supply chain integration can be found:

Exhibit 14.2 *Dimensions Impacted by IT on Supply Chain Integration*

<i>Dimension</i>	<i>Elements</i>	<i>Benefits</i>
Information integration	• Information sharing and transparency • Direct and real-time accessibility	• Early problem detection • Faster response • Trust building
Synchronized planning	• Collaborative planning, forecasting and replenishment • Joint design	• Reduced bullwhip effect • Lower cost • Optimized capacity utilization • Improved service
Workflow coordination	• Coordinated production planning and operations, procurement, order processing, engineering change and design • Integrated, automated business processes	• Fast response • Improved service • Earlier time to market • Expanded network
New business models	• Virtual resources • Logistics restructuring • Mass customization • New services • Click-and-mortar models	• Better asset utilization • Higher efficiency • Penetrate new markets • Create new products

- Information integration
- Planning synchronization
- Workflow coordination
- New business models

Information Integration

Information is the backbone of supply chain operation. Hence, integration of information from and to at various decision-making points is a must. In fact, information integration refers to the sharing of information among members of the supply chain. This includes any type of information that could influence the actions and performance of other members of the supply chain. The information may be pertaining to demand data, inventory status, capacity, production schedules, promotion plans and shipment schedules. Ideally, such information can be accessible by the appropriate decision-makers on a real-time basis. The information is essentially required for the following.

Planning Synchronization

Planning synchronization is a joint design and execution of plans for product introduction, forecasting and replenishment. In essence, planning synchronization defines what is to be done with the information that is shared; it is the mutual agreement amongst members as to specific actions based on that information.

Workflow Coordination

Workflow coordination refers to streamlined and automated workflow activities between supply chain partners. In this, shared information is used to sequence the activities like a relay race. The procurement activities from a manufacturer to a supplier can be tightly coupled so that efficiencies in terms of accuracy, time and cost can be achieved. In essence, for coordination supply chain partners rely on technology solutions (information flow) to automate many or all of the internal and cross-company workflow steps.

E-business Models

Adopting e-business approaches to supply chain integration promises many improvements in system efficiency. E-business allows partners to redefine logistics flows so that the roles and responsibilities of members may change to improve overall supply chain efficiency. A supply chain network jointly creates new products, pursue mass customization, and penetrate new markets and customer segments. New rules of the supply chain game can emerge as a result of integration process due to the Internet.

Information technology supports internal operations of a company and its supply chain partners. Using high speed data networks and databases, companies can share data to better manage the supply chain as a whole and their own individual positions within the supply chain. The effective use of this technology is a key aspect of a company's success. The model is used to classify and gain perspective on available technology and how it can be used.

All information systems are composed of technology that performs three main functions: data capture and communication; data storage and retrieval; and data manipulation and reporting. Different information systems have different combinations of capabilities in these three functional areas. The specific combination of capabilities is dependent on the demands of the job that a system is designed

to perform. Chopra and Meindl¹ define several kinds of systems that support supply chain operations, as follows:

- Enterprise resource planning
- Procurement
- Planning and scheduling
- Transportation planning
- Demand planning
- Customer relationship management
- Sales force automation
- Supply chain management
- Inventory management
- Manufacturing execution
- Transportation scheduling
- Warehouse management

These systems support daily supply chain operations. They provide capabilities to efficiently run the ongoing operations to keep track of inventory levels and stocking locations within a network and they support the actions needed to pick, pack and ship products to fill customer orders.

TECHNOLOGY—ROLE IN INFORMATION SYSTEM

Any DSS to work successfully require data (both internal and external) identification and input. The data may manually be identified and captured for further processing. The internal data refers to transactions associated with receiving material, staging, storage, location, picking, manufacturing, product status, etc. However, the external data refers to transactions in purchase, acquisition, movement and location of items in the supply chain. The manual data has some disadvantages in data capture. It is slow and prone to delays and errors. To overcome the disadvantages of manual data capture, automatic data capture techniques are developed. These techniques make the job of data capture, speedy and error free. Following are techniques based on the latest technology which are commonly in practice in business applications

Bar Codes

Bar codes are used for identification, handling, retrieval and storage of goods in warehouses and stores. It is the most popular identification technology in many applications. The individual inventory items, cartons or unitized packages are affixed with a bar code, which can be read by bar code scanner attached to the online computer system. Bar codes are assigned to particular inventory items to show its identity during storage, retrieval and dispatch. The bar codes are further used for communication of dispatched items for preparing bills by accounts departments and making periodic reports on inventory status and sales. The bar codes facilitate to track the item in warehouse during inventory audit or material pick up. It also helps in tracking the consignment during transportation and inspection at the customer end. The information, which may be required, is country

code, manufacturer's name, product details, date of manufacture, material content, etc. These details are required at user's end for inventory management. These details in machine-readable codes are in the form of bars and spaces.

Bar codes are seen on all types of goods today. It is sequence of parallel lines of different thickness with spaces in between. These bars are nothing but the items of information in the codified form, which can be decoded or read with the help of scanner. In other words, bar code is a type of Morse code to put information of the item in code language. The bar code facilitates data accuracy, real-time data availability, uniformity and easy usage, which are recognized universally.

The bar codes have found applications, as diverse as in automobile industry, logistics, retail chains, electronics, defence, pharmaceutical industry, banking, consumer goods, libraries, airlines, passports, etc. Today it is invariably used in all industries. The bar codes increase productivity in three ways such as speed, accuracy and reliability and offer the following advantages:

- Easy identification of inventory items in various logistics activities.
- Reduction in paper work and processing time leading to cost reduction.
- Elimination of human error.
- Increase in the productivity of warehouse.
- Facilitation of system automation.

The bar codes are described by the symbology used. Symbology means the pattern of lines and spaces which are used within the bar code to represent the number or the alphabet. The various bar code symbologies differ both in the way they represent data and in the type of data they can encode. Some symbols can encode numbers, others numbers and letters, and some letters, numbers and characters, that is, ASCII codes. The latest symbologies include options of multiple languages.

Radio Frequency Identification (RFID)

Radio frequency identification is currently one of the preferred forms of auto identification of the goods in the manufacturing, retailing and logistics industries. The identification is relying on storing and remotely retrieving data using device called RFID tag or transponders. The tag is an object that can be applied to or incorporated into a product for the purpose of identification using radio waves.

Most IRID tags contain at least two parts. One is an integrated circuit for sorting and processing information, modulating and demodulating a (RF) signal, and performing other specialized functions. The second is an antenna for receiving and transmitting the signal. The payment card with RFID can be recharged with cash at add-value machine or in shops, and can be read several centimetres from the reader. Delhi Metro uses this technology. High-frequency RFID tags are used in library book or bookstores, jewellery or pallet tracking, building access control , airline baggage tracking and apparel tracking, etc.

Geographical Positioning System (GPS)

The more accurate system in use in developed countries is based on global positioning system or geographical positioning system (GPS), wherein a vehicle could be traced accurately with the help of geo stationary satellites.

Box 14.2

RFID—Enhancing Productivity in the Laundry Industry

A 300-room Five Star Hotel has laundry service as one of the major and complex room service product to offer to its clients. The laundry department in such hotels typically handles around 600–1,200 pieces of laundry on any day depending on the season. The process consists of the following sequential activities:

- *Laundry pick-up from the rooms.*
- *Allocating bar code to each piece of laundry (based on; room number, men/ladies wear, laundry/press service, type of return service—next day, same day or express, type of cloth—Silk, cotton, synthetic, etc; wet clean or dry clean service).*
- *Transporting the day's laundry to the factory.*
- *Washing, drying and pressing.*
- *Packing room-wise.*
- *Distribution.*

Garments and linens go through dozens, if not hundreds of collection, cleaning, sorting and delivery cycles, in what are typically very labour intensive processes to enhance the productivity in laundry in developed countries. REID tags are being used currently. In laundry adds the harsh and high temperature environments and RFID has to perform to levels beyond most normal industrial settings.

The high-frequency 13.56 MHz RFID tags that are encapsulated in plastic to withstand the harsh temperatures and detergents found in commercial laundries. Tags are programmed with a unique ID number that can be read and linked to asset tracking and logistics applications. RFID system gives visibility on how much inventory is in the laundry's processing pipeline to schedule optimum washing and drying loads as well as work shifts. As clothes and linens move through the washing, drying, pressing and folding processes, RFID tags need to withstand extreme heat, moisture, pressure and harsh detergents. Opposed to bar code labels which require manual scanning of each item, and where printing can wear off or labels can become frayed, RFID presents a robust and durable alternative. The garments are then sorted by type, size and user and then they are then placed on hangers and packaged into orders for customer delivery. Using RFID orders may also be bundled according to delivery date and distribution route.

There are 12 stationary satellites in three layers with four in each layer, orbiting the earth. The fix from three of these satellites will determine the latitude and longitude to an accuracy of 1 metre. These devices are already been in use by mountaineers. Once the position of the vehicle is known, it can be transmitted to other centres, that is, consigner or consignee. The transmission network shall be a mobile telephone or the Internet. In GPS, a voice over facility for driver to speak and hear instruction and LCD (liquid crystal display) with keyboard, to receive and transmit text messages is possible. GPS helps in enhancing the effectiveness in logistics operation. The familiar GPS application is locating the vehicle location. It also determines the best route to any destination,

provides turn-by-turn navigation instructions, and estimates arrival time. In India, such systems are in use at limited places. The other system in use is dual mode wireless vehicle system, but usage is restricted to small geographical areas.

GPS receivers calculate geographic location from signals received from a minimum of 3 satellites. The satellite signals improve accuracy of location of the object. In combination with information from roadmap databases and user-entered destination locations GPS enables receivers to calculate speed, direction, distance travelled, time and distance to destination, and more. Via wide-area networking technology (for example, GSM/GPRS), GPS-equipped mobile computers can provide real-time location information to an enterprise's IT system., particularly those with operations in truck route accounting, direct store delivery, trucking, parcel delivery and postal services and field services.

GPS reduces fuel consumption and fuel waste, searching for locations or using inefficient routes. It improves route efficiency, allowing drivers to service more customers. It brings access to real-time traffic updates, helping drivers avoid congestion. It eliminates the need for paper maps, with routes determined more quickly and safely. It enhances safety, voice prompts enable drivers to keep their eyes on the road. In route auditing, GPS receivers capture the travel history of vehicles or personnel and the entire routes-information which can be easily stored for analysis at any time. This enables to document and verify all stops made on a route. It helps to identify opportunities for enhanced efficiency by analysing routes taken. It reduces unnecessary fuel usage and vehicle wear. It adds more stops per day by streamlining driver travel. It identifies and eliminates out-of-route travel.

In GPS, geo-coding can validate that a service was rendered; archive the location of assets placed in the field; identify landmarks or points of commerce; initiate a secondary business process with geo-coded items and automatically trigger an activity when in proximity to a specified location. Similarly in GPS, geo-fencing creates a virtual boundary around a geographic area. That makes it simple to track assets carrying a GPS device when they enter or exit that boundary (distribution centre).

Equipping a mobile computer with integrated GPS offers significant advantages over separate solutions for GPS capability and mobile computing. A single-device solution means less maintenance, and the inherent ability to use multiple technologies in combination (GSM/GPRS with GPS). Making the most of these synergetic technologies requires choosing a device with the right combination of attributes for effective use in one's application as GPS capability alone isn't enough. Considering these key factors the firm should choose the right solution to meet its needs such as durability and reliability of equipment, power management, GPS usability, versatile data-capture capability and effective GPS integration.

Geographical Information System (GIS)

GIS are the software tools for the visualization of spatial features and databases relating to geography. Data relating to special location of any entity on earth is stored in databases and GIS then integrates the different types of data in different databases into a detailed, clear visualization of data with respect to the entity. This could in terms of physical maps of the surface of earth, layout of inner surface of earth or a layout of streets or roads. Data can be collected and used through different tools such as demographics, market research, physical features, aerial photography, satellite imagery, etc. All these can be fed into a GIS and then can be visualized as per the needs of the user. GIS in integration with GPS is used in logistical operation for tracking and tracing of the consignment location to the extent of road or street in a particular city.

Electronic Data Interchange (EDI)

EDI is an emerging technology used for transfer of business documents from one computer to another. Traditionally the business document such as invoices, *challans*, cheques, drawings are sent through mail, courier or by fax. However, with EDI these documents are transferred electronically from one organization to another. In short, EDI is a drive towards paperless document transfer or transactions. EDI is emerged in strategic areas such as the provision of better levels of customer service and improved marketing competitiveness. EDI is computer-to-computer communication with the same language. To accomplish this, the two computers should have same communication standard. The standard includes:

- Matching of data transmission speed
- Data coding and decoding system
- Matching of hardware of the computers

Different industries have developed their own communication and message standard for their own use. However, for large-scale usage of EDI in the business, it is important that a uniform standard has to be developed. The usage of EDI in the area of supply chain between the manufacturer and supplier will ensure substantial saving on transaction cost for both the parties. Similarly, using EDI between manufacturer and the customer will ensure the reduction in order cycle time and inventory that will help the customer to enhance his competitiveness. Thus, sharing the benefits of EDI by both seller and buyer will result in creation of partnership spirit leading to high switching cost for the customer and creating barriers for competitors. Today EDI is widely used in banks, marketing, customs, logistics and finance companies and in international trade in the developed countries.

Business Intelligence System (BIS)

Because of the fast pace of change in markets and their supply chains, it is very important for people and organizations to stay current with events as they happen and understand what these events mean. Business intelligence systems help companies to understand what is happening within their own organizations and within the markets they serve. BI systems collect, store, and analyse data. They collect data using from many different sources. Data can be collected from sensors and RFID scanners. Data can be collected by BPM (business process management) systems or data can be obtained from the many transaction-processing systems in a company such as ERP systems, order entry systems, or CRM systems.

Simulation Modelling

Simulation modelling software is a category of software that is growing rapidly. Because of the fast pace of change in business, companies are faced with the need to make important decisions more often and these decisions have significant consequences on company operations and profitability. Companies are faced with decisions like where to build a new factory or distribution centre and what is the best way to layout and equip a new facility.

Simulation modelling software allows people to create a model of a factory or a supply chain or a delivery route and then subject that model to different inputs and different situations and observe what happens. A design that may seem good on paper could very well turn out to have problems that are not apparent until the design is modelled and its performance is simulated under a range of different conditions. It is much faster and cheaper to find this out through simulations than to find out the hard way through real experience.

SUMMARY

Information is power, which is used as one of the resources to develop competitive edge in the business. Due to advancement of technology in information processing and communication, the speed of the information flow has increased manifolds. The real time decision-making is possible due to availability of information at the click of a button. With the advancement of technology the cost of this important resource is varying in inverse proportion with respect to its capability. Obviously it is to the advantage of the users. An enterprise's responsiveness capability can be increased with appropriate design of the information system, which can satisfy the information needs for decision-making to reduce the element of uncertainty and risk. The starting point of the supply chain information system (SCIS) is the customer order. The SCIS operates for information needs at four level, that is, operating, tactical, control and strategic levels. The requirement of information in terms of the purpose, frequency, quality, format, variety, differs with its required level. At operating level, it is required for coordination of activities like order processing, order filling, procurement, production scheduling, dispatching and responding to customer order status queries. At tactical level, information support is required for inventory management, facility planning, vehicle scheduling and route selection, and for decisions on outsourcing. The strategic level information requirements are for service customization, capability and capacity planning and alliances. For controlling the supply chain operations, the information requirements are in the form of performance ratios such as supply chain system productivity, inventory turn over, ROI, customer satisfaction and asset utilization. Thus, IT is an essential ingredient for business survival and improves the competitiveness of firms.

REVIEW QUESTIONS

1. 'Information is a major factor in supply chain competitiveness'. Explain.
2. Which are the different information requirement levels of information system?
3. Discuss the role and benefits of technology in decision support system.
4. Efficiency and effectiveness of supply chain depends on the information requirement at operational and tactical level, respectively. Explain.
5. Discuss the role of IT in managing supply chain.

INTERNET EXERCISES

1. Visit <http://www.chainlinkresearch.com>, to find out how to manage information flow with accuracy in supply chain system
2. To study more on information system implementations in SC visit <http://www.supplychain-online.com>

VIDEO LINKS

1. Supply Chains and Information Systems, <http://www.youtube.com/watch?v=DwNfNHGUjuY>
2. Supply Chains and Information Technology, <http://www.youtube.com/watch?v=SXDvHgiRNDQ>

PROJECT ASSIGNMENT

1. Companies like Hindustan Unilever and Marico Industries have implemented information systems in various forms using latest information technology tools. With implementation of the information systems they have successfully reduced the inventory level to the level, which was beyond expectations earlier. Study these information systems, the challenges in implementations and benefits reaped.

NOTE

1. Chopra Sunil and Meindl, Peter (2001), *Supply Chain Management: Strategy, Planning and Operation*, Pearson, New Delhi, pp. 336–355.

PART-IV

STRATEGIC APPROACH

15 Alliances and Outsourcing

Chapter covers a framework for strategic alliance, issues and challenges. It further discusses the role and 3PL and 4PL service providers, their selection criteria and their role in distribution integration, system productivity and customer service.

16 Agile Supply Chain

In this chapter the role of agile supply chain in managing market volatility and customer responsiveness is discussed. It also covers the elements of agility in terms of customer service in supply chains strategy, customer service perception and service design.

17 Global Supply Chain

This chapter covers the discussion on global market forces and their influences on design of global supply chain design. It further covers the risk factors in managing global supply chain, strategic issues operational factors and IT role in managing global SC.

18 Reverse Supply Chain

Chapter covers the importance in today's context and divers to reverse supply chain design. Further the scope of reverse supply chain and various design considerations along with challenges and limitation are also discussed.

19 Supply Chain Optimization

This chapter starts with definition of supply chain optimization process. The chapter covers various optimization techniques used in supply chain, their benefits, challenges and issues in optimization process.

20 Re-engineering SC to Differential Strategies

This chapter covers the cost based and other differentiation strategies used in managing supply chain. The chapter further includes the discussion on manufacturing strategies, aligning SC to marketing needs, digital and cellular manufacturing process.

21 Supply Chain Integration Strategies

This chapter covers the discussion on role of information in supply chain integration. The chapter further includes the strategies of push, pull and push-pull system. The discussion covers the impact of Internet on supply chain strategies and methods of collaborative planning, forecasting and replenishment to integration and visibility in supply chain.

22 Cold Chain Networking

In this chapter the scope and drivers to cold chain infrastructure are discussed. The chapter further covers cold supply chain models in practice their problems and issues.

ALLIANCES AND OUTSOURCING

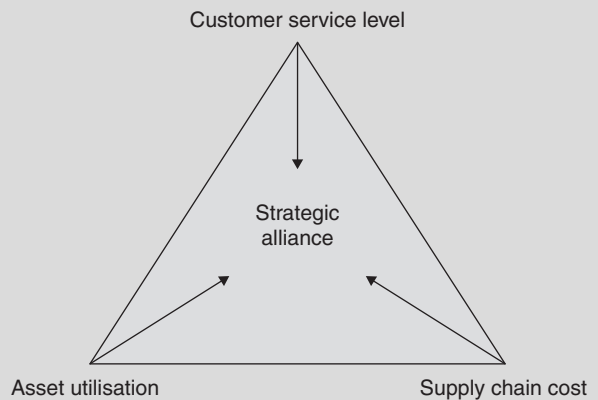
15

“Partnering is the quickest, most effective way to re-engineer a business
—Curtis E. Sahakian”

Chapter Objectives

To understand:

- A framework for strategic alliance, issues and challenges
- 3PL/4PL service providers
- Retailer–supplier partnership
- Distribution integration
- Alliance contact terms



Strategic Alliances Topology

Supply chain aims at delivering a set of goods or services to an end-customer by aligning the capabilities of supply chain partners who share common goals. Partners are not only the upstream suppliers but are also the downstream retailers or distributors. Supply chain is a collaborative chain wherein alliances can be leveraged to competitive advantage. There are many processes in the supply chain that partners can have alliance on. However, it is important that the firm should be clear about processes wherein alliance or collaboration is required. In general, these are the non-core processes wherein the firm does not have the expertise to perform these activities with efficiency and cost-effectiveness. These firms are the candidates for alliances.

Boeing 787—Global Suppliers Alliance Issues¹

Boeing is one of the leading aircraft manufacturers in the world. They are planning to launch a new passenger airliner—the 787 Dreamliner. The original launch was planned by May 2008 but till the end of December 2010 there was no sign of its launch. The delay was due to problems of the aircraft's extended global supply chain which slowed the progress on the assembly line.

Boeing tried to introduce an innovation in their supply systems at the same time they have designed innovation in products and assemblies for navigations. The main error in supply chain planning was that all systems and suppliers were quite away from their assembly lines plants, which resulted into a problem in the supply chain integration. It was in fact a three-tier suppliers' chain. This was an extended supply chain with many alliances.

For Dreamliner project, they outsourced sections of the aircraft to suppliers in Japan, Italy, South Carolina and Kansas and assembled in Washington. The problems were not just with suppliers in China, but the hiccups were in the assembly fitments in the US plant. The redesigns were taken up and suppliers were asked to rework the component sub- assemblies. In fact, outsourcing proved costlier in terms of delays due to lack of tight controls.

In the outsourcing process, Boeing lost balance between the 'flexibility' and 'control' as Boeing has no prior experience of outsourcing problems and challenges. In the Dreamliner project, Boeing outsourced design engineering, composites technology, and many other sub-assemblies, which resulted into the holdup. Boeing made the decision to outsource 60 per cent of the design and production—a radical move. With hardly any experience in outsourcing to a far away located alliance partner, Boeing contracted with more than 50 suppliers, 28 of them outside the United States. The airliner's wings were going from Nagoya, Japan, to Everett, WA. Fuselage Section 43 was going from Nagoya, Japan, to Charleston, SC, to Everett, WA. Fuselage Section 46 went from Grottagli, Italy, to Charleston, SC, to Everett, WA, and so on. In addition to the oversight, the buyer (Boeing) needed insight into what actually was going on in those vendors' factories that was missing in this project. Boeing could not pre-assess the risk of outsourcing and ultimately it proved a nightmare for them.

INTRODUCTION

Today, business organizations across the world are struggling for the competitiveness not only for growth but also for survival. The factors responsible for this are liberalized economies of the countries across the world, globalization of the businesses and recessionary trends in the markets. Moreover, the customer has become more demanding and looks for value-added services from prospective suppliers, as he wants value for the money he is spending. In such a situation, business organizations across the world have started reviewing their business processes and they realized that cost cutting and differentiation in value delivery process are solutions to the current problem. This can be done through outsourcing the non-core operations to experts in the field and concentrating on core business areas. The expert can do the job both cost-effectively and efficiently. Hence, the accepted trend observed in the industry is 'hollowing' of the corporations, that is, outsourcing non-core functional areas of the management and gaining operating efficiencies and effectiveness by engaging the services of the experts in that particular field.

Outsourcing brings about the responsiveness in supply chain operations as there is real-time information sharing amongst supply chain partners and customers. However, there are many other issues in supply chain collaboration with vendors and these are:

- As the decision-making centres in the downstream are more, decision-making is not uniform. Secondly, due to time lag in information progression in upstream from the market, there is lot of distortion in information. This is known as 'bullwhip effect', which results either in excess inventory or stock outs.
- If the demand information is not correct and reliable, the decision-makers in supply chain make their own guessing. In this situation, the product is manufactured in excess and the firm has no option but to resort to push strategy.
- In the absence of information sharing, demand forecasts and orders are often distorted.

The supply chain collaboration that can prove fruitful should be focused on information sharing. This requires coordinating vast amount of structured data using enterprise systems such as ERP, MRP, APS, SCM, EDI and many others.

STRATEGIC ALLIANCE

Supply chain management is involved in planning and management of procurement, manufacturing and distribution activities. IT coordinates and collaborates with channel partners. These are suppliers, intermediaries and third-party service providers.

There are a number of strategic alliances amongst the above mentioned players. A strategic alliance is a formal relationship formed between two or more parties to pursue a set of agreed upon goals or to meet a critical business need while remaining independent.

Partners may provide the strategic alliance with resources such as products, distribution channels, manufacturing capability, project funding, capital equipment, knowledge, expertise, or intellectual property. The alliance is cooperation or collaboration which aims for a synergy, where each partner hopes that the benefits from the alliance will be greater than those from individual efforts.

Box 15.1

LEVI'S: Closing Outlets and Collaborating with Wal-Mart

The most famous jeans manufacturer had good days for nearly 4–5 decades. Mr. David Burgen, Chief Executive Officer of Levi's noticed, there has been a continuous drop of business—had drop in sales close to 30 per cent and was looking for alternate sales areas. While studying the retail system of Wal-Mart with 100 million shoppers every week, (as against 1 million a year at Levi's), Mr. David Burgen straightaway came to a collaborative and transparent venture with Wal-Mart and started marketing his products in their retail outlet. Levi's gained in sales due to access to wider market due to alliance with Wal-Mart. The automated supply chain management and inventory systems of Wal-Mart helped Levi's to monitor and control its inventory and its related cost. Consequently, Mr. Burgen closed its 195 odd outlets, which could not generate enough sales to offset the fixed cost of the retail operations. With the association with Wal-Mart, Levi's once again touched 70 per cent share in the market within a year.

The alliance often involves technology transfer (access to knowledge and expertise), economic specialization, shared expenses and shared risk.

There are four types of strategic alliance in supply chain, which are as follows:

- **Third-party logistics—3PL:** It is the use of an outside company to perform all or part of the firm's material management and product distribution functions.
- **Fourth-party logistics—4PL:** It refers to the evolution in logistics from suppliers focused on warehousing and transportation (third-party logistics providers) to suppliers offering a more integrated solution. Amongst other services, fourth-party logistics providers include supply chain management and solutions, change management capabilities, and value-added services in their offering. The evolution of outsourcing can be seen in Figure 15.1.
- **Retailer–supplier partnerships:** It is the formation of strategic alliances between the retailers and their suppliers.
- **Distributor integration:** This appreciates the value of the distributors and their relationship with the end users and provides them with the necessary support to be successful.

Strategic alliances often bring partners the following benefits:

- Access to partner's distribution channels and markets
- Access to partner's technology and intellectual property
- Access to partner's capital

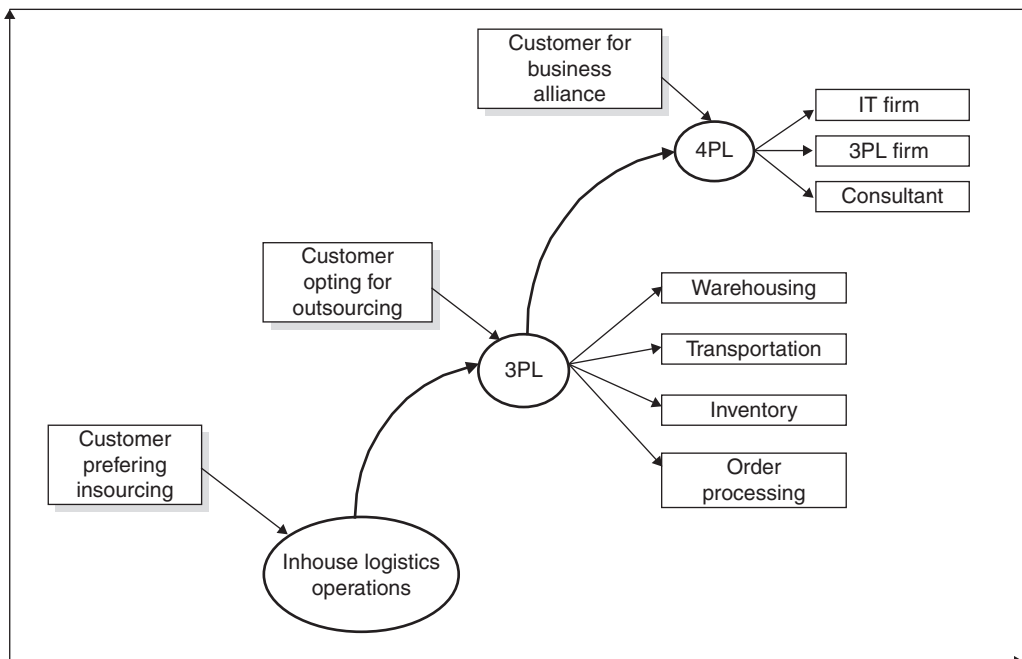


Figure 15.1 Evolution of Logistics

- Access to partners markets and their customers
- Increased brand awareness through partner's channels
- Speed to market due to joint efforts
- Reduced R&D costs and risks
- Rapid scale achievement, critical mass and momentum
- Access to management skills

Just-in-time inventory purchasing and supplying alliance between Wal-Mart and P&G has continued to prove successful over the years. Home Depot and Dell Computers have also built powerful alliances with their suppliers for cost saving through just-in-time inventory in similar applications. Below are some supply chain improvement areas available to firms through strategic alliance relationships:

- Management of supply channel conflict
- On-time product delivery
- Prompt response to complaints
- Greater quality and delivery consistency in inventory
- Improved handling of customer complaints
- Improved supply chain productivity
- Prompt response to quote requests and price problems
- Confidentiality of shared business strategy

THIRD-PARTY LOGISTICS

There are four categories of 3PL providers:

- **Standard 3PL provider:** This is the most basic form of a 3PL provider. They perform activities such as pick and pack, warehousing, and distribution—the most basic functions of logistics.
- **Service developer:** This type of 3PL provider will offer their customers advanced value-added services such as tracking and tracing, cross-docking, specific packaging, or providing a unique security system. A solid IT foundation and a focus on economies of scale and scope will enable this type of 3PL provider to perform these types of tasks.
- **The customer adapter:** This type of 3PL provider comes in at the request of the customer and essentially takes over complete control of the company's logistics activities. The 3PL provider improves the logistics dramatically, but do not develop a new service. The customer base for the 3PL provider is typically quite small.
- **The customer developer:** This is the highest level that a 3PL provider can attain with respect to its processes and activities. This occurs when the 3PL provider integrates itself with the customer and takes over their entire logistics function. These providers will have few customers, but will perform extensive and detailed tasks for them.

The trend of using strategic partnership in integrated logistics became the accepted practice in the industry. These firms are external to the company and provide one or more aspects of its entire logistics

service product portfolio. These services can be provided on stand-alone or integrated basis. The stand-alone operators are called 'Wholesalers' who extend only one type of service in which they have expertise. These services may be: warehousing, transportation, inventory management, packaging, etc. However, the one who provides entire logistics services and offers logistics solutions to the customer problems is called as an 'integrator'. The preferred trend in the industry is an integrated logistics solution provider to solve logistics problems from one source.

The growth of 3PL was witnessed in the 1990s when corporations around the world started concentrating on the entire supply chain and realized that logistics is a key to success in supply chain. They understood that the most important strategic utilization of logistics is improvement in customer service followed by gaining market share and reduction in cost. The most important reason for the rapid acceptance of 3PLs is due to the quick gains by the users of the 3PL services. As per the survey conducted by 3PL firms in the United States, during 1996 and 1998 the early users of 3PL firms reported the following benefits:

- Logistics cost reduced by 7.8 per cent
- Logistics assets fell by 21.6 per cent
- Order cycle time reduced from 6.3 to 3.5 days

Initially, corporations were outsourcing only warehousing and transportation to 3PL firms, but as the confidence level started going up and benefits started accruing, they were invited to provide services in the areas of traffic management, multimodal transportation services, freight consolidation, cross-docking, freight auditing and payment collections, etc. More and more companies have started using 3PL services as a source of strategic advantage with a view to achieve broader business objectives than cost saving and cycle time reduction. Some of the broader objectives, the corporations had in mind for going in for 3PL services are:

- Reduction in risk and liability
- Value-added services to customer
- Source of process improvement
- Wider market coverage

As this service industry is in the maturity stage in the developed countries, there is stiff competition in this sector and consequently they are offering customized services in the niche markets. They are providing value-added services such as consignment tracking, real-time data access and analysis, cross-docking, assembly, etc.

FOURTH-PARTY LOGISTICS

The next significant evolution in supply chain management after third-party logistics (3PL) is fourth-party logistics (4PL). Fourth-party logistics enables firms to respond to today's supply chain requirements more effectively. The fourth-party logistics provider is a supply chain integrator that assembles and manages the resources, capabilities, and technology of its own organization with those of complementary service providers to deliver comprehensive supply chain solutions.

The term 'fourth-party logistics' was coined and trademarked by Andersen Consulting. These are 3PL firms that have invested in technology, human resources and alliances in order to present a single point of contact for operation of a customer's supply chain. Those leading firms include

Menlo Logistics, Ryder System, Federal Express, UPS Logistics, GATX Logistics, Exel and Schneider Logistics.

In essence, the 4PL provider is a supply chain integrator of all 3PL service providers. 4PL service provider coordinates, assembles, aligns and manages the capabilities of 3PL vendors. 4PL is an asset-less vendor. 4PL operates on IT platform and deliver a comprehensive supply chain solution using capabilities of 3PL vendors.

Information technology plays a major role in logistics and supply chain management. Today the integration of logistics, which is a complex exercise, is totally dependent on IT support. Third-party logistics suppliers are providing logistics solutions to their clients, based on their domain knowledge they have developed or acquired over the years of experience in the logistics business. However, a new trend has emerged, wherein the IT firms are providing logistics solutions built around domain knowledge provided by 3PL companies. These new breeds of companies are called as Fourth-Party Logistics Service Providers or 4PL firms.

The dividing line between the 3PL and 4PL is very thin. The leading 3PL companies in the United States think that 4PL is a hype created by management and IT consultancy firms to take away the cream from logistics business which has been built by 3PL companies over the years through their efforts. The genesis of 4PL lies in forming a collaborative relationship amongst various logistics service providers based on IT (Information Technology) backbone. Hence, the network arrangement can be termed as 4PL, provided it fulfils the following requirements:

- Covers the entire supply chain of the customer
- Collaboration between two or more logistics service providers on the resources sharing basis to extent logistics solution to a common customer
- Alliances to be led by integrator with an IT-based and not an asset-based service provider
- Flexible arrangement

For example, a 4PL company of a large Indian FMCG manufacturer operating in Indian and overseas markets which needs to integrate its entire logistics operations handled by different 3PL firms in different geographical areas assigned to them, shall design and operate one single central information system instead of the different systems in different areas by different 3PL firms. The 4PL firm fulfils all the different needs of the client from single source instead of getting into multiple 3PL alliances to achieve different objectives.

Unlike traditional methods, which focus on reduction in operational cost and asset transfer, 4PL works in the following four ways:

- Increases revenue
- Reduces cost
- Reduces working capital
- Reduces fixed capital

4PL is the emerging trend and there are very few 4PL firms operating across the world. It is a complex model and offers greater benefit in terms of economies of scale. Recently, Hewlett Packard (HP) has appointed Circle International (CI) as their 4PL partner in Asia Pacific region. CI is operating from their central hub located at Singapore where it buys and stocks the HP's inventory requirements in the region. The network of warehouse hubs spread across the countries in the region takes care of the distribution. The local HP office in the country draws it inventory requirements by buying

from CI local hub. HP does not block its funds in inventory. Thus, 4PL provides logistics services by blocking its own money in someone else's products and components. 4PL supply chain solution includes three phases of work:

- **Reorganizing:** The most critical element in supply chain management is synchronization of supply chain planning and execution activities across all partners. Reorganizing in the areas of consulting skills, business strategy alignment and creatively redesign and integrate the supply chains of the participants.
- **Transformation:** Transformation efforts focus on specific supply chain function. Transformation covers in the areas of strategic planning, analysis, process redesign, organizational change management, and technology to integrate the client's supply chain activities and processes.
- **Execution:** It is an execution at the tactical level. An organization can outsource the entire range of its supply chain activities to a 4PL provider for execution.

4PL leverages a full range of service providers (3PLs) with their capabilities. The 4PL acts as a single point of interface with the client organization and provides the management of multiple service providers through a teaming partnership or an alliance.

Revenue growth is driven by enhanced product quality, product availability and improved customer service. With the 4PL focusing on the entire supply chain—not just the efficiency associated with warehousing or lowest-cost transportation—dramatic customer service improvements can be attained.

Operating-cost reductions of up to 15 per cent are driven through operational efficiencies, process enhancements and procurement savings. Savings are achieved through the complete outsourcing of the supply chain function and not just components and economies of scale. Synchronization of supply chain activities by supply chain participants leads to operating-cost reductions and a lower cost of goods sold, due to integration of processes and improved planning and execution of supply chain activities. The working-capital reduction is up to 30 per cent which is done through inventory reductions.

Fixed-capital is reduced with capital asset transfer and enhanced asset utilization. The 4PL's logistics service providers can take ownership of physical assets, thus freeing up client's assets. This allows the client organization to invest in its core competencies.

The key distinction between 4PL and current approaches to supply chain outsourcing is 4PL's unique ability to deliver value to client organizations across the entire supply chain. The 4PL approaches the concept of supply chain integration through four key drivers of shareholder value: increased revenue, operating cost reduction, working capital reduction, and fixed capital reduction. Traditional approaches have tended to focus only on operating cost reduction and asset transfer.

Revenue growth is driven by product availability and improved customer service. With the 4PL focusing on the entire supply chain—not just the efficiency associated with warehousing or lowest-cost transportation—dramatic customer service improvements can be attained.

RETAILER–SUPPLIER PARTNERSHIP

It's the formation of strategic alliances between the retailers and their suppliers.

Quick Response Strategy

Here suppliers receive point of sale (POS) data from the retailers and use this information to synchronize their production and inventory activities with actual sales at the retailer. In the strategy, the

retailer still prepares individual orders, but the POS data are used by the supplier to improve forecasting and scheduling and to reduce local time. In this strategy, the retailer has complete control over its inventory, but helps suppliers improve operations by providing POS data. Additionally, this type of partnership could be preferred if financial and personnel resources to develop a more integrated relationship are not available.

Continuous Replenishment Strategy

Here the vendors receive POS data and use these data to prepare shipments at previous agreed-upon intervals to maintain specific levels of inventory. In an advanced form of continuous replenishment, suppliers may gradually decrease inventory levels at the retail store or distribution centre as long as the service levels are met. Thus, in a structured way inventory levels are continuously improved. This type of partnership is a system between quick response and VMI, because suppliers and buyers together agree on target inventory and service levels. It involves less risk for retailers than VMI, and typically leads to a more stable and long-term relationship between suppliers and retailers than quick response does.

Vendor Managed Inventory (VMI) System

Here the supplier decides on the appropriate inventory levels of each of the products and the appropriate inventory policies to maintain these levels. This type of relationship is being used in Wal-Mart and P&G, whose partnership began in 1985. It has dramatically improved P&G's on-time deliveries to Wal-Mart while increasing inventory turns. This system is more integrated than the previous two systems, and requires a high level of trust between the supplier and the buyer. If implemented properly, VMI can lead to more overall system savings than the other two types of partnerships. However, VMI requires more commitment initially, and significant investment in information infrastructure, time and personnel. The VMI process brings benefits for both retailers and suppliers. The strategic alliance relationship depends on the following:

- Advanced information systems
- Top management commitment
- Trust amongst the partners
- Inventory ownership

For an agreement to be successful, performance measurement criteria must also be agreed to. These criteria should include non-financial measures as well as the traditional financial measures. For example, non-financial measures could include POS accuracy, inventory accuracy, shipment and delivery accuracy, lead times and customer fill rates.

When entering into any kind of strategic alliance it is important for both the parties to realize that there will be problems that can only be worked out through communication and cooperation.

DISTRIBUTOR INTEGRATION

Distributors form an essential link in the firm's marketing channel and are often the repository of vital market information. In the case of manufactured goods, distributors or dealers not only sell the product of the firm, but also perform vital functions like after-sales service, financing and promotion. The

traditional role of the distributor is now being challenged, however, as competition in the marketplace requires greater service level and a quicker response time.

The distribution strategies are based on the premises of making goods available to customers at the right place at the right time and at the least cost. These strategies aim at high level of service commitments on the part of organization. The modes of transportation for distribution are typically road for its flexibility, and door-to-door transportation is absolutely essential for distribution management. For the cost reduction firms choose rail or waterways as a mode of transportation. Air is also used in case of high-value item and for emergency deliveries.

In the distribution process, the role of warehouses cannot be ignored. The effectiveness of marketing decisions depends on the decision on warehousing. The capacity and the inventory level in the warehouse support in the success of distribution strategies. Today, the trends in distribution are driving shipment size even smaller and smaller with increasing frequency of delivery.

In general, firms adopt various strategies for distribution such as direct delivery to customer, delivery through warehouse or cross-docking. Each is having its positives and negatives. The product market configuration will decide the strategy to be adopted.

Distributor integration is one type of vertical integration, that is, forward vertical integration. In forward vertical integration, the company sets up subsidiaries that distribute market products to customers or use the products themselves. An example of this is an exclusive distribution channel. In distributor integration, parts are shared across the distributor network and specialized service requests are steered to appropriate dealers or distributors. For a successful distributor integration, the following points are necessary:

- Trust
- Commitment
- Manufacturer's support

Benefits of distributor integration are economics of scale and scope, cost reduction, competitiveness, higher degree of control over entire value chain which leads to reduction in threat from powerful suppliers and synchronization of supply and demand along the chain.

One of the best examples of vertically integrated companies is the oil industry (HPCL, IOC, Reliance Petroleum). They are active all the way along the supply chain from locating crude oil deposits, drilling and extracting crude, transporting it around the world, refining it into petroleum products to distributing the fuel to company-owned retail stations, where it is sold to consumers.

LEGAL CONTRACT

The firms outsource the supply chain operations to 3PL or 4PL firms for building competitiveness. In addition, other expectations in alliance are value addition in customer value delivery process, divert the risk and reduce liabilities. The alliance partner has to organize assets, add manpower, hire facility on lease or purchase it outright. To safeguard their interest and for smooth running of relationship, alliance partners usually enter into a legal contract or agreement, which is binding, on them. The following are the major points covered in such an agreement:

- Date and place of the agreement
- Names and addresses of the contracting parties

- Scope of service
 - Geographical coverage
 - Logistical services (in parts or full)
 - Management and usage of the customer's existing logistics assets
- Payment terms
- Delivery requirements
- Extra services from service providers
- Charges of the service offered
- Value proposition by service provider
- Damage liability
 - Loss (Due to fire, pilferage, rains, accident, floods, act of nature, negligence, ageing, etc.)
 - Insurance
 - Damage calculations basis
 - Demurrage due to delays in loading and unloading
 - Consequential damages
- Responsibilities
 - Employees
 - Assets management (office, facilities, equipment, etc.)
 - Warehouse management
 - Inventory management
 - Transportations
 - Communication
- Performance measurement criteria
- Risk sharing
- Termination of contract
- Notice for claims and filing suits
- Authority for settling dispute
- Notice period
- Governing laws
- Jurisdiction

In a nutshell, a strong and well-written agreement shall often encourage healthy relations between alliance partners.

ISSUES IN STRATEGIC ALLIANCE

The service providers help the business corporation to achieve two goals: to reduce the operating cost and to increase the revenue. As the service provider organizes the required assets, the investment in

owning the assets on the part of customer is reduced; this in turn allows the firm to invest in more productive activities and get more returns on the remaining assets and enhance the return on stock holders' investment. The alliance with the service provider will free the company's manpower for more productive work and concentrate on the area of core competence to get more returns. The firm gains on knowledge because of exposure and acquaintance to the best available practices and technologies, which are used by service providers. With these value propositions, the decision on logistics outsourcing can be justified.

To gain competitiveness, business organizations normally review their business processes to deliver the value to their customer and develop the competitive advantage. Many of them start outsourcing the non-core-competency areas to the experts who promised to do the job at reduced cost and simultaneously bring effectiveness and efficiency in the operation. As many of the Indian firms are in learning phase of outsourcing, they ignored the other critical issues which resulted into breaking the partnership with logistics service supplier within a short period and then they started looking for other options. The following are the few major issues² which need to be addressed and examined before deciding on 3PL or 4PL partnering:

- Switching cost
- Degree of control
- Human and electronic interface
- Tuning logistics services to the needs of channel partners
- Degree of outsourcing
- Legal aspects

Switching Cost

Outsourcing the processes results in reorganization of the existing assets (warehousing and transportation fleet if any) of the company to tune to the working methodology of the service provider. This includes:

- Management of existing assets fully or partly by the service provider.
- Deployment of the existing assets on lease to service provider.
- Divest the existing assets and fully switch over to the usage of logistics infrastructure provided by the service provider.

In each of the above case, a high element of risk is involved. Even though, outsourcing of services reduces the logistical cost substantially, but switching over to other service provider or to the original system in case of breaking away of partnership, will cause much more loss in terms of time in stabilizing the new system, customer service below the customers expectation level during transition period, erosion of customer base due to element of unreliability in the service experienced by the clients.

Degree of Control

The outsourcing firm has to be particular about its degree of control, on activities of service provider to get the service desired by the end user. Direct control over the activities of the service provider's

employees is not possible, but he should ensure timely availability of information to monitor the activities. It is better to develop a system to have the required information available without involvement of service provider's employee.

Human and Electronic Interface

For the smooth working of two organizations together in partnership, to achieve a common goal, the proper coordination through the intelligent interface is essential. As the working culture differs, there is a mismatch in degree of empowering the employees, speed and flexibility in taking the decisions, precision in operations, technologies used and access to confidential information. There are numerous examples of failures in other areas like acquisitions and mergers due to mismatch of culture and also due to some of the factors indicated above. Hence, proper interface between the employees of both the organizations is essential to resolve the issues raised out of misunderstanding, or miscommunication. For example, empowering the service provider to take decisions (with proper guidelines) on damaged goods during transit shall reduce delays in taking decisions through joint inspection by customer and manufacturer. This will help in building the confidence of the customer in service level of the supplier. However, the job of coordinators from both organizations is very crucial to formulate the policies and guidelines for the smooth operation of the outsourcing firm and service provider. The coordinator should be empowered to take spot decisions to resolve the issues immediately before it assumes ugly shape resulting into customer complaints.

Secondly, the problem may arise out of the mismatching of technologies being used at outsourcing firm and the service provider's end. For example, most of the third-party service providers of foreign origin in India are experts in EDI, but unfortunately the majority of Indian firms (barring from MNCs and large Indian business houses) do not have proper interface to take advantage of this latest technology. The result is performance below the expectations, which is promised at the time of alliance agreement. Similarly, the differentials in technologies used in communications, material handling, storage arrangements, inventory management may create delays, errors, and mistakes resulting into the performance far below the expected level.

Tuning Logistics to the Needs of Channel Partners

Logistics service standards are to be quantified as per the requirements of the channel members who in turn service the end users/consumers. For efficient channel management, logistics acts as a key enabler. In fact, channel and logistics management should go hand-in-hand for effective and efficient physical distribution system. Meshing of the two requires good coordination and intelligent interface. As pointed out by Bert Rosenbloom (1999)³, there are four major areas of interface between channel and logistics management.

- Defining logistics standards as required by the channel members
- Designing the logistics programmes as per the standards
- Implementation of the programmes
- Monitoring of the programmes

Any logistics programme beyond the demand of channel members makes mismatch in the system and invariably increases the cost of operations without any tangible benefits. The marketers or manufacturers should determine the type and level of logistics service required by each of its channel members such as wholesalers, distributors and retailers. For example, ITC, an Indian cigarette giant,

serves the pan shop (100,000 nos.) twice a day through its wholesaler, looking into his daily sales and financial capacity to purchase goods against cash. As against this, large distributors are served once a week with full truck loads from the factories. The proper implementation of the designed logistics programmes as per the standard required and defined will enhance the confidence level of the channel members in servicing his clients. Proper implementation of such programmes through the logistics service provider, who is invariably not so conversant with the company's channel culture and channel policy, requires an intelligent interface for the integration of the service provider with channel structure. At last, the monitoring of such a programme through periodic audits becomes necessary for synchronizing the operations to achieve the common goal. The mismatch shall lead to breaking the partnership.

Degree of Outsourcing

Business organizations resort to outsourcing in parts or in totality depending on the following.

- Existing infrastructure of the company
- Company's product portfolio
- Management policy for third-party involvement
- The anticipated benefits

Limited involvement of the service provider may be for only regional operations or for transportation or for warehousing and will come in the way of identifying the source of mistakes in the event of complaints at the customer end. In such cases, it is better to have an action plan for trouble shooting of anticipated problems. The areas of responsibilities and authority should be clearly demarked at both outsourcer's and service provider's end. Hence, with more degree of involvement of service provider, he should be given a free hand to resolve complaints and curtailment. He should be given access to the company's knowledge pool if so required; even the firm feels it is a guarded secret. If not done so, it will become a road block in customer satisfaction.

Legal Aspects

For reverse logistics of rejected goods at customer end during warranty period or goods wrongly dispatched, or return of goods damaged during transit, the policy guidelines have to be prepared to sort out the excise and sales tax problems. The responsibility and authority of the persons (from both the organizations) dealing with the government authority have to be clearly demarked to avoid the legal trap.

SUMMARY

Business world over is struggling for competitiveness in rapidly globalizing economy. Due to competitive pressure, corporations started going in for alliances and outsourcing non-core operations to outsiders who are experts in doing that job with efficiency, effectiveness and at reduced cost. The service provider is third party in the seller–customer link and is

capable of fulfilling the changing needs of the customer with his expertise and experience. These service firms have their core competency in logistics operation. The third-party service providers—3PL firms—are either ‘wholesaler’, that is, experts in one area of logistics, such as warehousing, transportation, inventory management, etc. or ‘Integrators’ who provide comprehensive logistics service solutions for the entire supply chain. The benefits of outsourcing are many: cost reduction, enhanced customer service, reduction in liability and risk, wider coverage, etc. The new emerging trend in logistics outsourcing is the ‘Fourth-Party Logistics’. Fourth-party logistics is the next generation of supply chain outsourcing. The activities in supply chain are very complex and information based. Due to globalization, companies have started globalizing their supply chain. With the complexity of supply chain, activities have grown considerably. This cross-border supply chain needs proper coordination and synchronization. To achieve this, a new generation of integration needs to be deployed, which is currently beyond the capabilities of traditional sourcing methods. This can be done by availing of the services of fourth-party logistics (4PL) vendors. 4PL service provider assembles and manages the resources, capabilities and technology of its own organization with those of 3PL service providers to deliver a desired solution. The genesis of 4PL lies in forming a collaborative relationship amongst various logistics service providers based on the backing of information technology. The other alliance arrangements are retailer–supplier partnership and distributors’ integration. The selection of alliance partner depends on the objectives to be achieved, service provider’s capability, expertise, past experience, customer base, infrastructure, technology base, reliability, etc. Partnership is formed with a written contract, which can be honoured in the court of law. The other critical issues to be considered while forming an alliance are switching cost, existing channel integration, degree of control and legal aspects. Management’s decision on outsourcing can be justified by value proposition.

REVIEW QUESTIONS

1. ‘Alliances in supply chain will considerably enhance the competitiveness of an organization.’ Explain.
2. What is ‘Third-party logistics’? Explain its role in the supply chain.
3. Discuss the various issues in strategic alliances.
4. ‘Fourth-party logistics is the extension of third-party logistics with value addition.’ Explain.

INTERNET EXERCISES

1. For studying the solutions, for logistics outsourcing visit <http://www.ups-scs.com/support/logistics-outsourcing.html>
2. Visit <http://www.findarticles.com>, for studying strategic alliances scope, challenges and issues.
3. Visit <http://www.logisticsviewpoints.com/category/logistics-outsourcing>. This site provides logistics professionals with clear and concise analyses of Logistics Trends, Technologies and Services.

PROJECT ASSIGNMENT

1. In the mobile telephony service supply chain, companies like Reliance, Vodafone, Tata, etc., enter into strategic alliances to gain competitive advantages. Take up any company and study the nature of alliance and how they leverage it to competitive advantage.

VIDEO LINKS

1. Supply chain collaboration—Professor Richard Wilding <http://www.youtube.com/watch?v=K68zKQFRmec&feature=related>
2. Logistics outsourcing—great insights <http://www.logisticsbureau.com.au/video-supply-chain-logistics.htm>

NOTES

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2. Sople, V. V. (2010), *Logistics Management*, Pearson, New Delhi, 2nd edition. pp. 196–198.
3. Rosenbloom, B. (1999), *Marketing Channels*, Florida, FL: Harcour Brace College Publishers, pp. 399–423.

AGILE SUPPLY CHAIN

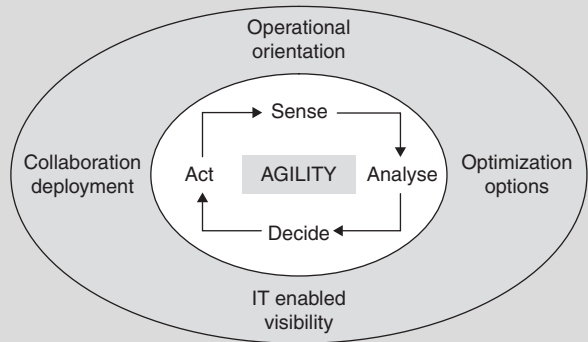
“Agility is the ability to cope with unexpected changes, to survive unprecedented threats of business environment and take advantages of changes as opportunities

—Sharif and Zhang¹”

Chapter Objectives

To understand:

- Market volatility
- What is agility?
- Agile supply chain
- Role of customer service in supply chain strategy
- Customer service perception and service design



Re-engineering to Agility

Today's global business environment is characterized by turbulent and volatile markets, shorter product life cycles, more demanding customer requirements and a variety of supply chain risks. In this environment, organizations seek new strategies to address risks particularly the volatility in the markets to enhance supply chain performance. The risk attached to long- and slow-moving logistics pipelines is very high. This forced organizations to look at how their supply chains are structured and managed. To address volatility, the supply chains are designed to be 'agile', which is described as being able to respond to sudden and unexpected changes in markets.

Re-engineering a Responsive Supply Chain

Ramon Cycles (RC) was a leading player in bicycle market since its inception in the late 1940s till 1990s. They produced a standard bicycle for local transport for low-end markets. However, due to economic liberalization policies, imports became cheaper from China and lot of local manufactures established in the markets over the years. Being an established player, Ramon Cycles had high overheads in the industry. As the bicycle became commodity product, Ramon's high-priced products contributed to the loss of its market share and poor financial performance in subsequent years.

The new CEO reviewed the situation and took note of the strengths in Ramon and decided to exploit these strengths to get over the problem. These strengths were: technical expertise, automated manufacturing setup, skilled workforce, strong brand name and a countrywide network of dealers. They surveyed the market and found that there is lots of demand for high-end bikes for sports, joy rides, trekking, health care and picnic segments. They focused on those segments and used the company's strengths to develop a responsive supply chain that could supply segment specific bikes. They could charge higher prices for these bikes. This strategy was for avoiding the high risk of commodity trap that resulted from their short-life cycle and low cut-throat pricing.

With the new strategy, customers would find numerous options for a bike covering: size, colour, components, special accessories for the particular customer segments, etc. The customer could make his choice at dealer's place through catalogues and models. The order then was placed to the factory, wherein with automated production set up bikes could be manufactured. The delivery of bicycle was ensured within 2 weeks. The numbers were less but margins were high to sustain and grow. Ramon, with the new strategy of make-to-order and delivery of 2 weeks for finished product, kept its inventory to a minimum as compared to its rival producing commodity (standard) product. Ramon re-engineered its supply chain to meet 2-week delivery time consistently. This is a good example of the responsive supply chain which tackled uncertainty in an innovative way.

INTRODUCTION

It was perception earlier that agility in the supply chain can be achieved through collaboration with suppliers. However, in practice collaboration with customers is also required. The enhanced responsiveness is a major characteristic to be exhibited by an agile supply chain. This characteristic is in addition to the cost-effectiveness, quality and smooth operations flow.

The agility is a capability of business that needs to be acquired through organizational structures, information systems, logistics processes and organization culture. A key characteristic of an agile organization is flexibility. The origin of agility as a business concept lies in flexible manufacturing systems. However, the automation in business may not make the business flexible. It requires the orientation of the people in that organization to work for flexibility. Thus flexibility is wider business concept.

The goal of the agile supply chain should be to carry inventory in a generic form, that is, standard semi-finished products awaiting final assembly as and when required by the market. This strategy is

called as postponement, which involves the differing of final configuration of product till order arrives. The advantages of postponement strategy are as follows:

- Inventory is held at a generic level resulting into a smaller number of SKUs.
- Generic inventory ensures flexibility in building final products varieties.
- Forecasting the inventory at a generic level is easy.

The large number of product varieties can be offered with generic level of inventory, which can be mass manufactured with accurate forecasting. This results into reduction in SKUs ensuring better flexibility in offering the final product variants in the markets. In other words, it is mass customization.

WHY AGILITY?

In the volatile markets, product innovation is an important factor for survival. Those companies will be successful that can implement innovative ideas quickly and profitably. For the company's growth new products (innovations) and new markets are the key factors. Globalization has greatly impacted the way firms do business today. With the advancement of technology the companies can collaboratively design, manufacture and market products anywhere in the world. This has created both new market opportunities and difficult challenges.

The importance of time as a competitive weapon has been recognized in the Industry. The ability to meet the demands of customers in the stipulated delivery times is of critical importance in this era of time-based competition. To become more responsive to meet the requirements of customers, companies require a high level of manoeuvrability in addition to inventory velocity. The firm today needs to be agile to counter variability (Exhibit 16.2). According to Martin Christopher:

Agility is a business-wide capability that embraces organizational structures, information systems, logistics processes and, in particular, mindsets. A key characteristic of an agile organization is flexibility. Agility is needed in less predictable market environment, where demand is unpredictable and requirement for variety (product SKUs) is high. Agility might therefore be defined as the ability of an organization to respond rapidly to changes in demand both in terms of volume and varieties.²

From the above definition it can be concluded that for an organization to be agile, it should be demand driven rather (Exhibit 16.1) than forecast driven. The information flow and sharing between the supply chain partners is a must to take real-time decisions to meet real-time demand. The information sharing between supply chain partners can be used for gaining competitive advantage through process integration. This will help in collaborative working between buyers and suppliers in different business processes.

Supply chain agility is an operational strategy to get velocity and flexibility in the supply chain. All organizations have supply chains of varying degrees, depending upon the size of the organization and the type of product manufactured. They have networks of supplies, service providers, distributors and customers. It is necessary to coordinate the activities of all partners in supply chain to have seamless integration to respond to the market requirement. To infuse agility in supply chain, it requires radical change in organization thinking and orientation to shift from traditional production orientation of economies of scale and EOQs.

Agile supply chains are market-driven. These organizations go for continuous product research and short product development and launching cycles. All activities are performed quickly and accurately. Here, the key success factors are flow (material and information) and time.

Exhibit 16.1 *Agile vs. Traditional Supply Chain*

<i>Traditional Supply Chain</i>	<i>Agile Supply Chain</i>
Multiple inventory decision points	Few inventory decision points
Forecast driven	Demand driven
Push strategy	Pull strategy
Limited use of IT	IT enablement
Functional approach	Cross-functional approach
Activity driven	Process driven

Exhibit 16.2 *Sources of Variability*³

<i>Variability in</i>	<i>Reasons</i>
Demand	Changes with customer tastes and preferences will impact demand in terms of product volumes and varieties. It creates discontinuity in production
Supply	Many suppliers at multiple location and having varied lead times
Product	New product introductions, product modification or substitute products
Equipment	Variations in equipment reliability and up times and maintenance scheduling
Manpower	Labour turnover, absenteeism and efficiency resulting into an uncertainty in production
Information	Disruptions in its journey from market to suppliers via manufacturing. Information reaching upstream of supply chain may be delayed and completely distorted

With this, the supply chain can respond to changing conditions arising out of market variability. Agility requires a company to be dedicated in its response to the changing needs of the market. Time-to-market is short. The agile supply chain is one which manages the variability. Information systems or technology play an important role in making the supply chain agile and responsive. Achieving agility starts with the physical flow of parts, from the point of supply, through the factory and shipment through agile distribution channels. It emphasizes bridging the gap between each point in the flow. It requires removing of non-value-adding activities/functions to induce velocity. The natural points of delay are eliminated and simplified.

The information chain is streamlined and electronically connected at every decision-making point, so that information flow is direct without interruptions and delays. This helps to take real-time decisions based on accurate and timely information.

BARRIERS TO AGILITY

The biggest barrier to agility is complexity of the supply chain. As companies grow the number of decision points increases leading to non-uniformity of decision-making due delays in information flow. The complexity is due to organizational structures and management processes that have grown up over time. It is also due to product complexity a design issues and excessive product varieties. The hierarchy

or levels of decision-making add further to complexity in the supply chain. The solution is BPR (business process re-engineering). BPR highlights non-value-adding activities that are inherent in traditional functionally-based business. These activities need to be eliminated to enhance agility. To bring agility, organizations require multi-skilling in HR force and cross-functional working approach. Team management will be an important facilitator for organizational agility.

CREATING AN AGILE SUPPLY CHAIN

Agility of an organization can be defined as the capability to produce products for volatile markets in response to the volumes and varieties. The firm's quick and effective reactions to changing markets require enablers to manufacturing process, which are:

- Collaborations with supply chain partners
- Internet-based business processes and information system

Agility brings competitiveness to the organization. The agile supply chain has four underlying principles: delivering value to the customers; being ready for change; valuing human knowledge and skills; and forming virtual partnerships. Agile manufacturing in addition to being flexible and responsive to market demands requires an adaptive capability to be able to respond to future changes. This includes development of internal capability and ability to configure the company's assets (human and capital) to take advantage of opportunities available today and also those that will surface in future. For creating 'agile supply chain' the firm needs a framework to improve supply chain performance by re-engineering existing industry processes and changing old practices.

For an agile supply chain, flexibility is a prerequisite and it should exhibit the abilities to:

- Reorganize and reconfigure assets in line with customer's need
- Change processes as demand changes adjust capacity

The logistics process flexibility involves alliances with third-party logistics providers and it should have ability to:

- Align to global requirements
- Provide differentiating services to each customers
- Vary warehouse space
- Vary transportation carriers
- Introduce product postponement
- Meet changing channel needs

To infuse flexibility in supply chain, firms have partnership with multiple logistics and channel service providers across the chain for delivery to specific customer locations. It should exhibit flexibility in product movement between nodes in the supply chain. Changing customer demand may translate into changing load and delivery requirements and these requirements may be in many forms such as shipping volume or schedules. As these requirements change, carrier cost and service time may also change resulting in the need to consider other carriers. Hence, a flexible supply chain, then, must be able to balance load and delivery requirements against carrier capabilities and costs.

Box 16.1

Tesco: Supply Chain Agility for Achieving Global Competitiveness

The business transformation of Tesco in the last four decades is one of the most remarkable in the British retailing industry. Changing with time, Tesco with appropriate planning infused achieved agility in supply chain to sustain and grow in international markets. From being a small retailer in local market, the company has become one of Europe's leading retail businesses, with retail operation in countries such as Ireland, Poland, Malaysia and Japan. In the United Kingdom, their 'loyalty card' and 'e-commerce' operations are considered to be world's leading and its expertise in this field.

The logistic and supply chain transformations have been the most important and impressive one, but they received less public consideration than the transformation in the location and format of Tesco. With passage of time, the company increased its services and range of products in the stores and online. In the past, the strategy of Tesco was to establish control over distribution. Tesco made its name by the operation of a 'pile it high; sell it cheap' approach to food retailing. Price competitiveness was critical to this and fitted well with the consumer requirements of the time. The company and its store managers were empowered to be individual entrepreneurs.

Tesco found a way with the 'operation checkout' to be closer to its consumers. Tesco adopted a new positioning, which helped itself to change its image and control its business. This retail transformation brought into sharp focus on the quality and the capability of Tesco supply systems and the relationships with suppliers.

In the 1990's, Tesco changed its business strategy. Earlier, Tesco focused on out-of-town superstores but later Tesco decided to begin a multi-format approach (hypermarket, supermarket, down town stores, etc.). The corporate brands have been strongly developed and Tesco decided to target the international market. All of this strategy success is due to the supply and the logistics management support. The distribution and supply of the products direct to Tesco's different stores has been the key to its success.

There have been many phases in the reconfiguration of the distribution strategy and operations. First, there was a period primarily of direct delivery by the supplier to the retail store. Second, there was the move to centralize regional distribution centres for goods and the refinement of that process of centralized distribution. Third, a composite distribution strategy was developed in the 1990s. Fourth, is the advent of vertical collaboration in the supply chain to achieve better operating efficiency.

With direct to store delivery process (DSD), suppliers and manufacturers went directly to the store to deliver. The store managers build their own relationships with the suppliers and manufacturers. The control was local. Tesco did not have central control and standardization. The same products were not found in all the Tesco's stores. The system could not contribute to the growth of Tesco. When the 'operation checkout' began, Tesco changed their supply chain process because the volume of products and quality was not enough with the DSD models. Tesco decided to centralize its distribution system.

The centralization of Tesco began with the diminution of the responsibility of the store managers. Tesco HO now had a control over products, pricing and stocking decisions. The centralized system became more and more efficient effecting delivery of all products within 48 hours. This strategy produced a more rationalized network of distribution centres linked by computer to stores and head office.

Tesco built its own electronic system to follow its supplier. This system based on the sales allowed Tesco to generate order for delivery to store within 24 hours. The change to the supply chain has been

concentrated mainly on the distribution centre to the retail store component. This change popularized Tesco as a model of supply chain and logistics expertise.

The supply chain model of Tesco is a success story. The way Tesco chose to graduate itself to be an international player was a very good approach. The credit goes to the timely reforms in supply chain management practices which were well planned. Tesco had always kept supply chain in forefront to build up competitive advantage leading to growth of the company.

As regards total flexibility in supply chain, both the product and the process are required to be modular in design and operations, respectively. Alternatively, the organization should adopt product standardization and keep stock of only few products.

The other flexibility required in supply chain is flexibility of network. It is the ability to meet the changing needs of customers. It requires changing the supply of product, including mix, volume, product variations, and new products. This further requires flexibility in sourcing product from raw materials to outsourced finished product. Supply flexibility influences agility. This covers the ability to add and remove suppliers. It addresses the ability of an organization to select suppliers who can add new products quickly, the ability to vary supplier relationships and the ability to have suppliers make volume changes.

The firm should select suppliers who can introduce new products quickly can add responsiveness to a supply chain. The most vulnerable aspect of product development in many companies is the failure to use the creative potentials of suppliers. Suppliers' involvement could contribute to new product success.

To bring about all the above changes, the organizational commitment and information systems capability is at the root. To achieve agility, employees should have right mindset greatly influenced by employee empowerment, organization structure and culture. Organizational designs that need to be considered with respect to supply chain flexibility are the organizational structure, the human resource practices and the workforce capabilities. HRM practices of the organization must support the competitive strategy of the organization and help develop the organizational capability to adapt to environmental changes. Culture can serve as a strategic resource in supply chains. Trends, such as globalization and technological innovation make business environment more challenging and further require the supply chains to act in a more integrative way.

Information sharing between the supply chain partners is crucial to flexibility. Information visibility within the supply chain means sharing information to manage the flow of products and funds in real-time between all supply chain members. The overall goal for the supply chain is to have visibility across the supply chain.

The cycle time compression is essential to flexibility. Hence, it is necessary to compress the four cycles such as procurement, manufacturing, replenishment and order fulfilment. Most organizations face a fundamental problem: the time it takes to procure, make and deliver the finished product to a customer is longer than the time the customer is prepared to wait for it. This difference in time is referred to as the lead-time gap. In the flexible supply chain design this needs to be taken care of and reduced. For the organization which matches demand with supply the process of forecasting and inventory management becomes redundant.

The biggest barrier to agility is the long lead-times from suppliers. This can be addressed by reducing supplier base and building a long-term relationship with the suppliers. The organization should consider the supply chain of supplier as an extension of his supply chain. M&M has reduced its suppliers' base to 500 from 1800 earlier and shared information with suppliers on a continuous basis.

To bring about these changes the traditional methods of procurements need to be replaced with e-procurement to bring speed and reduce transaction cost so as to remain competitive in product offerings. Hence, the organization should take a close look at the way things are procured. This can be done by outsourcing non-core activities to outside experts and focus only on core activities to use time and money productively.

Mapping the supply chain processes is the first step in understanding the scope for improvements in productivity through logistics process. In logistics, the non-value-added time is idle time. The product remains as inventory either as raw material, semi-finished product or finished product. In such a situation, if a product is produced early before the actual demand arises, then postpone the final configuration of the product. This would enhance the supply chain flexibility to meet the demand. However, for implementing postponement strategy, the organization should have a clear view of the demand in the marketplace. Hence, for enhancing agility the organization should be able to capture information on demand as close to the final point of consumption. This will help in managing the inventory at the optimum level to cater to the flexible demand.

A close connection with the marketplace through shared information enables a more responsive supply chain to be created. For example, at Cisco Systems, a fast-growing provider of network routing and switching equipment, orders are entered by customers directly through the internet. Typically these orders are for

Box 16.2

Addressing the Uncertainty

Sport Obermeyer is a leading US fashion skiwear producer and distributes its products through 800 specialty retailers located throughout the United States. Every year the firm has 95 per cent of its products which are new. Because of this reason, it constantly faces the challenges and risks of demand uncertainty. This resulted into stock-outs of running styles during the selling season and leftover inventory of old fashion at the end of the season.

To reduce the uncertainty, the company solicited early orders from important customers, that is the retailers. The company invited its 25 largest retailers to its factory each February to evaluate its new line. The company found that the early orders from these retailers helped to bring down forecast error to the level of 10 per cent. This helped them to get this information in advance but it didn't solve company's problem. The reason is long lead-time to commit itself to products at the time of planning.

As per their calculations, one day reduction in lead-time would have accrued saving of \$25,000. This is the amount they spent each day at the end of September shipping products by air from plants in Asia to reach in retail stores by early October when the season starts. To shorten the lead-time, they acted on improving information flow speed to get design information to the production manager in Hong Kong. Through such efforts, company was able to avoid uncertainty on half of its production by committing that production after early orders have been received in February.

To tackle the problem of uncertainty in styles, the company asked each of the six members of the committee responsible for forecasting to construct a forecast for all products, and used the average of the six forecasts as the company's forecast. After a year, it was found that the average forecast was accurate in the case when the six individuals forecast agreed. All these efforts resulted in cost cuts of over production and under production as a result profits increased by 60 per cent.

uniquely configured products which are assembled to order. Because Cisco's suppliers are linked directly to the information system they are immediately alerted of the requirement for components. At the same time, the third party logistics service provider is informed of the impending shipment requirements. As a result, customized products can be delivered and installed in much shorter time-frames.

These information linkages need to be based on, latest technology to share information across the supply chain partners. The key requirement is the willingness of all parties in the supply chain to act as partners and to recognize the need for open communication. The biggest improvements in supply chain can come through a change of attitude amongst the parties involved.

Dell computers has successfully introduced agility in supply chain. A customer initiates the sales process on the Internet. The Dell receives the order and begins manufacturing within hours. Each computer is custom-built and put through several hardware and software tests in less than one day. After an inspection, the computer is dispatched to a distribution centre to further transport customer. Thus, Dell computer keeps computers inventories to a minimum. This has helped the company to optimize the cost. Dell has rationalized the supplier base and suppliers are kept just functional to meet the need. The real-time information from customers enables Dell suppliers to change product mix and maintain their inventory velocity. Dell is transparent in sharing the customer demand information with suppliers. This is the root cause of Dell being an agile supply chain.

SUMMARY

The changed conditions in the global marketplace demand a much more agile response from the organization and its partners in the supply chain. True, competitive advantage is gained when an organization is able to consistently meet the needs of customers more precisely and in a more timely way than anyone else. Today, it is no longer companies competing against company but rather supply chain against supply chain.

In the dynamic environment, companies would survive based on their ability to keep up with continuous and unexpected changes. This has led to an increase in the importance of agility in supply chain. By being flexible, the supply chain can meet specific customer needs and influence the long-term commitment of customers. In terms of customer satisfaction, flexible supply chains should lead to improvements both in terms of service and responsiveness. Supply chain responsiveness factors involve velocity in execution.

For innovative products and firm with mass customized product, satisfying the customers will require a responsive supply chain that can manufacture closer to the time of the order. Customers today are demanding more variety, better quality and service, including both reliability and faster delivery. The flexible supply chain requires lower inventories at each node as product is produced closer to customer consumption and lead-times are reduced. Inventory in transit is reduced as supply chain partners choose modes of transportation that shorten lead-times.

One of the biggest challenges organizations are facing today is the need to respond to ever-increasing volatility. For a variety of reasons, product and technology lifecycles are shortening, competitive pressures force more frequent product changes and consumers demand greater variety than ever before. To meet this challenge, organizations need to achieve greater agility such that it can respond in shorter time-frames both in terms of volume change and variety change. In other words, it needs to be able to quickly adjust the output to match the market demand and to switch rapidly from one variant to another.

REVIEW QUESTIONS

1. Discuss the dimensions of 'agile supply chain'.
2. Compare and contrast 'lean' and 'agile' supply chains.
3. Explain, how 'agility' can be induced in the supply chain?
4. Discuss the various issues in agile supply chain.

INTERNET EXERCISES

1. To learn more on agile supply chain visit http://www.marcbowles.com/courses/adv_dip/module4/module11/module11five.htm
2. <http://www.logistics.about.com>; presents many articles and case studies on supply chain management in various industry segments. Visit the site to study techniques used for bringing agility in supply chain.

VIDEO LINKS

1. The agile supply chain http://www.youtube.com/watch?v=_Ct7azFYQw8
2. Cisco's agile aftermarket supply chain <http://www.youtube.com/watch?v=nbWaIgv3f7s>

PROJECT ASSIGNMENT

1. Consider two competing retail chains; Wal-Mart and Spencer or Pantaloon and Westside (Trent). Find out the factors, which decide the agility of supply chain for the retail chains. Measure these factors giving proper weightages/ratings on 1 to 5 scale for the companies you have chosen. Comment on the 'agility index' of the supply chains of these companies and suggest strategies to improve the same.

NOTES

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GLOBAL SUPPLY CHAIN

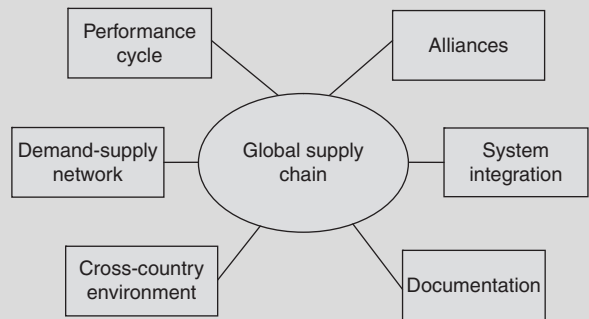
“The Dell theory of conflict prevention argues that the two countries that are both part of the same global supply chain will ever fight a war as long as they are each part of that supply chain

—Thomas Friedman”

Chapter Objectives

To understand:

- Global market forces
- Local vs. global supply chain
- Risks in global supply chain operation
- Strategic issues in global supply chain
- Operational factors
- IT role in global supply chain



Elements of Global Supply Chain

Global supply chain encompasses cross-border transactions and movement of goods. Due to globalization, for survival and growth, many local firms today have decided to go global. Hence, global supply has become the reality for these firms. The main thing one should keep in mind for global supply chain management is various cross-border rules and regulations, policies and bottlenecks (infrastructure, tariffs, etc.), to integrate the business processes and control them to cost-effectiveness and system efficiency. The major points to focus are integration of logistics activities, network facilities, various service providers and management of procurement, manufacturing and distribution.

R-Toys: Global Supply Chain Freight Consolidation

R-Toy is a worldwide leader in design, manufacturing and marketing of toy products. The company's core brands are Jennie, Race Car, G-Puzzle and My Doll. The company is having offices and facilities in 36 countries and sell products in 95 nations. They are catering to three main Asian and Asia-Pacific markets. The products are manufactured in their Malaysia (Race Cars), Indonesia (G-Puzzles), Italy (Jennie) and Mexico (My Doll) plants and shipped to Singapore, India and Japanese markets. The country requirement is sourced on the individual plants for the products they manufacture. The material is shipped in containers to the respective markets at each destination. Indian market receives 40 shipments (containers) from Mexico, 24 from Italy, 48 from Indonesia and 60 from Malaysia. Each plant on an average dispatches 120–160 containers covering 38–50 cubic metres of volume costing \$1,00,000–1,16,000 freight for each market. The cargo in most of the cases being less than a container load, the shipping charges works out to be 18–25 per cent higher than full container load shipping. For three markets, R-Toy is spending & 325,000 on freight. In a bid to reduce the freight, R-Toy planned for multi-country freight consolidation and reduced the number of shipments by consolidating the cargo for three different markets and shipping to a 'hub' and distributing it further to individual markets through the 'spoke' by deploying a Hub and Spoke system of distribution. R-Toy chose China as a destination hub (Figure 17.1) for the consolidated cargo and there from the individual markets requirements were fulfilled.

With the new arrangements, the shipments from each plant have reduced to 58 to the hub and that too with full container loads. This multi-country freight consolidation reduced the freight cost to \$265,000 and ensured a saving of \$60,000 per annum.

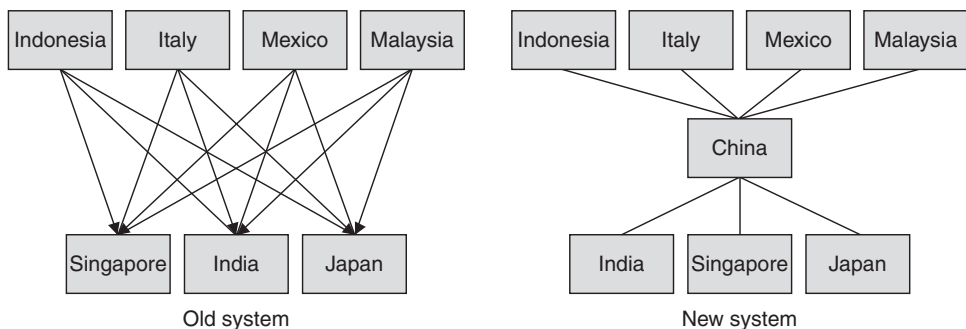


Figure 17.1 Old and New Freight Consolidation System

INTRODUCTION

Global operations of a business increase the complexity of the supply chain with an effect on cost of moving goods and servicing customers. With complexity, elements of uncertainty creeps in and that decreases the capability of control in movement of goods due to large distance, varied customer demand, diversity in working environment and multiple documentations. The global supply chain

must accommodate all the uncertainties associated with the above factors which are possible by aligning the capabilities of all multiple agencies having expertise in facilitating cross-board movement of goods.

In the present situation when there is cut-throat competition, one has to find out ways to provide the best service at a minimum cost. To make this possible the company has to find partners at proper locations to get cost advantages. For example, China is a manufacturing hub while Dubai and Singapore are for trans-shipment at the free trade zones (FTZ). Today the components/raw materials are brought from various global locations, assembled/processed in other countries and sold in international markets. Hence, the aspects which are important in global supply chain are speed, flexibility and cost-effectiveness. This needs capabilities of material suppliers and service providers.

Differentiation is the only strategy for survival and growth in the global market. The way to develop competitive advantage is alliances with service providers to make customized service offerings to customers. This can be done by understanding customer's needs and expectations. Here, the company has to focus on its internal operations to be more responsive to accommodate customer's needs.

Another concept for global supply chain is distribution channel, which helps firms make their goods available to the customers spread across the globe. Distribution channels are not alike in all countries. For example, in India and France the channels of distribution are not the same for a similar category of products.

In the global supply chain, smooth movement of goods is critical and is dependent on the transportation. In addition, customs rules, logistics regulations, government policies and tariffs, very much impact the effectiveness and efficiency of international supply chain. The importance of 3PL and 4PL service providers cannot be ignored in successful global supply chain operations.

DOMESTIC VS. GLOBAL SUPPLY CHAIN

With globalization, the trade barriers have reduced and the competition has changed dramatically. In order to remain competitive in global market, firms have begun to implement two strategies by seeking supplies and productions on a global scale and reducing operations in-house via outsourcing and strategic alliances. With global sourcing, companies take advantage of the unique conditions existing in different countries, such as low wages, raw material availability and proximity to markets. In addition, outsourcing strategy gives firms more flexibility to react to the increased volatility in marketplace. Global supply chain management usually involves dealing with many countries with both opportunities and threats. There are four major differences between domestic and international supply chain (Exhibit 17.1).

1. **Performance cycle:** In domestic markets, the transit times and total performance cycle varies in days. However, in global SC, the performance cycle varies in weeks or months. The reasons for a longer performance cycle are communication delays, financing requirement, special packaging requirement, ocean freight scheduling, long transportation times and custom clearance.
2. **Documentation:** International operations require multiple documents for goods clearance through customs.
3. **Systems integration:** This includes the ability coordinate various supply chain processes to fulfil the orders in agreed time frame.
4. **Alliances:** International alliances can provide market access and expertise and helps to reduce the risk in global operations.

Exhibit 17.1 *Global Logistics Challenges*

<i>Parameters</i>	<i>Domestics Operation</i>	<i>Global Operation</i>
Performance cycle	Shorter	Longer (due to greater distances, customs clearances, more intermediaries, more use of slow sea travel)
Documentation	Simple	Complex (for customs, banking and foreign exchange clearance requirements)
Alliances	Few	Multiple (on global front with logistics service providers, distributors and manufacturers)
Information flow	Simple	Multiple channels, varied standards, alternative languages

Box 17.1***International Trade—Port Operations***

The port operation starts with the ship reaching the port area. The ship has to call upon the port where it wants to dock to get the permission to anchor on the port. The ship will get the permission only on the submission of the Import General Manifest (IGM) to the customs (through port authorities). IGM is the list of materials which will be unloaded at the said port with details like the loading port, consignor, consignee, quantity, etc. After the ship submits the IGM, it has to clear the port dues, before it gets docking facilities. Normally the shipping company's agent does this before the ship reaches the port. The accounts department of the port gives the clearance to the traffic department, who will thereafter allocate berth to the ship. Thereafter the port gives anchorage permission to the ship and sends a pilot to the ship. The pilot is a person who actually manoeuvres the ship onto the shore. A pilot is required because it is difficult to navigate in the port area, which at times has rocks, etc., at the bottom, unlike the high seas. Also, there is problem of silting at the seabed in the port area, and only the pilot knows the channel from where the ship can be brought off the shore. Once the ship reaches the shore, the actual port operations start. Port operation is basically unloading and then loading the cargo from and into the ship. Stevedores are the people who handle the cranes and other equipments on the ship and these are supplied by the port authorities, to the unloading and loading operations. Once this is over, the shipping agent, on behalf of the shipping company, has to give an Export General Manifest (EGM) to the customs (through port authorities). This is the same as IGM except for the fact that this list for the cargo loaded on the ship for exports. Once the port gives clearance, the pilot will navigate the ship back into the high sea.

STRATEGIC ISSUES

Today, sourcing components or products from 'low-cost countries' have almost become a common practice across every segment of the industry. The competitive pressures on pricing are so demanding that the procurement department seeks to the lowest possible raw material component cost without compromising on the standard of quality.

The major strategic issues in global supply chain are:

- Available infrastructure
- Alliances for outsourcing
- Safety and security
- Skilled human capital
- Responsiveness
- Technology

Time is another big issue that should be addressed when dealing with global supply chain management. The productivity of the employees and the extended shipping times at the sourcing country may positively or negatively affect the company's lead-time. However, lead-times need to be taken into consideration into the company's procurement plan. The weather conditions on one side of the world often vary greatly from those on the other and can impact production and shipping dramatically. The customs clearance time and other governmental red tape can add further delays that need to be taken into consideration during planning stage.

Another issue that must be incorporated into a global supply chain management strategy is supplier selection. Comparing vendor bids from within the company's home-country may be difficult enough but comparing bids from global suppliers may be more complex. In addition, companies must make decisions about the number of suppliers to engage. Smaller supplier base may be easier to manage but could also lead to potential problems. If one vendor is unable to deliver as expected or if one vendor tries to leverage its supply power to obtain price increase then multiple suppliers may be a firm's choice. In general, the barriers to global supply chain are as follows:

- **Marketing/competition:** It includes entry restriction, information availability, pricing and competition. Entry restrictions limit market access by placing legal or physical barriers on importing. Pricing and the related topics of tariffs are other marketing-related barriers. International pricing is strongly influenced by exchange rates.
- **Financial barriers:** These mainly result from forecasting and institutional infrastructure. They result from major differences in facilitating intermediaries such as banks, insurance firms, legal counsellors and transportation carriers operations.
- **Distribution channels:** Infrastructure non-standardization at different parts refers to differences in transportation and material handling equipments, warehouse and port facilities, and communication systems. Trade restriction barriers can influence channel decisions such as the rules that restrict the volume of imports or increase duties once a specified volume has been reached.

To manage flow of inventory and information across the global supply chain is more complex than that of managing logistics operations within the country. It is because of the diversity in markets in terms of the following:

- Customer needs
- Economic and regulatory environment
- Logistical infrastructure

The dominant factors need to be identified during the planning stage and proper strategies to be evolved to overcome the barriers.

Internal Issues

Logistics planning For companies with global operations, logistics network planning is crucial for gaining competitiveness. The formulation of logistics network strategy will also depend on factors like unit value of the product, markets and competition. For example, when the firm decides on developing new markets and relocating facilities, the sourcing of raw materials becomes important as far as the delivery time frame, logistics cost and reliability is concerned. Therefore, formulation of logistics strategy should take into consideration the location of production facilities, sourcing of raw materials and components, and the product–market characteristics. For a truly global operation, the following are options for firms for planning their logistical operations.

- The firms may concentrate their manufacturing and sourcing at few locations despite of selling their products across the global markets. The examples are BMW, Thomson. These companies have major focus on the outbound logistics for distribution of their finished products in global markets.
- The other categories of firms are those having manufacturing at one place and selling the products in the global markets. These are the companies manufacturing very high-value products such as aircrafts and defence products. They source their raw material and components from all over the world. The major focus of these companies is on inbound logistics.
- The third categories are companies manufacturing mass-consumed low-value products such as soft drinks. Firms like McDonalds, Coca Cola and Pepsi adopt the strategy of supplying finished products to the markets from the local manufacturing plants. The products are sold within 200 kilometres of distance from the plants. Beyond a particular distance, the logistical cost makes the finished products uncompetitive.
- The forth category is sourcing and distribution from and to many locations. They focus on both inbound and outbound logistics. The examples are Japanese car and consumer electronics manufacturing companies such as Honda, Toyota, Sony, etc.

Inventory: make-to-order or ‘make-to-stocks’ The major shift in inventory planning is ‘make-to-order’ to deliver products directly to the customers to reduce the inventory levels. The approach to this is to consolidate the global production in a single or few focused factories to cater to the needs of various markets. The variations in needs of local individual customer or local markets are fulfilled through the strategy based on rationalization of product design. The local needs are taken care of through modular approach to product design, wherein the product can be configured to its final shape at the distribution centre catering to the local markets. This is a pull system wherein the execution is carried out after the customer places an order. The internal system is so flexible so as to gear for execution of orders for product volumes and variety.

Product variables The unit value of the product decides the reach of the logistical system. In the globalized marketing environment, the firms with low-unit value products invariably resort to a local manufacturing system for extending excellent customer service. For products like soft drinks, the distribution is mostly restricted within 200 kilometres of the bottling plant. The cost of serving the customer becomes uneconomical beyond a certain distance for such products. Configuration of the plants very much depends on the logistical reach of the product. However, in case of high-value products, the logistical reach is wider and the transportation cost as a percentage in total cost of product is insignificant and hence centralized manufacturing or distribution centre for better inventory control

can be thought of. This is an important consideration for consumer products sourcing and distribution network in the global markets.

Flexibility Invariably, the global players focus on the strategy of economies of scale for cost advantage. But this strategy has an element of inflexibility to respond to the dynamic market and demanding customers. The logistics system associated with the above strategy also becomes inflexible in responding to the changing distribution needs. For example, the emphasis on freight consolidation with few dispatch schedules has shifted to frequent and small consignments to reduce the inventory-related costs.

External Issues

Short lead-time In the global market, the emphasis is on responsiveness with lean supply chain. In such circumstances, the customers bank on shortest lead-time for the inputs going into his product manufacture to compress his performance cycle to extend superior customer service and simultaneously reduce overall inventory levels. However, in case of inflexibility in manufacturing system, the supplier needs to keep some buffer stocks to maintain the desired level of customer service sacrificing the benefits of lean inventory.

Transit time extensions and delays The freight cost is directly proportional to the speed of transportation mode. Air transportation may be obviously costlier than sea transportation. But inventory carrying cost over a longer period of sea journey will offset its benefits more, due to low freight charges. Moreover, this will pose constraints to the basic logistics principle of postponement. Documentation and customs clearance may further add to the cost of in-transit inventory shipped through slow-speed transportation modes.

ROLE OF IT SUPPLY CHAIN

IT-enabled supply chain management is a core strategy being adopted by leading business firms for developing sustainability. With the Internet, communication and information sharing amongst the partners is on time and seamless. High volumes of data can be transferred at low cost and with speed.

The Internet enhances the degree of coordination and communication between the partners. Internet-based technologies such as ERP, EDI, DRP, WMS, CRM, CPFR, etc., has improved the integration amongst the buyers and sellers many folds.

On the Internet platform, e-business has emerged enhancing supply chain efficiency and responsiveness by sharing real-time information regarding inventory, shipment movements, goods availability and demand between the partners. The speedier reach to relevant information in supply chain can help in reducing the costs both at the buyer and the seller ends.

Electronic Data Interchange

EDI is a method to automate the traditional paper-based, error-prone and time-consuming business transactions. EDI transfers the process and transaction documents. Just-in-time (JIT) implementation and its success depend on communication automation. The communication automation is possible with EDI. EDI also supports communication amongst the partners.

Intranet

Intranet is an organization's internal network for communication. It is done through LAN (local area network) or WAN (wide area network). Both the networks use Internet technology and communication

protected through company's firewalls. Intranet helps employees to share documents regardless of their location. Employees can share various things such as internal policies and procedures, training programmes, product catalogues, design drawings and manuals.

Extranet

Extranets have global reach. It is used for communication with two or more organizations. Extranet also ensures privacy, security and coordination amongst partners, suppliers and customers

B2B Collaboration

In the B2B markets, the organizations establish long-term relationships with customers. However, with the internet it has become easier to collaborate with partners more quickly than the traditional way. The internet facilitates online ordering and tracking, managing their logistics, sharing the forecast, demand planning, etc.

Trading (Internet Based) Hubs/Exchanges

Trading hubs are independently owned organizations (market places) bringing the buyers and suppliers together on a real-time basis. It enables the buyers and sellers to initiate and conclude the transactions or contracts. The internet hubs connect a large number of buyers and sellers according to the requirements of both. The advantage of the internet hub is enhanced speed and reduction of transaction cost. The membership of trading exchanges is by registration or by invitation.

Collaborative Planning, Forecasting and Replenishment

Collaborative planning, forecasting and replenishment (CPFR) is one of the fastest growing technologies for business firms to collaborate with its partners. CPFR involves working with network members to plan for forecast, production, sales, shipments, etc. CPFR is based on IT. CPFR aims at accuracy of demand forecasts and replenishment plans resulting into lowering inventory levels. CPFR success depends on the trust between partners. For it to succeed, (point of sales) partners must be willing to share their promotion schedules, POS data and inventory data. CPFR is proving to be a win-win situation for the partners involved in meeting the customers' demands, reducing inventory, lowering costs and improving the bottom line.

CROSS-BORDER LOGISTICS

As businesses continue to globalize, the attention has increasingly turned to logistics operation. The cargo movement needs to be done physically using the available means of transportation. The speed and efficiency of the movement of goods across the national boundaries of the nations depend on the available modes of transportation, their capacity and capability, inter-modal facility for movement, packaging and handling, and the logistical regulations in the countries where the buyers, sellers and carriers are located. In imports and exports business, for physical movement of cargo, the role of intermediaries (freight forwarders and customs-house brokers) are indispensable. The domain knowledge, connectivity with international cargo carriers and the documentations are the three crucial areas, which need focus in global logistics. As a consequence of globalization, managing logistics at a global level poses challenge of considerable complexity. The complexity of challenge varies

with business and the environment it is operating in. To face logistical challenges, there is no single standard solution available. Individual companies will have to understand both the strategic and management issues to bring in more clarity in knowing the implications of globalization of their business on the logistics.

Logistics is an important element of any business venture. If we closely examine the elements of the cost of a product, the raw material cost forms a significant portion, nearly 40–70 per cent of the total cost. Hence, materials must be procured and provided for processing in an optimal manner. The decisions like economic order size, reorder level, etc., are generally considered to minimize the cost of holding inventories. In addition to these dimensions, there is the cost of transportation of raw materials which in turn depends on the location of vendors and intermediate warehouse(s), if any in the system. Again, within the plant, the logistics play a major role to minimize the in-plant movements which help in minimizing time and cost of transportation. They will be dispatched to the market points either directly or through some intermediate finished goods warehouses. In international logistics system, the movement of materials/products will be in the following five stages:

- Vendors
- Raw material warehouses
- Plants
- Finished goods warehouses
- Markets

In the individual firm, logistics activities absorb a substantial portion of its operating costs. It has been estimated that in the recent years, physical distribution costs in India have skyrocketed to 30–40 per cent of the consumer prices of the products. As such, physical distribution has become the most important area for cost control. And savings on this account means a great deal to the Indian business firms operating under competitive conditions.

Although the cost of market logistics is high, a well-planned market logistics programme can be a potential tool in competitive marketing. Companies can attract additional customers by offering better services, faster cycle time, or lower prices through improvement in market logistics. Experts call market logistics the lost frontier for cost economies. Lower market logistics costs will permit lower prices to yield higher profit margins or both.

Effective management of physical distribution can help a company gain advantage over competitors through better customer service and/or can be the basis by which a firm gains and sustains a strong differential advantage. Without effective physical distribution no firm can survive in the market.

RISK ASSOCIATED

An international supply chain is conceptualized as a complex and dynamic system in which disruptions happens due to long shipping and lead-times leading to costs implications. There are four disruptions in the supply chain, which are as follows:

1. Demand-related
2. Supply-related
3. Manufacturing-related
4. Distribution-related

Demand-related disruptions are the most important source of instability in the supply chain. Production-related disruptions do not have a severe effect in terms of production shortfalls and demand fulfilment, although they do seem to create a need for significant international communication. Production-related disruptions decline over time, although demand-related disruptions may last for a comparatively longer period.

Distribution-related disruptions cause due unavailability of products at the place of demand. Here, the costs take the form of expedited shipping, inventory shortages and lower demand fulfilment leading to customer dissatisfaction, etc.

The supply (sources)-related disruption raises the costs, due to emergency purchases and also due to loss of opportunities in markets.

To understand the disruptions, a firm needs a much deeper understanding of the difficulties of various types of value chain activities. The impact of demand, production and vendor-related disruptions in one company is on the cost of integrating an international supply chain which is product and market specific. If certain product attributes make international production difficult, then we need to investigate more thoroughly how these attributes can be changed over time.

Perhaps the most neglected in area is the potential role of management in improving the performance of international supply chains. More sophisticated simulations could help management understand the trade-offs entailed in international supply chains and make better decisions concerning facility location, shipping mode, choice of vendors and inventory levels. Moreover, there is a need to examine ways in which managers can reduce the level of disruptions to the supply chain, thereby decreasing the costs and difficulties of international sourcing.

OPERATIONAL FACTORS

The major areas of logistical expertise required in cross-border cargo movement are, mode of transportation, cargo insurance, packaging, shipping documentation for customs clearance.

Transportation

Transportation mode plays a vital role in the movement of cargo within or between countries. Normally, cargo is moved using three modes of transportation: road, sea or air depending on the cost, urgency and the destination. However, for cross-border cargo movement mostly sea and air, as a mode of transportation, is preferred as most of the countries are connected well by air and sea. The road option is preferred when countries are connected by land and other options are either costly or not feasible. In the Indian subcontinent, the road is an important mode of cargo movement across India, Nepal, Pakistan, Bangladesh and Bhutan. In Europe, the railways are the important mode because of the availability of modern and efficient train systems.

For selection of the transportation mode, logistical manager should take the following into consideration:

- Location of the market
- Cost of transportation
- Speed of cargo movement
- Reliability of mode

The location of the market is the most important factor in deciding the transportation mode. For shipping goods from India to Middle East countries market, the best and most convenient mode of

Box 17.2

Global Logistics Hubs

Amongst the large distribution hubs in the world, Dubai and Singapore are heavily dependent on logistics. Dubai has recently commissioned the world's first truly multi-modal integrated logistics platform called Dubai Logistics City (DLC). Spanning over 25 square kilometres, DLC is the first phase of the new Jebel Ali Airport City and will become the world's first truly multi-modal facility for air, sea and road services. DCL will have a major impact on future air cargo growth in the region. It is to be the preferred location for businesses which require, or offer, logistics and multi-modal transport services to the countries covering Middle East, India, Africa and the CIS countries—a market of more than two billion consumers. Singapore's excellent infrastructure, strategic position in the Asia Pacific, and various goods and services tax (GST) relief schemes make it a natural logistics hub. Singapore is, essentially, a duty-free port. Except for liquor, petroleum, tobacco and cars, there are no customs duties on goods imported into Singapore. Its efficient customs procedures are major draws for companies using Singapore as a logistics hub.

Source: <http://www.dubaicity.com>, www.sedb.com

transportation is sea for bulk cargo (steel, food grains, ores, minerals and chemicals, automobiles, etc.) and air for high-value cargo (jewellery, electronic goods, office equipments, etc.).

The second factor is speed, where the cargo is required in shortest delivery time or urgently. Obviously, the cost of the air transportation is very high which should be evaluated and justified by the criticality of the need in terms of time and the opportunity cost. The floricultural and horticultural products, which are perishable in nature, are directly sent by air to the destination or to the place of consumption. The roses produced at various farms near Pune in India are sent to the Mumbai airport through temperature controlled vans for further dispatch directly to Holland and Middle East countries by air to reduce spoilage.

The third important factor is the cost of transportation in international travel. The cost to travel is directly proportional to the speed of travel. Air transportation is the costliest. However, the air cargo needs less packaging due to less handling, practically no exposure to hazardous storage, transit and handling travel conditions due to a short journey period. On the other hand, for sea cargo good packaging for cargo is needed to withstand the hostile storage, travel and handling conditions during a long journey. The packaging cost for sea cargo is very high. Due to speed (Air), the inventory in transit is less and importantly for a short period of time. This reduces the inventory level-related costs, resulting in faster inventory turnover.

In India, amongst all the transportation modes air accounts for less than 1 per cent of the total cargo movement (domestic), while rail contributes to 30 per cent, road 60 per cent and water 10 per cent. Each of the modes has its own disadvantages and there is no ideal mode of transportation. The choice depends on the urgency at the customer end and affordability for the costs involved.

The other advantage of air transportation is responsiveness. It can quickly respond to the urgent and unpredictable demand of parts or components. It imparts minimum transit damages to the cargo. In addition, the insurance cost is less than the other modes. Due to speed, the marketer can avoid the risk of opportunity cost.

Air transportation is traditionally confined to high-value density items, that is, the items having high selling price so that the transportation cost as a percentage of the price of the product becomes

insignificant. Secondly, the value of the cargo being high, the capital tied up in inventory in transit is released fast.

The air freight rates are charged on the basis of the weight and the sea freight rates are based on both weight and volume, whichever gives higher charges. The freight forwarding agents quote ocean freight rates. The rates quoted are based on both weight and volume. The sea freight rates are based such that higher revenue is generated.

The three types of shipping companies that undertake sea cargo transportation are as follows:

- Independent lines
- Tramp vessels
- Conference lines

The individual lines operate and quote rates individually and independently. They accept cargo from all shippers through the freight forwarding agents. Their services are limited to the route they operate on. However, they have fixed schedules on the fixed routes.

The second category is the tramp vessels which do not have any fixed route or schedules. They operate on charter basis. They are mainly involved in bulk cargo transportation.

The third one is the conference line, which is an association of shipping companies across the world. They join hands and have the common codes/rules for cargo movements, freight rates, shipping conditions, etc. The freight rates amongst the conference members are identical. They have a dual ocean freight rate system. They charge lower rates for contract shippers as allowed by the conference line. With the conference's permission, the member shipping company can operate its own vessel if the schedules are not available on the particular route for a longer time.

Sometimes it may so happen that under the regulatory condition of certain countries, the shipper may not have any option to shipping vessels other than stipulated by the authority. The mandatory use of a particular vessel is due to laws enacted to protect the interest of a particular country. In Japan, it is mandatory that all Japanese automobiles for exports to other countries should be shipped in vessels with a Japanese flag.

Insurance

Transit insurance covers property against loss or damage during its movement from one place to another. Insurance can be done for the goods transported in own vehicle or public transporter. The transport insurance policy indicates the type of transport being used for goods movement. For example, goods movement by sea transport is covered under marine insurance.

Cover Goods-in-transit policy will protect the insurer from:

- Theft during transit
- Loss of goods (physical) during transit
- Damage caused due to vehicle accidents

Insurance policies protect the insurer if goods are new but inferior, below standard, or damaged through inappropriate packaging or sub-quality goods are not covered under this policy. The insurance coverage are of two types: old-for-new items replaced at their current market value and indemnity cover-to-cover general depreciation.

Special features Insurance policies cover some special features. These special policies covers legal expenses, possessions in vehicle, food spoilage in freezers, garage cover and outbuildings cover. The special features incorporated in insurance coverage will increase the value of the premium. To reduce the insurance premium to a minimum, the insurer should leave off special features.

Level of risk The risk in insurance depends on the cost of goods-in-transit. If risk is very high the premium becomes expensive. If the past record shows a high level of risk then the consigner should take measures to increase the level of the security of goods.

Types of risks (INCOTERMS) These are the trade codes used in cross-border trade. These codes define the risk and cost obligations on the part of importers and exporters. It also explains about the transit insurance in international movement of goods. However, the obligation for insurance is at a minimum cover. The party can go for increased cover also depending on the needs.

In domestic cargo movements, the carriers take responsibility for transit loss or damage. However, in sea transportation the carriers do not take this responsibility because of the high degree of risk involved due to unavoidable perils during sea journey. The purpose of marine insurance is to protect the sea cargo against loss or damage in transit. The coverage of marine insurance is much broader as compared to domestic cargo insurance. The marine insurances are of two types:

- Open blanket coverage
- Special coverage (one time)

Open policy insurance coverage is for a specified period for estimated total value of the cargo that the shipper may dispatch. No record is kept nor are reports required for individual shipments made during that period. The premium is charged on the basis of the total estimated cargo that the shipper intends to move during the contract period of insurance policy. However, the shipper declares all the shipments to the underwriter, who insures the shipments at agreed insurance rate within the limitations of the insurance policy.

On the other hand, special insurance is a one-time policy and commands relatively high premium charges. It is specific to one shipment and cannot be spread over to number of shipments of the same order. It is adopted for infrequent export shipments.

The shipper while taking out the marine insurance policy has to specify the kind of protection required for providing the coverage. The policies are normally differentiated as 'particular average' and 'general average'.

Under 'general average' are risks shared by all the parties involved, that is, cargo owner, shipper and carrier, any loss to goods or cargo in sea transportation to reduce the impact of impending peril to save the lives of the persons on board, or to the save the ship, etc. Under this scheme, the risk is shared through voluntary sacrifice by all the concerned parties.

Under the particular average, the insurance coverage has to be specified against the particular peril. In this, only the shipper shares the risk of peril and gets the benefits of the policy. The various types of insurance coverage are:

- **Fire and sea peril:** This includes the losses caused by the unusual forces of nature while operating in and about navigating waters. For example, the perils are heavy leakages of seawater in the vessel due to opening of seams and causing damage to cargo, impact of water waves on the cargo, collusion with sea objects, iceberg, or other vessels, etc. If the insurance covers fire and sea perils,

the claims are paid only when vessel is stranded, sunk, burnt or collided and damage is caused by fire and sea perils.

- **Free of damage insurance:** This insurance covers the total actual loss of cargo. It does not cover the partial loss of the cargo.
- **Named perils:** The coverage includes fire and sea perils with average and additional perils which are to be named, such as hook damage, fresh water damages, non-delivery, breakages, thefts, etc.
- **Fire and sea peril with averages:** This covers fire and sea peril but it is not necessary that the vessel is stranded, sunk, burned or collided.
- **All risk insurance:** This is the most comprehensive insurance coverage but applicable to loss or damage of cargo from external causes during transit. The causes may be war, strikes, delays and inherent natures of the goods.

Similarly for air cargo, the goods are required to cover the various risks associated with air transportation. However, degree, variety and frequency of risk involved in air transportation is less, the insurance normally is not so comprehensive as compared to marine insurance.

Packaging¹

For overseas shipment, packaging (Figure 17.2) is more critical as compared to domestic shipment because of the nature and number of hazards the packaging has to face or undergo during its journey to the destination. The logistical packaging needs to withstand the varying storage, transit and handling conditions during transit and protect the packed material. Over and above this, the logistical packaging has to comply with the shipping regulations of the country of origin and destination. Hence, packaging for overseas consignment is a cost spinner. In many cases, the sea-worthy packing cost over 5–6 per cent of the price of engineering goods. In overseas shipments, following are the problems associated, which are to be taken care of by the shippers.

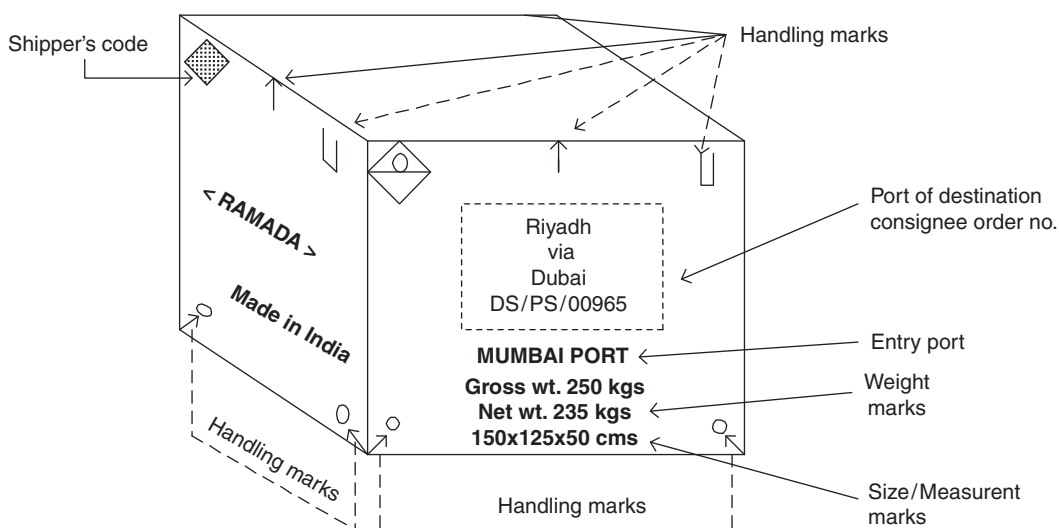


Figure 17.2 Export Packaging

Weight The weight of the packaging will add to the gross weight of the consignment and freight will be charged on the gross weight. Hence, the over designed/protected packaging will add to the transportation cost of the consignment more than it worth.

Transit damage/breakage Due to multiple handling and varying handling condition at multiple points of trans-shipments during journey the product and its packaging is susceptible to breakages and damages. In developed countries, advanced automated material handling equipment is used at all places like ports, nodal points and even at the customer end while in some countries, very primitive methods of material handling may be in use. Hence, the packaging should be designed to take care of such material handling conditions. To guard against this, it is recommended to use the unitizing or palletizing methods with shrink-wrapping for securing material in place during transit. A proper marking for safe handling is required to be printed at the appropriate places on the packaging. The shrink-wrapping may provide additional protection to cargo from heat, rain, dust, if the same is stored in uncovered places. For all these problems, containerization is the best solution.

Pilferage and theft Packaging alone is not the solution for avoiding thefts and pilferages. The effective way to reduce the same is a prompt pickup and delivery. The packaging should not be used for advertising the contents, particularly when high-value items are packed. The use of codes or marking should be encouraged, so that the pilferage and thefts may be reduced.

Containerization Today the majority of exporters are resorting to container shipments in international trade. It is increasingly becoming the popular method of shipment for domestic and exports cargo. Due to a variety of advantages of container shipment (Chapter 11), the share of container shipment in total cargo movement all over the world is increasing every year. In India, the container traffic has grown at a rate of 18–20 per cent during the last few years. The most common popular sizes of containers in use in international trade are 20 ft and 40 ft. However, the growth of containerization very much depends on the container handling facilities on the major ports in the country and the regulatory environment for facilitating multimode transportation within the country. The containers can take care of all the problems associated with traditional logistical packaging. The extra costs associated with container packaging can offset more than the losses due to traditional methods of logistical packaging. However, the job of a logistical manager will be to choose the right container for the right products because containers are available in varying sizes and types. In many cases, the containers may be provided with fixtures to accommodate or pack the particular products. For example, the high-value fashion garments are shipped in hung position in box containers using special types of hanging fixtures. Even cars are exported in containers using special fixtures.

The regulations issued by the government are implemented by the export inspection agency. These regulations are covered under Heavy Packages Act, 1951. The regulations include the guidelines for quality standards, protecting testing and packaging. The export consignment needs to have identification marks on the package. The information should include the name of the manufacturer, name of the consignor, name and address of the consignee, port of destination, country of origin, product type, quantity, shipping weight and handling instructions. The above information is required to be put in the languages of the country of origin and the country of destination (if required). The package should be duly inspected and stamped by the Export Inspection Agency for clearance through customs. For high-value cargo susceptible to pilferage or theft, blind or coded marking are recommended.

The instructions, markings, their contents and their positioning on the package should be as per the guidelines covered under the regulations. These are more specifically mentioned by the regulations

issued by International Maritime Organization (IMO) and International Air Transport Association (IATA) for dangerous and general goods cargo.

Shipments by air do not require heavy packaging. However, the high-value, delicate, perishable or fragile products are containerized and sent through air using smaller size light closed containers.

Intermediaries

The role of intermediaries is crucial and pervasive in cross-border trade. They are basically logistical service providers having expertise in customs clearance, export/import documentation, and the cargo movement from the place shipment to the destination. The cargo movement in the country of destination is coordinated with their business associates in that country.

The shipping line appoints agents at every destination (the port where the ship calls upon). The agents handle the commercial and operational matter and documentation on behalf of the shipping company in that particular location. On the commercial parts, these agents book cargo on behalf of the shipping companies. They do it themselves through freight forwarders. On the operational part, they file the IGM with the customs, get the port clearance, pay the port dues and pay taxes on behalf of the shipping line. For this they get a total commission of 5 per cent of the total export cargo and 2.5 per cent of the import cargo handled. They also arrange stevedores to handle the port operations like loading and unloading the cargo and moving it between the ship and specified area of the port.

Freight forwarder is basically a broker who books space on the ship for his customer—the shipper (exporter). The next person in the chain is the custom house agent (CHA). This person arranges for customs clearance of import and export cargo. But over the time, CHAs have extended their operations and have diversified into freight forwarding business. This is to provide their customers all the services under one roof.

After the people who directly deal with the importers and exporters, come the people who handle the activities at ports called stevedores. They handle the cargo coming in and out of the port and the loading and unloading of cargo. They hire people and equipments needed for the work.

There are two types of intermediaries in international trade:

Box 17.3

Business Process in Exports Trade

The exporter calls up on the freight forwarders or the shipping agents to book space on the ship going at particular destination. He contracts with the shipping company (through the shipping agent) which charges the lowest freight rate to that destination. If the exporter has less than the container load (LCL), then the freight forwarders will consolidate the export cargo from various small manufacturers exporting to the same destination and then ship it out. The exporter also has to get the shipping bills cleared by the customs department. This is done by customs house agents (CHAs). They prepare the document on the basis of the information given by the exporter in the invoice and packing list. They also arrange to get the export benefits from the customs department of director general of foreign trade (DGFT), on the basis of the bill of lading and shipping bill, in case of exports. For an importer they arrange for import license and also handle to get the material out of the port. They also arrange for trucks to carry the material to the importers factory or warehouse.

Freight forwarder The role of freight forwarder is forwarding the freight locally or internationally. They have expertise in the following areas of global logistics operations:

- Traffic operations, that is, choosing the right mode and the carrier for the given destination.
- Initiate or organize documentation (from shipper) required for cross-border shipments.
- Customs clearance of the cargo as per the regulations at the port of shipments.
- Customs clearance and documentations at the port of destination.
- Cargo movement and handling at the port of entry and destination.

Freight forwarders represent shippers in both ocean and air shipments, as the procedure and documentation required are the same. Freight forwarders are business houses licensed (by government) for doing the above task. They are compensated for the work they do for the shippers by way of commission or brokerage. As an exporter gets an enquiry from an overseas client, he contacts the freight forwarders for advice on the following:

- Freight cost
- Port charges
- Cost of documentation
- Insurance cost
- Forwarding charges
- Packaging requirements
- Documents required

Freight forwarders also assist the shippers in other areas such as reserving space on the shipping vessel, preparing ocean bill of lading, forwarding documents to customer's bank, etc.

Customhouse broker The counterpart of freight forwarder in import of shipment is called a customhouse broker. He is licensed to do a similar task of freight forwarders but for importers. His main responsibility includes clearing of importer's shipment through customs. He is compensated for this by way of brokerage. He assists the importer in processing of imports documentation, movement of cargo from shipping vessel to customs bonded warehouse, inspection of imported cargo, clearance of the cargo therefrom and moving the cargo to importer's place.

DOCUMENTATION

Most international trade transactions require transport, administrative, commercial and insurance documents (Exhibit 17.3). There are many documents that may need to be produced to complete an export/import transaction. Some estimates indicate over 100 different documents are used in international trade. However, in most export transactions, only a few documents are used (invoice, packing list and transport document). These documents serve the following functions:

- **Proof of contract:** Evidence of contract of sale and conditions stipulated therein.
- **Title to the goods:** Transport documents giving the right to the consignee to collect the goods from the carrier (just as a bank draft gives you the right to collect funds against the drawer).
- **Information:** Documents provide information on the price for the goods (invoice), the contents of package units (packing list), etc.

Exhibit 17.2 *Cross-Border Trade Documentations*

<i>Commercial Documents</i>	<i>Transportation Documents</i>	<i>Official Documents</i>	<i>Insurance Documents</i>
Commercial invoice	Export packing list	Certificate of origin	Insurance policy
Pro-forma invoice	Bill of lading	Export license/import license	Certificate of insurance
Export cargo shipping instruction	Sea way bill/Air way bill	Movement certificate	
Dock receipt		Inspection certificate	

- **Customs:** The customs authorities of the destination country require documents from the consignee to clear goods. These documents are required for charging appropriate taxes and duties.
- **Proof of compliance:** Certain documents serve as proof that the conditions stipulated in the contract of sale are complied with, such as date of shipment (transport documents), the origin of the goods (certificate of origin), etc.
- **Documents of origin:** The purpose of these documents is to show wherefrom the goods originated.

The type of documentation required often depends on the INCOTERM (international code for delivery) used for the contract. A freight forwarder will usually assume responsibility for export documentation, although it is still a good idea to understand the different types and procedures.

Commercial Documents

Commercial invoice It is a bill for the goods shipped. It includes product description, shipping marks, unit price, quantity, etc. The invoice helps the customs authority to assess the import duty.

Pro-forma invoice This is an invoice sent to the customer in advance of dispatch of goods to organize foreign exchange, an import license or letter of credit (LC).

Export cargo shipping instruction (ECSI) This document is issued by the exporter to the forwarding agent or carrier with a set of terms and conditions for the movement of goods and assigning responsibility.

Official Documents

Export license To be obtained by the exporter from the local government authorities. It is required for certain goods to be exported legally (e.g., dangerous goods, explosives, drugs, chemicals, etc.).

Certificate of origin (C/O) It is a certification indicating the place (manufacturing/shipping) of the goods.

Movement certificate This certificate ensures preferential tariff treatment for the goods exported from the EU to a country covered by the EU trade.

Import license Import licenses are required for certain countries for all goods or certain categories of goods. The customer should comply with the import procedures.

Inspection certificate It is required by the customs authority in the importer's country to confirm that the shipped goods meet their specifications.

Transport Documents

Export packing list This document specifies the weight, volume and type of cargo. This list is attached to the outside of the package to be shipped.

Bill of lading This document mentions the condition of the goods when transferred to the shipper. It is also a document of title transfer. It is a contract between the owner of the goods and the carrier.

Sea waybill It is a document issued by sea freight liner for receipt of goods from the consigner. It states about the possession of goods during transit. It serves the same purpose as the 'consignment note' issued by a road carrier.

Air waybill It is a document issued by airline cargo companies for receipt of goods from the consigner. It states the possession of goods during transit. It serves the same purpose as the 'consignment note' issued by a road carrier.

Insurance Documents

Insurance policy This document is issued by the insurance company covering for the full value of goods, during transit against damages.

Certificate of insurance This document details the degree of protection, and list the policy number and all relevant details (e.g., dates, destinations, transport method, route, description of the cargo and value for which it is insured).

FREE TRADE ZONE

Free trade zone, also called foreign-trade zone², formerly free port, an area within which goods may be landed, handled, manufactured or reconfigured, and re-exported without the intervention of the customs authorities. Only when the goods are moved to consumers within the country in which the zone is located, they are subjected to the prevailing customs duties applicable on normal imports.

Free trade zones (FTZ) facilitate the transactions and smooth physical flow of goods. These zones are earmarked by various countries, where no customs formalities are required for inward and outward movement of the goods. Invariably, these zones are the warehousing hubs for MNCs with global operation for distribution across world markets. To name a few, Jebel Ali in Saudi Arabia and Hong Kong in the east are two main free trade zones in Asia for international companies to route their trade.

FTZ is a special area earmarked wherein trade barriers like quotas and tariffs are not applicable. FTZs are developed in places which are in proximity to international airports and seaports. In the free trade zone, the components are imported, assembled and re-exported as finished products to different countries. No duties or taxes are applicable in these zones. These are manufacturing zones that provide employment and foreign exchange to the country. FTZ provides opportunities, business options and manufacturing options to businessmen. The aims of FTZs are to develop export-oriented units, increase the foreign exchange earnings and generate employment opportunities. It facilitates cross-border trade by removing obstacles imposed by customs regulations.

India was one of the first in Asia to recognize the effectiveness of the export processing zone (EPZ) model in promoting exports, with Asia's first EPZ set up in Kandala in 1965. With a view to overcome the shortcomings experienced on account of the multiplicity of controls and clearances, absence of world-class infrastructure, and an unstable fiscal regime with a view to attract larger foreign investments in India, the special economic zones (SEZs) policy was announced in April 2000.

SUMMARY

Global supply chain management usually involves a plethora of countries associated with a plethora of new difficulties that need to be dealt with appropriately. The movement of cargo between the countries is much more complicated as compared to transportation within countries. The major considerations in cross-border cargo movements are selection of transportation mode, insurance, packaging and documentation as per the laws of the land. The two most widely used transportation modes in international trades are air and sea. The sea route is chosen for bulk cargo with comparatively low-unit price while air route is preferred for high-value cargo. The major considerations for choosing the modes are cost, speed, urgency and reliability. To cover the risk of cargo breakage, loss or damage, the goods are insured under open or special insurance coverage. Due to varying conditions of storage, transportation and material handling during the journey of the goods, packaging needs special design consideration as far as cost and robustness are concerned. In the export and import of goods the role of freight forwarding agent and customhouse broker is crucial. They facilitate the smooth movement of goods by taking care of the documentations as per the regulations. Documentation is the most important part of cross-border trade. The important documents in foreign trade are export license, commercial invoice, bill of lading, air way bill and certificate of origin amongst others.

REVIEW QUESTIONS

1. Discuss the various issues in global supply chain.
2. How do global supply chain operations differ from their domestic counterparts?
3. What are the barriers in the global supply chain?
4. Discuss the role of IT in global supply chain.
5. Explain how 'free trade zones' and the 'trading blocks' influence global logistics.
6. What are the critical operational factors in global supply chain?

INTERNET EXERCISES

1. Visit <http://www.transworldeducation.com/articles/nlogistics.htm>, for free subscription and access to articles on global logistics and supply chain
2. To study the terms and conditions of the overseas shipping cargo visit <http://www.customgl.com/about.php>. Custom Global Logistics are full service transportation provider in international trade.

VIDEO LINKS

1. Global supply chain management <http://www.youtube.com/watch?v=ZuQ200JAViA>
2. Keeping the global supply chain moving <http://vodpod.com/watch/5356631-keeping-the-global-supply-chain-moving>

PROJECT ASSIGNMENT

1. Catalogue marketers like Lands End have global reach in garment markets in 105 countries. They have their main manufacturing facility in the United States. Find out the problem they are facing in managing their global supply in catering to the requirements of their customers spread wider in different geographical areas.

NOTES

1. Sople, V. V. (2010), *Logistics Management*, Pearson, New Delhi, 2nd edition, pp. 145–158.
2. <http://www.sezindia.nic.in/index.asp>

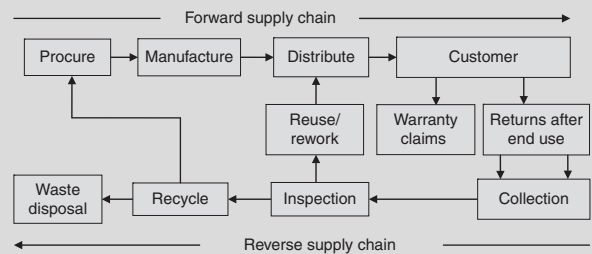
REVERSE SUPPLY CHAIN

“Improperly handled returns erode 32–35% of potential profits of the business organization¹
—Gartner 2004”

Chapter Objectives

To understand:

- Reverse supply chain (RSC) and its importance
- Drivers of RSC in today's context
- RSC design considerations and challenges
- Managing returns



Forward and Reverse Supply Chain

The consumer power today is exhibited by the product recalls of the brand. For keeping the image of the brand intact and to keep the customer satisfied, leading global players resort to mass product recalls through their reverse supply chain infrastructure. These firms use product recall for adding value to their competitiveness. The reverse supply chain is strategically used to allow forward channel participants—such as wholesalers and retailers—to reduce the risk of buying products that are slow moving. The resellers are allowed to return the goods within a reasonable time period using fine-tuned reverse chain which is in place. It can be used for immediate cleansing, if possible, of any purposefully tainted products (pharmaceutical, automobile) to gain customer confidence and keep the company's image intact.

Reverse Logistics Process Improvements

A leading world class apparel, footwear and sports accessories retailer was facing problems in 'returns' system. The returns processing system was facing three primary challenges. The first was returns from walk-in customers as the earlier system was designed for receiving return requests from customers over phone. The customer support representative would then collect the returned item from the customer's doorstep. As the existing system was not designed for walk-in-customer, it was causing delays and dissatisfaction for them. The second challenge was returns in instalments. The existing system generated two separate credit notes for the customer who returns in instalments. This created unnecessary load on the system. The third was 'less than pre-pack returns.' It is a common practice for a customer to open a pre-pack, use a few items and return the rest. The existing returns system did not have the capability to process less-than pre-pack returns. But, the customer has the right to get credits for the returned items. This created a customer-service challenge.

The retail consultants analysed the existing system and suggested process improvements in returns handling. The suggested processes were then mapped for feature enhancements in the existing returns processing system. The process improvements addressed each of the challenges faced by the client.

For walk-in customers it was suggested to implement a separate returns process. With this walk-in-customer's returns were to by-pass the carrier pick-up process and the inspection process. As soon as products were received, online credits were generated to integrate with the finance system to approve credit. The inspection process would be triggered after customer's credit generation. For 'returns in instalments' a windowing system was introduced to combine multiple returns of the same customer. For example, multiple returns from a customer at different dates can be added into the first return until it reaches the inspection complete process. For 'less than pre-pack returns,' the process improvement followed was that the operator performs rework by combining same products back in the official pre-pack quantities before updating the stock account. In situations where this is not possible, the system will automatically generate credit for the customer without updating the stock account.

The new system of returns accrued the following benefits to the retailer: reduced cycle time for returned product, financial savings through immediate reflection on inventory, availability of saleable stock, enhanced ability to trace a product anywhere in the returns cycle, increased customer satisfaction, developing a comprehensive warehouse management solution.

INTRODUCTION

Companies spend more time and money in fine-tuning their forward supply chains while ignoring their backward supply chains. However, in today's competitive business environment when there is both external and internal pressure, companies can no longer ignore reverse supply chains. Efficient reverse

supply chains bring many benefits to the companies. However, reverse supply chains are different from forward supply chains and most of the existing forward supply chains are not designed to handle reverse flow of materials.

Reverse supply chain process covers planning, implementing, and controlling the efficient, cost-effective flow of raw materials, in-process inventory, finished goods and related information from the point of consumption to the point of origin for the purpose of recapturing value or proper disposal. Many companies do not have an awareness of the reverse supply chain and the costs associated with reverse flow of material. These companies lack in awareness, defined processes and support of the system. The reverse supply chain handles varying returns in terms of quantity and time. Hence, such system should be designed keeping in mind the flexibility nature of the returns. In general, reverse product flow programmes are complex in nature. It involves lots of paperwork and tracking to the users.

WHY REVERSE SUPPLY CHAIN?

According to Devangshu Dutta, director of a supply chain solutions company, 'Nearly 20% of everything that is sold in America is returned'. This means it is imperative for the company to have a system to handle the reverse flow of materials.

The reasons for returns are varied. However, returns in business-to-business markets and business-to-customer markets are based on different reasons. Due to varied nature of reasons, the design of reverse flow network faces many challenges for which different strategies need to be evolved (Exhibit 18.1).

Traditionally, in the supply chain of an organization there is one directional flow of goods, that is, from manufacturer to the end user. Almost, all of a logistician's attention has been focused on their 'forward' logistics activities. Once the product is sold and delivered to the user, the manufacturer feels

Exhibit 18.1 *Returns and Reverse Supply Chain Challenges*

	<i>Business Markets</i>	<i>Consumer Markets</i>
Reasons for returns	• Product defects • Recall • Warranty • Discontinued product • Stock adjustments	• Product defects • Damages • Warranty and service • Recall • End of use
Challenges	• Non-uniform product quality and packaging which affects utilization and expenses • Outdated and possibly obsolete returned items	• Unpredictable demand and no advance notice of returns • High number of transactions necessitating efficient disposition
Scope	• Refurbishing, repairs, remanufacturing to reduce the need to purchase new parts • Detailed data required to perform root cause analysis	• Refurbishing, repairs, remanufacturing while delivering customer service • Seamless integration of supply chain partner for disposal • Customer-focused strategy

that there is an end to his responsibility. The manufacturers consider that their responsibility is limited to the extent of replacement of defective products covered under warranty period or products damaged during transit. What is happening to the used materials, packaging waste, disposable product waste generated out of the finished products supplied by them? The left out material and the wrappers cause environmental pollution and create problems in disposal for the civic authority. However, in wake of the growing concern about environmental pollution, developed countries across the world, have started passing legislations, asking the manufacturers to take care of products discarded by their customers after usage. The leading corporations across the world are taking this as an opportunity to develop a system for reverse material flow. They are focusing on the 'reverse' logistics activities to use it as a tool for competitive advantage.

Reverse logistics may be defined as a process of moving the goods from their place of use, back to their place of manufacture for reprocessing, refilling, repairs or waste disposal. It is a planned process of goods movement in reverse direction done in effective and cost-efficient manner through an organized network. It can be a standalone or integrated system in the company's supply chain. The drivers for reverse logistics are:

- Economics
- Legislation
- Corporate social responsibility (CSR)
- Consumerism
- Competitiveness

In competitive markets, to retain the existing customers, the manufacturers are creating switching barriers by extending value added service to the customers. To remain competitive and differentiated, more and more firms across the world are showing speed and reliability in service offerings such as:

- Replacing defective goods
- Repairing the used product
- Refurbishing the returned product
- Calling back substandard or harmful goods
- Disposal of product waste

These additional services are adding to the competitiveness of the company operating in regulatory environment and creating customer value by providing clean environment through reverse logistics services without any extra cost to the customer. Today, corporations across the world are leveraging reverse logistics, for growth by enhancing the level of customer satisfaction beyond the traditional boundaries of product and service supply.

REVERSE SUPPLY CHAIN

According to authorities Dr. Dale S. Rogers and Dr. Ronald S. Tibben-Lembke, reverse logistics is:

The process of planning, implementing, and controlling the efficient, cost effective flow of raw materials, in-process inventory, finished goods and related information from the point of consumption to the point of origin for the purpose of recapturing value or proper disposal.

The council of supply chain management professionals define reverse supply chain² as:

The process of planning, implementing and controlling the efficient, cost effective flow of raw materials, in-process inventory, finished goods and related information from the point of consumption to the point of origin for the purpose of recapturing value or proper disposal.

Wikipedia defines reverse logistics as:

As a process of removing new or used products from their initial point in a supply chain, such as returns from consumers, over stocked inventory, or outdated merchandise and redistributing them using disposition management rules that will result in maximized value at the end of the item's useful life.

To sum up, reverse logistics is the process of moving goods from their typical final destination for the purpose of capturing value, or achieving proper disposal to the satisfaction of the customer or consumer. Remanufacturing and refurbishment activities may be part of the procedure. Reverse supply chain include processing returned merchandise due to damage, seasonal inventory, restock, salvage, recalls and excess inventory. It also includes recycling programmes, hazardous material programmes, obsolete equipment disposition and asset recovery. For a practitioner, the disposition choice is determined by the most profitable alternative:

- **Reconditioning:** When a product is cleaned and repaired to return it to a 'like new' state.
- **Refurbishing:** Similar to reconditioning, except with perhaps more work involved in repairing the product.
- **Remanufacturing:** Similar to refurbishing, but requiring more extensive work; often requires completely disassembling the product.
- **Resell:** When a returned product may be sold again as new.
- **Recycle:** When a product is reduced to its basic elements, which are reused—also referred to as 'asset recovery'.

SCOPE OF REVERSE SUPPLY CHAIN

The reverse supply chain, though considered as a drain on the company's profits, but in today's competitive markets it can be leveraged as a tool for customer satisfaction. More and more manufacturing firms are thinking of incorporating reverse flow system in their supply chain process. The reasons for this are as follows:

- Growing public concern for environment pollution
- Government regulations on product recycling and waste disposal
- Growing consumerism
- Stiff competition

Reverse supply chain is important for asset utilization (re-utility), asset recovery (capture value), profit maximization, full environmental obligation (or regulatory compliance) and customer relations management. The reverse supply chain network can be used for various purposes such as refilling, repairs, refurbishing, remanufacturing, etc. Depending on the nature of the product, unit value, sales volume and distribution channel the reverse logistics can be organized and the system can be designed for the following activities.

Refilling

In industries such as soft drinks, wines, oil, LPG distribution, etc. reverse flow of used containers is integrated to their regular supply chain because of the reusable nature of the packages such as glass bottles, tin containers, metal cylinders, etc. In case of soft drinks the delivery van en route delivers the filled bottles to the retailers and collects the same number of empty bottles from them for delivery to the factory. No extra transportation costs are involved in the process as the delivery van originates and terminates its journey at factory where these reusable bottles are refilled for re-delivery to the customers. The arrangement deployed here is like hub and spoke distribution system in reverse flow system. For refilling of LPG cylinders, the similar arrangement is in use by the petroleum companies (Hindustan Petroleum, Bharat Petroleum). From the bottling plant, the truck load of filled LPG cylinders are dispatched to dealer's godown for delivery and on the return trip the same truck carries the empty cylinders for depositing at the bottling plant. In case of metal cylinders (due to the less damages and prolonged life), the recovery rate of the cylinders is to an extent of 99.9 per cent.

The logistics service suppliers who are supplying pallets³ on leased basis to their various clients for packing and moving the goods keep the pool of pallets at fixed locations. The customers draw the pallets for use and deposit the pallets at the assigned centre after usage. The damaged pallets are repaired or replaced regularly keeping the required quantity in the circulation. The return of the pallets is on the exchange basis between the buyer and seller with whom, the pallet supplier is a common link who co-ordinates the reverse logistics operation. This system is quite common in developed countries.

In India, the box containers used in multi-modal transportation system are leased by the container lease corporation Ltd. (CLCL), which keeps the inventory of empty box containers at inland container depots (ICDs) operated by container corporation of india (CCI) or private agencies. The customer requirements of containers for packaging and movement of goods in domestic or foreign markets are drawn from these depots. These containers are loaded at container freight stations operated by the CCI or others. The empty containers after de-stuffing of materials are deposited at the container depot near to the place of delivery for further reuse and reverse flow to the place of origin. Thus, the entire movement of containers in forward and reverse logistics is controlled by CLCL.

Shaw Wallace (leading wine manufacturer) has integrated the reverse flow of empty bottles with their regular forward distribution system. The empty bottles collect at the area distribution warehouse with the help of their dealers, who are in touch with hotels and scrap dealers who deploy his network to collect empty wine bottles from house holds, hotels, clubs, pubs, etc. The bottles thus collected at

Box 18.1

Reverse Supply Chain in Beer Industry

London Pilsner of UB group, a leading beer brand in Maharashtra, India, does filling of 3 lakh beer bottles per day at their plants. The recovery rate of empty glass bottles from the market is 92 per cent, as 3 per cent of bottles are broken and 5 per cent are lost or put to other usages. The cost difference of 20–25 per cent between new and old glass bottle contributes to the extent of 12–15 per cent in the total manufacturing cost of the product. Hence, the firm saves a lot by using the used bottles. They have developed a stand-alone reverse logistics system for recovery of the empty bottles through the scrap vendors who collect these bottles at throw away price from hotels, clubs, pubs, bars, etc.

field warehouses are sent back to the factory for refilling. The empty bottles recovery is to an extent of 85–88 per cent of the supply.

Repairs and Refurbishing

This is a regular feature for serviced-based products under warranty period, which has to be incorporated by the manufacturers in their product offerings. Almost all consumer durables such as television sets, audio systems, washing machines, fans, refrigerators, and other consumer durables and all industrial products need repairs on a regular basis. The refurbishing is done for the goods returned by the customers during warranty period due to damage, defective piece or performance below the promised level, etc. For consumer durables and engineering products manufacturers in general keep in place the reverse logistics system for extending the service within warranty period (free of cost) and beyond the warranty period on a chargeable basis. In addition to extending value added services to the customers, these systems are major revenue earner for the company. The system operates through the company's service centres where the repair and the refurbishing take place. The collection of the products is done through dealer's network. The collected products are dispatched to the nearest service centre for overhaul, repair or refurbishing. The documentation and payment collection are responsibilities of the concerned dealer. For large value industrial product locating and coordinating, the customer is not posing any problem, as the number of customer is less and customer is personally known.

Product Recall

Product recall is calling back the products from the customers or requesting the customers to return the products. This may be due to safety concerns or design defects or labelling error at the manufacturer's end. The drive for product recall is cut throat competition, concern for environment and growing consumerism. Today product recalls are not uncommon. Many brand leaders recall their defective product from the market to guard their image or avoid costly court litigations. The reasons for recalls may be as follows:

1. Damage control.
2. Safety concerned for all users or accidents.
3. Design defects in product which may deteriorate the image of the company.
4. Avoid litigations for costly settlements.
5. Consumers' life may be at risk.

Thus, product recalls can be handled and it is possible to build in a system for possible product recalls. Customers may demand damages through the courts resulting in financial implication for companies. The delay may be fatal for consumers in certain products like electric appliances, automobiles and food items/medicines. Many failures are due to poor design and ignoring safety aspects, like sharp corners in toys or loose parts or buttons on children garments during manufacturing. For normal defects, normal warranty procedures are adequate. However, product recalls are preventive in nature and can avoid costly litigations and financial and goodwill losses. For example, in 1977 Glaxo Laboratories (now Glaxo Smithkline Beecham), due to some problem in composition of baby milk powder in some batches, recalled the product from the market. This process of recalling products from market through channel partners or consumers is reverse logistics and normally results in heavy expenses. To call back widely distributed consumer products when they are out of marketing channels is quite cumbersome and

expensive. The other dimension is the time; after the product defect is identified the recall process has to be expedited to avoid further damage to the company's image and goodwill. Even though there was immediate threat to the life of babies, Glaxo's decision in product recall was due to concerns to the society.

In another case in 2007, Mattel Inc. USA, a leading toys manufacturer in the United States recalled millions of toys from the market because of a small detachable and swallowable magnet in toys and lead (in paint) taint, which are hazardous to children, in some toys manufactured for Mattel by Foshan, a China-based company. The recall ran into millions of dollars. A class-action lawsuit was filed against Mattel related to its recall of more than 1 million lead-tainted toys. Nokia Mobile phone batteries BL-5C (manufactured in China but for Matsushita, Japan) have been recalled (46 millions) for getting overheated and bursting during charging. IBM for its Lenovo brand computers recalled, in 2006, more than 5 lacks laptop batteries worldwide made by Sony Corporation because of a fire risk. In this case, both IBM and Sony Corporation announced a global replacement programme of the company's lithium-ion batteries. In 1994, Maruti Udyog India recalled Maruti 800 cars due to some problem in front axle components which endangered customer safety on road. This recall was for axle replacement only. For Maruti it was easy to do that as customers were in finite numbers and their database was available with the dealers/company.

Product recalls are done under certain situation. It may be because of process failure, faulty product or packaging and labelling goes to market. The safety-related aspect is known only after product is sold off to consumers and put to use. In 'product recall' products sold to customers or available with dealers are called back to the factory in emergency situation. The reasons may be as follows:

- Product not giving the guaranteed performance.
- Quality complaints from many customers.
- Defective product causing harm to human life.
- Products beyond expiry date.
- Products with defective design.
- Incomplete product.
- Violation of government regulations.
- Ethical consideration.
- Save the company's image.

This type of situation may arise due to the reasons of production short cuts, bypassing stage inspections, employee negligence, human error or negligence by management. The product recall in such situation puts huge financial burden on the company. For handling such emergencies, no organization design in advance and keep in reserve reverse product flow system in anticipation of such eventualities. However, to handle such emergencies many firms have shown great deal of organizing ability to mobilize company's resource to achieve the time-bound objective.

A leading Indian auto manufacturer, after launching multi-utility vehicle, in the Indian markets in the 1980s, received large-scale complaints from customers due to defective gearbox design. All the vehicles dispatched to the customers were called back within a short time span deploying their dealer's network and company's sales force, for refitting the gearbox with improved design. In the mean time, the redesigning programme of gearbox was completed on a war footing.

Johnson & Johnson Health Care, a U.S.-based multinational company introduced milk powder in South African markets, as substitute for breast-fed milk for newborn babies. However, due to large scale deaths of the babies by consumption of the powdered milk, Johnson called back the entire unsold

powered milk stocks in the market under a time-bound programme and gave away compensation to the victims on the ethical grounds. The cause of the death was contaminated milk prepared under unhygienic conditions and not the milk powder. The company failed to educate the mothers on a hygienic process for milk preparation during the sales campaign.

The scope and effectiveness of the recall process is dependent on the type of the product, its distribution network, consumption pattern and unit value. The industrial products of high unit values, which are directly dispatched to the customers and the number of customers are limited, the recall process is very effective. The customer being knowledgeable fully extends his co-operation in the process. However, in case of mass consumed products distributed through multi-level channels structure, identifying the product location becomes a problem. It is easier to trace the product location within the boundaries of the channel network. Once the product is sold and handed over to the user, the degree of effectiveness of product recall process reduces drastically because of the following reasons:

- Lack of customer database
- No motivation on the part of customer to return the product
- Product not meant for critical use

For service-based products like consumer durables which are accompanied with warranty cards the product can be located, provided the documentation is maintained at the point of sale. This makes the recall process easy.

In the wake of the stiff competition and growing consumerism, many manufacturers have built up a system for product recall as value-added service to build competitiveness. The recall of defective products during warranty period is now the most common feature of the customer service offers. In fact, it is a cost burden on the part of the manufacturers. However, many companies are considering product recall as an opportunity in increasing customer satisfaction and asset recovery operation. Nestle India, for example, gets back the 'Yoghurt' every day from their distributors after its expiry of 24 hours shelf life. Similarly, Monginis, a Mumbai-based leading confectionary producer, collects the cakes for their distributors after its shelf life is over. The van delivering the fresh products to the distributors collects by default the products, which are not sold, once their shelf life is over.

Recycling and Waste Disposal

The left out materials, used products, and wrapper wastes cause environmental pollution and create problems for disposal. Hence in developed countries, the governments are devising regulations to make manufacturers responsible for minimizing the waste by way of recycling the products. The Germany is the first country in the world to implement such regulations. Legally, the manufacturer is responsible for taking back pallets, cardboard boxes, wrappers, strapping, etc., which are used for protecting the products during transit. They have implemented three stage packaging ordinance. In the first stage, the wrappers or packaging wastes are collected from households by the retailers. In the second stage, retailers send it to manufactures, which in the third stage is sent to packaging manufacturers for recycling or disposal. The levy for recycling is indicated on the product by green dots. In Germany, the FMCG manufacturers have jointly promoted DSD (Dealers System Deutschland) with common funding for collection of packaging waste of FMCG products. They have reduced cost of retrieving and recycling the packing waste through economies of scale by joining hands together. In the United States there exists a law to take back from the customer car batteries, soda bottles, etc.

Box 18.2

'Green Dot'⁴ on Wrapper for Recycling

Germany's new packaging law, (1992) regulates the management for recycling of packaging waste. The law stipulates that all transport packaging are required to be accepted back by manufacturers and distributors. The packaging items covered under this legislation are crates, drums, pallets, foam containers, etc. German packaging law stipulates that the manufacturers, distributors and retailers will have to take back returned (of the used product) secondary packaging (Figure 18.1). The packaging items include cardboard boxes, blister packs, paper wrappers, etc. The manufacturers, who are participating in the 'green dot' programme, require to collection the used packaging materials from the consumers or from decided local collection centres. In Germany, it is a legal offense to either dump or incinerate such materials. Firms are free to engage third-party operators to fulfil the requirements of the new regulations on their behalf.

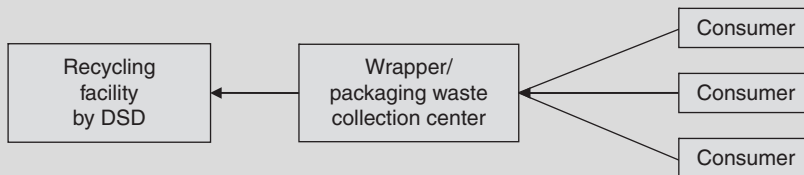


Figure 18.1 Reverse Supply Chain for Packaging Waste

The green dot symbol is adopted by the Duales System Deutschland GmbH (DSD—Dual System of Germany). DSD corporation was established in September 1990, with over 400 participating companies in Germany. The fee is collected from the participating manufacturers for the right to display the green dot symbol on their products. This revenue is used for packaging waste collection and recycling programmes. The green dot indicates to the consumer to deposit the used wrappers at collection containers which are further taken to the recycling centre. The responsibility of depositing the wrappers in the collection containers rests with customers who very well know that they have paid for the collection and recycling cost which is included in product price.

In Europe, BMW, a leading auto manufacturer, is the first company, which has effectively developed a car recycling supply chain system. The car is mostly an assembly of components made out of metal, which can be easily recycled. For product refurbishing, SONY, a consumer electronic giant operating in the United States, uses its regular dealers network for reverse flow of video or audio systems returned by the customers within warranty period.

In India, the supreme court is in the process of imposing a ban on charging and discharging of automotive batteries on the grounds of pollution, and the union government is coming out with a new legislation making it mandatory to return used batteries to manufacturers for recycling and disposal.

Remanufacturing

The manufacturers in developed countries are putting into practice a new concept of 'remanufacturing' emerged in the late 1990s. According to them, during usage the product undergoes wear and tear. The worn parts are replaced with the new ones and the performance of the product is upgraded to

the level of a new one. This is practised more in defence establishments, wherein the fighter planes after each flight are checked and their performance level is every time brought to the level of a new one without any compromise on the quality front. Similarly, the equipments sold in the markets, after use can be checked to qualify for remanufacturing process and brought to the remanufacturing unit. A leading cell phone giant in Europe has outsourced the remanufacturing activities to a few vendors in Europe and developed a separate network of reverse flow of the used or discarded products which have been used by the customer for a short period and discarded due to switching over to the latest models. They have identified a huge market for remanufactured products in developing countries. The investment in remanufacturing and related reverse logistics supply chain can be justified on economies of scale. Critics say that the remanufacturing will solve the problem of waste disposal in developed countries but create new dumping grounds in the developing countries.

The different activities in reverse supply chain process include collection, inspection and testing, sorting, reconditioning, disposal and redistribution. All these activities are required to be integrated to create value. And finally, the value needs to be delivered to the customers through the reverse supply chain network. Many good companies integrate the reverse supply chain network with its forward supply chain on costs consideration and assets utilization.

In a nutshell, the scope for reverse supply chain is immense and it can play an important role in publication houses, beverage industries, FMCG industry, pharmaceutical industry, automobile industry, petroleum industry, etc.

REVERSE SUPPLY CHAIN DESIGN

Irrespective of whether it is forward or reverse supply chain, the network design is of prime importance. In the reverse flow of material, the location of channel members, warehouses, company's manufacturing plant and their capacity to handle the reverse flow are the important determinants in its design. Many good companies handle the reverse flow of the material through their forward supply chain. The forward supply chain is designed in such a way to take up the reverse flow of the material. The structure of reverse supply chain can be divided into two segments. The first one is the convergent network wherein the companies collect the returned material at some convenient place or location and transfer the accumulated material to reworking/recovery facility. The other one is divergent network, wherein there is linkage with the customers to collect the reusable material. It is integrated with forward supply chain of the company. The strategic points in designing the reverse supply chain are as follows.

Collection of Returned/Used Products

The major cost component of reverse supply chain operations is product collection. In FMCG industry, the collection of defective material after it leaves the channel network becomes difficult. Locating the end user is difficult due lack of record/information. However, in case of high-value products, channel members have records to trace the end user and product which are under warranty claims. In reverse supply chain, retailers and distributors are deployed for collection of returns.

Testing/Grading Process

In this stage, only the products can be assigned for recovery opinion. Hence, the location of testing or grading operation is important as it decides the flow of goods. The cost of transportation may be significant as a percentage of value of the used product. In many cases, the testing and grading of recovered

product is done at some central location as specialized and expensive testing equipments and skilled hand are required for this operation. In many cases, the geographically decentralized testing facility is used to avoid transportation cost for unwanted useless returns.

Reprocessing

This facility requires capital investments. The specialized equipments are required for recycling or product recovery. The facility can work profitably only on economies of scale. The recovery facility should be integrated with regular manufacturing line of the new products to share labour and equipments.

Redistribution

The redistribution of reworked products can be done by integrating with the forward supply chain. Here, consolidation and responsiveness in transportation are required to achieve efficiencies. Redistribution can also be done along with new products.

Reverse SC is also put in practice for recovering value from products. Recovery is actually only one of the activities of reverse SC. Other processes include collection, inspection/testing selection and sorting. Collection refers to bringing the products from the customer to a point of recovery. At this point, the products are inspected, that is, their quality is assessed and a decision is made on the type of recovery. Products can then be sorted and routed according to the recovery decision. Thus, we have the following:

- Collection
- Inspection/testing
- Selection
- Sorting
- Recovery

The product exchange offers in marketing for consumer durables are examples of value recovery (Figure 18.2) or value proposition of used products in secondary markets.

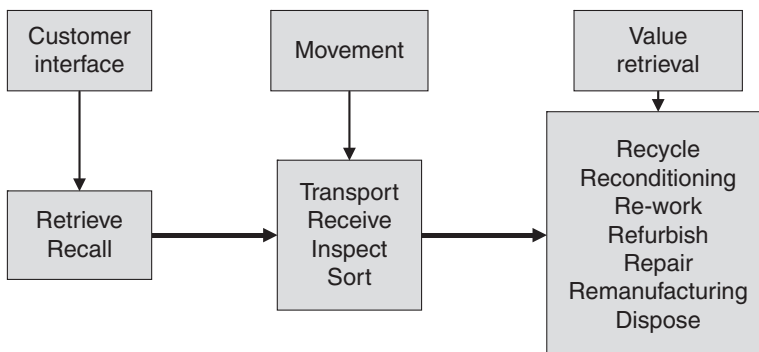


Figure 18.2 Reverse Supply Chain Process

To run reverse supply chain process essentially, the following players are the most crucial:

- Forward supply chain actors, (supplier, manufacturer, wholesaler, retailer and sector organizations)
- Specialized reverse chain players (such as jobbers, recycling specialists, dedicated sector organizations or foundations, pool operators, kabadiwala, scrap dealers, etc.)
- Governmental institutions (national and state governments)
- Opportunistic players

These players will have to play roles to manage, execute and accommodate.

The companies which are working with customers for product replacements shows their concerns for customers, exhibit efficiencies and reduce cost by avoiding product obsolescence. In effect, it helps in release of tied up capital and unproductive occupied storage space. Many companies are using reverse logistics as differentiator to gain competitive advantage. Key drivers to design and implement cost-effective reverse supply chain are:

- Customer satisfaction and retention
- Reuse (beer bottles/gas cylinders)
- Recycle (paper packaging)
- Defective goods returns (within warranty period)
- Mass recalls (defective products)
- Products beyond expiry/usage dates
- Asset recovery/restock
- Excess inventory with channel partners

In conclusion, traditionally almost all manufacturing firms focus their attention on 'forward' logistics activities. During the last few years, changes in environmental laws, increased consumerism and stiff competition in markets have raised the requirement for effective reverse logistics to be used as a competitive weapon. For many years, soft drink manufacturers practised reverse logistics in glass bottle refilling. The cost advantage in reverse logistics is possible through economies of scale. Today, many big corporations, across the world have developed either reverse logistics systems or integrated the same into their forward logistics systems. The system design takes into consideration factors like the product locating system, product collection mechanism, documentation, product recycling/disposal centres, cost implication and legal aspects. In developed countries like Germany and the United States, stringent environmental regulations have prompted corporations to develop three-tier product waste recycling/disposal through reverse logistics.

When it comes to completing the end-to-end processing, companies should do two things: take a full view and understand the total cost of the supply chain, which includes having and dealing with returns and realize the opportunity that an analysis of the return part of the system offers in terms of finding and solving problems, better satisfying customers and consumers, and discovering a way to increase revenues.

SUBSYSTEMS IN REVERSE SUPPLY CHAIN

The success of the reserve logistics system to achieve desired objectives shall be depending on the efficiency and effectiveness of the following subsystem.

Product Location

The first step in the call back process is to identify the product location in the physical distribution system of the firm. The product may be lying in company's mother warehouse, distribution warehouse, dealer's godown, retailer or with the customer. Locating the product becomes easier and quick if it is within company's warehouse, depot, or distributor where tracking can be easily done. But once it enters retail network, particularly the FMCG or mass consumed low-unit priced products, the task becomes difficult because of large number of retailers, wider geographical spread and lack of proper documentation.

Locating the product becomes more difficult after it is sold and handed over to the customer. The product location is bit easier in case of industrial or high value product due to limited number of customers and personal touch with the clients due to direct selling.

Product Collection System

Once the product location is identified, the collection mechanism gets into operations. This can be either done through company's field force, channel members or third party. However, proper guidelines or instructions have to be given to motivate the customer for returning the products. The customer is the biggest hurdle in retrieval of products as he does not want to part with something which he owns. The collection centres are to be located conveniently so as to ensure wider coverage and minimum collection cost. The company's intermediaries are the most effective centres in product call back as far as the cost and the coverage are concerned. With the right training, they carry out the preliminary inspection of the called backed/returned products before inducting the same into the system for further processing.

Product Repairs/Recycling/Disposal Centres

These may be company's manufacturing plant or the mother warehouse where from the finished products were dispatched to the market or some fixed location in reverse logistics network. The called back products are inspected before they are further processed for repair, refurbishing, re-manufacturing or waste disposal. The investments in facilities for these activities depend on the objectives of the system, cost implication, complexity of the operations, and the expected gains.

Documentation System

The tracing of the product location become easier, if proper documentation is maintained at each channel level. However, at the time of handing over the product to the customer, the information on name of the user, address, contact phone numbers, application, and other personal data if collected through proper documentation, can form a good database which can be used in case of product call backs. The product warranty card with product and customer detailed duly filled at the point of sale is quite common for the service-based products such as consumer durables or industrial products. But in most of the cases, the warranty cards are handed over to the customer to show it as proof of purchase, at the time of claims during warranty period. Except for high value industrial products, no records are kept with retailers for consumer durables or mass-consumed products. The documentation system in reverse logistics ensures maintaining of customer-product database at the point of sale. To ensure smooth running of the system, a procedure is to be introduced for entry, exit and flow of the called back products in the system for tracking and tracing.

Cost Implications

The reverse logistics system is a cost centre. However, these costs are incurred for achieving certain company objectives and can be broadly subdivided into the following activity heads:

- Product location (by company staff, channels member or third party)
- Transportation
 - Product collection (Customers > retailers > plant)
 - Disposal (plant > suppliers > disposal)
- Refilling, repairs, refurbishing, remanufacturing and waste generation
- Documentation (for product tracking and tracing during entry, exit and flow in the system)
- Communication

Due to cost implication, the manufacturers invariably integrate the reverse logistics into their forward logistics system with little modifications. The same network components are geared up to accommodate the reverse flow of the products with same efficiency and effectiveness. The investment and operating costs involved in tuning the existing 'forward' flow system to 'reverse' flow shall be less than a separate stand alone system devoted only for reverse flow unless it is designed on scale of economies.

Legal Issues

Under Indian regulations, the excise paid goods once sold by the manufacturer cannot be brought back to the plant without proper documentation and declaration to the excise authority. This is a very cumbersome and time-consuming process and non-compliance may put manufacturer in a legal trap. For resale of repaired or refurbished goods both the excise and sales tax authorities are involved and clearing of such goods require documentations, certifications and declarations. Hence, such activities are normally carried out at service centres or at dealer's premise.

In reverse logistics the critical strategic points in design of network are: acquisition/collection of returned/used products, testing and grading operations, reprocessing, and redistribution/disposal. For reverse logistics to be successful the collaboration of supply chain is very important. Technologies like bar coding or RFID helps in making reverse logistics more effective and responsive

REVERSE SUPPLY CHAIN CHALLENGES

To design and deploy reverse supply chain is a complex task. The reverse supply chain may be a stand-alone system or may be integrated to the forward supply chain operations. However, the management has to face the following challenges in running the reverse supply chain.

Channel Member—Manufacturer Conflict

In case of FMCG or consumer durable products which are sold through the retailers or dealers, there is a difference in objectives of manufacturers and channel members. Many times there is a disagreement on timeliness of claims, condition of item and its importance to customers. The lack of proper inspection system to accept returns may aggravate the situation. The products damaged during transit sometimes cause disputes due to conflict of interest of two parties. However, at the end the customer suffers. Therefore, a proper return policy in place may avoid such conflicts arising out of interests.

Responsiveness

Many companies in the past have shown lack of interest in handling returns. According to them, large investments in product returns handling system is not justified. However, of late due to competitiveness in market, pressure from environmental agencies, returns legislation and strong consumerism have started thinking about reverse supply chain system. In spite of all the company executives usually disagree with reverse supply chain issue resulting into lack of responsiveness. In many cases they are reactive (late) than proactive.

Information System

The information system in the organization is designed to handle forward flow of the inventory. However, for returns handling the information is not designed to data collection and reporting. As a rule, those firms that manage their information well manage their supply chain well. This applies to reverse supply chain too.

Problems Related to Returns

Dr. Richard Dave of the Fritz institute of international logistics identified the following six symptoms of problems in returns:

- Returns arriving faster than processing or disposal.
- Large amount of returns inventory held in the warehouse.
- Unidentified and unauthorized returns.
- Lengthy returns processing cycle.
- Unknown total cost of returns processing.
- Customers have lost confidence in suppliers repair activities.

TRENDS IN MANAGING REVERSE SUPPLY CHAIN

Looking into the environmental factors such as consumerism, pro-consumer legislation, eco-balance pressure there is considerable attention on reverse supply chain designs and implementation and to use it as a tool for competitive advantage to create switching barriers to the consumers. Some of the trends observed are as follows.

Reverse Supply Chain Need Recognition

Many firms have become aware of the importance of reverse supply chain. However, they are yet to realize its importance as a strategic tool for customer satisfaction leading to sustainability. The firms need to focus on the following aspects in reverse supply chain:

- Returns policy formulation
- Standardization of process
- Returns categorization
- Documentation

- Quick response
- Data management
- Assigning responsibilities

Product Life Cycle Management

In future, companies should manage products from cradle-to-grave. This means the company has to support and manage products throughout its life cycle. This requires companies to provide proper logistics and marketing support base during the life cycle of the product. The basis of product life cycle is that all the products have a finite life and they move through various stages such as introduction, growth, maturity and decline.

As the product travels through its life cycle, the inventory- and returns-related issues change. The company needs to design its reverse supply chain to address the related issues during the life cycle of the product.

Data Collection and Information System

Most reverse chain systems are paper intensive, resulting in delay in information flow and further slowing the responsiveness. The system on IT platform will make a difference. Many leading companies are making use of bar coding or RFID technology in supply chain operation for system automation and speedier information flow and its analysis. The use of bar code or RFID will make the reverse supply chain system more effective and efficient. Another important technology, that is, *EDI* (electronic data interchange) which is used in supply chain operations for document transfer may also help in reverse supply chain for similar application. Improved data collection system will always help the decision-makers to act proactively and positively. The product tracking and tracing technology on IT platform will improve the competitiveness of the reverse supply chain system.

Web-based System

To accept the returns from customers, details such as product manufacturing and date of sale are required at the time of complaints or while registering returns. A web-based system will help to know the product history and other details to decide whether to accept the returns or not depending on the complaints, expiry date or warranty period. With the web-based information system, the middlemen also can take decisions and further know the batch numbers and manufacturing dates of the returns.

Third Party Engagement

Many companies earlier had the returns handling activities at a centralized location or centres. People were made responsible and assigned to do both forward and reverse supply chain jobs. Over the years, the companies have realized that under dual responsibilities the employees were not giving justice to returns management. However, the major trend in supply chain today is to engage a third party or an expert to perform your non-core activity. Many companies are assigning returns handling jobs to logistics service providers, who can perform it efficiently and cost effectively for the company.

SUMMARY

Companies spend more time and money in fine-tuning their forward supply chains while ignoring their backward supply chains. However, in today's competitive business environment when there is both external and internal pressure, companies can no longer ignore reverse supply chains. Efficient reverse supply chains bring many benefits to the companies. However, reverse supply chains are different from forward supply chains and most of the existing forward supply chains are not designed to handle reverse supply chains.

Invariably, all of the manufacturing firms have 'forward' supply chain activities for inventory flow from suppliers to customers via manufacturing operations. However, due to changing environmental factors such as increased consumer awareness, growing consumerism, pollution control laws and growing market competition the manufacturers have felt the need to deploy reverse SC in an effective way to gain competitive advantage. The reverse logistics is practised in soft drink industry for many years for bottles refilling and similarly, in the LPG cylinder refilling too as the empty bottles and cylinders could be recovered from the consumers and refilled after due treatment. In today's competitive world many big companies have developed stand alone reverse supply chain for customer satisfaction and gaining competitive advantage. However, many companies integrate the reverse supply chain system with forward supply chain. For designing the reverse supply chain, the following factors are taken into considerations: product locating system, product collection mechanism, documentation, product recycling /disposal centres, cost implication and legal aspects. In countries like Germany and the United States, due to strict environmental regulations the manufacturing firms are mandated to develop product waste recycling/disposal system on their own. They put in place the reserve flow system for collection of the used products or the wrappers through reverse logistics. Reverse supply chain acts as switching barriers to their customers to offering value added services to build up the competitive edge and leveraging for customer satisfaction. In many industries, product recalls programmes are initiated to call back product with poor designs and poor quality to save company's image. Product recalls take place when large number of consumer's safety is at risk and company's image is at stake. However, to recall products from market requires a reverse supply chain system in place.

REVIEW QUESTIONS

1. Discuss the role of 'reverse supply chain' in a company's competitiveness in today's business environment and the barriers in its implementations.
2. 'Reverse supply chain competency is a result of concern to environment across the globe'. Explain.
3. How is 'reverse supply chain' used as a tool for competitive advantage?
4. Discuss the various design considerations for reverse supply chain system.
5. 'Reverse supply chain' is essential for implementing 'green marketing' concept. Discuss.
6. Explain how the manufacturer benefits from a reverse supply chain system.

INTERNET EXERCISES

1. With the growing economy, the increasing amount of disposed e-goods can induce environmental issues if they are not properly managed at product end of life. Investigate scope and design of reverse supply chain system for e-waste. For more information on e-waste in India visit <http://india.ewasteguide.info>
2. The Reverse Logistics Executive Council is a not-for-profit professional organization whose purpose is to develop best practice industry standards that take costs out of the system for consumers, retailers and manufacturers; provide on-going benchmark studies; return causal information in order to improve the entire reverse logistics process. Log on to <http://www.rlec.org/index.html> for more information on initiatives and practices in reverse supply chain in various industries
3. For live cases on reserve supply chain log on to <http://thirdeyesight.in/articles.htm>
4. Visit <http://www.supplychain-forum.com/index.cfm> to study latest developments in managing the reverse supply chain.

VIDEO LINKS

1. Reverse supply chain philips case http://www.youtube.com/watch?v=Xfdgk_CyPYI
2. E-waste recycling—A simple step for a better future <http://www.youtube.com/watch?v=Lh5Vft8Q0FY&feature=related>
3. A reverse logistics story <http://www.youtube.com/watch?v=h7tHKwYEZIk&feature=related>

PROJECT ASSIGNMENT

1. Leading automobile companies in general recall from the market the entire batch of vehicles with defective designs or sub optimal performance. Select any such firm and study their entire reverse supply chain, its management, cost implications and time bound schedules.

NOTES

1. Grant Woolf, 'Logistics in Customer Centric Service Chain', http://www.norcaloaug.com/seminar_archiev/2007_training-day-pres/2.05-woolf.ppt Integrated Reverse Logistics.
2. <https://www.cscmp.org/education/university/warehousing.asp>
3. Rick LeBlanc, (posted on 7/1/2007) 'Realm of Reverse Logistics Services Ripe for New Business Opportunities: Pallet Suppliers in Position to Expand', <http://www.palletenterprise.com/articledatabase/view.asp?articleID=2342>
4. <http://www.gruener-punkt.de/?L=1>

SUPPLY CHAIN OPTIMIZATION

19

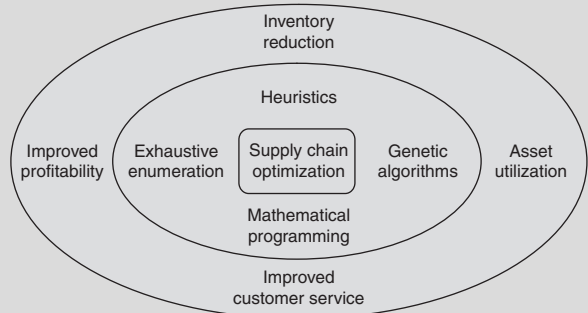
“In a general layman’s sense, it tends to mean ‘make it a lot better,’ or ‘as good as we can achieve under the present circumstances.’ As in, ‘We’re going to optimize our safety stock levels,’ or ‘We optimized the layout in our distribution center’

—Dan Gilmore”

Chapter Objectives

To understand:

- What is supply chain optimization?
- Benefits of optimization
- Optimization techniques
- Challenges in optimization process



Optimization methods have been around since post World War II, with the advent of operations research and management sciences. However, there has been a great interest in applying these concepts to supply chain during the last decade. Supply chain optimization is the application of process and tools to ensure the optimal operation of a manufacturing and distribution supply chain. This includes the optimal placement of inventory within the supply chain, minimizing operating cost (manufacturing, transporting and distribution). This is an application of mathematical modelling techniques using computer software.

SC Optimization at Lombard Pvt. Ltd (LPL)

Lombard Pvt. Ltd (subsidiary of auto manufacturer) distributes cars in economy, deluxe and luxury segments in India. The vehicles are distributed to a network of approximately 100 dealer sites through 15 distribution centres across the country. The distribution system was based on push philosophy. The push system has a limitation of choice to customers to select vehicles on colour and optional features. The customers were selecting vehicles from current inventory. The second limitation of the push system was the high system costs covering dealer's commission, cost of transportation, financing and storage. Vehicles were stocked at the point of sale, and it is the most expensive location in the distribution network.

LPL management thought of optimization in the current inventory level implementing the 'pull' strategy to serve the end user's requirements as its key performance measure. They established the following criteria for performance measurement:

- Maximize number of customers to receive their first choice of vehicles.
- Replenishment/deliver to the dealer from firm's inventory within 48 hours
- Reduce dependence on dealers
- Minimize inventory cost by optimizing the number of distribution centres

The basic idea muted was establishing supply points near to demand points to get the benefits to customer for his choice and inventory reduction by optimizing distribution centres and dealers network.

To achieve the above, the critical issue was the determination of location of a new distribution centre. Customers come to dealers, which were strategically divided into 20 major markets through distribution centres. If the vehicle is available at the dealer inventory, it is counted as a strong first choice hit. If not, the dealer attempts to satisfy the customer's choice from a distribution centres inventory. If it is not there, another dealer nearby is asked for a trade. Although the latter options incur extra costs, they still satisfy the customer's first choice. If none of these options works, the customer directly orders from the mother distribution centre/factory for the first choice vehicle. Consequently, the team measured the customer service using various counts such as first choice at dealers, distribution centres, and other dealers, second choice at factory orders, and last choice lost customers.

For the new vehicle flow cycle the total distribution cost was broken into three components: plant to processing centre cost, processing centre to distribution centre cost and distribution centre to market area cost. Finally, 10 distribution centres were fixed up based on the optimum profit with desired service level to customers.

INTRODUCTION

Supply chain managers efforts are towards maximizing the profits in the operation of their firm's supply chain. This includes measures like maximizing gross margin return on investment in inventory. This means balancing of the cost of inventory at all points in the supply chain with availability to the customer, minimizing total operating expenses (transportation, inventory and manufacturing),

or maximizing gross profit of products distributed through the supply chain. Thus, supply chain needs to be optimized to address the supply chain problem of delivering products to customers at the lowest total cost and highest profit. This includes trading off the costs of inventory, transportation, distributing and manufacturing.

Supply chain optimization has applications in all industries manufacturing and/or distributing goods, retail chains, industrial products, and fast moving consumer goods (FMCG). Supply chains have to satisfy the requirements of the customers with respect to the customized products as well as the corresponding services. The first step towards designing an effective supply chain is to assess the nature of the demand of the goods and services. For example, the shorter life cycle for innovative products exhibits dynamic demand patterns and companies have to periodically modify their supply chains to effectively utilize all the opportunities in the markets to achieve profit.

For gaining a competitive advantage a firm needs a highly effective logistics distribution network. Thus, logistics becomes an integral part of the product distribution that is being delivered to the customer. In addition, the critical element in the quality of the customer service is the lead time. Technological advances can be utilized to reduce the lead times. This can be by using EDI to improve information/document flow (for better knowledge of the customers' needs at each stage of the supply chain) to support the on time decision-making process.

Thus, in supply chain there are two polemic goals: to achieve customer satisfaction (which is cost driven) and reduce operational cost. Optimization means to achieve the trade-off between the above two goals by re-organizing the supply chain linkages.

OPTIMIZATION

The use of optimization in supply chain planning is extremely beneficial wherein supply chain process is complex in nature. The examples are supply chains in petroleum, retail and automobile industries. The use of optimization solutions are justified by such supply chain benefits as the following:

- Inventory reduction
- Improved customer service
- Increased asset utilization
- Improved corporate profitability

Optimization is beneficial in complex manufacturing environments, wherein there are many interrelated activities to take decisions. These include environments with many resource constraints and large numbers of products, plants, suppliers, and distribution centres. Planners in these complex environments need technology support to make optimized decisions. In strategic and tactical planning, the requirements of optimized solutions are more important than it is for low-level tactical and operational planning. In the former, the set for decisions is much larger, meaning there are more opportunities to make poor decisions. Also, these decisions have serious revenue and cost implications. In industries that experience material shortages, optimizing the use of these materials in the manufacturing process is very crucial. In make-to-order environments, especially in job order or customized manufacturing, optimized production schedules are crucial. For distribution intensive supply chains, planning must focus on optimizing manufacturing and distribution operations simultaneously. In general, optimization is more useful in mature, relatively stable manufacturing industries, where product demand and manufacturing processes are predictable. In such cases, realistic models can be constructed to support all levels of planning. However, in volatile product demand industries, optimization will be

less useful for strategic and tactical planning. In these cases planning process focuses on supply chain responsiveness rather than on operational efficiency. Optimization will be more useful when the level of uncertainty is substantially less. For example, in the paper industry, optimization leads to substantial gains by reducing waste paper from cutting operations. Optimization is used in the fruit beverage and oil industries, where raw materials are limited and prices are volatile. Materials must be efficiently blended to create products that meet specifications.

The goal of supply chain management is to replace physical stock with intelligent information and planning, with the ability to analyse what is possible in different parts of the chain and what is the best global/national plan.

The challenges of supply chain management vary by industry and customer segment. In simple terms, supply chains can be divided into three main classes depending on where the major constraints are: material-intensive, asset-intensive and distribution-intensive industries. In material-intensive industries, such as high-tech and machinery manufacturing, largest costs and constraints are related to the management of procured and manufactured materials. In asset-intensive industries, such as paper and metal, planners need to concentrate on optimizing machine capacity utilization and the product mix. In distribution-intensive industries, such as consumer packaged goods industry, the major challenge is to manage the distribution network and finished goods inventory from the factory to the end-customer.

AREAS OF OPTIMIZATION

Planning and Forecasting

Planning and forecasting allows businesses to create, modify and track their financial, manufacturing and promotional plans and communicate changes across their entire supply chain. The solution removes or averages out peaks in demand caused by promotions and declines generated by adverse business conditions. Firms can forecast confidently and adjust the plans quickly, meaning that inventory consumes less working capital while firms can meet customer demand.

Accurately predicting the impact of changing demand on key performance indicators such as margin, inventory turn, payment receipts and inventory flow increases sales, reduces working capital requirements and keeps customers happy. However, accurately targeting assortments to customer preferences and predicting the right inventory levels across all channels requires real-time access to critical information for key players across the supply chain.

Planning and forecasting provides the crucial front end to supply chain that takes business to a higher level. Planning includes everything a firm needs to predict and respond to changes in elements such as:

- Demand forecasting
- Channel planning
- Financial planning
- Assortment planning
- SKU planning
- Promotion planning
- Store cluster planning

Inventory

The small variances in inventory can have big impact in the supply chain. Too little inventory will have rush charges, express delivery fees and unhappy customers. On the other hand, too much inventory increases costs and risk write-offs for obsolete or expired goods. A view of the inventory across all channels ensures that the goods are where they are needed.

Inventory optimization enables firm to see inventory across entire network no matter how many channels, products or outlets firm may have. If the inventory data can be shared with suppliers so they can better manage production and shipments. The more extensive firm's supplier network and more intricate the supply chain be an, the more the firm will need inventory optimization. If inventory issues keep firm from doing the best for customers and firm's bottom line, inventory optimization can help to get it right, every time during:

- Replenishment across SC
- Vendor-managed inventory
- Collaborative efforts with SC partners

Order Fulfilment

Even for companies with only one channel or limited products, managing of orders can prove remarkably difficult. Surprise delays lead to stock-outs. Sudden drops in demand generate excess inventory. When a firm's supply chain is complex, managing incoming orders to ensure goods reach customers on time and as specified (without pushing inventory costs high) can seem nearly impossible.

Stock-outs make customers unhappy and cost businesses in terms of lost sales every year. With optimization in order fulfilment, firms can see orders across channels as they come in, manage procurement from suppliers as needed, predict and respond to impending shortages, and track deliveries to ensure that the firm's customers have the goods they want, where and when they want them. The enhanced visibility provides information a firm needs to satisfy customers without carrying excess inventory.

Transportation

Because the customers count on firm to deliver the right products to them when and where they are needed, firm cannot afford to be blindsided by shipping problems at any location in the supply chain. Managing deliveries is only one part of the job, though; firm also needs to manage costs. Firms need to strike a right balance between their own fleet and commercial carriers and negotiate the best rates in every mode. Transportation optimization provides help to reduce costs, re-route around trouble spots, coordinate deliveries precisely and deliver more, faster, for less money. In this, firms require holistic view of transportation across all modes.

Distribution

Firm's distribution network extends from supplier's factory to firm's customer's front door. When a firm is dealing with scores of suppliers on distant continents, third-party logistics companies, multiple distribution centres and customers around in the world, getting visibility into the full distribution process can be extremely challenging.

Distribution optimization solutions provide collaboration and communication with an extensive network of suppliers and partners to allow right products at the right time and at lowest cost.

OPTIMIZATION METHODS

For optimization, different methods or algorithms are used to find solutions. These methods are categorized into four groups as follows.

Mathematical Methods

Under these methods models are built using mathematical equations describing the limitations and objectives. Using a set of mathematical equations, an optimum solution can be arrived at following a certain algorithm. The common amongst mathematical programming techniques used is linear programming (LP). This method works only if all the constraints and a single objective can be expressed as linear equations. The optimum solution is tradeoff of maximum or minimum single objective which can be arrived at. The assumption in LP is that the decision variables can be expressed as regular and continuous numbers. If some decisions can only be expressed as an integer or whole number, LP does not work.

To handle integer or whole number the mixed integer programming (MIP) was developed. This method works only if all the equations are linear. In contrast to LP, however, while an optimum solution can be generated, it may take too long. Other mathematical programming methods include dynamic programming and nonlinear programming, which are not quite often used in supply chain planning optimization.

Heuristics

Under heuristic methods, improvements to a known feasible solution are attempted using the prescribed steps. Heuristics do not guarantee an optimum solution. As an illustration, a simple heuristic for maximizing an objective might follow a three-step approach:

- Freeze decision variables at zero value.
- Increase decision variables one at a time as long as the objective continues to increase.
- Stop when increasing all decision variables no longer increases the objective.

While this heuristic method might not lead to optimum results, but solutions will usually get better or stay the same. Heuristics methods are based on the logic which a reasonable person might follow in looking for an optimum solution.

Some optimization solutions for scheduling use heuristic logic based on the theory of constraints (TOC) of Eli Goldratt. These methods focus on 'bottlenecks' to develop a schedule. The TOC approach revolves around a drum, buffer, and rope concept. First, TOC uses the critically constrained resources to develop a master plan or drum that the plant or system 'beats to' or to which the pace is set. Buffers, such as work-in-process inventories and surplus time in the schedule, are put in place to ensure maximum utilization of the critically constrained resources which ensures that they do not sit idle. Lastly, all non-critically constrained resources are 'tied' together according to the drum, creating the so-called ropes that pull work through the system.

In addition to TOC, there are many types of heuristic methods that are proprietary knowledge of the vendors. Some of these are based on known, published approaches such as simulated annealing and repair-based scheduling methods.

Genetic Algorithms

These are based on a biological selective breeding concept of survival of the fittest. In this, an optimized solution is derived from a large set of possible solutions by comparing them and selecting the best ones of the group. The ones that survive this test are then mutated or crossbred to establish another set of solutions. This search method continues testing from generation-to-generation for some duration of time, thereby developing an optimized solution. These methods work well when a baseline schedule or plan exists. For example, sequencing a number of orders through a single assembly-line operation to maximize on-time delivery or to minimize changeover is an optimization task where genetic algorithms could be used.

Exhaustive Enumeration

In the exhaustive enumeration method, all possible combinations of decisions are evaluated to find the best combination. This method is used when there are relatively few decision-variable combinations to consider. For example, a job shop with 2 machines and 15 orders to sequence would generate multiple potential combinations for evaluation. This can be done manually. However, this can be done with ease on computer.

PROBLEMS IN OPTIMIZATION

Optimization solutions are used where there are resource constraints and decisions need to be made in a limited resource environment. Most supply chain optimization problems require matching demand and supply when one or both are limited. In supply chain, the most important limited resource is time. It is required for procuring raw material and components, manufacture finished products and deliver the same to the customer. This needs to be done under constraints. As the rate of procurement, production, distribution, and transportation resources is limited, demand cannot be immediately satisfied. It takes some lead time to fulfil the demand. In addition other resources, such as warehouse storage space or a truck's capacity, may pose as constraint in meeting demands. An optimization solution may have the following dimensions:

- Schedules of material procurement to match the production capacity and schedules for order fulfilment.
- Schedules of the finished product to ship to a customer or distribution centre with limitations on the supply plan.
- Supplier's capacity to produce for a specified number of hours per day and a worker that must only work so much overtime.
- A distribution centre's capacity to handle and process receipts.

The constraints in an optimization process may be categorized into two types. The first ones are those which are hard to change. These include fixed working hours in a shift or the maximum loading capacity of a truck. These must be accepted and adhered to. The second types are the ones, which can be

Exhibit 19.1 *Types of Solutions*

<i>Feasible Solution</i>	<i>Optimum Solution</i>	<i>Optimized Solution</i>
This satisfies all the constraints of the problem	This is the best feasible solution that achieves the objective of the optimization problem	This is a solution that partially achieves the objective of the optimization problem. It is not the optimum or best solution, but it is a satisfying or reasonable one. This is usually one of the best feasible solutions

subjected to change. These cover finished products delivery due dates or warehouse space limitations. Customer due dates can be changed or a product may be squeezed into a warehouse temporarily, making constraints less stringent. The cost penalties are imposed if these constraints are not met. For example, missing a customer delivery date is a more important concern than inventory updates in a warehouse. Hence, the objectives of optimization are:

- Maximizing profits
- Minimizing supply chain costs
- Minimizing cycle times
- Maximizing customer service
- Minimizing dates
- Maximizing production throughput
- Satisfying customer demands

Once a problem for seeking optimization is formulated, the firm needs to determine the best course of action using a set of logical steps or algorithms/computer programmes to search for a solution that achieves the objective. Three types of solutions towards optimization can be developed as shown in Exhibit 19.1.

A large amount of money is wasted due to time and cost inefficiencies at various points in the supply chain. This is due to lack of integration of processes in the supply chain system. In many places manual techniques are used to information analysis and exchange. The time and cost wasted by such manual processes results into loss of orders, late deliveries, high inventory costs and low customer satisfaction.

BENEFITS OF OPTIMIZATION

The supply chain optimization (Exhibit 19.2) improves demand visibility and allows manufacturers and distributors to have forecasting, manufacturing and stocking decisions with high degree of accuracy based on the timely information. The result is reduction in inventory carrying costs. On the marketing side, the decisions on pricing and availability can be taken on real-time information. The errors in orders and product dispatches are less resulting into fewer product returns. Process automation in the flow of supply and demand information up and down the supply chain helps in reducing the operating costs associated with manual information flow. The speed and accuracy level communication gets enhanced many times.

The Internet is driving a global business transformation as people increasingly conduct business online. An e-business brings about a wealth of opportunities for companies of all sizes. Being online

Exhibit 19.2 *Earlier vs. Optimized Supply Chain Network*

<i>Earlier Network</i>	<i>Optimized Network</i>
Local or ad-hoc decision-making	Coordinated decision-making
Sub-optimal production locations	Optimal production locations
Many interplant transfers	Minimized 'emergency' transfers
High transportation costs	Optimized transportation
Inconsistent delivery performance	Predictable delivery performance

they can extend their reach into new market segments and achieve speedy communications with employees leveraging the Internet and integrating all business processes across the entire supply chain. IT enables to have advanced planning in supply chain based on complete view of customers across the entire supply chain to analyse, optimize and execute on a common set of information platform. This uniform execution enables organizations to fulfil customer demands in the shortest time, at the lowest cost and with maximum profits.

VALUE ENHANCEMENT

It is important to have an understanding of the three dimensions of supply chain. These are time, transparency and trust. Each one is these are interrelated and dependent on the others. The time dimension of the supply chain helps firms to have transparency of activities across the entire. This in turn builds confidence amongst the all supply chain partners and creates trust between them.

All the dimensions are interrelated. If anyone of these is missing the integration in supply chain is broken resulting into loss of transparency, trust and an understanding of time. Of all, 'time' is a starting point and it creates transparency and trust in process and partners, respectively, resulting in gaining the competitive advantage.

Time dimension decides the velocity in the process. In general, in many organizations there is a velocity without direction. This means that the partners and employees are doing things which are not aligned with the goals of the organization.

The major objective of supply chain is to maximize the customer perceived value of the product/service. The velocity will further add value by way of customer satisfaction at the right time and place. To have velocity in supply chain time compression is required in all processes such as procurement, manufacturing and distribution. Time is a common measure across all supply chain partners, whereas cost and price data is open to a variety of interpretations. The time is the only focus to get rapid bottom line results.

The time dimension is applicable to non-value added activities (Figure 19.1) in supply chain process. These can be categorized into queuing time, rework time and time wasted due to decisions or indecisions. Sometimes the high safety stocks to enhance customer service level may prove excessive and can be reduced by understanding the time dimension of the non-value added activities.

In supply chain, the transparency of cost is of critical importance. Hence, it is critical for the supply chain manager to understand that the total cost is not the sum total of cost of sourcing, converting and delivering. There are some hidden costs which can only be revealed through ABC (activity-based costing) technique.

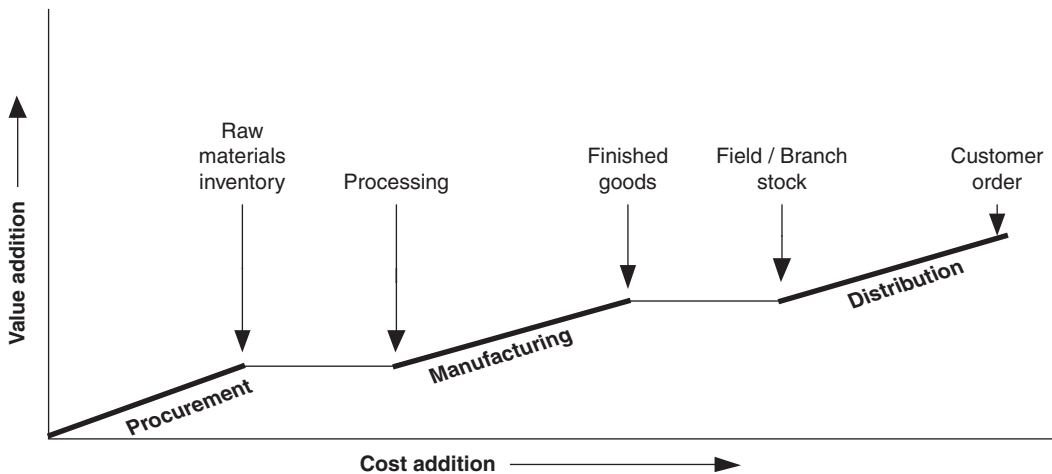


Figure 19.1 Activities: Value and Cost Addition

The third dimension of supply chain is the trust, which is required for seamless integration and collaboration of all supply chain processes and partners to achieve the common objective of maximizing customer value and reducing costs. The role of HRM (human resource management) is very critical in building the dimension of trust between organizations. Trust helps in gaining competitive advantage. Trust is required for co-operation, co-ordination and finally collaboration amongst supply chain partners to make the supply chain competitive. All the above three dimensions if dealt with properly helps supply chain process to be:

- Responsive
- Efficient
- Cost-effective

Thus, dilution in time dimension may result into an excessive inventory. It may further lose transparency in supply chain processes and inventories and consequently may result in loss of trust amongst supply chain partners and the employees.

Supply chain complexity has a major impact on supply chain performance, that is, inventory cost, inventory turns, flexibility, responsiveness, etc. In today's environment, businesses are facing many challenges that are increasing the supply chain complexity such as increasing numbers of mergers, acquisitions, and joint ventures to achieve geographic coverage and market penetration, broadening supply base spanning different geographic areas, rising rate of new product launches to combat commoditization and meeting increasing customer needs and their demand for higher service levels in terms of speed in delivery, reliability and flexibility.

In a nutshell, there are three main drivers of supply chain complexity. These are operational configuration and structure (the physical attributes of a supply chain), operations management processes, and systems and products/services managed by the operational processes. Hence, for reducing complexity and to achieve performance the supply chain needs to be optimized.

The supply chain optimization is nothing but a strategic supply chain design which involves three steps: defining supply chain strategy, design and optimize a simplified supply chain, and manage change with a roadmap.

With the network-wide supply chain optimization, considerable process efficiency can be gained to balance distribution cost vs. capital expenditure, global sourcing and assembly vs. local sourcing and assembly, design for growth vs. design for flexibility, and service levels vs. inventory holding costs.

OPTIMIZATION STRATEGY

A complete supply chain strategy is essential to drive a company forward. The ability to achieve improved marketplace performance begins with a focus on the business needs and supply chain possibilities of an enterprise. As a point of entry into supply chain optimization, companies must first decide whether to go with the latest Internet technology. They must then conduct a strategic assessment of the enterprise to establish a best-fit solution combining business and supply chain strategies, process and technology.

Leverage the Internet

Optimizing and integrating physical processes to enhance growth is not possible without technology. In order to maximize IT investment, companies must align the supply chain strategy with the business strategy to maximize process capabilities, deploy the technology to support the process and connect more potential trading partners more quickly, effectively and inexpensively than ever before.

Build Value Networks

The precise mix of capabilities needed to build a high-performance supply chain must be brought together. This is done by adding value to the operations of each supply chain partner, developing joint efficiencies and removing non-performing assets and generating more trust and confidence in both the ongoing and new supply chain relationships.

Develop Service Collaborations

By developing service collaboration through franchise route, companies can make a stronger service commitment and thereby gain competitive advantage by recognizing and meeting the needs of each individual customer.

Promote Customer Leadership

Company's sales, marketing and supply chain personnel must work together to understand a complete view of the enterprise, including consumer behaviour, customer profitability, product financial performance, promotion effectiveness and segmentation of marketplaces to operate and sell more effectively. The supply chain must be demand-driven rather than forecast-driven.

Deliver Value to Customers

Today's Internet-based supply chain model is designed to increase companies' bottom line by enhancing collaboration between businesses and their customers, suppliers and trading partners, increasing speed and flexibility of delivery. Customers today demand reliable and accurate delivery of competitively priced products and services that meet their unique requirements. They also want real-time product availability, pricing and delivery before they place an order. They want to design products to suit their unique needs and 24×7 access to track order status, payment and delivery information. A company's

ability to respond to customer needs and fulfil customer orders, faster than its competitor is important for an organization to survive in the fast dynamic business environment.

OPTIMIZATION AND SIMULATION

In general, mathematical optimal solutions are not used in practice. Heuristics are used as a pre-process to break the problem down to make it easier for the optimizer to find a solution. Optimization with suitable solution for implementation is used for supply chain network planning, transportation planning, inventory optimization, factory scheduling, etc. In optimization, mathematical techniques like linear programming give the best result.

When using optimization processing over a very large data set, such as a complex global supply chain network or huge transportation plan, heuristics are often used to reduce the size of the problem that the optimizer is working against. This enables it to complete faster, or to ensure it does not produce theoretically optimal, but practically impossible solutions.

Optimization is generally based on some fixed estimates of demand over a given time frame. You can alter that demand estimate and run a different scenario to compare the impact on the recommended solution, but optimization, in general, is not good at handling highly variable demand or system inputs.

In optimization, it is hard for the user to really understand the interplay of various factors in supply chain system. That is where simulation comes into play. In simulation, a model of the system is built and scenarios are tested to get the desired results.

Running the simulation allows the analyst to see the behaviour of the supply chain system over time, as these inputs change. It may allow bottlenecks to be identified that would be missed in an optimization programme. The benefits of simulation are better ability to understand the impact of dynamic events, better total system understanding and risk mitigation. But these benefits come at some cost. However, optimization and simulation can be used together in supply chain network design.

SUMMARY

The benefit of supply chain optimization is a smoother and a more streamlined operation that saves costs and enhances profitability. The savings come from reduction in inventory, order entry points, shipping locations, manufacturing plants, etc.

To achieve supply chain optimization, a structured and flexible framework is needed to implement complex changes and balance polemic goals (cost and service) and diverse priorities. Then assessment and integration of infrastructure and information technology, management systems and mechanisms, organization, jobs and skills, and culture must be performed to make business processes better.

Supply chain optimization can increase revenues by accessing new customers in global markets, promote customer loyalty by delivering what the customer needs, and save money by reducing inventories, cycle times and procurement costs. Supply chain optimization is an effective method to improve a company's effectiveness and profitability. It requires commitment from the corporation with a sense of mission.

REVIEW QUESTIONS

1. Explain the different methods/techniques used in supply chain optimization.
2. Which are the different areas in supply chain wherein optimization is applied?
3. Describe the scope of optimization and enumerate the major benefits of optimization.
4. Compare and contrast optimization and simulation.

INTERNET EXERCISES

1. Manhattan Associates helping companies streamline their supply chains to achieve lower costs, higher profits and happier customers. Visit their website <http://www.manh.com/> to understand the scope and dimensions of supply chain optimization solution. Click the diagrams on website for an interactive approach to understanding SC optimization solution.
2. Applied optimization develops customized, enterprise solutions to improve performance at each link in the supply chain. Their solutions are built around rigorous mathematical models of the customer's process(es), using state-of-the art programming platforms. Visit <http://www.applied-optimization.com/CaseStudy1.aspx> to study more.

VIDEO LINKS

1. Supply chain optimization <http://www.supplychainbrain.com/content/videos/2011-videos/scl-summit/total-supply-chain-optimization/>
2. Tackling challenges of SC optimization <http://www.supplychainbrain.com/content/videos/2010-video/operation-stimulus/tackling-the-challenge-of-supply-chain-optimization/>

PROJECT ASSIGNMENT

1. Take up a company in FMCG sector where product availability is the key metrics for customer satisfaction and firm's growth. In general, due to the nature of product purchases by the customers the company in FMCG sector locates one distribution centre in each of the major markets in the region. Study the current inventory levels and find out if there is any scope for optimizing the number of distribution centres and inventory level without compromising the service levels.

NOTE

1. Dan Gilmore, *Supply Chain Optimization vs. Simulation*, *Supply Chain Digest*, 31 May 2007.

RE-ENGINEERING SC TO DIFFERENTIAL STRATEGIES

20

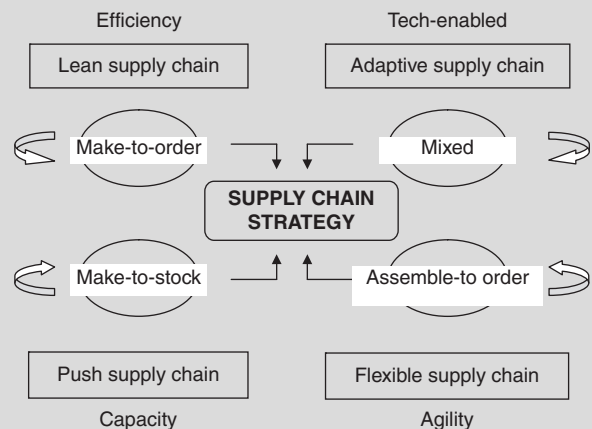
“Supply chain excellence cannot be achieved without synthesized supply chain.
Supply chain synthesis required design, planning and execution

—James A. Tompkins”

Chapter Objectives

To understand:

- Supply chain constraints
- Manufacturing strategies
- Aligning SC to marketing needs
- Digital and cellular manufacturing
- Design of SC to manufacturing strategies



Supply Chain Differential Strategies Guideline

Manufacturing is a process of value addition. The effectiveness of manufacturing process is judged by the cost per unit in processing of the product. The firm may adopt make-to-stocks, make-to-order or assemble-to-order manufacturing strategies depending on its product characteristics and marketing strategies. But cost control and flexibility can be achieved more by its material requirement planning, scheduling and capacity to produce and capability to get produced. Supply chain design depends on the degree of requirements of an interface between logistics, procurement, and manufacturing strategies to address the market needs in volumes, varieties, speed and quality.

Streamline.com¹: Differential Supply Chain Strategy

Streamline.com is a grocery retail chain in the United States. They have a re-engineered supply chain for differential strategy. Innovation is needed in the area of ordering physical goods such as groceries and fulfilment of the same. Streamline studied the consumer behaviour of their regular clients and found that most of them want the convenience of shopping from home. Many of them have shown their willingness to pay extra for home delivery. The extra charges are for extra services to achieve economies by the retailer.

They had two options to service the client at home. The first one is to fulfil the order from the nearest shop as and when the customer orders grocery. The second one is to maintain inventory at customer's home by making periodic deliveries by Streamline employees. The first one may help retailer to save site cost but they may take longer time to deliver the material to customers. Streamline took a second approach with a service called 'Don't Run Out'. Every week, for every customer, Streamline automatically fulfills a customized standing order for a selection of standard grocery products, such as milk, juice, pet food, and diapers. It makes home deliveries from a selection of 10,000 groceries to customers in three metropolitan areas (Boston, Chicago, Washington). In addition to the basic groceries, Streamline provides other services to its customer, including photo-development, dry-cleaning and UPS-pickup. Streamline also installs a free refrigerator in customers' houses in order to deliver fresh food items when the customer is not at home. This adds value to Streamline's product-service offerings. This has given competitive edge to Streamline's supply chain. With this winning formula, the customers increase their switching cost if they migrate to a competitor. Only disadvantage with this unique service is that many customers might be reluctant to give home access to Streamline employees. The other one is the fixed service fee for deliveries at home on weekly/monthly basis creating a deterrent effect on the part of customers.

INTRODUCTION

The business organizations world over have realized that the business processes to satisfy the demand of the customer are of critical importance to them. They realized that functional approach to business process would no more help in developing competitiveness. Due to this recognition, there was paradigm shift in their outlook and thinking. They experienced that, today, the competition is shifted from products to supply chains, that is, supply chain of one company competing with that of the other, to deliver superior customer value. Hence, to effectively and efficiently manage any business, supply chain has become of critical importance to manage and offer superior product proposition to its customers. One of the biggest challenges organizations are facing today is the need to respond to ever-increasing volatility in the market with cost-effectiveness. To tackle various supply chain issues, firms have successfully adopted innovative strategies to unleash the hidden potential through the supply chain linkages.²

Manufacturers add value by converting raw materials into consumer or industrial products. The value creation is by producing and marketing product service bundles to users or middlemen in supply chain. The range of products a firm manufactures evolves from its technological capability and

marketing strategy. Firms overcome manufacturing competencies based on market opportunities and willingness to take risk. A trading partner will view a firm initiating manufacturing operations to produce automotive parts differently from one that produces soaps and detergents. While the products produced are clearly different, the real differentiator between firms is found in competencies related to knowledge, technology, process and strategy. The capabilities and competencies that are exhibited by a manufacturing firm are dynamic and are continuously modified in the eyes of supply chain partners as it conducts business and develops new products, and performs promised value-added services. Thus the combination of products, services, capabilities, and competencies represents a firm's value proposition and provides dimensions to its supply chain opportunities.

MARKETING NEEDS

Brand Support

Many manufacturers spend a great deal of promotional money to create brand awareness and acceptance among prospective buyers resulting into being identified by their product brands. The measure of a customer's purchase preference based on a manufacturer's reputation, product quality, and supply chain capabilities is known as brand equity. However, the stronger a firm's product brand image among buyers, the more leverage the manufacturing organization will have in determining supply chain structure and strategy. For instance, Intel dominates how chip is sold, distributed and used for. As the products are independent of customer acceptance, a firm that brands and markets a particular line of products may not, in fact, be engaged in either the actual manufacturing/assembly or in the performance of supportive logistics services. Today, it is common practice for an organization to outsource some or even all manufacturing and logistics operations required to market a specific product. However, the type of manufacturing process, cost in operations and distribution determines the attractiveness of outsourcing. Logistical requirements in terms of inbound materials and finished product distribution are created by the location of manufacturing operations and those of suppliers and customers. However, the power to determine the range of value-added services, physical product movement requirements, timing and characteristics of flow along the supply chain is directly related to the demand of the brand in the market.

Flexibility

The effectiveness of manufacturing processes can be defined in terms of the relationship of cost per unit to volume of output. The traditional perspective is to treat volume in terms of the well-established principle of economy of scale. Economy of scale results from efficiencies generated by specialization of process, workforce, fixed asset utilization, procurement economies and limited need for process changeover. Economy of scale is extremely important in manufacturing situations, wherein fixed cost is high in machinery to convert raw materials to finished products. For example, the continuous processing industries like refineries, paper and steel gain competitiveness through economies of scales. In such industries, high capital investment coupled with high cost of change over tends to results into long production runs. In terms of logistical support, two considerations related to volume, influence supply chain design. First, supply chain operations must accommodate the frequent production runs and tune both inbound and outbound logistical requirements accordingly. Secondly, plan the warehousing and material handling needs to the lot size produced during a specific manufacturing schedule.

Variety

In addition to volume production to achieve economies of scale, other requirement from manufacturing set up is flexibility in product varieties. Here manufacturing processes are characterized by relatively frequent product runs and high repetition of small lot sizes. The manufacturing processes that feature variety, can rapidly switch production from one product to another while retaining efficiency. These processes are referred to as having economy of scope. It means that a manufacturing process can use varied combinations of materials, equipment and labour to produce a variety of products.

Variety refers to the product variations that the firm can manufacture in a given manufacturing process. The achievement of economy of scope in such situation is directly related to the speed and cost of changeover from one product to another. In terms of logistical support, product varieties calls for small manufacturing lot sizes, and flexible material requirements. The product variety through flexible manufacturing setup influences the requirements of transportation and warehousing services to accommodate the production output.

SUPPLY CHAIN CONSTRAINTS

Product volume and variety determines the nature of logistical support. However, the three primary constraints that influence manufacturing operations are capacity, equipment and setup changeover.

Manufacturing Capacity

Manufacturing capacity decides how much product can be produced per unit of time. The relevant measure of manufacturing capacity is a firm's ability to reach and maintain a specific level of quality output in a predictable time period from a (installed) process or machine having fixed rated capacity. However, in the 'flexible demand' market situation responsiveness can be achieved through flexibility in a combination of manufacturing, procurement, and logistical operations of the firm. The major constraints will be the basic maximum capacity beyond which the increase may not be possible.

Equipment constraints are related to flexibility concerning the use and sequencing of specific machines to perform multiple manufacturing tasks. However, some manufacturing requirements are more easily accommodated across a family of machines and by using variable work sequences than are others. In many situations, a specific machine or work task tends to constrain or act as a bottleneck to the overall manufacturing process. Likewise, logistical capability to accommodate different patterns of equipment utilization may serve to increase or reduce the flexibility of the manufacturing process.

The manufacturing setup constraints are directly related to product variety. Today manufacturing management is progressed to speed up both process changeover time and the time required to reach required capacity through modular manufacturing units.

Lead Time

Manufacturing lead time indicates an elapsed time between release of a work order to the shop floor and the completion of all activities to get products ready for dispatch. The manufacturing process consists of both operational and inter-operational time. Operational time is the combination of setup and running or actual production time. In any manufacturing situation, the greater the amount of

total lead time accounted for by actual production, the inherently more efficient is the conversion process. Manufacturing processes also encounter unexpected losses of time. Manufacturing efficiency is negatively impacted during periods when a process, line, or machine is idle due to queuing, waiting, or under breakdown. On the other hand, logistical delay on the part of a raw material or component supplier can result in manufacturing failure to meet the planned output. As a general rule, firms that compress manufacturing lead times and control or eliminate unexpected performance variance (in manufacturing or logistical operations) can achieve flexibility to accommodate customer requirements with low-cost manufacturing.

Logistical operations committed to supporting manufacturing can impact operating efficiency in a variety of ways. Lot-size efficiencies related to manufacturing frequency and repetition are dependent on reliable logistical support. The decision to produce large manufacturing lot sizes directly creates a need for logistical support. Economy of scale drives procurement best practice and average inventory investment across the supply chain. The decision to focus on 'variety in manufacturing' impacts the logistics requirements by adding complexity of frequent changeover. Hence, logistics is critical in achieving high levels of lead time performance or logistical failure can increase manufacturing lead time by introducing unexpected delays. Thus, the management of raw material and finished inventory stocks is a prime responsibility of logistics.

SUPPLY CHAIN DIFFERENTIALS

Many firms having diverse product range, but the supply chain for all the products is same. The question is how a single supply chain can deliver different value propositions for different products in different markets. The marketing people have different strategies for different brands and markets, while in majority of the firms, for other functions like production, finance, purchase the entire product range is undistinguished by the portfolio of products. This may not work as the products in the different stages of product-life cycles need different support from the supply chains. For example, a premium product with the high image building promotional support and uncertainty in generating volumes will certainly require a different supply chain configuration than for a low-priced high-volume product targeted to generic markets. The high profile building product certainly requires support of product availability. The demand generated due to huge spendings on promotion will be a loss for the company if product is not available in the distribution channels. Thus, in the above two cases the approaches required by the company for two different products meant for two different markets will differ. Hence, differential supply chain strategies come into picture. In other cases, for successfully marketing a consumer durable product in a small niche market, a 100 per cent service support is a must to keep the customer satisfied to endorse a word-to-mouth certification. The firm cannot afford to lose the customer as the opportunity cost will be very high. Under these circumstances, the setting up of a higher buffer stock inventory in the supply chain is a must. However, this level will be decided considering the low volume sales of the product. In addition for higher degree of responsiveness the firm will have to engage customized transportation, which may cost more than the normal truckers. Further, down the supply chain the procurement and production department should take special care to schedule the procurement and production for this product. Thus, the premium brands will get both differential and preferential treatment as compared to other brands. The supply chain differentials may be based on the various strategic approaches, such as business-driven, product-driven, customer-driven or may be a combination of these to achieve the desired objectives in the given business environment.

Customer Segmentation

Customers are traditionally segmented on industry, trade and usages, demographics, etc. and the same level of service is extended to all. The new approach should be segmentation specific based on the need or benefits or value of the customer and supply chain should be appropriately designed to service each segment. For example, TOYOTA has different supply chains for standard and customized models to satisfy the needs of different customer segments. In non-differentiated industrial product like steel sheets, used for fabrications in a variety of industries, the steel maker offers a product differentiation by applying rust-proof coating on the sheets and sells it at a premium price. But this creation of separate supply chain for such premium product is dictated by the long-term profitability of the particular segment and the investment pay backs. Some firms adopt to innovative logistics solutions to provide cost-effective delivery schedules to its customers or offer VMI (vendor-managed inventory) solution to their profitable customer, may call for configuring of a separate supply chain on the part of the supplier to support the customer to reduce his inventory level.

Postponement

This is nothing but product differentiation closer to the customer. A leading PC manufacturing company was catering to global markets having different voltage standards for powering electric equipments for different countries. The supply chain was configured in such a way that at the parent manufacturing location in Europe, only bare PCs without power packs were made and at the warehouses in the destination countries, the PCs were fitted with power packs suitable for the voltage system prevailing there. Thus, by configuring suitable supply chain the parent company could reduce the inventory over 50 per cent and could bring flexibility in production to counter the demand fluctuations in different markets.

Benetton, a major supplier of knitwear in Europe, having longer manufacturing lead time for manufacture of sweaters to address the issue of responding to changing tastes of consumers, revised the manufacturing process by postponing the dyeing of the garments until after the sweater was completely assembled. Thus, the colour choices could be delayed until after more forecasting and sales information were received, resulting into lower surplus inventory.

Strategic Procurement

This calls for drastic change in the role of the purchase function; it traditionally has to fulfil the requirements of the production department. Under the new role, the purchase department to achieve its major goals requires the following:

- Reduce the cost of raw material
- Reduce the cost of transaction
- Ensure acceptable quality of supplies

This is only possible by developing partnership with the suppliers. Build up supplier base with few partners through stringent vendor development process. Help them technically and financially to reap out long-term benefits of cost reduction, supply consistency and quality reliability. Treat them as extension of your manufacturing facility and share both benefits and losses. In short 'help to get helped'.

Forecasting and Planning

In the major supply chain initiatives, there is a need for drastic changes in planning philosophy. Traditionally, forecasting was used for production scheduling, which invariably used to result into frequent changes in manufacturing plans and procurement schedules due to demand fluctuations and changes in customer requirements. The outcome used to be build up of unwanted inventories, shortages of required material and delayed replenishments to dealers leading to customer dissatisfaction. The firm had to face losses on both the fronts: fund blockages in unwanted inventories and opportunity loss in sales to customers. For an effective demand management, successful firms across the globe have to adopt the following guidelines in their planning process:

- Annual product forecasting to be used for capacity planning.
- Quarterly forecasting for resources planning (materials and manpower).
- Monthly forecast to be used for resource commitment.
- Weekly forecast for production scheduling.

This was the road map for companies who had gone for supply chain initiatives and subsequently succeeded in planning.

Technology Strategy

Basically supply chain is about management of three flows: information, inventory and cash. The complexity of these flows depends on company's product portfolio, supplier base, width and length of product distribution channel, customer base and market-product segments. Under such circumstances, these flows need to be managed by using latest information and communication technologies. IT enabling will help the company to implement the supply chain strategies effectively.

This includes the use of enterprise resource planning (ERP), materials resource planning (MRP), distribution requirement planning (DRP) and other supply chain planning algorithms. With advent of the Internet and web-based technologies, some of the firms could enhance their responsiveness to markets (both customers and suppliers end) many fold and this has helped them to reduce the planning cycle time to 40 per cent of what they used be earlier.

Box 20.1

Asian Paints: Manufacturing as Differentiator

Asian paints, a paints giant in India has manufacturing facilities in 23 countries and is also the largest exporter of paints in India. Its major brand offerings are in decorative (interior and exterior), automotive wood and metal paints segments. It has an overall market share of 45 per cent in the Indian market.

To strengthen its manufacturing process, Asian Paint implemented i2 suite which comprises of modules like demand planner (for forecasting accuracy to reduce inventory), supply chain planner (to rationalize the deviation between the forecast and distributed production capacities), factory planner (was to ensure optimal utilization of available capacities) and rhythm planner (does plant level sched-

uling). Asian paints, for supply chain visibility along with IT enablement has also taken the following initiatives in manufacturing operations:

- **Overall asset effectiveness:** Introduced automation for high order responsiveness and throughput improvement, adopted right first time capability and stability factor by doing it right in the first time, and removed bottlenecks by eliminating the robust measurement systems, restructuring operations and improvement in availability by addressing non-value adding operations.
- **People:** A new approach of optimization of the available human resources, creation of enabling work culture and improvement in HR processes emphasizing on the improvement of the skill set of employees.
- **Environmental initiatives:** Drafting of environmental policy in the year 2001 and committed to pollution prevention, statutory compliance, and continual improvements towards achieving a 'clean environment'.
- Quality systems through TQM.
- **Safety:** Introducing working safety measures through employee training.

With the above initiative, the company benefited in production schedule compliance by 95 per cent, WIP inventory reduced by 44 per cent, blocked production capacity released by 32 per cent over 4 years, effluent load decreased by 31 per cent over 3 years, cost per tonne of the final product was reduced by 3.5 per cent, re-work reduction by over 90 per cent, safety in workplace improved (number of accidents on the shop floor reduced by 85 per cent and addition of 51 per cent in production man days). As a result Asian Paints achieved their goal of manufacturing excellence by both IT enablement and other operational initiatives.

Waste Identification, Mapping and Elimination

Many firms use waste removal strategies to gain cost advantages. To remove the waste in the supply chain, it is essential to understand the type and the location of the waste across the supply chain of the firm. The firm may use different techniques to understand the waste. However, Cannon used 'waste workshop' method for waste mapping. The starting point is to identify the generic waste that exists across the supply chain. As per this method, a number of personnel are selected from across the value stream and asked to complete a workshop which covers:

- Waste power and energy
- Waste human potential
- Environmental pollution
- Unnecessary overhead
- Inappropriate designs
- Department culture
- Inappropriate information

The waste is mapped, analysed and interpreted to take appropriate steps to eliminate or reduce it. Scores are given to each waste to understand its gravity in terms of its impact on supply chain effectiveness and

efficiency. Methods are then identified to eliminate the wastes. The programme is then scheduled with fixed time table.

In India, a lot of service firms in hospitality industry have taken energy wastage very seriously and have gone in for reducing energy consumption for making their service operation supply chain better cost-effective. Why all this? For an improved competitive position such as 'going green' is good for business and attracts more customers.

The Orchid, a leading five star hotel in Mumbai, Maurya Sheraton in New Delhi, and Taj Group's Lake Palace Hotel deploy waste reduction strategies to reduce supply chain operating cost and attract more customers to eco-friendly environment.

Performance Control

Without any performance control measures in place, for a system however great it is, the entire planning will be a futile exercise. For smooth operation of any system, the deviations have to be noted on a regular basis and actions initiated to avoid the loss. However, this responsibility lies with persons involved in operating and managing the supply chain. Hence, the firm should design performance measurement system based on fair appraisal, incentives, rewards and penalties for those who are involved in supply chain operation and management. For collaborative supply chain operations, the rewards and penalties should be fairly shared to have a win-win situation.

Transparency and Information Sharing

When 'outsourcing' is the buzzword today, multiple partners are involved to bring effectiveness and efficiency in supply chains. The success of the supply chain largely depends on the healthy relationship amongst the collaborating partners. The relationship in turn depends on mutual trust and transparency in sharing information to manage the three flows of the supply chain. The firms needs to resolve the sensitive issue of information sharing which varies with the industry, technologies used, intellectual property rights, etc.

MANUFACTURING STRATEGY

Both the marketing and technological forces are critical elements in manufacturing strategy formulation. A firm's prevailing marketing practices may serve to ground manufacturing strategy in terms of customer acceptability. Technology drives strategy to a manufacturing process that is competitive. For example, a manufacturer having a process dominated by economy of scale may desire to improve process flexibility. But for that significant investment will be required to increase frequency and repetition.

The changing nature of the market and available technology serve to alter a firm's existing strategic posture. For example, the steel industry, which was long dominated by processes, is highly dependent on economy of scale. However, after witnessing market acceptance of a wide range of new steel-based materials combined with value-added services, the nature of basic steel production has also undergone a change. New process methods are being perfected that has reduced long-time dependence on high-scale manufacturing processes. The combined impact of this change in market and process has shifted the old mindset of steel producers. The manufactures are now shifting to new technology and method such as 'cellular manufacturing' and 'digital manufacturing'.

Cellular Manufacturing

In a traditional manufacturing firm, all the workmen are housed under one roof to put maximum hours to get targeted output. For mass production there was a need for skill and facilities to speed up product flow. However, with advancement in technologies the need for work force with right skills coupled with knowledge was felt. This gave rise to a new concept with coordinated approach of people, processes and products in places of production.

With the focus on managing each part of the factory methods to design, deploy, and evaluate the efficiency of each individual manufacturing process became the norm. The concept of plant within a plant and product-specific cells got evolved.

In the new factory design, the focus of manufacturing cells is to reduce throughput times and lessen inventories to meet market responses. The other benefits are improved resource utilization, better quality control, reduction space and improved work enrichment. In a cellular manufacturing design, the cells are classified as follows:

- **Product-focused cell:** A single dedicated product focused cell with multiple sequential processes to have large volumes.
- **Group technology cell:** These are 'mixed model cells' with multiple processes catering to groups of similar products, if the process is not sequential.
- **Functional cell:** These are functional cells, which handle a huge variety of unrelated process/parts. The examples are painting or heat treatment processes.
- **Project cell:** The cells that produce a wide variety of products using a wide variety of processes. They handle processes with permutation and combination to produce desired output. These cells are having both the product and process focus. The examples are a tool and die shop.

The key factors that affect the design of an efficient cell can be grouped broadly as the number of processes required, number of products that fit into the cell, market segments/requirement, and degree of automation.

There are other components, which are considered during the design of the work cell, namely manufacturing performance, product demand (also referred to as market/customer requirements), relation between demand variability and operational flexibility and asset care (equipment characteristics and improvement).

The important element contributing to the success of cellular manufacturing initiative in an organization is the human factor. The influence of morale, dedication, and acceptance to change of the personnel involved in and affected by changes is wholly acknowledged by the community at large. Cellular manufacturing is not much different in this case and personnel management plays a prime role in sustenance. The human factors that influence the design and deployment of work cells are leadership, commitment, change adoption, teamwork and interpersonal skills, and total employee involvement.

Cellular manufacturing lays importance on operator skill enhancement as operators may be required to perform fixed, floating or circulatory tasks. If the operations follow the fixed task (one operator one equipment), the cell needs to be inherently balanced in terms of workload to maintain parity across processes. The more commonly accepted tasks are the floating (one operator, several equipments) and the circulatory (one operator, one product, namely all operations). Operator mobility and ergonomics are also considered to eliminate wasteful movement and fatigue. The existing supervisors focus more on scheduling and availability of resources.

During the implementation of cellular manufacturing systems, care must be taken to maintain trusted and dedicated team and maintain a positive attitude towards the work in spite of the ups and downs encountered in implementation. An operator in a work cell is generally more skilled and knowledgeable and thereby the compensation could be restructured accordingly. Incentive programmes that recognize or reward incremental improvements help in sustaining continual improvement.

The benefits of a well-planned and implemented cellular manufacturing are improved asset utilization, reduced lead times, improved product quality, better space utilization, lesser blocked capital in inventory and improved work space. Cellular manufacturing initiative makes the plant operations lean and flexible. These small focused cells work in unison towards the common company goal of increasing profitability, reducing lead times, enhancing product quality and thereby attaining customer satisfaction.

Digital Manufacturing (DM)

Digital manufacturing is a software technology. It optimizes manufacturing processes, manage production process information, and support effective collaboration among engineering disciplines by using software. It takes into consideration a holistic view of product and process design. It enables product design to be sensitive to process constraints and capabilities.

It encompasses OEMs and the suppliers. It cuts excess planning time and being first to market helps companies to be competitive. Organizations using digital manufacturing are achieving substantial benefits and return on investments. It supports process planning, factory modelling, visualization and collaboration, simulation of operations, ergonomic and human factor analyses and optimizes the manufacturing process design. Digital manufacturing software supports functions cover the following:

- Translating designs to manufacturing platforms
- Process planning
- Production operations and process planning
- Assembly sequencing
- Task detailing
- Quality measurement and reporting
- Manufacturing documentation and shop floor instructions

DM mitigates risk, provide virtual plant tours, deploy machinery, validate processes before release to manufacturing, reduce floor space and redesign of equipment layout, identify bottlenecks, collisions, and worker issues before they happen, improve resource utilization, programme machines and cells offline, eliminate prototypes, and reduce rework or scrap. DM solution providers often combine individual functional capabilities to provide full industry-oriented solutions for manufacturing operations. DM solutions are used in automotive operations, automotive power train, airframe assembly, shipbuilding assembly and electronics.

Digital manufacturing is a software for the virtual process of designing, optimizing, simulating and executing a manufacturing process. The process is executed with high degree of predictability. The product design is also done with digital manufacturing software. The processes which performed in silos give unpredictable results. However, with digital manufacturing these can be performed sequentially and seamlessly. Currently, digital manufacturing is used by leading global players in automotive

sector for simultaneous product and process development to achieve reduction in cost and time. In the entire process, at every step there is digital access to global capacity and capability resulting into getting predictability. The benefits received are as follows:

- Sustained profit
- Reduction in product launch costs and time
- Re-use of existing processes
- Re-use of best practices
- Compress cycle
- Continuous improvement in product design

Digital manufacturing is a virtual process tool helping manufacturers to bring products to market faster with improved quality and reduced costs of operation. With a 3D vision of the entire manufacturing process the firms create products, define production process, simulate the factory, and test results before committing to resources. It helps to:

- Reduce the number of tool design changes
- Shorten manufacturing planning process
- Reduce inventory
- Optimize plant layouts
- Save facility costs
- Improve labour utilization
- Reduce number of machines
- Reduce tools and work centres

A key to success in digital manufacturing lies in comprehensive integration, process re-engineering, and worldwide availability of data. In a typical automobile manufacturing factory, due to digital manufacturing, the production time was cut by 40 per cent and overall production cost reduced by 13 per cent.

Digital manufacturing accrues many benefits to the manufacturer. These include process acceleration, product improvement and scrap reduction. However, it is difficult to quantify the above benefits, but it has positive impact on a company and its products. Digital manufacturing is the next industrial revolution and it will change the basic premises of manufacturing.

STRATEGY TO MARKET REQUIREMENTS

The varieties of marketing strategies are being evolved ranging from mass, segmental and focused, or one-on-one. These strategies are differentiated in terms of the required degree of product and service offerings. In mass marketing, limited product/service differentiation is offered. However, the one-on-one marketing strategy is built on unique or customized product/service offerings for each and every customer. The marketing flexibility and agility to accommodate specific customer requirements is directly related to manufacturing capability. Thus, a firm's manufacturing capability drives the feasible range of effective marketing strategy. Hence, for competitiveness, a manufacturing firm must be able to integrate manufacturing capability into its customer value proposition.

Inventory

The most common inventory strategies are make-to-stock (MTS), make-to-order (MTO), and assemble-to-order (ATO). MTS strategies are used to exploit economy of scale that results from long production runs of standard repetitive products. Large finished goods inventory is manufactured to meet future customer requirements. The logistical requirement to support MTS is warehousing capacity to store finished product and to facilitate product assortment for specific customers. When flexible manufacturing is introduced to speed up switchover, the inventory lots produced are smaller in quantity. However, warehouses are still required for temporary storage and to facilitate product assortment, as in MTS the products are not directly shipped to customers.

In contrast, MTO manufacturing strategies seek to manufacture products to customer specification in relatively small quantities. Logistical capacity may be required for temporary storage and to achieve outbound transportation consolidation, but most products produced in a MTO environment are shipped directly to customers.

In ATO strategy, the products and components are manufactured in anticipation of future customer orders; however, the products are not fully assembled or customized until a customer's order is received. Such final assembly reflects implementation of the principle of postponement in manufacturing. In ATO products, finalization is being performed in distribution warehouses. Hence, for implementation of an ATO strategy the warehouse operation needs to be integrated in the manufacturing process to perform customizing and assembly operations.

Supply of Critical Parts

Speed to market is a key supply chain differentiator, especially in the supply of critical parts, where customers expect delivery within hours. This is required by OEMs to speedily settle warranty claims with their end customers, for protections against downtime losses. In addition, to face pressures to reduce inventory levels and associated investments, OEMs and other companies need greater visibility, not only to parts availability, but also to pipeline supplies. It is a great challenge when supply networks are numerous and geographically spread.

Total Cost Concept

The marketing and manufacturing strategies of a firm drive the logistical service requirements. For example, MTO manufacturing strategies typically require less finished goods inventory than MTS and ATO strategies. However, MTO strategies typically require component inventory support and may result in high-cost market distribution. In light of such cost trade-offs, the design of a logistics support system should be based on the total cost concept. Total cost of operations consists of production/procurement, inventory/warehousing, and transportation. All of the above costs are impacted by manufacturing strategy. As such, total cost management represents the foundation for formulating a market distribution strategy.

The cost of manufacturing and procurement declines as quantity increases, reflecting economy of scale associated with MTS. Inventory and warehousing costs increase, reflecting the impact of larger manufacturing lot sizes. Transportation cost per unit decreases as a result of shipment consolidation. In contrast, MTO strategies reflect high per unit manufacturing and procurement costs which are, in part, offset by lower inventory and warehousing costs. In the MTO strategy, transportation cost per unit is higher, reflecting small shipment and/or premium transportation. The total cost management

results from functional integration of manufacturing, procurement and logistics. From the perspective of integrated management, it is important for manufacturing firms to design a supply chain strategy that achieves lowest total cost across the entire process.

Interfaces

The efficient and effective coordination of manufacturing strategy with the procurement of materials and components ultimately relies on logistics. Resource inputs must be procured and made available when needed for manufacturing operations. Whether the manufacturing strategy is MTO, ATO or MTS, logistics links the supplier base with manufacturing processes. Clearly, the more seamless the interface, the better the opportunity is for achieving lowest cost of ownership and, ultimately, lowest total cost of manufacturing. Such operations only emerge when there is high-level supplier integration in both operations and in design. Just-in-time, materials requirements planning, and design for logistics represent three approaches in achieving desired coordination.

The key to JIT operations is that requirement for components and materials depend on the finalized production schedule. Once the production schedule is established, just-in-time arrival of components and materials can be planned to coincide with those requirements, resulting in reduced handling and minimal inventories.

In JIT strategy, it is necessary to deal with suppliers who have high and consistent levels of quality, as their components will go directly into the finished product. Absolutely reliable logistical performance is required, which reduces the need for buffer stocks of materials. JIT generally requires more frequent deliveries of smaller quantities of purchased inputs, which may require modification of inbound transportation. For JIT to work, very close cooperation and communication between a manufacturer's purchasing organization and suppliers is essential. In JIT operations, companies gain the benefits of backward integration without formal tie up of ownership and thus keep away from investments.

Originally, JIT was applied to manufacturing processes characterized as MTS, since the effective functioning of the system is dependent upon a finalized production schedule. However, as manufacturing strategies have evolved with more emphasis on flexibility, reduced lot-size production quantities, and quick changeovers, JIT concepts have evolved to accommodate ATO and MTO manufacturing as well. In many situations, lead suppliers are used by manufacturers to sort, segregate and sequence materials as they flow into assembly operations to reduce handling and facilitate continuous supplies.

Today, some manufacturing organizations are going further and accepting VMI system to get the full benefits of JIT. The suppliers are empowered to use the customer's premises to keep and maintain the inventory of components they supply. Vendors have full access to production schedules, and have responsibility for scheduling arrival of materials to ultimately reduce lead times and cost. In VMI, the ownership of inventory rests with a supplier.

MATERIALS PLANNING

In complex manufacturing organizations a process known as materials requirements planning (MRP) is frequently used, which is an interface between the purchaser and supplier. MRP systems attempt to accrue the following benefits to manufacturers:

- Reduce inventory
- Ensure high utilization of manufacturing capacity
- Co-ordinate delivery with procurement and manufacturing activities

Implementation of MRP systems requires a high level of technological sophistication. Software applications such as advanced planning and scheduling systems have been developed to deal with the complexity of information required, such as lead times, quantities on-hand and on-order, and machine capacities for literally thousands of materials across multiple manufacturing locations.

The logistics interface with procurement and manufacturing, as well as with engineering and marketing, can be greatly enhanced by considering 'logistical requirement' right from product development to distribution stages in supply chain. At least, product packaging and transportation requirements of the product need to be incorporated into the design process. For example, if inbound components are packaged in containers with a standard quantity of 25, but only 15 components are needed to meet production requirements, then, waste will occur. Additionally, product and component design must take into consideration the transportation and internal materials handling methods to ensure that cost-efficient, damage-free logistics performance can be achieved. Similar design considerations must be made for the finished product itself.

Managing supply chain requires an interface between logistics, procurement, and manufacturing strategies. The primary concern of procurement and manufacturing is a superior quality product, a prerequisite for any firm that desires to be a global competitor. In fact, product quality has several different dimensions. World-class companies have implemented total quality management (TQM) programmes in all their activities in their efforts to achieve quality from their customers' perspective. In fact, with TQM in logistics, procurement and manufacturing operations of the firm helps in enhancing its competitiveness in the market place. Procurement in an organization is charged with responsibility for obtaining the inputs required to support manufacturing and operation. Ultimately, modern procurement professionals focus on the 'total cost of ownership' of acquired resources, not just the purchase price of those inputs.

Procurement strategies today involve consolidation of the volumes purchased into a smaller, more reliable, and number of suppliers. They include efforts to build up relationship to integrate supplier and buyer operations to achieve better and lower-cost logistics performance. Supplier integration in new product design represents another strategy to reduce total ownership costs.

SUPPLY CHAIN VISIBILITY

Today's supply chain is the primary processing mechanism of every manufacturing company. Its multifaceted, multi-company, multinational structure makes it the most complex management challenge found in any enterprise. Supply chain management does not mean that the right resources and the right materials move to the right place at the right time. Today, it also means ensuring that the entire chain of events involved in producing goods and distributing them to customers satisfies customers, minimizes costs and maximizes profit.

Managing a supply chain requires information. However, pushing the information about partners or products into a report will not achieve the goals. What today's supply chain managers need is information visibility. Supply chain visibility is the ability to know the location and status of all physical components, from raw materials to finished goods, as they move from suppliers through various stages of production to delivery to customers.

In addition, most business decisions today require collaboration, so visibility also requires that the information be shared among partners and colleagues. The visible system offers users an analytical framework within which they can work with their information. The analysis enhances visibility by providing an additional context for the information.

Today's international, multifaceted, multiple-partner supply chain makes creating a visible information environment to support it both difficult and necessary. The complexity of supply chain structures and the amounts of data they generate create the need to implement a visible supply chain.

Other pressures to control supply chain processes come from both external and internal sources. Externally, one source of that pressure is customer demand, particularly for products sold worldwide in markets that are very competitive. When operating at that scope and in such markets, visibility is essential, but gathering and managing the information that enables demand to drive the manufacturing and distribution processes is a complex job because it is dispersed widely.

Internally, cost pressures make finding low-cost suppliers and managing their participation in the supply chain a business imperative that can be executed most efficiently when the processes are managed through a visible supply chain. Implementing other cost containment programmes such as scrap minimization, efficient transportation systems and inventory reduction similarly require a visible supply chain, as do creating and managing initiatives such as distribution programmes that meet delivery goals, component quality initiatives and effective target marketing programmes. Then there is the issue of aligning the supply chain itself with corporate strategies. To maximize the contribution that the supply chain makes to overall enterprise performance means that supply chain decision-making has to be both timely and accurate, which is another ongoing source of pressure.

A visible supply chain can be implemented by using technologies that are readily available and in many cases are already in place in manufacturing enterprises. Most supply chain data is managed in a combination of the ERP system and specialized software for supply chain management (SCM). However, to plan and implement visible supply chain, key technologies must be added to the resource mix. Modern technology will make the supply chain visible, enabling production and distribution managers the ability to see both numerically and graphically what is happening in their areas of operations. This is also called dashboard technology. In dashboard technology, the decision-makers need data. No matter what methods are used to gather the data, it must be available on the manager's dashboard whenever they need it.

DIFFERENTIATION THROUGH POSTPONEMENT

Postponement is one of the strategic moves to infuse agility in companies supply chain. Amongst the many options for postponement in most cases, companies go in for manufacturing postponement. In this the final configuration of a finished product is delayed to customize the goods at the place of demand (distribution centers). With this strategy, the company postpones some value-adding supply chain activities until a confirmed order is received from the customer. The companies go in for postponement strategy because of the following reasons:

- Too many SKUs are extremely difficult for planners to plan and handle accurately.
- Demanding customers are not brand loyal and hence companies are forced to be responsive.
- Product in maturity stage of PLC and hence competition leading to market offering a variety of products.
- High level of product obsolescence, leading to inventory buildups.
- Customer service level requirements are extremely high due to competition.

The products for postponements are as follows:

- Product with short life cycles
- High variety/options to customers

- High value profile
- Standardized components and modular design

There are three design elements for implementing an effective postponement strategy such as choosing a right product, adopt standardization or modularization in products/components and find out a right stage in value chain for customer decoupling point. The success stories for postponement strategy which lead to differentiation are Dell, Mattel Toys, Sony, IKEA, Toyota, HP and Asian Paints.

SUMMARY

The design of supply chain depends on the manufacturing strategies. And the manufacturing strategies depend on the product–market characteristics. In fact, the firm has to achieve two polemic goals: cost control and manufacturing flexibility to achieve the desired customer value proposition. The firm may adopt MTS, MTO or ATO manufacturing strategies depending on its product characteristics and marketing strategies. But cost control and flexibility can be achieved more by its material requirement planning, scheduling and capacity to produce or the capability to get produced by third party in the event of an outsourcing. Managing supply chain requires an interface between logistics, procurement, and manufacturing strategies. The pervasive role of TQM, logistics, procurement and manufacturing operations of the firm cannot be denied as it helps in enhancing its competitiveness in the market place. Today, the procurement professionals focus on the ‘total cost of ownership’ of acquired resources, and not just the purchase price of those inputs. In addition, supplier integration in new product design represents another strategy to reduce total ownership costs during the life cycle of the product.

For improving competitiveness, companies adopt differential strategies in one or more of the operational areas of the supply chain depending on the target product–market combinations. In today’s competitive market, where competition is between the supply chains of the firms and not between the products, the dependence on a single supply chain may not help to differentiate its offerings in different stages of its product life cycle. In fact, it will be suicidal. The firm may not think of having separate supply chains for different offerings, but certainly will have to have smaller trajectories or mini supply chains within its main stream of supply chain to service different market–product combinations to implement its business-driven, product-driven or customer-driven strategies.

REVIEW QUESTIONS

1. What are the three primary constraints that influence manufacturing operations in general?
2. Explain MTO, MTS and MTA strategies in manufacturing and their relevance to supply chain effectiveness and efficiency.
3. ‘Design of supply chain depends on manufacturing strategies.’ Explain with illustrations.
4. Explain what is ‘MRP’ and how it helps in reducing the inventory in supply chain.
5. Marketing strategies have an influence on designing of supply chain. Explain.

INTERNET EXERCISES

1. Log on to the following websites and find out how sub-systems of SC get impacted with manufacturing strategies?, <http://www.scmr.com>, <http://www.cio.com>, and <http://www.theprogress-group.com/vsc/manufacture.htm>
2. Visit <http://www.ventanaresearch.com> to study supply chain visibility to ensure end-to-end optimization

VIDEO LINKS

1. Supply chain strategy—Professor Richard Wilding http://www.youtube.com/watch?v=RFtbI_c-cZc&feature=related
2. Wal-Mart's supply chain strategy <http://www.joc.com/logistics-economy/video-wal-arts-supply-chain-strategy>

PROJECT ASSIGNMENT

1. Select a company manufacturing non-differentiated industrial product such as (a) cements, (b) lubricating oils, (c) industrial fasteners and (d) adhesives and study their supply chains and suggest a few supply chain strategies leading to differentiation.

NOTES

1. <http://www.streamline.com>
2. Martin Christopher (1994), 'New Directions in Logistics', *Logistics Distribution and Planning*, edited by James Cooper, Kogan Page, p. 18.

SUPPLY CHAIN INTEGRATION STRATEGIES

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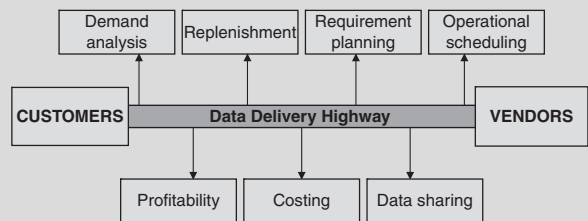
“Integration is prerequisite for effective sharing and utilization of information between different companies in the chain

—Mike Raymond”

Chapter Objectives

To understand:

- Supply chain integration—why and how?
- Role of information in SC integration
- Push, pull and push–pull system
- Collaborative planning forecasting representing
- Impact of the Internet on SC strategies
- Information visibility and SC integration



Elements of Supply Chain Integration

Supply chain is no more confined to a firm's internal operations. It has expanded outside the boundaries of the firm covering customers, suppliers and service providers. Firms are adopting seamless integration approaches between its supply chain partners. Firms are developing close relationships with profitable clients and improving relationships with its loyal suppliers. To work effectively together, manufacturing partners have developed tools for communicating with each other and managing inventory across the supply chain. Information technology has enabled companies and their supply chain partners to share information and integrate their business processes resulting in more efficient supply chain operations. The supply chain integration may be done manually but as the complexities increases, the use of technology makes the job easier and speedier.

Nita Paints: Information Visibility to Integrate Distribution Process¹

Nita Paints, a Mumbai-based paint (water-based) producing company in India, was facing a problem with dispatch of its products sequential to the manufacturing dates. The company has eight SKUs with different sizes of containers stored in the warehouse. Even though the date of manufacturing was printed (in small letters) on the bottom side of the containers, the workers dispatched containers without noticing the dates. This created a problem for the company, as many times the products produced last got dispatched first. This resulted in two problems:

- *Pile up of old inventory*
- *Customer complaints about the date of manufacturing*

The warehouse had limited storage space and keeping the continuously incoming stocks as per the manufacturing dates could not be done properly due to large order frequencies and delivery urgencies. This resulted into not taking into consideration manufacturing dates written on containers at the time of order filling and dispatches thereafter. In many cases, they were following LIFO (Last-in-and-first-out) unintentionally without noticing.

The company wanted to implement FIFO (First-in-and-First-out) system. The problem was studied by a consultant and a cost-effective solution was offered, which was accepted and implemented by the management immediately. The consultant advised to have a visual indication on the containers, that is, colour strips, to identify the period of manufacture. The warehouse workers would pick up those containers with particular colour code for order filling in that period. The solution was to have four different colour stickers (to be put on the containers) for goods manufactured in four different quarters of the year. The warehouse workers would then pick up those containers with particular colour stickers till the inventory of that SKU is used up and then switch over to other containers with other colour scheme. The management would periodically inform to all concerns about colour of the sticker and the quarter of manufacture.

INTRODUCTION

The development and manufacturing of products has become increasingly complex in today's global economy. Multiple suppliers are involved in product design and development, and these suppliers are spread over larger geographical areas. Without a central location or an integrated system of communication, it is difficult for decision-makers to identify problems in the production line or respond quickly to a change in customer demand.

To work effectively together, manufacturing partners should have some tools for communicating with each other and managing inventory across the supply chain. With a closely-managed inventory, the delivery of products can be automated based on predicted rates of end-customer sales and achieve greater efficiency and profitability. In such circumstances, supply-chain integration is a must to build a strong partnership. Manufacturers can reduce time to market, decrease costs, and manage inventory turns through supply-chain integration. The supply chain integration may be done manually. However,

as the complexities increase the use of technology makes the job easier. Companies like SAP, Oracle, JD Edwards, Accenture and i2 have built software applications to help manufacturers collaborate with their supply-chain partners. These software help manufacturers to achieve the following:

- **Total asset visibility:** Inventory movement across the supply chain.
- **Enhanced system integration:** Integration amongst manufacturing, trading, and logistics service providers.
- **Effective collaboration:** Collaborative planning, forecasting, and replenishing inventory along the supply chain.

The objective of supply chain management is maximizing services to customers of choice at the lowest total cost. However, this requires a strong commitment to close relationships among trading partners. The integration amongst supply chain participants helps in lowering the operation cost and enhances service level. The integration comes through collaborations amongst the supply chain participants (both internal and external).

ROLE OF INFORMATION IN SC INTEGRATION

The emerging business models are based on inter-firm collaboration and integration. And in today's competitive marketplace, with increasing consumer power, it is competition between supply chains which matters. This understanding calls for an integrated approach towards management of critical supply chain functions and information is key to this integration. Information transparency is a prerequisite for optimum performance of the supply chain, because it enables firms to take decisions, which are aimed at superior supply chain performance.

One major obstacle to integration is the way in which firms are traditionally managed and owned. Traditionally, firms were inward looking, with business processes aligned towards the corporate goal of profit maximization or greater market share. Additionally, as firms grow in size, the organizational structure gets complicated with different functional silos, often in the forms of departments, developing within the firm. The usual consequence is a very fragmented organizational structure with very little inter-functional communication or synergy.

PUSH STRATEGY

In push strategy, the inventory is produced in advance based on past data about the market demand. By the time the inventory is produced, the actual demand changes and then manufacturers push the inventory through middlemen offering more incentives. Hence, long-term forecast is at the base of a push strategy in the supply chain. The production and distribution decisions are made on forecasts. Hence, the accuracy of decision is dependent on accuracy of forecast. The production planning in general is done by manufacturer based on the orders he receives from the middlemen. His schedules are dependent on demand forecast. As the manufacturer has no direct contact with markets, he gets market information belatedly. If the supply chain is push based, he takes longer time to respond to the market. The demand forecast with its following characteristics has great impact on supply chain operations.

- Forecast is rarely correct because it is futuristic; hence, it is difficult to match supply and demand.
- Deviations are large with longer forecast horizon meaning that it is difficult to predict customer demand for a long period of time.

- Frequent updates by decision-makers based on his/her assessment of market forces him/her to update inventory levels, safety stock and order quantities. The non-uniformity amongst all decision points across supply chain either leads to shortfall or excess inventory commonly known as bullwhip effect. This means that variability of orders received from middlemen is much larger than variability in customer demand.
- Collaborative forecast tends to be more accurate. This is due to uniformity in inventory related decision at all decision-making points across the supply chain.

PULL STRATEGY

In a pull supply chain, manufacturer produces inventory based on actual demand rather than demand forecast. In a pull system (Exhibit 21.1), the firm produces inventory to the order thus reducing the impact of bullwhip effect and increasing service levels. However, for a manufacturer it is very difficult to implement pull strategy. In pull system, the information accuracy and speed play a key role in its implementation and success. For example, Wal-Mart used AITS (automated inventory tracking system) using satellite communication. The inventory movements are conveyed to suppliers on real-time basis so as to manufacture and replenish the inventory with least lead time. Here, availability of inventory plays the key role in successful implementation of pull strategy. The manufacture of inventory is based on real-time demand information.

As the production decisions are based on actual and real-time demand, the firm cannot take advantage of economies of scale in pull strategy. Hence, the firm would go for batch production and not full truckload deliveries. This in turn leads to high costs. Looking into the pluses and minuses of both pull and push strategy, companies are looking for via media solutions or new supply chain strategy to take advantage of the best of the both. It may be called as two-in-one or push-pull strategy.

PUSH-PULL STRATEGY

This is a two-in-one strategy using both push and pull strategies in different stages of the supply chain. The back end processes in supply chain are operated in a push-based manner while the front end processes are operated in a pull-based manner. However, the firm has to decide the boundary between these two depending on its internal strength and product market configuration.

Typically, a TV manufacturer who builds to stock makes all production and distribution decisions based on forecast. Then liquidate the stocks through dealers' network using push strategy. But in push-pull strategy, manufacturer assembles the final configuration of TV to order. This means he procures

Exhibit 21.1 *Push-Pull Supply Chain*

<i>Push Type</i>	<i>Pull Type</i>
Initiated by supplier	Initiated by customer orders
Demand is not known and forecasted	Demand is known with certainty
Respond to the speculated forecast	Reactive to customer demand
Require information for material planning and master production	Information on actual demand is transmitted throughout the entire chain

the components and parts based on forecasted demand. However, the final assembly is done in response to the request of a customer. In this example, the push part is a supply chain prior to assembly, while the pull part starts with assembly which is based on customer demand (time and specifications). The logic behind using combine push–pull strategy is that uncertainty in component demand is much smaller than uncertainty in finished goods demand. This helps the manufacturer to reduce the safety stock level.

Dell Computers is using push–pull supply chain strategy. Dell keeps inventory of components and only assembles when there is an actual order. In other words, the demand is pulled while the assembly is pushed. Many companies use the strategy of postponement, which is an example of a combined push–pull strategy.

The postponement strategy helps in reduction of finished goods inventory. This strategy is applicable, wherein the customer demand for a specific end product has a high level of uncertainty and product differentiation can be done in response to individual demand. Thus, for the front end of the supply chain starting from the time of differentiation, pull strategy is more appropriate. The push strategy is more applicable at the back end of supply chain during the process of procurement and creating sub assemblies.

The pull strategy is more appropriate to the portion of the supply chain where demand uncertainty and variability are high. Here, decisions are made in response to on-time demand. The aim of using push–pull strategy is to reduce inventory-related cost, as inventory is a real culprit in supply chain. However, the demarcation of boundary between push and pull territory has to be decided by the management.

Thus, from a whole supply chain viewpoint, deciding whether a particular supply chain is push or pull is often difficult and generally depends on the perspective of what constitutes the supply chain and where particular participants are placed in the chain. For example, the manufacture of Toyota automobiles is a leading example of a demand-driven supply chain. However, the mining of the iron ore or operation of blast furnaces that process the iron ore for ultimate manufacture of automobiles is not. At some point in most supply chains, in their widest sense, demand push meets demand pull, and at this point inventory accumulates. This point is referred to as the push–pull interface or supply chain decoupling point.

The advantages of operating to a pull model are very compelling for businesses because the planned level of production and/or service delivery is not dependent on forecasts with their inherent inaccuracies. A business operating in demand pull mode will strive to move the inventory decoupling point further upstream to its supplier or its supplier's supplier. The determinants of push or pull supply chain are economies of scale and uncertainty of demand. For example, designer custom-built car

Box 21.1

Pull Strategy at Asian Paints

Asian Paints manufactures only few major paints ingredients, which with combinations in right proportions create 1,000 shades. Asian Paints does this combination at dealers' place using computer controlled paint dispensing machines. The customer would get the desired paint shade (fresh) in the quantity he wants. Here, the manufacture (mixing) for final configuration of paint is based on pull strategy. The firm designs the product and the manufacturing process so that decisions about which specific product is being manufactured can be delayed till the demand is created. Thus, with postponement strategy Asian Paints could reduce its implant finished goods inventory by 35–40 per cent.

is made to order in response to specific customer demand. Here, clearly the inventory is very low. However, for standard models a large number of cars are made to capture scale advantages but at the risk of uncertain timing of purchase, the company needs to discount some for sale. A '30-minutes pizza' at Dominos shop produces to specific demand pull to supply specific customized pizzas. However, there is an element of push in this instance. To enable responsiveness, the pizza bases and ingredients are built as inventory in advance of the customer order. The final assembly is delayed until there is a specific order. This is an example of postponement and is an excellent example of push and pull operating in the same business.

STRATEGY IMPLEMENTATION

Depending on the location of a push–pull boundary in its supply chain, a firm can implement push–pull strategy. The boundary of supply chain starts with suppliers and ends with customers. It is also called as time line. The company shifts its strategies along with this time line. The push–pull point is located along the time line where the firm switches from managing the supply chain using a push strategy, to managing it using a pull strategy.

There are potential locations for the push–pull boundary together with industries that implemented it at various locations. For instance, furniture manufacturers locate the boundary at the production point while online shopping and delivery services like Amazon.com and Peapod locate the boundary at their distribution centres.

Evidently, as products move along the supply chain time line, their value increases, that is, the value of components at the procurement stage is smaller than after assembly and this is again smaller than the value of the same product at the store.

This implies that it is better to keep inventory in earlier stages of the supply chain rather than in later stages, since the cost of holding one unit inventory increases as the unit moves in the supply chain. However, in a push–pull strategy, the closer the boundary to the beginning of the supply chain time line the longer the delivery lead-time, that is, the time that elapses from receiving a customer order until the customer receives the product.

Thus, locating the boundary at assembly time, as is done by Dell computers, is not appropriate for products with long assembly or transportation lead-times. In this case, it is more appropriate to move the boundary towards the end of the supply chain time line. Consider for instance the car industry, an industry known for its long transportation lead-times. Prior to its latest effort to change its business model to the make-to-order strategy, there were previous attempts in the car industry to implement a push–pull strategy. In 1998, M&M changed its distribution strategy. Dealers could order vehicle they did not have in their lot from the distribution centre and these would be delivered within 24 hours. Thus, M&M employed a push–pull strategy in which inventory at their regional distribution centre was managed based on long-term forecasts while delivery to dealers was based on realized demand.

The demand pull business should be developing business processes and infrastructure that allows it to produce products or services in direct response to customer demand in the shortest time. The demand pull business emphasizes technologies and processes and relationships that reduce time to respond to customer demand. The demand push business focuses on having tight business processes and a capability to respond to uncertain customer demand. They include demand planning and sales and operations planning (S&OP), planning technologies, excellent warehouse management systems and safety stock, and customer service policies. The demand pull strategy calls for reserve production

capacity and blending processes to manage uncertainty. Supply relationships tend to be most critical because often production is highly dependent on the quality and continuity of raw materials, whereas inventory buffers variability on the demand side.

The other element of integration is collaboration. It is a collaborative process of decision-making among interdependent parties in the supply chain. It calls for joint ownership of making decisions and collective responsibility for better results. The success of collaboration depends upon the ability and willingness of supply chain decision-makers to build meaningful relationships and create trust amongst the partners.

For collaboration at an operating level, the firms have to deviate from their traditional (secretive) thinking to more transparency in information flow. This collaboration requires significant change from standard business practices and the success depends on information exchange. This approach calls for sharing of data, operating plans, and even some financial information. This new approach is called as collaborative planning, forecasting, and replenishment (CPFR). In this, trading partners jointly develop long-term demand projections rather than relying upon separate, independently generated forecasts. The projections are constantly updated, based upon continuous flow of information from market. This helps in matching supply and demand with greater accuracy.

Cost-effectiveness plays a key role in gaining competitive advantage. The cost advantage is short-term as it can be copied by the competitors. Hence, the firm has to go for some differential strategies to outperform its competitors. Customized services with value addition would be the solution to gain competitiveness. This can be done through collaboration with service providers. The internal core competency coupled with the expertise of external service providers helps firms to tailor service offerings to customer preferences and requirements. However, to achieve this, the firm has to be more focused internally, so that it may better respond to customer expectations and accommodate customer needs. Here, the quality of relationships with collaborators or service providers is key to the success of the push-pull strategy. The pre-requisites of collaborations are process synergies, transaction transparency and information sharing. Collaborations are known as an opportunity for improving operational efficiency in supply chains. Through better coordinated business processes across supply chain and increased sharing of more timely and accurate information, intercompany collaboration can help integrate information and material flows and reduce wastes.

The integration can bring cost savings to the entire supply chain, but the savings come at the expense of the buyer's firm. While in some situations the savings can be significant, there are also situations where it is better for all the firms to develop their production schedules independently. It was found that savings are more significant when (1) inventory holding cost is low, (2) supplier lead times are short, (3) forecasts are accurate and (4) the suppliers' flexibility of accommodating schedule changes higher.

High service levels require building inventory that is in direct conflict with the objective of reducing system wide cost. By contrast, a focus on cost reduction in the supply chain forces the firm to cut inventory and thus, may lead to low service levels.

Push-pull strategy allows the supply chain partners to achieve both objectives. Indeed, the push part of the supply chain is applied to the portion of the supply chain where long-term forecasts have small uncertainty and variability. Thus, service level is not a big challenge and the focus is on cost reduction. On the other hand, the pull part is applied to the portion of the supply chain where uncertainty and variability are high and hence the focus is on matching the demand and supply. Consider, for instance, a supplier of fashion skiwear such as Sport Obermeyer. Every year the company introduces many new designs, or products, for which forecast demand is highly uncertain. One strategy used successfully by

Sport Obermeyer is to distinguish between high-risk and low-risk designs. Low risk products, that is, those for which uncertainty and price are low, are produced in advance using long-term forecasts and focusing on cost minimization, a typical push-based strategy. But, decisions on production quantities for high-risk products are delayed until there is a clear market signal on customer demand for each style, that is, a pull strategy. Since fabric lead-times are long, the manufacturer typically orders fabric for high-risk products well in advance of receiving information about market demand and based only on long-term forecasts. Therefore, in this case, the manufacturer takes advantage of aggregate forecasts which are more accurate. Since demand for fabrics is an aggregation of demand for all products that use that fabric, demand uncertainty is low and thus fabric inventory is managed based on push strategy. The analysis thus implies that Sport Obermeyer uses a push-pull strategy for all the high-risk products and a push strategy for all low-risk products.

INTERNET AND SC INTEGRATION

On the Internet, not only buying and selling but also servicing customers and collaborating with business partners is done with ease and speed. Internet technology has helped business firms to redefine their business models to improve the enterprise performance. The Internet has had a great impact on supply chain performance. It has helped companies to move away from the traditional push strategy employed by most supply chains.

The appropriate supply chain strategy depends on the industry, the company and individual products. If uncertainty in customer demand is higher, then its supply chain should be managed based on a pull strategy. Similarly, the higher the delivery cost, relative to the unit price, the more important it is that parts of the supply chain be managed based on a push strategy.

In the computer industry, there is high uncertainty and low delivery cost. In this industry, the larger part of the supply chain is managed on a pull strategy. However, for products with low demand, uncertainty and high delivery costs, for example, in grocery industry (soft drinks, staple food, etc.) a pull strategy is not appropriate. In this case, a push strategy works well. The reason is managing an inventory based on long-term forecast does not increase the inventory holding costs, while delivery costs are reduced due to economies of scale.

In some product categories, there is a mismatch between the demand uncertainty and delivery cost. In such cases, uncertainty suggests the supply chain to have 'push' structure while delivery cost suggests a 'pull' structure. Books and CDs fall in this category. There are products and industries for which uncertainty in demand and delivery costs are high.

To stay competitive, companies have gone for greater coordination and collaboration among supply chain partners in a bid to integrate the supply chains. The growth in Internet and information technology has played a key role in the supply chain integration process. The Internet has helped in redefining the way the processes such as product design and development, procurement, production, inventory, distribution, after-sales service support and marketing are conducted. It has also helped in redefining the roles and relationships between partners in supply networks.

IT had a significant impact on supply chain integration. New business models are continuously being developed based on Internet technology. The companies have resorted to e-business approaches for supply chain integration. They have realized that with e-business approach they would get efficiency improvements, better asset utilization, speed to market, reduction in total order fulfilment time, enhanced customer service and responsiveness, higher return on assets, and ultimately lower operational cost with improved customer service.

E-BUSINESS SC INTEGRATION

There are four key dimensions by which e-business has impacted supply chain integration. These are: information integration, planning synchronization, workflow coordination and new business models.

Information Integration

This means sharing of information among all partners of the supply chain. The information sharing has great impact on the performance of other members of the supply chain. For example, M&M allows its suppliers to have access to the production schedules available on M&M's website to plan their production (parts, components and sub-assemblies) on a real-time basis.

Synchronized Planning

It includes joint design and execution of plans for product introduction, forecasting and replenishment. Planning synchronization means with shared information the supply chain partners initiate specific actions simultaneously. This helps SC partners to have their procurement, manufacturing, replenishment and distribution plans coordinated to meet the common objective of cost reduction and customer service.

Workflow Coordination

This means all activities in supply chain should have coordination amongst them like an Olympic relay race. The supply chain should have streamlined and automated workflow activities between supply chain partners. The coordination between activities can be organized in a better way using the technology platform. In fact, many internal work flow steps can be seamlessly coordinated with automation so as to reduce the cycle time.

New Business Models

Many companies are designing new business models based on Internet technology. With the new models they are creating new business opportunities which were not previously possible. With joint efforts of partners and using technology platform, companies are redefining logistics flows so that the roles and responsibilities of members may change to improve overall supply chain efficiency. With the supply chain network companies may jointly create new products, pursue mass customization, and penetrate new markets and customer segments. Here to achieve success, channels of communication must be well defined and maintained, with roles and responsibilities clearly articulated. In addition, performance measures for partners in the supply chain also need to be clearly defined in quantitative and qualitative terms and should be monitored on a continuous basis.

COLLABORATIVE PLANNING, FORECASTING AND REPLENISHMENT

The positive value of information sharing in supply chain management is well appreciated by leading firms and this has led to various industry initiatives to encourage firms to develop a framework for collaboration for mutual advantage. One such industry initiative is known as CPFR (collaborative

planning forecasting and replenishment) systems. This is a software system which allows retailers and manufacturers to share information on past sales data and future price, and promotion measures. This is a relatively new development but is gaining popularity across the industry.

One of the main issues inhibiting increased supply chain performance today is the lack of visibility into down-stream demand. This lack of demand and inventory visibility leads to lost sales and high inventory levels for both retailers and manufacturers.

Most companies forecast future demand based on historical customer orders or shipment levels and patterns. However, actual consumer demand may be very different from the order stream. Each member of the supply chain observes the demand patterns of its customers and in turn produces a set of demands on its suppliers. But the decisions made in forecasting, setting inventory targets, lot sizing and purchasing distort the demand picture. Thus, demand supply mismatch creates a bullwhip effect. The distortion of the demand picture imposes high supply chain costs in the form of sub-optimal customer service levels, high inventories and low returns on asset. By providing business partners the visibility into inventory on shared forecast of customer demand, supply chain partners can positively create value across supply chain processes.

CPFR represents an innovative business model that extends vendor-managed inventory principles by taking a holistic approach to supply chain management among a network of trading partners. CPFR has the potential to deliver improvement in sales, system efficiency, cash flow and return-on-assets (ROA) performance.

The CPFR process model represents voluntary guidelines aimed at structuring and guiding supply chain partners in setting up their relationship and processes. This business scenario shows how enterprises can carry out collaborative supply chain planning activities with their business partners over the Internet. By sharing information over the Internet, the buyer and seller develop a single dynamic forecast. The result is more accurate forecasting with lower inventory levels. These benefits save time and money.

SUMMARY

Information is a vehicle for supply chain integration. Information has been identified as a key enabler of successful implementation of supply chain strategies. The gain from information integration is seen to be positive for all aspects of supply chain management and for all firms, provided a mechanism for sharing the benefit can be established. The value of information is generally well perceived and sharing of information has enabled firms to reduce the level of inventory and to reduce lead time.

Many examples can be quoted to show success of creating customer demand through pull supply chains. There are certain products the pull is quite appropriate. There are many advantages the firm can have switching over to a pull supply chain. However, all supply chains cannot exclusively pull or push. Many times it is a combination of both. The demand pull supply chains reduce waste and inventory and are intrinsically more attuned to meeting end customer needs. However, some supply chain participants will shoulder a burden of inventory, particularly if remotely located from end customers. Therefore, these businesses adopt supply chain strategies appropriate to the circumstances, including if possible one with right elements of push and pull.

The Internet is an enabler to supply chain integration. Data computing and information communications can be done with speed and accuracy on the Internet. The execution of both front end and back end business processes can be done properly to drive supply chain integration. The Internet gives global visibility to the business across its extended network to respond quickly to market needs. In business on digital technology platform, the supply chain integration is based on four critical dimensions: information integration, synchronized planning, coordinated workflow and business models. With the IT platform, businesses can be seamlessly integrated in its supply chain processes to get benefits of reduced costs, increased flexibility and speedier response times.

REVIEW QUESTIONS

1. What is supply chain integration and how is it achieved?
2. What are the different barriers to the supply chain integration and how are these overcome?
3. Explain why coordination and cooperation are essential in supply chain management and how information helps achieve them.

INTERNET EXERCISES

1. The supply chain management information portal provides access to focused, concise and unbiased information on various SCM topics. This portal is sponsored by the Supply Chain Resource Cooperative (SCRC) at North Carolina State University as a service to the supply chain field. Visit <http://www.scm.ncsu.edu/public/about/>, for tutorial on SC topics such as CPFR, inventory management, SC integration, procurement, etc.
2. Identify the role of information in CPFR, visit <http://www.cpfr.org>

VIDEO LINKS

1. Supply chain integration http://www.youtube.com/watch?v=S_yMW2b0kNk&feature=related.
2. Integration through technology in Ford Manufacturing Supply Chain <http://www.youtube.com/watch?v=qyO9QSo0FjU>

PROJECT ASSIGNMENT

1. Supply chains of companies in FMCG and retail sectors are quite complex. Select a firm in one of the above industry sectors and study the supply chain integration strategies implemented by the company. Find out the benefits the company is enjoying after implementing SC integrative strategies.

NOTE

1. Mayuri Gupta, 'First-in-and-First-out' (FIFO) Solution to a Paint Unit, *Supply Chain Pulse*, Vol. 1, Issue 2, Jan-Mar 2010.

COLD CHAIN NETWORKING

22

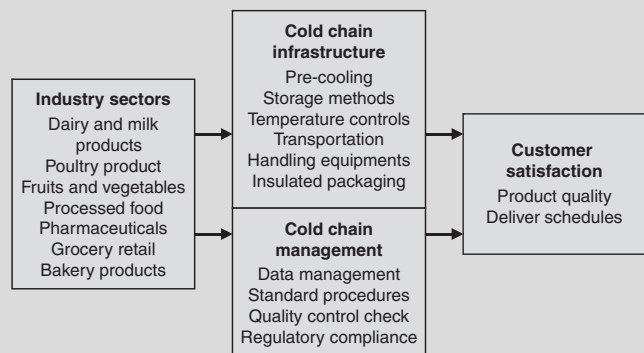
“Unlike hard goods, the food industry must work within seasonal constraints. For perishable foods, there are supply adjustments necessary to get products closer to consumption. Perishability creates its own demand cycles, on top of the ongoing demands that people will eat. Food is going to move one way or another

—Petersen”

Chapter Objectives

To understand:

- What is cold chain?
- Drivers to cold chain infrastructure
- Cold supply chain models
- Cold supply chain problems and issues



Cold Chain Networking

Cold chain is a logistic system for movement of perishable products (vegetables, fruits and other food items) under controlled temperature from the point of origin to the point of consumption. The cold chain helps to reduce product spoilage and retains its quality. The cold supply chain is designed and built in an integrated manner. It used for storage of primary produce in agriculture, horticulture, fisheries and aquaculture sectors. The intermediaries like the manufacturers who process the food for ready-to-eat or cook format together with the packaging companies make use of cold chain. The retailers, wholesalers and caterers are in the last stage of the cold supply chain. The other industries that need a well synchronized cold supply chain are pharmaceuticals, biotechnology, health care, blood transfusion, seafood, frozen foods, meats, dairy and hospitals

Snowman Frozen Foods¹: Providing Cold Chain Solution

Snowman Frozen Foods Ltd. is a Bangalore (India) based company founded in 1997 having a joint venture between Gateway Distriparks Ltd. of Singapore, Mitsubishi Corporation (global trading company) of Japan, and Nichirei Logistics Group Inc (temperature controlled logistics service provider) of Japan.

Snowman with its capabilities is a single window temperature controlled logistics solution provider offering the entire range of integrated logistics services. Snowman cold storage maintains the air temperature for storage at -20°C in order to retain the temperature of product stored at -18°C which is international standards for frozen food. The freezer storage has been designed for operation at -25°C with flexibility of incoming materials up to -15°C . The material storage arrangement is on standard two deep racking systems, which can store materials on Euro pallets of size $1,000 \times 1,200 \times 1,500$ mm with vertically high arrangements. The pallet movement is with the help of battery operate electric forklift reach truck suitable for lifting the material from fourth level and second depth of the rack. The suitable dock levellers are used to loading and unloading of the materials from or into the refrigerated truck.

Transportation through the Snowman reefer vehicle ensures temperature reliability and prompt delivery. The company also possesses a well connected communication network which provides immediate information to the client on the movement of goods. Snowman current fleet size is 100 in number ranging from 6 ft to 24 ft trucks equipped with thermo-packing and carrier refrigeration units. Frozen food transportation covers around 100 cities in India which offers fixed schedule operation.

The Snowman stores are built to meet international standards of temperature and hygiene for frozen foods, vegetarian and non-vegetarian products. The cold storage facilities provide for computer controlled inventory management system and have palletized storage system, racked storage system and concept of sharing the cold store. The company has a comprehensive MIS for monitoring payment collection from the market and sales tax administration. Snowman's distribution network covers more than 75 cities throughout the country to reach all major markets quickly.

Snowman has over five fruits and vegetable processing centres at major fruit farm produce areas to cover the maximum geography. Major activities involve pre-cooling, washing, sorting, grading, waxing, packing, weighing, ripening and quality testing. Snowman supports and maintains the cold chain from post harvest till it reaches to the customer. Snowman's IT-enabled information service has always resulted in innovative services and solutions in pre- and post-harvest fruits and vegetable transportation and processing.

INTRODUCTION

As per Meckanz reports, about 30 per cent of the fruits and vegetables grown in India get wasted annually due to non-availability of cold chain. There are no adequate infrastructure of cold storage facilities in close proximity to farms and refrigerated transportation facility. In addition to fruits and

vegetables, cold chain infrastructure can be deployed for storing chocolates, ice cream, dairy products, pharmaceuticals, meat, fish and poultry products. The cold chain unit is very essential and important as to reduce the post harvest losses of commodities during transit from field to market. The cold chain is defined as:

A cold chain is basically a logistics system, which helps in maintaining and providing a series of facilities for ensuring ideal storage conditions for the perishables from the point of origin to the point of sale. A well developed and efficiently organized cold chain reduces wastage, spoilage and helps keeps the perishables intact thereby helping to maintain the quality of the harvested food products ultimately making the whole system cost effective to the farmers and that which ensures top-notch quality to the end user.²

The cold chains play a significant role in retail food market and pharmaceutical markets as well. In recent years, there is a significant growth in fast foods, ready-to-eat and frozen food markets. In storage of these foods there are several food temperature levels required to suit different types of products. The success of implementing cold chain management involves continual monitoring of product temperature throughout distribution of the products. A well designed and maintained cold chain helps to:

- Reduce costs
- Maintain product quality and freshness
- Reduce wastages
- Reduce returns of expired stocks
- Increase customer satisfaction

The cold chain logistics infrastructure generally consists of pre-cooling facilities, cold storages, refrigerated transportation vans, thermal packaging, warehousing, and efficient information systems to trace and track the product.

The customers growing demand for affordable, high quality agricultural produce and perishable products throughout the year is putting tremendous pressure on the entire supply chain. As the perishable products like fruits, vegetables, meat, poultry, seafood and bakery products are highly susceptible to quality loss based on time, temperature and humidity, the infrastructure of cold chain is a must. There are many challenges in implementing cold supply chain infrastructure such as to source products from vendors, having it delivered to the shelves in optimum conditions, sell it at a price that is reasonable to the customer, all the while maintaining consistent product quality without loss of flavour and taste.

Cold chain management, in other words, is a specialized element of supply chain for perishable and temperature-sensitive products. In every stage of the supply chain there is loss of product quality. The great damage to perishable products results from a breakdown in the cold chain and/or poor handling. However, by monitoring and analysing the cold supply chain partners, the product loss can be minimized and the bottom line maximized.

With the well designed cold chain, the damage to the quality of the perishable products can be greatly reduced. The cold chain infrastructure in general covers the following:

- Pre-cooling facilities
- Cold warehouses
- Temperature controlled transport
- Insulated packaging
- Information management systems
- Traceability

There are lots of logistics challenges in food (cold) chains, which is a very cost-conscious industry. In addition, there are several governmental regulations forcing manufacturers and retailers to maintain hygiene and product quality.

DRIVERS TO COLD CHAIN

In recent years, due to growth in Indian economy and the consumer needs, preferences have changed drastically. The food habits and product preferences have forced industries to change the ways they are reaching out their products, which need controlled environment and have short life cycle with expiry dates. This changed market scenario is driving the cold chain industry to grow. The few other drivers adding to the industry growth are as follows.

Growth in Organized Retail

The retail chains markets lot of perishable products. These products need temperature controlled logistics networks and system. Due to multi-location procurement and distribution the retails sector needs nationwide cold chain to store the products during different stages in goods movement in supply chain. As the organized retailing is set for high growth in India, the existing cold storage capacity is grossly inadequate. The booming retail sector has set off growth in the cold chain segment. It is a highly specialized service and caters to time-sensitive and perishable items.

Shift Towards Horticultural Crops

India in the last decade has become the second largest producer of fruits and vegetables in the world, next only to China. India currently produces about 190 million tonnes of fruits and vegetables. Diversification and value addition are the key words in the Indian agriculture today. These changes along with the emerging food retail sector, especially with the opening up of foreign direct investment, have brought in a lot of opportunities in the domestic sector. Market liberalization and increasing shift in consumer demand have also offered attractive opportunities for agricultural exporters from developing countries. However, one of the biggest constraints witnessed in the horticulture sector has been the extent of losses, which is huge and the wastages occur due to poor cold chain infrastructures. The World Bank and other studies have also pointed to the development and strengthening of the integrated cold chain as the most essential intervention in infrastructure, if India has to reap the advantages gained through low cost production of perishable.

Growth in Processed Food Sector

India has tremendous potential to become the primary source of processed food products in the world. It has about 53 per cent of arable land (land available for cultivation of crops as compared to a measly average of about 11 per cent for the rest of the world). India is the second highest fruits and vegetables producer in the world, sixth largest producer of fish, but still doesn't make up to the top 10 food exporting countries. However, the vision 2015 document outlines the Indian government's goal of Rs 13,500,000 million size of the processed food industry to go by 2014–15. This translates into an overall growth target of 10 per cent p.a. in this sector. The government has announced lot of incentives in food producing industry sector. There are many private players who have jumped into the sector. The need for cold chains is triggered.

Demand from Pharmaceuticals Sector

Storage condition of pharmaceuticals goods is becoming a public issue. Majority of the medicines needs to be stored under cool conditions (below 35° C) and the day time temperatures in India cross above 40° C making the efficacy of the medicines questionable. Hence, chemists will have to have their outlets under temperature controlled to maintain the efficacy and shelf-life of the medicines. For stockists, the cold storage facilities are a must. As the pharmaceutical sector is growing, the requirement of cold chain is growing at a rapid pace. In addition, FDA (food and drug authority) is tightening the implementation of cold storages for pharmaceutical products distribution down chemists' level.

Government Initiatives

To reduce the losses of perishable products, the government of India (GOI) has given priority to the cold chains sector. To increase the cold chain infrastructure the government has lifted restrictions on imports of cold storage machinery and the FDI (foreign direct investment) is increased to 51 per cent. In addition, NHB (national horticulture board) is granting 25 per cent subsidy for cold storage projects. Under the tenth plan period, government of India has recommended to create additional cold chain facilities. The government has also encouraged public-private investment participation in cold chain creation. The government has taken lots of steps to support the farm sector by way of developing system covering drip irrigation, food warehousing, cold chain, etc. for better pre-harvest and post-harvest crop management.

Creation of Agri Export Zones (AEZs)

The government has planned and created special agri-export zones to increase international trade in agri-commodities. This is an approach to encourage trade in specific commodities located in contiguous areas (Tamil Nadu would focus on grapes, mangoes and chikkoo; Kerala on vegetables; Punjab and Haryana on kinos, wheat and rice; Karnataka on vegetables and flowers; Maharashtra on mangoes, grapes and flowers; Gujarat on bananas, mangoes, castor and garlic; and Uttaranchal on litchi and medicinal plants).

Private Sector Initiatives

The private sector has also taken lot of initiatives in the food processing and service sectors. In past few years, there is lot of initiatives and investments in cold chains by private sectors in India. To name a few, they are McDonalds-India, Amul-GSMDC, E-choupal by ITC, which are the success stories of procurement of produce from small farmers. Snowman has established a nationwide cold chain to serve the manufacturers of perishable goods and commodities.

COLD SUPPLY CHAIN

The cold chain process activities starts at the farm and ends with the consumer. The consumer could be ultimately served through home delivery or through a retailer or a neighbourhood pop and mom retail shop. An international consumer could be served through retail chains or food malls. The product is required for both local and export market as well. It is to be produced locally and delivered at the point of consumption which may also be other country. Hence, the cold supply chain

planning process needs to be in place. The demand to be forecasted for each of the channels to serve the customer with expected lead-time. The pricing and the quality of service are the two prime factors for repeat purchase.

Box 22.1

Cold Supply Chain for Ice Cream

Supply chain in the ice cream industry has its unique dynamics, owing to its distinct product nature. This calls for efficient cold-chain maintenance as well as sound inventory management. Right from the time ice cream comes out of the factory till it reaches consumer, the entire period requires delicate handling, with focus on temperature and quality. Unlike frozen meat, ice creams cannot sustain their temperature in case of failure of a refrigeration unit. Hence, the chances of wastage are also very high. Here the critical aspect is the time factor. The raw materials and finished products are perishable too and hence quick delivery is of utmost importance. To ensure this, many ice cream manufacturers purchase cream only from dairies lying within a vicinity of 40–50 km from its plant. Hence, ice cream SC involves a closely integrated network that connects procurement, production, transportation, distribution and retail.

Amul, the market leader (37 per cent market share) in Indian ice cream industry, has nine manufacturing plants all across the country. Milk is procured from village-level cooperative societies and stored in bulk chillers. From here, it is transported to the production units. Once the ice cream is ready, it is sent in refrigerated vehicles, with temperature constantly maintained at -20°C , to 40 depots (cold rooms) at different locations in India. Further, the ice cream is dispatched to Amul's 1,500 distributors across the country, who maintains cold storage facilities at their warehouses. Finally, the product is dispatched to around 70,000 retailers, including exclusive outlets and retailers.

For maintenance of desired temperature level at retail points, Amul has come up with an innovative idea of Hamara Apna freezer scheme, and persuaded retailers to buy the refrigeration units from selected vendors at a negotiated price. This has ensured that retailers selling Amul products are using standardized equipments for maintaining the desired temperature levels. Amul also goes for periodic inspection of cold chain equipments at distribution and franchisee points.

Amul uses the packaging material with properties that protect ice creams against moisture and air. For primary unit packs, where the material is in direct contact with the product, food grade materials such as BOPP (bi-axially oriented polypropylene) films, poly-laminated material, papers are used. At the plant, ice cream is dispatched from freezers into cup-filling machines, which fill the ice cream in half litre and one litre cups. These cups are semi-automatically put in 4-litre mini or inner cartons made of duplex board and stored in cold rooms. From here, 12 such mini cartons are set in a master or outer carton made of craft paper and these are then manually loaded in vehicles in air-locked conditions, with temperature at $1-4^{\circ}\text{C}$. Time is another aspect that governs most of the logistics operations in an ice cream cold chain and impacts the product quality. In general, at Amul factory they ensure that loading 1,500 bulk packages in a vehicle should not take more than 45 minutes.

Ice creams cannot be stocked for more than 6 months and hence proper inventory management is crucial. The same holds true for perishable raw materials. The average gap between milk procurement and putting it into ice cream production process should not be more than 6–8 hours. However, for

meeting sudden rise in demand, they store ingredients that have conserved properties, such as cream and milk powder. These can be used during peak demand. Amul keeps stocks of (2 months consumption) other ingredients, such as dry fruits, syrups, colours and chocolates. They do daily analysis of market demand to match production with demand to avoid stocks piling up. Amul depots can store from 10,000 to 1 lakh litres of ice cream. During peak seasons, it goes upto 50,000 litres.

The risk of temperature abuse calls for close monitoring at all points: transportation, distributors' and retailers' ends. Amul for recording temperatures during dispatch and receipt of ice creams at transfer points, use data loggers fitted in vehicles and depots and set to record temperature at regular intervals. Later, a graph is prepared to monitor temperature variations, if any, and corrective action is taken. Thus, staying cool in ice cream SC is a challenge. It calls for quick decisions, high flexibility and a focused approach throughout.

Source: Mohan Das, Staying Cool in Scorching Heat, Fastrack, Issue 17, Apr–Jun 2008

In formulating the policies of farming, production, processing, distribution and retailing and also in financing these activities the governments play leading role. At present this plays great importance in view of the globalization of the food industry.

There are several regulatory measures handled by many government departments. These need to be streamlined for the cold supply chain to be productive. The industries that need a well synchronized cold supply chain are pharmaceuticals, biotechnology, health care, blood transport, seafood, frozen foods, meats, dairy and hospitals. These cold chains require maintaining temperature controls in the most stringent terms due to the nature of the products (for human consumption) being handled.

The suppliers of food items face challenges in creating and preserving the unique characteristics of their products and conveying information about those characteristics to consumers. This calls for a well designed cold chain information system. The information system should supplement sharing and the coordination of activities, product monitoring and quality assurance, and efficient allocation of costs and returns.

In cold supply chain, product-temperature configuration forms the biggest challenge. Quality and safety of chilled foods during storage are largely determined by time-temperature tolerance. The storage life is directly dependent on storage temperature. The other threat is potential growth of microorganisms such as *Listeria*, *Yersenia* and *Aeromonas* at chill temperatures.

With the multiple trans-shipment and handling, the distribution process of perishables has become very complex. Fresh vegetables and fruits transportation from farm to retail stores requires temperature controlled vehicles. In this complex chain, the product is often mishandled. Regardless of what point a perishable product is in transit, it is always susceptible to temperature abuse. It may so happen that in the long journey the perishables remain too long in an open air at a distribution centre or above an acceptable control temperature setting at the wrong section of a multi-refrigerated trailer. To avoid this, the following measures suggested to control of product quality:

- Deploying temperature controlled vehicles
- Maintaining temperature during transport.
- Defined pick-up and delivery times.
- Better handling procedures.
- Mechanized equipments to reduce product damages in handling.

A process-oriented approach towards cold chain needs to develop an end-to-end cold chain visibility solution. This is based on IT-enabled network that can give managers a reliable and precise overview of what a perishable product has endured on its journey from the farm all the way to the shelf.

To ensure that Standard Operating Procedures are followed to take care of the health of the general public. Within the cold chain, it has been common practice to embed temperature sensing devices inside boxes, on pallets, and within trucks moving toward distributors, restaurants, grocers, and other recipients as well to make the decision of whether or not to accept or reject the shipment.

SUMMARY

Cold chain refers to the storage and movement of temperature-sensitive products from the point of origin to end customers in the best possible quality. Such products require temperature-control and can only accept a certain range of temperature fluctuations during transportation and storage. Maintaining the temperature within the desired range for every shipment is the challenge of cold chain logistics. Close monitoring at periodic stages is essential and such audits help to identify areas of weakness where logistics professionals can implement and devise the right tools to support and maintain the cold chain.

Humidity, temperature and environment monitoring in the present-day technological devices can drive full monitoring platforms, which include the management of requirements such as humidity, temperature and environment measurement from wall-to-wall and floor-to-ceiling. In addition, remote monitoring is possible via global positioning systems (GPS), along with online or Web-based systems for rapid alert functions. Such technologies, of course, do not come cheap.

Global regulatory requirements for the handling, storage and distribution of temperature-controlled goods and commodities have emphasized the importance of assuring product quality and integrity. These requirements are not compromised in the distribution channel. The guidelines on cold chain outline a comprehensive view of cold chain management across the supply chain, including manufacturers, warehouses, distributors, transporters and retailers. The cold chain is a highly capital-intensive industry. Even for the cold chain of large capacity a payback period is very long.

With the change in the consumption patterns of Indians, the demand for frozen food has gone up drastically. Even next door grocery stores are stocked with imported fruits. These factors are contributing to the huge demand for cold chain infrastructure in the country.

REVIEW QUESTIONS

1. What is cold chain and its scope in various sectors?
2. What are the drivers to cold chain infrastructure in India?
3. Discuss the various challenges in building cold chain infrastructure in India.

INTERNET EXERCISE

1. Sensitech offers complete cold chain solutions for monitoring temperature-sensitive products enabling customers to track and monitor assets from beginning to end of the cold chain. For studying white papers and articles on cold chain, visit <http://www.sensitech.com/about/index.html>

VIDEO LINKS

1. Transport of perishable goods <http://www.youtube.com/watch?v=2IAXwqAFiM4&feature=related>
2. One of South Africa's leading cold chain specialists, Dr Malcolm Dodd, of the fresh produce exporters forum, talks to Fruitnet TV about managing the costs of supplying fresh produce from around the world <http://www.youtube.com/watch?v=818CjQyuMOW>

PROJECT ASSIGNMENT

1. National Dairy (Safal) or Reliance Fresh are the leading retail chains for fruits and vegetables in the country. Study their cold chain infrastructure, standards and procedures they follow to deliver fresh farm products to the end customers at affordable price and simultaneously keeping the cost of delivery low.

NOTES

1. <http://www.snowman.in/services.html>
2. Rajiv Kumar, *Cold Chain and Frozen Food Distribution System: A Brief*, http://www.streetdirectory.com/food-editorials/meals/food/cold_chain_and_frozen_food_distribution_system_a_brief.html.

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PART-V

MEASUREMENTS AND CONTROLS

23 Quantitative Techniques in Supply Chain

This chapter covers the scope and role of quantitative techniques in managing supply chain. The different quantitative techniques used in supply chain decision making are also covered. The techniques discussed are: linear programming, modeling, simulation, queuing theory, inventory models, forecasting methods, etc.

24 Supply Chain Risk Management

This chapter starts with the discussion on supply chain vulnerability and types of supply chain risks in general encountered in managing the supply chain. The chapter further includes the risk mitigation strategies, approach to risk mitigation, success factors in risk management and risk management models.

25 Pricing, Costing and Financial Decisions

This chapter covers product pricing decisions, costing techniques and methods and financial metrics in supply chain. The SCM and SCOR model are discussed in the chapter. The effect of supply chain performance on EVA is also covered.

26 Performance Measurement and Controls

In this chapter, the concept of benchmarking and gap analysis are discussed. The performance measurement metrics along with concept of balance scorecard to effectively manage supply chain are also covered here.

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QUANTITATIVE TECHNIQUES IN SUPPLY CHAIN

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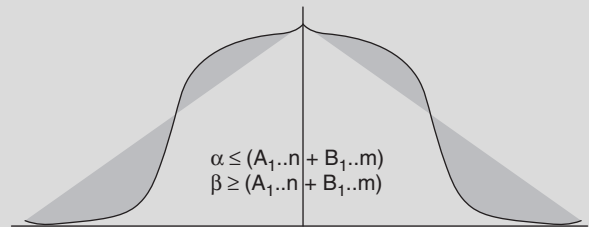
“You can use all the quantitative data you can get, but you still have to distrust it and use your own intelligence and judgment

—Alvin Toffler”

Chapter Objectives

To understand:

- Role of quantitative techniques in supply chain
- Different QTs used in decision-making in SC
- Linear programming (LP)
- Modelling and simulation
- Queuing theory
- Inventory models



QTs in Supply Chain

Quantitative techniques (QT) play an important role in managerial decision-making process. QTs are used in cost reduction, servicing customers and in business process optimization. It provides for analysis and enables proper deployment of resources at optimum level. As supply chains become increasingly complex due to growth of the business, the decision about resource allocations in variable situations often prove inadequate. This makes the use of formal mathematical tools necessary. Quantitative methods are now-a-days increasingly being used for solving complex problems in transportation, storage, inventory and marketing.

Linear Programming in Octagon Petroleum

Octagon Petroleum Ltd (OPL), to optimize their refinery operations, thought of using linear programming techniques to bring efficiency and effectiveness in their entire refinery supply chain operations.

The supply chain of OPL involves a wide range of activities, starting from crude purchase and crude transportation to refineries, refining (cracking) operations, product transportation and finally delivering the product to the end users. The nature of the value chain is such that all the operations and processes are interdependent and economies are interlinked.

The main operations of OPL are crude selection (indigenous and imported crudes which cost 90 per cent of input cost), crude transportation (transportation cost from the crude suppliers to refineries), crude processing (to get maximize yield), making available product mix as per market demand and physical distribution of the products to demand/sales points.

OPL used LP techniques in the areas like selection and evaluation of crude oils and raw materials, long-range and short-term operations planning, capital investments evaluation for process equipment, analysis of the profitability, evaluation of processing agreements and product exchange contracts, evaluation of new process technologies, control of the refinery performance, product blending control, down-time planning, inventory management and physical distribution of the products to end customers at demand points.

OPL looked into available LP modelling software to match to their refinery configuration. At OPL, the refinery model represented an LP matrix with 1,500 rows, 3,500 columns, 1,500 equations, 1,500 constraints and 5,000 variables. The model covered different time period variants to meet different planning objectives associated with annual planning (1×4 quarter), quarterly planning (1×3 months) and monthly planning (1×4 weeks). The key to the successful implementation of LP in OPL is an integrated approach that addresses the following issues:

- *Updates and maintenance of LP model on a continuous basis.*
- *Integrating different software solutions for desired results.*
- *Well set documented procedure for planning exercise.*

Petroleum refining in the new millennium will continue to be an extremely competitive business. The application of LP methodology in OPL has shown huge margin benefits in terms of rupees per barrel crude refined. The linear programming technique has provided an efficient and effective method to quickly evaluate and quantify the impact of internal and external changes on overall refinery profitability.

INTRODUCTION

Quantitative techniques are increasingly being used to find solutions to complex problems. These are used in decision-making in marketing, logistics, supply chain, operations and even in finance, when the variables are dynamic in nature, manual methods to find the solutions becomes time-consuming

and error prone. For example, in the past the control of an urban expressway in Japan, has been dependent upon skilled operators' judgement and decision. However, in 1970, the Hanshin Expressway Public Corporation started operating an automated traffic control system to maximize the total traffic flowing into its expressway network. The system relies on two control methods. The first is to limit the cars coming onto the expressway at each entrance ramp to avoid congestion in any section. The system calculates the maximum allowable inflows by solving a linear programming problem once in every 5 minutes using data from detectors installed along the expressway and at all ramps. The second method is to give drivers the most recent and accurate traffic information about the expressway and its vicinity, including expected travel times and accidents, through various information channels so that they can decide what routes to take. This system is greatly appreciated by the corporation because it is extremely cost-effective and also benefited by drivers, who now consider it an indispensable service. The linear programming using fuzzy logic solved this complex traffic congestion problem for ever.

In the process industries, a given raw material can be made into a wide variety of products. For example, in the oil industry, crude oil is refined into gasoline, kerosene, home-heating oil, and various grades of engine oil. Given the present profit margin on each product, the problem is to determine the quantities of each product that should be produced. The decision is subject to numerous restrictions such as limits on the capacities of various refining operations, raw-material availability, demands for each product, and any government-imposed policies on the output of certain products. These problems can be solved using QTs.

The application of quantitative techniques is predominant in the area of distribution, for example, there are m factories that must ship goods to n warehouses. A given factory could make shipments to any number of warehouses. Given the cost to ship one unit of product from each factory to each warehouse, the problem is to determine the shipping pattern (number of units that each factory ships to each warehouse) that minimizes total costs. This decision is subject to the restrictions that demand at each factory cannot ship more products than it has the capacity to produce.

FORECASTING

Forecasting is about predicting the future as accurately as possible. It is usually based on information available including historical data and knowledge of future events, which will have an effect on forecast. Forecasting is a statistical exercise in business, where it helps to take decisions about scheduling of production, transportation and personnel, and provides a guide to long-term strategic planning.

Modern organizations require short, medium and long-term forecasts, depending on the specific application. Short-term forecasts are needed for scheduling of personnel, production and transportation. Medium-term forecasts are needed to determine future resource requirements in order to purchase raw materials, hire personnel, or buy machinery and equipment. Long-term forecasts are used in strategic planning. Typically, businesses use relatively simple forecasting methods that are often not based on statistical modelling.

There are two broad forecasting methods such as qualitative and quantitative. 'Quantitative' techniques use the past data to project the future. However, the 'qualitative' forecasting techniques are based on the judgement of experts to forecasts. The qualitative methods are used in situations, where historical data is not available. Three qualitative forecasting methods in use are: Delphi technique, scenario writing and the subject approach.

QUALITATIVE FORECASTING

- **Delphi technique:** In Delphi technique, consensus of the panel experts is used in developing forecast. The methodology here is that each expert separately is asked to respond to a questionnaire. The opinions of experts are reviewed again in light of opinions of others. This process ends after some degree of consensus among experts is reached.
- **Scenario writing:** In this method, based on different sets of assumptions likely scenario of the business outcome are drawn out. The decision-maker has to decide which scenario is most likely to happen.
- **Subjective approach:** This approach calls for brainstorming exercises for employees. It allows employees to participate in the forecasting exercise. The forecasts are based on their feelings, ideas and personal experiences. New ideas to solve complex problems are thus developed. In this subjective approach, many times a survey of the company's sales people is taken.

QUANTITATIVE FORECASTING METHODS

Quantitative forecasting methods are based on an analysis of historical data. The independent variable is time and the dependent variable may be, for example, sales or profits. In this, the past data is analysed to find trends and same trend is used to forecast the future for dependent variable. In this type of forecasting method, time series on past data of the variable is used. Hence, this is called time series methods.

In the other method of forecasting along with historical data the cause-and-effect relationships of the variables is used to work out the future values of dependent variables. These forecasting techniques are called as causal methods. These methods of forecasting are used in business planning to organize resources. A few mathematical methods used for forecasting are as follows.

Correlation analysis

Correction analysis is used to measure the strength of association between quantitative variables. For example, we could measure the degree of a relationship between sales (Y) and the corresponding and spent (X). The strength of a relationship between two sets of data (sample) is usually measured by the correlation coefficient, r , where

$$r = \frac{N \sum XY - (\sum X)(\sum Y)}{\sqrt{N \sum X^2 - (\sum X)^2} \sqrt{N \sum Y^2 - (\sum Y)^2}}$$

$$N = \text{sample size} \quad \text{and} \quad -1 \leq r \leq 1$$

A relation is said to be perfect positive correlation when $r = 1$ and perfect negative correlation, when $r = -1$. The correlation analysis is one way in measuring the variance of a simple linear regression model. There are three different types of association between variables: positive, negative or no correlation.

The following data (Exhibit 23.1) about promotional expenses and sales required to be checked for correlation.

The correlation coefficient with the above formula works out to be 0.96, which means two variables highly correlated.

Exhibit 23.1 *Promotional Expenses and Sales*

<i>Year</i>	<i>Promotional Expenses</i>	<i>Sales</i>	<i>Year</i>	<i>Promotional Expenses</i>	<i>Sales</i>
2001	19	70	2006	44	119
2002	22	76	2007	49	132
2003	25	80	2008	51	144
2004	29	89	2009	54	155
2005	40	102	2010	60	161

Time Series Forecasting Models/Trend Exploration

Under these methods the historical data is explored to find out its cyclic nature or trends, if any. Mathematical techniques are then used to extrapolate to find out the future value/trends. The most common mathematical models in use are:

- Last period model
- Moving average model
- Weighted moving average model
- Exponential smoothing model
- Simple linear regression model (least-squares method)

Time series in general have four components such as trend, cyclical, seasonal and irregular. If these four components are combined, it provides specific values for the time series. The periodicity varies with the measurement taken at the interval of every hour, day, week, month or year.

The most popular model is linear regression model, which is suitable for the time series data that contains a trend. Linear regression is a statistical technique that expresses the forecast variable as a linear function of some independent variable. In time series modelling, the independent variable is the time period. The model (least-squares regression equation) is

$$\hat{y} = \hat{a} + \hat{b}\bar{x}$$

where

$\hat{y}$ = forecast for dependent variable, y

$\bar{x}$ = independent variable, x , used to forecast y

$\hat{a}$ = estimated intercept term for the straight line

$\hat{b}$ = estimated slope coefficient for the straight line

$$\hat{a} = \bar{y} - \hat{b}\bar{x}$$

(x_i, y_i) = observed values for time period i

$\bar{y}$ = average y value

$\bar{x}$ = average x value

n = number of observations

Once the equation of the straight line is obtained, the forecast value y can be calculated by putting in values of x .

Multiple regression is another common mathematical technique used for analysis of time series data. This model looks at the relationship between one independent variable and two or more dependent

variables. Multiple regression helps to understand how a group of variables (working in unison) affect another variable.

Cross-impact matrix method In this method, the assumption is that the occurrence of an event can, in turn, effect the likelihoods of other events. For analysis, probabilities are assigned to reflect the likelihood of an event in the presence and absence of other events. This method helps decision-makers to look at the relationships between system components, rather than viewing any variable as working independently of the others.

Scenario The scenario is an expected forecast that is likely to happen. It describes the interrelationships of all system components. The events considered under scenario are new technology, population shifts, and changing consumer preferences. The scenario analysis enables the decision-makers to organize resources and decide course of action in advance.

Decision trees Decision trees method is graphical representation of the structural relationships between alternative choices. The alternatives are represented as a series of yes/no choices. The decision tree structure with feedback loops is used as foundation to computer flow charts. In this, choices are assigned with probabilities to the likelihood of any particular path. The expected value of the variable/event is useful for decision-makers as it is the most likely value based on the probabilities of the distribution function.

LINEAR PROGRAMMING

A linear programming problem may be defined as the problem of maximizing or minimizing a linear function subject to linear constraints. The constraints may be equalities or inequalities. Here is a simple example. Find numbers a_1 and a_2 that maximize the sum $a_1 + a_2$ subject to the constraints $a_1 \geq 0$, $a_2 \geq 0$, and

$$a_1 + 2a_2 \leq 4$$

$$4a_1 + 2a_2 \leq 12$$

$$-a_1 + a_2 \leq 1$$

In this problem, there are two unknowns, and five constraints. All the constraints are inequalities and they are all linear in the sense that each involves an inequality in some linear function of the variables. The first two constraints, $a_1 \geq 0$ and $a_2 \geq 0$, are special. These are called non-negativity constraints and are often found in linear programming problems. The other constraints are then called the main constraints. The function to be maximized (or minimized) is called the objective function. Here, the objective function is $a_1 + a_2$. Since there are only two variables, we can solve this problem by graphing the set of points in the plane that satisfies all the constraints (called the constraint set) and then finding which point of this set maximizes the value of the objective function. Each inequality constraint is satisfied by a half-plane of points, and the constraint set is the intersection of all the half-planes.

Linear programming models are flexible. They adequately describe many realistic problems arising in modern industrial settings. These models take advantage of computational linear algebra that has been developed during the last 50 years. Linear programming models are largely used in logistics, transportation, finance, and warehousing, etc.

Many practical problems in operations research can be expressed as linear programming problems. The examples are network flow problem, multi-commodity flow problems which are considered

important enough to have generated much research on specialized algorithm for their solutions. LP is profusely used in microeconomics and business management, either to maximize income or minimize costs of production scheme. Some other problems are food blending, inventory management, portfolio management, resource allocations for human and machine resources, planning, advertisement campaigns. LP will have the following structure:

- Defined objective with a set of decision variables.
- Defined constraints (e.g. a limited supply of resources).
- Infinite number of solutions).
- Objective and constraints are in the form of linear equations.

In linear programming, there are maximizing problems. The problem is stated with defined constraints. The problem is usually expressed in matrix form then becomes maximized. Other LP forms are minimization problems, problems with constraints on alternative forms, as well as problem involving variables can always be written into an equivalent problem in standard formats and solved through LP. LP model can be used for solving product mix problem of a firm, investment problem, scheduling problem, transportation problem and assignment problem.

Assignment Problems

Assignments problems are the ones that involve the allocation of some resources to different tasks. The objective of assignment problem is to determine the optimum assignment of given tasks to a given number of workers so that the task can be completed in minimum time. The algorithm here seeks to assign the tasks to workers in such a manner that each task will be assigned to one worker and all tasks are done in minimum time. These problems can be solved by:

- Completely enumerating all possibilities and choosing the best one.
- Drafting and solving the problem as linear (integer) programming problems.
- Solving transportation problems.
- Hungarian assignment method.

The problem is put in a matrix form. A classic example is that of airline crew members who have to be assigned to various flight schedules taking into consideration their duty hours, minimum cost of overtime and availability of flights. This problem can be solved by assignment method. These can be solved using the Hungarian method.

Another example is that of allocation of jobs to various workers for completion in optimum time.

Transportation Model

Transportation problem is a special type of linear programming problem and typically involves situation where goods are required to be transferred from some sources or manufacturing plants to some distribution centres, markets or warehouses at minimum cost. Typically in such problem a matrix is given where sources are given row wise, destinations are given column wise. The unit cost of transportation from each source to each destination is provided. The purpose is to work out dispatch schedules to reduce the shipping cost within the limitations of supply and demand quantities. The transportation

can be used in other areas such as inventory control, employment scheduling and personal assignment. The transportation model can be represented as follows:

x_i —quantity at supply point

y_j —quantity at demand point

a_{ij} —transportation cost per unit from source to destination

b_{ij} —quantity shipped from source to destination

There are m sources and n destinations as shown in Figure 23.1. The general form of transportation problem in linear programming is as follows:

$$\text{TC min} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ij}$$

$$\sum_{i=1}^m a_{ij} = y_j \quad \text{for } j = 1, 2, 3, \dots, n \text{ (Demand constraints)}$$

$$\sum_{j=1}^n a_{ij} = x_i \quad \text{for } i = 1, 2, 3, \dots, m \text{ (Supply constraints)}$$

$$a_{ij} \geq 0 \quad \text{for } i = 1, 2, 3, \dots, m \text{ (Non-negative constraints)}$$

$$j = 1, 2, 3, \dots, n$$

The other methods for solving transportation problems are

- Northwest corner method
- Least cost method
- Modified distribution method (stepping stone method)

Northwest corner rule method According to this rule, you start allocating quantities in matrix to cells from top left-hand corner cell. Allocate the maximum possible quantity in this cell to make allocation for a row/column complete. From this cell, move to the next row/column. Keep on allocating maximum possible quantities till the allocation is complete. Total number of ‘occupied cells’ must be $m + n - 1$, where m is the number of supply centres and n , the number of demand centres.

If the number of occupied cells is less than $m + n - 1$ the solution is said to be ‘degenerate’. In such case, assume ‘o’ (zero) allocation to a suitable cell to make occupied cells equal to $m + n - 1$.

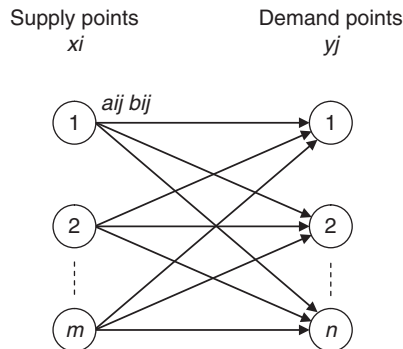


Figure 23.1 Transportation Model

Exhibit 23.2 Demand-Supply Allocations

Problem					Supplies	Solution				
Factories Workers	W/H 1	W/H 2	W/H 3	W/H 4		Factories Workers	W/H 1	W/H 2	W/H 3	W/H 4
F1	48	60	56	58	140	F1	140			
F2	45	55	53	60	260	F2	60	200		
F3	50	65	60	62	360	F3		120	240	
F4	52	64	55	61	260	F4			10	210
Demand ▶	200	320	250	210						

In the above example (Exhibit 23.2), the demand of four warehouses is fulfilled from four factories. The cost of transportation is indicated in the matrix. Logistics manager has to find out a solution for sourcing the right quantity from different factories for fulfilling warehousing demands at the optimum transportation cost. The solution is also indicated above.

Row minima method Make maximum possible allocation to the minimum cost cell in a row.

Column minima method Make maximum possible allocation column by column to the minimum cost cell in each column.

Matrix minima method Make maximum possible allocation to the minimum cost cell and proceed in the same manner for remaining allocations.

Vogel's approximation method (VAM) For every row and every column, find the difference in cost between the least-cost cell and next best least-cost cell. This difference is the penalty for failing to make an allocation to the least-cost cell. Make maximum possible allocation to the row or column, where penalty is maximum. Cancel row or column for which allocation is complete and proceed in the same manner till all allocations are complete. Vogel's approximation method gives a near-optimal initial feasible solution.

Distribution method In this method, an arbitrary initial allocation is made, and it is improved upon step-by-step till the optimum schedule is reached.

Modified distribution method (modi method) In 'modi' method, the test of optimality is simplified. In this method, a set of dummy row numbers and a set of dummy column numbers are decided, and any one number is arbitrarily chosen as zero. This cell value, as mentioned earlier, is the increase in cost per unit of material transported, by making an allocation to the cell. Once the cell values are known, one can make an allocation to the cell giving maximum saving and adjusting the other allocations accordingly. The process for finding dummy numbers is a part of the test for optimality, and it is repeated for every new allocation. The negative cell value indicates incremental cost per unit, if an allocation is made to that cell. This helps in analysing the costs, if for some reason one is constrained to make an allocation to a particular cell. If the costs of making supply from a supply centre to a demand centre changes, the cell value will change accordingly. Thus, one can compute how much increase in cost per unit is allowed without making a cell-value negative. Similarly, the effect of an increase or decrease in capacity of a supply centre and effect of increase or decrease in requirement of a demand centre can be worked out by analysing changes in allocations and cell values.

ROUTE PLANNING

In a supply chain, maintaining a low-cost route structure that meets both business constraints and customer service requirements is critical to the success. In inventory movement, route planning needs to create the most efficient logistical routing plans to obtain both low cost and higher customer service. Route planning helps companies to:

- Reduce transportation costs
- Improve customer service
- Reduce cross-over miles

Route planning offers sophisticated optimization, analysis and scheduling tools for choosing among available options. It allows you to determine the optimal route mix through route schedule construction, route schedule enhancement, asset minimization, zone design, vehicle events, service technicians and dynamic sourcing.

The routing problem refers to the problem of selecting a sequence of links on a network in a particular order. In route planning, the constraints (capacity of vehicles) can be expressed as linear conditions on some appropriate variables, but can equally well be imposed on proposed solutions during a search procedure.

The route problem in logistics is commonly known as vehicle routing problem (VRP). VRP seeks to allocate some vehicles, starting at a manufacturing plant/warehouse/depot to a set of demand locations each, and minimize costs, while satisfying other constraints (typically total length of each tour or capacity constraints on vehicles). For these problems, one basic idea is to construct reasonable tours and then modify them, based on some savings by interchanging locations on the tour. This works well for many practical problems in logistics applications.

The VRP, with capacity constraints, is quite frequently encountered in practice, for example, in the weekly despatching problem in multi-product, multi-location environments, where a good service frequency is desired. It is quite often used in local or secondary movement of goods, where frequent despatches, which combine several locations, is a cost-effective option. This is very common in delivery of perishables such as milk, ice-cream, vegetables, courier operations, etc.

The other common constraint in VRPs is the time window constraint, which gives a time interval during which a certain node must be visited. This makes optimal routing more difficult to obtain.

A routing problem for which an exact solution is easily available is the so-called shortest path problem. This is to find a sequence of nodes (from a given origin to a given destination) on a network for which the total cost is minimum. This can be done by a constructive procedure such as the well known Dijkstra's algorithm. Finding the shortest path for all origin-destination pairs turns out also to admit an efficient procedure (e.g., Floyd's algorithm).

A problem which is perhaps peculiar to India is the Indian truck routing. The simplest version of this is to find a set of paths so that each path begins at a given root node, all the nodes in the network are covered by the paths and the sum of all the path lengths is minimized. The root node refers to the factory or the dispatching points and the other nodes are the demand locations.

QUEUEING THEORY

The queueing theory is an important tool used to find solutions to supply chain problems. It is used to study situations in which customers (or orders placed by customers) form a line and wait to be served by a service or manufacturing facility. Clearly, long lines result in high response times and dissatisfied

customers. The queuing theory may be used to determine the appropriate level of capacity required at manufacturing facilities and the staffing levels required at service facilities, over the nominal average capacity required to service expected demand without these surges. Any business process where lines are a matter of fact, queuing theory is used. Examples are as follows:

- Doctors' clinics
- Restaurants
- Server load in a network environment
- Fulfilment/distribution centre or warehousing
- Project management
- Call centres

Queuing solution is suitable for solving a service-oriented problem where customer arrives randomly to avail of the service. The queuing model is relevant in service-oriented industries such as logistics, transportation, shipping, hospitality, banking, etc. The key elements of the process are given here.

Source Customers

Source involves a finite or limited number of customers. In a telephone system there is a large source population. When there are more than 100, the source can be treated as if it is infinite in size and the number of potential customers will influence the arrival behaviour of customers for the service system.

Arrival Process

The customer will reach the service system as per their needs. Many customers may arrive in batches (e.g. family) or individually. Customers may also arrive on a scheduled basis (e.g. appointment with manager). The customer arrival process describes the behaviour in which customers reach the service system. Arrival process is measured either by arrival rate numbers per hour or inter arrival, for example, each visiting every 15 minutes. When service is provided on a scheduled basis, the arrival rate or inter-arrival time is fixed. In unscheduled situations, however, customers arrive in a random pattern. The random pattern in most queuing situations follows the Poisson distribution.

Waiting Line

Customers do not physically form a queue but the queue is formed in the booking at their arrival in the system. The important factors to consider in a queuing system are size or capacity of the waiting area (customers may turn away if it is full), queue length (customers may refuse to join if it is too long) and queue organization (customers may refuse to join the queue if the queue is not organized).

Queue Discipline

The method in which customers from the waiting area are selected for service is referred as the queue discipline. The following are queuing disciplines which can be followed:

- First-in-first-out (FIFO)
- Last-in-first-out (LIFO)
- Priority scheme

Service Process

The simplest case is a single service facility while others may consist of multiple servers in one stage or multiple stage servers. Following are some examples:

- Single server, single stage
- Multiple servers, single stage
- Multiple parallel, non-identical servers, single stage
- Single server, multiple stage
- Multiple server, multiple stages

Regardless of the design configuration, it will take time to perform the service at each server. There are two ways to describe this service process, as follows:

- Service rate—The number of customers served per unit of time, for example, 30 per hour.
- Service time—The time taken to serve a customer, for example, 2 minutes per service.

Service time may be constant (e.g. machine processing) or may fluctuate (e.g. check out point in a supermarket) within some range of value. We can use probability distribution (e.g. negative exponential distribution) to describe the service process if it is fluctuating.

Departure

Most customers may return to the queuing system after servicing while others may never return again.

There are several measurements that should be considered when measuring the performance of a queuing system. However, the average value of the following listed measurements for a system is in a steady state which needs to be calculated.

- λ —Average arrival rate
- μ —Average service rate
- ρ —System utilization
- LS—Average number of customers in the queuing system
- L_q —Average number of customers in the waiting line
- WS—Average time a customer spends in the system
- W_q —Average time a customer spends in the waiting line
- P_n —Probability of there being n customers in the queuing system

There are two types of queuing systems: transient state (where probability of the number of customers in the system depends upon time) and steady state (where probability of the number of visiting customers in the system is independent of time).

A single queue single server model is represented ---- ($M / M / 1 / \infty / \infty$)

Wherein:

- Poisson arrival rate— M
- Exponential service time— M
- 1 server—1
- Infinite waiting capacity— ∞
- Infinite source— ∞

The assumptions here are: unlimited customers, unlimited waiting area. First-come-first-served, single server, arrival rate follows a Poisson distribution, and service time follows a negative exponential distribution. The formula calculations are:

$$\text{System utilization } \rho = \frac{\lambda}{\mu}$$

$$\text{Probability of no customer in the system } P_0 = 1 - \rho$$

$$\text{Probability of } n \text{ customers in the system } P_n = \rho^n P_0$$

$$\text{Average number of customers in the system } \mu \text{ LS} = \frac{\lambda}{\mu - \lambda}$$

$$\text{Average number of customers in the queue } L_q = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

$$\text{Average time a customer spends in the system } WS = \frac{1}{\mu - \lambda}$$

$$\text{Average time a customer spends in the queue } W_q = \frac{\lambda}{\mu(\mu - \lambda)}$$

For example, a petroleum company distributes its products by tankers loaded at its only loading station. Both company's and contractor's tankers are used for this purpose. It was found that on an average every 7 minutes one truck arrived and the average loading time is 5 minutes; 40 per cent trucks belong to contractors. Using queuing model and by making certain assumptions, one can find out the probability that a truck has to wait before loading, etc.

A queuing system is a model with the structure such as customers arrive and join a queue to wait for service given by n servers, after receiving service, the customer exits the system. A fundamental result of queuing theory is 'Little's law'. For a queuing system in steady state, the average length of the queue is equivalent to the average arrival rate multiplied by the average waiting time.

For example, consider a warehouse with 5,000 pallets of product that turns @4 times per year. Does it have enough labour to support these transactions? We get

$$5,000 = \rho (0.25 \text{ year})$$

$$\rho = \text{System utilization} = 20,000 \text{ pallets/year}$$

Assuming a 10-hour shift per day of about 250 working days per year, there is roughly 2,000 working hours.

$$\text{then, } \rho = 10 \text{ pallets/hour}$$

The above analysis helps to estimate the labour force required to move pallets, receive product, move product, and get work done.

In addition to the above, in general there are many more applications of queuing theory which show the interplay between the arrival rate and the service rate, which both reveal the characteristics of the queue and, ultimately the customer experience.

SIMULATION

Simulation is used to study situations characterized by uncertainty. Simulation involves the creation of a model of a system based on specific assumptions about system behaviour and information about probability distributions associated with various variables. By running simulations, it may be possible to determine the best values of various system parameters, in context with underlying assumptions. The main advantage of simulation is that it can be used to study extremely complex systems that cannot be easily modelled by using other mathematical tools. According to T. H. Taylor:

Simulation is a numerical technique for conducting experiments on digital computers, which involves certain types of mathematical and logical relationships necessary to describe the behaviour and structure of a complex real world system over extended period of time.

Simulation is a procedure whereby one can draw conclusions about the behaviour of a given system by examining the behaviour of a corresponding model whose cause and effect relationships are similar to those in the actual system. Simulation provides trial and error method towards optimal solution. Simulation is a quantitative method, wherein a mathematical model would represent the interrelationships between the variables involved in the actual situation in which a decision is to be taken. Then, a number of experiments are conducted with the model to determine the results that can be expected when the variables assume various values.

Simulation can serve as a 'pre-test' to try out new decision rules for a system. Simulation can anticipate problem and bottlenecks that may rise while operating a system. There are a few basic concepts which must be understood before applying the simulating technique.

- **System:** It is that segment to be studied or understood to draw conclusions. In the product-market system, the market for the products together with the firm's production process constitutes the relevant system. The variable can be identified only after defining the system. These are the variables, which interact with one another in the system and establish their relationships mathematically.
- **Decision variables:** Decision variables are those variables whose value is to be determined through the process of simulation.
- **Environmental variables:** These are the variables which describe the environment. In marketing, competitors' average price, consumer preferences and demand, etc. are the environmental variables.
- **Endogenous variables:** These variables are generated within the system itself. In marketing, quantity sold, sales revenue, total cost and profit are endogenous variables.
- **Criterion function:** Any variable can be used as the criterion function for evaluating the performance of the system. In marketing, profit is used as the criterion function.

Mathematical modelling requires the setting up of mathematical relationships which would represent the system. While some relationships can be expressed as equations, other relationships or constraints on the criterion function may be expressed only as inequalities (as we have seen in linear programming). If the mathematical model set up could always be optimized by the analytical approach, then, there would be no need for simulation. Only when interrelationships are too complex or there is uncertainty regarding the values that could be assumed by the variables or both, we would have to resort to simulation. In the complex interrelationship situation, 'Monte Carlo' method is a technique that involves using random numbers and probability to solve problems. The term 'Monte Carlo' method was coined

by S. Ulam and Nicholas Metropolis in reference to games of chance, a popular attraction in Monte Carlo, Monaco.

Computer simulation has to do with using computer models to imitate real life or make predictions. When you create a model with a spreadsheet like Excel, you have a certain number of input parameters and a few equations that use those inputs to give you a set of outputs (or response variables). This type of model is usually deterministic, meaning that you get the same results no matter how many times you re-calculate.

Monte Carlo simulation is a method for alliteratively evaluating a deterministic model using sets of random numbers as inputs. This method is often used when the model is complex, non-linear, or involves more than just a couple uncertain parameters. A simulation can typically involve over 10,000 evaluations of the model, a task which in the past was only practical using super computers.

By using random inputs, you are essentially turning the deterministic model into a stochastic model. Monte Carlo simulation is categorized as a sampling method because the inputs are randomly generated from probability distributions to simulate the process of sampling from an actual population. So, we try to choose a distribution for the inputs that most closely matches data we already have, or best represents our current state of knowledge. The data generated from the simulation can be represented as probability distributions (or histograms) or converted to error bars, reliability predictions, tolerance zones and confidence intervals.

GAME THEORY

Game theory (GT) is a powerful tool for analysing situations in which the decisions of multiple agents affect each agent's payoff. As such, GT deals with interactive optimization problems. GT may be used to model negotiation processes in supply chains and to develop insights into the balance of power in supply chains. Enterprises can be seen as players in a game defined by a common goal, but has separate constraints and conflicting objectives. In order to obtain acceptable trade conditions, a certain form of negotiation turns out to be necessary. GT provides a mathematical background for modelling the system and generating solutions in competitive or conflicting situations. The basic rationality principle of game theory states that each player acts to optimally accomplish his/her individual goal, taking into account that the others play in the same manner. However, if the individual goal of each player is uniquely to maximize his gain or to minimize his loss, the agreements obtained by negotiation may be fragile and will not generally guarantee enterprise optimality for the whole supply chain, particularly when external demand is stochastic. For these reasons, much effort has been recently devoted to conceiving contracts strengthening the commitments of partners through risk, profit or cost sharing, and/or moving the equilibrium state of the game towards a better global performance. Examples of contract parameters that can be used to achieve coordination, are quantity discounts, returns (buy backs), quantity flexibility, and the use of subsidies/penalties.

Cooperative GT can be of great help to design a supply chain or a virtual enterprise by selecting an optimal coalition of partners. But a non-cooperative (also called strategic) approach is certainly more appropriate to determine the set of equilibrium points that can be reached in trade conditions. A case of particular interest is when there exist decisional states from which neither player has interest to depart. Such cases, called Nash equilibria, have received a considerable attention both in theory and applications of GT. A particular property of a Nash equilibrium point is that, if the internal models of the players are known it is immediately reached at the first iteration of the game. Therefore, existence of Nash equilibrium points reduces the negotiation process to a one-shot exchange of information. In many real

situations, the equilibrium is not unique, and the first player imposes the outcome of the game. The particular equilibrium reached in such an asymmetric game is called 'Stackelberg equilibrium'. Outside the full information context, the outcome of a game generally depends on who plays first and how the players negotiate.

In enterprise networks, decentralized decisions are generally less efficient than a centralized mechanism maximizing a global utility function. In particular, when decisions are decentralized with different utility functions, a system with a dominant actor usually leads to a Stackelberg game in which the leader gets the maximal value of his/her utility function while the followers are maintained at their minimal acceptable satisfaction level. However, in spite of this unbalance, such an equilibrium may correspond to global optimal conditions, provided that the contracts between the partners allow for a shift of local equilibria toward globally optimal values. In this respect, different contracts have been shown to possess this property delay adjustment mechanisms for retail prices, delay dependent prices for delivery from the supplier.

Thus, GT has become an essential tool in the analysis of supply chains with multiple agents, often with conflicting objectives.

MODELLING

A model usually refers to a particular 'level in the managerial decision making' context and the decisions of the higher levels. For example, an operational model will have certain constraints imposed by a tactical level decision. Mathematical modelling is probably most useful for tactical decision-making, with a certain time horizon and where certain level of aggregation of data is possible. For higher level decision-making, wherein data is uncertain and intangibles are many, the hard modelling is not useful for the most part. Other ones, such as soft systems modelling, some models of discrete decision analysis and cognitive maps are possibilities for strategic decision-making. In all modelling the quantification of certain key elements of cost, or quantifying the attributes of various scenarios, is often very useful in strategic decision-making process.

Models are useful to describe the inter-relationships between different quantities of interest and sometimes to derive certain optimal or good policies. The models based on mathematical programming, inventory theory, routing and scheduling theory can directly address a sharp decision area, or can be part of the quantitative assessment for more aggregate decisions. For many strategic decisions, a generalized approach called 'alternatives assessment', which works through the consequences of a limited number of possibilities or alternatives is good enough.

Inventory Models

Inventory (stocks) is an inevitable integral part of every business operation. Inventories occur in all forms and for the most diverse purposes. In applications involving inventory, manager must answer two important questions: How much should be ordered? When to order?

The EOQ or economic lot size model is referred as Wilson–Harris square root formula since it was devised by Ford Harris and R. H. Wilson independently. The assumptions under this model are as follows:

- Demand is known. It is constant and over spread uniformly over a period of time.
- No lead-time for re-supply of material and material is supplied instantaneously.
- The cost of ordering per unit is same irrespective of the lot size.

In this model, inventory-carrying cost may be taken to be proportional to the average inventory held during a period. Therefore, by reducing the inventory, its carrying cost can be reduced. On the other hand, the smaller lot size will increase the number of lot sizes per year to cover the annual demand and hence, the cost of ordering will be more. Hence, economic lot size will have to balance these two opposite costs.

The mathematical formula for economical lot size is

$$Q = \sqrt{\frac{2 (\text{Annual usage in units}) \times (\text{Order cost})}{(\text{Annual carrying cost per unit})}}$$

$$Q = \sqrt{\frac{2DS}{HC}}$$

where Q = Economic order quantity in units

S = Cost of placing an order in Rs

D = Average annual consumption in units

H = Percentage of inventory cost vis-à-vis unit cost

C = Cost per unit

The formula is based on certain assumptions, which in practice is not relevant. These are:

- Standard pack size available in the market forces buyer to modify EOQ.
- EOQ may be modified if seller offers the quantity discount.
- The ordering quantity in practice is based on funds availability and may compel the buyer to buy less than EOQ.
- Higher quantities may be ordered to take care of the future shortage of material in the market.

The EOQ formula is a guideline for the decision-makers and can be used with deviations in light of the practical limitation.

Square Root Law

This law was formulated by Maister in 1976 to minimize the risk in inventory carrying at multiple placing. It is done for risk pooling and inventory centralization. This law in the form of a formula calculates the benefit obtainable by inventory centralization. The formula is

$$X_2 = X_1 \sqrt{\frac{n_2}{n_1}}$$

where n_1 = number of existing facilities

n_2 = number of future facilities

x_1 = total inventory in existing facilities

x_2 = total inventory in future facilities

Exhibit 23.4 demonstrates how average inventory level increases as the number of stocking locations increase.

Exhibit 23.4 *Stocking Locations and Average Inventory*

<i>Stocking Points</i>	$\sqrt{n_2/n_1}$	<i>Total Inventory (Numbers)</i>	<i>Percentage Change</i>
1	1.000	5,000	–
2	1.414	7,071	141
4	2.000	10,000	200
8	2.828	14,140	283

SRL is based on the following assumptions:

- No inventory transfer between stocking locations.
- No variations in the lead-time.
- Uniform customer service level.
- Demand at each location is normally distributed.

The limitation to SRL is its effectiveness, when the demands across the markets are served by the various stocking points. SRL will be most effective when markets have demands, which are negatively correlated and there is little or no benefit from consolidation when demands faced by the various stocking points are positively correlated. Additionally, the more the uncertainty of demand at each location, measured by the coefficient of variation, the better the SRL of inventory works.

Warehouse: Cost vs Area

Bowman–Stewart formula explains the relation between the cost of warehouse operation and optimum area served by the same. The thumb rule is that the supply point be near to the demand points. This means the warehouse location should be in proximity to the market. There is no limitation to the area of the warehouse for serving the particular market. However, the logistics cost goes up with the distance between warehouse and the place of delivery. As per Bowman–Stewart formula, the cost of delivery is proportional to the square root of the area covered by the stocking point. This can be expressed as:

$$\text{Cost 'C'} \sim \text{Stocking Area 'S'}$$

The choice of location is decided on the cost of delivery. The other variable affecting the cost is sales off-take. If cost of delivery is, say, A and B is sales volume for a given time period, then A/B is cost per unit sales, which is part of total cost element of the warehouse. Considering the fixed cost of the warehouse as F, the total cost of warehouse operation shall be expressed as:

$$C = F + A/B + c \cdot \sqrt{S} \quad (1)$$

The formula can be further refined by introducing the sales density 'D', that is defined as sales per unit area (geographical) served by the particular warehouse.

$$C = F + A/DS + c \cdot \sqrt{S} \quad (2)$$

For optimizing the cost, the total cost function has to be differentiated with respect to warehouse area S .

$$dC/dS = 0 = 0 - A/D \cdot S^{-2} + \frac{1}{2} \cdot c \cdot S^{-1/2} \quad (3)$$

$$A/D \cdot S^{-2} = \frac{1}{2} \cdot c \cdot S^{-1/2} \quad (4)$$

$$S = (2A/cD)^{2/3} \quad (5)$$

The warehouse service area can be found out by knowing the variable cost related to sales volume 'A', the sales density 'D' and the cost of 'c' delivery. For higher cost of warehouse operation, the warehouse service area needs to be enlarged, and in case of increase in servicing cost, servicing the area needs to be reduced for optimization.

LIMITATIONS OF MATHEMATICAL MODELS

Any mathematical model of a supply chain rests on assumptions about the supply chain. The outcome obtained is only as good as the assumptions and the model. The recommendations obtained from any mathematical model must be tested for sensitivity to the underlying assumptions and the underlying assumptions must be thoroughly tested for reasonableness.

It is also important to realize that mathematical models can only provide answers to questions posed by a decision-maker. The task of asking the right questions and identifying the underlying problems facing a firm cannot be performed by a mathematical model. For instance, a sophisticated procedure for the routing of vehicles will not significantly benefit a firm that is selling the wrong set of products or is using the wrong set of suppliers. Supply chain mathematics cannot act as a substitute for intelligent decision-making by managers.

SUMMARY

Quantitative techniques are used by corporations to solve their problems on cost and inventory optimization with enhanced customer service. SCM requires extensive decision support tools for the effective monitoring, control and management of the supply chain, that is, tools for channel design, transportation and distribution planning, inventory control, etc. Various analytical and quantitative methods form the core of these decision support system(s). The quantitative models used in SCM are in general large linear programming models viz. model(s) for job scheduling, transportation and distribution, warehouse/facility location, etc. All these models have one intrinsic limitation, they are, more often than not, single objective/criteria optimization methods. But, it is very rarely, in real life, that one encounters single criterion problems, by default all real life problems are multiple criteria decision-making (MCDM) problems.

Supply chain analysis involves the use of quantitative techniques in, strategic and tactical planning of the organization while still giving importance to the sphere of operational research. The quantitative analysis approach for decision-making is the process consisting of steps such as formulations of problem, determining assumptions, model building, collect data, solve the model, interpret results, validate model and implement the solutions. The quantitative techniques in supply chain are used for route planning, transportation, trans-shipments, task assignments, inventory management, capacity planning and replacement models, etc. Forecasting is used in business planning to organize and then commit resources to achieve business goals.

As the environmental forces change continuously, the parameters that have effect on the position of an organization in the market need to be forecasted for various planning process. A few mathematical models such as correlation, time series models are used for forecasting. Linear programming models are abundantly used in logistics, transportation, finance and many other practical applications. Assignment and transportation problems are solved with linear programming technique. The routing problem refers to the problem of selecting a sequence of links on a network in a particular order. In determining the sequence of locations visited by a distribution vehicle, the routing problem is best dealt with as a discrete problem, since the many constraints to do with precedence and vehicle coverage are simple to express as constraints on routes constructed in a finite dimensional search domain. For many organizations, inventories are a major investment. Inventory management is an important function in many organizations even in the Internet age. The fundamental questions in inventory control are when to order and how much to order. The inventory ordering is resolved through EOQ model under the condition of certainty and uncertainty. When decisions are to be taken under conditions of uncertainty, simulation can be used. Simulation as a quantitative method requires the setting up of a mathematical model which would represent the interrelationships between the variables involved in the actual situation in which a decision is to be taken.

REVIEW QUESTIONS

1. Find out the different application areas of quantitative techniques and discuss the role of quantitative techniques in supply chain.
2. Discuss the applications of 'linear programming (LP)' in logistics.
3. Explain with illustrations how queuing theory is used in logistics supply chain optimum servicing rate and optimum cost.
4. What is modelling? Explain how it is useful in supply chain optimization.

INTERNET EXERCISES

1. Read more on forecasting—advantage, type, cost, practical applications, methods, <http://www.referenceforbusiness.com/small/Eq-Inc/Forecasting.html#ixzz0oZxuww7f>
2. For queuing problems and solutions visit <http://www.shmula.com/91/queueing-theory-part-1#ixzz0n2ns47MK> and <http://www.shmula.com/117/queueing-theory-part-2#ixzz0n2oLyyQP>
3. Visit <http://www.profit.com> and study 'Optimizing Routing and Transportation' using software.
4. Visit <http://www.opsresearch.com> for studying more on vehicle routing and modelling.
5. Find out if there is relationship between the 'economic batch quantity' and the 'Kanban' quantity.

VIDEO LINKS

1. Transportation problem <http://wn.com/Profribasmat217>
2. Linear programming optimisation <http://wn.com/Profribasmat217>

PROJECT ASSIGNMENT

1. Manufacturing companies are maintaining multiple warehouses in different states in India and due to current CST rules to reduce tax incidence. With new GST implementation, the companies will reduce the number of warehouses to reduce the operation cost. Select a company with multiple warehouses and use appropriate QT to decide the optimum number of warehouse to extend present level of service level.

NOTE

1. Tsuyoshi Yoshino, Tsuna Sasaki and Toshiharu Hasegawa, The Traffic-control System on the Hanshin Expressway, Interfaces, Vol. 25, No. 1, *Franz Edelman Award Papers* (Jan.–Feb., 1995), pp. 94–108, <http://www.jstor.org/stable/25061977>, Tsuna Sasaki and Takamasa Akiyama, Traffic Control Process of Expressway by Fuzzy Logic, Fuzzy Sets and Systems, Vol. 26, No. 2, May 1988, pp. 165–178.

SUPPLY CHAIN RISK MANAGEMENT

24

“Supply chain disruptions can arise from many sources—and more often than not without warning

—Jack Anderson”

Chapter Objectives

To understand:

- Supply chain vulnerability
- Types of supply chain risks
- Risk mitigation strategies
- Approach to risk mitigation
- Success factors in risk management
- Risk management models

Demand	Environment	Supply
External risk factors		
Supply chain risk management Identify, assess, manage and monitor		
Internal risk factors		
Process	Contingency	Controls

Supply Chain Risks Sources

Supply chain risk refers to an uncertainty or unpredictable event affecting one or more of the parties within the supply chain or its business setting, which can (negatively) influence the achievement of business objectives. Supply chain risk management is a structured and synergetic process throughout the supply chain, which seeks to optimize the totality of strategy, processes, human resources, technology and knowledge. The aim is to control, monitor and evaluate supply chain risk, which will serve to safeguard continuity and maximize profitability.

Ashwini Healthcare: Reducing SC Risk Using Technology

Ashwini Healthcare (AH) is one of the leading chains of Hospitals, which is benchmarked to International standards. AH are concerned about the growing number of patients who are misidentified before, during or after medical treatment. In addition, medical facilities were under-utilized.

Blood, has critical demand in various situations like operations, transfusions, accident treatments, etc. There are many issues attached to supply chain of blood. There are many instances, where the blood received from a blood bank has either passed the safe date of usage, or has resulted in complications since the relevant details were not available. The absence or negligible scale of computerization in information system of blood bank operations (to reduce errors, facilitate tracking and to allow monitoring of recipients) has been one of the major reasons for blood bank-related errors.

To eliminate this type of risk, AH planned and implemented RFID-based blood bank management solution and it provided real-time information on the availability of blood in the nearest bank, which is critical in emergencies. This solution provides the ability to uniquely tag (identify) each bottle/bag, the various parameters associated with the specific bottle, the log of certification, the exact movement details of the bottle, etc. The key benefits with this SCM solution were:

- Online information to users regarding blood availability.
- Faster actions during emergencies.
- Automated updates at blood inventory.
- Reduced manual handling resulting in reduced errors.
- Real-time information on identities of people handling blood samples.
- Tracking of the blood sample from its storage to its departure.
- Quick reports on blood movement and history.
- Quick checks for shortages or wastage.
- Elimination of paper work on donors, blood sample and stocks.

The application of RFID was to track and manage mobile assets and enhance healthcare workflow and business processes was another application. This helped in enhancing assets utilization. RFID-based hospital management solution would now track patients, doctors and expensive equipment in hospital. RFID tags were attached to the ID/bracelets of all patients, or just patients requiring special attention, so their location can be monitored continuously. The key benefits are continuously to track each patient's location, real-time tracking of doctors and nurses location in hospital, track location of critical instruments and equipments, restrict access to drugs and critical areas by unauthorized staff, and review and updates of patient's information through RFID tag. This resulted into high system productivity and elimination losses both on material, equipment and human fronts.

INTRODUCTION

Supply chain management is the buzzword in today's competitive economy, where competition is no longer among enterprises but among supply chains. However, measures to operate the supply chain more efficiently lead to increased supply chain vulnerability. Single sourcing saves cost, but production continuity depends upon the single supplier. Outsourcing leads to more complex transactions in supply chain leading to increased probability to disruptions. Just-in-time production and inventory reduction eliminate the buffer to rely on when unexpected events occurs. In addition, uncertainties are becoming more frequent leading to disruptions in supply chains. Hence, supply chain risk management has become imperative.

Supply chain risk management is concerned with removing the ambiguity or uncertainty in supply chain planning, implementation and operations. Risk is pervasive in all parts of the supply chain. The size of the risk depends on the potential loss in terms of money or value for the event. In logistics activities, the risk is more visible closer to the end customer. International logistics can bring in added network complexities and supply chain risk considerations. Assessing and managing cross-border logistics risk is more complex issue as compared to domestic operations.

Today's marketplace is turbulent and clouded with uncertainty. Demand is more volatile than in the past. Product and technology life cycles have shortened. Larger number of products introduced in the competitive markets makes demand projections very difficult for the business firms. The compressed product life cycle makes the supply chain more prone to risks. It is not only the effect of external events such as natural disasters, strikes or terrorist attacks, but also the impact of changes in business strategy leads to the vulnerability of supply chains to disturbances. For example, strategy of 'lean' practices, leads to outsourcing and a general tendency to reduce the size of the supplier base. However, this move potentially increases supply chain vulnerability.

Supply chain risk management area is likely to grow as supply chains become more and more exposed to risks. Accordingly, supply chain risk and disruption management will continue to attract more attention as supply chains will become more and more complex.

SUPPLY CHAIN VULNERABILITY

As a result of trends and developments within supply chain, organizations are continually faced with new or changing uncertainties because of the following:

- 'Hallowing' of corporations or outsourcing
- Integration
- Cross-border logistics
- Demanding customers
- Dependency on technologies
- Spread of e-commerce
- Legislation

Vulnerability to sudden supply chain disruptions is one of the major threats of today's companies. To stay in control over the resulting consequences, increasing emphasis is laid on flexible supply chains. It is not only the effect of external events such as natural disasters, strikes or terrorist attacks, but also the impact of changes in business strategy such as the move to outsourcing and shortening the supplier base potentially increase supply chain vulnerability.

There is always a move to improve the efficiency of the supply chains and minimize losses due to stock outs, delayed deliveries or excess stocks. A need is always felt to speed up material movement through the supply chain. This brings with it a set of unforeseen problems which can disrupt the activities and increase the vulnerabilities of a supply chain to external and internal risks impacting the company's performance. A large number of events can affect the operations of a supply chain. These operations which have very intricate and complicated intertwined processes and dependencies generate their own risks and give rise to new ones.

Due to developments within supply chain, organizations are continually faced with new or changing uncertainties. These uncertainties crop up due to the following developments.

Firstly, due to reducing or outsourcing activities, which involve a certain amount of risks. These activities are not directly profitable and are closely related to the core business. It creates an increasing dependency of organizations in the supply chain. Secondly, downsizing or relocation of production locations and/or sales outlets, with an emphasis on greater efficiency and economies of scale. Thirdly, due to shorter product life cycles due to greater product variety and substitutes, as well as an emphasis on continuous innovation and flexibility. The growing value of market information also plays an important part here. Fourthly, due to increasing use of IT and communication technologies to serve the customer efficiently and effectively and to streamline the organization itself. Fifthly, increasing legislation on several areas is not only affecting individual companies but also entire supply chain, forcing supply chain members to collaborate. In order to become or remain compliant to different laws and regulations, areas like product liability, food legislation and strict environmental policies require new ways of finding the middle course between being compliant and remaining competitive. Sixthly, due to vulnerability to sudden supply chain disruptions is one of the major threats of today's companies. Terrorism, theft and disasters are some of the possible causes of these disruptions. To stay in control over the resulting consequences, increasing emphasis is laid on resilient supply chains.

MEANING OF RISK

As a layman understands, risk is an issue which leads to unpleasant reactions. In commercial terms, risk is a threat that a disruptive event may occur which will adversely affect the functioning of its normal activities. Risks may occur because of the complicated cause and effect due to interdependences of activities and the cascading effects each have on the processes impacting the final outcome. For example, due to the global recession the automotive industry has been adversely affected. The hardest hit will be the ancillary upstream units. As a result of recession, people have lesser money and demand for automobiles has drastically fallen forcing automotive manufacturing majors such as Tata Motors to shut their factories for a specified period because of lack of demand. As far as supply chain risks are concerned, at any point in the supply chain, from initial raw material sourcing to consumers, risks can disrupt the supply of goods and material or demand for products. They can cause sudden peaks in demand or collapses. Effects of risks can be short-term. Long-term risks are strategic in nature. The short-term risks are borne out of operational processes. There is a third one which is period-related risk. It is a tactical risk which deals with risks associated with budgeting and planning of activities for a specified period.

Changes in global economy, market demand, customer expectations and resource availability create supply chain risks, which need to be attended to, in order to be effective in the supply chain process. Risks arising in the value chain must be identified and contained before they create a crisis which could be harmful for a company which may have its working capital blocked in its supply chain.

However, as per regulatory requirements, all listed companies have to identify their risks and publish it as a paragraph in the chapter titled management discussion which forms a part of the director's

disclosures. Identification of corporate risks and highlighting them is a mandatory disclosure under SOX 409 of SEBI and under clause 49 of SEBI.

In order to ensure that unpredictable events do not give rise to undesired outcomes on the company's performance, each critical activity in a supply chain process needs to be studied in detail and assessed. This is a risk and needs to be studied in detail and needs to be contained properly. The cascading effect of each risk (cause and effect) needs to be studied in detail to understand the root cause.

CATEGORIES OF RISKS

Supply chain management is responsible for the movement of material and information through the supply chain. Supply chain risk appears as an event which affects the movement of material in a supply chain and disrupts the planned flow of material and money starting from initial suppliers to the end consumer. These risks may prevent the ultimate objectives of the supply chain from being achieved and could even force abnormal termination of certain key activities. The supply chain risk management can be defined as:

the systematic identification, assessment and quantification of potential risks along the supply chain and an attempt made to understand and preempt such risks from creating a negative impact on the supply chain performance.¹

Supply chain disruptions can arise from external or internal risks. External risks include raw material shortage or cost inflation, variability in market demand, problems with trading partners, economic downturn, etc. Internal risks include poor quality, unreliable suppliers, machine breakdowns, uncertain demand, poor forecast, financial risks, etc. Management of these risks includes development of continuous strategies designed to control, mitigate, reduce or eliminate such risks.

There are four categories of risks: three related to the planning time span of activities and one related to regulatory risks in the supply chain. Strategic risks are aligned with the long-term goals and the vision, mission of a company. The risks due to acquisition, mergers, expansions, joint ventures need to be assessed properly. Most of the strategic process failures are due to inadequate due diligence or risk assessment.

Tactical risks involve inadequacies of supply chain configurations and inadequate innovation of products. Operational risks are those risks which are borne out of transactional activities such as inventory stock outs, operational continuity issues, etc. The operational risks involve failure of operational processes and deeply impact the financial results. Reporting risks involve weaknesses in reporting of internal controls and inability to prevent transactional inaccuracy.

As mentioned earlier, a supply chain process by its very nature is full of risks as there are several internal and external factors which impact the success of supply chain activities. Supply chain risks are numerous but critical supply chain risks have a high impact financial health of company. The other supply chain risks are economic risks, contractual risks, reputational risks, demand risks, market risks, flexibility risks, obsolescence risks, failed partnership risks, logistic risks, plant operations risks, outsourcing or sourcing risks. These risks originate either from the supply side or demand side.

SUPPLY CHAIN RISK MANAGEMENT

Supply chain risk management is a pro-active approach to ensure effective management of all potential risks throughout the supply chain. The risk can be controlled throughout the supply chain in the most effective and efficient manner by establishing internal and external accountability to stakeholders

with regard to risks and their management by monitoring current performance in supply chain risk management and the results to be achieved.

The scope of supply chain risk is wide, covering procurement, inbound logistics, operations, planning, manufacturing, transport and outbound logistics, as well as support functions. Risk in relation to a business entity falls into two categories of control which are as follows:

Risks Within the Company's Control

Design, planning, procurement, warehousing, inventory management, payment, investment recovery, asset management, contracts, people, supplier selection, internal auditing, data integrity, procedures and policies, taxation and rebates, and transportation.

Risks Outside of the Company's Control

Weather, innovation, marketing pricing, inflation/deflation, political policies, supplier business structure are the external sources for supply chain risks. The sources of risks either are natural disaster or internal events. Disruptions can also result from attempts to create a more efficient, cost-conscious supply chain environment. Hence, it is essential on the part of the organization to understand risk within and outside to the organization.

The organizations need to analyse the key risks spanning supply network connecting suppliers, manufacturers, distributors, retailers and customers. Risk analysis results in a better understanding of the potential sources and, most importantly, the potential costs of a disruption. Thus, business organizations can more effectively address risks arising out of:

- Shifting manufacturing facilities overseas.
- Not being able to address variability in consumer demand.
- Not having real-time inventory information.
- Managing every customer as if they were equal.
- Operational disruptions at suppliers or distributors.

In today's complex environment if the risk element in the supply chain is addressed properly, the financial performance and competitive advantage of an organization can be enhanced considerably.

MODELS OF SUPPLY CHAIN RISK MANAGEMENT

As a result of the heightened risk, organizations will need to develop appropriate programmes to mitigate and manage the risks. For this lots of studies are conducted, different models are developed some of them are explained as follows.

Model 1: Deloitte's Fully Integrated SCRM Approach

Deloitte's fully integrated SCRM approach comprises of four main phases:

- **Step 1: Identify Risk**—Recognizing relevant supply chain risks for the organization, that is, risk from suppliers and customers and from internal business environment.
- **Step 2: Evolving SCRM Strategy**—Considering all the options to evolve a suitable SC risk mitigation strategy.

- **Step 3: Implementing Strategy**—Execute the action plan by educating people who are involved in implementation process.
- **Step 4: Monitoring the SCRM Process**—Continuously evaluate and improve SCRM strategy and actions. Monitoring can be focused on change management (input), the current level of risk management (output) and the secondary effects of the strain put on the company objectives.

Model 2: A Review by Christopher S. Tang

According to this model, supply chain risks can be managed through co-ordination or collaboration among the supply chain partners so as to ensure profitability and continuity. Risk management can be thought of along two dimensions. The first one is managing operational risks and the second one is mitigation approach. The first dimension addresses the risk level of certain events. Operational risks are uncertainties such as uncertain customer demand, uncertain supply and uncertain cost. Disruption risks are disruptions caused by natural and man-made disasters such as earthquakes, floods, hurricanes, terrorist attacks, etc., or economic crises such as exchange rate fluctuations or strikes. In most cases, the business impact associated with disruption risks is much greater than that of the operational risks.

To mitigate the impact of supply chain risks, there are four basic approaches, supply management, demand management, product management and information management, that every firm goes in for a collaborative mechanism. Each of these four basic approaches is intended to improve supply chain operations via co-ordination or collaboration as follows.

First, a firm can coordinate or collaborate with upstream partners to ensure efficient supply of materials along the supply chain. Second, a firm can coordinate or collaborate with downstream partners to influence demand in a beneficial manner. Third, a firm can modify the product or process design that will make it easier to make supply meet demand. Fourth, the supply chain partners can improve their coordinated or collaborative effort, if they can access various types of private information that is available to individual supply chain partners.

Model 3: SAP Supply Chain Risk Management

The growing amount of globalization, supplier dependency and variability of demand has created an increasing vulnerability of supply chain networks when disruptions occur. While operating the supply chain through planning and execution, sensing deviations and responding to them lays the foundation of adaptiveness. To identify risks, develop efficient response strategies and monitor the actual processes, a disciplined approach is necessary. Managing supply chain risks requires taking a specific perspective on the supply chain and the involve business partners, locations and dependencies on different levels in the value chain.

To mitigate the ill effect of sudden rise of risk and disruption of supply chain thereafter, the organization can go systematically to address the problem. The organization may adopt the process of identifying, measuring, managing and monitoring the risk.

ROLE OF IT IN MANAGING RISK

IT plays a very important role in today's global supply chain. IT tools can help company's aggregate baseline data and create supply chain profiles. Enterprise-wide software solutions can also help traditional supply chains evolve from linear and sequential operations to adaptive networks

capable of adjusting to changing economic and market conditions. The same functionality that supports supply chain network visibility, cooperation, and analytics can also enable supply chain risk management. Companies can synchronize supply to demand, and sense and respond to supply chain events through an adaptive network with real-time distribution, transportation and logistics capabilities.

Implementing enterprise-wide software solutions can be crucial because in many cases companies actually possess the data they need, but disconnected systems come in the way to make the right data accessible to the right people at the right time. With an end-to-end software solution, the overall result is a more transparent and responsive supply chain network that supports organizational efficiency as well as flexibility and resilience. With IT in supply chain, the risk in operation gets mitigated because of the following reasons:

- Increased degree of collaboration and transparency between partners.
- Reduction in inventory due to real-time information flow.
- Error-free handling of huge data.
- Information availability at anytime, anywhere in the right format.
- Technology increases visibility, coordination and integration.
- JIT implementation becomes possible.

SUCCESS FACTORS IN RISK MANAGEMENT PROCESS

In view of the growing complexities of a company's operations, a need is felt for a very elaborate risk management practice. This has a close look at all the activities that constitute key processes and having assessed the various risks involved, formulate mitigation procedures to minimize the impact of such risks. The risk management tool should be able to predict the down trends of events in processes before it turns into a crisis. An elaborate business continuity plan should be included as a part of an effective enterprise risk management process (Exhibit 24.1). This enables the company to plan for alternatives in the event of process risks surface. Risk management and business continuity processes should be elaborate and cover intra- and inter-company risks.

Exhibit 24.1 *Risk Management Process*

<i>Identify Risk</i>	<i>Measure Risk</i>	<i>Manage Risk</i>	<i>Monitor Risk</i>
<i>Find out the factors that influence risk management in strategic, tactical and operational areas.</i>	<i>Assign probability of the occurrence using information sources of service providers.</i> <i>Assign impact levels through KPIs and establish control systems as sensory networks.</i>	<i>Create plans for risk mitigation. It covers hedging and trading mechanisms.</i> <i>Document risk and responses.</i> <i>Develop process and scenarios</i> <i>Translate risk mitigation strategy into actions.</i>	<i>The behavior of risk is dynamic in nature. Hence, the managing process needs constant monitoring. Activities need to be adapted and adjusted to the real world.</i>

Risk assessment and management process is a very seldom practised process today and companies are waking up to the fact that they are carrying veritable risks with them, which could seriously undermine their chances of success at the marketplace. Very few companies have a well documented risk policy in place and further even fewer employees understand the risks entailed in their function. Similarly, very few companies with strong supply chain dependencies actually are verbose in listing the key issues involved in their respective supply chains which could pose as serious risks to the companies. Supply chain risk assessment and management process should be embedded within the performance management system of a company.

BUSINESS CONTINUITY PLANNING

Supply chain activities are critical in nature. Any disruption in these activities is catastrophic for the outcome of a supply chain process. Hence, every company which depends on the supply chain as a strategic process should have an inbuilt business continuity to take care of each of its critical activities such as sourcing, outsourcing, manufacturing, logistics, etc. Business continuity planning seeks to prevent disruption of critical activities. The activities are designed and tested to use for restoring operations in the event of disruptions. Key activity in business continuity planning includes finding viable alternatives for performing critical supply chain processes in the unlikely event of their failure. Various components of business continuity planning include crisis planning, contingency planning, business recovery planning and continuity planning.

APPROACH TO MITIGATE RISKS

The following approaches may help business organizations to reduce the degree of risk in supply chain operations.

Box 24.1

Supply Chain Risk Assessment

As per clause 49 of the listings agreement of the stock exchange, it is mandatory for all listed companies to publish their risks in the annual report under the title—Management Discussion: Issues of Risks and Concerns. To illustrate Asian Paints and Pantaloon Retail, respectively, disclosed the risks along with the impacts. According to Pantaloon, there is an impact of economic slowdown on their business resulting in decreased spend power. Secondly, inadequate upward communication from retail outlets to upstream of supply chain on agility and flexibility resulting in mismatch between demand and supply. As per Asian Paints, threat from very strong grey market especially in the decorative paints segment leading to stiff competition in pricing and, hence, reduced margins. Thirdly, power shortage in manufacturing plants leads to either loss of continuous production or extra capital spend by way of erection of a captive power plant and subsequent additional expenditure for fuel. Fourthly, there is a threat from competitors leading to diminished market share. Fifthly, loss of opportunity at retail outlets due to stock outs created by inadequate forecast leading to customer attrition. Sixthly, pile of stocks of finished goods due to lack of demand due to slow down in infra structure projects. Seventhly, there will be loss of key technical people and lastly, there will be inadequate commitment by distributors.

Box 24.2

Hewlett Packard–Approach to Supply Chain Risk

At HP risks are categorized into two types: risks through natural phenomenon (floods, earthquakes, etc). Under high probability risks commodity costs are considered as high probability with high impact. This risk is addressed through its procurement risk management.

HP developed a methodology and evolved a set of tools to address risk within HP. They have created algorithms using various assessments of the market, suppliers and procurement patterns. This resulted into a well drafted procurement contract with suppliers that shares risk between HP and the supplier and ensures a smooth flow of product under fluctuating market demand scenarios. This is helping HP suppliers in their commitments upstream, effectively smoothing the supply chain. Through this risk mitigation approach, HP has lowered cost up to 20 per cent. For this exercise, they require three basic inputs which are demand, cost and uncertainty. They have extended this approach beyond procurement into energy, spare parts, and logistics.

Strategic Planning

A company with a strategic planning process will avoid conflict of cost vs risk and bring value improvement, speed to market, and innovation. Cost efficiency can reduce supply chain risk, provided that cost-efficient processes focus on core trading partner relationship management. The tool for managing this relationship is the contract, which can be written to include service level agreement for transaction compliance. This arrangement increases the ‘visibility’ of supply chain partners’ performance, thereby reducing risk.

Co-operation, Collaboration and Integration

To achieve a common objective co-operation with all supply chain partner is must. To close the skill gaps the collaboration between them is essential. However, for continuous operation, integration in assigned set of work is a must. For sorting out the issues involved in mitigating complex risks requires a greater degree of collaboration. The vulnerabilities arises out of lack in co-operation, collaboration and integration.

Tradeoffs

The inventory will address the issue of increased service cost. But it may create the issue of stock outs and opportunity loss. Hence, tradeoff is must to optimize the risk. The tradeoff solution in general should cover the following:

- **Assessment:** To identify key risks in supply chain to understand its potential economic consequences.
- **Analysis:** To arrive at an optimal future state risk profile for supply chain.
- **Roadmap:** Develop plans needed to implement the changes required to mitigate organization’s future risks.

Supply chain disruptions can arise without warning. These disruptions can be entirely external, such as a natural disaster, or they can be internal, rising from the failure to integrate all functions in a supply chain.

A supply chain must be fully integrated. Its inability to understand the potential vulnerabilities makes it weak to handle unexpected and sudden shocks. For being fully integrative supply chain organization need to embrace the extended supply chain (i.e., their customers and suppliers as well as their customers' customers and their suppliers' suppliers).

In an attempt to make supply chain more cost-effective and efficient, organizations adopt techniques like just-in-time (JIT) in their manufacturing operations. In fact, this adds to supply chain complexity and risk thereof. It will be prudent on the part of the organization to take a note of the potential risk in adopting these techniques and think about the solutions in advance.

The supply chain disruptions have a great impact on financial performance of the organization. However, the firm's ability to effectively manage the impact of potential risks can create a strategic advantage in a competitive marketplace.

To mitigate the financial impact due to supply chain disruptions, the organizations should look into the following:

- Understand the dependencies within the supply chain.
- Identify the weak links within supply chain.
- Identify the SC risks that may be mitigated, or eliminated.
- Incorporate the element of risk in decision-making.
- Assess the flexibility of SC to encash on opportunities.
- Capture an enterprise-wide risk profile.
- Use tools and skills to understand the financial impact of risks.
- Benchmark the activities that make up effective supply chain.

Finally to address risk, firms should understand the global business drivers that affect an organization's profitability and performance.

SUMMARY

Risk management is an opportunity to address the uncertainties that are an inherent part of supply chains. It provides an insurance against unpredictable, but not unlikely events. The challenges in implementing sound supply chain practices also apply to risk management. Managers will need a better understanding of the complex networks they use to provide products and services to their customers. They need to convince their staff to adopt new processes. This change management effort has to happen in the context of record-high task loads, globally dispersed teams, short-term metrics, and limited appreciation of the complexity and inherent delays.

Implementing risk management requires more than a top-down effort.

The vision of using risk management requires an ongoing organizational effort to make it a success in today's complex and uncertain business environment. Each company engaged in supply chain activities should have a strategic map of all its strategic, tactical and operational activities. They should do a risk assessment of such activities and devise mitigation plans for such risks.

Risk management is an opportunity to address the uncertainties that are an inherent part of supply chains. It provides support against unpredictable, but not unlikely events. The vision of using risk management requires an ongoing organizational effort to make it a success in our complex and uncertain world.

REVIEW QUESTIONS

1. What do you mean by supply chain risk?
2. What are the various types of risks associated with supply chain?
3. Discuss the various models available for mitigating supply chain risk.

INTERNET EXERCISES

1. AMR research provides advisory services and peer networking opportunities to supply chain, sustainability, and IT executives in the consumer products, life sciences, manufacturing and retail sectors. For studying SC risk mitigation strategies visit <http://www.amrresearch.com>
2. Visit <http://www.pwc.com> to study the cases on SC Risk Management.
3. Follow link [http://www.sdcexec.com/print/Supply-and-Demand-Chain-Executive/Benchmarking-Green-and8212-Supply-Chain-and-Product-Lifecycle-Strategy-Guide/1\\$10842](http://www.sdcexec.com/print/Supply-and-Demand-Chain-Executive/Benchmarking-Green-and8212-Supply-Chain-and-Product-Lifecycle-Strategy-Guide/1$10842), to understand how to execute programmes that mitigate supply chain disruption, comply with regulations, and enable sustained performance.

VIDEO LINKS

1. Supply chain risk management strategy, <http://www.youtube.com/watch?v=LbNgYYB7tV0>
2. Supply chain risk reduction—Professor Richard Wilding, <http://www.youtube.com/watch?v=QLZ6TyUaYpw&feature=related>

PROJECT ASSIGNMENT

1. In the event of natural calamities, the organizations with global reach, faces the disruptions in their supply chains. The disruption may discontinue raw material/components supplies or disruption in manufacturing operation. The global companies like CISCO, Hewlett Packard have successfully tackled this problem. Find out whether the same can be made applicable to Indian companies with global operations. Select any Indian global company in any industry sector of your choice for this exercise.

NOTE

1. 'Managing Risk in Your Organisation with SCOR Methodology', Supply Chain Council Risk Research Team, <http://www.supply-chain.org>.

PRICING, COSTING AND FINANCIAL DECISIONS¹

25

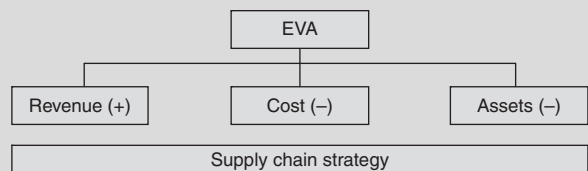
“Financial managers strive to obtain borrowed funds at the lowest cost, to select projects that offer the best returns, to balance the financial risks taken with investors’ expectations of returns and to keep the business liquid

—James K. Keebler”

Chapter Objectives

To understand:

- Product pricing decisions
- Costing techniques and methods
- Financial metrics in supply chain
- SCM and SCOR models
- Effect of supply chain performance on EVA



EVA Components and Supply Chain²

Supply chain management is the backbone for the success of a business. Financial issues are very important for supply chain. The challenge of every business lies in allocating the resources required to effectively and efficiently meet the expectations of its various constituencies. To enhance its financial health, a firm should enhance its functional capability and process assessments and should benchmark its best practices to achieve competency. For that, the supply chain needs an integration of its processes to manage the business cycle time and align itself to the capabilities of its service providers to get maximum benefits in terms of service and savings. The financial metrics are keyed to economic value added (EVA), a widely accepted set of financial performance measures.

ITC Managing Cash Flow with No Credits to Customers

ITC is one of India's leading private sector companies with a turnover of \$6 billion. It has a diversified presence in cigarettes, hotels, paperboards and specialty papers, packaging, agribusiness, packaged foods and confectionery, information technology, branded apparel, personal care, stationery, safety matches and other FMCG products.

In 2009–10, the revenue from cigarette division was Rs 18,110 crores contributing to 65.5 per cent of the total sales revenue of the company. ITC caters to the Indian cigarette market, with around 47 SKUs to reach 48 million customers (smokers). The company ensures freshness of 2 months for cigarette products.

For cigarettes, ITC has a wider distribution network of 1 million retailers, 60,000 wholesalers (with 9,000 vans/cycle salesmen), 800 wholesaler dealers and 23 C & F agents. The company also has four manufacturing plants and six contract manufacturers.

ITC's ultimate customer is the retailer who has very few or no financial standing and who has investment capacity of ½ days. Hence, ITC offers no credits to the retailer but ensures high-frequency supplies. The cycle salesman collects the money against the delivery of even very small quantity (one pack of cigarette). The company has 17 cash collection centres and simultaneous fund transfer facility. No credit at any stage of supply chain is encouraged. Remittance from wholesale dealers follows the same pattern. Even though the margins to wholesaler dealers are very low, the ROI is high because of high stock/cash turnaround, which is 30 times a year. The major task of field sales force is follow-up on payments/cash flow at all stages of the supply chain. This credit policy helps ITC to generate funds for payment of excise duty, which is very high. In view of high incidence of excise duty, there is tight control on inventory. Due to the branch manager's forecasting and direct ordering (on daily basis) from factories, there is no time lag, resulting in better inventory management. The pipeline inventory (excise paid—stock on hand/in transit) is kept within 4 days.

As per the company's past experience, even a 1 day delay in transit time costs the company Rs 26 million by way of interest. The transport mode is truck and not the rail. Full truckloads of deliveries are made to destinations without trans-shipments. The company has a bonus–penalty system with transporters. With its tight finance policy, ITC is maintaining its trade receivables within 3.6 days (last year 1.4 days) and has no major financial issues in its supply chain.

INTRODUCTION

Every business has a common challenge of allocating the required resources to effectively and efficiently meet the expectations of its various stakeholders. These are investors looking for ROI (return on investment), employees expecting right compensation, customers requiring value for money they spend on the products and services they purchase, channel partners expecting help from manufacturer, suppliers expecting timely payment, community expecting socially responsible behaviour, and government expecting payment of taxes and fair business practices, as well

as regulatory compliance. The challenge for business firms is to seek out ever-improving ways to satisfy all the above areas.

Technology is a pervasive factor in supply chain and can be leveraged for cost and service leadership. Nowadays many firms are making efforts at creating accurate and integrated cost measures to increase the visibility of costs within the chain using activity-based costing method. Ultimately, restructuring the supply chain to exploit efficiencies also requires a mechanism capable of identifying and equitably allocating costs and benefits between the partners as changes are implemented. The investigation of financial issues in supply chain management has raised fundamental questions about competency and confidence levels of supply chain managers in their financial skills, adequate interface between the financial function and the supply chain functions, and the financial implications to partnering, strategic alliances and outsourcing. The supply chain operations reference model (SCOR), developed by the Supply Chain Council is the base for supply chain performance metrics used herein. The decisions on items such as transportation modes, sourcing and replenishment are based on operating expenses. If an enterprise-wide perspective is not adopted, the real value of SCM will not be achieved. SCM drives key financial metrics such as revenue growth, percentage cost of goods sold and days in inventory. The financial metrics are keyed to economic value added (EVA), a widely accepted set of financial performance measures. Improving most SCM professionals' financial acumen is a critical factor in making the financial-SCM connection. SCM professionals must understand how SCM business processes and strategies impact key financial metrics and contribute to returns to shareholders.

Since 1990, supply chain and supply chain management, have become pervasive terms in the business and academics. Supply chain concept has virtually replaced the predecessor terms, 'business logistics', which was quite common earlier. This is because the effective management of the supply chain can help boost a firm's performance; in particular, they realized that SCM can provide a sustainable competitive advantage and improve profitability by taking out the costs. A well-managed SC can help to succeed on four dimensions of competitiveness—cost, quality, response time and flexibility.

Supply chain management spans all movement and storage of raw materials, work-in-process inventory and finished goods from point-of-origin to point-of-consumption. SCM has been significantly successful in enhancing the control of operational activities through the application of supply chain planning and supply chain execution solutions. However, most executives still find it difficult to interact with SCM solutions in answering many questions and have doubts as to whether the drivers of this operational activity really execute financial and strategic objectives. The major challenge for businesses is to construct integration between these operational elements and systems in view of the strategic and financial goals of a business.

The supply chain performance metrics used here are based on supply chain operations reference model (SCOR),³ developed by the Supply Chain Council. The financial metrics are keyed to EVA, a widely accepted set of financial performance measures.

PRICING DECISIONS IN SUPPLY CHAIN

To manage demand and to assure the desired order flow in supply chain, companies like Wal-Mart have adopted the strategy of 'everyday low pricing.' This strategy helps in reducing variations in demand and enhances the capability in inventory availability leading to better customer service at lower operating costs. The pricing strategies are also evolved and implemented to clear slow-moving items and build order flow resulting in increasing revenues.

The major tools used in supply chain demand management are pricing and promotion. They temporarily discount selected items below the average price level to clear slow moving inventory or increase retail store traffic. To exhibit image and high quality, sometimes prices are set slightly above their competitors' prices. It is observed by Pashigan and Bowen (1991) that retailers who reduce the initial price after the first period are significantly more profitable than those who maintain a fixed price. From the supply side, the manufacturing level of the supply chain, which often must adjust capacity, as retail orders rise and fall over time need to be lined with pricing strategy. The costs, which include overtime expenses like hiring, training and layoff costs, as well as subcontracting and inventory carrying costs are reflected in the unit prices that manufacturers charge the retailers. However, retail pricing strategies that help smooth demand swings helps reduce production change costs, increasing margins and perhaps total profit for the supply chain. The general experience is that the items with seasonal demand patterns (consumer durables, clothing, toys and sporting goods) are not affected by low pricing strategy and this strategy will do little to stabilize their demand. If the firm is implementing low pricing strategy during seasonal demand, the cost arising out of production change will be absorbed by the supply chain. This will obviously reduce the margins and supply chain profits.

Pricing and Profits

Low pricing strategy is appropriate for supply chains that produce and sell items with stable demand, but it cedes most control over demand. However, with this strategy, the profitability is low. The price changing policy is an appropriate strategy when demand is seasonal. There are other environmental factors such as demand elasticity, promotion cost and production change costs that impact decisions on pricing strategy, keeping in mind the goal of performance criterion is profit. To understand pricing dynamics, we consider a three-level supply chain model for a single product. The model enables us to compare alternative pricing strategies in a hypothetical supply chain. This model includes retail and manufacturing sectors, and spans multiple periods. Key parameters include retail demand characteristics (seasonality, price elasticity), retail-level cost characteristics (promotion costs, ordering costs, holding costs) and manufacturing-level cost characteristics (labour costs, production change costs and limitations, and production rates). Decision variables include retail prices, replenishment orders to the manufacturer, and the manufacturers' short-term capacity decisions to meet those orders.

The role of the retailer is to set retail prices and to periodically order sufficient product to satisfy forecasted demand. These decisions influence retail demand, total revenue, promotion expenses, and inventory costs. EDLP (every day low price) retailers maintain a constant price over the planning horizon, and as a consequence avoid additional promotional expenses. It is assumed that the EDLP is chosen to maximize profit for the entire supply chain. HLP (high and low prices) retailers may change prices periodically, incurring a promotion expense proportional to the monthly revenue when the discount price is in effect. The high and low prices will be chosen optimally to straddle the single optimal price for EDLP retailers, as will the timing of any change in prices.

The manufacturer, with short- and intermediate-term capacity constraints, must produce sufficient quantities to satisfy the temporal demand manifested in the retailer's replenishment plan. The manufacturer uses a combination of production planning strategies such as overtime/undertime, workforce level changes, and inventory to fill the retailer's orders at minimum cost. To maintain adequate service levels, backorders are not allowed here.

Supply Chain Profit Model

Here assumption is that both retailers (EDLP and HLP) optimize their offering price(s) with respect to total supply chain profits. The EDLP (retail price) is held constant throughout the planning horizon. The HLP retailer may switch between the optimal regular price and the optimal discount price as desired, incurring a promotion cost each period when the discount price is in effect. These decisions determine monthly demand and total revenue for the supply chain. Additional costs incurred by the retailer include ordering costs and inventory holding costs. Here, for optimizing total supply chain profits, the manufacturer's transfer prices are at cost.

COSTING DECISION IN SUPPLY CHAIN

One of the most important and also most difficult parts of supply chain management is the supply chain cost analysis. This is an area incredibly important and extremely difficult. Irrespective of the type of organization, firms need to make sure to know supply chain costs. Companies can lower their supply chain costs by using new management strategies to reduce their inventory safely. First, they must stop treating transportation as a management priority which is separate from inventory management. Second, companies should install technologies that provide detailed inventory visibility throughout their supply chain. These are the prerequisites for lowering inventory and achieving supply chain cost reductions.

Inventory levels, not transportation costs, drive supply chain savings. For most manufactured products, transportation is only 2–5 per cent of total cost. In contrast, raw materials, components, and sub-assemblies typically constitute 55–75 percent of total cost. This reality helps explain why large cost savings result from complete supply chain solutions that reduce raw material and finished goods inventories.

There are many opportunities to reduce total cost in supply chains, which are responsible for many unnecessary overhead costs to generate forecasts, count inventory on-hand, generate purchase order inputs through MRP (material requirement planning) systems, place purchase orders, wait for parts to arrive, expedite those that are late, receive (and maybe inspect) materials, warehouse, group into kits for scheduled production, and distribute within the plant. These costly and time-consuming steps can be avoided with a spontaneous supply chain, which is able to pull in materials and parts on-demand.

There are three basic ways to reduce the supply chain costs and they are:

Simplification The prerequisite to cost savings, in supply chain management, is simplification, which corresponds to the basic lesson of industrial engineering, that is, 'Simplify (process and product) before automating or computerizing'.

Product line rationalization To eliminate or outsource the most unusual products that has the most unusual parts.

Standardization To reduce the variety of parts and materials to a point where they can be pulled into the factories.

Selective vertical integration To eliminate any remaining long-lead time parts and materials, the SCM costs metrics attempt to segment cost centres into a process-based activity. Below are listed suggested level two and three metrics with brief definitions.

- Order management cost
 - Customer service cost
 - Finished goods warehouse cost
 - Outbound transportation cost
 - Contract and programme management cost
 - Installation planning and execution costs
 - Accounts receivable cost
- Material acquisition cost
 - Purchasing cost
 - Raw material warehouse cost
 - Supplier quality cost
 - Component engineering and tooling cost
 - Inbound transportation cost
 - Accounts payable cost
- Planning cost
 - Demand planning cost
 - Supply planning cost
 - Supply chain finance control cost
- IT cost
 - Application cost
 - IT operational cost for supply chain
- Inventory carrying cost
 - Opportunity cost
 - Obsolescence cost
 - Shrinkage cost
 - Taxes and insurance cost

In order to perform a supply chain cost analysis in a company, one needs to primarily find out the accounting method that is used in the organization. If firm is using an activity-based accounting, then it is a good sign as this generally divides all sorts of costs into proper categories and firm can hand pick which ones are associated with the supply chain. The data gathering for activity-based accounting supply chain cost analysis is very easy.

TYPES OF SUPPLY CHAIN COSTS

Below is a list of supply chain cost centres and categories that firm might want to find out and track individually:

- Transportation costs (inbound, in-plant and outbound)
- Warehouse costs (fixed and variable)
- Packing and packaging costs
- Working capital costs (blocked in inventory)
- Insurance costs
- Costs for mistakes (invoicing, delivery, orders, etc.)

Comparing these specific costs across the organization could reveal inefficiencies in some parts of the company, it could reveal the best practices to be adopted in some areas of the supply chain, and it could provide direct actions to reduce costs in organization to realize savings.

Box 25.1

NANO: New Approach to Supply Chain Costing

The secret of designing the Tata Nano is a concept called 'target pricing or target costing,' taking into consideration the features and functions customer want in a small car. To cut the cost, Tata Motors have focused on their supply chain, particularly on upstream (suppliers). Setting the price and working backwards also required a fundamental shift in the way the car was designed, since many costs are fixed once the design is set. To accomplish this, Tata Motors worked in collaboration with suppliers very early in the process. They set the functional goals for many parts rather than technical specifications. This approach tapped the ingenuity of the supply chain, delivering parts that met the functional requirements and at extremely low prices

To reduce cost, they look at supply chain upstream rather than downstream. Tata brought in suppliers such as Bosch, a German maker of appliances and motors, and Delphi, a world leader in automotive parts in early-stage design, challenging them to be full partners in the Nano innovation by developing lower-cost components. Tata built partnerships with a limited number of suppliers and put everyone in the same room to work through problems and make suggestions and that created enormous value.

Tata has taken a three-pronged approach. The first one is alternative design and materials with low form factor in design. Second one is low cost manufacturing and logistics allowing the company to capitalize on the region's comparatively lower cost R&D, manufacturing talent and single market approach. The third one is strategic sourcing approach, extending the use of its e-sourcing (online auctions for 45 per cent of material cost) and competitive online bidding methods with fewer suppliers, and improving supplier development and capacity. With an aggressive timetable for savings, Tata used a mix of sourcing automation and external sourcing and spend category expertise to better control its direct materials costs.

SUPPLY CHAIN AND FINANCE

The financial supply chain parallels the physical supply chain and represents all transaction activities related to cash flow, from the buyer's initial order through reconciliation and payment to the seller. Until recently, the financial supply chain went virtually unnoticed. The amount of time required to process transactions was unthinkable long because the manufacturing supply chain had inefficiencies. Nonetheless, companies prospered, in part due to lack of global competition and, in part, because all domestic companies were saddled with the same poor quality demand forecasting limitations, inefficient distribution and lack of supply chain visibility. They used the same strategy to compensate for these problems and hedge against uncertain demand, excess inventory, excess capacity and surplus labour. Since the cost of capital was low, reserve positions were routinely used.

Box 25.2

ABD Bank—Financing the Supply Chain Operations

Supply chain financing is one of the fastest growing segments in the banking sector these days. A more advanced and sophisticated version of the age-old working capital financing (WCF) called supply chain—financing (SCF) has been created and implemented by ABD bank.

Earlier, ABD bank looked at each entity on its own merit, and went by fairly standard and rigid 'permissible bank financing' kind of formulae. Now, ABD bank is looking at value chains around large manufacturing companies. ABD Bank is looking at them like original equipment manufacturers (OEMs). ABD is now saying that they are financing the suppliers and distributors of OEMs.

The supplier's financing is considered safer for the ABD bank, as suppliers can be decent-sized companies (compared to a distributor); and more important, the credit risk is in a way, backed by the original equipment manufacturer (OEM). ABD bank opened a line of credit for a supplier in consultation with the manufacturer (OEM), who is the bank's customer. This line of credit is very specific to the product that the vendor supplies to the manufacturer. An OEM gives a guarantee to its vendors for a specified off take of parts over the year. On the basis of this guarantee, the vendor approaches the ABD bank which is usually recommended by the OEM and obtains better financing interest rates and services. In exchange, ABD bank finances the working capital requirements of the vendor. The recovery of the loan is linked to the vendor supplying to the OEM on time schedule. After the vendor supplies the OEM's, payment is routed through an escrow account, to give further comfort to the bank. In this process, there is little chance of the vendor diverting funds somewhere else. In case there is a cash discount to the OEM from the vendor, the bank may even pay the vendor upfront, and collect from the OEM later. All this also involves, the bank keeping tabs on production, dispatch, payment, and financial health data of the vendor.

It is more like a performance risk on the part of bank. The vendor benefits as he gets cheaper financing than it may get from traditional working capital routes. The OEM benefits in two ways: better supply from the vendor, and also the OEM reduces its own working capital needs. Vendors can supply just in time because their finances are more robust now. This enables the OEM to keep lesser inventory. In other words, the WCF is shifting from OEM to vendors by this route.

On the distribution side, ABD bank provides working capital to selected distributors allowing them to lift output from the manufacturer. Often there is provision of an escrow account kind of mechanism where the sales of the distributor get credited to an account with the bank. The OEM has access to this bank account, and can make payments to himself from this account. So the OEM now has the comfort of getting payments. The bank can monitor sales and receipts of the distributor. In this, the bank also saves on a lot of charges for collection, processing and security.

Today the typical financial supply chain remains somewhat fragmented, complex and not integrated with the physical supply chain. Goods move faster than money, and disparate parties are involved. Moreover, even in today's world, the financial supply chain is partially composed of paper-based processes. To stay competitive, 'just-in-time' working capital management should be the goal. Financial supply chain solutions enable CFOs and treasury managers to accurately manage their receivables or payables, forecast their company's financial future and reduce their working capital needs.

Customers are impacted most directly by supply chain and shareholders more so by corporate financial performance. There is a synergy between the two. For example, customer may expect consistent on-time delivery to support JIT operations metrics that focus on inventory and transportation activities in the supply chain which will determine how effectively the firm is satisfying a customer's need for JIT support. Getting those activities right may improve the firm's competitive position by yielding higher revenues through greater market share. Financial supply chain management is the most effective approach to optimize working capital levels, streamline accounts receivable processes, and eliminate excess costs linked to payments. The lack of synchronization between the financial supply chain and the material supply chain creates excess working capital, or non-productive float, which is a hedge against the lack of visibility into the supply chain itself. A better way is to optimize the financial supply chain, integrate it with the material supply chain and substitute information for float.

FINANCIAL—SCM

Despite SCM's potential, relatively few companies utilize SCM as a tool to drive financial performance. Failure to make the financial—SCM connection is driven by several factors. Firstly, many executives continue to view SCM as a tactical cost-centre activity. Secondly, most SCM professionals do not speak the language of finance. Hence, they lack the ability to link SCM to key financial metrics and to articulate in enterprise language. Therefore, SCM strategic and tactical decisions cannot be made in a vacuum. Yet most scorecards and analyses of SCM initiatives are incomplete, as they are not from an enterprise-wide perspective. Decisions on items such as modes of transportation, sourcing, and replenishment are often based solely on operating expenses. If an enterprise-wide perspective is not adopted, the real value of SCM will not be achieved.

FINANCIAL METRICS

SCM drives key financial metrics such as revenue growth, percentage cost of goods sold, and days in inventory (DII). The values of the gaps may be based on benchmarks from competitors, industry aggregates, historical performance, and aspirations derived from business intelligence tools. The value of the gaps can be measured using a variety of value-based financial measures such as free cash flow, economic profit and stock price.

The values of the gaps are an effective means to communicate to the organization the need for change and the potential value of improved SCM. For example, communicating that closing a 10-day gap in inventory is worth Rs 100 crores in cash flow and can potentially add Rs 100 per share in stock price is likely to generate greater motivation for change compared with the strategy of simply announcing the need to reduce inventory.

Gaps in financial metrics are driven by unique organizational factors such as sales mix, pricing strategies and outsourcing. The next step in the top-down approach is to link gaps in financial metrics to SCM-related business processes and strategies. A gap in profitability related to percentage cost of goods sold, for example, can be mapped to an SCM-related process such as distribution and logistics, which, in turn, is linked to a key activity such as warehouse management. Warehouse management is related to tasks such as receiving, put-away, pick, pack, and ship, to key performance indicators such as labour costs, average time per pick and pick accuracy. This mapping provides a better understanding of the cause-and-effect relationships between SCM business activities and financial performance.

The information provided above that provides a methodology for identifying specific areas of opportunity in supply chain. It also provides a disciplined approach for estimating the monetary benefits and understanding the critical success factors and risks of SCM solutions required for building a business case. Improvements in SCM business processes and strategies typically cannot completely close financial performance gaps. But for many companies, it can make a significant contribution.

Using SCOR model, it can be seen that supply chain integration and specific related measures of supply chain performance have a direct impact on a firm's financial performance. The supply chain metrics used by operating personnel focus attention on the day-to-day activities in the supply chain. By paying attention to the collective details of these daily activities, companies can boost their overall supply chain performance. And this in turn, will be manifested through a positive impact on over all metrics of financial performance.

STRATEGIC ALIGNMENT

The link (Figure 25.1) between corporate performances and supply chain performance must begin with strategy. For this, the overall business (or business unit) strategy and SCM strategy needs to be aligned. With this the impact of SCM on corporate performance can be effectively measured.

Linkage approach incorporating EVA and the SCOR model is another way of adopting the 'balanced scorecard'. This tool identifies performance drivers and outcomes measures. The elements of the SCOR model are important SC-related performance drivers, while the corporate performance metrics represents important outcomes. Understanding the link between performance objectives and outcomes measures is fundamentals to achieving improved financial performance.

ECONOMIC VALUE ADDED (EVA)

EVA, developed by Stern Stewart, is recognized as a comprehensive measure of value creation. EVA provides a more comprehensive measure of profitability than traditional measures because it indicates how well a firm has performed in relation to the amount of capital employed.

EVA emphasis and isolates activities that help to drive value creation. Those activities may be generally categorized as revenues, costs and assets. By understanding the EVA drivers, managers become more aware of the impact of innovation, cost reduction, technology improvement, and capital base reductions on value creation. As such, the EVA drivers align well with the performance drivers in SCM.

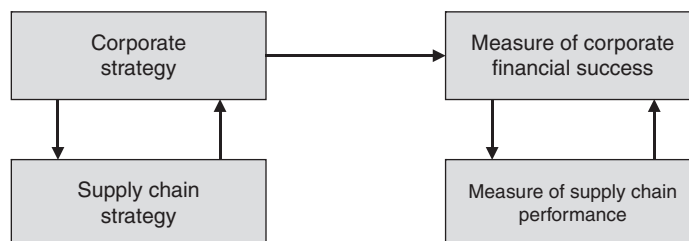


Figure 25.1 *Linking the Measures*

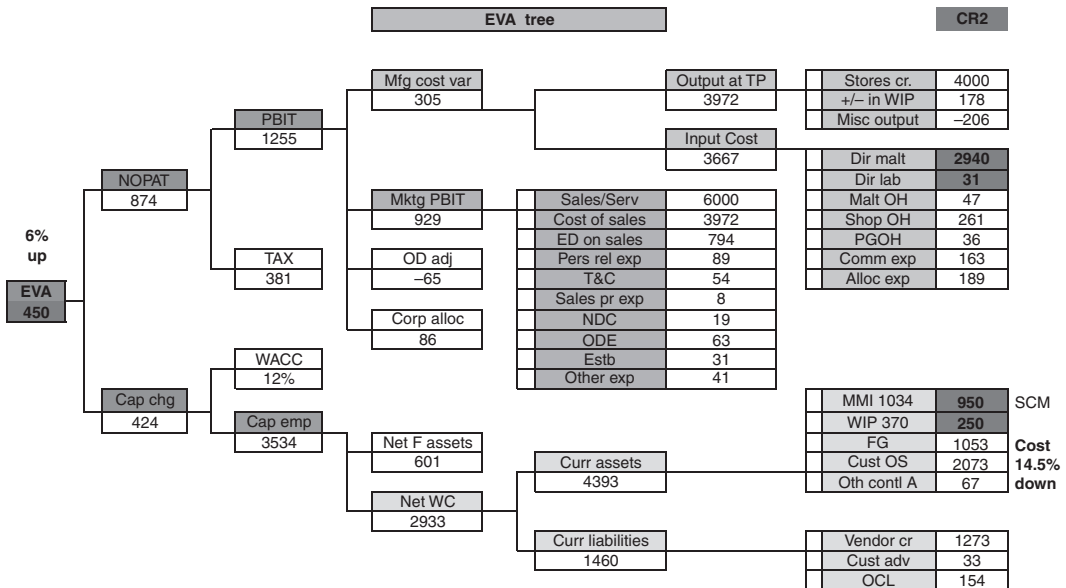


Figure 25.2 Effect of Supply Chain Performance on EVA

An EVA orientation forces a firm to define its supply chain broadly. Aberdeen group, for example, found that integrating the firm's product development efforts with a supplier's engineering department through e-design technology may reduce time-to-time cycles by 10–15 per cent. In addition, this real-time collaboration may contribute to cost reductions by minimizing re-design time and uncovering opportunities for standardization. Since response time to market and cost control are key factors in market success, effective supply chain management on the input end of the chain may help to boost a firm's revenue through larger market share while lowering costs. These are the two key elements in improving economic value added.

The action that can improve the profitability and value creation in the supply chain are generally under the firm's control. Better management of supply chain should have a major impact on revenue growth, cost reduction and asset turnover. The cost of capital is determined by outside forces and is essentially a given over which management has little control. Therefore, one of the keys to boosting EVA for the firms is to efficiently and effectively manage its supply chain (Figure 25.2).

SCM AND SCOR MODEL

While organizations may have existing process mapping techniques that could make the linkages, whereas the SCOR model provides an especially dynamic and flexible process for assessing supply chain effectiveness. The SCOR approach of standardizing process analysis works both within an organization and across multiple partners in a supply chain.

SCOR model can bring a measure of organization to the supply chain measurement process and provide the link to overall corporate performance. The model identifies the need for corporate level strategies, objectives and business plans as well as a starting point for identifying best practices

concepts and tools as well as selecting appropriate metrics. As such it helps to overcome the disconnect between the supply chain metrics and overall corporate performance.

While the SCOR model is a useful tool which demonstrates the direct connection between the SCM and overall corporate financial performance, the model itself does not make that connection. Instead, the SCOR model focuses on hierarchy of planning and business goals, with requirement of objectives, strategies and metrics at multiple levels in the business structure.

SCOR Card Performance Metrics

These are the strategic supply chain metrics. These metrics include performance attributes that have direct impact on customer (customer facing) and the firm (internal facing). SCOR emphasizes the need to continuously focus metrics on the requirements on (first two rows indicate revenue, next five rows indicate cost, the last three rows indicate assets) the next higher level in the process to ensure continuity and support for corporate competitive performance targets (Figure 25.3).

		Performance attributes				
		Customer facing			Internal facing	
	SCOR level 1 strategic supply chain metrics	Reliability	Responsiveness	Flexibility	Costs	Assets
REVENUE	Perfect order fulfillment	★				
	Order fulfillment cycle time		★			
COST	Upside supply-chain flexibility			★		
	Upside supply-chain adaptability			★		
	Downside supply-chain adaptability			★		
	SCM cost				★	
	Cost of goods sold				★	
ASSETS	Cash-to-cash cycle time					★
	Return on supply chain fixed assets					★
	Return on working capital					★

Figure 25.3 SCOR Card Performance Metrics

EVA Component

SCOR model level 1 supply chain objective is consistent with customer needs is 'perfect order fulfilment' (reliability on delivery performance). Examples of objective to support 'perfect order fulfilment' is 'on-time delivery in full' and 'order shipped to schedule'. High performance here can help the selling firm establish a competitive advantage leading to increase in revenue through capturing greater market share. This impacts on the revenue component of economic value added. The internal-facing level 1 metrics impact the cost and asset components of EVA.

Using economic value added as the measure of overall financial performance, the SCOR model process provides for linking the supply chain metrics to corporate financial goals. The focus is on the components of EVA typically under a firm's control—revenues, costs and assets.

Figure 25.4 shows the relationship between SCOR model metrics and the revenue components of EVA of a business firm. One strategy for increasing revenue is to improve customer satisfaction. The SCOR model provides the process to prioritize and select the most significant supply chain support strategies and performance indicators to achieve these goals.

Perfect order fulfilment is a key attribute in SCOR model to support EVA which has many components. Therefore, a decision must be made on the appropriate second level supporting metrics. A metrics is 'on-time, in full,' which is critical in industries operating in a JIT environment. The number of processes that support 'on-time, in full' (level 2) can prove to be significant. Continuing with decomposition process to level 3 to ensure that the efforts of those who impact level 1 perfect order fulfilments goals are properly focused. One of the level 3 performance indicators which are important is 'order shipped to schedule,' an important component of perfect order.

In addition to identifying the specific SCM measures, companies need to establish goals for each based on the desired level of performance. Goal setting at each of the levels in the SCOR model plays a major role in establishing the level of effort and creativity required to achieve the goal through continuous process improvement initiatives.

Hence, properly selected and linked performance measures will focus resources on contributing to overall corporate performance.

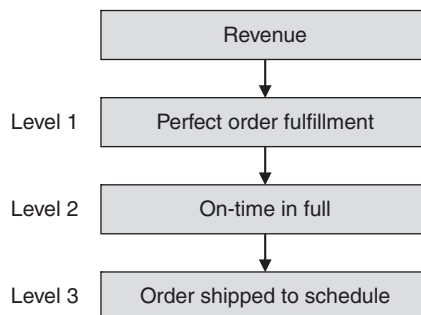


Figure 25.4 EVA Component: Revenue
Source: SCOR model

Cost Component

SCOR model can be linked to metrics in support of the cost component of the EVA (Figure 25.5). SCOR model identifies total SCM costs as the level 1 objective with supporting metrics in levels 2–4. ‘Total SCM Costs’ objective is the sum of the costs of a number of significant functional operations. Here, only three types of costs (supply management costs, acquisition costs, receiving costs) have been mentioned and supply management is used as the level 2 performance measure.

The model focuses the management’s attention on supply chain management to achieve the above objectives of reducing total SCM costs. The saving on transaction costs is a great deal of additional contribution to profit-boosting EVA.

The decomposition to levels 3 and 4 isolates supply management-related activities that are not included in the actual expenditure for goods and services. Here, product acquisition cost is shown as a level 3 metrics. Those costs may include salaries for all those involved in the purchasing process and other transaction costs. For example, one of those costs relates to the receiving activities, a component of acquisition costs. This is a level 4 metrics. There is a link between actions taken to reduce costs in receiving at level 4 to the effort to reduce total SCM costs at level 1.

Analysing each layer of the supply chain process for every function would produce a very complex network of linked process metrics. Hence, it is important to prioritize the opportunities. The goal is to focus on those factors that will most effectively support the improvement in the EVA cost elements.

Asset Component

A number of factors impact the level of assets a firm puts to use for delivering value to the market. To have a positive impact on EVA, the firm needs to minimize the asset levels used to deliver that value. Factors such as capital utilization, cash velocity, inventory turns, and cycle time reduction will impact how effectively the firm is managing its assets. The planning level of the SCOR model focuses on identifying a balance of supply chain resources necessary to meet supply chain requirements. One of the model’s key performance attributes is asset level. The goal here is to provide just the right amount of asset to meet requirements. This balance will improve the model’s ‘cash-to-cash cycle time’ metric of the model.

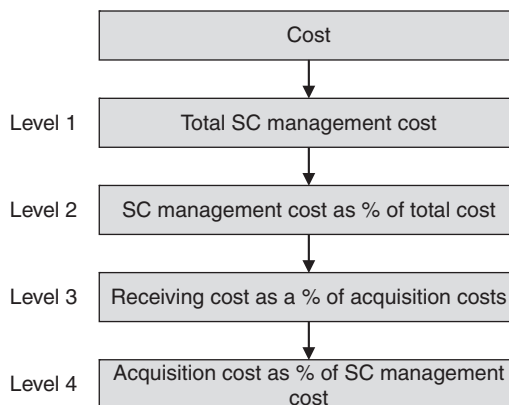


Figure 25.5 EVA Component: Costs

Source: SCOR Model

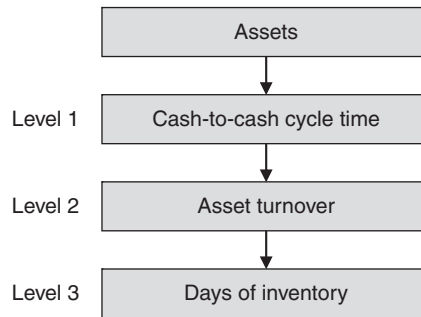


Figure 25.6 EVA Component: Assets
Source: SCOR Model

Figure 25.6 identifies cash-to-cash cycle time as a level 1 metric. This metric represents the average number of days that elapse between paying for raw materials from suppliers and receiving payments for final products from customers. Cash-to-cash cycle time is, in turn, directly impacted by asset turnover at level 2, which itself is impacted by the days of supply in inventory at level 3. Management intent on improving EVA through the asset component that can use the cash-to-cash cycle time metric to force attention on asset turnover by setting objective for the days of supply in inventory. A reduction in inventory should increase turnover at a consistent levels of sales. Ultimately, these actions are reflected in a reduction in cash-to-cash cycle time. In the end, the asset components of EVA are reduced assuming no increase in asset levels. Thus, as in the first two EVA components the asset component is clearly linked to the overall measure of corporate financial performance.

The benefits of the hierarchical mapping process described for the revenue, cost, and asset components of EVA are significant. The process helps to tie the goals and strategies at the operational level to measure the overall organizational performance that demonstrates the impact on shareholder value. The SCOR model provides a hierarchical framework. The overall performance measure of EVA provides the link to shareholder value. Hence, business firms must develop an understanding of how the day-to-day activities of managing the supply chain relate to corporate financial performance leading to healthy EVA.

SUMMARY

Supply chain management helps to provide higher returns to shareholders money. However, in general, a small percentage of companies use SCM to manage overall financial performance. The CFO must take a leadership role in making the financial-SCM connection. CFO should make this connection using a three step, top-down approach: first benchmarks key financial metrics and values gaps; second maps gaps to SCM business processes, activities, and KPI's; and third uses this information to explore and prioritize initiatives to improve SCM business processes. The valuation of gaps in key financial metrics should be extended to SCM-related information found in many business intelligence tools. These tools often provide detailed information at the activity and KPI level. Finally, improving most SCM professionals' financial understanding is a critical factor in making the financial-SCM connection. SCM professionals must understand how SCM business processes and strategies impact key financial metrics and contribute to returns to shareholders.

REVIEW QUESTIONS

1. Discuss various financial issues in managing supply chain.
2. What is EVA (economic value addition)? Explain how supply chain can improve EVA.
3. Explain how supply chain performance is keyed to EVA.
4. Explain through SCOR model the implications of asset component on EVA.

INTERNET EXERCISES

1. Visit http://www.valuebasedmanagement.net/methods_eva.html and find out how EVA is calculated and its application area.
2. Visit <http://www.accessmylibrary.com/article-1G1-133836607/evaluating-internal-operations-and.html> for evaluation of SC performance using EVA.
3. Business Management is information sources on European business information. Visit <http://www.bme.eu.com/article/The-value-of-supply-chain-management/> for information on financial issues in SC.

VIDEO LINKS

1. Activity-based costing <http://video.google.com/videoplay?docid=8333373775159231485#>
2. Contracts—pricing mechanisms <http://www.logisticsbureau.com.au/video-supply-chain-logistics.htm>

PROJECT ASSIGNMENT

1. Select a leading player in the automobile or FMCG or retail sector and map all the business process to understand the cause-and-effect relationships between SCM business activities and financial performance.

NOTES

1. Sople V. V. 'Financial Issues in Managing SC and Its Linkage to EVA'. Presented in IMRC 2008 Conference at IIMB on 22–24 December 2008.
2. Monczka, Robert, ISM Conference on E-procurement, Scottsdale, AZ, June 2000.
3. The Supply Chain Operations Reference Model (SCOR) is a process reference model that has been developed and endorsed by the Supply Chain Council as the cross-industry standard diagnostic tool for supply chain management. SCOR enables users to address, improve and communicate supply chain management practices within and between all interested parties.

PERFORMANCE MEASUREMENT AND CONTROLS

26

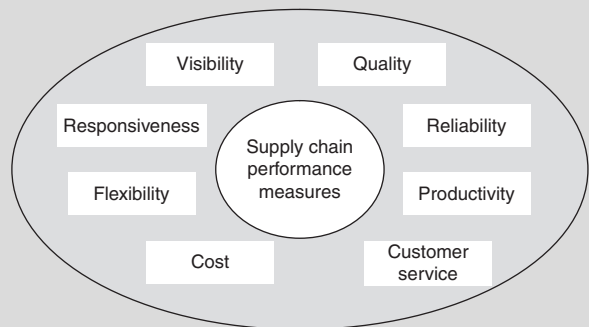
“If management is a science of measurement, adequate supply chain measures will be one of the keys to successful supply chain management

—Dr. Octavio Carranza”

Chapter Objectives

To understand:

- Concept of benchmarking
- Gap analysis
- Bench marking for best practices
- Performance measurement criteria in supply chain
- Performance levels and controls
- Balance scorecard



Supply Chain Performance Measures

Supply chain performance measurement process keeps track of the health of the supply chain system. For total integration and coordination of supply chain operation, a well-designed performance measurement and control system is a prerequisite. The performance measures have two broad perspectives: internal, which includes financial and non-financial perspectives and the external, which includes customer service and innovation perspectives. The measures normally followed are productivity, operating cost, customer service, delivery performance, asset utilization, etc. Audit is used as a tool for measuring capability and quality of operation of a firm. Irrespective of the size and complexity of the supply chain operations, all business firms need performance measurement and controllership for the efficient and effective back-end processes.

C-Auto Tyres: Performance Measures for Supply Chain Health

C-Auto Tyres management, together with a leading supply chain consultant undertook a complete redesign of their supply chain. The firm placed the supply chain division as the link between manufacturing and marketing to ensure seamless effort towards meeting the objectives of the organization. They ensured that there is continuous flow of information internally. IT is extensively used to ensure that information flow is always available. C-Auto Tyres has heavily invested in information technology towards this end.

C-Auto Tyres wants to ensure that right quality product is available at the right place at the right time for its customer. In the tyre industry, product quality is a poor differentiator; C-Auto Tyres wants the customer to make his choice based on the most effective service, who gives him what he wants when he needs it. Dealers too, will stock a lesser quantity of the product if they are assured that they will receive goods as and when they require them. The dealer, in this sense, is the customer who, in turn, services the end customer.

C-Auto Tyres designed a performance measurement system for dealers that if 100 units were sold in a month, the minimum stock to be held would be 4 units a day. If, at any time in any of the selling locations, the stock falls below 4 units a day, that day would be captured as a sale lost. The firm kept the measurement parameters of sales loss to reach 5 per cent. For finished goods inventory the performance level placed at a desired level of 25 days.

Firms introduced a very simple measurement tool—CLIP (committed line item production). The CLIP score was kept as 95 per cent. A monitoring system also has the advantage of making the transporter more accountable, particularly at the time of negotiating a yearly contract. The other performance measure parameters were: stock at < 25 days, factory compliance to >75%, despatch time < 24 hours, transit delay < 5 per cent, sales skew < 42 per cent WK 4. In the end count, organizations must realize that what they measure is what they will get. Simple measurement tools can be worked out for various operational activities, which will clearly indicate where you are and where you want to move. At C-Auto, reports are generated on the health of the supply chain on the basis of a weekly CLIP measurement.

INTRODUCTION

The fundamental performance measure of a supply chain operation is to produce products to match customers' demand cycle to deliver maximum value to the customers. The competitive environment calls for speedy, cost efficient, accurate and reliable supply chains. Hence, a number of technologies and managerial attention has gone into improving supply chain performance. Supply chain management, today, is a strategic issue calling attention of top-level management. The supply chain is a source of competitive advantage and creation of customer value. Hence, the measurements around the supply chain performance are critical of importance to the organization to keep its health intact. To measure the performance, organizations have to develop key performance indicators (KPI) to track progress and to achieve desired goals in supply chain. The control of the supply chain system is not possible through one measure; it requires both financial and non-financial measures to direct the operations for efficiency and effectiveness.

The system performance is judged by its output with respect to the inputs. The higher output to input ratio is a measure of the system's efficiency. However, the above ratio is also to be reviewed with respect to the time period. In today's competitive environment, industries are being forced to focus on the effective and efficient use of the scarce resources. For that, the companies have to deploy lot of resources for movement of goods and information across their supply chain. Hence, it becomes necessary to monitor, control and improve the performance of the supply chain operation so as to attain the set goals of cost reduction and customer satisfaction. The performance measurement system ultimately helps in either reducing or eliminating the non-value added activities from the system. In today's competitive business scenario, the importance of accurately measuring and controlling the supply chain performance cannot be denied as the level of competency has become more critical in enhancing the competitiveness of the business process.

PERFORMANCE MEASUREMENT SYSTEM

Measurement is the first step that leads to control and eventually to improvement. If the firm cannot measure performance, it can not be controlled improved upon. The performance measurement system shall have the following objectives in general.

Decisions on Metrics

The first objective in designing a performance measurement system is to decide upon the parameters, which are to be continuously or periodically reviewed and controlled. With the known parameters for performance monitoring, the benchmarking task becomes easier.

Monitoring

With a monitoring system in place, the reporting on the current status of the operations can be made available to decision-makers. For example, the monitoring system in marketing will report on the total orders booked, orders cancelled and the orders completed (sales effected). The information may be monitored on a daily, weekly or monthly basis depending on the volumes and the criticality. In case of supply chain, the monitoring system reports on the cost breakup across the various activities like procurement, manufacturing, warehousing, transportation, packaging, material handling, etc. It also indicates the service level parameters such as delivery status, transit damages, wrong dispatches, etc.

Controlling

The controlling metrics compares the actual performance with the set standard or objectives. It reports the deviations from the set goals. For example, the control system will report on the current order fill rate in warehouse as 75 per cent and will indicate the deviation of 25 per cent with the set standard of 100 per cent, so that corrective action can be initiated for inventory replenishment. Similarly, the percentage increase product damages during handling, will prompt the management to change the logistical packaging or go in for mechanization in product handling as against the manual one, which is being currently practised. The decrease in warehouse productivity, shall help in identifying the cause and help to improve on the storage layout, material handling methods and the product unitization, etc.

Directing

The major objective of the performance measurement system is to motivate individuals in the system to enhance individual performance resulting into improvement in overall system performance. The performance measure system will help management to evolve the incentive scheme for the operating people who have crossed the targeted productivity level. For example, the workforce engaged in dispatches may be motivated through rewards for crossing the targeted tonnage of goods dispatched in the assigned time frame or reducing wrong dispatches (due to errors in dispatch documents) in the current year as compared to the last year.

The metrics that are used in performance measurement and improvement should be those that truly capture the essence of organizational performance. A measurement system should facilitate the assignment of metrics to where they would be most appropriate. For effective performance measurement and improvement, measurement goals must represent organizational goals and metrics selected should reflect a balance between financial and non-financial measures that can be related to strategic, tactical and operational levels of decision-making and control. As firms begin to define the metrics, a proper documentation of measurement is essential. The definition for the metric should be agreed upon by senior management. The agreed upon definition should be published in an easy to access method. The definition should spell out:

- Purpose of the measurement
- Measurement parameters
- Mathematical techniques for measurement

The performance metrics defined should be validated. The importance of validation cannot be over emphasized. Validation is a critical step in creating metrics. The performance measurement metrics should be measurable (quantifiable), attainable, realistic and time bound.

WHY BENCHMARKING?

Benchmarking is one of the most powerful tools a business that can be used to manage and measure profitability across the supply chain. Properly applied within the appropriate parameters, benchmarking can enhance firm's competitiveness, increase shareholder value and provide the business with a long-term strategic focus. It can also identify opportunities for improving key components in the supply chain and optimize the allocation of resources.

In today's business world, supply chain management has a crucial role to play. It is a source of gaining competitive advantage. It creates the right environment to serve the customers better by providing material, information and on time services. To attain competitiveness, there is a need to develop a model against which company could benchmark their own supply chain performance parameters. With benchmarking, the firm has to achieve newer heights and quality in the supply chain management. Benchmarking is defined as follows:

Benchmarking is the practice of being humble enough to admit that someone else is better at something, and being wise enough to learn how to match them and even surpass them at it.¹

There are three processes in benchmarking: comparison (of performance levels, processes and practices), learning (from the benchmarking partners to introduce improvements in your own organization) and improvement (which is the major objective of any benchmarking study).

Benchmarking is a process used in management and particularly strategic management, in which businesses use industry leaders as a model in developing their business practices. This involves determining where you need to improve, finding an organization that is exceptional in this area, then studying the company and applying its best practices in your firm. Benchmarking systematically studies the best firms amongst the industry, then uses their best practices as the standard of comparison, a standard to meet or even surpass.

Benchmarking is a process of setting of level of quality practice and do all that is required to achieve it. It is an ongoing process. It involves cross-firm, cross-industry or cross-regional benchmarking. It is a process for a firm to continuously search for best practices in the best firms around the world. The goal is to become an exceptional company in terms of function or task of firm's business through benchmarking. The benchmarking may be done from all the functional areas such as production, marketing, purchasing, information technology to customer service.

The benchmarking can be categorized into 'best practices benchmarking', 'process benchmarking' or 'competitive benchmarking'. The 'competitive benchmarking' needs competitor analysis. It should choose the best amongst the competitors firms for benchmarking the practices. Benchmarking is conducted in a separate project whose individual objective is to improve one of the organization's business processes. There are a number of models describing the different steps that constitute a benchmarking study. One such model is the so-called benchmarking wheel (Andersen, 1995). The procedure adopted is:

- Identify problem areas
- Identify organizations for best practices
- Study their best practices
- Implement the best
- Repeat

Benchmarking is an expensive process, but most companies find that it more than pays for itself. The cost-benefit analysis will prove its worth after gaining competitiveness which will help in business growth. In benchmarking studies there are three main types of costs: visit costs, time costs and benchmarking database costs. With benchmarking project organization can create and maintain a database of best practices and the companies associated with each of the best practices.

Benchmarking in supply chain means deciding firm's performance indicators, which need to be benchmarked against the best ones of other companies. In the benchmarking process, the firm uses a performance scorecard which helps the firm to understand where the favourable and unfavourable gaps exist in the current operations. This information will help the firm to determine which improvement initiatives will yield the most strategic value to the company and which will provide the firm with the highest return on the investment.

BENCHMARKING PROCESS

In most companies, two major and very interdependent issues are commonly addressed. The first one is delivering products with customer-acceptable quality, with very short lead times, at a customer-acceptable cost and the second is keeping inventories throughout the supply chain at a minimum level.

Benchmarking is a process used in management and it is strategic in nature. In benchmarking process, businesses use industry leaders as a role model in developing their business practices. This process involves the following four things:

- Determining the area where the firm needs to improve
- Finding an organization that is exceptional in this area
- Studying that organization for the best practices
- Applying its best practices in the firm

Benchmarking exercise studies the best firms and their best practices. It is then followed by imitating the best practices as the standard of comparison. Benchmarking project is the process which is based on the following beliefs. First progressive firms must seek to continuously improve all facets of their operations. Therefore, their attitude should be one of fixing or improving a work method before it breaks. Second, best practices should be identified and studied which typically means searching outside one's own enterprise.

Supply chain performance is a key factor in a company's competitiveness. The leading companies around the world, in general, benchmark most of their supply chain processes for gaining competitiveness through productivity, cost leadership and system efficiency. By adopting the Internet tools the process of benchmarking can be expedited as data collection and analysis can be done with speed. In supply chain benchmarking, the following components of needs are to be examined with respect to the performance of the leading companies.

- On-time delivery, fill rate, lead times, etc.
- Supply chain costs and cost saving initiatives
- Demand/supply chain initiatives
- Relationship between vendors and customers

With benchmarking studies, organizations can better understand how to measure performance, close gaps and select productivity initiatives to maximize effectiveness and improve the corporate bottom line with respect to the leaders.

Strategic and operational are two components of benchmarking. As the name suggests, the strategic benchmarking is concerned with basic strategic issues like direction of the company, its global involvement, its structure in terms of centralized or decentralized and core competencies. Strategic benchmarking requires a highly structured and focused effort. Strategic benchmarking is more focused on internal aspects rather than external competition. Operational benchmarking quite evidently focuses on the operations, that is, the activities and the processes. This can be further divided into four types. They are as follows:

Working task benchmarking This covers the single works performed in logistics like palletization. As per this, activities like loading, scheduling pick-ups, etc. are the main areas of concentration.

Function wide benchmarking As the name suggests this involves analysing the function as a whole and not its components. Thus, if warehousing is benchmarked then all tasks under warehousing are analysed and benchmarked.

Management process benchmarking In this type of benchmarking management aspects like quality, rewards system, etc. are earmarked for benchmarking. This is more complex as it is cross-functional and process-oriented.

Total operation benchmarking In this type, the entire logistics operations or the supply chain on the whole is benchmarked.

STEPS IN SUPPLY CHAIN BENCHMARKING

A flow depicting the stages of benchmarking is as follows:

Identify benchmark potential Firms should look for the very best practices in the industry. It should further conduct a research amongst customers, suppliers, financial analysts, trade associations, and magazines to determine which companies are worthy of study.

Target and study focus areas Once the benchmark has been decided, the next priority is to evaluate and analyse in detail. Supply chain consists of many components like suppliers, distributors, etc. so it is necessary to analyse only those components which are the weak links.

Form benchmarking team The team should be consisting of two to five (max) employees closest to the issue. The team should be given responsibility with power to plan, organize and implement the entire project.

Develop measurement metrics Performance metrics will enable the team to perform gap analyses and identify areas needing improvement. The following are the key metrics for supply chain management:

- **Delivery performance:** On-time delivery, delivery consistency, fill rate, and return rates.
- **Cycle time:** Order fulfilment cycle, replenishment cycle, manufacturing cycle and procurement cycle
- **Inventory:** Inventory days of supply, inventory turns.
- **Cash management:** Day's sales outstanding, day's payables outstanding and cash-to-cash conversion.
- **Supply chain cost:** Order management costs, inventory carrying costs, supply chain finance and planning costs, supply chain IT costs, procurement cost.

Design benchmark approach The approach will have to analyse what are the components which need to be studied. It should be either strategic or operational.

Study and evaluate target areas It can be done either through direct study wherein the people who are involved in the benchmarked supply chain help you to analyse their strengths and weaknesses. Another approach is to study the supply chain process of the company independently or maybe with the help of a consultant.

Adapt and implement findings The responsibility should be delegated to benchmarking team members with due authority to implement the benchmarking project. The team should be necessarily cross-functional. The measurable goals need to be set with specified time frame.

PERFORMANCE DIMENSIONS AND MEASURES

The key dimensions of supply chain performance can be defined in terms of cost/financial, quality, time, productivity, flexibility, reliability and customer service which are described in Exhibit 26.1.

Exhibit 26.1 *Performance Dimensions and their Scope*

<i>Dimension</i>	<i>Description</i>	<i>Scope</i>
Cost	Shows effectiveness of cost control	Material cost, inventory cost, manufacturing cost, inbound logistics cost, outbound logistics cost, power/fuel cost, rework/rejection cost, labour cost, information carrying cost, and demand/supply planning cost
Quality	Conforms to the specified standard	Accuracy of forecast, supply planning, capacity planning, on time delivery, product quality, service quality, number of returns/rejection rate
Time	Measure how fast an activity is completed	Covers judging the performance from time dimension are cycle time, down time, supplier response time, lead time, number of days of inventory, and on time delivery
Productivity	Ability of a supply chain to use the assets as profitably as possible	Return on investment, capacity utilization, work in process inventory, value added, wastage/scrap/rework, output, efficiency, finished goods inventory, raw material inventory
Flexibility	Usage of internal and external business changes	Measures such as number of channels, supply sources, product variety, time to adapt to change in demand, raw material availability, source flexibility, upside production flexibility, and delivery flexibility
Reliability	Ability of a system to perform under stated conditions for a specified period of time	Number of plans that meet schedules, on time delivery, incoming material quality, percent of defect items, responsiveness to urgent deliveries, order fill rate, service reliability, number of returns/rejection rate, in process failure rate, degree of information sharing with partners, and stock out rates
Customer service	Ability to satisfy the customer and meet the expectations	Customer satisfaction index, on time delivery, fill rate, customer complaints, stock out rate, service reliability, reject rate/number of returns, percent defect items, product quality, service quality and customer retention

There are two levels of performance measures: First one is internal measure and the second is the external measure (Figure 26.1). As per Vijay Govindarajan and Robert Anthony,² firms must strike a balance between the external measures such as customer satisfaction and the internal measure such as productivity, cost, quality, and asset management because the companies normally ignore the external results in the belief that the internal measure are enough or sacrifice internal development for external results.

The performance measures must be evaluated through the various perspectives before they are evolved and implemented. In general, the performance measure may be viewed through the four perspectives such as financial, non-financial, internal and external perspectives.

Financial Measures

There are several fixed and operational costs associated with a supply chain. Ultimately, the aim is to maximize the revenue by keeping the supply chain costs low. Costs arise due to inventories, transportation, facilities, operations, technology, materials and labour. The financial performance of a supply chain can be evaluated by looking into the following items:

- Cost of raw material
- Revenue from goods sold
- Activity-based costs (material handling, manufacturing, assembling)
- Inventory holding costs
- Transportation costs
- Cost of expired perishable goods
- Penalties for incorrectly filled or late orders delivered to customers
- Credits for incorrectly filled or late deliveries from suppliers
- Cost of goods returned by customers
- Credits for goods returned to suppliers

The operating cost is the most important performance measure, which reflects the efficiency and the effectiveness of the supply chain system. The operating cost element covers warehousing cost, freight, material handling equipment running and maintenance cost, labour cost, cost of return goods, inventory carrying cost. The cost is measured in terms of percentage of total sales or per unit volume or per unit weight of the inventory moved through the system.

Return on investment (ROI) has both financial and non-financial perspective. This is an important financial performance measure to indicate whether the investment made in logistical assets such as warehouse, equipment, storage system, transportation vehicles is paying you the dividends by way of the profit enhancement through cost reductions. The numerator in the ratio is the income and the denominator is the investment. It also indicates whether the investment is properly utilized so that it generates the revenue as desired.

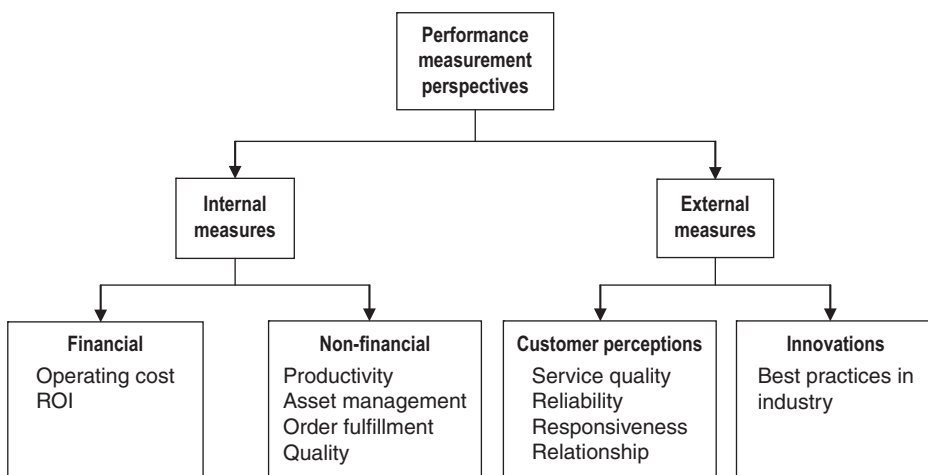


Figure 26.1 *Performance Measurement Perspectives*

Non-financial Measures

The non-financial measures include productivity, asset utilization, order fill rate, quality and customer perception.

The productivity of supply chain system is judged by its output with respect to the input into the system. It is an important performance measure of finding the efficiency of the system. For example, in the organizational context, the incremental increase in head count in the marketing team should bring additional sales revenue as per the norms set by the firm. A firm may decide on productivity norms (based on past experience), so that incremental sales generation per sales person will justify the addition of new recruits to the marketing department. In logistics, the productivity performance measures are:

- Units loaded per person per hour on to the transport vehicles.
- Cases filled and packed per person per 8 hour shifts.
- Idle time of the handling equipment.
- Through put in the warehouse per hour.

The productivity measures are sometimes difficult to compute as the outputs are hard to measure and inputs utilization is difficult to spread over and match the stipulated time frame of output.

Asset utilization means the utilization of the capital invested in the assets such as warehouse building, storage system, and material handling equipment and also funds tied up in the inventories. The utility of the investment is measured through inventory turn over ratio, returns on investment, and inventory stock levels in number of days.

Inventory turn over ratio indicates the rotation of the given value of inventory with respect to the value of sales in a set time frame. Normally, the time frame is 1 year. The higher turn over ratio indicates the faster cash rotation in the business cycle and higher utilization of the assets. This ratio has both financial and non-financial perspective.

Order fulfilment is one of the important criteria to judge the level of the customer service of the firm. This is the relative ability of the firm to satisfy the customer. This requires a close coordination of all the functional areas of the organization. The various sub measures covered under the order fulfilment are:

- Cycle time (order processing, replenishment, procurement and manufacturing, distribution)
- Delivery (on-time/delayed)
- Fill rate (order fill, case fill, product fill)
- Stock out frequency
- Shipping errors (wrong delivery, incorrect invoice, material shortage)

For the firm to enhance its order fulfilment capability, a close coordination of all the arms of logistics is a must. The system needs a real-time information support and exceptional report generating and alarming capability for proactively initiating the corrective steps.

The perception of the supply chain process quality is created through 'near-to-the-perfection' in order fulfilment process. This may be interpreted as error-free order delivery. The impression of quality logistics service is created through initiating and controlling the following performance measures:

- Transit damage frequency
- Value of the total damage
- Frequency and cost of goods returned from customers (damaged/sub quality)

- Material shortages frequency
- Deliver commitment deviations—frequency and tolerance

Quality measure is basically evaluation of the entire supply chain process rather than individual activities involved in supply chain operation. It speaks of the degree of effectiveness of the entire order fulfilment process to deliver error-free service.

Inventory velocity is speed of inventory movement through the supply chain. It is also measure of the efficiency and cost effectiveness of the supply chain operations.

The most important external supply chain performance measure is customer perception. For initiating the steps to achieve competitiveness in supply chain operations, the firm needs to take a regular feedback from the customers on the existing service levels and its gaps. Customer feedback may give the firm a comparative analysis of the service levels and value-added service offerings of the competitors operating in the markets. The feedback on delivery performance, reliability, responsiveness, and the relationship initiatives and practices of the firm vis-à-vis competitors needs to be regularly obtained towards the updates on external performance measures and controls thereafter. The firm should regularly conduct or sponsor customer surveys to get such measures for mapping the customer perceptions for improvement in the logistical services and enhancing competitiveness. These surveys may be conducted or organized through firms own marketing and service team, or through channel members or through consultants.

For calibrating the supply chain performance measures, firms go in for innovative techniques like benchmarking with excellent supply chain practices in the industry. The benchmarking may be with competitors, firms in related industry or firms in non-related industry. The benchmarking performance measures help the firms to gain competitive edge. For example, Xerox Copr, USA benchmarked I. L. Beans Corp for distribution procedure for improving on its lead time in exports sales.

The benchmarking may be done based on the published information, alliances with firm to be benchmarked or appointing a consulting firm/expert depending on the criticality of the results.

Federal express has point system for measuring results. Every day points are accumulated and conspicuously displayed (Exhibit 26.2) to front line workers so they can understand system performance and take measures for improvement in performance.

Exhibit 26.2 *Key Performance Indicators (KPIs) as benchmarked*

No.	Attribute	Typical	Benchmarked
1	Asset utilization	68%	80%
2	Delivery performance	60%	90%
3	Fill rate	72%	98%
4	Inventory in stocks	25 days	15 days
5	Inventory turns	9	18
6	Logistics cost to sales	8.5%	3.5%
7	Order fulfilment	65%	95%
8	Production flexibility	35 days	15 days

SUPPLY CHAIN OPERATIONS REFERENCE (SCOR) MODEL

The supply chain operations reference model also known as SCOR model, helps manufacturers to build best-of-class operations. It is a cross-industry standard diagnostic tool for supply-chain management. The SCOR model is a process reference model endorsed by the Supply Chain Council. It enables users to address, improve and communicate supply-chain practices within and between all interested parties. It is a management tool spanning from the supplier's supplier to the customer's customer. It is a toolkit that can be used to define measure and manage supply chain processes. The SCOR model is a pyramid of four levels (Exhibit 26.3) that represents the path a company takes on the road to supply chain improvement.

It is a method of benchmarking and measuring improvements in supply chain performance. SCOR model contains standard process definitions, standard terminology, standard metrics, supply-chain best practices, and enabling information technology. The SCOR model defines common supply chain management process, and matches them against 'best practices'. SCOR model was designed to enable companies to communicate, compare and learn from competitors and companies both within and outside of their industry.

SCOR includes all customer interactions from order entry through paid invoice, all product transactions (whether physical or service) and all market interactions from understanding demand to fulfilling it at each individual order level.

The SCOR model is a process reference model that expands to analyse processes involving cross-functional activity. For instance, the plan process involves sales, marketing, manufacturing, finance, logistics and others. It draws attention to process gaps rather pointing to specific departments' performance which can help the company communicate without ambiguity and help measure, manage and refine processes.

Metrics can include a wide variety of performance measures. The commonly used parameters to measure the performance of a supply chain are as follows:

- Cycle times
- Fill rate

Exhibit 26.3 *SCOR Model—Pyramid of Four Levels*

<i>Steps</i>	<i>Road Map</i>
Level 1	It provides a broad definition of the plan, source, make and deliver process types and is the point at which a company establishes its supply chain competitive objectives.
Level 2	It defines the 17 core process categories that are possible components of a supply chain. A company can configure both its actual and ideal supply chain by selecting from these core processes.
Level 3	It provides a company with the information it needs to successfully plan and set goals for its supply chain improvements. Planning elements include process definitions, target benchmarks, best practices and system software capabilities to enable best practices.
Level 4	It focuses on implementation, when companies put specific supply chain improvements into play. It defines practices to achieve competitive advantage and to adapt to changing business conditions.

- Inventory turnover
- On-time shipping and delivery
- Perfect order
- Transportation measurements

However, the limitations of SCOR Model are that it excludes sales and marketing, research and technology development, product development and some elements of post-delivery customer support.

PERFORMANCE MEASUREMENT AND CONTROLS

The supply chain performance metrics are developed on various parameters. The key parameters amongst them are:

Cycle Time

There are four cycles involved in supply chain process. These are:

Order fulfilment cycle It is the time period required to supply the product after it is ordered. In case of retail chain, the order cycle time is zero with regards to the customer. Customer will pick up the product from retail shelf if it is available there. For other products, it is the time till the product is in customers hand after he places an order.

Inventory replenishment cycle time It is the time taken to replenish the product at retail store or dealer showroom after it is delivered to the customer. The replenishment is done from company's depot or regional distribution centre or from manufacturing plant depending on the physical distance between source and the place of delivery.

Manufacturing cycle time It is measured from the time the firm planned order until the final production is done. It is time taken for converting raw material and components into finished product in usable form.

Procurement cycle time It is the time required for getting the raw material and components after it is ordered to vendors. For imported material the procurement cycle time is longer.

Cash-to-cash cycle time The number of days between paying for raw materials and getting paid for the product. It is calculated by inventory days of supply plus days of sales outstanding minus average payment period for material.

Fill Rates

The fill rate indicates the availability of inventory. It is a measure of shipping performance expressed as a percentage. The following are the variants in fill rate measurements:

- **Line fill rate:** This means amount of order lines shipped on the initial shipment versus the amount of lines ordered. This measure may or may not take into consideration as per the requested delivery date. For example, a customer orders 5 products one each from 5 product lines. The manufacturer dispatches 4 products from 4 product lines. In this case, the line fill rate is 80 per cent.

- **SKU fill rate:** This represents the number of SKUs (stock keeping units) ordered and shipped as against the ordered. For example, manufacturer has 100 SKUs. The customer ordered for 80 SKUs. The manufacturer could have only shipped 60 SKUs in full quantity. The SKU fill rate in this case shall be 75 per cent.
- **Case fill rate:** This represents the amount of cases shipped on the initial shipment versus the amount of cases ordered. For example, a customer orders 5 products that total 100 cases. The manufacturer ships out 85 cases within 24 hours of the order. The fill rate for this purchase order is 85 per cent.

INVENTORY TURNOVER

Inventory turnover indicates the number of times that a company's inventory turns over per year. A frequently used method is to divide the annual cost of sales by the average inventory level. The results vary by industry, a typical manufacturing company may have 6 inventory turns per year. High volume/low margin companies (like grocery stores) may have 12 inventory turns per year or more.

Inventory turnover measures the number of times inventory is sold or used over in a period such as a year. It is equal to the cost of goods sold divided by the average inventory. The low turnover rate may point to overstocking, obsolescence, or deficiencies in the product line or marketing effort. A high turnover rate may indicate inadequate inventory levels, which may lead to a loss in business. An item whose inventory is sold (turns over) once a year has higher holding cost than one that turns over twice, or three times, or more in that time. Stock turnover also indicates the briskness of the business. The purpose of increasing inventory turns is to reduce inventory for three reasons. Increasing inventory turns reduces holding cost. The organization spends less money on rent, utilities, insurance, theft and other costs of maintaining a stock of good to be sold. Reducing holding cost increases net income and profitability as long as the revenue from selling the item remains constant. Inventory with higher turn over helps to increase responsiveness to change requirements while allowing the replacement of obsolete items. Inventory turns of a few leading global corporations are indicated in Exhibit 26.4.

Exhibit 26.4 *Inventory Turns of Leading Global Corporations*

<i>Company</i>	<i>Inventory Turns</i>
Wall-Mart	7.5
Pantaloon Retails	3.9
Mark and Spencer	20.8
McDonalds	96.5
Apple Computers	50.8
Toyota Motors	10.1
Nokia	17.9
Pepsi	8.2
IBM	18.9

Source: AMR Research Inc 2007, <http://www.supplychainbrain.com>

Ontime Shipping and Delivery

It is a calculation of the number of orders shipped on or before the agreed/requested date of delivery.

Other Order Measurement Metrics

The other order measures calculate the error-free rate of each stage of a purchase order fulfilment. This measure captures every step in the life of an order. It measures the errors in the order line. These are generally calculated as shown in Exhibit 26.5.

Defects Per Million Opportunities (DPMO)

DPMO is a six sigma calculation used to indicate the amount of defects in a process per one million opportunities. DPMO is sometimes used instead of defect per unit to allow for comparison between processes with different levels of complexity.

Transportation Measurements

Freight cost per unit shipped This is calculated by dividing total freight costs by number of units shipped per period.

Transit time It is measured in terms of hours/days from the time a shipment leaves the facility to the time it arrives at the customer's place. It is also called as lead time.

Claims/Damages It is calculated by dividing total loss and damage claims by total freight costs. A high number, generally indicates packaging problems or material handling process.

Truckload capacity utilized It is calculated by dividing the total tonnage of goods shipped in a truck and total theoretical shipping capacity of a truck.

Truck turnaround time This is calculated by measuring the average time elapsed between a truck's arrival at the company's facility and its departure. This ratio is important for the companies having their own transportation fleet. The companies like Pepsi or Coca Cola having large fleet of delivery vehicles which carry filled bottles to deliver to retailers on the particular route and collecting the empty bottles. The vehicle turn around time is critical for the company towards asset utilization.

On-time pickups It is calculated by dividing the number of pick-ups made on-time (by the freight carrier) by the total number of shipments in a period. This also indicates the reliability of the transportation mode.

Exhibit 26.5 Other Order Measurement Metrics

<i>Measure</i>	<i>Unit</i>
Order entry accuracy	99.5%
Warehouse pick accuracy	96.0%
Error-free invoicing	98.9%
Delivered damage free	99.6%
Overall order measure	98.5%

SUMMARY

Managers today want to measure the performance of the supply chain and the results of improvement efforts across supplier, company and customer operations. Supply chain measurements or metrics such as inventory turns, cycle time, DPMO and fill rate are used to track supply chain performance. The commonly used SCM metrics can help you to understand how your company is operating over a given period of time. Supply chain measurements can cover many areas including procurement, production, distribution, warehousing, inventory, transportation and customer service. However, good performance in only one part of the supply chain is not sufficient. The solution is to focus on the key metrics in each area of supply chain.

Measurement is the first step that leads to control and eventually to improvement. If you cannot measure something, you cannot control it and if you cannot control it, you cannot improve it. However, a well-designed performance measurement system supported by information will make cross-functional coordination and integration easier.

REVIEW QUESTIONS

1. 'Performance measurement system tracks the health of supply chain of a company'. Explain.
2. Discuss the various performance measurement metrics with regards to supply chain.
3. Explain what is benchmarking and its role in achieving supply chain competency.
4. In an organization the supply chain 'performance measurement system' will fail due to lack of 'information system'. Explain.

INTERNET EXERCISES

1. Visit <http://www.supplychainmetric.com/> to study further on measurement metrics used in improving the supply chain performance.
2. For studying the articles on SC performance measurement visit http://www.iimm.org/knowledge_bank/1_performance-measures-scm.htm
3. The Balanced Scorecard Institute, a strategy management group company, helps organizations succeed through improved strategic focus and performance. Visit <http://www.balancedscorecard.org/BSCResources/PerformanceMeasurement/tabid/59/Default.aspx> to study how the balance score card system works.
4. To get insights in SC performance measurement metrics, visit http://www.scmr.com/article/330620-True_Measures_of_Supply_Chain_Performance.php

VIDEO LINKS

1. Measuring Supply Chain Performance. Supply chains are tasked with being effective, efficient, and adaptable. All the companies have supply chains, it is only via performance measurement that managers can truly prove the strength of their supply chain, <http://www.youtube.com/watch?v=R4HPYYR5iLw&feature=related>

2. Benchmarking Your Supply Chain, <http://www.logisticsbureau.com.au/video-supply-chain-logistics.htm>
3. Supply Chain Key Performance Indicators—KPIs, <http://www.logisticsbureau.com.au/video-supply-chain-logistics.htm>

PROJECT ASSIGNMENT

1. Supply chain management, metrics can help firms to understand how company is operating over a given period of time. Find out the key metrics of different areas in supply chain in the following industry sectors:
 - a. Automobile
 - b. Food products
 - c. Hospitality

Visit <http://www.supplychainmetric.com> to get more details.

NOTES

1. *American Productivity and Quality Center: Basics of Benchmarking*, APQC 1993, Houston, Texas.
2. Robert Anthony and Vijay Govindrajana (2000). *Management Control Systems*, Tata McGraw-Hill, p. 259.

PART-VI

SUSTAINABILITY PROCESSES AND PRACTICES

27 Sustainability Practices and ‘Greening’ of SC

This chapter begins with the concept and practice of corporate social responsibility. It further discusses the role of green supply chain in corporate social responsibility. The drivers of green supply chain along with prevailing green practices in procurement, manufacturing for firm’s journey towards sustainability are also covered.

28 Regulatory Compliance

This chapter covers impact of regulatory environment on SC operations. The various legislation on multimodal transportation, goods and service tax, warehousing, VAT and CENVAT, contract manufacturing and transfer pricing, regulations on dangerous goods, TIR carnet and INCOTERMS are discussed here.

29 Ethical Orientation

This chapter starts with concept and practice of business ethics. The ethical practices in purchase, manufacturing and distribution are further covered in this chapter.

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SUSTAINABILITY PRACTICES AND 'GREENING' OF SC

27

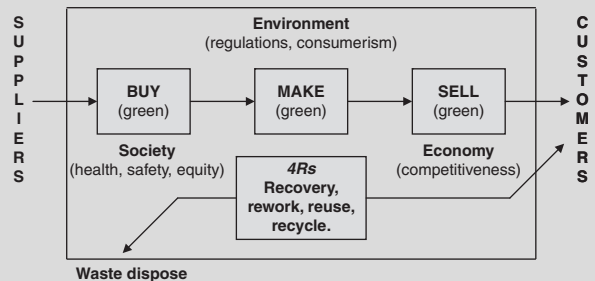
“Fortune favors the prepared mind

—Louis Pasteur”

Chapter Objectives

To understand:

- Corporate social responsibility (CSR)
- Supply chain role in CSR implementation
- Green supply chain and its drivers
- Green procurement, manufacturing and marketing practices
- Greening strategies



Green Supply Chain Model

Role of CSR in supply chain is now well recognized in the industry. Many CSR issues are concerned with social and environmental areas. These issues can be dealt with by the purchaser using his purchasing power to demand an environmental performance from the supplier in the supply chain. 'Greening supply chain' refers to large business corporations' requests to their vendors to practice environmental responsibility in their business operations covering procurement, manufacturing, marketing and distribution. It is a powerful tool in the hands of large corporate buyers to influence their suppliers to adopt socially responsible practices. 'Going green' is about more than just reducing packaging or your carbon footprint. It involves making environmentally wise choices in supply chain design and execution, including managing the reverse supply chain to efficiently manage and remove waste.

Hindustan Pencils Ltd—Journey Towards Sustainability¹

Hindustan Pencils Limited is one of the leading primary school stationery products manufacturing company in India. Since its inception in 1958, they have mastered the art of making quality stationery products which covers pencils, sharpeners, erasers, scales and pens. All the products conform to European CE EN-71 safety standards. The brands 'Apsara' and 'Nataraj' are quite popular and have emerged as household names. They are having their presence in global markets exporting the products to over 30 countries.

HPL is making conscious and continuous efforts in improving processes and maximizing utilization of inputs to prevent degradation of the environment. All ingredients used by the company are non-toxic in nature. Wood is one of the main inputs for pencils. They have their own wood plantation to ensure renewal of this precious natural resource. They have switched over to eco-friendly water-based lacquers and PVC-free formulations.

The commonly used raw materials in pencil manufacturing are 40 in number. Apart from wood, the others include graphite and clay. Hindustan Pencils strongly believe in a greater responsibility towards society and in the betterment of the quality of life of stakeholders. Their commitment to this responsibility is realized through their contribution to the preservation of trees. Instead of thoughtlessly plundering trees for pencil manufacture, HPL strives to protect Indian bio-diversity and to maintain ecological balance. For this, they had gone for renewable sources of timber by procuring their requirement from those who either grow trees on their farms, land or in the courtyard of their residential premises, rather than relying on forest grown timber.

As a part of HPL commitment to India's national policy of increasing the forest cover, they have developed their own timberland. They have adopted scientific practices of cultivating tree crops and selected suitable tree species, which can grow within a cycle of 7–10 years, which grow on climatic and ground water resources. The wood farming lay emphasis on planting superior, disease resistant and fast-growing clones, which improve land productivity, quality timber yield and maintain ecological balance on a sustainable basis.

INTRODUCTION

The term 'greening the supply chain' describes a wide variety of actions that companies are presently conducting to reduce bad impact on the environment. Greening the supply chain initiatives are aimed at achieving improved environmental, health and safety performance. Greening supply chain helps increase efficiencies in the use of energy, water or other natural resources or raw materials; reducing the environmental and societal impact of business operations upon people around and the biosphere. It is only after the Second World War that business organizations felt the pinch of material shortages and started showing concern for the environment. Organizations then started recycling materials for reuse and also reduced the wastages in operations. Consequently, the environmental concerns had led to the development of environment-friendly practices. Corporations today feel that

by adopting environment-friendly practices, they can become competitive and improve the financial performance by reducing cost of wastages. Public awareness about environment forced the governments of many countries to regulate operations/practices of business organizations to adopt 'greening' more seriously.

Today, the government regulations towards controlling the environmental pollutions are becoming harsher. In developed countries like the United States, government imposes huge fines on producers of substances responsible for depletion of ozone layer. Governments have also introduced a law for recycling contents in the products. In Europe, laws are quite strict for solid wastages because of limited land availability for dumping solid wastes. In addition, shortages of raw materials are forcing manufacturers to recycle the scarce raw materials. The pressure from the consumer groups for greener environment compels the manufacturers to look into their supply chain practices and to be in line with the need of the hour. The product design decisions, cost control, manufacturing, planning and control, and distribution have a major effect on the environmental performance of an organization. In short, green supply chain takes a serious call for environmental impact by the supply chain processes and practices of business organizations.

CORPORATE SOCIAL RESPONSIBILITY

Corporate social responsibility (CSR) is a theme that engages business, academics, social and environmental activism, politics and media and increasing awareness about social and environmental inequities. An understanding of CSR suggests that business organizations should be responsive to social and environmental issues. Many organizations active in propagating CSR have tied CSR to sustainable development suggesting companies to contribute to securing sustainable development. In general, following are the elements of CSR:

- Management
 - Corporate governance
 - Compliance
- Product Responsibility
 - Supply chain management
 - Quality
- Employees
 - Diversity
 - Work-life balance
- Environment
 - Climate change
 - Take-back and recycle
 - Hazardous substance

In a bid to address the issues of CSR, many leading companies are taking the route of greening the supply chain. The micro level initiatives by business corporations will largely mitigate the environmental issues at a macro level.

Box 27.1

IBM Suppliers to Meet Environmental Goals

With the move to go green and sustainability IBM has asked its 28,000 suppliers spread across 60 countries to establish environmental goals and measure energy conservation, greenhouse gas emissions, and waste management/recycling practices. All the suppliers have been asked to report about their plans, their implementation and their results. IBM's requirement for suppliers are based on corporate responsibility and environmental management system, measuring performance, setting goals and disclosing results. IBM asks its suppliers to define, deploy, and sustain a corporate responsibility and environment management system that identifies significant aspects of the supplier's intersections with these matters, including those articulated in IBM's supplier conduct principles and the electronic industry citizenship coalition code of conduct. They were further asked to establish programmes (within the management system) to control operations that intersect with these matters and confirm compliance with applicable law, regulation and any particular contractual requirements. The suppliers are further asked to measure performance associated with environmental aspects such as energy conservation, greenhouse gas emissions and waste management, and recycling. The objective is to achieve 'greening' of supply chain as an important step towards social corporate responsibility.

Source: <http://www.ibm.com/ibm/environment/ems?>

SUPPLY CHAIN RESPONSIBILITY

It is likely that the company at the end of the chain (retail outlet) will feel more compelled to fulfil its responsibility in the supply chain. A manufacturing company in the middle of the chain is very concerned with its supply chain responsibility if it is involved with product stewardship. This company finds it difficult to fulfil its responsibility since its products (semi-finished products) are used for many different consumer products. For this company, it is almost impossible to control all life stages after delivery.

In the food sector, supply chain responsibility is more critical due to consumer protection. This is due to increased requirements of large retail chains in the field of food safety. In the agricultural sector, where labour conditions and environmental issues in India give rise to CSR awareness, there seems to be little attention from Indian companies for CSR in the supply chain. Most of these companies work through a middleman who has actual contact with the farmers. The companies do not instruct middleman on CSR and consider CSR to be the concern of the middlemen and the farmers. They consider their own responsibility to be very limited.

The supply chain responsibility of a bank is quite different than that of manufacturing companies. For example, in chemical industry the companies look into CSR issues relevant to its own sector. However, banks have very broader CSR responsibility as they are involved with many different sectors and diverse CSR aspects due to their financing activities. Thus, financing institutions consider CSR as their largest challenge, since their financing activities and the supply chains behind that can yield much more CSR benefit than in-house measures.

Consumer attention can also influence small companies. Even the companies which do not have an explicit CSR policy or no measures taken with respect to supply chain responsibility, make one exception, that is, they do not engage child labour. Some companies ensured CSR in its supply chain by

Box 27.2

GUCCI's CSR Initiative Across Its Supply Chain

GUCCI, the world's leading brand has voluntarily initiated the certification process for CSR across supply chain. It has launched its worldwide eco-friendly programme to reduce the dependence on materials and use 100 per cent recyclable paper in packaging. As this 90-year-old company is approaching its 90th anniversary in 2011, it has taken this important action towards environmental responsibility. The shopping bag is 100 per cent recyclable and Forest Stewardship Council (FSC) certified. FSC certificate verifies that all trees used to create paper bags and boxes come from well managed forests. All stages in production process comply to FSC standards. In addition, GUCCI is also rationalizing its packaging to limit the excess. For example, shoes will be packed in one flannel instead of two. Gift boxes will only be given out when requested. All mannequins will be replaced by 100 per cent recyclable material. GUCCI took a leadership position in industry by voluntarily initiating process of certification for CRS (SA 8000).

Source: Nandini Raghavendra, GUCCI goes green with eco-friendly packaging., The Economic Times. Mumbai, 06 June 2010

Box 27.3

BERGER Paints India's Green Supply Chain Initiative

Berger Paints has switched its entire formulation in synthetic enamel to non-lead formulations. There only water will evaporate and not the chemical reducing the carbon footprint of the paint. Berger is launching a paint which will insulate the building from the heat of sun, thereby reducing the energy cost of the air-conditioning.

Source: Business Brands, The Economic Times, 06 June , 2010

paying their suppliers well. They have clear terms and conditions for their suppliers. These companies perform audits on vendors with regard to pension fund, safety measures, hygiene, wages, and has a check on its dealers on corruption and pricing.

The research shows that the attitude of MNCs towards their business partners and suppliers is quite ambiguous. On one hand, they incorporate CSR standards in the contract or attach their code of conduct to the contract. On the other hand, they consider it to be the primary responsibility of their business partners to solve their own CSR dilemmas. This implies that these contracts do not provide a guarantee for supply chain responsibility. SMEs do not incorporate CSR requirements in their contracts with their suppliers. Other methods to promote CSR in the supply chain could be by providing information on the CSR policy to suppliers and subcontractors, for example, by providing documentation or information on the Web site, by providing training, etc. Some companies (with no formal CSR policy) take into account ethical considerations in the selection of business partners. This is done in two ways: discussions with potential partners and inquiring with other companies and governments.

With the concerns of environmental issues and global warming, consumers demand more explanation about the products they are purchasing. Companies will have to answer questions about how green their supply chain processes are, their carbon footprint and how they recycle. However, the leading business firms have seen this as an opportunity to convert public's interest in all things

green into increased profits. The manufacturing companies have found that there is a link between improved environmental performance and financial gains. For example, General Motors reduced disposal costs substantially establishing a reusable container programme with their suppliers. By these cost reductions they have identified and complemented the company's commitment to the environment. Thus, by redesigning the firm's supply chain—from purchasing, planning and managing the use of materials to shipping and distributing final products—savings are identified resulting into indirectly implementing green policies.

GREEN SUPPLY CHAIN MANAGEMENT

Green supply chain management (GSCM) has emerged as a key approach for enterprises seeking to become environmentally sustainable. The notion of GSCM implies the insertion of environmental decisions within the traditional concept of supply chain management.

'Greening' the supply chain means integrating environment thinking into supply chain process, including product design, material sourcing and selection, manufacturing processes, delivery of the final product to the consumers, and end-of-life management of the product after its useful life. In green supply chain design, the green process improvement approach (as shown in Figure 27.1) is normally adopted. It addresses the following five fundamental areas of the supply chain: upstream, downstream, within the organization and logistics.

- **Procurement activities** of the organization include evaluation criteria of suppliers' environmental performance.
- **Distribution activities**, usually comprises those activities related to the disposal and sale of excess stock, including recovery and recycling opportunities.
- **Design activities** include involvement with 'design for the environment' with suppliers.
- **Manufacturing activities** include green production causing no pollution and wastages.
- **Logistics activities** such as just-in-time or quality management intertwine with environmental criteria.

Green supply chain accrues the following benefits to the organization:

Judicious utilization of resources GSCM helps in efficient and cost-effective utilization of available resources of the organizations. Organizations will purchase 'green' input resources for environment-friendly production process to produce desired outputs.

Cost reduction Green practices reduce transaction costs and promote recycling and reuse of raw materials. The generation of waste and hazardous by-products is reduced or eliminated, helping the firms complying with the regulations. Consequently, the relevant handling and operational cost involved can be further reduced and, in the mean time, the efficiency of using resources can be enhanced.

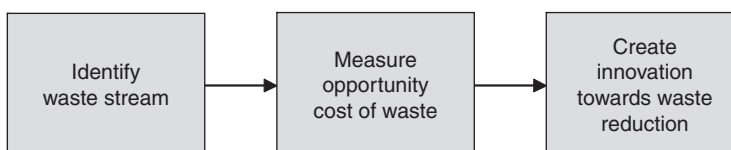


Figure 27.1 *Green Process Improvement Approach²*

Gaining competitive advantage As the issue of environmental concern is addressed successfully, differential positioning of its product in market is possible. Besides attracting new profitable customers for organization, it will give competitive edge in the marketplace. It will strengthen the brand image and reputation in the marketplace.

Risk mitigation Organizations adopting GSCM practices can reduce the chances of being prosecuted for anti-environmental and unethical practices.

Ease in product adoption Environment-friendly products are adopted without any hesitations by the consumers. This will enhance the brand image and brand reputation in customers' minds.

Logistics is the integrated management of all the activities required to move products through the supply chain. For a typical product, this supply chain extends from a raw material source through the production and distribution system to the point of consumption. The logistics is also used for reverse flow of material from customers to manufacturers. While moving the inventory across the supply chain, logistical activities comprise freight transport, storage, inventory management, materials handling and all the related information processing. The main objective of logistics is to coordinate these activities in a way that meets customer requirements at minimum cost. In the past, this cost has been defined in purely monetary terms. As concern for the environment rises, companies must take more account of the external costs of logistics associated mainly with climate change, air pollution, noise, vibration, fuel/energy consumption and accidents.

In logistics, shippers need to seek out carbon-efficient carriers. Two of the biggest sources of CO₂ emissions can be found during the manufacturing and transporting process. If companies can openly share information, the logistics service provider can analyse existing practices and make proposals to optimize the supply chain and reduce both CO₂ emissions and costs. The green practice include better stuffing practices which means better container utilization, converting to different modes of transport, especially to sea from air, and to rail from road, and shipping with a more CO₂-efficient ocean carrier. Hence, for 'greening', supply chain logistics should achieve a more sustainable balance between economic, environmental and social objectives. The green supply chain best practices can be best summarized as follows:

- Align supply chain goals with business goals.
- Evaluate supply chain as single life cycle stream.
- Use green supply chain analysis as catalyst for innovation.
- Focus on source reduction to minimize waste.

Thus, green supply chain management takes into account all suppliers in various stages of the manufacturing process, transportation of finished goods and services, and the integration of reverse logistics. A green supply chain utilizes recyclable material, streamlines distribution processes, reduces redundancies, minimizes waste to provide cost benefit to society, people, nature, business and economy ensuring pollution reduction, environmental stewardship, reduced operational costs, streamlined processes and better supplier relationships.

COMPETITIVE SUSTAINABILITY

In green supply chain initiative the firm would use environment-friendly inputs to transforming input products which can be recycled to conserve scarce resources and reduce the environmental pollution. The output of process used would be reclaimed and reused at the end of life cycle of the

Box 27.4

Herman Miller—Competitive Advantage through Sustainable Practices

A U.S.-based furniture manufacturing company uses 'Cradle-to-Cradle' protocol for designing of the range of its office chair brands which are sold at 20–30 per cent more prices than its competitors. The C2C protocol means, 'Development that meets the needs of present without compromising the ability of the future generation to meet their own needs'. The principles underlying C2C protocol: eco-effectiveness vs. eco-efficiency; waste equals food; biological and technical nutrients; green, yellow, orange-red list; disassembly; recyclability; recycle contents. As per C2C, the DfE (design for environment) index, which is index derived using weighed average to each of the above individual parameter should be more than 50 per cent to qualify for C2C design protocol.

Box 27.5

Starbucks—Sustainable Practices in Supply Chain³

Starbucks Coffee has initiated coffee and farmer equity (C.A.F.E.) practices programme, which is a supplier programme in line with UN's global compact principles into business. This programme helps to promote improvements in product quality, economic transparency, social responsibility and environmental leadership at the producer level. In response to the socially conscious consumer and other important stakeholders, Starbucks developed a wide range of sustainable practices in its supply chain. One such initiative is Starbucks' responsible sourcing programme called the Coffee and Farmer Equity (C.A.F.E.) practices. In 2000, Starbucks established an agreement with Trans Fair USA, a nonprofit organization member of the Fairtrade Labelling Organizations International (FLO) and a third-party certifier of Fair Trade products, to start selling Fair Trade certified coffee. It contributed to sustainable development by offering better trading conditions to, and securing the rights of, marginalized producers and workers with fair compensation. Since launching of Fair Trade Certified (FTC) coffee in 2000, Starbucks is committed to corporate social responsibility through its sustainable supply chain initiatives.

product. The sustainable supply chain aims at reducing the cost of operation and keeps the environment pollution free.

There are many facets of the sustainable supply chain. The starting point is product design and process of manufacturing which needs to be focused and addressed properly for making the product and process environment-friendly with reduction in cost. On the other hand, many companies are also focusing on packaging and transportation to reduce environmental issues. For example, by proper designing of packaging, the amount of cardboard or filler material used can be reduced for it and will help to reduce the loading (volume) on the truck. Agarwal Packers and Movers have modified their trucks body to increase the loading capacity by 30–40 per cent. This helped in reducing the unit freight cost and also the unit fuel consumption for the goods transported.

As shown in Figure 27.2, organizations are adopting a variety of tools, both tactical and strategic in nature, with short- and long-term perspectives. With these tools in mind, corporations plan and deploy various programmes to gain competitive advantage. General Electric deployed an 'Ecomagination'

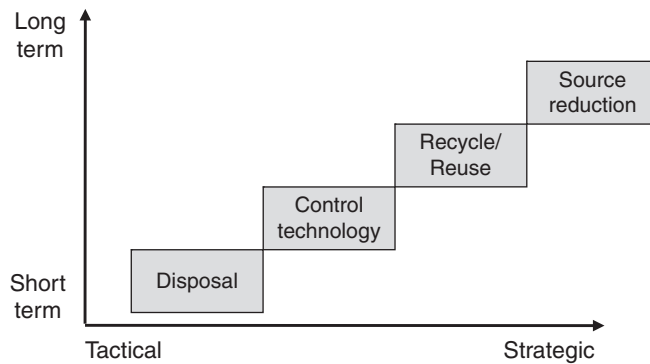


Figure 27.2 Pollution Prevention Hierarchy Used as GSCM Tools⁴

programme to have revenue stream of \$20 billion by 2010 from environment-friendly products. This shows that they recognized the opportunity associated with saving the environment.

The United States Environmental Protection Agency (EPA) published a written guideline called the 'The Lean and Green Supply Chain'. It is a practical guide for materials managers and supply chain managers to reduce costs and improve environmental performance. It is a four-step decision-making process for greening supply chain:

- The first step is to identify environmental costs within a company's process.
- The second step is to find opportunities for reducing environmental impact.
- The third step is to calculate the benefits of proposed alternatives.
- The fourth step is to decide, implement and monitor solutions.

The benefit of implementing a green sustainable supply chain is that profitability of a company can be improved and it helps to keep the environment green. Thus, towards a environment-friendly supply chain practice, the following are identified as potential areas:

- Product design, packaging and labeling.
- Transportation of hazardous raw material.
- Manufacturing processes for waste reduction/elimination.
- Supplier selection.
- Waste/scrap disposal.

GREEN PROCUREMENT

The awareness about the protection of the global environment is growing stronger in people across the globe. Hence, companies are promoting environmental measures in all phases of product development and activities in its factories and work sites. In order to make these activities more effective, companies must reduce environmental burdens and avoid environmental risks by reinforcing protection activities. Companies should work together with their business partners in the purchase of materials that impose minimum environmental loads. For example, the RoHS (European standard for hazardous substances) directive restricting the use of certain hazardous substances in electrical and electronic equipment came into effect in February 2003 in the EU (European Union). RoHS prohibits the use

Box 27.6

Procurement Policies Towards Sustainability

Bayer India⁵ has developed a supplier code of conduct which addresses social, ecological and ethical issues. The principles which are expressed in the supplier code of conduct comprise an important component of supplier selection and evaluation. Bayer Group expects its suppliers to share:

- **Ethics:** Suppliers are expected to conduct their business in an ethical manner and act with integrity to meet social responsibilities.
- **Labour:** Suppliers are expected to protect the human rights of their employees and to treat them with dignity and respect.
- **Health, safety, environment and quality:** Suppliers are expected to provide a safe and healthy working environment and to operate in an environmentally responsible and efficient manner. Suppliers will integrate quality into their business processes.
- **Management systems:** suppliers are expected to implement management systems to facilitate adherence to all applicable laws and to promote continual improvement with respect to the expectations set forth in the supplier code of conduct.

This supplier code of conduct is made available to suppliers with the goal of strengthening the mutual understanding of how sustainability should be practised in day-to-day business.

of 6 substances, are lead, mercury, cadmium, hexavalent chromium, polybrominated biphenyl (PBB), and polybrominated diphenyl ether (PBDE) after July 2006.

In line with environmental policies established under the basic environmental philosophy, the Sharp Group (Japan) has framed its 'Charter of Corporate Behavior' and the 'Sharp Code of Conduct' accordingly. Sharp is pursuing the reduction activities of environmental loads in all aspects of its business activities. Similarly, Konica Minolta Group offers products with low environmental impact. They believe that in order to achieve this goal, it is necessary to develop and design environmentally conscious products, while at the same time procuring materials, parts, and components with low environmental impact. For the Konica Minolta Group, green procurement is reducing environmental impact. They have added a category of the ecology to their traditional evaluations criteria (quality, cost and delivery) of suppliers, in order to give preference to procuring parts, materials, and so on with low environmental impact, from suppliers with active commitments to the environment.

In short, to ensure the greening of supply chain the following practices are adopted by corporations with regards to procurement:

- Suppliers' participation on industry-wide environmental panel.
- Environmental risk assessment.
- Judicious resource management.
- Usage of reusable packaging and shipping material.
- Disclosure of the composition of material used on the label.
- Careful analysis of material impact on environment before its usage.
- Reclaim hazardous material.

In the chemical industry, environmental aspect of supplier evaluation is very important. For example, at Dow Corning of Midland Michigan, the following criteria are used to address environmental issues to decide upon the suppliers:

- How the cost of operations of supplier is compared with his competitors?
- What are the past records of supplier on environmental mess ups?
- Has the supplier ever violated environmental regulations?
- What is the degree of employee exposure to safety?

The important criteria in chemical company are supplier selection by understanding and assessing the environmental risk associated with the chemicals to be purchased from the supplier.

GREEN MANUFACTURING: A STRATEGIC PRIORITY

Green manufacturing cuts across every aspect of manufacturing, including information decisions, process technologies, energy consumption, material selection and material flow. A lot of the decisions manufacturers make are related to cost, function and quality. Now they are adding another dimension, which is environment.

Many large, multinational companies are aware of environmental regulations in developed countries and growing consumer demand for a new generation of environment-friendly products. To answer this, they have embraced the notion that green products and production techniques are a competitive weapon.

In California, most automobile companies are complaining about meeting new clean air emission limits. Yet, Honda and Toyota have developed and produce new engines to meet and exceed requirements. For many of these companies, it is part of their business strategy. May be auto companies across the globe will have to adopt manufacturing process to comply their products to 'EURO III' or 'Bharat III' emission standards (for Indian companies).

MNCs like General Electric, DuPont and Toyota have adopted green practices since long. These companies work with the government to help develop this policy. The manufacturers who have foresight of environmental regulations improve their chances of producing compliant, cost-effective products that meet customer demands. They are proactive to regulatory framework with local and national policy makers. For example, Toyota for material handling in October 2006, introduced a new line of lift trucks that not only surpasses 2007 federal emission standards but also meets California's stricter 2010 standards.

GREEN MARKETING

On the distribution side, there are many marketing activities of a firm that are intended to create a positive impact or to lessen the negative impact of a product on the environment. These activities will answer consumers' concerns about environmental issues. The activities cover everything from design to using recycled material in making a product to claim in advertising or on package labels. Environment-friendly distribution attempts to respond to the criticisms about pollution and waste. To counter to the environmental regulatory and consumer pressures, the firms have started the 'cradle-to-grave', that is, life cycle approach to its products. The life cycle of the product typically include the following:

- **Stage I:** Process of product development which includes design, acquisition of raw materials, components and consumable materials from the suppliers including 'green practices', etc.

- **Stage II:** Product manufacturing with reduced wastages, minimum environmental pollution, safety to workers with reduced health hazards, minimum consumption of energy during manufacturing, etc.
- **Stage III:** Consumption stage includes minimum packaging, recyclables packaging and minimum energy consumption during product usage and maintenance friendly.
- **Stage IV:** This is the last stage which covers ease in disposal, recycling, reuse or re-manufacturing.

The recent example of products under fire was 'Aseptic boxes', which were introduced in the 1980s. It was called as the most significant food-science innovation in 50 years. It is made of multiple layers of plastic, paper, and foil. Nearly 3 billion of the boxes are purchased every year in the United States. They consume less energy and raw materials to produce, fill, and ship, than any other ready-to-drink package. Aseptic boxes are 'shelf-stable', which means they are able to maintain a drink's freshness for a long time without refrigeration. They are safer than glass, and they keep their content cool for hours after refrigeration. A typical aseptic box is 96 per cent beverage and 4 per cent package, creating one-twelfth the waste of a glass container. Despite all of these features they are not environment-friendly. Their multiple-layer construction makes them more difficult to recycle than 'single material' packaging such as glass or aluminium. These boxes were subsequently banned through legislation in the United States.

Sometime back disposable diapers have become the topic for debate. A large portion of disposable diaper is plastic, which is not bio-degradable. Over 20 states in the United States have considered legislation banning or placing special taxes on disposable diapers. The alternative is cloth diapers, which can be washed and reused. However, when the entire life cycles (from manufacturing to disposal) of cloth diapers and disposables are compared, it is not clear which does the least damage to the environment. Two major manufacturers of disposable diapers, Kimberly-Clark and Proctor & Gamble, have found ways to reduce the bulk of diapers and their packaging by as much as 50 per cent through recycling.

Due to pressure from consumers and thereafter from the U.S. government, McDonald's replaced its clam shell packaging with waxed paper because of concern relating to polystyrene production and ozone depletion. Xerox introduced a 'high-quality' recycled photocopier paper in an attempt to satisfy the demands of the firms for less environmentally harmful products.

Bubble or blister plastic packaging is standard in health-and-beauty aids, hardware, electronics components and cosmetics. However, as they accounted for nearly 1 per cent of all solid waste, blister packs have been targeted by environmentalists as unnecessary and wasteful. An advantage of blister packs is that they are strong enough to protect a product while still allowing the customer to see it. A major criticism is the size of many packages.

Many companies for conforming to environmental regulation adapt products and/or packages to make them more environmentally sound. This green marketing strategy requires more substantive effort. A firm might replace an offensive ingredient with a more acceptable one, change production or distribution methods to reduce pollutants and waste, or design a product for easier recycling, closable, zip-lock bag that consumers can reuse when the popcorn is gone. For example, Johnson and Johnson packages shampoo in a pouch that is made from 80 per cent less plastic than a conventional shampoo bottle. Other firms have gone beyond packaging. Marcal Paper Mills in New Jersey takes tonnes of undelivered third-class mail from all 18 Long Island post offices and recycles it into toilet paper.

Indraprasta Gas Ltd., a New Delhi-based company, launched CNG, a clean fuel for automobiles towards forcing the petroleum companies to launch a clean fuel that is lead-free and has low sulphur content.

To appeal to environmentally conscious consumers or respond to legal mandates 'Estee Lauder' has developed a line of cosmetics called 'Origins' that is made entirely from flower and plant essences. In other case, 'Chemlawn', a lawn-care company, has formed a special unit that markets to homeowners,

who want less hazardous materials, used on their lawns. In response to government regulations that ban the use of CFCs as a refrigerant by 1995, refrigerator manufacturers have developed a substitute, which is less an ozone-depleting substance. Dow Chemicals, U.S.-based chemical giant developed special chemical carriers on roads to minimize fumes contamination to environment

Green marketing is not limited to manufacturers. Loblaw, a Canadian supermarket chain with 340 stores, has introduced a line of what it calls 'environment-friendly' products under the 'Green' brand name. The line consists of products, which include:

- Disposable diapers made from non-chlorine-bleached pulp.
- Phosphate-free detergents.
- Alar-free apple juice.
- Recycled motor oil.
- Rechargeable batteries.
- High-efficiency light bulbs.

Wal-Mart has also adopted a green marketing strategy. The chain puts special green tags on items it deems environment-friendly. In addition, it has set up recycling centres in the parking lots of its 1,500 stores and encourages recycling in its TV commercials. BMW has put up used car dismantling plant to address the environmental concern. The dismantle parts are either sent for recycling or upgrading and finally reuse.

Green marketing as a strategy will work only if enough consumers find it appealing. The factors which are responsible for making green marketing so important are as follows:

- Government pressure to produce better society and consumers.
- Pressure from consumer groups.
- Competitive pressure for differentiation leading to competitive edge.
- Sense of social responsibility amongst the corporates.
- Opportunities to reduce cost.

Consumers can affect the environment at two points, by either buying or rejecting environmentally unsound products, and by recycling products or discarding them in the garbage. Right now, recycling is the more common pro-environmental behaviour. If green marketing is legislated as in the case of CFCs, all firms in an industry will be forced to conform to a standard. As a result, it cannot be considered strategic. However, if a firm voluntarily chooses to implement a particular green marketing programme, it may serve as a basis for an appeal to customers for product differentiation. As described earlier, the strategy may entail everything from a statement on a label to a new line of products.

Towards the greening of supply chain, DHL has extended the carbon-neutral 'Go Green' logistics service in Asia. Under this system, a shipper can choose to ship the cargo/shipments under go green by paying a 3 per cent premium charge over the usual charge; in return DHL undertakes carbon reduction projects in its system: better trucks, hybrid trucks, route mapping and so on; the core idea being to reduce DHL's carbon footprint and users get that much carbon credit which in turn can be used in their reporting and such like. DHL offered to ferry all materials and equipments using its go green service and also put two biogas powered vehicles for use by people there. In India, DHL is using CNG vehicles, which come under this go green initiative as well.

Another option is to make it easy for consumers to behave in an environmentally responsible way. Many companies are trying to make them environmentally responsible through green certification

mark. 'Green seal' may act as certification to ensure environmental standard products and create trust in the mind of the customers. If a company's product meets the standard, it can use the green seal of approval on its packaging and in its advertising. An indicator of environmental concern like the green seal may make it easier for consumers to act on their beliefs.

SUMMARY

Green supply chain management is a relatively new concept for the majority of Indian corporations. Adding the 'green' component to supply chain management involves addressing the influence and relationships of supply chain management to the natural environment. Green business practices that maintain and sustain good environmental quality are increasingly becoming a vital component of business organization. As per General Motors, 'greening the supply chain' refers to buyer companies requiring a certain level of environmental responsibility in core business practices of their suppliers and vendors. Many businesses have internal standards, policies and/or environmental management systems that govern their own environmental performance and efficiency. If suppliers do not abide by these same standards, the buyer company may not be buying and using products that do not meet their own standards. A supply chain can be complex, with environmental issues occurring at the second- and third-tier supplier levels. Some companies may also attempt to work the other way by educating their customers on the environmental benefits of their products. However, by adopting 'green' concept in their supply chain, leading organizations have, in fact, reduced product costs, reduced wastes and increased safety in usage of products by the end consumers by reducing their health hazards.

REVIEW QUESTIONS

1. What are the drivers of 'greening' supply chain initiatives in large companies?
2. How green supply chain has an implication on sustainable practices of the firm?
3. What dimensions of an internal and external environment might be affected by green supply chain initiatives?
4. How would strategic marketing planning apply to green supply chain initiatives?
5. Explain 'cradle-to-grave' and 'cradle-to-cradle' concepts in green supply chain.

INTERNET EXERCISE

Log on to the following websites and find out various initiatives corporations should take towards greening the supply chain of their organizations

1. <http://www.greensupplychain.org>
2. <http://www.environment-agency.gov>
3. <http://www.greenscn.com/>

PROJECT ASSIGNMENTS

1. Visit a large size chemical company in your area to study their supply chain and find out their 'green' practices. Also suggest the areas where further improvements are possible.
2. Many good companies are practising a concept of 'Cradle-to-Grave' management for their hazardous products. Find out how an automotive battery manufacturing company (Exide) is practising this concept in this supply chain operation.

VIDEO LINKS

1. Greening the supply chain <http://www.youtube.com/watch?v=pOjdJr0NHPQ>
2. Challenges of sustainable SC <http://www.supplychainbrain.com/content/videos/2011-videos/scl-summit/challenges-of-a-sustainable-supply-chain/>

NOTES

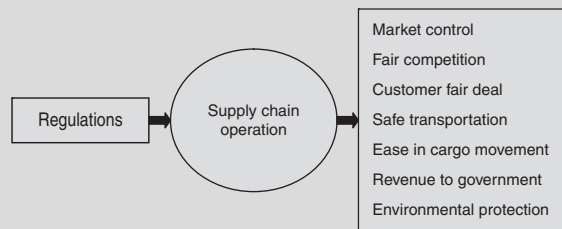
1. <http://www.hindustanpencils.com>
2. M. Porter and Van der Linde, 'Green and Competitive', HBR, Sept–Oct 1995.
3. 'Company Fact Sheet', Starbucks Coffee Company. Nov 2006. 5 Feb. 2007 < http://www.starbucks.com/aboutus/Company_Factsheet.pdf >
4. U.S. Environmental Protection Agency.
5. <http://www.bayer.com>

“The growing need for companies to comply with a wide varieties of regulatory requirements creates new and difficult challenges that can distract an organization from adding value to its products, (and) yet achieving compliance is essential to protecting the value of its innovations
—Jim Brown”

Chapter Objectives

To understand:

- Impact of regulatory environment on SC operations
- Multimodal Transportation Act, 1993
- Goods and service tax
- Warehousing Act
- VAT and CENVAT
- Contract manufacturing and transfer pricing (TP)
- Regulations for dangerous goods
- TIR carnet and INCOTERMS



Regulatory Framework

To run a business house, companies today have to be aware of the legal and regulatory framework that impacts their operations. Supply chain efficiencies are affected by regulatory compliances requirements. These are customs, taxation, immigration services, security, health and safety legislations, food and environment legislations and transport and warehousing regulations. Regulatory control is not new in international trade. In managing international supply chain, tariff and exchange fluctuations remain the major challenges for operation. Operational efficiency of supply chain and regulatory control need be at odds with each other. The challenge is to combine the needs of regulators with supply chain management principles.

ABL-Transfer Pricing

ABL Ltd. manufactures goods for Rs 1,000 in China, but its Indian-based subsidiary buys it for Rs 1,200, and then sells it for Rs 1,500. By doing this, the company's taxable profit in India is substantially decreased. At a 30 per cent tax rate, the company's tax liability in the India is only Rs 9 {i.e. 30% of Rs 300 (1,500–1,200)} as compared to Rs 300 (i.e. 30 per cent of Rs 1,000 which should have been the case). The various areas, wherein the concept of transfer pricing is reflected are the Central Excise Act, 1944, which speaks of levying of excise duty for transactions between 'related party' determined by value at which it sells goods to an unrelated party (Section 4 (3)(b)). The valuation rules under the Customs Act, 1962, recognize the principle of ALP (arm's length price) in dealing with TP. Under the customs rules, for ABL, the 'assessable value' is the invoice value (i.e. 'Transaction Value' under Rule 4(2)) which is Rs.1,000. TP is valued under S.14 of the Act read with S.2(2) Customs Valuation (Determination of Prices of Imported Goods), Rules, 1988. While provisions in the Companies Act, 1956 such as S.211 which deals with the form and content of balance sheet and the profit and loss account requires financial statements to provide the true and fair picture of the state of affairs of the company. Similar provisions relating to disclosure requirements and financial statements have indirect implication on TP. The Accounting Standards (AS-18) framed by the Institute of Chartered Accountants in India (ICAI) deals with transactions between a reporting enterprise and its related parties. S.8 of Foreign Exchange Management Act (FEMA) provides for provisions relating to ALP. Computation of transfer pricing using ALP has been done on related entities using these provisions.

The various provisions under Income Tax Act, 1961 (as amended by Finance Act, 2001) and Rules 10 A to 10 E effectively deals with TP in India. S.92 (1) provides that 'any income arising from an international transaction should be computed as per the arm's length price. An associated enterprise as per the Income Tax Act has to comply with (a) maintaining a prescribed documentation and (b) obtaining an accountant's certificate. While the MNE is free to determine the TP but it is the duty of the authorities to see that it is in ALP. Thus, where the market prices are not reflected in prices set by related parties, the Tax authorities will have the power to adjust profits so that they represent an arm's length result. It is here where the issue of strict compliance comes into the picture.

INTRODUCTON

The purpose of the regulation is to ensure equality and justice in society by framing laws, acts and regulations. As the labour laws of various nations differ, an authority is required to regulate matters relating to employment and labour at the international level. The International Labour Organization has laid down many laws and guidelines to regulate labour matters in the international trade.

India has a number of laws dealing with labour matters. These laws aim at providing legal and corporate status to workers, protecting their health, safety and welfare, giving them proper benefits in case of sickness, maternity or any injury, to ensuring that they receive regular and prompt wages and other benefits. All those countries (including India) which have strict laws are now liberalizing the norms to suit the changing market conditions.

Intellectual property rights are given to those who have invented a new method or technology useful to the entire society or to those who have created an original artistic or literary work. Trade-related intellectual property rights (TRIPS) are framed by the WTO to regulate issues relating to intellectual property at the international level. Intellectual property rights cover: trademarks (marks which help customers identify a genuine product from a fake one), copyrights (the rights given to any original literary or artistic work in tangible form) and patents (given for any new invention or innovation which is beneficial to society).

License is an official sanction given by the government to import certain goods which are under the trade block. The regulations related to product promotion differ from country to country and from commodity to commodity. International marketers should thoroughly understand the norms and regulations of the foreign country before entering its market. The culture, beliefs, and language of the people should also be taken into consideration apart from the laws of the country.

The Competition Act looks into the anti-competitive policies in India. It is framed to ensure fair competition and prevent monopolistic abuses in the market. In the United Kingdom and United States, the business environment is regulated by the European Union and U.S. antitrust law.

The objective of legislation is to streamline business growth, protect the firms from unfair business practices, protect the consumers and check malpractices. Today, groups like environment protection groups and consumer forums are putting more pressure on government to restrain businesses organizations. The deregulation of transportation, privatization of ports, rationalization duties and tax structure, and enforcement of environmental protection laws are a few measures that have had an effect on this industry. The new policies and regulations are being evolved and enforced, keeping in view the objectives of globalization, privatization and liberalization with an additional regulatory dimension for environmental protection. The regulatory environment has an impact on the operation of logistics and supply chain operations too.

MULTIMODAL TRANSPORT OF GOODS ACT, 1993

The multimodal transport covers the door-to-door movement of goods by a single transport operator. The origin of the concept was developed with the container revolution initiated in late 1951 by Malcom McLean in the United States and his trucking operations. The driver for multimodal transportation was the emerging container technology and growing international trade. In fact, the basic premises for successful trading very much depend on efficient transport services.

In cross-border trade, the movement of goods is done by using various transport links. These may be air, water, surface and rail. The storage and handling conditions also vary from country to country. This situation created many problems over the years. Hence, to avoid the involvement of multiple agencies, a concept of handling transportation by a single agency emerged. The variations in cultures, languages and commercial practices at buyer and seller ends result in complexity. It becomes very difficult to align the various members of the chain. The solution is to let one qualified operator be responsible and accountable for the entire transport chain.

To take advantage of the potential offered by the new technologies such as the Internet and EDI, the international trading community updated its uniform commercial practices. These are regarding trading terms, letters of credit and multimodal transport documents. Multimodal transport implies the safe and efficient movement of goods, where the MTO accepts the corresponding responsibility from door-to-door. The purpose of the Multimodal Transport of Goods (MMTG) Act, 1993, is to

streamline multimodal transportation in India. The regulating authority for MTG is directorate general of shipping, which will regulate the unorganized MTOs also. This act came into force in April 1993. As per this act, registration of multimodal transport operator is mandatory. The MTO registration is valid for a period of 1 year and needs to be renewed for a further period of 1 year from time to time.

The MMTG Act gives exporters a sense of security in transporting their goods. The basic concept of the act is a door-to-door delivery using multimodal transportation. The cost competitiveness (logistics) is one of the important aspects of multimodal transportation.

Under the MMTG Act, three types of companies are eligible to be registered as MTOs, and shipping companies, freight forwarding companies, and companies which do not fall in either of the above two categories, but are related to transportation.

FREE TRADE AND WAREHOUSING ZONES ACT 2005

Under the foreign trade policy 2004–09, the Government of India had announced to set up FTWZ (free trade and warehousing zones). FTWZ is a trade-related infrastructure to carry out the transactions related to import and export of goods and services in free currency. On 23 June 2005, the Parliament of India passed the Special Economic Zones Act, 2005, and on 10 February 2006, the same was notified by the Government of India.

In essence, FTWZ is a ‘deemed’ foreign territory for carrying on business. FTWZ’s are created to act as ‘international trading hubs’ and are designed to provide ‘world class’ infrastructure for warehousing handling and transportation equipment, commercial office space. All related utilities such as telecom, power and water are made available to the companies to facilitate imports and exports of goods. It provides a single window clearance of import and export of goods. FTWZ is a key facility linking global supply chains with logistics. The customized facilities will be available for chemicals, food, electronics and oil products. FTWZ will have both dry and refrigerated storage facilities, transportation facilities, information system for cargo tracking and office space support facilities with medical, canteen and business services.

For development of FTWZ, 100 per cent foreign direct investment is permitted. Minimum size of the warehousing stipulated at 1 lakh sq mtrs (0.1 mn sq mt). The FTWZ is under the administrative control of the development commissioner (DC). For the key benefits of FTWZ in supply chain operations, refer to Table 28.1.

For companies with global supply chain operation, by availing of FTZ facility and by proper planning, the international freight cost can be reduced considerably with enhancement in response.

GOODS AND SERVICES TAX (GST)

Goods and services tax (GST) is a part of the proposed tax reforms aimed at an efficient and harmonized consumption tax system in the country. In the union budget for the year 2006–07, the finance minister proposed that India should move towards national level GST that should be shared between the centre and the states. He proposed to set 1 April 2010 as the date for introducing GST.

The introduction of GST to abolish taxes such as octroi, central sales tax, state level sales tax, entry tax, stamp duty, telecom license fees, turnover tax, tax on consumption or sale of electricity, taxes on transportation of goods and services, and to eliminate the cascading effects of multiple layers of

Exhibit 28.1 *FTWZ Benefits*

<i>Areas</i>	<i>Benefits</i>
Fiscal and Regulatory	• Custom duty deferment • Income tax and service tax exemptions • Excise duty exemptions
Infrastructure	• Warehouses (for single product or shared with other products and equipments)
Administration	• Reduction in custom formalities • Improved delivery period • Effective support services • Availability of banking and insurance facilities
Region	• Flow of foreign investments • Creation of employment potential • Boosting up ancillary industries • Enhanced economic activity

taxation. GST has facilitated seamless credit across the entire supply chain and across all states under a common tax base. It has offered significant benefits to industries, the government and the consumers. They are as follows:

- **Logistics re-arrangement:** With GST implementation, sales tax differentials across the different states have been eliminated. This has helped the marketers in reducing the number of warehouses to store the goods and distribute the goods with no state boundaries. With the reduction in number of warehouses, most of distribution are going through the hub and spoke distribution model operating on economies of scale leading to substantial cost reduction.
- **Tax credits:** Under GST, manufacturers are entitled to input tax credit of all inputs and capital goods purchased from within the state as well as inter-state. With GST, there is no cascading effect of taxes and has resulted in a reduction in the cost of doing business.
- **Inventory costs reduction:** With GST implementation, tax paid on inventory is fully recoverable immediately against input tax credit. This has resulted in reducing the inventory financing costs.
- **Cash flow benefits to middlemen:** Dealers and distributors have started to collect GST from customer but are required to remit it to the government only at the time of filing their returns. This will help them to avail of extra cash.

The GST rate is around 14–16 per cent. After GST rate is arrived at, the states and the centre will decide on the central GST and state GST rates. Before GST, services are taxed at 10 per cent and the combined indirect taxes on most goods are around 20 per cent.

VAT AND CENVAT

CENVAT is the new name for MODVAT. Basically they are the same. These are related to central excise. CENVAT means, tax on value addition on the goods manufactured according to the central excise and customs act definition.

VAT is an excise duty collected by the government (of the state in which the final consumer is located)—which is the government of destination state on consumer expenditure. CENVAT stands for central value added tax. These are a component of the tax structure employed by many countries across the countries in the world. The key objective of designing ‘CENVAT’ and ‘VAT’ is to minimizing a cascading effect of tax incidences on goods and services at various transaction points.

India already has a system of sales tax collection, wherein the tax is collected at one point (first/last) from the transactions involving the sale of goods. VAT would, however, be collected in stages from one stage to another. The mechanism of VAT is such that, for goods that are imported and consumed in a particular state, the first seller pays the first point tax, and the next seller pays tax only on the value-addition done—leading to a total tax burden exactly equal to the last point tax.

VAT which was earlier introduced was replaced by MODVAT (modified value-added tax) and now being replaced with CENVAT to remove some limits on the type of taxation that could occur on goods used in the manufacturing process of finished consumer products. CENVAT encourages the production of goods within the country, rather than outsourcing the production to countries where the economic and tax climate is more favourable. CENVAT would reduce the tax burden for the end user of the materials.

WAREHOUSING (DEVELOPMENT AND REGULATION) ACT, 2007

This act proposes to regulate the warehousing business. However, till the government issues a notification, the existing law will be operative. Currently, the business of warehousing is governed by the law of contracts and it is left to the contracting parties to decide the terms and conditions of the warehousing arrangements.

One significant aspect of the business of warehousing is the status accorded to the warehouse receipt and whether such receipt is transferable by endorsement and delivery along with the property in the goods represented by such receipt. Under section 2(4) of the Sales of Goods Acts, 1930, a warehouse receipt is also included in the definition of the documents of the title of goods. But law does not provide sanctity to such receipt as negotiable document and provide for rights and obligations of warehouseman and holder of warehouse receipt. Since it is not a negotiable instrument or document, it is not possible to deal in the document title for trading in commodities. And raise finance against the security of warehouse receipts. The new law fills these gaps and provides a new regulatory regime for warehouse business. It facilitates the issue of negotiable warehouse receipts by any person carrying on business of warehousing. The salient features of this act are:

- The object of the act is to make provision for the development and regulation of warehouse, negotiability of warehouse receipts and establishment of a warehouse development and regulatory authority.
- A person who delivers goods to the warehouse men for storage is defined as a depositor and the goods are defined as all tangible goods excluding actionable claims, money and securities whether fungible or not.
- Definition of negotiable warehouse receipt recognizes negotiability of such receipt and provides for effect of transfer of goods represented thereby when the receipt is endorsed and delivered (negotiated).
- The law applies to any person who is registered as a warehouseman with the authority for carrying on the business of warehouse, that is, maintaining warehouses for storage of goods and issuing negotiable warehouse receipts.

- The law recognizes a warehouse receipt in electronic form issued by warehouseman or by his duly authorized representative.
- Any person desirous of undertaking business of maintaining warehouse issuing negotiable warehouse receipt has to obtain registration. Any person, firm, cooperative society or any association of persons, whether incorporated or not can be warehouseman.
- Existing warehouse business has to apply for registration within 30 days from the date of commencement of the act.
- The law makes detailed provisions about particulars to be included in the warehouse receipts, their negotiability by endorsement and delivery, warranties on negotiation of warehouse receipt and issue of duplicate receipts.
- The law provides for registration of accreditation agencies for issue of certificate of accreditation to warehouse issuing negotiable warehouse receipts.
- Any existing or future warehousemen not intending to issue negotiable warehouse receipts will not be governed by this law. Such a person can continue his business activity without obtaining registration.

This law has distinct advantages for agriculturist as well as commodity traders. Negotiable warehouse receipts will facilitate trade in commodities without physical handling of the stocks of commodities and other goods. Warehouse receipts being documents of title to goods can be easily offered as security for any loan. Development of modern warehouse would also facilitate storage of surplus crops and sale at a future date when there is demand for goods/commodities.

TRANSFER PRICING

The Indian TP regulations have been enacted with a view to provide a regulatory framework which is capable of computing reasonable, fair and equitable profit after tax in India so that the profits chargeable to tax in India do not get diverted elsewhere by altering the prices charged. The Indian transfer pricing regulations which became applicable on 1 April 2001 is largely based on the OECD guidelines.

A contract manufacturing arrangement is similar to outsourcing of manufacturing operation. Under a contract manufacturing arrangement, the firm enters into a contract with the other manufacturing company, to manufacture products according to the specifications under a buy-sell arrangement. By manipulating a few book entries in the accounts books, multinational corporations are able to transfer huge profits with practically no actual change in the business process. For instance, a company may manufacture goods in one country and buy it in other country at different price and sell it at quite different price. Taking into account the differential tax rates, the company may raise or lower the purchase price and thus avoid the real tax incidence by hiding the real transaction price that would have been otherwise.

Thus, multinational corporations derive several benefits from transfer pricing. Since each country has different tax rates, they can increase their profits with the help of transfer pricing. By lowering prices in countries where tax rates are high and raising them in countries with a lower tax rate, such organizations can reduce their overall tax burden, thereby boosting their overall profits. Indeed, one often finds that corporations located in high tax countries, in fact, pay very little corporate taxes. Hence, the transfer pricing regulation and its compliance is a must on the part of enterprises in buying-selling transaction.

Indian transfer pricing regulations are drawn from OECD guidelines. Indian transfer pricing regulations list the methods that may be used in the determination of the arm's length price, and requires the use of the most appropriate methodology from among the available and commonly practised ones.

The Finance Act, 2001, introduced the detailed transfer pricing regulations (TPR) in India with effect from 1 April 2001 corresponding to the assessment year 2002–03. The rules for transfer pricing have been notified on 21 August 2001. The sections and rules under the Income Tax Act, which deals with TP regulations are sections 92 to 92F and rules 10A to 10E and sections 271(1)(c), 271AA, 271BA and 271G. According to Section 92, the transfer pricing is defined as:

- Any income arising from an international transaction shall be computed having regard to arm's length price. [Section 92(1)].
- The allowance for any expense or interest shall be determined having regard to the arm's length price while determining income under sub-section (1) above. [Section 92(2)].
- The cost or expense allocated or apportioned between two or more associated enterprises shall be at arm's length price. [Section 92(3)]. This is a case of cost contribution arrangement or cost sharing arrangement between intra group entities.

Similarly, any transaction between two permanent establishments in India would be covered as both will be non-residents of India. These regulations cover transactions of borrowing and lending. There are various methods for computing arm's length price or transfer pricing:

- Comparable uncontrolled price method (CUP)
- Resale price method (RPM)
- Cost plus method (CPM)
- Profit split method (PSM)
- Transactional net margin method (TNM)
- Such other method as may be prescribed by the board.

Sub-section (2) provides that where more than one price may be determined by the most appropriate method, the arm's length price shall be taken to be the arithmetical mean of such prices. This is a unique concept in India.

The relevant rules envisage determination of ALP by applying margins of each comparable company to the appropriate base of the enterprise. The regulations further provide that, where more than one price is determined by the most appropriate method, the ALP shall be taken to be the arithmetical mean of such prices. An alternative practical approach to arrive at such ALP is to compute the arithmetic mean of margins of comparable companies and apply the same to the appropriate base of the tested party to determine the ALP.

In India, the provisions in computation of arm's length price are in line with the international guidelines and laws of various countries. In view of the fact that initial burden of maintaining records is with the tax-payer and this would be justified in selecting the most appropriate method and maintaining the records for the same. Rule 10C provides that the most appropriate method shall be the method which is best suited to the facts and circumstances of each particular international transaction, and which provides the most reliable measure of an arm's length price in relation to the international transaction.

Arm's length price (ALP) is the internationally accepted transfer pricing standard which must be applied for tax purposes by MNE's and tax administration. Even, Indian TP regulation recognizes

determination of pricing between associated enterprises on an arm's length price basis. The term "arm's length price" is defined to mean a price applied in uncontrolled conditions. In other words, it refers to the market value of a particular transaction ignoring the impact on pricing due to existence of special relationship between associated enterprises.

Penalty for concealment of income or furnishing inaccurate particulars thereof is 100–300 per cent of the tax sought to be evaded. Penalty for failure to keep and maintain information and documents in respect of international transaction is 2 per cent of the value of each international transaction and Penalty for failure to furnish report under section 92E is Rs 100,000.

IATA GUIDELINES FOR CARGO MOVEMENTS

For transportation of dangerous goods by air The International Air Transport Association (IATA) establishes and maintains a set of regulations. This regulation is used as a standard to harmonize the transportation of dangerous goods by air.

The purpose of this regulation is to 'promoting safe, reliable, secure and economical' air services. Its provisions cover the classification, marking, packing, labelling and documentation of dangerous goods shipped by air. The dangerous goods are defined as:

According to the 42nd edition of the IATA DGR, as reported by EMO Trans (a company specializing in international freight forwarding), dangerous goods are "articles or substances which are capable of posing a significant risk to health, safety or to property when transported by air and which are classified according to Section 3". Section 3 of the DGR provides a set of criteria used to determine if the term "Dangerous Goods" applies.

As per the rule, the goods meeting the definition of dangerous goods, while being transported, in order to ensure the safety of the shipment, the shipper should provide an adequate level of protection for workers exposed to the shipment, and enable the best practices in preventing events, which may harm the environment.

The explosives, toxic gases, substances that may spontaneously combust, water-reactive substances, oxidizing substances and organic peroxides, poisonous infectious substances, and radioactive materials are banned from shipping by air by The IATA DGR. As per rules, the transport company should impart proper training to individuals involved in preparing goods for shipment by air.

In 2009, IATA DGR, (50th edn), IATA incorporated several changes to the DGR, including revisions of the list of dangerous goods in order to harmonize the list with the UN transport numbers. The new list includes fuel cell cartridges and lithium-ion batteries, etc.

Box 28.1

Air India—Cargo Regulation

Carriage of cargo by Air India services is governed by conditions of contract under international and non-international carriage of cargo (other than baggage and mail). The conditions of contract are printed on the consignee's and consignor's copy of air waybill and are also available in Air India/agents offices. Shippers are required to complete and sign the instructions for dispatch of goods which are available at Air India and their agents.

Air India accepts the various shipments/commodities with advance arrangements such as shipments of unusual shape and size, human remains, perishables, valuable cargo, live stock, dangerous goods, shipments of concentrated weight exceeding aircraft floor limitation, any other shipment requiring special handling/care. These shipments are required to be packed so that they do not injure or damage persons or other cargo, properties and to ensure safe carriage with ordinary care in handling. The dangerous goods are required to be packed in accordance with IATA Dangerous Goods Regulations. The articles must be listed with proper shipping name. The shipper is also required to sign shipper's certification as required under IATA Dangerous Goods Regulations. The shipments containing liquids (wet cargo) are required to be packed in containers strong enough to prevent breakage or damage in transit or even caused by pressure or altitude. The shippers are required to write legibly and durably the consignee/consignor's name, city, address, contact telephone number (if any) on each shipment. For the carriage of dangerous goods, the shipper shall have to attach to each package containing such articles the appropriate label(s) as shown in IATA Dangerous Goods Regulations. Where space permits, labels shall be adjacent to the consignee's address. Remove and/or obliterate the old labels. Contract of carriage as evidenced by air waybill will automatically be terminated once the delivery is effected. The shippers may charter entire commercial capacity of aircraft subject to applicable charter rules, regulations and tariff. The shipper has to pay minimum charges for any consignment. There are rates established for cargo in general, also known as 'general commodity rate'. There are specific commodity rates (SCR) for carriage of specifically designated commodity from/to 'Specifically designated' points to promote the growth of 'specified commodities'. There are class cargo/rate(s) for live animals, valuable cargo, newspapers, magazines, books, periodicals, human remains, etc. In addition, there are valuation and documentation charges.

Source: <http://www.ia-cargo.com/regulations3.php>

'TIR CARNET' FOR TRANS-BORDER LOGISTICS

To overcome the problems of cross-border shipments through road transportation, UN (United Nations) road transport protocol called 'TIR 1 (transport internationaux outier) convention' can be adopted for business trading. A lot of trade potential that exists between India and countries like China, Afghanistan, Iran, Pakistan, Bangladesh and CSI countries is not being exploited to mutual advantages because of high logistics cost. In the land route trade, there is always double handling of goods at border (due to customs formalities) causing damages and delays. The utilization of land route was made possible in Europe for cross-border logistics using TIR Carnet (transport internationaux routier) System, which is administered by International Road Transport Union (IRU). This organization is partnered by all inter-governmental organizations under UN fold, having its head office at Geneva. TIR carnet is a customs transit document used for international transit operation of goods. TIR carnet is a guarantee for safety of revenue given by IRU to customs organizations of nations through which freight transit takes place across border. This guarantee is executed by Apex national transport organisation, which becomes the general member of IRU. It also sets standards of road and vehicles safety. As many as 40 conventions have been adopted by UN on road movement in the last 60 years. With this convention and TIR carnet instrument, the truck can move/transit through the nations eliminating the border clearance by customs by way of opening packets or

offloading of goods. This system eliminates the necessity of an army of customs officers at the border and also the need of necessary infrastructure of warehouses. Ghana and Pakistan have taken steps for adopting TIR carnet for increasing cross-border trade. In this system, IRU issues a guarantee of \$6 bn for a day to the customs. With the adoption of this system, countries can save huge cost due to transit damages and delays in cross-border logistics. It is also applicable to buses and passenger traffic. It can be also applied to multimodal transport including rail transport. This could be a great process tool when Asian railway materializes.

CERTIFIED CARGO SCREENING PROGRAMME

For exporter to the United States, the 9/11 Commission Act of 2007 requires that all cargo carried on passenger aircraft in the United States had to go through a security screening. To ensure the movement of cargo, TSA (Transportation Safety Administration) developed the Certified Cargo Screening Program (CCSP). This is a voluntary programme designed to move some of the screening process to shippers, third-party logistics providers (3PLs), air forwarders, and independent screening services. Cargo screened at a CCS facility and then transported through a secure chain of custody will not need to be inspected at the airport. This is for avoiding the delays in transportation to the customer.

In CCSP system, there is a rigorous tracking of the chain of goods custody. It uses tamper-evident technology to assure that after the cargo has been screened, the cargo remains secured in transit to the aircraft. Under CCSP, airlines have the ultimate responsibility to ensure that cargo has been screened prior to flight. As per CCSP, certified cargo screening facilities must adhere to a set of rules set by the TSA. This includes security requirements mandated by the TSA's security programme such as to maintain the integrity of cargo through chain of custody, permit onsite validations and periodic inspections and screening of cargo at the piece (not each pallet) level.

INCOTERMS

International commercial terms, known as 'Incoterms', are internationally accepted terms defining the responsibilities of exporters and importers. 'Incoterms' are reviewed and published by the International Chamber of Commerce and are used by businesses throughout the world. 'Incoterms' are used to make international trade easier. 'Incoterms' were introduced in 1936 and they have been updated six times to reflect the developments in international trade. The latest revision was in 2000. There are thirteen 'Incoterms' that are used by businesses and are used in four different areas.

'Incoterms' define the transfer of liability involved in their international sales and the arrangement of shipments. Under these terms the ownership or the transfer of title of goods is not covered. The type of term used will have an impact on the costs and responsibilities involved in shipping, insurance and tariffs.

Exporters, while negotiating with importer should determine the 'Incoterm' that works best for their company and be prepared to quote on those terms. For example, in 'Ex Works' term an exporter's responsibility ends at their facility's loading dock. This means that importer should pick up goods from exporter's facility for forward delivery to the destination. In this, cost of transportation and insurance will be deducted from the importer's account. The term negotiated needs to be incorporated in all the documents related to that transaction.

The most burdensome Incoterm for the exporter is delivered duty paid (DDP). Under this term, all arrangements and costs are borne by the exporter. 'Incoterms' are categorized into four groups (E, F, C, D)

and are listed below in order of increasing risk/liability to the exporter. Some Incoterms are applicable to ocean/inland transport only. There are 13 Incoterms as discussed as follows.

GROUP E—Departure

It covers only one Incoterm, that is, EXW.

Ex works (EXW) It is followed by a name of a place. For example, EXW, Mumbai; under this term seller's responsibility is to make the goods available at the seller's premises. The seller is not responsible for loading the goods on the vehicle provided by the buyer, who then bears the full cost involved in bringing the goods from there to the desired destination. EXW represents the minimum liability to the seller. Risk and expenses are borne by the buyer, including payment of all transportation and insurance costs from the seller's door. EXW is used for any mode of transportation.

GROUP F—Main Carriage Not Paid by Seller

It covers three terms such as FCA, FAS and FOB. Here seller pays for pre-carriage at origin but does not pay for main carriage.

Free carrier alongside (FCA) It means free carrier alongside and is followed by a name of a place. For example, FCA Mumbai: The seller fulfils its obligation to deliver when it has handed over the goods, cleared for export, into the charge of the carrier named by the buyer at Mumbai. Here, risk passes to buyer, including transportation and insurance costs on the buyer's collecting vehicle, it is the buyer's obligation to receive the seller's arriving vehicle unloaded.

Free alongside ship (FAS) It means free alongside ship and is followed by a named port of shipment. For example, FAS Mumbai; this means the seller is responsible for the cost of transporting and delivering goods alongside a vessel in a port in his country. FAS should be used only for ocean shipments. In this, the risk and responsibility of shifting goods rest on seller till the goods reaches of the ship's crane. Buyer makes payment of onward transportation and insurance costs, once delivered alongside ship by the seller. FAS is used for sea or inland waterway transportation, wherein the obligation of export clearance rests with the seller.

Free on board (FOB) This term is most commonly used in this group. It means free on board and is followed by the named port of shipment—for example, FOB Mumbai. Under FOB terms, the goods are placed on board of the ship by the seller at a port of shipment indicated. Here, the buyer bears the risk of loss or damage to the goods when the goods pass the ship's rail. The seller pays the cost of loading the goods.

GROUP C—Main Carriage Paid by Seller

This group covers four terms. Here, the seller arranges and pays for the main carriage but does not assume risk.

Cost and freight (CFR) Cost and freight is followed by a named port of destination, for example, CFR, New York. Under this the seller has to pay the costs of goods and freight necessary to bring the goods to the named destination. The risk of loss or damage to the goods, as well as any cost increases, is transferred from the seller to the buyer when the goods pass the ship's rail in the port of shipment. Insurance is the buyer's responsibility. Under CFR the risk and insurance cost pass to buyer when

delivered on board the ship by seller, who pays the transportation costs to the destination port. This term is used for sea or inland waterway transportation.

Cost, insurance and freight (CIF) Cost, insurance and freight and is followed by a named port of destination. For example, CIF, Sydney. Here seller purchases insurance against the risk of loss or damage to goods. Seller pays for all transportation charges till the ship reaches the destination port. Here, risk passes to buyer when delivered on board the ship by seller, who pays transportation and insurance costs to destination port. Used for sea or inland waterway transportation.

Carried paid to (CPT) Carried paid to is followed by a named place of destination. For example, CPT, Dubai. Under this, the seller must pay the freight for the carriage of the goods to the named destination. The risk of loss or damage to the goods and any cost increases transfers from the seller to the buyer when the goods have been delivered to the custody of the first carrier, and not at the ship's rail. Here, risk and insurance costs pass to buyer when delivered to carrier by seller, who pays transportation costs to destination. Used for any mode of transportation.

Carriage and insurance paid to (CIP) Carriage and insurance paid to is followed by a named place of destination, for example, CIP, Damascus. CIP has the same 'Incoterm' meaning as CPT, but in addition, the seller pays for the insurance against loss of damage. Here, risk passes to buyer when delivered to carrier by seller, who pays transportation and insurance costs to destination. Used for any mode of transportation.

GROUP D—Arrival

The seller assumes the most cost/risk because goods must be made available upon arrival at agreed destination.

Delivered at frontier (DAF) Delivered at frontier is followed by a named place. For example, DAF, Nairobi. Here, the seller's responsibility is complete when the goods have arrived at the frontier but before the customs border of the country named in the sales contract. The buyer is responsible for the cost of the goods to clear customs. Here, the risk and responsibility for import clearance passes to buyer when delivered to named border point by seller. Used for any mode of transportation.

Delivered ex ship (DES) Delivered ex ship is followed by a named port of destination, for example, DES, Madagascar. Here, the seller shall make the goods available to the buyer on board the ship at the place named in the sales contract. The cost of unloading the goods and associated customs duties are paid by the buyer. The risk and responsibility for vessel discharge and import clearance pass to buyer when seller delivers goods on board the ship to destination port. Used for sea or inland waterway transportation.

Delivered ex quay (DEQ) Delivered ex quay is followed by a named port of destination. For example, DEQ, Johannesburg. DEQ means the seller has agreed to make the goods available to the buyer on the quay at the place named in the sales contract. Here, risk passes to buyer when delivered on board the ship at the destination point by the seller, who delivers goods on dock at destination point cleared for import. This term is used for sea or inland waterway transportation.

Delivered duty unpaid (DDU) Delivered duty unpaid is followed by a named place of destination. For example, DDU, Moscow. The seller has to bear the costs involved in shipping the goods as well as the costs and risks of carrying out customs formalities. The buyer pays the duty and has to pay any additional costs caused by its failure to clear the goods for import in time. Here, risk and responsibility of import clearance pass to buyer when seller delivers goods to named destination point. Buyer is

obligated to import clearance. Seller fulfills their obligation when goods have been made available at the named place in the country of importation. This is used for any mode of transportation.

Delivered duty paid (DDP) Delivered duty paid is followed by a named place of destination, for example, DDP, Bakersfield. The seller has to pay the costs involved in shipping the goods as well as the costs and risks of carrying out customs formalities. The seller pays the duty and the buyer has to pay any additional costs caused by its failure to clear the goods for import in time. DDP should not be used if the seller is unable to obtain an import license. Here, risk passes to buyer when seller delivers goods to named destination point cleared for import. Used for any mode of transportation.

SHIPPING TRADE PRACTICES BILL

The Shipping Trade Practices Bill is being introduced to bring uniformity in maritime logistical trade practices in India. The bill aims at bringing transparency in maritime logistics practices in the industry. As per the bill, the registration of service providers with the government is a must. The service providers will now have to publish their rates, mode and manner of fixing tariff. They are required to display all the above information on their website.

Currently only containerized shipments through maritime logistical service providers are covered under the scope of the bill. With this bill the exporters and importers, who are exploited by unregistered service providers, would benefit with the new enactment. They will (consigners/exporters) now be able to estimate their logistics cost before they take any decisions. This legislation will be regulated by a separate authority which will govern and have dispute redressal power.

With growth in world trade, the maritime industry in India would be charting new waters, with lines introducing e-commerce, electronic bills of lading, etc. Hence, it necessitates formation of a strong independent body with sufficient authority to handle complex maritime legal issues and deliver quick judgments.

SUMMARY

Government of any country across the world monitors and controls the growth of the industry and commerce in that country through legislations. The regulatory environment in the country is normally in line with the policies adopted by the government for the economic growth of the nation. The logistical industry in India is no exceptions for the same. The current regulations are in line with the liberalization measures adopted by the Indian Government since 1991. The transportation industry except railways is privatized long back. Private investments are welcome for infrastructure development projects such as roads, airports and seaports, which are necessary for the growth of logistical industry in the country. The various acts passed by the government are for curbing the mal-practices in the industry and exploitation of gullible customers by the organizations that are having financial power. The regulatory environment today is much more conducive for the industry growth in country as compared to the one which existed before liberalization of Indian economy in 1991. The deregulation of transportation, privatization of ports, rationalization of duties and tax structure, enforcement of environmental protection laws are few measures which had positive effects on this industry. The new policies and regulations are being evolved and enforced, keeping in view the objectives of globalization, privatization and liberalization with the additional regulatory dimension for environmental protection.

REVIEW QUESTIONS

1. Explain the implications of regulations on logistics operations in cross-border supply chain.
2. What is 'transfer pricing'? How does TP regulation impact contract manufacturing in SC operation?
3. Discuss how the GST regulation helps in reducing the number of warehouses.
4. Explain the various 'INCOTERMS' and their importance in international trade.
5. What is the 'TIR Carnet System' and its role in cross-border goods transportation?

INTERNET EXERCISES

1. Read more on; IATA's Dangerous Goods Regulations; visit <http://www.ehow.com>
2. For more details on 'Incoterms' chart of responsibility; visit <http://www.i-b-t.net/incoterms.html>

VIDEO LINKS

1. Shipper compliance with 10+2 security regulations <http://www.joc.com/maritime/di-nucci-shipper-compliance-102-security-regulations-video>
2. International tariffs http://www.export.gov/logistics/eg_main_018130.asp

PROJECT ASSIGNMENTS

1. Prepare a comparative study report on the regulations in countries like the United Kingdom, United States and other European countries, impacting the operations of cross-border supply chains of exporting Indian companies.
2. Take up a manufacturing company having multiple warehouses in various Indian states due to differential sales tax system and find out how the number of warehouses (without compromising on service level) can be reduced if GST is enforced in future.

NOTE

1. Ganpati S. I., 'Transborder Logistics: Removing Hurels', *SEARCH*, Jan 2007, http://www.iru.org/index/en_iru_carnet.

ETHICAL ORIENTATION

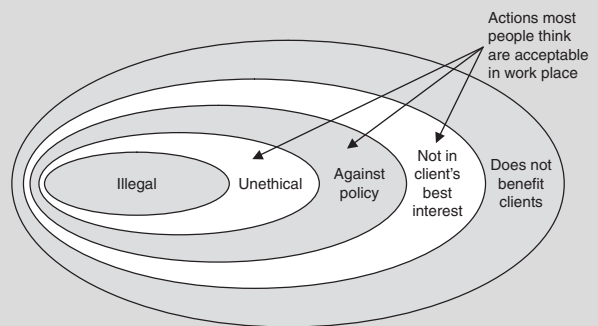
“Commerce is as a heaven, whose sun is trustworthiness and whose moon is trustfulness

—Baha u llah”

Chapter Objectives

To understand:

- Business Ethics
- Ethics in purchase, manufacturing and distribution
- Supply chain vulnerability
- Conformance to applicable laws and industry practices
- Confidentiality and proprietary information



Five stages of ethical erosion wherein one would likely to get caught doing¹

Good business ethics should be a part of every business. Many global businesses, including most of the major brands that the public use, can be seen not to think too highly of good business ethics. The eye-catching headlines like Mattel does damage control after Recall, Del Monte Pet Products recalls food items, Toshiba Laptop Batteries Pose Fire Hazard, and Dole Pre-packaged Salads Recalled are a real-world example of a leading brand taking the hit for ethics or compliance breaches committed by suppliers. Money is the major deciding factor. If a company does not adhere to business ethics and breaks the laws, they usually end up being fined. Billion dollar profits blind the companies to their lack of business ethics, and finally the money wins.

Mattel Toys: Lapses in Quality Checks

In the supply chain, companies move things, make things, store things and throw things away. It is a part of an organization's environmental footprint. Supply chain management is the integration of both ethical and operational practices of the company. It is a proven way to reduce a company's impact on the society and environment while improving business performance.

On 14 August 2007, Mattel announced the product recall decision. It involved toy cars coated with lead-based paint on them. They also called back Barbie Doll, Polly Pocket and Batman toys, which were fitted with small, powerful magnets that could, if swallowed, harm children. This was followed by a recall of the Fisher-Price toys coated with lead-based paint.

The problem surfaced due to supply chain quality control inefficiencies in Mattel. Mattel had to face this problem because they engaged low-cost Chinese suppliers. This was a lesson to the company that quality should never be compromised and that it should operate ethically to keep up the image. Mattel's example shows that it is very difficult for a multinational company, despite its best efforts, to keep tabs on all suppliers around the globe.

This happened due to negligence from the sub-supplier. The main supplier of the cars to Mattel Toys was Early Light Industrial, who subcontracted the painting to another company, Hong Li Da. The subcontractor was supposed to use paint supplied by Early Light. However, due to negligence, Early Light was asked to procure paint. The paint quality was compromised over low cost. The paint used contained potentially poisonous lead. This is a case of supplier cutting corners.

About 65 per cent of Mattel's toys were made in China. The balance 35 per cent procured from vendors in Thailand, Malaysia, Indonesia and Mexico. Mattel used to randomly test-finished toys. This quality lapse missed safety checks and resulted in recall of 4,36,000 toy cars contaminated with lead paint. This is a hard way to learn from mistakes and that the business ethics is not only limited to the company and its one-tire suppliers but is further extended to two-tire suppliers too. The path to sustainability of supply chain is quite arduous, where there is no place to compromise on quality as it will have implications on the image of the company in the eyes of the government, in particular, and the customer in general.

INTRODUCTION

Being an ethical company is not enough anymore. These days, leading brands are judged by the company they keep and trust they build. Consumers, investors, business partners, regulators and media organizations now expect a company and its entire supply chain to be ethical. The supplier-generated ethics scandal is probably one of the biggest (and least foreseen) business risks most leading companies face today. The damage can be great, and protective measures can and should be adopted immediately. The good news is that feasible, affordable solutions exist and can be implemented relatively quickly and painlessly.

Today the business organizations are facing pressures from various sources (government, consumer forums and competition) to improve upon the working conditions in their supply chains.

The stakeholders are focussing more on the ethical issues, corporate social responsibilities and environmental issues. Hence, managing the issues in supply chain is one of the indicators of how well a company is ethically run. The pressure on companies is increasing and the media is gunning on business practices.

The poor labour standard in many companies is coming in their way of progress on establishing good labour practices and implementing ethical codes. However, the organizations that are taking these issues seriously are at various stages of the process and are out of public and government focus.

The eighteenth-century economist Adam Smith demonstrated how in a free market the self-interest of producers and consumers will produce an outcome desirable to all concerned. He further indicated that the market can also lead to inequality of income, wealth and market power. According to him, monopoly suppliers can exploit consumers, monopoly buyers can exploit supply firms and worldwide inequality of income can result in unethical practices such as the child labour. The prominent ethical issues in the society are involvement in the community, honesty, truthfulness and fairness in marketing, use of animals in product testing, agricultural practices (intensive farming), degree of safety built into product design, donation to good causes, etc. The real ethical issue in the business is the extent to which a business accepts its alleged responsibilities for mishaps, spillages, leaks and selling of addictive products. Some of the ethical issues arising from the industry practices are as follows:

- Treatment of customers
- Organization's loyalty to employees in difficult economic conditions
- Working conditions and treatment of workers
- Supply firms' business practices
- Child labour

PROCUREMENT—ETHICAL PRACTICES

Purchasing is a strategic function in the organization. It has broadened the scope of purchasing. Procurement is the process of acquisition of goods and services in a responsible manner for conducting a business. It has also changed the responsibilities of the purchasing managers who are empowered to spend a huge amount of money in purchasing the goods and services for the organization. This responsibility has presented them with ethical dilemmas involving questionable purchasing practices.

The approaches by institutions/organisations to 'ethical procurement' often concentrate only on supplier standards and practices rather than on the institution's/organisation's own policies and practices. According to the UK's Institute of Business Ethics,² ethical issues arise mainly in three areas of procurement practice and all three should be considered as follows:

Procurement Conduct

This refers to the way the firm's staff does the business. The procurement conduct is the manifestation of the organization culture. It encourages and supports the employees to act with integrity and in line with the firm's ethical values in establishing and maintaining vendor relationships. The suppliers are treated with dignity and paid on time. The corporate gifts are officially registered. The buying firm provides adequate transparency during tendering or engagement in contract terms.

Selection of Suppliers

In the process of supplier selection, there are set criteria and guidelines rather than the subjective decision-making base of likes and dislikes and personal gains. In the process, the buying firm seeks to further its corporate responsibility objectives through its purchasing activity and supplier relationships.

Supplier's Practice

In this, the firm will go to the extent of understanding the supplier's supply chain and also their procurement practices. The buying firm imposes social and environmental standards on their suppliers (e.g., health and safety).

In today's competitive business, the importance of sustainable supplier relationship cannot be denied. The reliability of a firm's suppliers can be considerably enhanced through mutually beneficial terms, fair practice and trust. The risk with regards to the firm's reputation very much depends on its suppliers' integrity and reputation.

Buying firm's ethical business policies imbibes ethical values and responsibilities in their employees. A standard of ethics covers issues not covered by law concerning a specific function—like procurement. Once commitments and expectations have been established, it is important to ensure that they are met and continued to be appropriate. The procurement staff will need training, particularly in how to resolve dilemmas and competing priorities. For establishing ethical procurement practices in business the following steps need to be carried out by the firms:

- Establish a code of conduct in purchasing
- Integrate ethical purchasing into existing management systems
- Use rewards and performance review system
- Ensure top management support for establishing ethical code
- Understand the business environment and culture of the countries wherefrom goods are sourced
- Develop skills and knowledge appropriate to ethical buying
- Develop a communication strategy for all stakeholders
- Provide ethical training for purchasing staff
- Ensure that labour standards are a key criterion when selecting suppliers
- Visit factories and workplaces of suppliers for ethical audits

MANUFACTURING—ETHICAL PRACTICES

This area of business ethics usually deals with the duties of a company to ensure that the manufacturing processes do not cause harm to employees and the society. In manufacturing, to define a degree of permissibility depends on the changing states of preventative technologies or changing social perceptions of acceptable risk. It deals with defective, addictive and inherently dangerous products and services, (e.g., tobacco, alcohol, weapons, motor vehicles and chemical manufacturing), ethical relations between the company and the environment, ethical problems arising out of new technologies and product testing ethics (animal rights and animal testing, use of economically disadvantaged groups as test objects).

Box 29.1

Recall of 'A-Star' by Maruti—Ethical Practice

Maruti Suzuki India (MSI) had launched its fifth global model 'A-Star' on 19 November 2008. The 998 cc K10B petrol engine car is manufactured only in India at its Manesar facility. Maruti Udyog Ltd spotted an anomaly in the fuel tank in some of the A-Star models which were already in the market. These vehicles belonged to a lot made till 22 August 2009. In case fuel is filled to the brim, beyond the fuel auto cut off position, a possible fuel leakage from the fuel pump mounting area may take place. No problem has been reported by customers. However, as a matter of caution, the company decided to check all vehicles from the lot and the fuel pump gasket and ring were being replaced for around 100,000 cars including those exported overseas.

The leading business organizations in general follow the following ethical policies:

Forced labour The firm will not make use of forced labour in the manufacturing operations. The employment in the organization including overtime shall be on a voluntary basis.

Child labour The firm will not put children (below 16 years of age or younger than the age for completing compulsory education in the country) to work.

Working hours and compensation The applicable national and local wage and working hour laws will be followed consistently by the firm. Overtime work if necessary will be conducted with the consent of the workers. The workers will be given adequate compensation for overtime work as per the prevailing standards.

Health and safety The firm will provide a clean, safe and healthy place to work and all facilities to promote an awareness of health and safety issues to their employees. The firm will give proper training to their employees for fire prevention, emergency evacuation, proper use of safety equipment, basic first-aid and the proper use and disposal of hazardous waste materials.

Abuse and discrimination The firm will not discriminate employees on any ground. The treatment to all employees will be with dignity and respect. No employee shall be subject to abuse, cruel or unusual disciplinary practices.

Communication of principles All employees will be communicated properly through oral and written way.

Environmental impact The firm takes care of the environmental issues through proper programs and practices. The emphasis is mainly on reduction and recycling of waste. The manufacturing facilities comply with all applicable laws relating to the environment pollution.

Freedom of association Firms allow employees to choose (or not) to associate with legally sanctioned organizations or associations. The firm will not make any unlawful interference in their rights.

Monitoring The management conducts periodic on-site visits of working and living conditions. They initiate audits of production records and practices and of wage, hour and payroll information maintained by facilities, to review and ensure compliance with these principles.

Compliance with applicable laws The manufacturing facilities would comply with the laws of the country in which they are conducting business.

Fair dealing Employees would endeavour to deal fairly with the company's customers and suppliers and each other. No one should take unfair advantage of anyone else through manipulation or misrepresentation of material facts.

Quality Products that meet the quality standards are essential to the enterprise success. Everyone in the company is responsible for product quality and must be committed to ensuring the effectiveness of the quality management system.

Confidentiality Employees, officers and directors would maintain the confidentiality of information entrusted to them by the company, its customers, and its vendors and suppliers, except when disclosure is authorized or legally mandated. Confidential information includes all non-public information.

MARKETING—ETHICAL PRACTICES

In marketing, the common ethical issues are a lack of clarity in pricing, selling at a loss to increase market share and destroy competition in order to subsequently raise prices, price fixing cartels, encouraging people to claim prizes when they phoning premium rate numbers, “bait and switch” selling, high pressure selling to the elderly, counterfeit goods and brand piracy, copying the style of packaging in an attempt to mislead consumers, deceptive advertising, irresponsible issue of credit cards and the irresponsible raising of credit limits, unethical practices in market research and competitor intelligence. The ethical issues relating to products are selling goods abroad, which are banned at home, omitting to provide information on side effects, unsafe products, built in obsolescence, wasteful and unnecessary packaging, deception on size and content, inaccurate and incomplete testing of products, and treatment of animals in product testing.

Ethics is the set of rules or standards that govern the conduct of a person or members of a profession. Selling is a profession that has been widely criticized for the unethical dimensions associated with it. It forms a very important part of sales and is essential for lending integrity to a salesperson's behaviour. Social responsibility covers ethical behaviour and is defined as an individual's or institution's concern for the consequences of his/its actions as these might affect the interests of others in the society.

In general, the companies doing business with no regard to social responsibility run the risk of attracting the attention of environmental groups, earning negative publicity, and losing the goodwill of society. Therefore, companies try to instil a sense of ethics in their employees and conduct business in a socially responsible way. Indian companies are now more concerned with corporate social responsibility than ever before. Companies in the past were concerned more about making profits than anything else.

All companies are expected to imbibe values pertaining to corporate social responsibility in their mission and make CSR a part of the organizational policies. There is a much wider scope for corporate social responsibility in the Indian corporate scenario with its coverage extending to human rights, labour standards as well as environmental issues. Companies in the current scenario are operating in an intensely competitive environment due to globalization.

In organizations, a sales manager faces ethical issues that cover the ethical dilemmas of his sales people as well as the ethical aspects pertaining to his decisions regarding hiring and evaluating of salespersons, assigning territories, etc. The ethical issues facing a sales manager may be studied with regard to the sales manager's relationship with his subordinates, the company, customers and competitors.

On the other hand, the ethical issues facing a salesperson pertain to the salesperson's accountability to the top management, and his relationship with other salespeople and with customers. The following are the areas in marketing wherein the firms should have an ethical code of conduct for exhibiting its responsibility towards corporate social responsibility.

Product safety should be a top priority for a firm. The firm should stand behind every product it makes and takes product safety very seriously. Product safety and quality assurance systems are based on a comprehensive and stringent review process at every stage of product development. Products shall be reviewed from the design stage to the point where they are ready for shipment to the store shelves.

All products should be reviewed by quality assurance reliability engineers in the development stage and safety and reliability specifications are issued and a test plan for each product is required and is a must. The test plan should specify all design and manufacturing criteria that the product must meet. The specifications also include labelling and any warnings required on the packaging, the product or the instructions. Following production, there are also regular sample audits made periodically throughout each shift by both factory and quality assurance personnel. Testing by independent outside labs could be used as part of the quality assurance process. No product should be shipped until testing is complete to the satisfaction of the firm's quality assurance department.

In the areas of environment, health and safety (EHS) at both the corporate and industry levels, the firm should have longstanding commitment to sustainability. The firm's commitment should include specific short- and long-term goals at manufacturing sites, including solid waste reduction goals, paper and chipboard recycling goals, plastic recycling and emissions reduction goals. Some of the EHS compliance should include:

- Industrial fire protection and emergency preparedness guide.
- Environmental protection agency (EPA) programme
- EPA waste wise programs (US).
- ISO14001 environmental management system registration
- Occupational safety and health association (OSHA)
- Voluntary protection program (VPP)
- STAR site certification of manufacturing facility

To oversee all of EHS activities, a firm should have an EHS committee designed specifically to address issues of sustainability.

DESIGNING AND IMPLEMENTING ETHICAL CODE

The ethical code is a set of principles governing morality and acceptable behaviour. It covers personal behaviour, for example, when dealing with customers and suppliers, corporate behaviour, for example, when negotiating deals, behaviour towards society, for example, when recruiting, and behaviour towards the environment, for example, when deciding on a process. Institute of Business Ethics (UK)² suggests for good ethical practices, which, according to them, organizations should issue statements of ethical practice in respect of:

- Relations with stake holders
- Environment and society

- International business
- Mergers, acquisitions and takeovers
- Governance

To check the implementation of ethical codes, the companies should conduct ethical audits. This is an audit of the firm's activities with the purpose to check that ethical principles are being pursued or check the extent to which actions are consistent with the organization's stated ethical intentions.

ETHICAL ISSUES IN THE SUPPLY CHAIN

Codes and compliance Many companies have purchasing codes in purchase manuals. Having a code is not enough to make change happen. The effectiveness of a code depends on how serious the management is to implement it. It all depends on the management's commitment to compliances.

Limits of audits Audits are used by firms to monitor supplier performance. However, audits are not commonly used to improve labour standards. Audits may not find hidden problems of discrimination and harassment. Audits are inappropriate for casual/contracted workers and often do not reach beyond the first tier of suppliers.

Challenges The future challenges for companies engaged in ethical supply chain management include: aligning commercial and ethical agendas, building suppliers' capacity to manage the issues for themselves and engaging with local governments and organizations to widen the impact of activities. Due to the limitations of audits, companies are faced with various challenges such as improving the impact of audits by focussing on continuous improvement, providing effective follow-up, improving the quality of audits through staff training.

Governance structures Most organizations have a wider network of vendors, suppliers, customers, distributors and business partners. Understanding and leveraging the relationships with all partners is a key success factor in managing supply chain.

Suppliers, business partners and customers come in many varieties. Many times, institutions underestimate the importance of suppliers to the viability of their enterprise. Understanding who composes supply chain is step 1 in understanding how to leverage it.

SUPPLIER ETHICS MANAGEMENT (SEM)

Supplier ethics management is the management of suppliers and supply relationships with strategies, programs and metrics that better align supplier business conduct with purchaser standards, with the goal of reducing the purchaser's overall risk of corporate integrity failure in the supply chain. 'Corporate integrity failure' embraces any enterprise-level scandal involving a violation of compliance, ethics or corporate responsibility standards. The corporate leaders believe that their duty to ensure ethical business conduct ended at the four walls of their organization and did not extend beyond their own employee base. The companies inserted clauses into their standard procurement contracts, requiring suppliers to represent and warrant that they would comply with all applicable laws, statutes and regulations. In short, corporate defence to allegations of wrongdoing in the supply chain has historically boiled down to suppliers.

When a supplier-generated ethics scandal occurs, this kind of defence is no longer enough to cool down angry consumers, investors, business partners, regulators, advocacy groups or media organizations. The leading brands have to do something more in order to keep the trust of the marketplace.

In most large companies, procurement executives are responsible for the company's supplier relationships. In order to have any type of positive impact on these supplier networks, legal and compliance officers know they must cooperate and collaborate with their procurement and purchasing colleagues. In fact, the legal and compliance officers should proactively offer the following to colleagues in the procurement department in the following five major areas:

- Compliance and ethics—key factor in supplier selection and evaluation
- Segmenting suppliers by commercial importance and ethical risk
- Creating and maintaining compliance histories of important suppliers
- Assigning a compliance personnel to important supplier relationships
- Conducting regular assessments of supplier risk profiles

This type of internal collaboration is inexpensive and sensible from an organization's perspective. Increased procurement compliance interaction will position the organization to declare that it is not the ethics problem in the supply chain. And it stands a good chance of preventing the problem from actually occurring in the first place.

SUMMARY

The problem of unethical behaviour is common in good times and bad too. Companies do not always understand the impact of their actions, and if consumers are unaware of how products get to market, there will not be a price to pay for cutting corners.

The situation is changing, however, as companies do a better job of auditing their operations and working environments. They might be motivated in part by the desire to act ethically, but there are more immediate reasons for their change. Consumers today are becoming better educated about working conditions, and the unfair treatment of workers and that can seriously harm brand image. The public's growing awareness of environmental issues also puts pressure on companies to build more sustainable supply chains. The law is yet another powerful motivator. Industry groups have formed to address the issue, as governments clamp down on violations of labour standards. And when one company is caught, others scramble to re-evaluate their own policies and practices. Ethical behaviour is good business. Companies that mistreat their employees often experience huge losses from internal fraud and theft by a demoralized workforce. The less time a corporation spends taking care of those employees, the more likely they are to have those kinds of problems.

The supply chain encompasses many if not most of the processes that are subject to unethical behaviour. For gaining competitiveness, the firms have to conduct their business in accordance with high ethical and business standards. The firm should seek to have its vendors, suppliers and licensees to conduct themselves ethically. Firms should have concerns about the quality and character of working conditions in their establishments. Implementation of the ethics principles enables firms to ensure that manufacturing facilities involved in the production understand and adhere to the customer and industry requirements. The main areas of concern for unethical behaviour would be purchasing, manufacturing or distribution in supply chain. The supply chain is a prime location for taking action. It covers many if not most of the processes that are subject to unethical behaviour.

REVIEW QUESTIONS

1. What is business ethics? Discuss the various ethical practices a business organization should adopt to run a successful business.
2. Explain 'ethical code' and its scope.
3. Discuss the various ethical issues and challenges in the supply chain.
4. Explain, 'suppliers ethics management'.

INTERNET EXERCISES

1. To study the 'Manual for managing responsible business enterprise' issued by the US Department of Commerce visit http://www.trade.gov/goodgovernance/adobe/bem_manual.pdf
2. *Business Ethics* is a magazine of corporate responsibility; visit <http://business-ethics.com/> to study the real-life business ethics cases.
3. The Institute of Business Ethics, UK, was established in 1986 to encourage high standards of business behaviour based on ethical values. Visit <http://www.ibe.org.uk/> to learn good practices in business ethics.

VIDEO LINKS

1. Ethical supply chain video podcast, <http://vimeo.com/2847927>
2. Ethical supply chain video podcast Series, Episode 3, Hewlett-Packard Interview, <http://businessassurance.com/ethical-supply-chain-video-podcast-series-episode-3-hewlett-packard-interview/>

PROJECT ASSIGNMENTS

1. The business house like 'TATA' in India is known for their ethical business practices. Get an in-depth view of the 'code of conduct' for Tata Organisations and for its implementation visit http://www.gryphonshafer.com/blog/2008/08/i_happen_to_be_a.html
2. Kodak, Hewlett Packard and Bayer have developed their 'code of conduct' for ethical operations in all functional areas. Write a report on their ethical business practices and its implementation.

NOTES

1. http://www.gryphonshafer.com/blog/2008/08/i_happen_to_be_a.html (with permission)
2. Institute of Business Ethics, UK, <http://www.ibs.org.uk>

PART-VII

CASES

- CASE 1** **Palas Acqua**
Seeking Solution to Lower Distribution Cost
- CASE 2** **Arihant Consumer Products**
Harnessing the Power of IT in Managing Supply Chain
- CASE 3** **Rem Chemicals**
Cost Reduction Through New Procurement System
- CASE 4** **Mask Packaging**
Supply Chain Integration Initiative
- CASE 5** **Zapak Ltd**
Service Supply Chain Delivery Optimization
- CASE 6** **Indian Paints**
Innovative Supply Chain Practices
- CASE 7** **Mann Diesel Sales and Services**
Supply Chain Initiative in Spare Parts After Market Support
- CASE 8** **Padmini Motors**
A Lean Supply Chain Through 'Just-in-Time' System
- CASE 9** **Digital Storage (India) Ltd**
Leveraging Innovative Supply Chain Practices to Competitiveness

CASE 10 **Zara**
Global Supply Chain Keeping Ahead of Competition

CASE 11 **Econ Industry**
Strategic Procurement Initiatives

CASE 12 **Afcon Watches**
Tuning the Supply Chain to Changing Times

CASE 13 **ZICO**
Supply Chain Initiative—Localization

CASE 14 **La Vogue**
Supply Chain Leadership Through System Approach

CASE 15 **TRIKA Exports**
Seeking Supply Chain Solution to Garment Exports

PALAS ACQUA

Seeking Solution to Lower Distribution Cost

Water is the most important necessity for life. The drinking-water needs for individuals vary depending on the climate, physical activity and the body constitution, but for an average consumer it is estimated to be about 2–4 litres per day. Looking into the customer need of pure drinking water, Palas Acqua (PA) established its mineral water plant in central India at Nagpur. The growing number of cases of water-borne diseases, increasing water pollution, urbanization, scarcity of pure and safe drinking water, etc. has made bottled water business just like any other consumer item.

PA is a leading brand in mineral water market having presence in western and central India. Over the years many local brands have cropped in the market and gaining some share in market, which is growing at 10 per cent per annum. The mineral water is a commodity product and the demand is price elastic.

The supply chain manager of PA was preparing the next year's plan to find out changes in distribution system, which would help the company enhance its profitability. PA management wants to use the resources judiciously and release the funds to use elsewhere.

PA started as a regional bottled water company in Maharashtra and spread its market to two other states. Today the company's 'Palas' brand is popular in Maharashtra, Gujarat and Goa. As market demand grew, the company increased the volume it sold within the existing area and expanded its area of supply. As PA grew, it faced problems of profitability and customer service. The company had a large number of enquiries from new customers in the other states like Andhra Pradesh and Madhya Pradesh, it was uncertain as to how these could be profitably exploited. The problem with the company was the steadily rising distribution costs (trade discounts and logistics) in servicing more distant market areas.

Manufacturing

PA had a plant in Nagpur (Maharashtra) 550 kilometres from Mumbai, the commercial capital of India. Nagpur is well connected with other cities in India through road and rail. The mineral water met all the government regulations concerning purity and it is certified by the public health authorities. Clear plastic bottles with attached labels are used for filling the mineral water. These plastic bottles are sealed with tamper-proof opening. For bacteriological inspection, a sample from each production batch is removed and tested. Bottles are packed in corrugated paper cartons, which are placed on belt conveyors to move to warehouses. The cartons are stored in warehouses. These are stacked onto pallets and stored on racking storage system using fork lift trucks. The variable costs of production are Rs 1.3 for each bottle.

The advantage of PA's location is its excellent connectivity to all markets and thus lower manufacturing cost. The only concern was high distribution cost in comparison (logistics plus channel cost) to other local mineral water manufactures. All three states had a number of local producers of mineral water,

who are confining their distribution in their home states. Overall fixed cost of the operation, including all manufacturing labour and bank interest on the outstanding loan, was Rs 12.5 crore.

Markets

The market for mineral water is steadily growing over the years. The reasons are growing awareness of not-so-pure (with contamination) water from municipal sources. With the increase in disposable income and growing consciousness about health, the market for pure bottled water is growing very fast. The market growth is 10 per cent per annum. However, consumption pattern is cyclic with seasons.

PA is largely catering to three major market segments. These are retail shops (small ones), large retail chains and restaurants. The high-end restaurant sector was dominated by premium brands but PA has a small share in that market also. The mid- and low-end restaurants were stocking PA and other local brands. In this segment, PA has considerable market share. In the stand-alone retail shop segment, PA is priced little higher than other local brands and consumers were also prepared to pay a higher price for PA. The retail market, which made up the bulk of the total volume sold was characterized by price discounts and distribution elasticity. The large retail chains were banking on price concessions and availability due to their bulk and continuous purchases. They were not willing to pay higher prices for PA products and would purchase local products rather than buying PA. In the price-sensitive retail market a large part of sales occurred during special offer periods. Purity of water with the certification was vital to customers and it would help the manufacturer to achieve a reasonable level of sale. The most preferred and moving pack size was 3.6 litres. In addition, large pack sizes were also preferred.

The retail market was served by many small and a few large players. The price offered was Rs 12 per bottle. Heavy discounts were common in the industry. Average discount offered was 18–20 per cent. The target price of PA was Rs 13 per bottle. Due to available price elasticity of Rs 8–10 for the manufacturer, the pricing and cost control was vital to business profitability.

As per research, customers are interested in the source of the water and the certification. Packaging (clear plastic bottle) is also perceived as another part of the product benefit, confirming the quality of the water and its purity. Bottled water generally have a long shelf life of over 8 months. However, with plastics bottle the taste of product alters, if exposed to strong sunlight over a long period of time. This reduces the shelf life by approximately 5 weeks. The second important element of the product benefit is attitudinal. In the restaurant sector, the attitudinal component is sophistication of the product. However, in the retail sector the traditional value of purity was more important. The name PA was associated with clear and pure water.

Retail shops (pop and mom) accounted for 20 per cent, retail chains 30 per cent and restaurants 50 per cent of PA's sales in the three states. The majority of outlets could hold 2–3 days' stock but not more. Most outlets also demanded that suppliers support their products. PA had 40 depots spread across the three states. There was one more segment emerging in metro cities, that is, door delivery to households. Market surveys suggested that this part of the market was likely to grow rapidly in the B grade cities. Retail margins were typically around 15 per cent of the retail price.

Current Distribution

At present, PA is using trailers to distribute products in the three states. The distribution is done through 40 depots. The market size, the road distance in miles (one way), and the rail-link in miles (if in existence) are provided in Table 1.

Each truck carries 9,500 kilos of water—about 2,500 bottles. PA had sales of Rs 6 crore last year. The overall annual fixed costs of the current operation were Rs 1.98 crore.

Future Plans

The company had identified two new markets for expansion. The statistics of these areas are in Table 2. The problems of supplying these areas would be acute. On the basis of the sales force volume forecast, additional vehicle mileage would be in excess than that of last year.

Two distribution alternatives had been identified to reduce the cost. The first option was the use of rail. However, the rail speed was 50 kmph, but the network could, and did, operate over 24 hours. Rail cost was around Rs 1.2 per tonne per km of track. Because the company would have to deliver to the local railhead near the factory, and then transport product from the railhead at the destinations, with additional costs—storage and onward shipment. Storage handling and collection would cost around Rs 70 per tonne, by rail.

The second option was subcontracting the road transport operation. Three road transport companies quoted price based on full truck loads to the various distribution centres. Their quotes are given in Table 3.

PA had ambitious plans to increase its level of promotional investment in the following year. In the previous year, the company had spent Rs 1.2 per bottle on average in sales promotion, and in the current year planned to increase this to Rs 1.5 per bottle and with this, the company hopes to increase its market share by one point.

Table 1 *PA's Market Analysis*

<i>Centre</i>	<i>Sales (Rs/Mn)</i>	<i>Deliveries (No. of Trucks)</i>	<i>Avg. Road Distance (Kms)</i>	<i>Avg. Rail Distance (Kms)</i>
Maharashtra	36.0	1200	250	200
Gujarat	19.4	740	300	220
Goa	09.6	320	180	110

Table 2 *New Centres for Development*

<i>Centre</i>	<i>Market (\$million)</i>	<i>Deliveries (No. of Trucks)</i>	<i>Distance (Kms)</i>	<i>Rail-link (Kms)</i>
Andhra Pradesh	150	300	140	120
Madhya Pradesh	130	250	120	100

Table 3 *Quotes Submitted by the Three Transport Companies*

<i>Markets</i>	<i>TPL 1</i>	<i>TPL 2</i>	<i>TPL 3</i>
	<i>(Unit Rs per ton-per km)</i>		
Maharashtra	2.4	2.5	2.2
Gujarat	2.5	2.3	2.4
Goa	2.5	2.4	2.5
Andhra Pradesh	2.6	2.3	2.4
Madhya Pradesh	2.5	2.4	2.5

REVIEW QUESTIONS

1. Analyse different distribution alternatives for PA.
2. Discuss the pros and cons of different transportation modes PA has to choose from.
3. Choose the most cost-effective transport company from the available options.

ARIHANT CONSUMER PRODUCTS (ACP)

Harnessing the Power of IT in Managing Supply Chain

CASE 2

Arihant Consumer Products (ACP) is having a product mix of toiletries, hair care, household care, fabric care and body care. ACP plants are located at Malanpur (M.P.), Guwahati (Assam), Silvassa (U.T.) and Baddi (H.P.). ACP organization is harnessing the power of IT in supply chain management, dealer management, customer relationship management and sales force management.

ACP had gone for IT support in its business operations in 2000, and decided to adopt ERP across the enterprise. In 2001, ACP decided to look beyond ERP and leveraged the Internet to the maximum possible extent. ERP helped ACP significantly to gain benefits within the organization. The company's new initiatives were to connect the business partners outside the organization. ERP has a de-centralized architecture. This standard ERP did not support B2B, dealer and supplier networks. Hence, ACP hired the services of another service provider for the next 5 years.

This requirement gave rise to a new initiative in the form of connecting distributors and another initiative to connect to the suppliers. ACP also has an initiative known as 'ACP-IT'. This is a portal for employees of the organization. ACP-IT is the company's intranet and is like a virtual workplace encompassing all the workflows like performance appraisal, leave and reimbursements into it. It also helps the employees located away from head office to know what is happening in the organization. It also helps in sharing best practices, learning, events, and updates across the organization. All the above initiatives and the current IT investments at ACP align with the strategy document on IT submitted by external service providers.

DISTRIBUTOR-IT is one of the most successful IT implementations at ACP. It is a combination of more than one product. One of the challenges the company faced 3–4 years back was to bring distributors into the IT infrastructure. The need came because of the non-visibility of stocks at the distributor end. ACP had to entirely depend upon their estimates, intuition and knowledge of the field force.

However, bringing the distributors into the IT architecture was a big challenge. They were suspicious and apprehensive about the new IT solution. ACP needed a solution to connect all concerned and align them to the business logic. After initial resistance, distributors realized the benefits from the system and have started participating actively in it. ACP has around 1,000 distributors of which around 375 are A class distributors contributing 70–75 per cent of the total business. The remaining 750 distributors are B and C class distributors.

The underlying business concept behind this IT initiative was based on replenishment. It monitored the inventory levels at distributor's end and based on that it suggests a replenishment plan. The distributor has to furnish information on sales to carrying and forwarding agents (CFA). The information consists of current stock levels and goods-in-transit flow from CFA to distributors and also to ACP. The system considers all the above three things to determine the order level by considering the replenishment logic.

The initial challenge was that the distributors were not open to sharing the above data. Also the technical challenge was that most of the distributors were using some software or the other. The distributors felt that ACP will get insights into their data and they had an apprehension that their data will be shared with other distributors. ACP overcame this, through various dealer's education programmes.

The IT team at ACP had a great role in pushing the distributors. Each of the 375 (A Class) distributors was extensively trained by a special IT team of ACP. This resulted in benefits such as a daily visibility of the retailing information to ACP and the distributors benefited from much lower inventory carrying costs. This implementation has re-organized the push-sales tactic followed before. Now, the distributors invest smaller amount of capital, for shorter time periods, leading to reduction in product stagnation on the distributor's end and fewer chances of scarcity of a particular brand in the marketplace. Invariably, this translated into better brand visibility and higher profits. Though this project took some time to reach wide acceptance, all distributors were finally happy about the benefits it accrued.

SUPPLIER-IT was another initiative towards the suppliers. CFA generated orders and informed regional warehouses on the quantity that was to be shipped to the CFA. Thus, CFA inventory level was maintained at minimum. The data from the regional warehouse was shared with the plant warehouse and, based on it, the plant warehouse decided on the quantity that was to be shipped to the regional warehouse.

Based on this, the manufacturing plants derived the quantity that was to be shipped to the plant warehouse. This information helped ACP to finally decide on the quantity of different products to be produced at each of the plants. Based on this, the suppliers were informed of the quantity that was to be dispatched.

SUPPLIER-IT was thus an extension of **DISTRIBUTORS-IT**. Using these two IT initiatives, the production was done on the basis of demand. Production now matches the demand. Thus, the entire supply chain has optimum inventory.

Using these two programmes, replenishment for A class distributors is done on a daily basis, and for B and C class distributors it is done on a biweekly basis. For B and C class distributors ACP did not want to give the ERP software. This was because it is not easy to maintain the software at so many distributed sites. Also, these are small distributors and most of them do not even have PCs. These distributors are also located at remote places and providing them technical support would not have been easy.

For 750 distributors, ACP thought of a different solution so that the investment was also minimized. These small distributors call CFA two times a week and read the sales and stock data over the phone. CFAs key in the data by logging on the Web site, and then the data is taken backwards like in the case of A class distributors. Thus, ACP is now connected to all distributors.

In the third stage, ACP planned to connect to the retailers through the **RETAIL-IT** initiative. This initiative has started recently. In this, a very large number of retailers are connected, extending to the tune of 60,000–75,000. The system is based on replenishment logic again. With this new initiative, the retailers come into the current IT architecture. From an organizational perspective, the implementation is almost complete.

ACP uses PDAs and mobile applications for capturing retailing and non-IT distributors' data. Earlier, field force used to visit the retailers to get orders, but now they will be getting stock details using PDAs from retailers. The distributors will then synchronize the data taken from the field force with **DISTRIBUTOR-IT**. Each A class distributor can thus serve 200–350 retailers.

Thus, it is a trio of distributor, supplier and retailer IT initiatives giving the company a real-time visibility of inventory across end-to-end supply chain.

On an overall basis, inventory level has come down by 50 per cent and sales have gone up. Distributors reported a higher ROI—some distributors have reported about 20–30 per cent increase in their sales because of IT. There is a higher visibility of material flow and data flow across the supply chain.

The inventory turnover ratio (net sales/inventory) improved. While the inventory turnover ratio was 6.74 for FY 2000, it has improved to 13.56 in 2005. Overall, there has been a reduction of inventory holding days by 60 per cent and a reduction of stock-out situations by 65 per cent. The other business benefits from each of these IT implementations were improvement in quality of service, accurate data of market variations, on-time dispatches from CFAs to distributors, efficient delivery enhancing order fulfillment from 76 per cent to 91 per cent, reduction in lead times in replenishing a distributor, faster responses than competitors via promotions, release of capital by distributors and overall improvement in profits and cash flow, and improvements in customer support and average response time to the distributors.

REVIEW QUESTIONS

1. Why did ACP feel that it needed planning and implementing IT in supply chain?
2. Explain the IT initiatives taken by ACP.
3. Discuss the various challenges faced by ACP during implementation of trio IT packages.

REM CHEMICALS

Cost Reduction Through Improved Material Procurement System

Rem Chemicals (RC) is a large manufacturing organization having 5 large- and 12 smaller-capacity plants in various locations. They are the leading market players and manufacturers of polyester, petrochemicals and polymer products in the country.

The regular store requirements like gaskets, valves, welding electrodes, fasteners, pipes and pipe fittings, and electrical and instrumentation consumables, were procured directly from various stores across the sites by raising POs after calling quotations from 'registered' regular suppliers and placing orders as and when required. The management observed that this procurement process caused inconsistency and variability in purchase pricing, resulting in overall higher procurement costs. They also observed the variations in product quality at different points and times of purchases. The management decided to review the prevailing procurement process and initiated the following actions:

- Assess the annual requirements of each of the above items at all locations.
- Raise the requirements (purchase requisitions) for a longer period of requirement under the ARC (annual rate contract) system.
- Review the number of vendors supplying these quantities.
- Reduce the number of PRs (purchase requisitions) and POs (purchase orders) to reduce transaction cost and paperwork.
- Review the cost in procurement process.
- Reduce the lead time for these regularly required items.

After studying these factors, the management reduced the suppliers' base of the company by 45 per cent. RC established the ARCs with the suppliers which reduced considerably the inconsistencies over the price of the consumables in all the plants. As the orders increased, a strong business relationship was developed with the suppliers. Buying in bulk reduced the costs. By avoiding repeat buying through tendering at various locations, lead time between PR and PO was reduced by 63 per cent. These benefits accrued; a few mechanical items listed in Table 1 is illustrative of the new procurement system. Similarly, Table 2 is illustrative of reduction in lead time after the implementation of ARC. The savings achieved after implementation of the new procurement system is listed in Table 3.

A major reason for lead-time reduction was reduction in tendering time. As the vendors were aware in advance of the possible orders, they were prepared for supply. Due to the implementation of the ARC system, the suppliers directly got the electronic copy of PO from the company leading to reduction in time as well as transaction costs.

Table 1 *Suppliers Base Reduction*

<i>Item</i>	<i>Old Suppliers' Base</i>	<i>New Suppliers' Base</i>	<i>Reductions in Suppliers' Base (%)</i>
Gaskets	490	439	9
Ball/butterfly valve	65	21	68
GGC valve	75	31	58
Welding electrodes	61	22	66
Safety valves	41	18	50
Pipes	61	42	31
Bearings	68	44	27
Fasteners	241	219	3
Refrigeration gases	9	1	86
Industrial gases	21	16	24
Steam straps	4	2	50

Table 2 *ARC Implementation: Benefits*

<i>Item</i>	<i>Pre-ARC Lead Time (Days)</i>	<i>Post-ARC Lead Time (Days)</i>	<i>Reductions in Lead Time (%)</i>
Plug valve	170	75	56
Pipes	165	31	81
GGC valve	159	102	35
Flanges	129	24	80
Transmission belts	119	39	62

Table 3 *New Procurement System: Savings*

<i>Items</i>	<i>Savings (Rs in Lakhs)</i>
Valves, pipes, steam straps and seals	115
Mechanical consumables	227
Electrical items	68
Mechanical spares	90
Instrumentation consumables	226
Fire and safety	32
Bearings	112
Gases	15
Total	885

REVIEW QUESTIONS

1. Review and discuss the old procurement process in Rem chemicals.
2. Explain the changes the management has made in old procurement process.
3. What is annual rate contract (ARC) and what should a firm look into while going in for ARC?
4. Discuss the benefits accrued to Rem by adopting a new system of procurement.

MASK PACKAGING

Supply Chain Integration Initiative

Mask Packaging (MP), a leading player in the packaging industry was facing challenges of fulfilling the rush orders, which was causing logistic problems. Hence, it was necessary to look into the order management process to improve upon. MP is a Mumbai-based packaging manufacturing company and has been in the business for the last 10 years. It deals with the manufacturing of paper packaging required in the FMCG industry. Since inception, it has expanded to five factories. It had 250 employees and had a turnover of Rs. 175 crore (1750 mn). It makes products for fast-moving consumer goods, food industry, pharmaceuticals, health-care products and cosmetics and the agriculture industry.

The company has two divisions, that is, paper making and packaging. The paper division produces paper for packaging based on waste paper. The specialized paper machines make corrugated paper, liner paper, tubes and core paper. The Packaging Division produces corrugated board, solid board, tubes and cores. They also make packaging for the products such as flowers and bulbs. The customers of MP are leading players in industries such as FMCG, food, pharmaceuticals and cosmetics, and the agriculture.

During the last few years, it has changed its strategy to encompass not only the production of packaging, but also a range of value-added services around those materials. MP wanted to move from just providing commodity products to focusing more on the service they provide to customers. In addition to producing paper or corrugated board, this requires the ability to deliver products at the right moment, in the right quantity based on the customer's needs.

To support that strategy, Mask needed to cut costs while at the same time increasing speed and accuracy in its operations—a challenging task. The company as a whole had grown over the years 10 times leaving it with a wide variety of software spread across its various units; this made it difficult to get an integrated view of the business. Moreover, supply chain processes in its paper division—which manufactures products for use in the packaging division and for sale on the open market—were fragmented, driving up costs and delays.

To solve the problem of fragmented business, MP thought of an integrated communication tool using the e-business solution so as to give customers a good service. Using the new technology they faced the challenges of buyers markets as against the sellers markets in which they were operating in the past. In the past decades, customer expectations as well as customers themselves have considerably changed. Customers now tend to be more marketing driven, with dynamically evolving requirements, which requires a higher service level (e.g. in terms of variety in products, just-in-time delivery and smaller lots). It was clear that customer-wise solutions would become an important leverage for MP to stay competitive with other major players in the packaging industry. With the changing environment the company was confronted with a need for the following:

- Shorter lifecycle for its products
- Higher variability in demand
- Shorter lead-times

Initially MP reacted to this changing trends using flexibility measures in its production process including blind stock buffering, which ultimately resulted in huge holding costs, high administration costs and, more often than not, in investments in the wrong stock. Even with this strategy, they could not do away with rush orders. The end result was poor performance.

To overcome this, they invested in a software solution which enabled Mask to integrate its supply chain. It allowed MP to have a strategic collaboration with its customers through the information sharing and exchange. This helped MP to move from a product-oriented approach to a customer-oriented approach. The software covered the following:

- Tracking of customer's specific stock holdings (VMI)
- Optimizing customer's orders through advanced planning and periodic analysis of stock needs of customers
- Exchanging information on stock level

MP aimed at optimising both their own supply chain and that of their customers (mostly B2B). In essence, MP created a virtual organization, characterized by openness on the supply chain from both sides. The new approach with software solution created a trusted, open relationship between the company and its customers. The information sharing helped MP organize its material requirement planning within 2 weeks. Now, MP promises to optimize the customer's stock and demand, while improving the delivery reliability. MRP (material requirement planning) is reflecting the customer's needs for the coming weeks with respect to all the products MP is offering; these are sent online to MP on a daily basis. MP then processes the incoming MRPs and thus constantly improves the match between orders, production and stock levels.

On the basis of this data exchange, MP responds to the short-term needs of customers with products available in stock (Replenishment Proposal 1). To avoid shortages, the production division of MP brings the stock back to its prior level (Replenishment Proposal 2). Finally, production takes care of the long-term needs of MP customers, based on forecasting. An important advantage of being online is the agreement on common standards between the client and MP. Thus, both sides communicate in MRP files. Since last year, MP has started using a portal Web site, where customers can log in and place their orders.

The preparation phase took about 1 year and included planning and working on proposals for how the Web portal online concept should look and when it could be operational.

They run a pilot project to test the new concept, involving the paper division of MP packaging, a very large FMCG company and one of Mask's major customers. Periodic evaluations (every 3 months) were made to reveal possible weaknesses of the system and to suggest improvements. The pilot project was successful and it was fully implemented. This resulted in reduction in the number of rush orders, while the level of accuracy with respect to production and deliveries had increased. Moreover, instead of placing separate orders for each of more than 80 different MP products, the software has to send an updated version of its MRP to MP. Stock administration on both sides was simplified, especially for the customer. Apart from some communication problems at the beginning of the pilot project, no serious difficulties emerged during the implementation process.

MP and FMCG customer were very pleased with this new way of working and decided to continue the project. After the successful start-off, MP exported this concept to the number of customers that got involved in the growing project. MP adopted a step-by-step approach to this expansion instead of pursuing strategy.

The online integration concept created a win-win situation for both parties. MP experienced significant benefits as follows:

- Reduced need for working capital
- Decrease in stock value
- Lower operating costs from simplified administrative procedures
- Enhanced views of order history
- Fewer rush orders
- Increased visibility
- Increased delivery reliability
- Reduced stock shortages

Online integration helped MP to reduce the complexity in planning production processes and linking them with the supply chain. Production divisions are now facing fewer rush orders. The number of stock shortages decreased, which means that delivery reliability increased. Stock rotation periods have gone down from more than 8 weeks to 2–4 weeks, and the need for storage capacity subsequently decreased. Holding costs decreased and working capital requirements dropped by more than 30 per cent.

Although now maintaining less stock, the company has improved service levels. Delivery reliability (time and quantity) increased from 95 per cent to 99.5 per cent. Thus, for MP online integration has become a key variable in building long-lasting relationships with customers.

Furthermore, administrative burdens decreased by about 15 per cent and resulted in a more streamlined and responsive administration. The overall cost for implementing of online integrator is small relative to the total company costs. In addition, MP reported that their overall IT total cost of ownership (TCO) had decreased from more than 1 per cent to less than 1 per cent of revenue. This is believed to be very good in comparison with the TCO performance of other companies within the packaging industry. This case study of MP illustrates the potential online technology solutions to improve SCM through trusted open exchange of data with the customer via the Internet.

REVIEW QUESTIONS

1. Discuss the various supply chain problems faced by mask packaging.
2. How has MP overcome the above problems?
3. What is an 'online integrator' and how has mask packaging implemented it?

ZAPAK LTD

Service Supply Chain Delivery Optimization

Zapak Ltd is the global leader in verification, testing, inspection and certification services. It has excellent geographical network with an employee strength of 42,000 spread over 845 offices, 338 laboratories located in 102 countries. Zapak is basically a service provider.

In 2007, Zapak was involved in the replacement of 700 laptops and desktops in their offices spread over different locations in India. The new machines were to be procured from Lenovo, Singapore, as per the global contract. The first consignment of 200 machines was dispatched to Chennai in July 2007 and the same was to be shipped to 34 offices throughout the country. Figure 1 illustrates the material flow in distribution network.

The courier dispatched the machines through rail instead of road as agreed. Due to transportation by rail most of the boxes were damaged and they arrived at the offices with a foul smell of fish. Fortunately, none of the machines were damaged.

But it is obvious that the consignment was subject to risk during travel. Hence, corrective action was a must. Time taken by train travel was also more, which resulted in delay in material supply and availability of machines. The following corrective steps were taken to avoid the above.

The courier company was hauled up and rates were renegotiated as per rail transportation, which was almost 50 per cent less than road transportation. Two other logistics service providers were approved, and all consignments were to be handled by these three service providers.

As the consignments were to arrive every quarter for the next 2 years, Zapak entered into a long-term contract with all the three parties on special terms. The schedule for dispatch for each one was also worked out. With the new arrangement the flow illustrated in Figure 2.

The logistics companies picked up the consignments directly from the Chennai port and delivered to various sites by road and saved 6 days in total. This resulted in damage-free delivery and saving of time by 6 days.

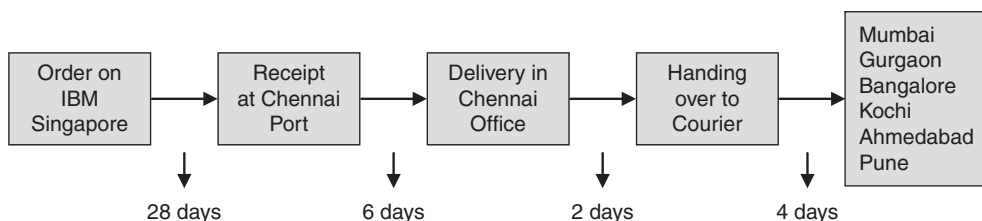


Figure 1 Material Flow – Old Arrangement

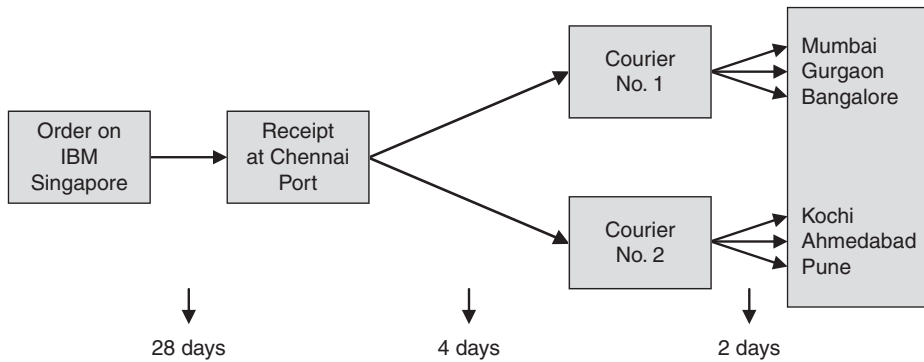


Figure 2 *Material Flow – New Arrangement*

REVIEW QUESTIONS

1. Explain the term 'optimization' in connection with supply chain.
2. How did Zapak achieve optimization in delivery of laptops?
3. What are the benefits Zapak can derive in delivery optimization?

INDIAN PAINTS

Innovative Supply Chain Practices

Indian Paints is a Rs.11,000 crore sales turnover company, with five manufacturing locations and over 60 sales branches spread across India. They have an extensive distribution network of over 14,000 dealers and a product range of over 2,500 SKUs (stock keeping units) and is supported by the world's first state-of-the-art IT solution to SCM (supply chain management) from i2 Technologies, and an ERP (enterprise resource planning) solution from SAP. Indian Paints is a supply chain pioneer in India in paint industry.

IP manufactures and markets a wide spectrum of coatings which include decoratives, production paints and heavy-duty coatings. The manufacturing facilities of the company for paint products are currently spread over four locations in Maharashtra, Gujarat and Andhra Pradesh. Indian Paints offers the widest range of paints in terms of products and shades, as well as pack sizes. IP manufactures and markets more than 2,500 items of paints. IP has four popular brands.

Distribution Strategy

In the decorative paint segment, a wide product range in terms of colour and pack size was a crucial factor for success. IP has a competitive edge of offering the widest range of products. It also created the distribution channel for reaching the wide range of products to customers in every nook and corner of the country. IP has used the new technology to offer any colour shade to the customer through mixing at the retail outlet using an automated machine installed at the retail outlet. The desired shade is prepared by mixing different shades and stainers in the required proportion. IP is now maintaining only half-a-dozen basic colourants with retailers; mixing can create the other variants. The new arrangement helped IP to manage with a narrow range of paint ingredients. Thus, the number of SKUs handled has drastically reduced resulting into cutting down on inventory holding costs.

The new technology reduced the importance of the physical distribution task in the business, compared to the position in the earlier years. In the earlier years the physical distribution and channel management task was the most crucial one in paint marketing. Even now, physical distribution and channel management continue to be crucial functions in the business.

Distribution Challenge

Both physical distribution and channel management posed formidable challenges in paint distribution. In the earlier years, the distributors concentrated on big cities, where they could make the sales without much investment in distribution infrastructure and market development. The distributors controlled the paint business and were making it impossible for a new paint company like IP to enter and establish in the business. IP studied the then scenario correctly and formulated a unique distribution strategy of entering in rural market which was not tapped. IP bypassed the bulk buyer segment and went to individual consumers of paints. IP went slow on urban areas and concentrated

on semi-urban and rural areas. IP went retail. IP went in for an open-door dealer policy. IP went for nationwide marketing and distribution.

Challenges

IP's decision to serve the individual consumers resulted into maintaining a wide product range. The individual consumer segment involves a very wide choice in terms of products, materials, shades and pack sizes. This forced IP to maintain a large inventory. Naturally, distribution was becoming more complex and expensive for IP. IP had to go in for more dealers in order to serve the scattered semi-urban and rural market.

An extensive network of dealers and a matching physical distribution infrastructure play a crucial role in the decorative paints segment. This is essential for ensuring easy access of the product to customers. In this, IP scored over its competitors with a massive network of 14,000 dealers spread over 3,500 towns across the country.

IP established a large chain of company operated depots/stock points throughout its vast marketing territory, from where the retail dealers could conveniently pick up their requirements. After all, IP did not have any wholesale distributors to whom the responsibility for operating the stock points could possibly have been assigned. IP established a network of 30 company-run depots, spread throughout the country and serviced its retailers from them. The number of depots varied from city to city. For example, Bangalore had just one depot while Mumbai had four. The depots typically supplied material to about 200–300 dealers. These depots were controlled by 4 regional sales offices, 35 branch sales offices and a large number of sales supervisors and sales representatives spread all over the country.

With the extensive distribution network, IP's service level is more than 85 per cent whereas that of other large paint companies falls between 50 per cent and 60 per cent. This meant a further rise in IP's physical distribution costs. A high service level in physical distribution—in transportation, warehousing, order processing and inventories necessarily means a high level of costs. While following a totally customer-oriented distribution strategy, IP could not afford to ignore the cost angle.

Effective Inventory Management

Effective inventory management is the first major component of IP's strategy on distribution cost control, and IP achieved high efficiency in this regard using IT solutions. IP's average inventory level equals only 28 days sales, while the industry average is 51 days sales. This right away provided a 45 per cent edge in inventory costs to IP compared its competitors. IP's stock of finished goods was just 7 per cent of its net sales while for the others in the industry it was nearly twice that level. IP could provide a high level of service in product delivery to its customers and dealers as compared to its competitors and yet kept the inventory costs down by 45 per cent compared to its competitors.

IT Initiatives in Cost Control

IP's IT initiatives in respect to distribution-inventory control and control of receivables helped control distribution costs without lowering the service level. IP went in for a fully computerized distribution system. IP did this not only with an eye on distribution cost control, but also for the sake of distribution effectiveness per se. The 2,000 different items of paints, manufactured at four different plants, had to be distributed to 15,000 dealers in 3,500 towns spread all over the country, through 55 depots. IP accomplished this, maintaining the average service level at 85 per cent, a clear 25 per cent above that

of competition. The IT initiatives ensured prompt billing, accurate customer accounting and effective control of credit outstanding.

IT solution enabled IP to process recent sales data for the 100 fastest moving SKUs. This analysis was used to project sales of specific products, which helped plan production and raw material purchases. With computerization, IP was able to analyse past trends to arrive at a 90 per cent accurate sales forecast. Production was thus evened out month-to-month. IP linked all its factories and depots through the IT network resulting in a streamlined distribution. IP has implemented the supply chain management software and also an ERP solution from SAP.

IT-enabled Supply Chain

Indian Paints deployed an extensive IT infrastructure to integrated supply chain using i2 technologies. They used the enterprise resource planning (ERP) solution from SAP. IP has streamlined procurement, manufacturing and distribution processes for its decorative paint business. They used i2 technology solution for production planning, distribution planning and demand planning for 2,500 SKUs. The i2 technology solutions helped IP to manage the following business processes:

- Master production planning
- Material planning
- Distribution planning

The i2 Factory Planner was designed to enable IP to quickly react, synchronizing factory production plans and schedules to meet business challenges. It also assisted IP in performing efficient material planning by providing extensive modelling capabilities to solve problems in the entire material planning process domain. The i2 Factory Planner enabled IP to perform the following:

- Immediately resolve potential supply chain problems
- Create optimal manufacturing plans against capacity constraints
- Provide accurate due-date quoting to improve customer service
- Create powerful time-based report summaries
- Create powerful time-based report summaries and time to value

The i2 demand planner helped IP to estimate the demand in a more accurate way. The i2 supply chain planner provided a way to model the entire supply chain—from suppliers and outsourced manufacturers to factories, warehouses and distribution centres. The supply chain planner helped IP to optimize inventories, production, distribution and transportation. The supply chain planner was designed to support master planning, allocation planning, distribution planning, inventory planning and profit optimization.

IP benefited by using the i2 supply chain planner by way of reduction in lead times for responding to demand variability, improvement in service-level performance, customer satisfaction, improved usage of resources, improved inventory management and reduction in overall logistics costs.

REVIEW QUESTIONS

1. What are the supply chain issues and challenges faced by Indian Paints?
2. Discuss the various innovative supply chain strategies evolved and implemented by Indian Paints.
3. Explain the role of IT in Indian Paints supply chain.

MANN DIESEL SALES AND SERVICES

CASE

7

Supply Chain Initiative in Spare Parts After Market Support

Mann Diesels Sales and Services (MDSS) manufactures a variety of engines from 60 to 2500 HP capacity range. Mann Inc. serves its customers worldwide through more than 500 company-owned and independent distributor locations in 100 countries and territories and a network of 4,500 dealers and has facilities in 45 countries.

MDSS provides after-market support to customers for all Mann products that includes: in warranty and post warranty, engine repairs and re-con exchange of engines, re-con components and repairs of engine components, supply and maintenance of spares under operations and maintenance (O & M) contract and lastly the supply of allied products such as monitoring and power generation systems. MDSS has worked continuously to fulfill the needs of customers.

MDSS serves its customers through a network of 5 zonal offices, 15 field offices, 8 spare parts depots and 75 dealers having 120 service locations. The services extended are warranty claims, engine repairs, re-con exchange of engine, spare parts supply, annual maintenance contract (AMC), monitoring power generation system and supply of allied products.

MDSS has three input channels having different types of supply behaviour, lead times and capabilities. It received around 80 per cent of the supply in the form of manufactured (35%) and locally purchased spare parts (44%) with the rest of the 21 per cent being imported by local and overseas suppliers. Further MDSS has 4 different output channels with a value of 24 per cent, 52 per cent, 13 per cent and 11per cent supplying to 8 depots, 75 dealers, direct customers and MDSS service/region, respectively. Dealership stocking exhibited the worst supply performance. Also, supplier-delivery performance was low, which led to frequent stock-outs at the main warehouse. The responsibility of customer satisfaction was on dealers and depots. Prior to implementing the SC initiative, MDSS made an inventory of 109 days. The supply was unpredictable and inconsistent. The customer faced longer breakdowns of equipments, longer waiting period for the arrival of spare parts, higher investment in inventories and poor service level. These factors contributed to the loss of customer base to competitors.

To resolve the problem, a team of representatives of MDSS and channel partners was formed. The team included executives from inbound sourcing/materials, manufacturing, planning, warehouse and field office from company's side, and dealers and customer representatives as external participants.

MDSS then segmented spare part items (9,500 SKUs) based on volume, value of sales and service level. They categorized the parts into A, B and C types. They have also re-defined the service level as ex-stocks or against order. Customer break points, that is, all regular maintenance and running repair parts, should be available ex-stock from dealership point and they can be ordered 7–15 days in advance. In addition, the supply chain is expected to service emergencies within 2 days. New process design suggested by SCM team, that is, weekly sourcing and shipment to dealers for A and C category of all parts, as per the segmentation was implemented. A framework was made to set target inventory equivalent to 3 weeks of average consumption at each dealership point, where dealers were expected

to report about inventories on the specified day of the week, creating POS (point of sales) information available at MDSS. They used weekly bucketed information of sales for generating a forecast from desktop tools with lead time for A, B and C category parts. The safety stock levels for all parts were calculated every week, considering the variability during the last 30 weeks. For releasing firm orders, lead time was scheduled in the target week. Also, the next 4 weeks' forecast was worked out providing visibility for 4 weeks. They started using a new measure called CLIP (committed line item performance). CLIP is a binary indicator of whether or not a line has been fulfilled to the extent of 90 per cent plus target quantity. KPI measuring various CLIP parameters related to suppliers, order processing, dispatch and forecasting were fixed up.

In the warehousing and logistics, they started scheduling allocations on a daily basis, and agreeing on fixed delivery printing time played a major role in putting in place the picking and packing in right boxes and movements to shipping area. They introduced automated storage and retrieval system (ASRS) at the warehouse to improve warehouse efficiency.

In outbound logistics MDSS undertook a Six Sigma project to capture the increased opportunity in logistics and distribution network. The project investigated and analysed the total cost of shipment, time taken for covering the distance and consistency of performance. The company felt that transportation by a few agencies performing nationwide deliveries was a better option than individually nominated transporters.

In addition to the above, MDSS installed ERP as a part of its capability enhancement efforts. This system was multi-organizational and MDSS was one of the organizations included in the programme. ERP was launched for MDSS while MI was facing a dual challenge of implementing ERP as well as scaling up of SCM solutions.

REVIEW QUESTIONS

1. Map the spare parts supply chain of MDSS.
2. Discuss the problems faced by MDSS in its spare parts supply chain.
3. How were the problems sorted out through the supply chain initiatives?

PADMINI MOTORS LTD¹

A Lean Supply Chain Through 'Just-in-Time' System

After the liberalization of the Indian economy in 1991, the auto sector was one of the first few industries to face the stiff competition from the foreign MNCs, which started putting their foothold on the Indian soil to tap the green market. The auto manufacturers lost their government protection umbrella and had no other alternative but to remain competitive by developing cost-based or differential-based competitive strategies. Within the short span of 5–6 years, the market was flooded with world-class passenger car models and multi-utility vehicles produced in India by the world auto giants like, Ford, Mercedes, Hyundai, Toyota and so on.

Company Profile

Padmini Motors Ltd (PML) was amongst the old Indian veterans having started manufacturing auto vehicles way back in the 1950s having its first plant in Mumbai manufacturing a sturdy vehicle jeep in a single model to suit the poor road conditions prevailing then. Padmini had a virtual monopoly in jeep markets commanding 85 per cent of market shares till 1991. After liberalization the scene changed. It became buyers market rather than sellers' market, which was prevailing prior to 1991. One world-class manufacturer and one old Indian player entered in Indian market with a large number of multi-utility vehicle models, giving more choice to the Indian customers. In the GPL (globalized, privatized and liberalized) economic environment Padmini lost its monopoly and its share reduced to 55 per cent. Two other players shared the balanced market equally. In the changed competitive environment, Padmini observed the following:

- Heightened customer expectations
- Faster new product introduction
- Shortening of product life cycle
- Intense competition race for market share

To remain competitive and arrest the declining market share, Padmini adopted both cost-based and differentiation-based marketing strategy. However, for the cost-based strategy Padmini decided to focus on its supply chain to take cost out so as to make supply chain more lean.

Padmini is currently manufacturing around 60,000 multi-utility vehicles per annum. The product range covers 150 models. The firm has three main assembly plants of which two are located in Maharashtra and one is located in Uttar Pradesh. The assembly plants are supported by four sub-assembly plants exclusively dedicated to engines (two), axle and transmission gears. The firm has a vendor base of over 900 suppliers for 7,000 parts. The vendors are located in and around, Mumbai, Delhi and Vapi. Padmini is marketing its products through 20 area offices, which are controlling the network of 160 Padmini dealers spread across the country.

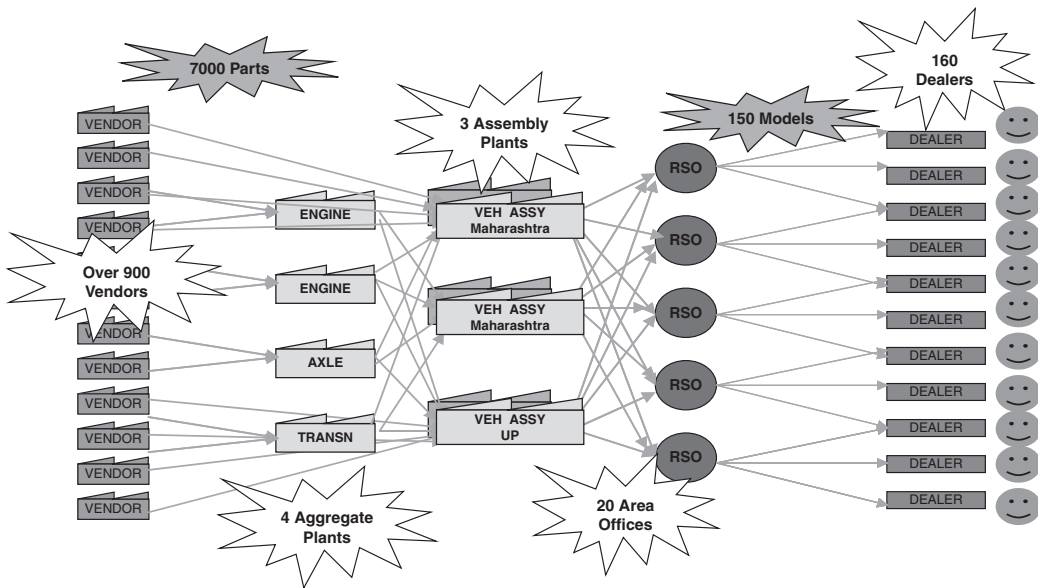


Figure 1 PMCL Supply Chain

Padmini employees a local work force of about 2,000 skilled personnel intensively trained in their areas of specialization. The company endorses equal opportunities and encourages women's participation in the organization's growth. The company has exports contributing to 10 per cent of its sales turnover. The Padmini products are well accepted by the local and international buyers because of its world-class quality and the back up after sales service offered by the company to its clients.

Paradigm Shift

The company has gained its reputation over the consistent efforts in making available to its clients the quality products and a variety of models to suit their requirements. The real efforts started after the liberalization when the company got the first jolt by way of substantial reduction in its market share in subsequent years. To counter the competition and remain differentiated, the company focused on its supply chain (Figure 1). It adopted the concepts of JIT (Just-in-Time) based on the 'Toyota Production System'.

Traditionally the company was planning its production on the forecasts, which was based on the customer demand trends observed in the past. The prevailing practices were as follows:

- Monthly requirements of sales based on forecast
- Monthly production plan based on requirements of sales
- Production plan translated into schedules
- Vendor schedules based on production plan
- Frequent changes in production plan due to demand variations
- Corresponding changes in vendor schedules

To Create a Demand Pull

The push system, which was based on forecast, proved wrong due to change in the customer's taste, and the demand pattern and competition. The production planned on the basis of forecasts was either inadequate or in excess of the actual demand. This resulted in excessive inventory to meet the desired level of customer service. The products thus manufactured were pushed through the dealer's network to sell. To get over this problem the company decided to adopt the principle of 'sell one make one' and go in for the pull system so as to let production be demand driven rather than forecast driven. They integrated its distribution, manufacturing and procurement operations by using latest IT tools and connectivity equipments based on the latest technologies. The firm now had all the supply chain participants on line with each other with proper information sharing security policy. The current practices under the newly introduced pull system are as follows:

- Replenishment action to start from dealers
- Dealer sales replenished by regional sales office
- Sales from regional sales office replenished from the fresh production

The firm had reviewed its product range and regrouped its products into two categories, that is, 33 models contributing to 88 per cent of the company's sales and 117 models contributing to 12 per cent of sales (Exhibit 1).

It was further decided that only FMG would be carried in the buffer stocks, while the SMG models will be strictly manufactured against the firm orders. The buffer quantity for replenishment worked out, considering the following:

- Forecasted average retail sales
- Lead time variable factor (considering logistics slipups)
- Demand variable factor (seasonal, short-term and regional pattern)

Thus getting daily orders from the dealers for FMG and SMG models created the demand pull. With the buffer maintained at the depots, the dealer's requirements for FMG models were fulfilled within 48 hours from the regional offices. The weekly orders (firm) from regional offices were sent to plants on both FMG and SMG models for replenishment.

Planning and Scheduling

To support the field requirements, Padmini stressed on the planning and scheduling of the manufacturing activities. The firms adopted a system of preparing and rolling out the quarterly sales forecast

Exhibit 1 *Product Groups*

	<i>No. of Models</i>	<i>% of Total Variety</i>	<i>% of Annual Sales</i>
Fast-moving group (FMG)	33	22	88
Slow-moving group (SMG)	117	78	12
Total	150	100	100

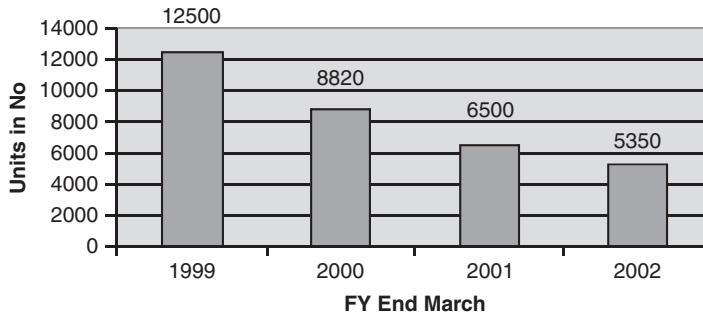


Figure 2 *Finished Goods Inventory*

for planning its manufacturing capacity and resources and preparing its monthly production plans. The firm has clear distinctions between the manufacturing planning and scheduling. The planning was for capacity booking and organizing resources while the scheduling was for resources allocation and actual physical production of the product on the shop floor. From the traditional practice of monthly production schedule Padmini had shifted to weekly production schedules to meet the replenishment at regional sales offices. The weekly schedules were further refined by the daily replenishment requirement from the regional sales offices, which were further getting the replenishment requirement from the dealers on a daily basis.

Flexible Manufacturing

To support the pull system, Padmini had further adopted and implemented the concept of 'let demand drive the production' with the following objectives:

- Hundred per cent order fulfilment
- Reduction in order-to delivery time
- Minimum pipeline inventory

To attain the objectives, the company had identified four key thrust areas such as manufacturing flexibility, synchronous production of aggregate, increase in frequency of ordering, and speed, frequency and automation of the information-processing system. The company heavily invested in latest IT tools like EPR and installed SAP for supply chain coordination on real-time basis. The company had gone for manufacturing flexibility by regrouping its critical assemblies/components (Exhibit 2) into two categories, that is, stocks and make-to-order.

Exhibit 2 *Critical Assemblies/Components*

<i>Assembly/Component</i>	<i>FMG</i>	<i>SMG</i>	<i>Total</i>
Axel	9	29	39
Transmission	9	24	33
Engine	14	22	36

Exhibit 3 *Vendor Base for Milk Run*

<i>Zones</i>	<i>Vendors</i>	<i>Parts</i>
Nashik	27	325
Delhi	30	180
Bangalore	18	176
Chennai	20	165
Pune	17	90
Daman	10	65
Total	110	1,194

For all the assemblies and components the delivery lead time was planned as 24 hours. However, the buffers were maintained for FMG assemblies and components and for SMG, it was decided to manufacture the same against actual orders. The firm went for daily order processing for faster fulfilment of orders as against the weekly earlier. The KANBAN system was introduced to have the speedy replenishment at the assembly line from the sub-assembly and components stores.

Material System

To support the flexible manufacturing, the demand pull was further extended to vendors. The material system was devised to ensure the availability of all materials to meet the scheduled production of the sub-assemblies and vehicles with simultaneous reduction in acquisition costs. Padmini used the latest tool for inventory planning and KANBAN for material scheduling. Over 90 per cent of material was put on the KANBAN replenishment system for speed, accuracy and reliability.

Padmini has a wide base of 900 vendors spread across the country. For the reliable and speedy deliveries the entire inbound logistics activities were outsourced to 3 PL firm for transportation and storage at various points. The firm divided its vendors into seven zones (Exhibit 3) for its plants. The 3 PL firm was responsible for the daily milk run (in each zone) to collect the material as per the schedule to either send it to the plant warehouse in case of the full truck load or keep it at the hub warehouse in the zone for getting the truck load filled for dispatch to plants.

The limits for stock levels at the hub was always 1 day and daily shipments were planned from hub to plants with limitation on transit time (5 days from Delhi to Nashik or other plants) to meet the JIT requirements at the plant.

Achievements

As a result of supply chain initiative Padmini achieved its strategic objective of cost leadership. The firm's products are price competitively in the market and command comfortable margins as compared to the competition. The inventory-related cost is reduced to more than half as compared to the level 4 years back. The average finished goods inventory level has come down to 5,350 in 2002 from 12,500 in 1999 (Figure 2). For differentiation-based objective, the firm has a very wide product range of 150 models for the customers to choose from and is supported by the network of the countrywide service centres for the after sales service. With the enhanced service level backup, the customer satisfaction

level is much above as compared to the nearest competitor, which is half of that enjoyed by Padmini. Even though the market share of Padmini did not grow, the supply chain management initiative helped the company to improve its profit margins and customer service.

REVIEW QUESTIONS

1. Compare and contrast the SCM and traditional approach of Padmini to service the market.
2. Discuss the role of 3 PL partner in meeting the strategic objectives set by the company.
3. What are the system inputs Padmini had planned for the SCM initiative to be successfully implemented?
4. For flexible manufacturing what are the challenges and limitations?

NOTE

1. This case is adopted from Sople, V. V. (2010), *Logistics Management*. Pearson Edition, New Delhi, 2nd edn., pp. 444.

DIGITAL STORAGE (INDIA) LTD

CASE 9

Leveraging Innovative Supply Chain Practices to Competitiveness

Digital Storage (India) Ltd (DSIL) is an Indian manufacturing company principally engaged in designing, developing, manufacturing and selling of a variety of storage media products such as compact disc, magnetic disc and storage units for its clients across the globe. The company is also in the manufacturing of various solar products such as crystalline silicon cell and modules. The company has its operations across 82 countries in the world including the United States, Japan, Netherlands, Taiwan, India, Cyprus, Slovakia, Germany, etc. DSIL has its headquarters in New Delhi, India.

The company reported revenues of Rs. 21,769 million during the fiscal year ended March 2008. Digital Storage was established in 1983 and successfully developed cutting edge technologies to become the world's second largest manufacturer of optical storage media like CDs and DVDs. The company also emerged as the first to market the next generation of storage formats like blu-ray discs and HD DVD.

DSIL has the distinction of being the preferred supplier to all top global OEM brands. DSIL's products are manufactured at its three state-of-the-art manufacturing facilities. It has over 6,000 full-time employees and multiple manufacturing facilities in the suburbs of New Delhi.

DSIL owns a comprehensive logistics and supply chain system enabled by an in-house IT system, which is integrated with freight forwarders, major shipping lines, customers and suppliers at the back-end. DSIL leverages this integration to respond with celerity to changes in customer expectations and market conditions. As one of the major containerized cargo shipping companies, DSIL has reduced its shipment lead time by more than 20 per cent in the last few years; DSIL also brought down the instances of delays in shipments to less than 0.1 per cent of the total despatches. Currently the company has a market share of 14 per cent and enjoys a long-standing relationship with global majors that has enabled the company to penetrate into over 40 countries.

Supply chain is the area where DSIL focused its attention to cut costs and enhance customer satisfaction. As DSIL was rapidly expanding its presence overseas, in 2001 it increasingly felt the need for a robust and efficient supply chain to ensure that all its products are delivered to its overseas customers on time. DSIL's IT-enabled SC has helped to overcome problems pertaining to its deliveries. IT-enabled supply chain is critical in bringing value to DSIL's customers through on-time and correct deliveries. In 2002, DSIL's growth made it imperative to develop a robust and integrated SC management. The purpose was to ensure that there were no errors in documentation and deliveries to customers. These systems were meant to add value and differentiate between the services provided by the firm and that of other competitors.

Today IT-enabled SC has helped DSIL to overcome problems pertaining to SKU identification and value-added tax (VAT) management. It has also facilitated zero-error deliveries to customers and efficiencies of operating teams, thereby providing a platform that has been used to integrate the system with those of customers. The system after its implementation drastically improved the working of SC.

DSIL has another initiative called 'the round robin movement'. This initiative was aimed at synchronizing inbound logistics and outbound logistics. DSIL's exports (80%) were routed through the Rotterdam port. Imports of shipments originating in Europe and being shipped through Antwerp and Rotterdam were carried out in 20 ft shipping containers. Due to a substantial market share in Europe, above shipping lane was critical to DSIL's distribution operation.

MBIL optimized its exports by using 40 ft containers with two-way traffic, moving on to same trade lane. It identified an opportunity where combined volumes and assured freight could be leveraged. DSIL had another supply chain initiative that involved a unique concept called 'landing legs'. Being an export-oriented unit, DSIL's exports were comprised of 90 per cent of their production with a majority of inputs being imported. DSIL's facilities are located in Noida, UP, about 1,500 km from the nearest gateway port. DSIL used to operate its export and import transactions through an inland container depot at Tughlakabad near New Delhi. However, since the new ICD at Dadri (UP) was closer to the DSIL plant in Noida, DSIL decided to shift all its imports and exports operations here.

The transportation of containers from ICD at Dadri to the DSIL plant became a challenge. DSIL's solution was 'landing legs on trailer'. It is a unique system that involves carrying its own fleet of trailers. Though DSIL faced teething trouble during the implementation of this novel concept, the project has been successful. DSIL is continuously working on innovative practices in SC to cash in on the opportunities to leverage its competitiveness.

REVIEW QUESTIONS

1. Discuss the various supply chain initiatives at Digital Storage (India) Limited.
2. How did innovative logistics practices leverage DSIL's competitiveness?

ZARA

Global Supply Chain Keeping Ahead of Competition

CASE 10

Zara is a clothing brand of garments for men, women and children. The brand is present in 77 countries and has a network of 1,608 shops (in 2009). Zara relies on its creative team of more than 200 professionals.

Zara gives high importance to the customer. Each employee pays attention to the customer's desire and tries to answer it as soon as possible. Also, the brand is able to adapt the product to the demand in a very short time compared to competition. A specific system enables this flexibility, that is, the vertical integration. The Spanish brand controls all the steps of the fashion process: design, manufacture, logistics and distribution to its own shops, which are also managed by the firm. Concerning the design, the Zara flexibility system allows adapting the offer to the demand rapidly. Over 200 designers are able to provide new models for each season based on information received from the shops. On an average, they design more than 40,000 products each year of which 10,000 are produced. That is why the designers' teams are constantly working on both current (updating of the old designs) and future collections. The designing process is very unique. Designers draw out a new product by hand and then talk about it with their colleagues (other designers, market specialists, planning and procurement people). It helps to have a global view of the product and to get a feedback of it. Then this product is drawn with a software, which allows making some changes and adjustments, to adapt materials and colours. Some samples are then made and shown in the office so that everyone can give suggestions and the problems can be noticed.

Each market specialist deals with a specific area of stores and should establish a personal relationship with store managers because they are constantly in touch with them to discuss new trend and orders. The final decisions on the products and new designs are taken by the group of designers, market specialists and buyers. Finally, the buyers are in charge of the total order fulfilment process which covers planning procurement, production requirements, monitored warehouses inventories, allocated production to different factories and third-party suppliers and track of shortages and oversupplies. The major part of the production (50 per cent of its products) is made in the network of 22 Spanish factories. They manufacture the most creative garments. The other half part is procured from 400 suppliers (in Europe and Asia). The group takes care of the fabric supply, marking and cutting and the garment's finishing.

Logistics is a key element in Zara strategy. The distribution of the production to a logistical centre is made according to the origin. Each store receives the products twice a week that allows providing new models to customers. Zara has four distribution centres in Spain: Arteixo (La Coruña), Onzonilla (Leon), Meco (Madrid) and Zaragoza.

Each product passes through the major distribution centre located in La Coruña, which employed 1,200 people. The own team designed the logistic software, which enables to deliver a product with a delay of 24 hours in Europe and a maximum of 48 hours in America or Asia. Each order of a store is

packed in one box and is ready 8 hours after the order receipt. For Zara, distance is measured not in kilometres, but in time.

At Zara 3,00,000 SKU's are made available in the market every year. Eighty per cent of the production is fashion garments, while the rest are more basic products. The major part of the transportation in Europe is made by trucks. International transportation is made by plane from La Coruna airport or Santiago airport.

The retailing is a key process in Zara. The stores have to place orders at a specific time; for example in Spain and in Southern Europe it is done on Wednesdays before 3 pm; in the rest of the world on Tuesdays by 3 pm and Fridays by 6 pm. If they miss their deadlines they have to wait for the next week. In fact, Zara keeps this strategy in order to minimize the risks of oversupply by retaining a low production volume at the beginning of the season and then during the season, reacts quickly to order. Concerning the pre-season inventory commitment, Zara keeps the level of production and procurement (in July for fall/winter season) between 15 per cent and 20 per cent, while in the industry it is around 45–60 per cent. And similarly, the in-season commitment at Zara is 40–50 per cent and in the industry it is a maximum of 20 per cent.

Zara's strategy in the supply chain is to work with relevant and precise information and forecast in order to react rapidly. The point of sale is also an important place for Zara, because it allows giving a feedback to the designers and to have some information on the trends asked by the customers. Zara does not have a lot of return of clothes from the client. Everything is done to make customers comfortable within the store to enable a link between the garments and the clients. The design and the organization of the store are changed every 4 years to be more creative and innovative.

Managers decide of the location of the garments inside the store in order to sell some products in priority. Zara uses stores and word-of-mouth to do the advertisement of the brand (only 0.3 per cent of the sales turnover is allocated on advertising). Concerning prices, this decision is taken by the head office which decides for prices applied in all the stores. Zara is one the garment supply chain wherein they have a low level of inventory as compared to its competitors. This means that the store's entire stock is on display. The stores have a benefit of rapid replenishment of new garments from the factories. This has a positive effect on the sales because it influences the customer to buy the product on the spot.

Zara tries to work by using a vertical supply chain. This is an uncommon strategy in the fashion industry. Vertically integrated business undertakes a variety of activities from designing, manufacturing, sourcing and distributing to retail stores around the world. When a company works with a vertically integrated system, it has a control over its business activities. When its competitors outsource, Zara favours local production. The brand, therefore, controls its entire production chain (design, production, logistics and sales) with the aim to increase flexibility and responsiveness. Eighty per cent of products are manufactured in Europe, 50 per cent in the 14 factories. The rest of the manufacturing is outsourced from cooperatives or plants located nearby. Zara is implementing an integrated production strategy, that is, dyeing and printing of manufactured parts are held in the factories of the brand at the last moment, according to fashion and demand (personalization). This is also called postponement strategy.

The heart of Zara's SC strategy is the logistics centre of Corunna. Every year, more than 120 million units are shipped from this warehouse. Quality control, ironing, packing and labelling items is done here. At HO, a team of 120 designers works on 11,000 models per year. Marketing actions are done via a team located in La Coruna and connected to the team of stylists. The communication aspect is very less; it represents 0.35 per cent of net sales of Zara, against 3–4 per cent on average for large retail clothing. Thus, the brand does not work on advertising and does not provide services for online sales.

Zara has established a direct communication system between all business functions through a fully automated information system. This communication allows real time information from all the stores on sales figures and customer feedback. The production is managed by the La Coruna centre and is directly related to the creation and a quick delivery of products between stores and the creation of models to meet customer requirements.

The supply chain of Zara is pervasive and directly linked to the creation of value. The brand continues to distinguish itself from its competitors by identifying emerging trends in the world of fashion, and making available quickly the new models at an affordable price. The whole culture of organization and information system of the company is geared towards this goal. Daily needs of production and allocation between stores and promotions are well adjusted, while the design team adapts models (colours, details trends) based on sales.

Real-time information coupled with innovative supply chain strategies is key to Zara's global supply chain which transformed Zara into a major international fashion company with the brand known world over. The strategy developed by Zara is to renew the shelves of shops every 15 days. For this, the company controls the entire supply chain. The organization of Zara is highly centralized at the Headquarter of La Coruna, with a minimum of functions distributed in each country, where the brand is present. Zara's business model also relies on a strong vertical integration from production to distribution, through to logistics and marketing. The entire value chain is controlled in order to control costs and increase responsiveness.

REVIEW QUESTIONS

1. Map Zara's supply chain and explain how it is helping Zara to create value for its customers.
2. Discuss the various practices in Zara's supply chain.
3. What are the differentiating factors in Zara's supply chain?

ECON INDUSTRY

Strategic Procurement Initiatives

Econ Industry (EI) is a leading parts and accessories manufacturing company supplying to the automobile majors in India. Econ produces parts made out of engineering plastics. The company's strength is its R&D wherein they spend a large part of their revenue for developing plastic parts. They procure raw materials, that is, chemicals for formulation for metal substitutes. Econ is purchasing raw plastics in the form of nodules and with the right process they mould the parts as per the requirements of the customers (OEMs).

Econ is facing a lot of competition from small-scale industries. However, because of its high-quality products and delivery performance, it is a preferred supplier to all automobile manufactures in India. To be more competitive and cost-effective, Econ has gone for new supply chain initiatives. They first focused on the procurement and subsequently move to the manufacturing and finally to the distribution process. The aim of new SC initiative was to implement JIT in its SC process.

Initially, they applied Pareto's law, that is, 80/20 rule to its inbound inventory and finally decided to concentrate on five of the inputs which are high-value and also critical in nature during manufacturing. These items are required for faster moving products and resulted in higher inventory velocity. Econ has a wide network of suppliers; the critical materials were sourced from a limited number of them. In those case contracts, bids were a ritual, with the winner known well in advance. Econ reviewed the service performances of the suppliers and wanted deliveries within 10 days from the date of order. Econ decided to have on-time delivery with plus or minus 2 days from the scheduled delivery date. This decision meant a 5-day service window. Econ wanted a minimum service within the 5-day window over 93 per cent fulfilment.

The manufacturing department wants a mixture of plastics as raw material. Econ's manufacturing process requires five major raw materials. The company's current procurement policy is to source each of these compounds from major suppliers through an annual rate contract process. Econ traditionally considered the best offer on the lowest price. But due to the new guidelines, the suppliers are to be selected on the basis of lead time on delivery performance and order fulfilment rates. The analysis of past data on suppliers' performance is shown in Exhibit 1.

Econ has now evolved the performance criteria of delivery within 4 days with a margin of 2 days further with the minimum fill rate over 93 per cent. Econ has a very flexible manufacturing process and takes up batch processing of different components without any time delay.

In order to avoid material shortages and thereby maximize production, Econ is maintaining a 6 days' inventory for each of the raw material. Econ wants to slowly change over to the JIT system. The company has recently implemented the ERP system. Marketing/sales department receive the customer orders online. This allows the manufacturing department to monitor the incoming materials' shipments as well as schedule the production runs. Under this system, most of the customer orders are produced within 5–7 days of the order. The material, after being manufactured, is sent to the distribution department for onward dispatch to the respective customers.

Exhibit 1 Suppliers' Performance

<i>Suppliers</i>	<i>Ingredients</i>	<i>Criteria</i>	<i>Raw materials</i>			
		<i>"α"</i>	<i>"β"</i>	<i>"μ"</i>	<i>"π"</i>	<i>"∞"</i>
Supplier A	Share	55%	55%			15%
	Delivery and fill rate	3–7 days 93%	2–6 days 94.5%			4–5 days 92%
Supplier B	Share		25%	40%	45%	
	Delivery and fill rate		2–4 days 97.4%	2–3 days 98%	3–5 days 96.7%	
Supplier C	Share	45%	20%			25%
	Delivery and fill rate	2–4 days 97.5%	2–3 days 98%			3–4 days 97.8%
Supplier D	Share			60%	55%	
	Delivery and fill rate			4–5 days 96.5%	2–3 days 97.5%	
Supplier E	Share					60%
	Delivery and fill rate					2–4 days 98.3%

Three to four days normally go by from the time the material leaves the manufacturing plant till it is shipped from the warehouse. The 80 per cent of Econ's customers are within 250 km from the plant. Econ now wants to select the right suppliers for the raw materials, keeping in mind the new selection criteria.

REVIEW QUESTIONS

1. Map Econ's supply chain indicating the value-added activities.
2. Using the new performance criteria, select the suppliers and allocate the quantities to be procured from them.
3. Discuss the scope of performance improvements at suppliers' end.

AFCON WATCHES

Tuning the Supply Chain to Changing Times

Afcon started in 1980 and today, is one among the top five watch brands in the world, manufacturing well-known brands of personal accessories, jewellery, sunglasses and prescription eyewear as well. Afcon has been instrumental in taking an Indian product of superior quality and trendy design to the international market.

To be successful, Afcon had to tune its supply chain to keep pace with the changing dynamics of the consumer durables industry. The company had to overcome many complexities in its supply chain for watch components and assembly, with more than 3,000 variants of watches, more than 15,000 finished and semi-finished components, multiple manufacturing locations and multiple distribution channels. The multiple distribution channels comprise the World-of-Afcon retail stores, re-distribution stockists, more than 11,000 multi-brand watch outlets, over 250 large format stores like Shoppers' Stop, Life Style, etc., and more than 100 direct dealers through more than 30 warehouses across the country. By continuously reviewing and refining sourcing, manufacturing, supply, warehousing and distribution processes, Afcon maintained its strategic position. However, in order to maintain this, the company adopted unique strategies to handle vendors and the complex supply chain requirements.

Procurement

Afcon is having a huge base of 600 suppliers and around 70 key suppliers of critical components like dials and straps, components and various raw materials. This covers stainless steel, leaded and lead-free brass for in-house manufacturing and specialty watch movements. It was critical for Afcon to facilitate an effective communication channel and a platform to analyse the capacity of vendors to plan its production schedules. In order to streamline information transfer and communication, the company implemented a vendor portal which acts as the main interface between the vendors and the sourcing team at Afcon. They have weekly delivery schedules. The vendors are intimated about the orders through the vendor portal itself. Besides simplifying the communication process, Afcon ensures that its vendors and suppliers meet the quality parameters set by the company. The quality control team visits the vendors periodically and holds training sessions for vendors' employees from time to time.

Distribution

Afcon is serviced by 33 clearing and forwarding agents (CFAs) and is also regularly upgraded by the timekeeper. The network up to the CFA level is managed by an integrated supply chain group, which is responsible for making right products available at the right time, at the right locations, within constraints of costs, distance and capital.

Demand

Afcon implemented the SAP-ERP solution in the year 1999 for mapping the demand projection of 3,000 watch variants to the available manufacturing capacity. To manage this task, the company integrated all transactions on one ERP platform ensuring better visibility of information. This helped the company to check rising inventory levels and costs thereof. The ERP application helped Afcon to integrate business, transactions, improve alignment to the market and reduce inventories. In addition to the above during 2005, Afcon implemented an advanced planning system.

With SAP-ERP at the back end, Afcon implemented a solution to improve its demand forecast accuracy, product availability, feasible production scheduling and plant and supply capacity. The ERP solution is mostly used for transaction solutions. SAP advanced planning and optimization (SAP APO) handles the complexity of production, capacity constraints and a multi-tier distribution channel. Considering the inventories at various locations and production plans in the pipeline, and the critical in-house as well as vendor capacities, the SNP (supply chain network planning) optimizer created the weekly production plan. With the SNP function, Afcon has successfully created a feasible, synchronized and optimized supply plan in a short- to mid-term horizon. Its production plans have become feasible because capacity constraints are being considered, resulting in higher alignment with the execution plan.

Apart from SAP-APO, Afcon has also installed other technologies to streamline its supply chain processes. Afcon has implemented a deployment optimizer for dispatching finished products, a process that is run daily. There is a higher availability of finished products with the 33 CFAs across the country.

Manufacturing

Afcon's integrated manufacturing plant deals with very small and tiny components. In the watch division, the main concerns are identification, critical handling of delicate material and keeping a count of the number of stock-keeping units (SKUs). Being a small component industry, the parts are so minute that it is not possible to handle them manually. For instance, the accounting of minute components like crowns or screws or wheels is done using the ratio counter.

Most of the minute module components are wrapped in volatile corrosion inhibitor (VCI) papers with silica gel to protect the components from rust formation and corrosion. The company has a well-organized finished part stores with custom-made mobile racks, storage facilities and exclusive packing methods based on the need of individual components. Besides the existing temperature control arrangements, the company allocates separate storage areas for each component as well as the finished products. The finished watch store is divided for storing watches separately for the export and the domestic market. Separate storage is necessitated due to difference in packing methods.

Technology

In order to increase the responsiveness of the warehouse to changing demands, Afcon also uses the warehouse management system (WMS) solution. Using the RFID technology along with the barcoding technique, the WMS server gets updated with information in real time. The information is automatically updated in the WMS server specifying the product and the location. The WMS is a niche application and developed specifically to meet the requirements. This tool, improves the efficiency of the finished watch stores at all the other assembly plants and each 33 CFAs.

The WMS supports the process by giving the exact location of the product. With the help of its dynamic allocation of storage location, the warehouse can optimally use the available rack space and

identify the storage area of the product. WMS also helps in easier retrieval of components and helps in reducing person dependency.

Today, the market share of Afcon is around 10 million watches, around 60 per cent of the market share among the organized sector. This has been possible due to its streamlined supply chain operations, strong and dedicated vendor base, robust in-house manufacturing processes with assured quality and effective distribution system, well backed by marketing and retailing initiatives.

REVIEW QUESTIONS

1. What are the problems faced in the supply chain of wrist watches?
2. How did Afcon transform its supply chain to tune to changing times?
3. Discuss the role of technology in Afcon's supply chain success.

ZICO

Supply Chain Initiative—Localization

ZICO was established in 1951 and today is one of the leading manufacturers of diesel fuel injection equipment in the world. In India, ZICO is having manufacturing facilities at three places. ZICO undertook SCM initiatives to achieve the overall profitability of the business by reducing the cost. Through the SC initiatives, ZICO integrated the three individual business processes of procurement, manufacturing and distribution.

Procurement

Procurement is one of the major cost drivers in ZICO's supply chain. The procurement cost is influenced by the following factors:

- The way procurement decisions are made
- Procedures adopted in the procurement process
- Relationship with the suppliers
- The firm's credibility
- Market intelligence

To control the procurement-related cost, ZICO went for e-Sourcing which streamlined workflows, enhanced flexibility and created transparency in the buyer-seller relationship. The e-procurement process improved the accuracy and availability of information on the supply and demand side, facilitating collaboration as well as control and compliance. E-sourcing freed up purchasing personnel to focus on more strategic concerns such as supply base development and relationship management, linking suppliers into up-front innovation processes and value chain restructuring. E-sourcing also helped in reducing the time to the market for the finished goods. ZICO took steps like supplier base optimization which resulted in various improvements in operational quality. The advantages were seen in reduced scheduling work, follow-ups and complexity. To reduce the cost of imported components, ZICO went in for 'localization' which resulted in huge cost benefits and reduced inventory levels of imported items. It helped in meeting the delivery requirements and reducing inventory cost substantially.

Processing

For developing a lean supply chain the emphasis was not on curtailing the processing/manufacturing cost through economies of scale, but on reducing the huge inventory-carrying cost resulting from mass production ahead of demand. In the past, the emphasis was on building mega-capacity factories to produce standard products in millions to reduce manufacturing cost and flood the market with low-priced products. The stress was on focused factories to manufacture standard products for the

global consumption. ZICO initiated indigenization/ancillarization whose agenda was to indigenize suppliers and outsource non-critical operations. The company was interested in expansion for which this was very helpful. The company reviewed its make/buy decisions for retaining its core competence. It was good to outsource the non-critical accessories since the in-house cost of manufacturing was very high and further investments were not allowed. Product cost reduced considerably and production was doubled without employing extra labour and investments. Due to the implementation of indigenization and ancillarization, the employment growth increased by almost 50 per cent each year and also the sales turnover increased resulting in high productivity. One very important factor was optimum allocation of space, which was done very efficiently after implementing ancillarization. ZICO also tried to save on the cost of materials. As 70 per cent of ZICO's cost came from the supply chain, the supply chain contributed significantly in reducing the cost of the materials. As per the suppliers suggestion ZICO used alternate materials and better processes. These reduced the material cost considerably. ZICO saw a very good reduction of 8 per cent in material cost after applying all innovative measures and means resulting in increasing the firm's profitability.

Distribution

Traditionally, the role of distribution in the business process is warehousing transportation. However, in the supply chain model the major task of distribution relates to the management of demand, that is, to make available the right product at the right place and at the right time and at the least cost. ZICO faced the challenge of bringing in world class suppliers quickly and also tracking and shrinking the life cycle of purchased parts. Quick product development was a must to survive in the fast changing world. ZICO took different steps to reduce the time to the market like supplier involvement, etc. The other benefits of supplier involvement include improved access to supplier technology, aligned technology road maps, and more innovative products. At ZICO, simultaneous engineering and rapid prototype development were being practiced to meet the 'time to the market'. The company managed to reduce lead time for development from 90 to just 38 days, which is a significant achievement leading to gaining of market share by 15 per cent. Earlier the share was 8 per cent, which increased to 23 per cent.

Role of Logistics

ZICO has focused on the areas like inventory management, material handling and storage systems, and packaging. ZICO has outsourced its outbound logistics and warehousing to reduce costs and overheads. Further, ZICO monitored its inbound and outbound transport costs and made efforts to keep it at the desired minimum level by proper vehicle planning and selecting appropriate transporters. ZICO has also introduced the VMI (vendor managed inventory) which again reduces the transportation and the handling cost. ZICO introduced milk run to reduce the freight from the suppliers to the warehouse. Inventory was controlled using 3PL. This helped in reducing the cost of logistics to just 15 per cent of the total cost. The methods included outsourcing of inbound logistics, milk run, better vehicle planning and optimization of logistics chain in India. Recyclable bins and containers have led to additional cost reductions and enhanced customer satisfaction.

Results

The key business goals were achieved and SCM improved top line and bottom line as well. Further, ZICO set new benchmarks to achieve further reduction in cost and improvement in supply chain

productivity. With the available infrastructure and all the endeavours of manpower in the organization, ZICO could excel and put up the name 'Made in India' in both domestic as well as international markets.

REVIEW QUESTIONS

1. Discuss the various supply chain initiatives of ZICO.
2. Explain how the manufacturing strategy evolved by ZICO helped in enhancing profitability.
3. Explain the role of logistics in ZICO's supply chain strategy.

LA VOGUE

Supply Chain Leadership Through System Approach

La Vogue is one of India's leading retail chains having 15 department stores countrywide. La Vogue has progressed from being a single brand shop in 1991 to becoming a fashion and lifestyle store for the family. La Vogue is a professionally managed and a system-driven organization with strong focuses on customers. The timely supply chain initiatives and committed work force have enabled it to achieve success and scale up to achieve aggressive growth targets. It has been successful in strengthening its brand position by offering their customers a top of the world, class shopping experience and superior merchandising. To support its pre-eminent position, La Vogue found it very critical to enhance the efficiency of its supply chain and to integrate both vendors and service providers to the operational and service requirements of the customers.

La Vogue revamped its systems and process and implemented the warehousing module of JDA in 1999, which was followed by initiatives like Auto Replenishment and Auto Purchase Order system. La Vogue launched its B2B connectivity with vendors to improve the communication. It displays a broad range of branded apparels, footwear, perfumes, cosmetics, jewellery, leather product accessories, home products, electronics, books, music and toys.

In the past, La Vogue did not have a good ERP system in place. This created problems for inventory control resulting in high inventory, shutting down of all operations for 48 hours for annual stocktaking and inefficient use of exertive time. To overcome this problem, La Vogue implemented the state of art ERP applications. It appointed KSA Technopark, a specialized management consulting firm and a leader in fashion and textile consulting in India, to offer solutions covering the entire value chain.

To improve business relations with partners, La Vogue took the following initiatives.

Business-to-business(B2B) software La Vogue provided their vendors' delivery authorization through suppliers' connect. The organization used electronic data interchange (EDI) to get details of the supplier's stock for raising a purchase order and getting an invoice detail of vendors; this application also makes vendors' payments electronically and on a timely basis.

Sources tagging To reduce the turnaround time of the stocks reaching the stores, the price tags were attached to the merchandise at the vendor's premises rather than tags being put at the La Vogue distribution centre.

Receipt and payment confirmation On receiving goods at the distribution centre, the receiver's confirmation report is prepared and sent to the finance and account department at the La Vogue corporate office. La Vogue releases payments of the vendors against the receiver's confirmation report. This led to 100 per cent reconciliation of receipts, ensuring timely payment to vendors.

Auto style set-up concept La Vogue along with Arvind brands developed a standard template which covers all the new style codes. Vendors send the details in the form of text files, which automatically

uploaded on ERP at La Vogue. The buying and merchandising team (at La Vogue) is required to upload the quantities and give it to the production team.

Due to these supply chain initiatives, La Vogue got many benefits, like reduction in operation cost resulting in profitability that substantially went up, improvement in the customer satisfaction index (CSI) from 62 to 65 and faster transactions (due to B2B, auto purchase order and source tagging). With an immense amount of expertise and credibility gained over years, La Vogue has become the highest benchmark for the Indian retail industry. It maintains strong a distribution and logistics network to operate 300,000 SKUs around approximately 400 vendors in the country. This strong network connected online with vendors enables La Vogue to deliver merchandise to the stores within 24 hours of notification.

The initiatives taken by La Vogue for systems comprise of the following points.

JDA retail merchandise management system (MMS) La Vogue implemented an end-to-end solution for managing merchandise throughout the chain, right from purchase orders to sales, to replenishment, to market down and inventory management. MMS is an integrated online system connected with the back end and front end of the store and the distribution centre. La Vogue was the first retailer in Asia to implement JDA.

Warehouse management system (WMS) This is one of the key modules of JDA MMS that assisted La Vogue revamp its system and processes and implement the warehousing module of JDA. The entire process is controlled electronically and has reduced the turnover time to 48 hours instead of 96 hours. This enables the distribution centre to store the exact number of units and merchandise needed; moreover, the system generates a put-away document, which allows a specific area to each product category. When it transfers to the store, it generates a pick list indicating where the merchandising is located; this helps in identifying the exact location of the merchandising at the SKU level in the rack and bin system

Auto purchase order It was initiated along with the warehousing modules of JDA MMS. The buying and merchandising team sets a pre-determined level of minimum stock for core style. APO is generated and sent electronically to the specific vendor. This order authorized the vendor to get the merchandise ready for delivery. This resulted in reduction of lead time, increase in merchandise time availability and improvement on hit rate from partners.

Auto replenishment The auto replenishment module in the JDA triggers off an automatic stock transfer command at the SKU's level for the merchandise in question and replenishes the same from the distribution centre attached to the particular store, with the ensures of replenishment in 24 hours and 100 per cent availability.

Perpetual inventory control system The distribution centre scans one department every day. All departments are scanned in quarters to check for pilferage, identify variances every day and provide accurate stock status. This system is a major tool for reducing shrinkage, resulting in the La Vogue cost saving. For improving logistics, La Vogue took the following initiative like delivery against purchase order. To ensure the right quantity of merchandise intake in the distribution centre, goods are accepted against the purchase order and the quantity mentioned therein. This results in elimination of overstock at the distribution centre. Based on the sales forecast for 10 days, La Vogue issued a delivery authorization number for every purchase order given to each vendor. This helped them in avoiding excess inventory stocking. In addition to the above, La Vogue also implemented the concept of intake consolidation, which involves the appointment of a national transporter conducting a milk-run for private label brands. The transporter sends his truck to the

vendor, combines the merchandise location-wise and sends a consolidation load of shipments to the respective distribution centres

In conclusion, SCM initiatives undertaken at La Vogue have mainly resulted in a decline in shrinkage from 0.66 per cent in 2002 to 0.54 per cent in 2003 to 0.40 per cent in 2004. Customer satisfaction index (CSI) improved from 62 to 65 through these initiatives. Distribution and logistics expenses decreased from 1.06 per cent in 2001–2002 to 0.83 per cent in 2003–2004 (per cent of turnover).

REVIEW QUESTIONS

1. Discuss the various supply chain initiatives taken by La Vogue.
2. What are the constraints faced by La Vogue in implementing the various technology-based solutions?
3. Explain the 'perpetual inventory control system' implemented by La Vogue.

TRIKA EXPORTS

Seeking Supply Chain Solution to Garment Exports

TRIKA is one of the leading exporters of garment from the Indian to the US and the European markets. The present set-up allows a production of up to a million pieces but the capacity of the back-end supply chain is limited thus slowing the entire process flow. Another impediment is the fabric or the cloth for the garment. If the fabric is locally sourced then the time to procure the cloth would take anywhere between 15 and 45 days, depending on the source. If it is imported (e.g. Fleece from Taiwan), it would mean another 15–30 days. So the sourcing process takes about 30–75 days.

Both logistics and supply chain play a very important role in the garment export business. Logistics is critical because most international retailers work on tight delivery schedules. TRIKA has to ensure that the goods are shipped at the right time. This implies that they have to manage their supply chain to ensure that raw materials are delivered quickly to their factories. The garment supply chain is very fragmented and of a small scale in nature. To ensure the quality and timely availability of all the materials is a great task. Moreover, depending on the order, the exporters also need to import some fabric and accessories from the international market. Supply chain issues, therefore, form the main link in the garment industry.

For embellishments like buttons, threads, zippers and so on, procurement takes between 7 and 25 days, high by any standards for the value of the items but, again, limited back-end supply chain capacities create a hurdle. Due to the nature of the demand pattern, TRIKA does not stock excess inventories as the fashion cycle keeps changing and exporters purchase the items based on the order.

Once the raw material are sourced, production, which comprises cutting, sewing, finishing, packaging and shipping, is usually completed in around 15 days. A typical garment export order fulfilment cycle is shown in Figure 1.

TRIKA is highly labour oriented, though in areas such as cutting, embroidery and trimming, mechanization has taken place. Overall, however, the industry still depends on labour for undertaking some of the major functions like sewing.

Cost Break-up

Material cost accounts for about 60 per cent, trims 10 per cent and labour 15 per cent of the total expenses (Figure 2).

Evidently, it is the material supply that has to be managed. However, here, the issues of capacities and sourcing have to be considered. TRIKA sources the fabric from the textile units and also from a fabric manufacturing unit in Daman. Even here, the production of the fabric takes around 30–45 days, as TRIKA has to develop the garments based on specifications provided by the retailer. The captive facility is quite small and can meet only 35 per cent of its requirements. The rest of the fabric has to be outsourced.

Fabric manufacturing involves a large capital expenditure in looms, cutting, finishing, printing and bleaching facilities. TRIKA has invested in a weaving facility with jacquard and dobby looms that cost around Rs. 50–55 crores.

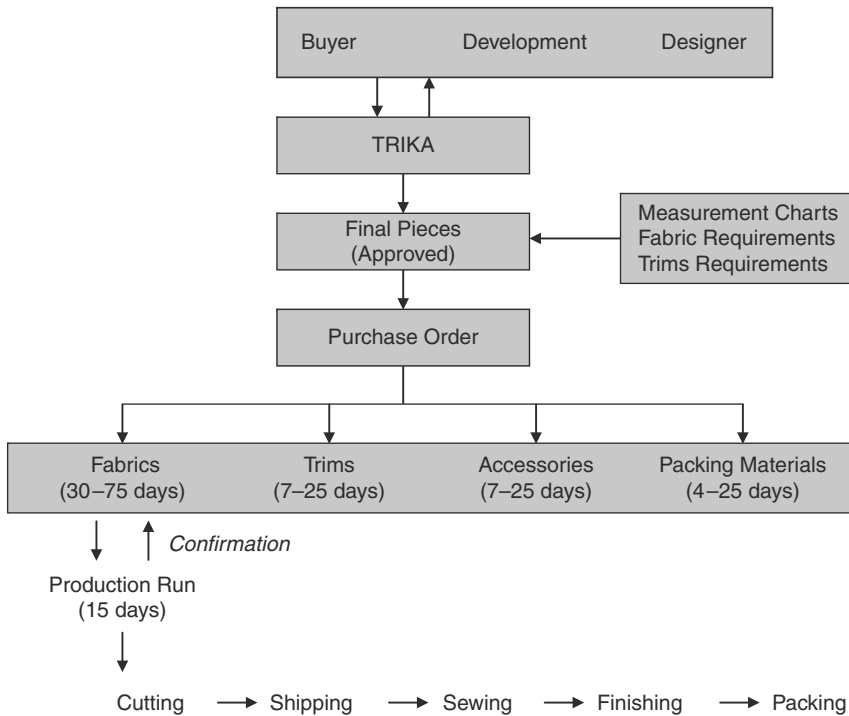


Figure 1 *Garment Export Order Fulfilment Cycle*



Figure 2 *Cost Break-up of Garments For Exports*

TRIKA made efforts to improve efficiency by bringing down the inventory. They planned their inventory management in 4 stages as follows:

- Sales forecast clarity by understanding customers needs
- Standardization of production mix
- Standardization of vendors for purchasing dyes and recipes
- SAP implementation

The other major hurdle in supply chain is the logistical infrastructure, which is evident from Tables 1, 2 and 3.

Table 1 Clearance Time for Cargo

Ports	No. of Days	Airports	No. of Days
Mumbai	2	Mumbai	1
Chennai	2	Delhi	1
Kolkata	2		
Tuticorin	3		
Cochin	3		

Source: ETIG survey

Table 2 Transport Lead Time for Finished Goods

By Road	To Ports			To Airports		
Apparel manufacturing centres	Mumbai	Kolkata	Cochin	Mumbai	Delhi	Chennai
Delhi/Ludhiana	3–4	–	–	–	1	–
Mumbai	–	–	–	–	–	–
Tirupur	–	–	2	–	–	2
Bangalore	–	–	2	3	–	2
Calcutta	–	–	–	–	–	–

Source: ETIG survey

Table 3 Transport Time From TRIKA to Ports by Rail

By Rail	To Ports				
Apparel Manufacturing centres	Mumbai	Chennai	Kolkata	Tuticorin	Cochin
Delhi/Ludhiana	2	–	–	–	–
Mumbai	–	–	–	–	–
Tirupur	–	3	–	–	2
Bangalore	–	2	–	–	–
Calcutta	–	–	–	–	–

Source: ETIG survey

TRIKA could get productivity improvement at each stage cumulating to 80 per cent. Even though TRIKA is internally strong with its system, there are more external factors which are bothering them much. An interesting feature of the garments industry is that they have a very short product life cycle but long lead times. Volatile fashion trends, low supplier reliability and longer supplier lead times characterize this business. This is because in this industry, supply chain is yet not fully integrated and forces the manufacturers to bear high inventory carrying and marking down costs.

With the global economic recession, supply chain management assumes utmost importance for TRIKA Exports. Value addition is another significant area and TRIKA Exports have to make efforts on this front. India has a huge base of designers several of whom have made it big in the international markets. TRIKA can introduce innovative designs to capture global markets. Innovations are needed in both fabrics and designs.

REVIEW QUESTIONS

1. Discuss various issues in garment supply chain.
2. What are the major supply chain problems faced by TRIKA?
3. Suggest some strategy to overcome external factors faced by TRIKA.

GLOSSARY

Activity-based costing Costing method that breaks down overhead costs into specific activities (cost drivers) in order to more accurately distribute the costs in product costing.

ABC inventory classification A method used to categorize inventory into groups based on certain activity characteristics. It is the process of classification of products as per the level of importance in terms of their relative criteria such as purchase or sales volume. ABC classification are used to develop inventory planning policies, set count frequencies for cycle counting, slot inventory for optimized order picking, and other inventory management activities. It can be described as a technique that is used in a business sense for denoting a categorization of large volume of data into groups. These groups thereafter can be marked as A, B and C. This means that activities that are considered high on priority are classified as A, those with a lesser priority are grouped under B, and the group of activities that are last on the list of priority are classified C.

Advanced planning and scheduling (APS) It is a real-time tool combining material and capacity planning at the same time, as opposed to separate batch material and capacity runs. APS tools use finite scheduling and material availability data to schedule orders and demand. They can typically be used to evaluate the effect of adding single new orders into an existing schedule, and are often used as a short-term tool in conjunction with longer-term family level planning.

Aggregate planning Sales, revenue, inventory and production planning done at total organization, facility or family levels. Typically done as part of the long-term production planning and budgeting process to identify overall production, facility and personnel requirements.

Airway bill (AWB) This refers to a bill of lading that is related to air transportation which is receipt for the shipper or consigner. It indicates that the carrier company has agreed to the listed goods and under an obligation to carry the products that are delivered in the form of a consignment to the destination airport in accordance with the conditions specified to the consignee and consignor. AWB is non-negotiable document.

Assemble to order (ATO) A manufacturing environment where the final product is assembled based on the receipt of a customer order (instead of to stock). The assembly is normally performed using standard components, modules and sub-assemblies that are already stocked based on forecasts developed from past usage history. An ATO environment allows each customer order to specify a custom combination of previously defined standard options.

Assortment In a retail environment, the selection of merchandise stocked or offered at a retail outlet or distribution centre. The target customer base and physical product characteristics determine the depth and breadth of an assortment and the length of time it is carried.

Automated storage and retrieval systems A system of rows of rack, each row having a dedicated retrieval unit that moves vertically and horizontally along the rack picking and putting away loads using programmed system. These are ASRS, AS/RS, unit-load ASRS and mini-load ASRS.

Backlog Booked (accepted) but unshipped customer orders or interplant orders. The backlog represents all open orders, not necessarily a past due status.

Balanced scorecard A business performance measurement and management system developed by Robert S. Kaplan and David P. Norton that analyses organizational success by reviewing the combination of financial, customer, internal business process and employee learning and growth perspectives. A balanced system includes both leading and lagging measures, and aligns individual and department goals with overall corporate strategic objectives.

Bar code An automated identification technology that encodes information in an array of parallel, rectangular bars and spaces that varies in width.

Benchmarking The practice of establishing goals and targets for process performance levels and identifying required improvement areas based on the published or known performance of direct competitors or a relevant industry.

Bill of lading (BOL) A document created for a given shipment that indicates the contents and destination, and forms a contractual basis for claims or resolution with the carrier if required.

Bill of material (BOM) It is a document which lists materials (components or ingredients) required to produce an item. Multilevel BOMs also show sub-assemblies and their components. Other information such as scrap factors may also be included in the BOM for use in materials planning and costing.

Bonded warehouse The warehouse meant for goods under customs clearance. Imported goods before clearance for distribution to local depots or customers is kept in bonded warehouses and released after import duty is paid and customs formalities are completed. Similarly, goods meant for exports are kept in bonded warehouse, before their clearance by customs authority for loading onto the ship or aircraft. In short, it is a facility or a dedicated portion of a facility where imported goods are stored prior to customs duties and taxes being paid. Bonded warehouses are licensed by the government.

Break-even analysis An examination of changes in fixed and variable costs based on varying revenue and production levels that identify a break-even point where revenues are equal to costs. It highlights the profit results from alternative levels of operation (syn: cost-profit-volume analysis).

Brokers Brokers are firms who bring buyers and sellers together. They specialize in products that are at the end of their sales life (scrap broker). They are often willing to deal with any product, in any condition, given a low enough price.

Business process mapping Techniques that specify the steps, control points and resources involved in current state business processes, identify desired changes and risk/reward tradeoffs, and create the methods to implement cross-functional systems that support the desired new processes.

Capacity requirements planning Process for determining amount of machine and labour resources required to meet production.

Carrying cost A measure, usually expressed as a percentage, of the costs incurred to hold inventory as a result of the cost of the item itself, insurance, obsolescence, theft, damage, use of storage facilities and other causes.

Cash-to-cash cycle The average elapsed time between payment to a vendor for materials or services and the receipt of payment from a customer for shipments.

Consolidation The combination of shipments from multiple points at an intermediate facility, reducing the number of individual shipments to end locations.

Continuous replenishment A method typically used between retailers and distributors or manufacturers that uses point-of-sale inventory and sales data to trigger automatic replenishments, with the vendor assuming responsibility for initiation and fulfilment.

Contract manufacturing A third party that performs one or more production operations for a manufacturer who will market the final item under their own name. They often charge on a per-piece or per-lot basis for the labour required for their services while using components or materials supplied and owned by the final item manufacturer.

Containerization It means putting the goods in boxes or trailers that are easy to transfer between two-transportation modes. They are used in multi-modal transportation system. Containerization refers to using standardized containers for the storage and transport of materials within a manufacturing facility as well as between vendors and manufacturers. The benefits of containerization include reduced product damage, reduced waste (by using reusable containers), less handling and greater levels of inventory accuracy by simplifying counting processes.

Contract warehouse A contract warehouse handles shipping, receiving, and storage of products on a contract basis. Contract warehouses will generally require a client to commit to a specific period of time for the services. Contract warehouses are for specific product storage. For example, in India the pharmaceutical companies store their products at various distribution centres in contract warehouses. These are temperature-controlled and FDA-approved facilities. Fees for contract warehouses may be transaction and storage-based, fixed, cost plus or any combination thereof.

Cross-docking Cross-docking is the action of unloading materials from an incoming trailer or rail car and immediately loading these materials in outbound trailers or rail cars, thus eliminating the need for warehousing (storage). Cross-docking is done in warehouse or at transportation hubs. Many 'cross-docking' operations require large staging areas where inbound materials are sorted, consolidated, and stored until the outbound shipment is complete and ready to ship. As per Wal-Mart, cross-docking should be done within 48 hours.

Cycle stock The portion of inventory incrementally consumed by demand and periodically replenished when a resupply is triggered by reaching a predetermined level. The portion of total inventory is over and above safety stock.

Cycle time The total time required to complete a transformation from one status to another. Total cycle time is composed of many elements, often broken into active (running or operating) time and idle (queue or wait) time.

Distribution centre A facility that stores inventory and ships finished goods orders to customers for a specific geographic area. Its stock is replenished on a periodic basis by manufacturing plants or other distribution centres, and it may process customer returns but does not have production capabilities.

Distribution network The description of distribution facilities and organizations serving as supply and replenishment points for given product lines and specifying the lead times and transportation methods between those points.

Distribution requirements planning (DRP) It is a process for determining inventory requirements in a multiple plant/warehouse environment done with the help of computer software. DRP may be used for both distribution and manufacturing. In manufacturing, DRP will work directly with MRP. DRP may also be defined as distribution resource planning which includes determining labour, equipment and warehouse space requirements.

Distributor A third party that purchases, stocks and resells products manufactured by another organization. While classified as a customer to the manufacturer, it is normally useful to track distributor shipments to the end consumer to evaluate demand patterns and changes.

Economic value added (EVA) A financial performance measurement developed by Stern Stewart & Company that gauges the creation of shareholder wealth, and evaluates tradeoffs in opportunity costs, by determining net operating profit after tax deductions.

Economic order quantity (EOQ) It is most cost optimum (cost-effective) quantity to order (purchased items) or produce (manufactured items). The formula basically finds the point at which the combination of order cost and carrying cost is the least. The difficult part of implementing the formula is getting accurate values for order cost and carrying cost.

Electronic data interchange (EDI) It is a system for business-to-business electronic communication. It is inter-company computer-to-computer exchange of business documents in standard formats.

Eighty/twenty (80/20) rule Rule based on Pareto's law stating that 80 per cent of an end result (quality problems, inventory valuation, distribution of wealth) comes from a source of 20 per cent (items, number of people, etc.).

Enterprise resource planning (ERP) It is a software system designed to manage most or all aspects of a manufacturing or distribution enterprise. ERP systems are usually broken down into modules such as financials, sales, purchasing, inventory management, manufacturing, MRP, DRP, etc. The modules are designed to work seamlessly with the rest of the system and should provide a consistent user interface between them. ERP systems usually have extensive set-up options that allow you to customize the functionality to the specific business needs of the enterprise.

Expert system/artificial intelligence The system based on technologies aimed at making computers imitate human beings.

First-in-first-out (FIFO) In warehousing, it is a method of rotating inventory to use the oldest product first.

Fill rate Sales order processing measurement that quantifies the ability to fill orders. There are various ways of measuring fill rate. Line fill compares the number of line items shipped complete to the total number of lines ordered (95 line items shipped complete, out of 100 orders received would result in a 95 per cent line fill rate). Order fill compares the number of orders shipped complete to the total number of orders shipped.

Flexible manufacturing system (FMS) A group of computer-controlled independent workstations or machines linked by material handling systems that are able to accommodate wide variations in types and quantities of products.

Free on board (FOB) Trade contract terms that specify when title passes to the buyer, typically either when the shipment is initiated or when it is received.

Full truckload (FTL) Shipments rated on the use of a truck's entire capacity based on weight or volume.

Free trade zone (FTZ) It is similar to a bonded warehouse in that it has a special status that allows products to be imported into it without taxes or duties being paid. However, FTZ actually has fewer restrictions placed upon it than a standard bonded warehouse and activities such as manufacturing can occur within an FTZ.

Fulfilment It is an activity of processing customer shipments. Though most manufacturing and warehouse operations will process customer shipments, this term usually refers to the operations that ship many small orders (usually parcels) to end users as opposed to operations that process larger shipments to other manufacturers, wholesalers or resellers.

Green dot A marking on the package of the product sold in the Germany, showing that product waste is recyclable through Duales System Deutschland.

Hedge inventory Inventory created to protect against a possible future event or disruption in supply, such as a strike, major vendor shutdown, prospective trade or government programme change, or similar situation.

Inbound logistics The systems and tracking methods used for incoming materials in the process of being shipped from customers or vendors to a manufacturing facility or distribution centre.

INCOTERMS Trade contract terms established by the International Chamber of Commerce, for example, FOB (free on board) and DDP (delivered duty paid) etc. There are thirteen INCOTERMS.

Incremental cost The cost added to an existing product or project status by processing the next operation or performing an additional activity.

Integrated logistics management The logistics concept that emphasis team work, both inside the company and along marketing channel of organization to maximize the performance of the entire distribution system.

Integrator The logistics service provider offering complete range of logistic services and solution to customer's problem.

Intermodal transportation It describes the use of multiple modes of transportation for a shipment. Ocean containers that are picked up by a truck, delivered to port, transported by ship, and then picked up by another truck are common examples of intermodal transportation. In the trucking industry, intermodal usually refers to the combination of trucking, rail and sea transportation.

Inventory Any quantifiable item that one can handle, buy, sell, store, consume, produce or track can be considered inventory. This covers everything from office and maintenance supplies, to raw material used for manufacturing, to semi-finished and finished goods, to consumables to power equipment used in the business.

Inventory turn It refers to a number of times inventory is consumed or sold during a 1 year period. It is calculated by dividing the average inventory level (or current inventory level) into the annual inventory usage (annual cost of goods sold). Inventory turn over ratio indicates the rotation of the given value of inventory with respect to the value of sales effected in a set time frame. Normally, the time frame is 1 year. The higher turnover ratio indicates the faster cash rotation in the business cycle and higher asset utilization of the assets. This ratio has both financial and non-financial perspective.

Inventory velocity The speed with which inventory moves through a defined cycle (i.e. from receiving to shipping).

Just-in-time (JIT) Term usually thought of as describing inventory arriving or being produced just in time for the shipment or next process. JIT is a process for optimizing manufacturing processes by eliminating all process waste including wasted steps, wasted material, excess inventory, etc.

Kaizen The Japanese process of continuous improvement using problem-solving and analysis techniques that may include the use of fishbone diagrams, control charts, affinity diagrams and other tools.

Kanban It is used as part of a just-in-time production operation, where components and sub-assemblies are produced based upon notification of demand from a subsequent operation. Historically, *Kanban* has been a physical notification such as a card (*kanban* cards) or even an empty hopper sent up the assembly line to the previous operation. *Kanban* is actually a simplistic means of both signalling the need for inventory as well as controlling the inventory levels.

Key performance indicator (KPI) Indicators used to provide measurements of the defined priority and key success factors of a project or system.

Last-in-first-out (LIFO) A usage rule and accounting and other systems that assumes the last item brought into inventory (or into a queue) will be the first one used. During periods of rising prices, it has the effect of increasing charges to the current period instead of subsequent periods.

Lead time Amount of time required for an item to be available for use from the time it is ordered. Lead time should include purchase order processing time, vendor processing time, in transit time, receiving, inspection and packing time.

Lean manufacturing An alternate term used to describe the philosophies and techniques associated with just-in-time (JIT) manufacturing. This means the total inventory across supply chain needs to be near to zero level.

Lean supply chain When movement of the inventory across the supply chain is so fast that firm holds very little or near to zero inventory on hand.

Letter of credit A letter generated by a bank to a seller that authorizes a draft payment for a specific amount based on meeting a predefined set of terms (an exact description of the goods, services and quantities).

Life cycle cost The analysis of cost issues over the entire life cycle of a given product, rather than a specific fiscal period. It includes factors based on product life cycle stage (introduction, growth, maturity and decline) and all research and development, design, marketing, production and logistics costs. A cross-functional approach used in investment and facility decisions.

Load unitization The process of grouping the large number of products in the convenient pack for ease in storage, handling and movement is called as unitization.

Logistics The process of planning, implementing, and controlling efficiently cost-effective flow of raw materials, in-process inventory, finished goods, and related information from the point of origin to the point of consumption for the purpose of conforming to customer requirements.

Logistics mix It is a set of functions encompassed by the logistics. These are order processing, warehousing, transportation, inventory control, packaging and information flow, etc.

Logistics network The linked set of resources used in moving and storing material at the required location and time, which can be classified as external (vendors, customers, distribution centres and transportation providers) and internal (production, material movement and storage).

Maintenance, repair, operating (MRO) items Items such as maintenance supplies, equipment repair parts, etc. that support the operating process and are not issued to production orders or included in product standard cost.

Make-to-order (MTO) A manufacturing method in which commonly-used raw materials and components may be stocked based on previous demand history, but further processing into higher-level items is not done until receipt of a customer order. Variability in customer demand will not allow stocking upper-level assemblies and major subsystems or modules prior to receipt of an order, as done in assemble-to-order (ATO) environments.

Make-to-stock (MTS) A manufacturing method in which finished goods are produced and stocked prior to receipt of a customer order. It uses a forecast based on past demand history to initiate production of end items when inventory has fallen below desired levels, instead of waiting until a final quantity and configuration is described on a customer order.

Manufacturing lead time The total length of time used to process raw materials and components through all upper levels in the bill of material into an end item. It specifies the total of all individual elements of lead time such as order preparation, queue, setup, run, inspection, etc. used and indicates a projected availability date for an end item if all lowest level raw material is on-hand.

Material The raw materials and purchased components used in the production of other items. The classification of any lower level item, whether purchased or manufactured. A sub-assembly that has material, labour and overhead content may be considered strictly as material in the context of being used in an upper level item.

Material requirements planning (MRP) The process that uses the MPS as demand, explodes the bill of material for specified MPS items, nets those requirements against on-hand inventory and existing open orders, and recommends placing new production and purchase orders and changes to existing orders based on parameters that consider safety stock levels, lot sizes and lead times. It is based on the idea of dependent demand (demand for an item is due to its usage in other items) and starts at the highest level in the bill of material, stopping at the first level where the quantity of inventory plus incoming receipts is adequate to meet demand and it is not necessary to send a requirement to the next lower level.

Master production schedule (MPS) Production schedule specifying specific items, quantities, and dates at which production is expected to take place. The primary purpose of an MPS is to manage capacity when at some time periods demand is expected to exceed capacity.

Milk run A milk run consists of a pickup and/or delivery route where several stops are made. Usually it refers to a regularly run route, but it may also refer to a one-time run where several stops are made. Some consider a milk run to mean a route where outbound shipments are delivered and inbound materials picked up in the same run.

Monte Carlo simulation A method that estimates possible outcomes from a set of random variables by simulating a process a large number of times and observing the outcomes.

Order cycle It is also called replenishment cycle. Order cycle refers to the time between orders of a specific item. It is calculated by dividing the order quantity by the annual demand and multiplying by the number of days in the year.

Outbound logistics The processes and network used to pick, ship, track and store if necessary items ordered by customers, distribution centres and other supply chain partners.

Outsourcing The ongoing use of an external third party instead of an internal resource which may include production of a single operation, a product or an entire line, shipping and order fulfilment, product design, network infrastructure support and many other functions. Outsourced functions are normally outside an organization's core competencies and are done to reduce cost, reduce lead time, improve quality or achieve some other stated goal.

Performance cycle A time period required for executing the order from the date it is received and confirmed by the supplier or manufacturer.

Pipeline inventory Inventory in transit after it is dispatched from the factory and before it reaches the destination for changing ownership.

Point of sale (POS) It is the point where ownership of the product transfers to the customer.

Postponement It is a manufacturing/distribution strategy where specific operations associated with a product are delayed until just prior to shipping. Storing product in a generic state and then applying custom labels or packaging before shipping is an example of postponement.

Product life cycle It is a marketing concept, which is depicted graphically showing the product sales history across the four stages of the cycle such as introduction, growth, maturity and decline.

Proforma invoice It is a draft invoice created to show a buyer what the details of the actual invoice will look like. Proforma invoices are commonly used with international transactions to provide the buyer with the information how much they will need to pay for the goods.

Public warehouse A business that provides short- or long-term storage to a variety of products, usually on a month-to-month basis. A public warehouse will generally use their own equipment and staff, however, agreements may be made where the client either buys or subsidizes equipment. Public warehouse fees are usually a combination of storage fees (per pallet or actual sq. footage) and transaction fees (inbound and outbound). Public warehouses are most often used to supplement space requirements of a private warehouse.

Pull system A system where the production or movement of inventory items is initiated as required by the using department or location, or to replace items removed from an authorization queue. Production and inventory moves are not done based on a predetermined schedule, but only when consumed or signalled by the point of usage.

Purchase-to-order (PTO) The method used to respond to demand in which an item is purchased only when required by a customer order that specifies that item, and not for processing or assembling into a different item number.

Purchase-to-stock (PTS) The demand response method where an end item or service part required as is for an expected customer order is purchased and stocked against a forecast. The forecast represents independent customer or interplant demand, and not usage in an upper level item.

Push system A system where the production and movement of inventory items is determined by a pre-existing schedule that authorizes a material issue or transfer, or the start of a production operation. The using department or location receives material when determined by the feeder location and not based on their own generation of a replenishment signal.

Quick response Systems that attempt to provide quick turnaround and reduced lead time based on the location and level of inventory, and linked communications systems that quickly translate and evaluate changes in customer requirements based on usage and POS data.

Radio frequency identification device (RFID) It refers to devices attached to an object that transmit data to an RFID receiver. These devices can be large pieces of hardware the size of a small book, like those attached to ocean containers, or very small devices inserted into a label on a package. RFID has advantages over barcodes, such as the ability to hold more data, the ability to change the stored data as processing occurs, does not require line-of-site to transfer data and is very effective in harsh environments where bar code labels will not work.

Reorder point The inventory level set to trigger reorder of a specific item. Reorder point is generally calculated as the expected usage (demand) during the lead time plus safety stock. Fixed reorder point implies that the reorder point is a static number plugged into the system. Dynamic reorder point implies that there is some system logic calculating the order point.

Return on investment (ROI) It is an important financial ratio of profits to the investment in the assets. It indicates whether the investment made in logistical assets such as warehouse, equipment, storage system, and transport vehicles is paying you the dividends by way of the profit enhancement through cost reductions.

Reverse logistics Reverse Logistics covers activities related to returned product, returned pallets and containers, returned materials for disposal or recycling.

Routing The detailed description of the requirements to produce a given item, which includes the operations performed, the order in which they are done, the labour or machine resources required at each operation, and the rate those resources process the item at each operation. It may also include tooling and process or other specifications, and serves as the basis for manufacturing lead time calculations, detailed capacity planning and the cost standards associated with production. Released production orders normally include the standard routing and bill of material, unless modified for a specific order.

Safety stock Quantity of inventory used in inventory management systems to allow for deviations in demand or supply. Safety stock calculations will take into account historic deviations and use a required service level multiplier to determine the optimal safety stock level. See article on safety stock.

SCOR model The supply chain operations reference model was developed by the Supply-Chain Council to measure total supply chain performance. It includes delivery and order fulfilment performance, supply chain response time, production flexibility, warranty and returns processing costs, cash-to-cash cycle time, inventory and asset turns, and other factors in evaluating the overall effective performance of a supply chain.

Seasonality Large variations in product demand that reoccur during the same approximate time-frame on a yearly basis and are not due to a trend or promotion. Products that exhibit a high degree of seasonality normally require an inventory build based on forecast prior to the high demand period, or the flexibility to greatly vary production and supply.

Simplification The philosophy and methodologies that seek to reduce product and process variation, quality problems and cost by identifying and eliminating non-value added tasks, and standardizing component and resource usage.

Simulation The use of models and logic tools to test the outcomes of a proposed group of inputs and processes, prior to or in place of their implementation in a live system.

Six sigma A quality measure and improvement programme developed by Motorola that focuses on the control of a process to the point of $\pm$ six sigma (standard deviations) from a centreline, or 3.4 defects per million items. It includes identifying factors critical to quality as determined by the customer, reducing process variation and improving capabilities, increasing stability and designing systems to support the six sigma goal.

Skew A data distribution that is not symmetric, or that shows distortion in a positive or negative direction.

Stock keeping unit (SKU) It refers to a specific item in a specific unit of measure. For example, if you distributed shirts in two colours, you would maintain the inventory as two SKUs even.

Standardization The methods used to reduce or eliminate custom, one-time and seldom-used components and processes that introduce variability and potential added costs and quality problems. Standardization techniques include rationalizing product line offerings and performing cost studies to determine the true costs associated with designing, documenting, performing, etc. a custom or variable process.

Standard operating procedures (SOP) Instructions and methods used for a specific process or situation. They document the normal or accepted methodology and help form the basis for conformance evaluation.

Supply chain It is a cross-functional approach to controlling the physical flow of products and services from the suppliers to the end users by coordination of the activities of the suppliers, manufacturers and the end customers. It aligns the capabilities of the suppliers, manufacturers, channel partners, service providers and the customers to develop a sustainable competitive advantage.

Supply chain mapping The breakup of supply chain operation into smaller elements to know the value-added and non-value-added activities, the time frame for performing that activity and its cost implications.

Supply chain optimization The coordination of linked resources across all or part of a supply chain in eliminating or reducing manufacturing and logistics bottlenecks and creating optimized schedules based on shared inventory and order information.

Supply chain planning The set of supply chain activities that focus on evaluating demand for material and capacity and formulate plans and schedules based on meeting that demand and the company goals. System functions often involved in the planning cycle include MPS, MRP, ERP, DRP and advanced planning and scheduling (APS).

Synchronize To bring to the same level, rate or status, as in the synchronization of linked production resources to the same timing and volume, or in distributing the same set of data to remote databases.

Third-party logistics (3PL) An independent logistics service provider who performs any or all the functions of logistics to get his clients products to the market. It describes businesses that provide one or many of a variety of logistics-related services. Types of services would include public warehousing, contract warehousing, transportation management, distribution management, freight consolidation. A 3PL service provider may take over all receiving, storage, value added, shipping, and transportation

responsibilities for a client and conduct them in the 3PL's warehouse using the 3PL's equipment and employees, or may manage one or all of these functions in the client's facility using the client's equipment, or any combination of the above.

Tier 1, 2 supplier A tier 1 supplier is the immediate or primary set of vendors directly used by a company, and tier 2 is a vendor to tier 1. In some industries, the final customers or dominant chain partners are consolidating (reducing) their number of tier 1 suppliers, and requiring proof of the communications and fulfillment capabilities between tier 1 and tier 2.

Total quality management (TQM) The comprehensive set of principles that focus company-wide attention on the aspects of design, production and logistics that lead to quality conformance and customer acceptance. It maintains a cross-functional view of the processes involved, and stresses that quality is not the responsibility of a separate quality control department.

Transfer pricing Prices used in the transfer of materials between units of the same company, often at standard cost or standard plus transportation costs. Used when transfers are not considered sales and purchases that generates receivables and payables.

Unit load It is a material handling term that describes any configuration of materials that allow it to be moved by material handling equipment as a single unit. While smaller manually handled configurations could be considered unit loads. The term generally defines larger configurations that would be moved by a lift truck such as palletized loads, crates, bales, etc.

Value-added services These are the unique or specific activities the supplier is performing to deliver superior value to the customer.

Value-added tax (VAT) A general consumption tax for the production and distribution of goods and services, added as a percentage charge based on price that reflects the incremental value added by a given activity.

Value chain The linked set of activities within a supply chain that actively add value to the end product, as opposed to support or reporting activities.

Value stream mapping A technique used in lean manufacturing that maps the flow of material and data, and associated time requirements from initial supplier to end customer for a given business process. Used to define improvement areas and sources of waste.

Variability The characteristic of a product or process in which parameters fluctuate to a significant degree but do not typically trend in a specific direction. Reduction of variability is a priority in systems that attempt to ensure consistent quality and reduce lead times.

Vendor-managed inventory (VMI) This is the process of a supplier managing the inventory levels and purchases of the materials he supplies. Vendor-managed inventory reduces internal costs associated with planning and procuring materials and enables the vendor to better manage his inventory through higher visibility in the supply chain. Vendor-managed inventory may be owned by the vendor (consignment inventory) or the customer. The ownership of the inventory is with the vendor. The customer pays immediately after the inventory is used.

Visibility The degree of insight into future requirements, or the key factors that will impact system performance.

Warehouse management system (WMS) Software package designed for managing the movement and storage of materials throughout the warehouse. WMS covers allocation, putaway, replenishment and picking. WMSs are often set up to integrate with data-collection systems.

Waste Any activity or process that does not add value to the goods or services required by the customer. Examples of waste include move time, counting inventory, inspection, the production of defective material, rework, etc. Waste is considered to cause increased cost, lead time and quality problems while not adding value, and may be created by vendors, personnel, equipment, incorrect process parameters and many other factors.

Wholesaler A stand-alone type of logistics service provider who offers only one type of service such as warehousing, transportation, inventory management or packaging.

Work-in-process (WIP) Inventory that is currently being processed in an operation, or inventory that has been processed through one operation and are awaiting another operation. WIP is actually an inventory account that represents the value of materials, labour and overhead that has been issued to manufacturing but has not yet produced a stockable item.

Zero defects A quality philosophy based on the idea that a level of perfect quality, as in zero defects, is achievable and should be a company-wide goal. It emphasizes the examination of all factors that lead to quality problems versus a system that builds in an average or acceptable quality level.

Zero inventory A term initially used to represent the optimum stock level in a just-in-time system and the idea that inventory is a liability instead of an asset.

SUPPLY CHAIN-RELATED INFORMATION SOURCES

A. Organizations

Aberdeen Group	http://www.aberdeen.com
Alberta e-Future Centre	http://www.e-future.ca
Australian Freight Councils Network	http://www.freightcouncils.com.au
Australian Logistics Council	http://www.ozlogistics.org
Chartered Institute of Logistics and Transportation	http://www.ciltuk.org.uk
Confederation of Indian Industries, Institute for Logistics Specialization	http://www.ciilogistics.com
Council of Logistics Management	http://www.clm1.org
Delta Nu Alpha (DNA) is an international organization of transportation and logistics.	http://www.deltanualpha.org
Discussion forum on logistics and supply chain	http://www.supplychaindaily.com
E-procurement	http://www.epiqtech.com/e-procurement.htm
Georgia Tech Supply Chain and Logistics Institute	http://www.scl.gatech.edu/resources
Global logistics and supply chain news, articles and links	http://www.supplychainbrain.com
Good source transportation and logistics information	http://www.trafficworld.com
Inbound Logistics Magazine	http://www.inboundlogistics.com
Industry Week	http://www.IndustryWeek.com
Information in the transportation sector	http://www.dot.gov
Institute for Supply Chain Management	http://www.ism.ws
Institute of Logistics Management	http://www.logistics-edu.com
International Supply-Chain Council	http://www.supply-chain.org
Just-in-Time Solutions	http://www.justintime.com
Latest news and special reports about the air cargo industry.	http://www.aircargoworld.com

Logistic Association of Australia	http://www.laa.asn.au
Logistics and Supply Chain Management Information	http://www.logistics.about.com
Logistics Articles	http://www.findarticles.com
Logistics Information	http://www.bestlogisticsguide.com/logistics-information.html
<i>Logistics Management Magazine</i>	http://www.logisticsmgmt.com
Logistics Network	http://www.logisticsnetwork.com
Logistics on sea freight	http://www.look4logistics.com
<i>Logistics Today</i>	http://www.logisticstoday.com
<i>Logistics World</i>	http://www.logisticsworld.com
Material handling online	http://www.mhmonline.com
Mckinsey Quarterly	http://www.mckinseyquarterly.com
SCM Institute	http://www.scm-institute.org
Stamford Global Supply Chain Management Forum	http://www.stanford.edu/group/scforum
Supply Chain Council	http://www.supply-chain.org
<i>Supply Chain Digest</i>	http://www.scdigest.com
Supply Chain Excellence through Technology	http://www.ascet.com/welcome.asp
<i>Supply Chain Management Review</i>	http://www.scmr.com
The Integrated Logistics Network (ILN) Australia	http://www.dotars.gov.au
Trade association for the material handling and logistics industry in USA	http://www.mhia.org
US Department of Transportation, National Transportation Library	http://www.dot.gov
US Department of Transportation	http://www.ops.fhwa.dot.gov

B. Management Journals–Logistics and Supply Chain

<i>ICFAI Journal of Supply Chain Management</i>	http://www.icfaipress.org/ijscm.asp
<i>International Journal of Business Insights and Transformation</i>	http://www.ijbit.org
<i>Indian Journal of Transport Management</i>	http://www.cirtindia.com
<i>International Journal of Integrated Supply Management</i>	http://www.inderscience.com
<i>International Journal of Logistics Management</i>	http://www.ingentaconnect.com
<i>International Journal of Logistics Research and Application</i>	http://www.tandf.co.uk/journal

<i>International Journal of Logistics Research and Applications</i>	http://www.tandf.co.uk
<i>International Journal of Logistics System and Management</i>	http://www.inderscience.com
<i>International Journal of Manufacturing Technology Management</i>	http://www.inderscience.com
<i>International Journal of Physical Distribution and Logistics Management</i>	http://www.ariel.emeraldinsight.com
<i>International Journal of Production Economics</i>	http://www.ingentaconnect.com
<i>International Journal of Production Planning and Control</i>	http://www.tandf.co.uk/journal
<i>Journal of Business Logistics</i>	http://www.cscmp.org
<i>Journal of Business Logistics</i>	http://www.seal.psu.edu/cscr/jbl_toc.html
<i>Journal of Operation and Production Management</i>	http://www.emeraldinsight.com
<i>Journal of Purchasing and Supply Management</i>	http://www.elsevier.com
<i>Journal of Supply Chain Management</i>	http://www.napm.com
<i>Journal of Transportation and Statistics</i>	http://www.bts.gov/program/jtc
<i>Logistics and Supply Chain Management</i>	http://www.inboundlogistics.com
<i>Supply Chain Forum: An International Journal</i>	http://www.supplychainforum.com
<i>Supply Chain Forum: An International Journal</i>	http://www.supplychainforum.com
<i>Supply Chain Management: An International Journal</i>	http://www.ingenta.com
<i>Supply Chain Management Review</i>	http://www.manufacturing.net/scm
<i>Supply Chain Pulse</i>	http://www.supplychainpluse.org

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